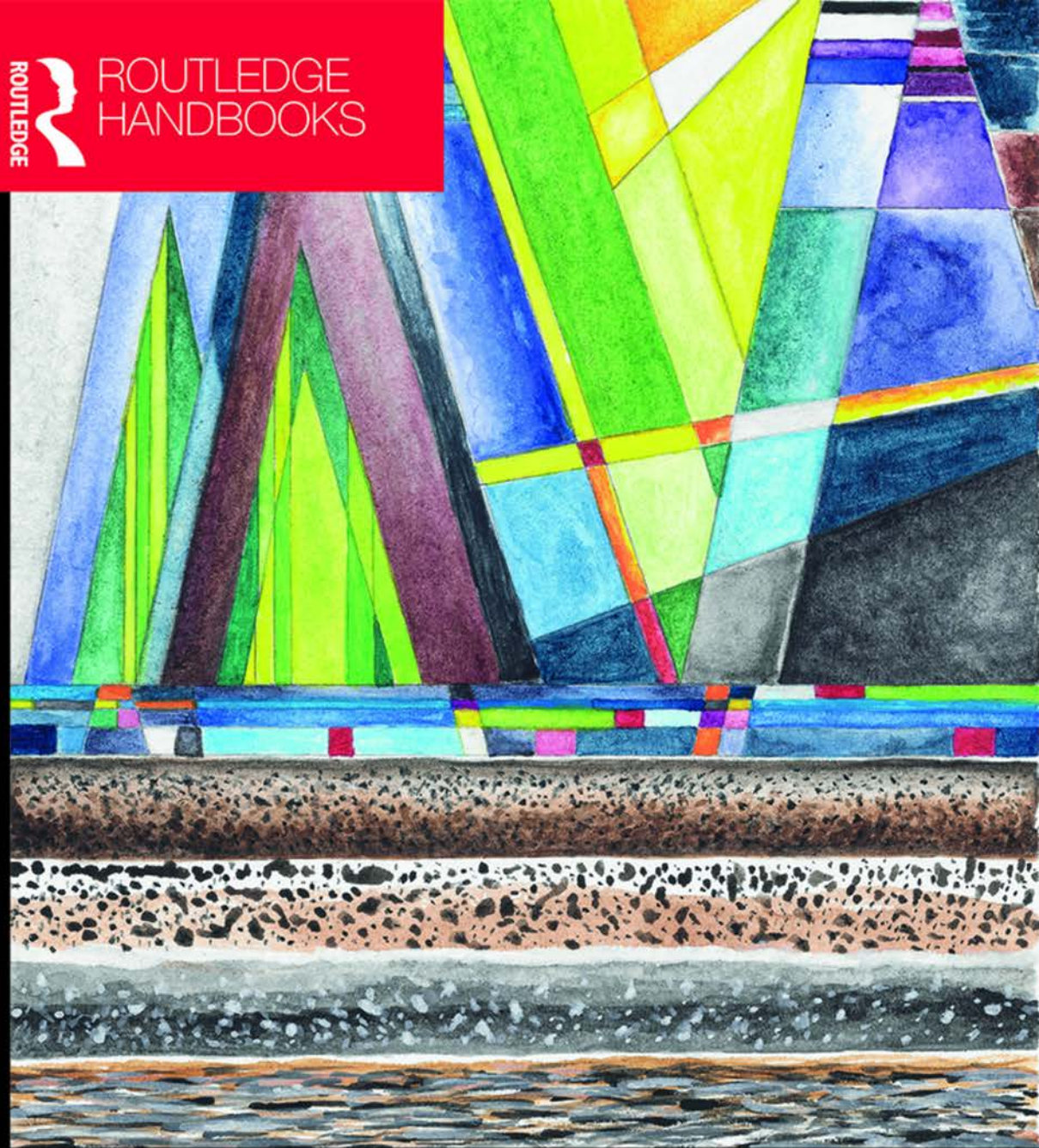




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THE ROUTLEDGE HANDBOOK OF METAPHYSICAL GROUNDING

Some of philosophy's biggest questions, both historically and today, are *in-virtue-of* questions: *In virtue of what is an action right or wrong? In virtue of what am I the same person my mother bore? In virtue of what is an artwork beautiful?* Philosophers attempt to answer many of these types of *in-virtue-of* questions, but philosophers are also increasingly focusing on what an *in-virtue-of* question is in the first place. Many assume, at least as a working hypothesis, that *in-virtue-of* questions involve a distinctively metaphysical kind of determinative explanation called "ground." This *Handbook* surveys the state of the art on ground as well as its connections and applications to other topics. The central issues of ground are discussed in 37 chapters, all written exclusively for this volume by a wide range of leading experts. The chapters are organized into the following sections:

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- II. Explanation and Determination
- III. Logic and Structure
- IV. Connections
- V. Applications

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Michael J. Raven is Associate Professor of philosophy at the University of Victoria and Affiliate Associate Professor of philosophy at the University of Washington, Co-Chief Editor of *Metaphysics*, and a founding member of the Canadian Metaphysics Collaborative. His research focuses on metaphysics, philosophy of language and mind, and epistemology.

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THE ROUTLEDGE HANDBOOK OF METAPHYSICAL GROUNDING

Edited by Michael J. Raven

First published 2020
by Routledge
52 Vanderbilt Avenue, New York, NY 10017

and by Routledge
2 Park Square, Milton Park, Abingdon, Oxon, OX14 4RN

Routledge is an imprint of the Taylor & Francis Group, an informa business

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Library of Congress Cataloging-in-Publication Data
A catalog record for this book has been requested

ISBN: 978-0-8153-6649-2 (hbk)
ISBN: 978-1-351-25884-5 (ebk)

Typeset in Bembo
by Apex CoVantage, LLC

Justin Zylstra (1987–2019), author of the chapter “Essence”, died shortly after this volume was delivered to the publisher. Justin was a promising and prolific young philosopher with a bright future. Many of the other authors had the good fortune to know him and his work. The imprint left by Justin’s insights and his thoughtful, gently tenacious commitment to philosophy will continue influencing us, even while his presence is sorely missed. This volume is dedicated with admiration and affection to Justin.



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ACKNOWLEDGMENTS

My interest in ground (at least by name) was sparked by Kit Fine's brilliantly original and enormously influential article "The Question of Realism" during my first academic year as a graduate student at New York University (2003–2004). Kit was on leave that year, and I was busy writing seminar papers on other topics when he returned. But around 2005, my focus on ground intensified. When I started writing my dissertation a year later, ground played a prominent role at the outset. At the time, few were aware of ground as such, and little had been published on it. Even in those lonely times, I often felt excitement that we in the know were on the cusp of something important (and sometimes—when the distinctive pessimism of a graduate student dug in—that we were on a precipice staring down at the abyss). I could not have imagined back then the torrent of interest in ground that would soon follow. And I certainly did not envision later editing an entire volume on the topic. I am grateful to Daniel Nolan, Routledge, and especially Andrew Beck for the opportunity to do so. My initial plans for the volume benefitted greatly from Andrew's advice as well as the advice from three anonymous referees. My thanks to them all.

Since first getting interested in ground, I have learned a great deal from many others also interested in ground, whether as friends or as foes. Although they are too numerous for me to list or recollect fully, I still wish to express my gratitude to them collectively. I was fortunate enough to recruit many of them individually to contribute to this volume. Each of the volume's contributors deserves my great thanks. Their essays are testaments to their considerable philosophical energy and insight. And the timely publication of this volume is a testament to their diligence. It was a pleasure and a privilege working with them all.

The volume not only benefitted from the individual labors of each contributor but also from two joint efforts. The first of these was the Hamburg Workshop on Metaphysical Ground. This workshop united most contributors and helped coordinate the preparation of their essays. Robert Schwartzkopff organized this workshop. Organizing is often an invisible and underappreciated task. My sincere thanks to Robert for doing such a terrific job. Funding was provided by Kit Fine's Anneliese Maier Research Prize from the Alexander von Humboldt Foundation, by a Connection Grant from the Social Sciences and Humanities Research Council of Canada, and by the University of Victoria and its Philosophy Department. I am grateful for their financial support.

Acknowledgments

The other joint effort also benefitted many of the volume's essays. Kit Fine taught a master class on this volume at the Università della Svizzera italiana (Lugano). Kit and his students discussed many of the essays and provided authors and me with detailed comments and advice. On behalf of these authors, I am grateful to Kit and his students for their feedback.

Writing the volume's introduction was a challenge. On the one hand, it was exciting to reflect on the state of the art and to frame the volume's essays. On the other hand, it was nerve-racking trying to accessibly and fairly characterize issues and controversies still in development. The introduction benefitted greatly from encouragement and feedback from Kit Fine, Jon Erling Litland, Colin Marshall, Conor Mayo-Wilson, and Ryan Tonkin.

Ryan Tonkin also served as an editorial assistant in the preparation of the manuscript. Many improvements stemmed directly from his diligence, sensible advice, and keen eye for detail. I would like to thank Ryan for his excellent work.

I would also like to thank Rose Choi for the beautiful cover art. Her creative mind allowed her to conceive of it merely from my vague rambles about levels, foundations, hierarchies, and structures. I am grateful for the hard work she put into its design and for the much harder work of being my partner.

In addition to the debts already owed to Kit Fine, there are still two more to add. One debt is personal. Kit has been a continual source of encouragement, inspiration, and support to me during the editing of this volume as well as throughout my academic career. Thanks, Kit. The other debt is philosophical. Without the background of Kit's pioneering work on ground, one struggles to imagine there being an art in a state worth surveying.

INTRODUCTION

Michael J. Raven

Some of philosophy's biggest questions are *in-virtue-of* questions. Examples include:

In virtue of what is a society just or unjust?
In virtue of what does a sentence mean what it does?
In virtue of what does someone possess knowledge?
In virtue of what is an action good or bad?
In virtue of what is a creature conscious?
In virtue of what is an artwork beautiful?
In virtue of what is something in motion?

Many more examples could easily be given. These *in-virtue-of* questions are also among philosophy's oldest. Attempts to answer them stretch back to antiquity. And *in-virtue-of* questions are also among philosophy's most enduring. Philosophers still try to answer them today. Their focus is usually on answering particular *in-virtue-of* questions rather than on the kind of question being asked. But in the new millennium, philosophers have increasingly focused on what an *in-virtue-of* question is in the first place. Philosophers increasingly assume, at least as a working hypothesis, that *in-virtue-of* questions involve a distinctively metaphysical kind of determinative explanation. This determinative explanation is sometimes expressed using phrases like 'because' or 'consists in'. But the phrase that is currently most fashionable is 'metaphysical ground', or just 'ground' for short.¹ Interest in ground swells; not just for its indirect role in formulating some of philosophy's most venerable questions but also for the interest in questions that are directly about it.

The purpose of this *Handbook* is to survey the state of the art on ground. There are already a few anthologies focusing on ground and related topics.² But their emphasis is on original research, and their target audience is experts. By contrast, this *Handbook* focuses on surveying the literature on ground for newcomers and experts alike. The target audience thus includes students at the advanced undergraduate and graduate levels as well as professional philosophers.

The *Handbook* adopts a dual perspective. It looks backward at where the literature has gone, and it looks forward to where it is going. Looking backward, the literature has rapidly expanded in the last decade or so. It shows no signs of slowing. Even experts have trouble keeping pace. There is thus a need for a backward-looking resource cataloging where the literature is and has been. Looking forward, the literature is in a state of rapid, ongoing development. Although some

consensus is emerging, many matters remain in flux. Terminology is unregimented. Formulations vary. Controversies abound. There is thus a need for a forward-looking resource assessing where future research should be directed. The *Handbook* is meant to provide both resources.

This introduction has two aims. The first aim is to provide a primer on ground. **Section 1** gives a crash course on how ground is commonly conceived. Although many aspects of this conception are controversial, it nevertheless prevails in the literature. Those already familiar with the literature should have no trouble recognizing this conception from it. And those unfamiliar with the literature should get enough from the primer to profitably read the essays to follow. The second aim is to provide an overview of the *Handbook*. **Section 2** gives a brief description of how the *Handbook's* 37 original essays are organized around five major themes. It also gives a brief history of the origins of the *Handbook* and how collaborative research among its contributors produced it.

1. Primer

Some philosophical topics are familiar from everyday life. These topics include *justice, meaning, knowledge, goodness, consciousness, beauty, and motion*, to name just a few. Topics like these can be easily seen in the history of philosophy (even when their histories are murky). But things are different for the notion of ground. Unlike the familiar topics above, ground is unlikely to be familiar outside of academic philosophy. And at least until recently, ground—at least under that name—was unfamiliar even within academic philosophy. What is this notion of *ground*?

We may distinguish two broad approaches to answering this question. An approach *from below* collects examples of ground's potential applications and extrapolates from them what ground is. An approach *from above* surveys ground's theoretical role together with its connections to other theoretical notions and triangulates from them what ground is. It needn't be supposed that these are the only approaches to characterizing ground. But they seem to be the most prevalent.

Approached from below, ground is extrapolated from its potential applications to various topics. The possible topics and the potential applications to each are too numerous to list exhaustively. So a smaller sample must be selected from the many options. This runs the risk of extrapolating from a poorly chosen sample. But the risk can be mitigated by selecting a broad and diverse sample. To illustrate, here are nine possible topics, each with a representative claim of ground:

Identity The identity of two sets is grounded in the identities of their members.

Law of Metaphysics Instances of the laws of metaphysics are grounded in the laws of metaphysics.

Law of Nature The laws of nature are grounded in patterns of local, qualitative matters of fact.

Logic The ball is red and round because it is red and it is round.

Mind Someone's being in pain is grounded in their firing c-fibers.

Normativity The rightness of an action is grounded in its maximizing utility.

Physicalism All nonphysical facts are grounded in physical facts.

Semantics That a fact obtains grounds the truth of a sentence expressing that fact.

Social Entities That a person is a citizen is grounded in their occupying a certain social role.

Each of these statements makes a claim about what grounds what. None of these ground claims is uncontroversial. But each illustrates a potential application of ground to some topic. None of these potential applications is uncontroversial either. Indeed, there is a great deal more to say

about each of these potential applications (**Part III**). Yet despite their many differences, we may still recognize an abstract similarity. Each seems to be asking a question of explanation. Not just any sort of explanation is at issue. It is not, for instance, causal explanation. Instead, the sort of explanation seems to be distinctively *determinative* or *constitutive*. What is explained is determined by or constitutively depends on what explains it (**Chapters 7 and 8**). Extrapolating provides a conception of ground from below.

Approached from above, ground is triangulated from its role and its connections to other notions. These connections form a vast and intricate landscape of similarities and differences. Zooming in provides a detailed view of the specifics local to a connection. Zooming out provides a panoramic view of the general contours of the global landscape. These contours are shaped by a grand organizing idea characteristic of ground. It is a stratified or hierarchical picture of reality. The details of the picture can be developed in various ways. An especially common version of the picture divides reality into levels. Some levels depend on or otherwise derive from others (**Chapter 7**). Perhaps the psychological level derives from the neurochemical level, or the chemical level derives from the physical level. Perhaps there is even a foundational level that derives from no others but from which all other levels derive. Or perhaps there is no foundation, just levels all the way down (**Chapter 16**). Whichever it is, ground's characteristic role is to stratify what levels there are. For the psychological level to derive from the neurochemical level is for the facts of the neurochemical level to ground the facts of the psychological level, and so on for other levels. There may and often will be causal connections between levels. Neurochemical causes might have psychological effects, and perhaps vice versa. But even when these causal connections induce stratifications of their own, they will not be the characteristic stratifications induced by ground. These are supposed to be induced by the distinctive kind of determinative explanation ground provides (**Chapter 8**). The result is a hierarchy of grounds. This hierarchical picture exerts at least some influence on ground's connections to other notions. To illustrate, here are ten such notions often taken to be connected to ground in various ways and to various degrees:

- Analyticity** Truths that hold solely in virtue of the meanings of the words used to express them.
- Cause** An event or agent making happen another event.
- Emergence** Novel facts, items, properties, or phenomena that somehow arise from others while being irreducible to them.
- Essence** What it is for some item to be the very item that it is, or the properties that such an item has by its nature.
- Fundamentality** What is basic and that from which the nonbasic derives.
- Modality** Possibility, impossibility, necessity, contingency, and related notions, such as supervenience.
- Ontology** What there is or what exists.
- Realism** What is real or holds in reality, as opposed to what is merely apparent.
- Structure** The distinguished or "joint-carving" aspects of reality.
- Truth-making** The relation between a true representation and what it represents and in virtue of which it is true.

None of these potential connections is entirely straightforward or uncontroversial. Indeed, there is a great deal more to say about each (**Part IV**). But together these connections help to locate ground's position on the theoretical landscape. Triangulating provides a conception of ground from above.

The dual approaches differ in their trajectory. On the approach from below, ground is extrapolated from its potential applications. On the approach from above, ground is triangulated from its theoretical role and connections. One might reasonably wonder whether these trajectories converge on the same target notion. Perhaps there are various distinct ground-like notions (**Chapter 13**). But even if there are various ground-like notions, it might still be that the trajectories converge on one of them. Just as philosophers have become accustomed to a single³ notion of *cause* involved in how some events (or agents) produce others, so too it is perhaps not much of a leap to accept a single notion of *ground* involved in how some facts consist in or hold in virtue of others. Much of the literature assumes, at least as a working hypothesis, that there is indeed a notion of ground at the convergence point of the dual approaches from above and from below.

The dual approach makes ground clearly visible in the history of philosophy. Questions of ground trace back to antiquity (**Chapter 1**). Our contemporary notion of ground has much in common with Aristotle's notion of *demonstration* (**Chapter 2**). Questions of ground continued to be discussed in the medieval and modern periods (**Chapter 3**), perhaps most prominently as they engaged with the principle of sufficient reason (**Chapter 4**). Even so, it has been only relatively recently that ground has been treated as a topic of investigation in its own right.⁴ Bolzano's pioneering work on ground was ahead of its time (**Chapter 5**) and influenced ground's role in the Austro-Germanic phenomenological tradition (**Chapter 6**). The anti-metaphysical sentiments of much of the 20th century resulted in little work directly on ground. But recent interest in ground was kickstarted by a trio of influential articles (Fine 2001; Schaffer 2009; Rosen 2010) soon followed by an anthology (Correia & Schnieder 2012). Since then, ground has become a hot topic. Articles are regularly published exploring it and its applications. And ground has suffused into many parts of philosophy.

The dual approach also provides guidance on what ground's characteristic features might be. For we may identify many of ground's characteristic features by attending to its intended role in its potential applications and from its potential connections to other notions. None of these features are beyond all controversy. Indeed, some have been challenged vigorously in the literature. So these features should not be regarded as *definitive* of what ground is. Even so, they may still be used to help illuminate what ground is supposed to be. These features may collectively aim at their intended target, even if the target does not have them all. This way of fixing on a topic is not altogether unfamiliar. It is perhaps somewhat like how a definite description ('the person drinking the martini') may be used to refer to an individual that does not satisfy the description (there is only water in the martini glass).⁵

With this as our inspiration, we may introduce ground's characteristic features by developing an extended example. A bartender fills a mixing glass with ice. After measuring some ingredients (gin, dry vermouth), they are poured and mixed in the mixing glass. The result is poured into a coupe glass and garnished with an olive. The bartender has made a beverage, a martini. A martini is a distinctive kind of beverage. Unlike fresh-squeezed orange juice or straight gin, a martini is a mixed drink. Unlike other mixed drinks such as a lassi or a latte, a martini is alcoholic. These two features determinatively explain the martini's being a cocktail:

- (1) This martini is a cocktail because it is alcoholic and mixed.⁶

We may say this using the terminology of 'ground' instead:

- (2) That this martini is alcoholic and mixed grounds the fact that it is a cocktail.

Call a claim of the form '*... grounds ...*' a *ground claim*. What fills in the '*...*' are the *grounds*. What fills in the '*...*' is the *grounded*. Ground claims are usually taken to be *many-one*. Any nonnegative

number of grounds may fill in the ‘...’, but only a single grounded item may fill in the ‘___’. There is a less common and more general *many–many* (or *bicollective*) formulation which allows a non-negative number of grounds or grounded items on *both* sides (Dasgupta 2014; Litland 2018). But our focus, along with most of the literature, will be on the usual *many–one* formulation.

Although a ground claim is not a conditional in the usual sense, it does state a kind of *determinatively explanatory sufficient condition*. The grounds determinatively explain, and *therefore suffice for*, what they ground.⁷ This suggests that the terminology of ‘antecedent’ and ‘consequent’ familiar from conditionals may be adapted to ground claims. Thus, given the ground claim ‘A, ... grounds C’, we may refer to the grounds collectively as the *antecedents* (A, ...) and to what is grounded as the *consequent* (C). This terminology is nonstandard. But it allows handy mnemonics (‘A’ for antecedents, ‘C’ for consequent) while avoiding alliterative awkwardness (cf. “the antecedents ground the consequent” vs. “the grounds ground the grounded”). Nevertheless, both terminologies have their place. I will switch between them as convenient.

The literature contains several competing notations for ground claims. The three most common use three different symbols for ground: a reverse arrow ‘←’ (Rosen 2010), an inequality symbol ‘<’ (Fine 2012), and an oversize arrow ‘→’ (Fine 2015). The following three statements are alternative symbolizations of the ground claim (2):

- (2a) This martini is alcoholic and mixed ← This martini is a cocktail
- (2b) This martini is alcoholic and mixed < This martini is a cocktail
- (2c) This martini is alcoholic and mixed → This martini is a cocktail

A minor but distinctive feature of the inequality notation for ground is that it is reversible like the numerical inequality sign. Uniformly reversing the numerical inequality sign and numerals in ‘2 < 3’ yields the equivalent ‘3 > 2’. The same analogously holds for ‘<’ but not the other notations. Compare (2a) through (2c):

- (2a’) This martini is a cocktail → This martini is alcoholic and mixed
- (2b’) This martini is a cocktail > This martini is alcoholic and mixed
- (2c’) This martini is a cocktail ← This martini is alcoholic and mixed

Neither (2a’) nor (2c’) are usually taken to be ground claims: (2a’) looks like a conditional, whereas (2c’) is an essentialist claim (in Fine 2015’s notation).⁸ But (2b) and (2b’) are the same ground claim, differently expressed. Thus, only ‘<’ reverses in the way ‘grounds’ reverses to ‘is grounded in’. There are other differences between the notations that might be contrasted as well. Be that as it may, Fine’s original notation ‘<’ has become standard. So it will be the notation used in this *Handbook*. We may thus write ‘A, ... < C’ to express the ground claim ‘A, ... grounds C’.

Before proceeding to ground’s characteristic features, we must first consider more carefully how ground claims are interpreted. The two main options were already implicit in our examples. The first example (1) had *sentences* flank ‘because’. Grammaticality required nominalizing these sentences in the second example (2). The two examples guide us toward the two most common interpretations of ‘<’: as a *sentential connective* or as a *relational predicate*.⁹ This grammatical choice may at first seem fussy. But it directly impacts the exact formulation of ground’s characteristic features. It also raises some delicate philosophical issues about ground. So it will be worthwhile to elaborate on the grammar of ground and its ramifications.

The connective interpretation reads ‘<’ as a *sentential connective* (or *operator*). Just as a truth-functional connective like ‘and’ connects some sentences to yield their conjunction, so too ‘<’ connects some sentences to yield another. Then the antecedents ‘A, ...’ list the *sentences* stating

the grounds, and the consequent ‘C’ is a *sentence* stating what is grounded. The ground claim is just another sentence formed by connecting sentences stating the grounds to the sentence stating what is grounded.

The predicate interpretation reads ‘<’ as a *relational predicate*. Just as the relation of loving denoted by the predicate ‘loves’ relates lover to beloved, so too the relation of ground denoted by the predicate ‘<’ relates the antecedent grounds to the grounded consequent. Then the antecedents ‘A, . . .’ list the *relata* that are the grounds and the consequent ‘C’ is the *relatum* that is grounded. The ground claim is a sentence stating that the relation of ground holds between its relata, the grounds and the grounded.

The connective and predicate interpretations have complementary virtues and vices. The predicate interpretation’s distinctive feature is its reification of a relation of ground and its relata. This has the advantage of directly securing ground’s relevance to metaphysics. Stratification, for instance, results from ground’s relata standing in relations of ground. But reification also has disadvantages. These stem from having to answer what the relata of the ground relation are. The relata are most commonly restricted to *facts, propositions*, or something of the sort. A less common and more permissive view imposes no restriction, allowing the relata to be objects, properties, facts, propositions, or anything else.¹⁰ Permissivists may view the restrictions as selectively attending to merely some of what they admit, whereas restrictionists will likely not comprehend what permissivists admit. Regardless, the dominant trend in the literature seems to be the restricted conception. Suppose so, and that the restriction is to facts. Then we must now engage the metaphysics of facts. This raises hard questions. Are facts just true propositions, or the truths that true propositions represent? Do facts have constituents? If so, how closely do they track the constituents of the sentences used to express them? If not, how can talk of constituents be sustained? Perhaps questions like these may be postponed while one is focusing on addressing other matters. But they cannot be postponed forever. Once it is time to answer them, various risks arise. There is a risk that we might mistake constituents of our *representations* of the facts with the constituents (if any) of the fact themselves. There is a risk of equivocation between those with different conceptions of facts. And there is a risk of making ground irrelevant to those who refuse to reify facts.

The connective interpretation’s main virtue is mitigating or altogether avoiding the risks of reification. There is less of a risk mistaking constituents of sentences with the constituents of what the sentences represent because our focus is on the former. Nor is there a risk of equivocation between different conceptions of facts because no conception is assumed. Nor is there a risk of irrelevance because no position is taken on whether there even are any facts. Of course, those who adopt the connective interpretation might ultimately wish to reify a relation of ground and its relata anyway. This is perhaps analogous to how necessity and possibility may be construed as *sentential connectives* (‘□’, ‘◇’) even while giving a semantics of them in terms of *relations* across worlds. But nothing about the connective interpretation itself requires or prevents this. That in turn allows compartmentalizing the metaphysics of facts away from ground. This has the potential disadvantage of not directly securing ground’s relevance to metaphysics in the way the predicate interpretation could. But it has the compensatory advantage of liberating ground from the risks of reification.

Nevertheless, it is difficult in practice to avoid the predicate interpretation. This is largely because it enables us to easily express ground claims that the connective interpretation struggles with. To illustrate, suppose we are also considering in virtue of what this Martinez is a cocktail:¹¹

(3) That this Martinez is alcoholic and mixed < That it is a cocktail

Now consider the conjunction that this martini is a cocktail and that this Martinez is a cocktail. Suppose that we wish to explain in virtue of what this conjunction obtains. The answer seems to be that it obtains in virtue of its conjuncts. We might then try to express this answer as:

- (4a) That this martini is a cocktail and that this Martinez is a cocktail < That this martini is a cocktail and this Martinez is a cocktail

The answer this gives to what grounds the conjunction is itself a conjunction. Indeed, the conjunction given as an answer just is the conjunction to be grounded. But we did not mean to express that this conjunction grounds itself. And so our manner of expression has failed us. The problem stems from our expressing the grounds conjunctively by using ‘and’. So we might try to avoid using ‘and’ by listing the conjuncts instead:

- (4b) That this martini is a cocktail, That this Martinez is a cocktail < That this martini is a cocktail and this Martinez is a cocktail

Now the problem is that this appears ungrammatical. This frustrates our attempt to state just what the grounds of the conjunction are in particular. It also frustrates any attempt to state that the conjunction even has any grounds in general. Perhaps the most natural way to say this would be to existentially generalize over the *list* of conjuncts:

- (5a) There is . . . , such that: . . . < That this martini is a cocktail and this Martinez is a cocktail

But English does not provide any natural way to generalize in this way over its own sentences. Indeed, it is an open question whether there is any natural way to do so.

The predicate interpretation neatly avoids these expressive difficulties. Rather than stating the sentences expressing the grounds or the grounded, we refer instead to the truths or facts corresponding to them:

- (6a) The truth that this martini is alcoholic and mixed < The truth that it is a cocktail
(6b) The fact that this martini is alcoholic and mixed < The fact that it is a cocktail

These statements, especially the second, roll off the tongue. More importantly, expressing claims in terms of facts allows us easily to distinguish the *two* facts, fact A and fact B, from the *one* conjunctive fact, fact A and B. This in turn allows us to express the conjunctive ground claim as intended:

- (4c) The fact that this martini is a cocktail and the fact that this Martinez is a cocktail < The fact that this martini is a cocktail and this Martinez is a cocktail

What’s more, English does provide a natural way to generalize over truths or facts. So we may express that the conjunction has grounds as:

- (5b) There are some facts X, . . . such that: X, . . . < The fact this martini is a cocktail and this Martinez is a cocktail

The ease of expressing ground claims in terms of facts likely explains why the literature often implicitly adopts the predicate interpretation, despite the problematic reification it seems to require.

What began as a seemingly innocuous question of the grammar of ‘ground’ ramified into deeper and more intricate matters. Choosing between the connective and predicate interpretations affects not only the exact formulations of ground’s characteristic features but also the methodology for discerning them. The connective interpretation investigates ground’s characteristic features in abstraction from broader metaphysical questions about facts and the like. By contrast, the predicate interpretation allows these metaphysical matters to influence what ground’s characteristic features are.

The literature shows some awareness of the significance of choosing between the connective and predicate interpretations. Occasionally, authors will even explicitly commit to one. But it is more common for authors to dodge explicit commitment. One gets the impression that the literature is still trying to work out how to weigh the metaphysical neutrality distinctive of the connective interpretation with the expressive ease of the predicate interpretation. For these reasons, I have not imposed either interpretation on this *Handbook*. But I have also not attempted to maintain strict neutrality between them. Doing so would often require giving separate formulations for ground’s characteristic features or separate treatments of the justifications for them. Whatever benefits gained by this neutral comprehensiveness would be outweighed by the costs to readability and intelligibility. So, in an ameliorative spirit, I adopt the expedient of noncommittally sliding between the interpretations as convenient. Most contributors have also adopted this expedient. Usually it runs only a very low risk of distorting what’s at issue. In the rare cases where the risk is higher, contributors will have flagged the risk.

Returning finally to ground’s characteristic features, ground is typically taken to be *factive*. This means that a ground claim is true only if all the grounds and the grounded are true (if $A, \dots < C$, then $A, \dots$ and C). There is also a *nonfactive* notion of ground that does not require the grounds or the grounded to be true. We may think of nonfactive ground as a *hypothetical* version of factive ground. Roughly, some facts nonfactively ground another fact just in case these facts would factively ground the latter fact, were they all to obtain. It is, however, unclear whether this rough statement can be adapted into a satisfactory analysis of nonfactive ground in terms of factive ground. Although sometimes the nonfactive notion is at issue, it is usually the factive notion. It is generally safe to assume that factive ground is at issue when the matter is not explicitly addressed.

Ground is also often taken to have *modal force*. According to *necessitarianism*, the determinatively explanatory sufficient condition captured by the ground claim $A, \dots < C$ implies the *modally* sufficient condition captured by the modal claim that $A, \dots$ necessitates C .¹² One motivation for necessitarianism is that it helps close the explanatory gap between the grounds and what they ground: for given that the grounding facts obtain, then so too *must* the grounded fact. Although necessitarianism is widely held, *contingentists* reject it. They argue that there are counterexamples to necessitarianism and that ground can serve its explanatory role without it. The debate between necessitarians and contingentists remains controversial (**Chapter 10**).

We have so far focused on *full* ground. There are also, however, notions of *partial* ground. Roughly, a partial ground for a fact is a fact that *helps* ground it. The most common notion of helping to ground a fact is to be part of a full ground for it.¹³ Thus, the fact that this martini is a mixed drink helps ground the fact that it is a cocktail, and the fact that this martini is alcoholic helps ground the fact that it is a cocktail. Using our notation, we may represent these claims of partial ground using the precedence symbol ‘<’:

- (7a) This martini is a mixed drink < This martini is a cocktail
- (7b) This martini is alcoholic < This martini is a cocktail

Each claim on its own is a “proper” part of the full ground. Both claims taken together are an “improper” part of the full ground:¹⁴

(7c) This is a mixed drink, This martini is alcoholic $\prec$ This martini is a cocktail

Thus, this notion of partial ground implies that a full ground is always a partial ground. Generally, a fact A *partially* grounds (or helps ground) another fact C (or $A \prec C$) just in case A is part of some full ground of C's ($\dots, A, \dots \prec C$).

The sense in which a *full* ground for a fact gives a complete or total explanation of it does not imply uniqueness. Indeed, it is widely assumed that some facts have no unique full ground. For example, consider a disjunctive truth with both disjuncts true. Neither disjunct is the disjunction's *unique* full ground, although each is *a* full ground for it. For another example, suppose that psychological facts are fully grounded in neurochemical facts and that neurochemical facts are fully grounded in physical facts. Then if grounds chain, neither the neurochemical nor the physical facts will *uniquely* fully ground the psychological facts, although each will be *among* their full grounds. Most of the literature regards these as cases of harmless overdetermination.

Ground is widely thought to have various structural features deriving from its determinatively explanatory nature. One of the least controversial of these structural features is *nonmonotonicity*. Ground would be *monotonic* were it so that whenever some facts grounded another ($A_1, A_2, \dots \prec C$), then adding any fact to the grounds would preserve their grounding the grounded fact ($B, A_1, A_2, \dots \prec C$). Even though adding a fact may preserve the grounding in some cases, it is not guaranteed to do so. For example, the first ground claim is true, while the second is false:

(8a) This martini is alcoholic and mixed $\prec$ This martini is a cocktail

(8b) Turtles have shells, This martini is alcoholic and mixed $\prec$ This martini is a cocktail

The second ground claim is false even though turtles do have shells. This is because the fact that turtles have shells does not help explain the fact that this martini is a cocktail. The nonmonotonicity of ground therefore captures the idea that the grounds of a fact must be explanatorily relevant to the fact they ground.

More controversially, there is an orthodox conception of ground which takes it to have structural features corresponding to a strict partial order (**Chapter 17**).¹⁵ We may first illustrate these structural features for partial ground. Partial ground is *irreflexive* just in case no fact helps ground itself ($A \nprec A$). Partial ground is *asymmetric* just in case some fact helps ground another just in case the other does not help ground it (if $A \prec C$ then $C \nprec A$). And partial ground is *transitive* just in case one fact helps ground a third if the first helps ground a second and the second helps ground the third (if $A \prec B$ and $B \prec C$, then $A \prec C$). Given that partial ground has these structural features and that partial grounds are parts of full grounds, it follows that generalizations of these structural features will also hold of full ground.

These three structural features are often taken to be consequences of ground's determinatively explanatory nature. Consider the following ground claim:

(9) This martini is a cocktail $\prec$ This martini is a cocktail

Of course, every cocktail is a cocktail, and so too for this one. But our aim was to explain why this martini is a cocktail. We do not explain this by repeating that it is a cocktail. So the ground claim is false. Generally, irreflexivity is motivated by prohibiting self-determination or self-explanation.

Similar considerations tell against accepting both of the following ground claims:

- (10a) This martini is alcoholic and mixed < This martini is a cocktail
- (10b) This martini is a cocktail < This martini is alcoholic and mixed

Of course, it is necessary that every alcoholic mixed drink is a cocktail, and vice versa. But our aim was to explain why this martini is a cocktail. We may do so by saying that it is an alcoholic mixed drink. But this explanation is ruined if we follow this up by explaining that it is an alcoholic mixed drink because it is a cocktail. Generally, asymmetry is motivated by prohibiting cycles of determinative explanation.

Just as mathematical progress relies on chaining past proofs together to help produce new ones, so too many have thought that explanatory progress relies on chaining explanations together. Analogous considerations seem to hold for ground. Consider these claims of partial ground:

- (11a) This martini is alcoholic < This martini is a cocktail
- (11b) This liquid contains C_2H_6O < This martini is alcoholic
- (11c) This liquid contains C_2H_6O < This martini is a cocktail

The first claim links mixology to the culinary arts. The second claim links mixology to chemistry. The third chains the first two to link mixology to chemistry. Generally, transitivity is motivated by the capacity for determinative explanation to chain.

Chaining also helps illustrate a distinction between *mediate* and *immediate* ground. To illustrate, consider the fact that this liquid contains C_2H_6O . This fact only *indirectly* or *mediately* helps ground the fact that this martini is a cocktail. It grounds this fact only by “first” helping to ground the intermediate fact that this martini is alcoholic. Generally, a fact A mediately helps ground another fact C when there is some intermediate fact B in a chain of partial grounds between them ($A < \dots B \dots < C$). By contrast, there is no such intermediate fact between the fact that this martini is an alcoholic mixed drink and the fact that it is a cocktail. So the first fact *directly* or *immediately* grounds the second.

Another controversial question is whether ground is *well-founded* in the sense that there are no infinitely descending chains of ground (**Chapter 16**). One often hears the slogan “All explanations must come to an end”. Those under the influence of the slogan might take it to apply to determinative explanation—to ground—as well. Just as other explanations must come to an end, so too must chains of ground terminate in facts that themselves have no grounds. A fact that has no grounds is an *ungrounded* fact. Ungrounded facts are to be sharply distinguished from facts that have an unusual sort of ground: nothing. A fact grounded in nothing is *zero-grounded* or *null-grounded*.¹⁶ A zero-grounded fact is grounded in nothing, and so grounded. An ungrounded fact is not grounded, and so not grounded in nothing. No zero-grounded fact is ungrounded, and no ungrounded fact is zero-grounded. But either would serve as a kind of terminal endpoint for chains of ground. If U is an ungrounded fact, then there is a clear sense in which U is a terminal endpoint of any grounding chain $U < \dots < A < B < C < \dots$ in which it occurs. For given that U is ungrounded, then the grounding chain ends with U. If Z is a zero-grounded fact, then there is also a clear sense in which Z is a terminal endpoint of any grounding chain $Z < \dots < P < Q < R < \dots$ in which it occurs. For given that Z is grounded in nothing (and nothing else), then the grounding chain effectively ends with Z. It remains controversial, however, whether there are any ungrounded or zero-grounded facts. And even if there are, it is no less controversial whether any or every grounding chain must terminate in ungrounded or zero-grounded facts.

One of the most vexing questions about ground concerns whether ground claims are themselves grounded. Once we start asking what grounds what, we are presumably willing to accept that there will indeed be some facts about what grounds what. These facts, however, are more grist for the mill. For we may then ask what, if anything, grounds these facts about what grounds what. This is the question of *meta-ground* or, alternatively, *iterative ground* (**Chapter 9**). If facts about what grounds what are ungrounded, are they then fundamental and inexplicable? If these facts are grounded, then what grounds them? Or are these facts somehow exempt from the question of ground entirely?

An important goal of systematic theorizing about ground is to investigate the various logics that might govern it (**Chapter 14**). A *pure* logic of ground concerns only the logical features of ground itself without regard to the identities of the grounds or the grounded. An *impure* logic of ground considers the logical features of the grounds and grounded as well. An important choice point in devising any logic of ground is whether it concerns the logic of ‘<’ construed as a connective or as a relational predicate.

Another important choice-point concerns ground’s *granularity* (**Chapter 15**). On a *fine-grained* (or *representational* or *conceptual*) approach, facts are individuated (almost) as finely as the sentences used to express them. On a *coarse-grained* (or *worldly*) approach, facts are individuated more coarsely in terms of what they demand of the world, not how it is represented. To illustrate the contrast, whereas a fine-grained approach may allow that A grounds its self-disjunction $A \vee A$, a coarse-grained approach will disallow this because it rejects any worldly difference between A and $A \vee A$. The question of granularity impacts the logic of ground as well as its connections and applications.

Perhaps the most disquieting choice-point concerns how to respond to various logical puzzles of ground (**Chapter 18**). These puzzles are somewhat like the more familiar semantic paradoxes. Just as the semantic paradoxes raise deep challenges for the logic of truth and reference, so too the puzzles of ground raise deep challenges to its logic. And just as few take the semantic paradoxes to cast doubt upon the very coherence of truth and reference themselves, so too have few so far taken the puzzles to cast doubt upon the very coherence of ground itself.

But other skeptical doubts have been raised against ground. One source of doubt stems from a broader skepticism of the intelligibility of metaphysical notions in general. More common sources of doubt stem from specific challenges to ground. Some doubts were of ground’s intelligibility. This sort of “old-school” skepticism has largely been superseded by a “second generation” of skeptical doubts about ground’s fruitfulness in contributing anything beyond what more familiar notions already contribute. Although a number of contributors to the *Handbook* express skeptical sentiments, two essays explicitly focus on skeptical challenges to ground and rejoinders to them (**Chapter 5 and Chapter 6**).

Dedicating just two of the *Handbook*’s 37 essays to skepticism about ground might seem as odd as relegating external-world skepticism to a supporting role in an anthology on knowledge. Indeed, skeptical challenges to ground have animated much of the recent literature. And yet here is a *Handbook* that gives little airtime to skeptical challenges. There is, however, a rationale for this. The guiding idea behind the *Handbook* was to survey the state of the art on what ground is *supposed to be*. Just as the external-world skeptic might wish to know what knowledge is *supposed to be* in order to argue for ignorance, so too the skeptic of ground might wish to know what ground is *supposed to be* in order to argue against it. Indeed, the project of exploring what ground is supposed to be might very well play into the skeptics’ hands. If the attempt to articulate what ground is supposed to be degenerates into a stable pattern of failures, then one might conclude with the skeptic that ground is unintelligible. Or granting ground’s intelligibility, if its intended applications to the big questions bear no philosophical fruits, then one might conclude with

the skeptic that ground is fruitless. Skeptics may use the *Handbook* to clarify the notion they ultimately wish to reject, whereas aficionados may use the *Handbook* to defend the notion they love. Either way, the *Handbook's* focus on what ground is *supposed to be* promises to help guide us in determining whether it lives up to its ambitions.

2. Overview

This *Handbook* contains 37 original essays on ground. They are organized into five parts, with each loosely focused on a theme. Each part begins with a short introduction describing its theme and the essays organized around it. The *Handbook* concludes with a glossary that originated at a meeting of groundhogs¹⁷ under the auspices of the Humboldt Foundation in Helsinki on June 7, 2016.

The five organizing themes of the *Handbook* correspond to what have already been or are expected soon to be “centers of gravity” in the literature:

Part I: History. The theme is the history of ground. Historical aspects include ground in ancient, medieval, and modern times, ground's connection to Aristotelian demonstrations, ground's role in the principle of sufficient reason, Bolzano's pioneering work on ground, and ground in the Austro-Germanic phenomenological tradition.

Part II: Explanation and Determination. The theme is the explanatory or determinative aspects of ground. Topics include dependence, explanation, meta-ground, necessity, skeptical doubts, anti-skeptical rejoinders, and varieties of ground.

Part III: Logic and Structure. The theme is logical or structural features of ground. Topics include formal logics of ground, the granularity of ground, infinitely descending chains of ground, whether ground is a strict partial order (irreflexive, asymmetric, and transitive), and various logical puzzles of ground.

Part IV: Connections. The theme is connections between ground and various other notions. These include analyticity, cause, emergence, essence, fundamentality, modality, ontology, realism, structure, and truth-making.

Part V: Applications. The theme is potential applications of ground to various topics. These include identity, laws of metaphysics, laws of nature, logic, mind, normativity, physicalism, semantics, and social entities.

These themes are not mutually exclusive. Some topics may fit under several themes. Nor are the themes mutually exhaustive. Some worthy topics have not been covered thoroughly or, in some cases, at all. Nevertheless, the themes can be readily discerned in the developing literature on ground.

The developing status of this literature provided an exciting opportunity for this *Handbook*. In this literature, one finds established scholars and emerging scholars alike grappling with ground side by side. My choice of authors for this *Handbook* reflects this. The result is a snapshot of a developing literature eager to explore uncharted territory.

But the developing status of this literature also posed a major challenge for constructing this *Handbook*. Even though questions of ground can be found in antiquity, the notion itself received little explicit attention until the last decade or so. Since then, there's been an explosion of interest in ground itself as well as an increased awareness of its influences on the questions posed in terms of it. Despite current enthusiasm for ground, the literature on it is young and very much still in progress. Indeed, there is a lack of consensus on what the central questions are and what methodologies should guide attempts to answer them. This lack of consensus increased the risks

of disunity and lack of cohesion in the central themes, topics, and approaches across the *Handbook's* 37 essays. It would not have been an appropriate solution to imperialistically impose unity and cohesion upon them. One of the major challenges was to minimize the risks of disunity and lack of cohesion.

A better solution would foster the organic emergence of unity and cohesion from the otherwise disparate perspectives of the diverse and numerous contributors. This solution was realized by organizing a workshop gathering together contributors to the *Handbook*. The goal was to provide contributors with a collaborative scholarly environment in which they could coordinate their research activities, critique their approaches, and focus the development of their essays.

Implementing such a workshop faced challenges of its own. The *Handbook's* diverse group of 37 contributors were scattered across four continents from countries including Australia, Canada, England, Estonia, Finland, France, Germany, India, Israel, Scotland, Singapore, Switzerland, and the United States. Gathering contributors from such geographically distant locations at a single workshop posed logistical and financial challenges. The logistical challenges were met by Robert Schwartzkopff, who expertly and efficiently organized the workshop. The financial challenges were met by Kit Fine's generous contribution from his Anneliese Maier Research Prize from the Alexander von Humboldt Foundation together with my own Connection Grant from the Social Sciences and Humanities Research Council of Canada and the financial support of the University of Victoria.

The workshop occurred May 25–27, 2018, and was hosted by the University of Hamburg. Twenty-four contributors attended. Each contributor presented their plans for their essay. Each of these presentations was followed by a discussion. The workshop culminated in a capstone session at which I collated the key points that had emerged. These included ideas on the directions of the essays, suggestions for coordinating topics and minimizing redundancies, signaling opportunities for cross-referencing, and more. This feedback was also shared with contributors who were unable to attend. All contributors were encouraged to consider this feedback when preparing their essays. The net result was an extended research collaboration on the state of the art on ground.

This collaborative effort did not and was not intended to result in a unified front on ground. The *Handbook's* 37 contributors do not speak with a single authoritative voice. Nor do they share a single vision of what ground's prospects are. But the contributors together represent the collective wisdom forged from the collaboration of a diverse group of philosophers who have each thought long and hard about ground. The essays they have produced are, I believe, an accurate survey of the state of the art on ground, looking backward to where it's been and forward to where it's going.

Notes

- 1 Actually, the literature divides into those who prefer 'ground' and those who prefer 'grounding'. My own preference is for 'ground'. But the matter is of little consequence. I have therefore made no attempt to legislate the matter in the chapters to follow.
- 2 These anthologies include Correia & Schnieder (2012), Jago (2016), and Bliss and Priest (2018).
- 3 Or at least a small number of notions of cause (cf. Hall 2004).
- 4 See Raven (2019) for a more detailed history of ground that relies on distinguishing between questions of ground and questions about ground.
- 5 This phenomenon, and the case of the alleged martini drinker, is from Donnellan (1966).
- 6 The expression 'alcoholic and mixed' is ambiguous. On one reading, it is a single *complex* predication (*being alcoholic+mixed*). On another reading, it *conjoins* two simple predications (*being alcoholic* and *being mixed*). Arcane questions arise for each reading and their relationship, especially in the context of ground. But throughout, I will only have the conjunctive reading in mind.

- 7 When A suffices for C in this sense, there is a corresponding sense in which C is necessary for A. But this sense of necessary condition needn't straightaway entail that necessarily if A then C whenever A grounds C. So the kind of sufficient condition at issue needn't prejudge the outcome of the debate over necessitarianism or contingentism about ground.
- 8 Whereas a ground claim states a kind of *sufficient* condition, an essentialist claim states a corresponding kind of *necessary* condition. Fine's oversize arrow notation is motivated in part to express this parallel between ground and essence.
- 9 A less common third option interprets '<' as a *meta-linguistic relation* (**Chapter 14**).
- 10 See Schaffer (2009).
- 11 The Martinez cocktail is a precursor to the martini made from Old Tom gin, sweet vermouth, maraschino liqueur, and bitters.
- 12 This modal claim may be represented as the strict conditional $\Box(A, \dots \rightarrow C)$. Strictly speaking, the expression ' $\Box(A, \dots \rightarrow C)$ ' is ill formed because the antecedent cannot be a *list* of sentences. But one might instead replace it with ' $\Box[(A \wedge \dots) \rightarrow C]$ ', which just takes the antecedent to be the conjunction of A, ...
- 13 There is another notion of partial ground that does not require a partial ground to be part of a full ground. It has received less attention in the literature.
- 14 Here I allude to the distinction between proper/improper parts: a proper part of a whole is any part that is not the whole itself, whereas an improper part is the whole itself.
- 15 The reason I say 'corresponding to' rather than 'is' is because we are maintaining an ambiguity between the connective and predicate approaches. If ground is a relation, then partial ground is a strict partial order. If ground is not a relation, then it cannot literally be a strict partial order.
- 16 The notion of *zero-ground* was first introduced in Fine (2012).
- 17 See entry 'groundhog' in "The Essential Glossary of Ground".

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PART I

History



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INTRODUCTION

Michael J. Raven

Questions of ground are often introduced by example. Perhaps the most common, as well as venerable, example is from Plato's *Euthyphro*. In that dialogue, Socrates and Euthyphro muse about the nature of piety. Socrates acknowledges that an act is pious just in case it is loved by the gods. But he still wonders whether the pious is loved by the gods because it is pious or whether it is pious because it is loved by the gods. It is nowadays common for Socrates's question to be interpreted as a question of ground. Is an act's being loved by the gods grounded in its being pious or is its being pious grounded in its being loved by the gods? Aristotle's notions of demonstration, the four *causes*, and their explanatory roles in the science of being *qua* being can be fruitfully interpreted in terms of ground. So there is a strong case to be made that interest in ground extends back to antiquity. Given the enormity of Plato's and Aristotle's influence on later philosophy, one might expect the history of ground to be a prominent chapter in the history of philosophy itself.

This impression, however, clashes with another portrayal of ground's history. In the literature, one can find authors who sometimes write as if ground was a more recent innovation whose emergence had to wait for the anti-metaphysical sentiments trending in much of the 20th century to fade. This portrayal acknowledges ground's historical roots. But the emphasis is instead on the recent surge of interest in ground, beginning with the trinity of Fine (2001), Schaffer (2009), and Rosen (2010), and followed soon after by a rapidly expanding, increasingly sophisticated literature.

It is not altogether clear how to reconcile the two portrayals of ground's history. On the one hand, ground has a venerable history stretching back to antiquity. On the other hand, ground is largely a recent innovation whose history is still being written. A natural attempt to reconcile the portrayals starts by distinguishing questions *of* ground from questions *about* ground. Socrates's question above, for example, is a question *of* ground. By contrast, whether ground is a strict partial order is not a question *of* ground but rather a question *about* ground. Questions *of* ground have engaged philosophers since antiquity. But questions *about* ground are, at least by comparison, more recent arrivals. This parallels how questions *of* possibility and necessity engaged philosophers since antiquity despite the comparatively recent focus on questions *about*

possibility and necessity stemming from the late 20th century's rapid developments in modal logic, modal metaphysics, and the like.¹

The main purpose of **Part I** is to survey some of the main historical antecedents of the recent surge of interest in it.

1. Phil Corkum's "Ancient" selectively surveys potential historical precedents to ground in the pre-Socratics, Plato's *Euthyphro*, and Aristotle's *Categories* and *Metaphysics*. Corkum's essay also asks a methodological question: when is ground a useful tool for ancient philosophy scholarship?
2. Marko Malink's "Aristotelian Demonstration" discusses Aristotle's treatment of direct and indirect demonstrations. Malink gives an account of Aristotle's influential thesis in the *Posterior Analytics* that all direct but not all indirect demonstrations proceed from premises that are prior in nature to the conclusion. The relevant notion of priority in nature can be viewed as a relation of ground. Malink indicates how Aristotle's thesis correlates with recent developments in the pure and impure logic of ground.
3. Margaret Cameron's "Medieval and Early Modern" observes that philosophers in the medieval, Renaissance, and early modern periods recognized the phenomena of metaphysical explanation, relations of metaphysical dependence, and relations of priority more broadly. To illustrate, Cameron analyzes the various and varied answers to the frequently asked question, "What is the subject matter of metaphysics?" Answers from Avicenna, Averroes, Aquinas, Suárez, Descartes, and Leibniz are analyzed. By examining the reasons for their answers to this question, one may get a broad understanding of how these philosophers understood the nature of metaphysical explanation and relations of metaphysical dependence.
4. Fatema Amijee's "Principle of Sufficient Reason" addresses three central questions. First, to what extent is the contemporary notion of metaphysical explanation continuous with the notion of sufficient reason endorsed by Spinoza, Leibniz, and other rationalists? Second, to what extent can the Principle of Sufficient Reason (PSR)—the principle that everything has an explanation—avoid the formidable traditional objections levelled against it if it is formulated in terms of metaphysical explanation? And finally, how might historical discussion of the PSR shed light on the contemporary notions of ground and metaphysical explanation?
5. Stefan Roski's "Bolzano" discusses Bernard Bolzano's views about ground. It briefly highlights historical influences on Bolzano's account and subsequently provides an overview of the most important aspects of it. A main focus of Roski's essay lies on relating Bolzano's ideas to positions familiar from the current debate about ground, including ideas that differ from current orthodoxy but may constitute interesting additions, challenges, or inspirations for the current debate.
6. Kevin Mulligan's "Austro-German Phenomenologists" surveys some of the main accounts of ground, dependence, foundation, and essence as well as some important applications thereof to philosophical questions which were put forward at the end of the 19th century and the beginning of the 20th century by the early Austro-Germanic phenomenologists and other heirs of Brentano (such as Meinong, Husserl, Reinach, and Scheler).

Note

- 1 This attempt at reconciling the two portrayals of ground's history is further discussed in Raven (2019).

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1

ANCIENT

Phil Corkum

Is there grounding in ancient philosophy? To ask a related but different question: Is grounding a useful tool for the scholar of ancient philosophy? These questions are difficult, and my goal in this chapter is not so much to give definitive answers as to clarify the questions. I hope to direct the student of contemporary metaphysics towards passages where it may be fruitful to look for historical precedent. But I also hope to offer the student of ancient philosophy some guidance on when drawing on the contemporary discussion of grounding may be beneficial.

Both issues hinge in part on how we ought to view grounding. Some theorists view grounding as the correlate of the *in virtue of* or *because* relation. Ancient philosophy abounds with such relations. Here's a sampler.

- The pre-Socratic philosopher Thales claims that water is the principle of all things. Subsequent philosophers offer other candidates. For example, Anaximenes views air as the principle and Anaximander the indefinite. One might view these claims as the suggestion that a given thing is so-and-so because water (or air or the indefinite) is the way it is. I discuss Thales in more detail in what follows.
- In Plato's *Euthyphro*, the character of Socrates wonders whether a thing is pious because it is loved by the gods, or is it loved by the gods because it is pious. Plato also aims to explain the behaviour of sensible, changing particulars by appeal to intelligible, unchanging Forms. For example, in the *Republic*, he claims that an act, agent or state are all called just because they participate in the Form of Justice. I discuss some of Plato's views in what follows.
- In his *Physics* 2.3, Aristotle discusses four kinds of causes. He writes that "we think we have knowledge of a thing only when we can answer the question about it 'On account of what?' and that is to grasp the primary cause."¹ Causes seem to be explanations why a thing is the way it is or why it changes in the way that it does. The material cause is "that out of which as a constituent a thing comes to be . . . for example, the bronze and the silver and their genera would be the causes respectively of a statue." The formal cause, "the account of what the being would be," corresponds to a definition of the thing. The efficient cause is the "source of the change" and seems to involve what initiates a process; Aristotle gives the example that the parent is a cause of the child. The final cause, "what something is for" includes the goal of purposive action, but also includes the state achieved or the product which results from an activity.

- In the *Posterior Analytics*, Aristotle lays out a rubric for scientific explanation. At the beginning of the Second Book, he describes his aims:

The things we seek are equal in number to those we understand. We seek four things: the fact, the reason why, if something is, what something is. . . . When we know the fact we seek the reason why (e.g., knowing that it is eclipsed or that the earth moves, we seek the reason why it is eclipsed or why it moves). (APo 2.1, 89b21–31)

We seek understanding. When some fact (*to hoti*) is known to obtain, we seek the reason why (*to dioti*) it is so, or an account of the fact. Aristotle continues:

As we said, to know what something is and to know the explanation (*aition*) of whether it is are the same; and the account of the fact that something is is the explanation (*aition*). This is either the same as it or something else; and if it is something else, it is either demonstrable or indemonstrable. If it is something else and it is possible to demonstrate it, then the explanation (*aition*) must be a middle term and the proof must be in the first figure; for what is being proved is both universal and positive. (APo 2.8, 93a3–9)

Scientific understanding is through an *aition*—in the above quotation, Barnes translates this as ‘explanation’ but in other contexts, the expression is often translated as ‘cause’. For example, this is how Charlton (1970) translates *aition* when Aristotle discusses his doctrine of the four causes in *Physics* 2.3, discussed above. Aristotle’s discussion of explanations yielding scientific understanding might strike one as employing a notion akin to grounding. Correia and Schnieder (2012: 2) tentatively suggest such a reading. Sandstad (forthcoming) argues that demonstrative explanations are not groundings. And Malink [Chapter 2, this volume] considers the influence of the *Posterior Analytics*.

- In the *Categories*, Aristotle views individual substances as separate from and ontologically prior to universals as well as individuals in nonsubstance categories. These latter entities arguably belong in Aristotle’s ontology in virtue of standing in a relation to individual substances. I discuss some aspects of Aristotle’s metaphysics in what follows.

If grounding is viewed sufficiently thinly, then it would be uncontroversial that there is grounding in ancient philosophy.

Many contemporary metaphysicians view grounding more thickly. Although there is little that can be said about grounding that is uncontroversial, here is a picture that perhaps provides a baseline for discussion. There are a variety of determination relations among facts that underwrite explanation. Some of these relations arguably bear sufficient similarity to group them together under a genus. Wilson (2014) calls the various determination relations small-g grounding relations and the genus capital-g Grounding; I will reserve ‘grounding’ for the generic notion and ‘grounding relations’ for the various specific relations. Talk of grounding is useful when we want to discuss the similarities among the grounding relations. And talk of grounding is also useful when we want to contrast such relations with relations that do not fall under this genus.

Here’s a few features shared by at least some of the various grounding relations. Grounding is a factive relation closely related to notions of structure and fundamentality. Grounding is a strict partial ordering and so yields structure on the ontology of facts. This structure is often thought of somewhat figuratively in terms of levels. So for example some physicalists might be characterized as holding that the physical facts ground the phenomenological facts and some

moral realists, as holding the physical facts ground the moral facts. When one set of facts grounds a second, the first set is relatively fundamental and the second relatively derivative.

And here's some of the contrasts between grounding relations, on the one hand, and other kinds of relations on the other. Grounding is distinct from causation, partly because grounding relations are typically (although not universally) viewed as synchronic relations among facts, but causation is typically (although not universally) viewed as a diachronic relation among events. For the relation between grounding and causation, see for example Wang [Chapter 20, this volume]. Grounding is distinct from ontological dependence, partly because grounding is often viewed as a relation among facts, but the relata for ontological dependence range over a broader variety of entities. Grounding, moreover, is closely associated with explanation, but ontological dependence need not play any explanatory role. For example, Tahko and Lowe (2015) note that the existence of water *depends* solely on the existence of hydrogen and oxygen, but the mere existence of hydrogen and oxygen does not underwrite a full *explanation* of the existence of water. For the relation between grounding and ontological dependence, see for example Schnieder [Chapter 7, this volume]; and for the relation between grounding and explanation, see for example Glazier [Chapter 8, this volume].

Let me flag some controversies in this picture. I have characterized grounding as a relation among facts. But some hold that entities other than facts can serve as relata in grounding; see for example, Schaffer (2009). Still others view grounding as an operator; see for example, Fine (2012). I have characterized grounding as irreflexive, asymmetric and transitive. Some disagree; for discussion see for example Raven (2013a) and Thompson [Chapter 17, this volume]. I have characterized grounding as underwriting metaphysical explanations, just as causal relations back causal explanations. Some view grounding as itself explanatory; see for example Raven (2015). Finally, I have characterized grounding as a genus grouping together a variety of grounding relations. Some are sceptical that there is a genus fruitfully grouping together these various relations. See for example Wilson (2014), Koslicki (2015 and [Chapter 11, this volume]). Others view grounding as not a genus but as a relation distinct from these other determination relations. For discussion see for example Raven (2017) and Maurin (2018).

For the historian, talk of grounding may conflate the thin and thick versions and so turn an unobjectionable ascriptive claim into a misleading anachronism. But such talk, when used judiciously, can shed light on the history of philosophy. In particular, talk of the generic notion of grounding can be fruitful when used negatively, to distinguish a view from notions such as causation or ontological dependence. It can be especially helpful when we can see that a historical figure is using some grounding relation or other, but it is difficult to offer a more specific interpretative proposal. However, grounding can be a blunt instrument. When a more specific relation can be identified, talk of grounding can bring little added value. Let's see these observations in action by discussing in more detail some of the candidate cases of grounding in ancient philosophy from our sample.

Thales

We begin with the Milesian philosopher Thales. Thales claims that the principle of all things is water. Is this a grounding claim on the thicker construal of grounding just sketched? On this line of interpretation, Thales is claiming not only that the water facts determine the nonwater facts and that the nonwater facts obtain in virtue of the water facts obtaining but moreover, that the water facts are fundamental, and the nonwater facts are derivative, within a structured factive ontology.

It is difficult to know what Thales' own intentions were. But Aristotle reads Thales somewhat differently from the interpretation just sketched. Aristotle sees the Milesians as attempting to explain change and diversity in the perceptible world by appeal to a single, material principle. As we saw in the sampler, Aristotle holds that matter is one of the basic kinds of causes or explanations. Aristotle appears to view matter as having several roles: an *originative* notion as that initial state out of which a thing changes, a *substrative* notion as the substratum that persists through changes, and a *constitutive* notion as those components out of which a thing is made. He goes on to develop an account of matter where its constitutive role dominates: for example, in the middle Books of the *Metaphysics*, Aristotle appears to view individual substances such as you and I as composites of form and matter. Matter in the *Metaphysics* retains a restricted role in explaining change: Aristotle occasionally treats matter as the substratum of substantial change, the coming into existence or passing out of existence of a substance. But when he introduces matter in *Physics* 1.7, it arguably appears to play a broader role as the substratum of any change whatsoever, including substantial change and qualitative change.

All three roles for matter are referenced early in the following discussion of the Milesians:

Of the first philosophers, then, most thought the principles which were of the nature of matter were the only principles of all things. That of which all things that are consist, the first from which they come to be, the last into which they are resolved (the substance remaining, but changing in its modifications), this they say is the element and this the principle of things, and therefore they think nothing is either generated or destroyed, since this sort of entity is always conserved, as we say Socrates neither comes to be absolutely when he comes to be beautiful or musical, nor ceases to be when he loses these characteristics, because the substratum (*to hupokeimenon*), Socrates himself, remains. Just so they say nothing else comes to be or ceases to be; for there must be some entity—either one or more than one—from which all other things come to be, it being conserved. Yet they do not all agree as to the number and the nature of these principles. Thales, the founder of this type of philosophy, says the principle is water (for which reason he declared that the earth rests on water), getting the notion perhaps from seeing that the nutriment of all things is moist, and that heat itself is generated from the moist and kept alive by it (and that from which they come to be is a principle of all things). He got his notion from this fact and from the fact that the seeds of all things have a moist nature and that water is the origin of the nature of moist things.

(*Meta.* 1.3, 983b6–27)

Aristotle seems to ascribe to Thales the view that water is an *ürstoff*, a basic substratum. There is no substantial change, since the *ürstoff* does not come into existence or pass away. All change is the qualitative change of water. On this theory, there are no birds or bees—that is to say, a bird is not a kind of substance, persisting through its own qualitative changes, but a quality of the underlying water, which, at some times and in some places, it possesses and, at other times and places, it does not. Compare this passage with the following:

[some Presocratics] make the underlying body (*sōma to hupokeimenon*) one—either one of the three [of the four elements] or something else which is denser than fire and rarer than air—then generate everything else from this, and obtain multiplicity by condensation and rarefaction.

(*Phys.* 1.4 187a12–16)

I have transliterated some of the Greek in the above two quotations to bring out that the expressions translated as ‘the substance remaining’, ‘substratum’ or ‘underlying body’ pick out the substrative role of matter: *to hupokeimenon*, ‘the underlying thing’, is traditionally translated as ‘substratum’.

We can perhaps gerrymander a grounding-based interpretation of Thales. But such an approach faces challenges. First, it is not clear that this line of interpretation would be successful. Thales, according to Aristotle, is not obviously concerned with the determination of all nonwater facts. Nor is his concern with a levels-style ontological structure. Second, the imposition of a factive ontology is arguably an anachronism. Suppose that these two challenges can be met. We may be able to describe Thales’ intentions in terms of grounding. And we may be willing to take on the cost of imposing an anachronistic framework if the approach yielded rewards. But if we can explicate the specific role that water plays, characterizing that role in terms of a small-g relation, then there is little added value in going on to describe that relation as grounding. Readers interested in the pre-Socratic philosophers might begin with Curd and Mckirahan (2011).

Plato

In the *Euthyphro*, Plato has the character of Socrates ask the question, “Is that which is pious loved by the gods because it is pious, or is it pious because it is loved by the gods?” (*Euthyphro* 10a, my translation, based on Cooper in Hamilton and Cairns (1961)). The *Euthyphro* question is commonly used when illustrating grounding: for some recent examples, see Raven (2013b), Bliss and Trogdon (2016), Schaffer (2016), and Maurin (2018), among many others. The question concerns at least the thin notion of grounding. Many grounding theorists view the question as concerning the thicker notion. For example, the mentioned authors seem to view the *Euthyphro* question as seeking to identify which is relatively fundamental, facts about which things are pious or facts about which things are god-loved, and which is relatively derivative.

Plato views the question somewhat differently. Socrates’ interlocutor, the character of Euthyphro, canvasses the answer that “it is because it is pious that it is loved; it is not pious because it is loved” (*Euthyphro* 10d), but Socrates rejects this answer: “Euthyphro, it looks as if you had not given me my answer—as if when you were asked to tell the nature of the pious, you did not wish to explain the essence of it. You merely tell an attribute of it, namely, that it appertains to piety to be loved by all the gods. What it is, as yet you have not said” (*Euthyphro* 11a). Socrates seeks a definitional account of piety. Being god-loved is an attribute of the pious, not the essence of what it is to be pious, and so is ill suited for supplying a *definiens*. When further attempts in the dialogue to define piety prove fruitless and merely circle back to the attribute of being god-loved, Euthyphro makes a quick exit, and the dialogue suddenly ends.

Plato’s concern in the *Euthyphro*, then, is to specify criteria for being a *definiens* and arguably does not concern a thicker notion of grounding. Is there other evidence for grounding in the *Euthyphro*? Correia and Schnieder (2012: 2–4), in a brief but engaging discussion of the *Euthyphro*, note that Plato’s argument draws on features associated with grounding. For example, Euthyphro endorses the claim that if something is pious, it is pious because it is god-beloved, and Socrates concludes that it follows that if something is pious, it is pious because the gods love it. This inference is licensed by the explicitly made claim that if something is god-beloved, it is so because the gods love it, and by the tacit assumption that ‘because’-clauses chain; as Correia and Schnieder put it, Plato assumes that grounding is transitive. But this and other features discussed by Correia and Schnieder are logical characteristics of ‘because’-clauses, and so are

features shared by thin and thick grounding. There may not be evidence for thick grounding in the *Euthyphro*.

Plato's response in other dialogues, to the definitional problem left unresolved in the *Euthyphro*, is to posit Forms. Forms provide *definiens* for ethical and aesthetic concepts such as piety, justice, and beauty. They play a variety of other roles and seem to be required for accurate perceptual judgements (as in the *Phaedo*), knowledge (*Republic* 5), causal explanation (*Phaedo*), and to provide referents for natural kind terms (*Cratylus*). And they serve some kind of explanatory role in explaining the behaviour of particulars. Where particulars are perceptible and changing, Forms are imperceptible, intelligible, and unchanging. Particulars approximate the Forms, or fall short of the Forms in some other way: for discussion, see, for example, Nehamas (1999). Particulars are sometimes characterized as mere likenesses of Forms and participate in them. For example, in the *Parmenides*, the character of Parmenides characterizes the Platonic position (ascribing it in the dialogue to the young Socrates): "You say you hold that there exist certain forms, of which these other things come to partake and so to be called after their names; by coming to partake of likeness or largeness or beauty or justice, they become like or large or beautiful or just" (*Parmenides* 130d–131a). By contrast with particulars, a Form is a paradigm of the feature for which it is named. And Forms appear to be universals. For example, Parmenides continues: "I imagine your ground for believing in a single form in each case is this. When it seems to you that a number of things are large, there seems, I suppose to be a certain single character which is the same when you look at them all; hence you think that largeness is a single thing" (*Parmenides* 131d–132d).

It is difficult to understand participation. One of Aristotle's criticisms of Platonic Forms is that talk of participation is uninformative. He writes that this "is empty talk; for 'participation' . . . means nothing" (*Metaphysics* 1.9 992^a28–9, my translation, based on Ross). Plato himself finds participation and other aspects of the Forms problematic. For example, in the *Parmenides* 132a–b, the character of Parmenides raises the worry that if the reason for positing a Form is a perceived commonality, and Forms are paradigms that exhibit the feature for which they are named, then the Forms fall into a regress.

Whatever participation might be, the relation between Forms and particulars occasionally looks like an explanatory relation yielding ontological structure. And the Forms themselves appear to exhibit levels-like structure. For example, consider the status of the Form of the good relative to the other Forms. In the *Republic*, Plato has the character of Socrates, speaking of the Forms of justice, piety, and so on, say that these "objects of knowledge not only receive from the presence of the good their being known, but their very existence (*to einai*) and essence is derived to them from it, though the good itself is not essence but still transcends essence in dignity and surpassing power" (*Republic* 509b). (To this the character of Glaucon comically responds, "Heaven save us, hyperbole can go no further".) This might suggest to the reader that the Forms themselves are structured hierarchically in a structure reminiscent of the ontological levels of grounding enthusiasts discuss.

However, there are reasons to hesitate to draw this inference. The quotation uses the Shorey translation from Hamilton and Cairns (1961), which translates the Greek *to einai* as 'existence'. This is a common translation; we'll see similar translation choices when we discuss Aristotle; and the Greek sometimes indeed carries existential force. But the Greek can also sometimes arguably carry predicative force. For discussion of the copula in ancient Greek philosophy, the reader might begin with Kahn (2009). On this alternative reading, what the Form of justice is, say, is derived from the Form of the good. As mentioned, Forms are essentially paradigms; and Plato seems to believe that the Form of justice is itself a good exemplar of being just. As something good, the Form of justice, like any other Form, participates in the Form of the good. Although

there is some sort of explanatory hierarchy among Forms, and between Forms and particulars, it is unclear whether such explanations have the characteristic marks of grounding.

Despite my wet blanketing, it may be fruitful to investigate whether there is a thick notion of grounding in Plato. Some contemporary philosophers hold that there are systematic connections between essence and ground, which could reinstate the *Euthyphro* question as relevant to grounding; for the relation between essence and ground, see Zylstra [Chapter 22, this volume]. For historical work in this direction, see the earlier-mentioned Correia and Schnieder (2012) and Evans (2012), who views the character of Socrates in the *Euthyphro* as attacking the view that beliefs can ground facts. And for a sympathetic reading of grounding generally in Plato, see Thomas (2014).

Aristotle's *Metaphysics*

Notions of ontological separation and priority are central to Aristotle's metaphysics. Aristotle characterizes ontological priority at *Metaphysics* 5.11 (1019^a1–4):

Some things then are called prior and posterior . . . in respect of nature and substance, such as those which can be without (*einai endechetai aneu*) other things, while the others cannot be without them.

G. Fine (1984) argues for the following connection between ontological separation and priority: one thing being separate from another and the second being inseparable from the first are jointly sufficient for the first to be prior in substance to the second. These notions play a role in a variety of issues. I will survey a few of these issues later in this section, but let me first note that notions of ontological separation and priority structure Aristotle's ontology.

In the *Categories*, Aristotle lays out a variety of ontological categories such as substance, quality, and quantity; let me call items in categories other than substance 'nonsubstances'. Aristotle also distinguishes individuals from universals within each category. This yields a four-fold ontology with individual substances such as a particular man or horse; universal substances, which include items picked out by species terms such as 'human' and 'horse'; universal nonsubstances, which include determinable qualities such as paleness; and individual nonsubstances—it is controversial what these last items are, and I will return to this controversy. Aristotle classifies these entities with two predicative ties, two ways in which an expression picking out an entity can be predicated of a subject. Universals are *said of* a subject, as when we say that Callias is human. And nonsubstances are *present in* a subject, as when we say that Callias is pale. I will also come back to this distinction momentarily.

Corkum (2008) argues that there is good reason to ascribe to Aristotle the following:

Primacy

individual substances are separate from, and prior to, both nonsubstances and universal substances, and both nonsubstances and universal substances are inseparable from, and posterior to, individual substances.

For example, Aristotle holds in the *Categories* that individual substances are prior to, and so separate from, universal substances. At *Categories* 5 (2^a11–19), he calls individual substances primary with respect to universal substances and universal substances secondary with respect to

individual substances. And generally, Aristotle holds that the ontological status of all other kinds of entity are somehow dependent on primary substances; for example, he claims at 2^a34–b7:

All the other things are either said of the primary substances as subjects or present in them as subjects . . . [C]olor is present in body and therefore also present in an individual body; for were it not present in some individual body it would not be present in body at all. . . . So if the primary substances did not exist it would be impossible for any of the other things to exist

(*einai*).

Notice the role here of the *said of* and *present in* predicative ties. I have followed the Ackrill (1963) translation in taking the Greek *einai* existentially. Is this the right translation? To answer this question, we must turn to the interpretation of ontological separation and priority in Aristotle.

PRIMACY suggests that ontological separation and priority concern some sort of dependency. And separation and priority in Aristotle have been standardly understood in terms of ontological dependence. One formulation of ontological dependence is expressed in terms of existence conditions. On this formulation, one entity ontologically depends on a second entity just in case necessarily, if the former exists, then the latter exists. G. Fine (1984) argues for the interpretative correlate:

Existential

A is *separate* from B's just in case A can exist without B's.

On a corresponding condition for priority, A is *prior* to B's just in case A can exist without B's but B's cannot exist without A. We can see EXISTENTIAL at work in the translation of *einai* in 2^a34–b7. EXISTENTIAL has met with criticism. Corkum (2008) argues that EXISTENTIAL fails to meet the demands imposed by the condition of adequacy, PRIMACY. For example, individual substances are separate from any given nonsubstance, but an individual substance cannot exist without its *propria*, its necessary but inessential properties. And Peramatzis (2011: § 10) argues that EXISTENTIAL is incompatible with the characterization of substantial priority from *Metaphysics* 5.11, quoted earlier.

Ontological dependence need not be cashed out in terms of existence conditions. K. Fine argues for an essentialist account of ontological dependence. On this account, one entity ontologically depends on a second entity just in case the latter is a constituent in the former's essence—equivalently, just in case the latter is a constituent in a proposition that expresses a real definition of the former. Frede and Patzig (1988), Spellman (1995), and Peramatzis (2008, 2011) ascribe to Aristotle an essentialist account of ontological separation and priority. Here is one way of cashing out that line of interpretation:

Essential

A is *separate* from B's just in case an account of what A is makes no reference to an account of what B is.

This approach has also met with recent criticism. For example, Corkum (2013a: 51–52) argues that the account captures the tie between a nonsubstance and substances. A definition of a

property such as whiteness might well make reference to the surface features of physical bodies. But it is less clear that the dependence of universal substances on individual substances is captured by ESSENTIAL. That is to say, ESSENTIAL violates PRIORITY. Those who defend ESSENTIAL might reject PRIORITY. And it is a deep and long-standing interpretative question whether Aristotle, in *Metaphysics* 7, rejects the fundamentality of individual substances in favor of universal substances. *Metaphysics* 7 is a dense text that rewards study; interested readers might begin to wade through the secondary literature with Burnyeat (2001). And for further discussion of ESSENTIAL, see also Witt (2012), Malink (2013), and Panayides (2014).

Corkum (2016) canvasses the suggestion that separation and priority in Aristotle are best characterized in terms of grounding. A way of cashing out this suggestion is as follows:

A is *separate* from B's if, for any given B, the fact that A has its ontological status (as an item in one of the categories, and so as a substance, a quality, and so on) is not grounded in the fact that A stands in some tie to that B

This condition conforms to PRIMACY. For example, individual substances exhibit the condition with respect to universals and nonsubstances. Individual substances are classified as substances independently of standing in any tie to anything else—independently, that is to say, of being *present in* or *said of* any other beings. Universals and nonsubstantial individuals fail to exhibit the condition. Although generosity, for example, does not depend on any particular individual, the property would not have the ontological status it enjoys were there no generous people whatsoever. Generosity falls within Aristotle's ontology in virtue of its being present in individual substances.

To assimilate separation and priority in Aristotle to grounding has some initial plausibility. Aristotle is concerned with relations in the world that underwrite certain kinds of explanations. These relations impose structure on ontology; and where facts about individual substances are fundamental, facts about universals and nonsubstances are derivative. Moreover, the generic nature of grounding offers interpretative advantages. Grounding is a big tent. To think of ontological separation and priority in terms of grounding allows for variation from one application to another. The specific way in which individuals underwrite explanations of universals can differ from the way in which substances underwrite explanations of nonsubstances: Aristotle suggests this difference in distinguishing the *said of* and *present in* predicative ties. The generic notion of grounding also allows the historian to make a negative point—separation and priority do not concern ontological dependence—without needing to specify each variation of grounding relation in play. When a more detailed account of these various grounding relations can be given, one could kick away the prop of generic grounding. Such an approach, however, may incur costs. For example, viewing separation and priority in terms of grounding arguably imposes an anachronistic factive ontology on Aristotle. Whether the interpretative benefits outweigh the costs requires careful judgement.

Here is a selection of scholarship on these issues. For criticism of Corkum, see Koslicki (2013: 8, n. 6; 2018: 141, n. 9) and Peramatzis (2011: 243, n. 11). Katz (2017) argues that ontological separation in Aristotle is asymmetric, not nonsymmetric. Mahlan (2019) discusses the substantiality of secondary substances in terms of ontological fundamentality.

Let me mention a few other interpretative questions and indicate how the interpretation of separation and priority terminology in Aristotle influences these topics. Our first topic: What sort of things are nonsubstantial individuals? Aristotle's examples, given at *Categories* 1a25–28, are a “certain item of grammatical knowledge” present in a soul and a “certain paleness”

present in a body. But the exact nature of nonsubstantial individuals is not clear from these examples. Call a property recurrent if it can be found in more than one subject and nonrecurrent otherwise. What is not clear from these examples is whether nonsubstantial individuals are recurrent or nonrecurrent. As we have seen, nonsubstantial individuals are present in a subject, and much of the debate on this issue has centered on the apparent definition of this notion at *Categories* 1a24–5. I will quote Ackrill’s (1963: 4) translation: “By ‘in a subject’ I mean what is in something, not as a part, and cannot exist separately from what it is in”. A commonly held assumption is that a nonsubstantial individual is nonrecurrent if it is inseparable from an individual substance. This assumption rests on an interpretation of separation along the lines of EXISTENTIAL; and we can see this interpretation at work in translation of 1a24–5. If a given nonsubstantial individual is incapable of existing independently from an individual substance, then it seems that the nonsubstantial individual must be nonrecurrent. The assumption that a property is nonrecurrent if it is inseparable from an individual subject thus drives much of the dialectic of the debated reading of 1a24–5. For recent discussion, see Erginel (2004), Corkum (2009), and McPartland (2013).

A second topic. Aristotle criticizes the Platonists—for example, at *Peri Ideon* 84.23–4, *Metaphysics* 1086b30, and *De Anima* 432a14—for separating Platonic Forms from sensible particulars. On EXISTENTIAL, Aristotle’s criticism is that Platonic Forms are wrongly viewed as capable of existing uninstantiated. On ESSENTIAL, the criticism is that Platonic Forms are wrongly defined without reference to sensible particulars. It may be a promising line of interpretation to investigate whether Aristotle’s criticism is that certain facts about Platonic Forms are wrongly not grounded in certain facts about sensible particulars. Of course, the proof is in the proverbial pudding, and an interpretation along these lines would require close textual study of the mentioned passages. Interested readers might begin with G. Fine (1993).

A third topic. Aristotle distinguishes between an active and a passive intellect. He characterizes the active intellect as separate at *De Anima* 3.5 430^a17–23:

And this [active] intellect is separate, unaffected, and unmixed, being in essence activity. For that which acts is always superior to that which is affected, and the first principle to the matter. Actual knowledge is identical with its object; but potential knowledge is prior in time in the individual but not prior even in time in general; and it is not the case that it sometimes thinks and at other times not. In separation it is just what it is, and this alone is immortal and eternal.

It is not clear what it is from which the active intellect is separate. But one option is that the active intellect is separate from the body. On EXISTENTIAL, Aristotle may be endorsing a *post mortem* existence of the intellect apart from the body. On ESSENTIAL, Aristotle may be claiming that an account of the active intellect does not make reference to the body. But Aristotle holds that the active intellect is not seated in an organ, and it may be a promising line of investigation to see whether Aristotle is claiming that certain facts about the active intellect, such as facts about the nature of its activity, are not grounded in facts about the human body. For discussion of the separation of the active intellect, see Corkum (2009), Miller (2012), and Cohoe (2018).

A fourth and final topic. Aristotle characterizes the mathematician as separating in thought the mathematical object from sensible and changeable substances. For example, at *Physics* 2.2 (193b31–34) Aristotle writes that “the mathematician, though he . . . treats of these things [i.e., the properties of the earth and the world], . . . [he] separates (*chōrizei*) them; for in

thought they are separable (*chōrista*) from motion.” And at *Metaphysics* 6.1 (1026a9–10) he writes: “it is clear that some branches of mathematics are considered as immovable and separate (*chōrista*).” On EXISTENTIAL, Aristotle might be read as making a claim about the existential status of mathematical objects. On ESSENTIAL, Aristotle may be claiming that an account of a mathematical object does not make reference to sensible and changeable substances. It may be promising to investigate whether Aristotle is claiming that certain facts about mathematical objects are not grounded in facts about motion. Corkum (2012) argues that Aristotle views the mathematician as treating something derivative, properties of sensible substances, as fundamental relative to certain mathematical properties. For further general discussion of Aristotle’s philosophy of mathematics, the reader might begin with Mueller (1970) and Lear (1982); and for a recent discussion of the ontological status of mathematical objects in Aristotle, see Katz (2018).²

Related Topics

The question of grounding in Ancient Philosophy abuts on several other issues discussed in this volume. Malink [Chapter 2] considers the influence of Aristotle’s *Posterior Analytics*. For the relation between grounding and causation, see Wang [Chapter 20]. For the relation between grounding and ontological dependence, see Schnieder [Chapter 7]; for the relation between grounding and explanation, see Glazier [Chapter 8]; for the characterization of grounding as irreflexive, asymmetric, and transitive, see Thompson [Chapter 17]; for scepticism about whether there is a genus of grounding, see Koslicki [Chapter 11]; and for the relation between essence and ground, see Zylstra [Chapter 22].

Notes

- 1 A note on the quoted primary literature. References to Plato are to the Stephanus pagination (for example, *Euthyphro* 10a). Translations are from Hamilton and Cairns 1961, except as noted. References to Aristotle are to the work and, when appropriate, book and chapter, with the Bekker pagination (for example, *Metaphysics* 6.1 1026a9–10). Translations are from the Clarendon Aristotle Series, except as noted.
- 2 Thanks to the participants of the Workshop on Metaphysical Grounding at Hamburg University and especially to Kit Fine and Mike Raven for written comments.

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2

ARISTOTELIAN DEMONSTRATION

Marko Malink

1. Aristotle on Demonstration

In the *Posterior Analytics*, Aristotle develops a theory of scientific knowledge (ἐπιστήμη). He starts from the assumption that, in order to have scientific knowledge of a truth, one needs to grasp the ground of that truth. Thus, Aristotle defines scientific knowledge as follows:

We think that we have scientific knowledge of something without qualification . . .
when we think that we cognize the ground (αἰτία¹) because of which the thing is, that
it is the ground of that thing, and that it cannot be otherwise.

(*Aristotle*, *Posterior Analytics* 1.2 71b9–12)

According to Aristotle, scientific knowledge is typically obtained through demonstration (ἀπόδειξις). A demonstration is a kind of valid deductive argument, one through which, when we possess it, we have scientific knowledge of the conclusion (71b16–19). As such, a demonstration has a number of features that distinguish it from deductions in general. For example, Aristotle maintains that the premises of every demonstration are true, that they are prior to the conclusion, more known than the conclusion, and grounds of the conclusion (71b19–31).

Aristotle distinguishes between priority ‘in nature’ and priority ‘to us’. Likewise, he distinguishes between being more known ‘in nature’ and being more known ‘to us’. In the *Posterior Analytics*, he regards the latter distinction as equivalent to the former, using the phrases ‘prior’ and ‘more known’ interchangeably.² He requires that the premises of a demonstration be both more known in nature than and prior in nature to the conclusion.³

Aristotle is clear that not every valid deduction from true premises constitutes a demonstration. For example, in *Posterior Analytics* 1.13, he distinguishes two kinds of deduction: those that specify the reason why the conclusion holds and those that do not. The former kind of deduction is illustrated by the following example:

The planets are near
Whatever is near does not twinkle
Therefore, the planets do not twinkle

In Aristotle's view, the premises of this deduction explain why the planets do not twinkle. By contrast, the premises of the following deduction do not explain why its conclusion holds:

The planets do not twinkle
 Whatever does not twinkle is near
 Therefore, the planets are near

Aristotle emphasizes that this deduction, though valid and sound, does not specify the reason why the planets are near. It is, as he puts it, 'a deduction not of the reason why (τοῦ διότι) but of the fact (τοῦ ὅτι); for it is not because the planets do not twinkle that they are near, but because they are near they do not twinkle' (78a36–8). The former deduction, on the other hand, is a 'deduction of the reason why, since the primary ground (αἴτιον) has been taken' (78b3–4). Thus, while deductions 'of the fact' are not demonstrations, those 'of the reason why' may constitute demonstrations. Given that the grounds of the conclusion indicate why (διότι) the conclusion holds, every demonstration is a deduction 'of the reason why'.⁴

Aristotle's distinction between deductions 'of the fact' and those 'of the reason why' was widely influential. Thus, when Bernard Bolzano sets out to develop his theory of grounding in his 1837 treatise *Theory of Science*, he writes:⁵

I have been strengthened in the view that there is a special relation of ground and consequence between truths by noticing that some of the most penetrating thinkers have been of the same opinion. It is known that Aristotle (*Anal. Post.*, book 1, chapter 13), and for many centuries after him the scholastics, distinguished two kinds of proofs, namely, those which show only the ὅτι, i.e., *that* something is the case, and those which show the διότι, i.e., *why* something is the case. Now, since truths which specify the διότι (the why) of another truth clearly stand to the latter in the relation of a ground to its consequence, we may conclude that the Stagirite was not unfamiliar with this relation between truths.

(Bolzano, *Theory of Science* §198 n. 1)

According to Bolzano, the premises of Aristotelian demonstrations stand to the conclusion in the relation of grounding. More specifically, Bolzano argues that the relation of priority in nature employed by Aristotle in the *Posterior Analytics* can be viewed as a relation of grounding.⁶ On this view, a scientific truth ϕ is grounded in scientific truths $\psi_1, \dots, \psi_n$ just in case the latter are prior in nature to the former.

The relation of priority in nature plays an important role in Aristotle's treatment of indirect demonstrations (i.e., demonstrations that employ the method of *reductio ad impossibile*). Aristotle argues, in *Posterior Analytics* 1.26, that direct demonstrations are superior to indirect ones in that the former but not the latter proceed from premises which are prior in nature to the conclusion. In what follows, I give an overview of Aristotle's argument and its historical significance (Section 2). The relevant relation of priority in nature is determined by the order of terms in acyclic chains of immediate universal affirmations (Sections 3 and 4). Given the deductive framework of Aristotle's syllogistic theory, it can be shown that all direct demonstrations but not all indirect demonstrations proceed from premises that are prior in nature to the conclusion (Section 5). I conclude by indicating how this fact correlates with similar results in modern versions of the impure logic of ground (Section 6).

2. Indirect Demonstration

In *Posterior Analytics* 1.24–6, Aristotle compares different kinds of demonstration. He begins by explaining why, in his view, universal demonstrations are better than particular ones and positive demonstrations are better than negative ones (1.24–5). He goes on, in chapter 1.26, to argue that direct demonstrations are better than those that proceed by *reductio ad impossibile*. In the opening sentence of the chapter, he writes:

Since positive demonstration is better than negative demonstration, clearly it is also better than that which leads to the impossible.

(Aristotle, *Posterior Analytics* 1.26 87a1–2)

Aristotle is here referring to the main result of the preceding chapter, that direct positive demonstrations are better than direct negative ones. Based on this, he seeks to establish that direct positive demonstrations are better than those by *reductio ad impossibile*. He does so by arguing that direct negative demonstrations are better than those by *reductio*.

To show that direct negative demonstrations are better than those by *reductio*, Aristotle argues that the former have more explanatory power than the latter in that they proceed from premises that are prior to the conclusion:

Given that a demonstration which proceeds from what is more known and prior is superior, and given that in both kinds of demonstration [i.e., direct negative demonstration and demonstration by *reductio ad impossibile*] conviction proceeds from something's not being the case, but in the one from something prior and in the other from something posterior, it follows that negative demonstration is better without qualification than demonstration leading to the impossible. Consequently, it is also clear that positive demonstration, which is better than negative demonstration, is better than that which leads to the impossible.

(Aristotle, *Posterior Analytics* 1.26 87a25–30)

Both direct negative demonstrations and those by *reductio* make use of negative propositions (i.e., propositions asserting that ‘something is not the case’). They differ from each other with respect to the priority relations that obtain between the premises and the conclusion. In direct negative demonstrations, the premises are more known than and prior to the conclusion; in demonstrations by *reductio*, on the other hand, the premises are posterior to the conclusion. Aristotle makes it clear that the sense in which the premises of direct negative demonstrations are prior to the conclusion is priority in nature (φύσει, 87a17). Accordingly, when he refers to these premises as ‘more known’, he does not mean that they are necessarily more known to us but that they are more known in nature.

By contrast, the premises of demonstrations by *reductio* are not prior but posterior in nature to the conclusion. In Aristotle's view, premises that are not prior in nature to the conclusion do not reveal the ground (αἰτία) of the conclusion.⁷ At the same time, Aristotle holds that one cannot have scientific knowledge of a truth without grasping its ground. Hence, demonstrations by *reductio* in which the premises are posterior in nature to the conclusion do not reveal the ground of the conclusion and cannot produce scientific knowledge. This does not mean that Aristotle takes these demonstrations to be invalid. There is no doubt that he regards them as valid deductive arguments capable of producing conviction (πίστις) of the conclusion.⁸ He denies, however, that they can produce scientific knowledge (ἐπιστήμη).

Aristotle does not explicitly state that the premises of direct affirmative demonstrations are prior in nature to the conclusion. Nonetheless, it seems clear that he is committed to this view. Having stated that direct negative demonstrations are better than those by *reductio* in that they proceed from premises which are prior in nature to the conclusion, he goes on to infer that direct affirmative demonstrations are better than those by *reductio* (87a28–30). This suggests that, in Aristotle’s view, direct affirmative demonstrations are better than those by *reductio* in the same respect as direct negative ones.

In *Posterior Analytics* 1.2, Aristotle requires that the premises of demonstrations be prior in nature to the conclusion. Given that demonstrations by *reductio* do not meet this condition, they are not genuine demonstrations. Instead, they are ‘demonstrations’ in a broader sense. On the other hand, direct demonstrations proceed from premises that are prior in nature to the conclusion. Hence, provided that they satisfy the other conditions laid down in Aristotle’s characterization of demonstration in chapter 1.2, they are genuine demonstrations. As such, they are capable of producing scientific knowledge.

Aristotle’s treatment of indirect demonstrations proved influential for centuries. For example, Proclus writes in his commentary on the first book of Euclid’s *Elements*:

When geometers reason through the impossible, they are content merely to discover the property of a given subject. But when their reasoning proceeds through a direct demonstration, then, if such demonstration is partial, the ground (τὸ αἴτιον) is not yet clear, but if it is universal and applies to all like things, the ‘why’ at once becomes evident.⁹

(Proclus, On Euclid’s Elements I 202.19–25)

According to Proclus, demonstrations by *reductio ad impossibile* serve to establish the fact that a given subject has a certain property. They do not, however, reveal the ground, or the ‘why’, of that fact. Instead, the ground can be revealed by means of a direct demonstration. Provided that the direct demonstration exhibits the appropriate level of generality, it succeeds in revealing the ground of the fact demonstrated. Thus, like Aristotle, Proclus holds that direct demonstrations possess an explanatory power that those by *reductio* lack. Since Proclus was thoroughly familiar with Aristotle’s logical works and wrote a commentary on the *Posterior Analytics*, it seems clear that he is here following Aristotle’s treatment in *Posterior Analytics* 1.26.¹⁰

The same view of indirect demonstration was endorsed by a number of thinkers in the early modern period.¹¹ For example, in his 1615 essay *De mathematicarum natura dissertatio*, Giuseppe Biancani denied that demonstrations by *reductio* proceed “from the ground” (*a causa*), citing the passage from Proclus just quoted.¹² In his *Lectiones mathematicae* from the 1660s, Isaac Barrow asserts that, as for reasoning by *reductio ad impossibile*, “Aristotle teaches, and everyone grants, that reasoning of this sort does not at all furnish knowledge that is very perspicuous and pleasing to the mind.”¹³ The point is made more explicit by Arnauld and Nicole in the *Port-Royal Logic*:

Those demonstrations which show that a thing is such, not by its principles, but by some absurdity which would follow if it were not so, are very common in Euclid. It is clear, however, that while they may convince the mind, they do not enlighten it, which ought to be the chief result of knowledge; for our mind is not satisfied unless it knows not only that a thing is, but why it is, which cannot be learnt from a demonstration by reduction to the impossible.

(Arnauld & Nicole, Port-Royal Logic IV 9.3)

Arnould and Nicole criticize Euclid and other geometers for giving demonstrations by *reductio* in cases when a direct demonstration is available. These geometers, they argue, “have not sufficiently observed, that it does not suffice for the establishment of a perfect knowledge of any truth to be convinced that it is true, unless, beyond this, we penetrate into the reasons, derived from the nature of the thing itself, why it is true” (*Port-Royal Logic* IV 9.1).

A similar view is expressed by Kant in the *Critique of Pure Reason*. In the course of specifying the methods of proof admissible in the discipline of pure reason, Kant excludes proof by *reductio ad impossibile*, or, ‘apagogic’ proof, on the following grounds:

The third special rule of pure reason, if it is subjected to a discipline in regard to transcendental proofs, is that its proofs must never be apagogic but always ostensive. The direct or ostensive proof, in all kinds of cognition, is that which combines with the conviction of truth insight into the sources of the truth; the apagogic proof, by contrast, can produce certainty, but cannot enable us to comprehend the truth in its connection with the grounds of its possibility. Hence the latter is more of an emergency aid than a procedure which satisfies all the aims of reason.

(Kant, *Critique of Pure Reason* A 789 / B 817)

This view of ‘apagogic’ proofs was common among theorists working in the Kantian tradition in the 19th century.¹⁴ In his *Theory of Science*, Bolzano argues that proofs by *reductio*, while they can produce conviction, cannot exhibit “the objective ground” of the conclusion.¹⁵ Similarly, Trendelenburg states in his *Logical Investigations* that “indirect proof, as Aristotle has already shown, possesses less scientific value than direct proof. . . . Indirect proof does not provide any insight into the inner grounds of the thing.”¹⁶ Among more recent authors, Lipton holds that indirect proofs “work by showing necessity but without providing an explanation,” and that they fail to be explanatory “because they do not show what ‘makes’ the theorem true.”¹⁷ Similarly, Poston maintains that “*reductio* proofs provide conclusive grounds *that* a claim is true without removing the mystery as to *why* the claim is true.”¹⁸

Unlike intuitionist logicians, these authors do not question the validity of indirect proofs. Instead, they doubt the explanatory value of such proofs. Of course, this does not mean that they agree with Aristotle, or with each other, on why it is that indirect proofs fail to be explanatory. Far from it. Nonetheless, they share a commitment to the same general thesis, which originates with Aristotle’s treatment of demonstration by *reductio ad impossibile* in the *Posterior Analytics*.

3. Priority in Nature for Universal Affirmative Propositions

In the *Posterior Analytics*, demonstrations take the form of syllogistic deductions. Aristotle focuses on deductions that consist of the following four kinds of categorical proposition:

AaB	A holds of all B	(or, All Bs are A)
AeB	A holds of no B	(or, No Bs are A)
AiB	A holds of some B	(or, Some Bs are A)
AoB	A does not hold of some B	(or, Some Bs are not A)

If a scientific proposition is fundamental in the sense that no proposition is prior to it in nature, Aristotle refers to it as ‘immediate’.¹⁹ Since demonstrations proceed from premises which are prior in nature to the conclusion, immediate propositions cannot be demonstrated but are indemonstrable principles of a science.²⁰

In any given science, the indemonstrable a-propositions determine chains of terms connected by immediate universal affirmations. Aristotle refers to these chains as ‘series’ (συστοιχίαι).²¹ In what follows, I refer to them as ‘a-paths’. For example, if the propositions AaB and BaC are immediate, they constitute an a-path from A to C through the middle term B. This a-path corresponds to a demonstration of AaC from the indemonstrable premises AaB and BaC.

Aristotle maintains that the relation of immediate universal affirmation is acyclic (i.e., that there are no chains of immediate a-propositions $A_1aA_2, A_2aA_3, \dots, A_naA_1$).²² Thus, a-paths can be represented by acyclic directed graphs such as the following:



Figure 2.1

This diagram represents an a-path from A to D through the middle terms B and C. Immediate a-propositions are indicated by arrows pointing from the predicate to the subject term. For example, the arrow from A to B indicates the immediate proposition AaB (or, “All Bs are As”).

If an immediate proposition AaB is part of an a-path from A to D, Aristotle regards AaB as an indivisible constituent of the demonstrable theorem AaD.²³ More generally, Aristotle appeals to a-paths to determine when scientific a-propositions are prior in nature to one another. For example, in *Posterior Analytics* 1.25, he considers a structure of a-paths in which AaD is prior in nature to AaE:

Let one demonstration show that A holds of E through the middle terms B, C, and D, and let the other show that A holds of E through F and G. Thus the proposition that A holds of D and the one that A holds of E are on a par. But that A holds of D is prior to and more known than [i.e., prior in nature to and more known in nature than²⁴] that A holds of E; for the latter is demonstrated through the former.

(Aristotle, *Posterior Analytics* 1.25 86a39–86b5)

In this passage, Aristotle considers a structure in which the terms A and E are connected by two distinct a-paths:

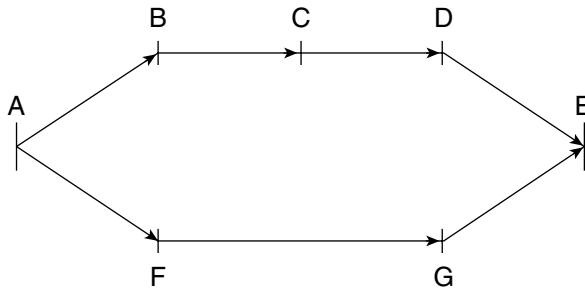


Figure 2.2

The propositions AaD and AaE are “on a par” in that both of them are demonstrable through the same number of middle terms (the former through B and C and the latter through F and G). At the same time, Aristotle maintains that AaD is prior in nature to and more known in nature than AaE. Specifically, AaD is prior in nature to AaE in virtue of the fact that the a-path from A to D is a proper part of the a-path from A to E. If there are multiple a-paths from A to D, then each of them is a proper part of some a-path from A to E.

More generally, on this account, an a-proposition AaB is prior in nature to an a-proposition CaD just in case (i) there is an a-path from A to B, and (ii) any a-path from A to B is a proper part of some a-path from C to D.

Clearly, this relation of priority in nature is transitive. Moreover, given that a-paths are acyclic, the relation is irreflexive. For, if an a-proposition AaB were prior in nature to itself, some a-path from A to B would be a proper part of another a-path from A to B, violating the condition of acyclicity. Hence, the relation satisfies Aristotle’s requirement that priority in nature be asymmetric.²⁵ In this way, structures of acyclic a-paths give rise to a well-defined relation of priority in nature among a-propositions.

If a scientific proposition AaB is indemonstrable, there is an immediate a-path from A to B; if it is demonstrable, there is a mediate a-path from A to B through one or more middle terms.²⁶ Thus, any scientific proposition AaB is underwritten by an a-path from A to B. For Aristotle, the only way to establish an a-proposition through a direct demonstration is by means of the syllogistic mood Barbara:

AaB, BaC, therefore AaC (Barbara)

If both premises, AaB and BaC, are scientific propositions, there is an a-path from A to B and an a-path from B to C. These two a-paths can be combined to form an a-path from A to C of which they are proper parts. More generally, any two a-paths from A to B and from B to C compose an a-path from A to C of which they are proper parts. Given the present account of priority in nature, this means that, if AaB and BaC are scientific propositions, they are prior in nature to AaC. Thus, whenever the syllogistic mood Barbara is applied to scientific propositions, the premises are prior in nature to the conclusion.

In the next section, we will see how this account of priority in nature can be generalized to cover all four kinds of categorical proposition countenanced by Aristotle. To this end, the framework of a-paths will be extended by introducing e-, i-, and o-paths. Readers less interested in the details of this extension may pass over to Section 5.

4. Priority in Nature for Negative Propositions

In addition to immediate a-propositions, Aristotle countenances immediate e-propositions in the *Posterior Analytics* (i.e., immediate propositions of the form “A holds of no B”).²⁷ If an e-proposition is immediate, it is an indemonstrable principle of the science under consideration. It is atomic in the sense that there is no middle term through which it can be demonstrated.

Demonstrable e-propositions can be established on the basis of the immediate ones. In Aristotle’s syllogistic theory, there are three ways to establish an e-proposition through a direct demonstration:

AeB, BaC, therefore AeC (Celarent)
 BeA, BaC, therefore AeC (Cesare)
 BaA, BeC, therefore AeC (Camestres)

If the two premises of Celarent, AeB and BaC , are immediate, they correspond to the following diagram, in which the immediate e-premise is indicated by a zigzag line:

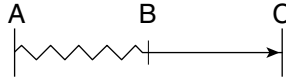


Figure 2.3

This diagram represents an e-path from A to C, composed of an atomic e-path from A to B and an atomic a-path from B to C.

Similarly, if the premises of Camestres are immediate, the e-path from A to C is composed of an atomic a-path from B to A and an atomic e-path from B to C:

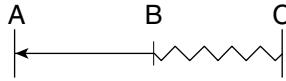


Figure 2.4

In the *Posterior Analytics*, Aristotle considers complex negative demonstrations in which an e-conclusion is derived by successive applications of Celarent, Cesare, Camestres, and Barbara.²⁸ In these demonstrations, the conclusion corresponds to compound e-paths such as the following:²⁹



Figure 2.5

This diagram represents an e-path from A to C, composed of an atomic e-path from B_3 to B_4 and a number of atomic a-paths.

Thus, an Aristotelian science determines a demonstrative structure that consists of a set of terms, T , equipped with the two binary relations of immediate a- and e-predication. In what follows, I will indicate these two relations by ' $\rightarrow$ ' and ' $\sim$ ', respectively. Thus, $A \rightarrow B$ just in case AaB is an indemonstrable principle of the science under consideration, and $A \sim B$ just in case AeB is such a principle. Moreover, we write ' $\rightarrow^+$ ' to designate the transitive closure of $\rightarrow$ (i.e., the smallest transitive relation containing $\rightarrow$). Thus, $A \rightarrow^+ B$ just in case there is an a-path from A to B (i.e., just in case there are $C_1, C_2, \dots, C_n \in T$ ($n \geq 1$) such that $A \rightarrow C_1$, $B = C_n$, and $C_i \rightarrow C_{i+1}$ ($1 \leq i < n$)).

Using this notation, Aristotelian demonstrative structures can be defined as follows:

DEFINITION 1: A *demonstrative structure* is a triple $\langle T, \rightarrow, \sim \rangle$, such that T is a nonempty set of terms, and $\rightarrow$ and $\sim$ are binary relations on T , such that for any $A, B \in T$:

- (i) If $A \rightarrow B$, there is no $C \in T$ such that $A \rightarrow^+ C$ and $C \rightarrow^+ B$.
- (ii) If $A \sim B$, both $A \neq B$ and $B \sim A$.
- (iii) If $A \sim B$, there is no $C \in T$ such that $A \rightarrow^+ C$ and either $B \rightarrow^+ C$ or $B = C$.
- (iv) If $A \sim B$, there are no $C, D \in T$ such that $C \sim D$ and $C \rightarrow^+ A$ and either $D \rightarrow^+ B$ or $D = B$.

In this definition, condition (i) ensures that the relation $\rightarrow$ is immediate as opposed to mediate a-predication. Moreover, it ensures that $\rightarrow$ is acyclic and irreflexive. Condition (ii) states that immediate e-predication, $\sim$, is irreflexive and symmetric. Condition (iii) guarantees that, if $A \sim B$, then no a-path starting from A reaches B or a third term reachable from B. Finally, (iv) ensures that $\sim$ is immediate as opposed to mediate e-predication.

In a demonstrative structure, an a-path from A to B consists of a set of pairs $\langle A, C_1 \rangle, \langle C_1, C_2 \rangle, \dots, \langle C_n, B \rangle$ such that $A \rightarrow C_1$, $C_i \rightarrow C_{i+1}$, and $C_n \rightarrow B$. Thus, since $\rightarrow$ is a subset of the Cartesian product $T \times T$, a-paths can be defined as subsets of $\rightarrow$ as follows:

DEFINITION 2: Let $\langle T, \rightarrow, \sim \rangle$ be a demonstrative structure. For any $A, B \in T$, an *a-path from A to B* is a set $\{\langle A, C_1 \rangle, \langle C_1, C_2 \rangle, \dots, \langle C_n, C_n \rangle\} \subseteq \rightarrow$ ($n \geq 1$) such that $C_n = B$.

An e-path from A to B consists of an atomic e-path from C to D together with an a-path from C to A (unless $C=A$) and an a-path from D to B (unless $D=B$). Thus, e-paths can be defined as subsets of $T \times T$ as follows:

DEFINITION 3: Let $\langle T, \rightarrow, \sim \rangle$ be a demonstrative structure. For any $A, B \in T$, an *e-path from A to B* is a set $\{\langle C, D \rangle\} \cup P \cup Q \subseteq T \times T$ such that:

- (i) $C \sim D$,
- (ii) either $C=A$ and $P=\emptyset$, or P is an a-path from C to A, and
- (iii) either $D=B$ and $Q=\emptyset$, or Q is an a-path from D to B.

This definition of e-paths allows us to extend the relation of priority in nature to scientific e-propositions. Specifically, an e-proposition AeB is prior in nature to CeD just in case (i) there is an e-path from A to B, and (ii) any e-path from A to B is a proper subset of some e-path from C to D. Likewise, AaB is prior in nature to CeD just in case there is an a-path from A to B and any such a-path is a proper subset of some e-path from C to D.

Consider an instance of Celarent in which the premises, AeB and BaC , are scientific propositions. Since both premises are scientific propositions, there is an e-path from A to B and an a-path from B to C. These two paths can be combined (by set-theoretic union) to form an e-path from A to C of which they are proper subsets. The same is true for any e-path from A to B and any a-path from B to C. Consequently, both AeB and BaC are prior in nature to AeC . Thus, whenever the syllogistic mood Celarent is applied to scientific propositions, the premises are prior in nature to the conclusion.

An i-path from A to B consists simply of two a-paths from A and from B that share a common endpoint, C:

DEFINITION 4: Let $\langle T, \rightarrow, \sim \rangle$ be a demonstrative structure. For any $A, B \in T$, an *i-path from A to B* is a set $P \cup Q \subseteq T \times T$ such that for some $C \in T$:

- (i) P is an a-path from A to C,
- (ii) Q is an a-path from B to C, and
- (iii) the sets P and Q are disjoint.

In this definition, condition (iii) excludes prolix i-paths that contain superfluous terms that are not needed to establish an i-proposition.

Similarly, an o-path from A to B consists of an e-path from A and an a-path from B that share a common endpoint, C:

DEFINITION 5: Let $\langle T, \rightarrow, \sim \rangle$ be a demonstrative structure. For any $A, B \in T$, an *o-path from A to B* is a set $P \cup Q \subseteq T \times T$ such that for some $C \in T$:

- (i) P is an e-path from A to C,
- (ii) Q is an a-path from B to C, and
- (iii) the sets P and Q are disjoint.

As before, condition (iii) excludes prolix o-paths that contain superfluous terms not needed to establish an o-proposition.

An i-proposition may be underwritten either by an i-path or by an a-path. Likewise, an o-proposition may be underwritten by an o- or e-path. If a path underwrites a proposition in a demonstrative structure, the former is called a ‘path for’ the latter:

DEFINITION 6: Let $\langle T, \rightarrow, \sim \rangle$ be a demonstrative structure. For any $A, B \in T$:

An a-path from A to B is a *path for* the propositions AaB , AiB , and BiA .

An e-path from A to B is a *path for* the propositions AeB and AoB .

An i-path from A to B is a *path for* the proposition AiB .

An o-path from A to B is a *path for* the proposition AoB .

A proposition is satisfied in a demonstrative structure if there is a path for it in this structure:

DEFINITION 7: Let $\langle T, \rightarrow, \sim \rangle$ be a demonstrative structure with $A, B \in T$. For any proposition AxB (where ‘x’ stands for ‘a’, ‘e’, ‘i’, or ‘o’): AxB is *satisfied in* $\langle T, \rightarrow, \sim \rangle$ just in case there is a path for AxB in $\langle T, \rightarrow, \sim \rangle$.

Aristotle holds that, for any terms A and B, either AaB or AoB is true and that either AeB or AiB is true. By contrast, Definition 7 does not guarantee that either AaB or AoB is satisfied in a demonstrative structure or that either AeB or AiB is satisfied in such a structure. Thus, Definition 7 does not capture the truth-conditions of categorical propositions. Instead, it captures their status as scientific propositions: a proposition is satisfied in a demonstrative structure just in case it is either an indemonstrable principle or a demonstrable theorem of the science under consideration. If neither AaB nor AoB is satisfied in a demonstrative structure, one of them is true, but neither is a scientific proposition of the relevant science.

Finally, we are in a position to give a general characterization of priority in nature between scientific a-, e-, i-, and o-propositions:

DEFINITION 8: Let $\langle T, \rightarrow, \sim \rangle$ be a demonstrative structure with $A, B, C, D \in T$. For any propositions AxB and CyD (where ‘x’ and ‘y’ stand for ‘a’, ‘e’, ‘i’, or ‘o’), AxB is *prior in nature to* CyD in $\langle T, \rightarrow, \sim \rangle$ just in case:

- (i) AxB is satisfied in $\langle T, \rightarrow, \sim \rangle$, and
- (ii) every path for AxB is a proper subset of some path for CyD .

Clearly, this relation of priority in nature is transitive. Moreover, given the acyclicity of $\rightarrow$, it is irreflexive and hence asymmetric.³⁰ As such, it constitutes a well-defined relation of priority in nature between scientific propositions.

5. Direct vs. Indirect Demonstration

In *Posterior Analytics* 1.26, Aristotle states that all direct but not all indirect demonstrations proceed from premises that are prior in nature to the conclusion. As we have seen, the term ‘demonstration’ is not used here in the official sense defined in *Posterior Analytics* 1.2, according to which the premises of every demonstration are prior in nature to the conclusion. Instead, a ‘demonstration’ in chapter 1.26 can be taken to be any deduction in which the premises are scientific propositions. On this account, Aristotle’s thesis in 1.26 is that all direct but not all indirect deductions in which the premises are scientific propositions proceed from premises that are prior in nature to the conclusion.

Direct and indirect deductions are studied by Aristotle in the *Prior Analytics*, in which he develops his theory of the categorical syllogism. In *Prior Analytics* 1.7, he presents a deductive system for this theory that consists of the following rules of inference:³¹

1. syllogistic moods: AaB, BaC, therefore AaC (Barbara)
 AeB, BaC, therefore AeC (Celarent)
2. conversion rule: AeB, therefore BeA (e-conversion)
3. rule of *reductio ad impossibile*

A direct deduction in this system is any argument in which a conclusion is inferred from given premises by means of the rules of Barbara, Celarent, and e-conversion. An indirect deduction is one which, in addition to these rules, employs the rule of *reductio ad impossibile*. The application of the latter rule is based on three laws of opposition between categorical propositions:³²

AaB is the contradictory opposite of AoB
 AeB is the contradictory opposite of AiB
 AaB is incompatible with AeB

This deductive system is strong enough to derive all syllogisms endorsed by Aristotle. Direct deductions consist exclusively of a- and e-propositions, and all deductions deriving an i- or o-conclusion are indirect. Thus, for example, AiC can be deduced from AaB and BiC by means of *reductio* using Celarent and the rule of e-conversion.³³

The deductive system just described is congruous with the relation of priority in nature introduced in Definition 8. As we have seen, the moods Barbara and Celarent have the special property that, when applied to propositions satisfied in a demonstrative structure, the premises are prior in nature to the conclusion:

THEOREM 1: Let $\langle T, \rightarrow, \sim \rangle$ be a demonstrative structure. For any deduction of the form Barbara and Celarent: if both premises are satisfied in $\langle T, \rightarrow, \sim \rangle$, then each premise is prior in nature to the conclusion in $\langle T, \rightarrow, \sim \rangle$.

This theorem holds because reasoning by Barbara from premises satisfied in a demonstrative structure amounts to extending a-paths in this structure, and reasoning by Celarent from such premises amounts to extending e-paths.

Since all scientific propositions are satisfied in a demonstrative structure, every instance of Barbara and Celarent in which both premises are scientific propositions proceeds from premises that are prior in nature to the conclusion. Given that a ‘demonstration’ is any deduction from scientific propositions that involves at least one application of Barbara or Celarent, every direct demonstration proceeds from premises that are prior in nature to the conclusion.

By contrast, this is not the case for indirect demonstrations. Consider, for example, an indirect deduction in which AoB is derived from AeC and BaC (*Posterior Analytics* 1.26 87a6–17):

1. AeC (premise)
2. BaC (premise)
3. | AaB (assumption for *reductio*)
4. | BaC (iterated from 2)
5. | AaC (from 3, 4, by Barbara)
6. AoB (*reductio*: 1, 3–5)

There are demonstrative structures in which the two premises, AeC and BaC, are satisfied while neither premise is prior in nature to the conclusion AoB, but instead the conclusion is prior in nature to one of the premises. Such a structure is specified by Aristotle in *Posterior Analytics* 1.26 as follows (87a3–12):³⁴

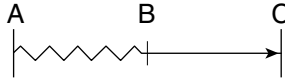


Figure 2.6

Both premises AeC and BaC are satisfied in this structure, since there is an e-path from A to C and an a-path from B to C. These two paths, however, cannot be combined to form an o-path from A to B. This is because the two paths overlap in BC and hence are not disjoint (see condition (iii) in Definition 5). The only path for AoB in the structure is the atomic e-path from A to B. Thus, the two paths for AeC and BaC are not proper parts of any path for AoB. As a result, the premises AeC and BaC are not prior in nature to the conclusion AoB. On the contrary, since the e-path from A to B is a proper part of the e-path from A to C, the conclusion AoB is prior in nature to the premise AeC.

Similarly, there are indirect deductions in which AiB is deduced from AaC and BaC such that both premises are satisfied in a demonstrative structure but the conclusion AiB is prior in nature to the premise AaC.³⁵ Thus we have:

THEOREM 2: There are indirect deductions such that both premises are satisfied in a demonstrative structure $\langle T, \rightarrow, \sim \rangle$ and the conclusion is prior in nature to one of the premises in this structure.

Given that a ‘demonstration’ is a deduction from scientific propositions, there are indirect demonstrations in which the conclusion is prior in nature to one of the premises. Thus, while some indirect demonstrations have premises prior in nature to the conclusion, many of them don’t. In this way, Theorems 1 and 2 account for Aristotle’s thesis in *Posterior Analytics* 1.26 that all direct demonstrations, but not all indirect demonstrations, proceed from premises that are prior in nature to the conclusion.

6. Priority in Nature and Grounding

Aristotle’s relation of priority in nature between scientific propositions can be viewed as a kind of grounding relation. Specifically, the account of priority in nature given in Definition 8

corresponds to the relation of strict partial ground described by Kit Fine in his semantics for the pure logic of ground. Fine characterizes the latter relation in terms of the idea of a fact verifying a proposition. Facts can be fused to form more complex facts of which they are parts. On Fine's account, if a true proposition ϕ is a strict partial ground of a true proposition ψ , every fact verifying ϕ is part of a fact verifying ψ .³⁶ Similarly, Definition 8 is based on the idea that paths underwrite scientific propositions. Paths can be 'fused' by set-theoretic union to form more complex paths of which they are proper parts (i.e., proper subsets). If a scientific proposition AxB is prior in nature to a scientific proposition CyD , every path for AxB is a subpath of a path for CyD .³⁷

The results stated in Theorems 1 and 2 have close analogues in modern theories of the impure logic of ground. These theories explore how the relation of grounding interacts with logical connectives and the rules of inference governing them in deductive systems. Consider, for example, a deductive system for the language of propositional logic using the connectives of negation ($\neg$) and conjunction ($\wedge$). The system contains four direct rules of inference, where ϕ and ψ are any propositions:

ϕ, ψ , therefore $\phi \wedge \psi$
 ϕ , therefore $\neg \neg \phi$
 $\neg \phi$, therefore $\neg(\phi \wedge \psi)$
 $\neg \psi$, therefore $\neg(\phi \wedge \psi)$

In addition, the system contains the following rule of *reductio ad impossibile*, where ϕ and ψ are any propositions and Γ any set of propositions:

$\Gamma, \neg \phi$, therefore $\psi \wedge \neg \psi$

 Γ , therefore ϕ

Taken together, these rules suffice to derive all the laws of classical propositional logic. The four direct rules are special in that, when applied to true propositions, the conclusion is not only a logical consequence of the premises but is also grounded in these premises. For example, if ϕ and ψ are true, then $\phi \wedge \psi$ is grounded in these two propositions (or, the fact that $\phi \wedge \psi$ obtains in virtue of the fact that ϕ and the fact that ψ). More generally, the relation of grounding is usually taken to obey the following laws corresponding to the four direct rules of inference:³⁸

If ϕ and ψ , then ϕ, ψ ground $\phi \wedge \psi$
 If ϕ , then ϕ grounds $\neg \neg \phi$
 If $\neg \phi$, then $\neg \phi$ grounds $\neg(\phi \wedge \psi)$
 If $\neg \psi$, then $\neg \psi$ grounds $\neg(\phi \wedge \psi)$

By contrast, there is no such agreement between the laws of grounding and the derived rules of inference that rely on *reductio ad impossibile* in the deductive system. Take, for example, the following derived rules, which are licensed by the above rule of *reductio*:

$\phi \wedge \psi$, therefore ϕ
 $\phi \wedge \psi$, therefore ψ
 $\neg \neg \phi$, therefore ϕ

When these rules are applied to a true proposition, this proposition does not ground the conclusion. For example, a true conjunction $\phi \wedge \psi$ does not ground either conjunct. Thus, while sound deductions instantiating the four direct rules proceed from grounds to what is grounded, sound deductions using the rule of *reductio* do not.

The four direct rules of inference are analogous to the syllogistic moods Barbara and Celarent in Aristotle's deductive system. When the former are applied to true propositions, the premises ground the conclusion; when the latter are applied to scientific propositions, the premises are prior in nature to the conclusion. In both cases, direct deductions proceed in the direction of increasing complexity. In the first case, the conclusion is syntactically more complex than the premises.³⁹ In the second case, the conclusion is path-theoretically more complex than the premises, in the sense that every path for a premise is a proper part of a path for the conclusion. In both systems, the direct rules of inference are especially useful in contexts in which preference is given to deductions that proceed from grounds to what is grounded, or from what is prior in nature to what is posterior in nature. However, these rules do not suffice to generate all deductive consequences of a given set of premises. To this end, one needs to employ the rule of *reductio ad impossibile* in the two systems. Once this rule is admitted, deductions from true propositions no longer follow the order of grounding, and deductions from scientific propositions no longer follow the order of priority in nature.⁴⁰

Related Topics

Ancient [Chapter 1], Bolzano [Chapter 5], Medieval and Early Modern [Chapter 3], Semantics [Chapter 36]

Notes

- 1 Here and in what follows, I translate Aristotle's terms αἰτία and αἴτιον as 'ground' (McKirahan 1992: 231–2). Alternatively, these terms might be translated as 'cause' (Bronstein 2016: 35–6) or 'explanation' (Barnes 1993: 89–93).
- 2 *An. Post.* 1.2 71b33–72a5; see McKirahan 1992: 30–1; Barnes 1993: 95–7.
- 3 See Zabarella 1608: 653f–654b; McKirahan 1992: 31; Bronstein 2016: 127–8.
- 4 See *An. Post.* 1.6 75a35, 1.13 78a25–6.
- 5 For Bolzano's theory of grounding, see Roski (Chapter 5 in this volume).
- 6 Bolzano 1837: §209 n. 1 (vol. II, p. 364); see also §198 n. 1 (vol. II, p. 341).
- 7 *An. Post.* 1.2 71b31, 2.15 98b17; see Zabarella 1608: 660c–d; Bronstein 2016: 128.
- 8 This is clear from the fact that Aristotle regards demonstrations by *reductio*, just like direct negative ones, as 'convincing' (πισταί, 87a26).
- 9 The translation of this passage follows the one given by Heath 1908: 150 n. 1.
- 10 For Proclus' reception of Aristotle's *Posterior Analytics*, see Helmig 2010: 27–9.
- 11 See Mancosu 1991: 15–36, 1996: 24–8, 58–64, 83–4, 100–15, 2000: 108–16.
- 12 Biancani 1615: 10, 12, 20.
- 13 Barrow 1860: 377. Accordingly, Barrow holds that reasoning by *reductio*, "as all acknowledge, is very obscure and ignoble" (1860: 357).
- 14 For example, Jakob 1800: 130; Krug 1806: 595–7; Kiesewetter 1824: I 149, II 492–3; Reinhold 1827: 409–10.
- 15 Bolzano 1837: §530, vol. IV, pp. 270–1, 278.
- 16 Trendelenburg 1870: 440. By contrast, Lotze (1874: 263–76) argues that some indirect proofs are explanatory in that they derive the conclusion from its grounds.
- 17 Lipton 2009: 48–9, 2011: 50.
- 18 Poston 2014: 79. Against this, others argue that some proofs by *reductio* are explanatory (e.g., Colyvan 2012: 80; Lange 2017: 234, 445 n. 9; Hanna 2018: 13–14).
- 19 *An. Post.* 1.2 72a7–8.

- 20 *An. Post.* 1.3 72b18–22.
- 21 *An. Post.* 1.15 79b7–11, 1.17 80b27, 81a21, 1.29 87b5–14. See Smith 1982: 122–6; Mignucci 1975: 341; Barnes 1993: 163.
- 22 *An. Post.* 1.22 83a36–b12; see Ross 1949: 578–9; Lear 1980: 31. Cf. also Smith 1984: 593–4, 1986: 62, 1996: 27.
- 23 *An. Post.* 1.23 84b19–85a1; see Malink 2017: 174–86.
- 24 See Burnyeat 1981: 127–8.
- 25 *An. Post.* 1.3 72b27–8; see also 2.15 98b16–21.
- 26 This is because, in Aristotle's syllogistic theory, the only way to deduce AaB is by means of a chain of premises of the form $AaC_1, C_1aC_2, \dots, C_naB$ (see Smiley 1973: 139–45; Thom 1981: 181–3).
- 27 *An. Post.* 1.15 79a33–b22, 1.23 84b24–31.
- 28 *An. Post.* 1.21 82b4–21, 1.23 85a3–7, 1.25 86b10–27.
- 29 See Smith 1996: 29, 2009: 57–8.
- 30 If a proposition AxB were prior in nature to itself, every path for AxB would be a proper subset of some path for AxB . But, given the acyclicity of $\rightarrow$, no a- or e-path for AxB is a proper subset of another a- or e-path for AxB , and no i- or o-path for AxB is a proper subset of any path for AxB .
- 31 *An. Pr.* 1.7 29b1–25. For an exposition of this deductive system, see Smiley 1973: 141–2.
- 32 The last of these laws, concerning the incompatibility of AaB and AeB , allows for the derivation of the rule of a-conversion: AaB , therefore BiA (*An. Pr.* 1.2 25a17–22). See Thom 1981: 39–41; Crivelli 2011: 156–8.
- 33 *An. Pr.* 1.7 29b1–19.
- 34 See Malink 2020.
- 35 To see this, consider a demonstrative structure that contains exactly two atomic a-paths: $A \rightarrow B$ and $B \rightarrow C$.
- 36 Fine 2012a: 71–3, 2012b: 7–9.
- 37 Thus, Definition 8 can be viewed as a special case of Fine's (2012b: 7–9) factual semantics for ground, when a 'fact' is taken to be any set of atomic a- or e-paths, and the operation of 'factual fusion' is taken to be set-theoretic union.
- 38 Fine 2012a: 58, 62–3; Correia 2014: 33–6, 2017: 530; similarly, Schnieder 2011: 449; Korbmacher 2018: 170.
- 39 See Correia 2014: 35–7.
- 40 Thanks to Mike Raven and Jacob Rosen for helpful comments on earlier drafts of this chapter. I am especially grateful to Kit Fine for detailed written comments, which led to considerable improvements.

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3

MEDIEVAL AND EARLY MODERN

Margaret Cameron

In his 2015 survey article, “Ground”, Raven notes the following about ground:

Indeed, one of ground’s distinctive hallmarks is how it links metaphysics to explanation. Somehow, ground is *metaphysical* because it concerns the phenomena in the world itself, but also *explanatory* because it concerns how some phenomena hold in virtue of others.

(Raven 2015)

However, Raven immediately points to what he describes as a “tension” and a “struggle”:

But these metaphysical and explanatory aspects seem to be in tension. On the one hand, it is supposed that metaphysics concerns phenomena in the world itself independently of the explanatory interests and goals of inquirers like us. On the other hand, it is supposed that explanations are sensitive to the explanatory interests and goals of inquirers like us. In light of this tension, how is the link supposed to be understood?

(Raven 2015)

Raven identifies two camps in the contemporary literature on ground: separatists, who separate ground from explanation, and unionists, who “link ground to metaphysical explanation by taking ground *to be* (a kind of) metaphysical explanation”. To be sure, medieval, Renaissance, and early modern philosophers did not have at their disposal the theoretical terminology afforded by the recent grounding literature, including the very term “ground” as it is currently used. It is, however, well known that ground has its roots in, broadly speaking, the Aristotelian metaphysical tradition. Given the fact that Aristotle’s philosophy, and specifically his metaphysics, provided the foundation for the subject during most of this time, it should come as no surprise that we find philosophers engaging in many of the same debates and discussions regarding questions of fundamentality, ontological priority, metaphysical explanation, and dependence relations. Raven himself was not unaware of this long tradition: “The debate [about whether ground is both metaphysical or explanatory or both] . . . appears to be entangled with profound questions concerning the link between metaphysics and explanation (more generally, epistemology). These questions have a long history” (Raven 2015). For many thinkers during the medieval and early

modern periods, too, discussions of dependence relations within the context of studying metaphysics often assumed this metaphysical/explanatory double life.

It is perhaps less than clear, however, that philosophers in the medieval and early modern periods recognized a “tension” or a “struggle” between metaphysics and epistemology, although they did certainly find themselves in multiple entanglements when trying to set things straight, especially when it came to the question of the subject matter of metaphysics. There are myriad ways to showcase this topic, but here we will focus on one particularly apt site of discussion and debate, namely, the various answers to the question, “What is the subject matter of metaphysics?”

John Duns Scotus (c. 1266–1308) opens the first question of his commentary on Aristotle’s *Metaphysics* in this way:

Concerning the object of this science [i.e., metaphysics], it has been shown above that this science deals with transcendentals.¹ However, it has likewise been shown that it deals with the highest causes. There are various opinions as to which of these ought to be its proper object.

Therefore our first question: Is the proper subject [matter] of metaphysics being qua being as Avicenna claims or God and the Intelligences as the Commentator Averroes assumes?

(John Duns Scotus 1997)

The expression “subject matter” translates the Latin “*genus subiectum*”, which in turn is a translation of the ancient Greek “*tò genos tò hypokeímenon*”: the underlying kind. To inquiry into the subject matter of metaphysics, which was characterized as First Philosophy, is to ask the question: “What grounds what, in the most fundamental sense?”

By this stage in the medieval tradition, it had become customary to start one’s commentary on the *Metaphysics*, which consisted in a close reading of the text usually organized around a series of questions, with this very question. Scotus summarizes two of the main answers debated amongst medieval and scholastic philosophers regarding the subject matter of the science of metaphysics. He traces these answers, correctly, to the Arabic-speaking philosophers Avicenna (Ibn-Sīnā, ca. 970–1037) and Averroes (Ibn Rushd, 1126–1198), the latter also known simply as “the Commentator” (while Aristotle was “the Philosopher”). That this question was a matter of debate is unsurprising: Aristotle himself gives multiple and varied answers in his *Metaphysics*. But what turns on answering this question? Simply put, the answer to the question of the subject matter of metaphysics establishes what is taken to be the foundation of all being. This is one answer to the question, namely, the ontological answer, i.e., the subject matter of metaphysics is being qua being. But there were two others: the explanatory, i.e., the subject matter of metaphysics is knowledge of the fundamental structure of being and its properties; and the theological, i.e., the subject matter of metaphysics is the sphere of the divine.

Before turning to the rival positions held by Avicenna and Averroes mentioned earlier, let us first take a look at what Aristotle said on the issue.² In *Metaphysics*, Aristotle holds both that the subject matter of metaphysics is being qua being as well as the sphere of the divine. First, metaphysics as being qua being:

There is a science which investigates being as being and the attributes which belong to this in virtue of its own nature. Now this is not the same as any of the so-called special sciences; for none of these others deals generally with being as being.

(Aristotle, *Metaphysics* 1003a)

This is the ontological characterization of the subject of metaphysics. Its remit—being qua being—is distinct from the special sciences, for example, from biology, which studies being qua *living* being.

The subject matter of metaphysics is, at the same time, according to Aristotle, explanatory, or epistemic. Specifically, Aristotle describes the “fundamental duty of the philosopher”, which is “to gain possession of the principles and causes of substance” (Aristotle, *Metaphysics* 1003b). Thus, the obligation of the philosopher is explanatory. Aristotle’s remarks characterize metaphysics as a branch of epistemology, that is, as having to do with the *knowledge* of what grounds what. Indeed, we will see that several of his commentators followed him on this point and identified the subject matter of metaphysics as being *knowledge* of the fundamental.

In another part of the text, Aristotle gives yet another opinion on the subject matter of metaphysics, describing it as “theology”:

Now all causes must be eternal, but especially these [i.e., the separable and immovable things]; for they are the causes of so much of the divine as appears to us. There must, then, be three theoretical philosophies, mathematics, natural science, and theology, since it is obvious that if the divine is present anywhere, it is present in things of this sort.

(Aristotle, *Metaphysics* 1026a)

This is the theological characterization of the subject of metaphysics. Aristotle argues that if there is an unchanging substance in the universe, i.e., the divine, then “the science of this must be prior and must be First Philosophy, and universal in this way, because it is first”.

Accordingly, for Aristotle, metaphysics is First Philosophy in three senses. It is most fundamental because what metaphysics studies, namely being, is what is most fundamental. It is most fundamental because the study of metaphysics is what is epistemically fundamental. And it is most fundamental because its subject matter is the first cause, namely, God.

These three opinions on the subject matter of metaphysics—namely, that the subject matter of metaphysics is both ontological, explanatory, and also theological—are found scattered throughout the *Metaphysics*, a treatise that was, famously, only posthumously assembled and so-named. However, Aristotle himself did not treat the question of the subject matter of metaphysics in an especially systematic fashion.³ It was Avicenna who first provided systematicity to the question, and he did so by applying the epistemological method suggested in Aristotle’s *Posterior Analytics* to his reading of the *Metaphysics*.⁴ The *Posterior Analytics* belongs to the collection of so-called “logical” treatises known collectively as “the organon”. It is a notoriously difficult but fundamentally important work in which Aristotle sets out in systematic fashion the structure of science and the ways of ascertaining proofs via syllogistic logic in its support. In a word, it is a treatise on how to acquire scientific knowledge, the characteristics of which were formulated by a demonstration whose premises are true, primary, immediate, better known, prior to the conclusion, and causes of the conclusion.⁵

Avicenna’s *The Metaphysics of the Healing* is not a commentary on Aristotle’s work, although it is deeply informed by it. It is a masterful work in its own right that advances a distinctive and highly influential theory of existence and divine causation.⁶ Avicenna’s application of the methods and principles drawn from the *Posterior Analytics* to the study of metaphysics was momentous: he at once argued that the study of metaphysics has a systematic grounding in an epistemological method, and by doing so, he fused or (as some might say) explicitly confused metaphysics and epistemology. Avicenna used Aristotle’s lessons to index a science to its subject matter: “It has also become known that for each science there is a subject matter proper to it”

(Avicenna 2005, 1.1(10)). The lesson drawn from *Posterior Analytics* A 10 (76b) is this: “Every science deals with a subject matter whose existence it takes for granted.” The subject matter of a science is made clear, according to Aristotle in *Posterior Analytics* A 7 (75a), in the act of providing a demonstration:

There are three things involved in demonstrations: one, what is being demonstrated, or the conclusion (this is what holds of some kind in itself); two, the axioms (axioms are the items from which the demonstrations proceed); third, the underlying kind [*genus subiectum*, i.e., subject matter] whose attributes—i.e., the items incidental to it in itself—the demonstrations make plain.

Since every science “takes for granted” the existence of its subject matter, it consequently does not have to provide a *proof* of its existence. Accordingly, Avicenna famously excludes God as the subject matter of metaphysics:

The existence of God—exalted be His greatness—cannot be admitted as the subject matter of this science; rather, it is [something] sought in it. This is because, if this were not the case, then [God’s existence] would have to be either admitted in this science but searched for in another, or else admitted in this science but not searched for in another.

(Avicenna 2005)

Avicenna rejects both of these alternatives, since none of the other sciences is capable of proving God’s existence, and neither is God’s existence self-evident. Avicenna concludes that “If, then, the inquiry concerning [God’s] existence is in this science, it cannot be the subject matter of this science. For it is not for any science to establish its own subject matter”, which is a lesson drawn straight from Aristotle’s *Posterior Analytics*.

Avicenna then considers whether the subject matter of metaphysics could be the four Aristotelian causes, glossed as “the ultimate causes for all existents”. These four causes are introduced by Aristotle in his work *Physics* and became known by the labels affixed by medieval authors, namely, the formal, material, efficient, and final causes.

Some of these (parts, for instance) are causes in the sense that they underlie a thing [the material cause]; others are causes in the sense that they constitute what a thing is (for instance, the whole, the compound, and the form) [the formal cause]. Then consider a seed, a doctor, a planner, and any other kind of agent: they are all causes in the sense that they initiate change or stability [the efficient cause]. Then there are things which are causes in the sense that they are the ends of other things, and are the good for which they are done [the final cause].

(Aristotle, *Physics* II.3, 195a)

Aristotle is concerned with the phenomenon of cause and identifies four different types (even when one and the same thing turns out to be multiply causal, e.g., in the case of biological reproduction). Avicenna, however, rejects the idea that Aristotle’s four causes can be the subject matter of metaphysics and presents a host of arguments to make this point, one of which has to do with the priority of what is investigated by metaphysics, which is knowledge of the absolute causes. According to Avicenna, “knowledge of the absolute causes comes about after the existence of causes for those things that *have* causes”. What underlies the causes is that which can be

subject to causation, and this is being. The science of being is subsequently perfected or completed by the discovery of the ultimate causes. In the end, by process of elimination, Avicenna concludes that metaphysics is the study of being *qua* being.

In chapters two and three of Book 1 of *The Metaphysics of The Healing*, Avicenna advances his positive case. Distinguishing metaphysics from the other theoretical sciences (natural science, mathematics, and logic), Avicenna states that metaphysics is about “investigating the state of substance inasmuch as it is an existence and a substance” (Avicenna 2005). It also investigates both material and immaterial bodies insofar as they are substances, as well as measure and number. Clearly, the focus of metaphysics is on substance and its accidents (e.g., quality, quantity, relation, and so on) as well as what are “akin to the proper accidents”, namely, the one and the many, the potential and the actual, the universal and the particular, and the possible and the necessary.

For Avicenna, metaphysics is First Philosophy because “it is the knowledge of the first thing in existence (namely, the First Cause) and the first thing in generality (namely, existence and unity)”. Knowledge of this is superior because it bears the mark of absolute certainty. What is known in First Philosophy is prior to everything else that can be known. Accordingly, First Philosophy is to the other sciences as a master to its servants, to use Avicenna’s own analogy. What does he mean by characterizing First Philosophy in this way? Avicenna embraces the double life of metaphysics as both ontological and explanatory. It is ontologically prior: it is ontologically prior to everything that is enmattered and in no way depends on the existence of material things for its own existence. And it is explanatorily prior: it is by knowing the principles of First Philosophy that one is able to validate knowledge of the other sciences. However, although First Philosophy has this foundational ontological and epistemic character, it is not the first subject to be studied. In fact, according to Avicenna, the study of metaphysics should be taken up only after a grounding in the study of natural science and mathematics is achieved.

As we have just seen, Avicenna rejects the idea that God (and the separate substances) can be the subject matter of metaphysics; in other words, the fact of God’s existence cannot be assumed as a principle with which metaphysical investigations can be undertaken. Rather, for Avicenna, one of the tasks of metaphysics is to prove the existence of God, which he assigns to the study of being (which can, within the purview of metaphysics as the study of being *qua* being, be assumed as a principle). Writing in response to Avicenna’s views on the subject matter of metaphysics, Averroes outright rejects the idea that God’s existence is what is to be proven by metaphysics. Consequently, Averroes rejects Avicenna’s claim that God cannot be the subject matter of metaphysics. To do so, Averroes relies on arguments drawn from Aristotle’s *Physics*, and especially arguments based on the theory of motion given there, to mount his arguments against Avicenna. Briefly, Averroes appeals to proofs given by physics in which causes can be inferred from effects, for example, that from the existence of movement can be inferred the existence of God as first cause.⁷ Moreover, it is within the subject matter of physics that the existence of the separate (divine) substances can be demonstrated, thereby permitting the subject of metaphysics to assume them as its subject matter:

What has become plain in physics with respect to the existence of separate principles is not superfluous in this science [of metaphysics], as Ibn-Sīnā maintains, but rather necessary, because it is employed in this science as a supposition.

(Averroes 2010)⁸

Given that Averroes’ views on this topic do not seem to have traction in the other authors surveyed here, let this brief mention of his view that the subject matter of metaphysics is God and the separate substances suffice.

Thomas Aquinas (1225–1274) also takes up the question of the subject matter of metaphysics and returns to the question on several occasions. It is explicitly addressed in the prologue to his commentary on Aristotle's *Metaphysics*. But of great value too is his answer to Question 9 in his commentary on Boethius' *On the Trinity*.⁹ Boethius' tractate was a subject of commentary for many reasons, not least of which is the fact that he characterizes and defends First Philosophy as theology, or the science of the immaterial, unmoved divine being.

Following Aristotle's own view that, in any investigation, we ought to proceed from "what is more intelligible and clear to us and move from there to what is clearer and more intelligible in itself" (Aristotle, *Physics* I.1, 184a), Aquinas too cites this methodological lesson in his preliminary discussion of the question of the division of the sciences and, ultimately, of the subject of metaphysics. In response to Boethius' protreptic opening remarks ("Therefore, come, let us delve in and examine each subject as far as it can be grasped and understood; for . . . it is a scholar's duty to try to formulate his opinion about each thing as it actually is"), Aquinas explains in his commentary on *On the Trinity*:

He uses these two words ["grasped" and "understood"] because the method of investigating anything should conform both to things and to us. If it did not conform to things, they could not be understood; if it did not conform to us, we could not grasp them.

(Aquinas 1986)

Thus, on Aquinas' view, there is a double conformation: our explanation of things must conform to the *things understood*, and it must conform to the way in which *we understand* things.

Addressing the question of the subject matter of metaphysics, Aquinas notes that the discipline is known by three names: metaphysics, divine science, and First Philosophy. In the commentary on *On the Trinity*, he focuses on the divine nature of the subject and, accordingly, explains why it is characterized in this way. It is divine because it does not deal with motion and it is abstracted from matter—not in thought but in existence. In the prologue to his commentary on Aristotle's *Metaphysics*, Aquinas elaborates on the priority of the subject more broadly. In providing a characterization of the metaphysician, i.e., the person with wisdom, Aquinas explains that the metaphysician is one who knows all things, knows difficult things, and knows them with greater certainty. The metaphysician chooses to know the subject for its own sake, and not for some other purpose. Finally, what the metaphysician knows must be "rather the more basic, i.e., nobler, than a subordinate science" (Lesson 2). As First Philosophy, metaphysics is the study of "the first causes of things" (Lesson 1), and "the more any sciences are prior by nature, the more certain they are" (Lesson 2). Thus, epistemic or explanatory certainty tracks, and is explained by, ontological priority.

Since it is well known now that Aquinas closely follows Avicenna's views with regard to the subject matter of metaphysics,¹⁰ here the focus will be on Aquinas' solution to an apparent paradox, which was also recognized by Avicenna. The paradox emerges with regard to metaphysics considered according to the order of nature and metaphysics considered according to the order of learning. The point is first raised in an objection to the question of whether speculative science should in fact be divided into the three parts: natural, mathematical, and divine. In his commentary on *On the Trinity*, Aquinas raises this objection:

That science on which others depend must be prior to them. Now all the other sciences depend on divine science because it is its business to prove their principles.

Therefore, Boethius should have placed divine science before the others [i.e., before natural science and mathematics].

(Aquinas 1986)

Like Avicenna, Aquinas explains that metaphysics is so called because it is to be learned after the study of physics, which is concerned with the study of material objects in motion. This establishes the pedagogical order. Moreover, according to Aquinas (explicitly following Avicenna again), the natural sciences “explain many things used by metaphysics, such as generation, corruption, motion, and the like” (Aquinas 1986). Yet Aquinas notes that metaphysics is characterized as First Philosophy because “all the other sciences, receiving their principles from it, come after it”. This establishes the natural or ontological order. The principles to which Aquinas refers are the most universal and have both ontological and explanatory priority (so in addition to generation, corruption, and motion, we can add place, time, and other universal axioms, such as whatever is moved is moved by another). The question is, then, does the subject matter of metaphysics, given that it is characterized as metaphysics both in the sense of being beyond physics and in the sense of being First Philosophy, contain a contradiction or a vicious circularity?¹¹

Aquinas’ resolution to the apparent paradox leans heavily on the distinction between priority with respect to us as knowers or inquirers (*quoad nos*) and priority with respect to the natural or ontological order (*naturaliter*). With regard to us, that is, to the order of learning, the natural sciences are prior, since it is by them and in them that many of the principles used by us when we engage in metaphysical speculation are proven. Aquinas adds that mathematics, too, should be studied before we study metaphysics because knowledge of mathematics is presupposed when studying astronomy, which must be learned before metaphysics in order to be able to understand separate substances.

So much for the order of learning. But how can it be the case that the subject of metaphysics is not viciously circular since, as First Philosophy, it both proves the very principles used by the natural sciences *and* presupposes the conclusions of proofs given by the natural sciences? Herein lies the heart of the paradox, and it does not seem that a mere appeal to the order of learning will help. Aquinas’ answer is both (unhappily) brief and dense:

The principles that another science (such as natural philosophy) takes from First Philosophy do not prove the points which the first philosopher takes from the natural philosopher, but they are proved by other self-evident principles. Similarly, the first philosopher does not prove the principles he gives the natural philosopher by principles he receives from him, but by other self-evident principles. So there is no vicious circle in their definitions.

(Aquinas 1986)

Let us unpack this explanation. In the order of learning, the natural philosopher provides the metaphysician with principles he has learned in the course of studying natural philosophy. When engaging in the study of natural philosophy, it is the job of the natural philosopher to use other, self-evident principles—that is, principles that within the scope of natural philosophy (and not metaphysics) are self-evident—to prove the principles that are given as ontologically prior to natural science. In the same way, when the metaphysician uses the principles acquired (according to the order of learning) from the natural scientist, the metaphysician does not use the proofs of those principles that were provided by the natural philosopher, proofs which, considered within the purview of natural science, were self-evident. Rather, it is up to the metaphysician to seek

proofs for the principles (acquired in the order of learning) that are self-evident within the ambit of the subject of metaphysics. Since natural science has as its subject matter material being in motion, proofs that are self-evident in this subject are not going to be self-evident within the scope of the science that studies separate being, i.e., immaterial being.

This is a resolution to the paradox, if one accepts that the self-evidence of proofs is indexed to the scope and subject matter of the different sciences. To accept this resolution demands that we recognize that ontological priority and explanatory priority are not just distinct but, in this case, inversely ordered to one another. This is, of course, an elaboration of Aristotle's general lesson mentioned earlier, namely, that we first grasp things as they appear to us and then move to an understanding of how things are in themselves. But what Avicenna and Aquinas explicitly recognize is that, if ontological and explanatory priority are not delineated and distinguished, then the very subject matter of metaphysics appears to be viciously circular.

Thus far, philosophers addressing the question of the subject matter of metaphysics are faced with embracing either the ontological interpretation, according to which metaphysics is the science of being qua being, or the theological interpretation, according to which metaphysics is the science of divine being. As a general rule, most thinkers throughout the 13th and 14th centuries preferred the ontological interpretation. However, early modern scholastic thinker Francisco Suárez (1548–1617) embraced the view that metaphysics has a triune character that is theological, ontological, and explanatory.

In his note to the reader in the prologue of his *Metaphysical Disputations*, Suárez unequivocally declares that theology is to be built upon a “solid metaphysical foundation”. He characterizes metaphysics as First Philosophy in this way:

Of all the natural sciences, that which holds primacy and has won the name of First Philosophy is most valuable for promoting sacred and supernatural theology. For among them all it approaches most closely to the science of divine things, and also explains and vindicates those natural principles which embrace the universe of being and in one way or other stand at the basis of all learning.

(Suárez online, “To the reader”)¹²

So far, Suárez's characterization is twofold: metaphysics is both theological and explanatory. It is theological in the sense that it “approaches most closely to the science of divine things”. It embraces the study of the divine because it is God who brings everything that is real into existence. But it is also explanatory, since it “explains and vindicates those natural principles which embrace the universe of being” and “in one way or another stand at the basis of all learning”. Accordingly, metaphysics explains the axioms that are shared by both metaphysics and theology.

In addition, in the First Disputation, there is no question that, for Suárez, metaphysics is interpreted as ontology. In answer to the question, “What is the (adequate) object of metaphysics?”, Suárez claims that metaphysics can be defined as the science in which being qua being is contemplated. By “being”, Suárez meant real—as opposed to actual (i.e., actually existent)—being, thereby including also the science of possible being. It is the study of all reality that has or can come about due to God's infinite power. Thus, in the end, Suárez embraces all three available characterizations of metaphysics as First Philosophy: it is theological, ontological, and explanatory. But it is also one single science with one, unified subject matter (following the lesson from Aristotle's *Posterior Analytics*), namely being.

Suárez's ability to embrace at once the ontological *and* the theological interpretations of metaphysics is inherited from, among other sources, lessons from John Duns Scotus. Scotus proposed a radical rethinking of the nature of being, considered as both divine and nondivine

(i.e., created) being. Although it would take us too far afield to enter into the complexity of Scotus' metaphysics here,¹³ suffice it to say that Suárez adopted Scotus' doctrine of the univocity of being, according to which being is univocally predicated of God and of the created universe. Previous thinkers had adopted an analogical theory of being according to which God's being, which is perfectly simple and transcendent, is related to created being by analogy. Thus, they could say that nondivine things have being only insofar as they imitate or participate in divine being.¹⁴ Armed with Scotus' radically simple, univocal concept of being, Suárez is able to bifurcate the subject of metaphysics into two, namely, general and special metaphysics, a distinction which holds throughout the early and later modern periods. General metaphysics studies being qua being and its principles, whereas special metaphysics studies God and other particular forms of being. General metaphysics is prior in explanation, since being and its attributes (the one, the true, the good) are prior in being to the beings studied in any of the special sciences.

Ongoing discussion about the subject matter of metaphysics can be found throughout this period, although none approaches Suárez's triune characterization of the science in its complexity. To close out this brief survey, let us look at just two more philosophers and their answers to the question of the subject matter of metaphysics: René Descartes (1596–1650) and Gottfried Wilhelm Leibniz (1646–1716). Neither Descartes nor Leibniz wrote commentaries on Aristotle's *Metaphysics*, and nearly all of their study of Aristotle's metaphysical lessons was gleaned through Scholastic sources and neo-Scholastic textbooks.¹⁵ Nevertheless, the question of the subject matter of metaphysics was very much on their minds, albeit in very different ways.

Most are familiar with this famous quotation from Descartes' *Principles of Philosophy*, published in 1644:

Thus philosophy as a whole is like a tree whose roots are metaphysics, whose trunk is physics, and whose branches, which issue from this trunk, are all the other sciences. These reduce themselves to three principal ones, namely medicine, mechanics, and morals.

(Descartes 2000)

In keeping with the long, Aristotelian tradition before him, Descartes glossed "metaphysics" as "First Philosophy". This arboreal image appeared in other neo-Scholastic textbooks at this time and so was not wholly original to Descartes. Nonetheless, the image of metaphysics as the roots of the tree makes it seem as if, for Descartes, the subject retains its foundational role, such that metaphysics remains the foundation of all knowledge. In a sense this is true, but not in the way we have previously seen.

For Descartes, the principles of metaphysics are not, in fact, ontologically prior, serving to underpin and thereby explain the principles of the special sciences. Rather, metaphysics is First Philosophy because it is *explanatorily* prior to the study of physics, medicine, mechanics, and ethics. In setting out the sequence of learning, after a propaedeutic study of logic and mathematics, the student can turn to philosophy proper. The first subject to be studied in philosophy is metaphysics, which contains

the principles of knowledge, among which is the explication of the principal attributes of God, the immateriality of our souls, and all clear and simple notions that are in us.

(Descartes 2000)

Thus, the study of metaphysics for Descartes grounds the study of physics and the special sciences and is to be learned first in the sequence of the proper subjects of philosophy. It is not ontologically prior.

But then how should the following set of claims by Descartes be understood? In *The Principles of Philosophy*, Descartes inquires after what it is and how it is that humans can acquire the “perfect knowledge” that is attainable by human beings (in contrast to God, who has absolutely perfect knowledge of what is). This knowledge is also to be instrumental, leading to knowledge of how to live the best life and how to have knowledge in the arts.

In order for this knowledge to serve these ends, it is essential that it should be derived from first causes; thus, to study to acquire it (what is properly termed philosophizing), we must begin with the investigation of these first causes, that is, the Principles. It is also necessary that these Principles should have two conditions attached to them (Descartes 2000).

The first condition addresses the very well-known criterion that the Principles are “so clear and evident that the mind of man cannot doubt their truth when it attentively applies itself to consider them”. Thus, this criterion addresses the epistemic certitude that provides a stable foundation for knowledge of other, less clearly and distinctly known things. It is strictly epistemic. Let us look at the second criterion in a bit of detail:

Second, it is on them that the knowledge of other things depends, so that the Principles can be known without these last, but the other things cannot be reciprocally known without the Principles.

(Descartes 2000)

This second criterion, which describes the principles which belong to what Descartes calls “First Philosophy” and are characterized as belonging to the study of metaphysics, establish a dependence relationship between them and “knowledge of the things that depend on them” (Descartes 2000). However, despite characterizing these Principles as belonging to the realm of the study of metaphysics, it is clear that their role is exclusively epistemological and that they are meant to service other types of explanation. They are not, in the classical or Aristotelian sense, ontological in nature.

According to Descartes, the subject that usurps the place traditionally reserved for metaphysics—first in the order of ontological explanation and last in the order of learning—is ethics:

By morals, I mean the highest and most perfect moral science which, presupposing a complete knowledge of the other sciences, is the ultimate degree of wisdom.

(Descartes 2000)

So-called First Philosophy for Descartes is entirely recast into the study of the principles of human knowledge. Those principles, which include (among many others) the proof for the existence of God, serve the epistemic role of guaranteeing against radical doubt and global scepticism.

This brief survey of historical answers to the question of whether metaphysics is the ground of all knowledge will come to a close with some of Leibniz’s views on the matter. Proponents of ground with an interest in the history of philosophy would be very well served by a more detailed analysis of Leibniz’s views on the role of metaphysical explanation and relationships of dependence found in his work.¹⁶ Above all else, Leibniz is a champion of, in Raven’s terminology, the unionist position. Since most specialists on the topic of Leibniz’s metaphysics recognize that his metaphysical views evolved, sometimes markedly, throughout his career, here the focus will be on just two of his works, written over a ten-year period. These are the *Discourse on Metaphysics* (1686, unpublished) and the *New System of Nature* (1695). (For reference and context, Leibniz’s culminating treatise on metaphysics, called *The Principles of Philosophy, or, the Monadology*, was written in 1714.)

The complete title of the first work is *A New System of Nature and Communication of Substances, and of the Union of Soul and Body*. Despite being written after the *Discourse on Metaphysics*, it is convenient and informative to begin with it due to its explicitly autobiographical character. It provides valuable insight into the reasons why Leibniz tried to defend the case for the primacy of metaphysics as the ground of all knowledge, in self-conscious opposition to the corpuscularian and Cartesian views that were fashionable at the time.

The self-consciousness of Leibniz's chosen position on the place of metaphysics and metaphysical explanation is admitted up front:

Since some important persons have desired to see my opinions further clarified, I have risked publishing these meditations, even though they are not at all popular, nor can they be appreciated by all sorts of minds.

(Leibniz 1989)

In this short piece, Leibniz makes a case for metaphysical explanation—over purely mathematical, physical, or mechanical explanation—and, accordingly, for the primacy of the subject of metaphysics. Specifically, a complete explanation of a natural scientific fact will be the facts that ground it. The basis for this return was multifold, as will be explained, but certainly Leibniz would have taken direction from the many university textbooks whose order and method were, broadly speaking, Aristotelian in orientation.

Leibniz admits that his own philosophical journey began with the study of the works of the Scholastics, but he subsequently “freed [him]self from the yoke of Aristotle”. Briefly seduced by the mathematical and mechanistic explanations advanced by the “moderns” (primarily the Cartesians), Leibniz registered a serious problem with mechanistic or purely physicalist explanations of phenomena. Recall that, for Descartes, matter turns out to be nothing more than three-dimensional extension. The familiar wax experiment from *Meditations* sets out this position: “Let us take, for example, this piece of wax” (Descartes 1985). Descartes describes the properties it seems to have as they are experienced by the senses. The wax has a particular scent, shape, color, size, and temperature. “In short,” Descartes claims, “it has everything which appears necessary to enable a body to be known as distinctly as possible”. But when this same piece of wax is brought near a fire, all of its perceivable qualities—its particular scent, shape, color, size, and temperature—change completely, although the entity retains its identity as a piece of wax. Descartes' conclusion to this experiment is this:

[T]he wax was not after all the sweetness of the honey, or the fragrance of the flowers, or the whiteness, or the shape, or the sound, but was rather a body which presented itself to me in these various forms a little while ago, but which now exhibits different ones. But what exactly is it that I am now imagining? Let us concentrate, take away everything which does not belong to the wax, and see what is left: merely something extended, flexible, and changeable.

(Descartes 2000)

Upon reflecting on this view, Leibniz came to realize that the study of extended mass cannot just operate at the level of a physicalist framework but must also “make use of the notion of force, which is very intelligible, *despite the fact that it belongs in the domain of metaphysics*” (Leibniz 1989, emphasis added). The role of force in Leibniz's metaphysics is disputed by scholars, but at the very least it is clear that it plays a foundational role in explaining the unity of entities and specifically of substances.¹⁷ Without consideration of force, the study of matter does not produce

the principles of a “true unity”. It only proves the existence of some kind of aggregate. This realization, Leibniz admits, led him to “rehabilitate” (his term) the theory of substantial forms.

In *Discourse on Metaphysics*, Leibniz is explicit about this rehabilitation effort:

I know that I am advancing a great paradox by attempting to rehabilitate the old philosophy in some fashion and to restore the almost banished substantial forms to their former place.

(Leibniz 1989)

The doctrine of substantial form plays a role not in any physicalist explanation; indeed, Leibniz claims that this is where the Scholastics failed, since substantial form plays no role in explanation at that level. Rather, the explanation for the unity of phenomena is robustly metaphysical and requires appeal not only to substantial form but, ultimately, to force (or power, as he sometimes calls it). To account for the unity of identity of entities through change, Leibniz maintains that an appeal to forces—which are both passive and active—is required. The details of Leibniz’s ontology of force are complex and deserve fuller treatment than can be given here.¹⁸ Suffice it to say for the purposes here, Leibniz’s recognition of the fundamentality of metaphysics and the appeal to the ontology of force to explain everything else restores (for a time) the primacy of metaphysics as the epistemic basis of knowledge.

Related Topics

Causation, explanation, priority, principle of sufficient reason, Aristotle, Leibniz.

Notes

- 1 Transcendentals were those things that had been identified as being the most common, or supracategorical, notions such as the one, the true, and the good, as well as thing (*res*) and something (*aliquid*). These were regarded as the properties of being and were, by and large, treated as co-extensional with being. For a general overview of the theories of transcendentals, see Aertsen 1996; Pich 2007; Aertsen 2012.
- 2 See Corkum [Chapter 1] in this volume.
- 3 The literature on *Metaphysics* is vast, but a good starting place for bibliographical information can be found in Cohen 2016.
- 4 For a detailed explanation of this application, see Bertolacci 2007.
- 5 This is not, it should be noted, a science of discovery. As Barnes 1993, xii explains, “Aristotle does not pretend to be offering guidance to the scientist—or, for that matter, to the historian or philosopher—on how best to pursue his researches or how most efficiently to uncover new truths. . . . In short, the primary purpose of [the method of] demonstration is to expound and render intelligible what is already discovered, not to discover what is still unknown.” According to Aristotle, “the cause why the thing is, that it is the cause of this, and that this cannot be otherwise” (*Posterior Analytics* I.2). For a different interpretation, however, that construes the treatise as primarily about learning, see Bronstein 2016. Also, see Malink [Chapter 2] in this volume for more on Aristotelian demonstration and ground.
- 6 On Avicenna’s metaphysics, see Wisnovsky 2003; Bertolacci 2006; Bertolacci 2007; Gutas 2014.
- 7 This summary of Averroes’ position relies upon the analyses of this thought given by Di Giovanni 2013; Bertolacci 2006.
- 8 Di Giovanni 2013, 71, explains that “supposition” “is used here as a technical term of logic . . . to indicate the position of a subject matter that is liable to scientific explanation”.
- 9 Anicius Manlius Severinus Boethius (c. 477–524), Neoplatonic commentator on Aristotle’s philosophy and a Christian theologian. *On the Trinity*, which belongs to Boethius’ five *Theological Treatises*, but which is rich in ontological theory, treats the apparently paradoxical question of how God can be said to be both three and one at the same time.

- 10 See Wippel 1984; Bertolacci 2006; Gutas 2014.
- 11 See Wippel 1984 for an extensive analysis of this issue as treated by both Avicenna and Aquinas.
- 12 All translations are taken from Sydney Penner's online resource "Suárez in translation": www.sydneypenner.ca/SuarTr.shtml.
- 13 See Frank and Wolter 1995; King 2002.
- 14 On medieval theories of analogy, see Ashworth 2017 and bibliography there.
- 15 See Gilson 1979; Ariew 2011.
- 16 See Amijee [Chapter 4] in this volume.
- 17 Whereas Garber 2009 holds the view that Leibniz's metaphysics take substances to be foundational (and modes of substances to be grounded in them), others have recognized the foundational role played by force. See Bolton 2008, 119; Whipple 2010, and more recently Jorati 2019.
- 18 See especially Jorati 2019 for a detailed analysis that pays close attention to the extent to which Leibniz's metaphysics of force can be compared to Scholastic doctrine.

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4

PRINCIPLE OF SUFFICIENT REASON

Fatema Amijee

According to the Principle of Sufficient Reason (henceforth ‘PSR’), everything has an explanation or sufficient reason. The PSR was a central tenet of rationalist metaphysics but has since gone out of vogue. Its unpopularity in recent metaphysics traces in part to the formidable consequences it is taken to have, including necessitarianism (the view that the world could not have been otherwise), the existence of God, the identity of indiscernibles (the principle that if two objects have all the same properties, they are identical), and the principle of plenitude (the principle that if an object can exist, it does exist).

But lately, interest in the PSR has been enjoying a resurgence. This has been due, at least in part, to the recent interest in grounding, and relatedly, metaphysical explanation. If the notions of ground and metaphysical explanation are in good standing, then we can formulate a version of the PSR on which ‘explanation’ picks out either ground or metaphysical explanation. Since this chapter figures in a volume on grounding, I will focus on the version of the PSR that says that every fact has a ground. My primary goal is to address three questions. First, how continuous is the contemporary notion of grounding with the notion of sufficient reason endorsed by Spinoza, Leibniz, and other rationalists? In particular, does a PSR formulated in terms of ground retain the intuitive pull and power of the PSR endorsed by the rationalists? Second, to what extent can the PSR avoid the formidable traditional objections levelled against it if it is formulated in terms of ground? And finally, how might historical discussion of the PSR shed light on the contemporary notion of grounding?

I proceed as follows. In §1, I briefly discuss the relationship between the intimately related notions of grounding and metaphysical explanation, and delineate features I take to be central to the former notion. In §2, I discuss apparent barriers to construing Spinoza’s PSR in terms of ground. In §3, I discuss apparent barriers to construing Leibniz’s PSR in terms of ground. In §4, I compare and contrast the notions of grounding and sufficient reason, with a focus on the role these notions play in arguments for the claim that the PSR entails necessitarianism. In §5, I discuss the strength of contemporary arguments for the PSR formulated in terms of ground and metaphysical explanation. In §6, I discuss how the early modern rationalist’s notion of a sufficient reason might shed light on grounding.

§1. Grounding and Metaphysical Explanation

Consider the fact that objects persist over time. In virtue of what do they persist? According to a four-dimensional perdurantist, an object persists over time in virtue of the fact that it is a collection of temporal stages and the fact that a temporal stage of the object at time t_1 is appropriately related to the temporal stage of that object at time t_2 . If this four-dimensionalist view is correct, these facts *ground* our original fact by being those in virtue of which the original fact obtains.

Grounding has taken center stage in contemporary metaphysics. But the notion is hardly new to philosophy. Even though the use of the term ‘grounds’—as introduced by Fine (2001)—is novel, the kind of explanation or dependence relation it is intended to pick out (the kind whose presence is often signaled by the locutions ‘in virtue of’ or ‘because’) is both familiar and ubiquitous.

Grounding is closely related to the contemporary notion of metaphysical explanation, where a metaphysical explanation for a fact tells us what makes it the case that the fact obtains. But what is the precise relationship between metaphysical explanation and grounding? Some insist that a single metaphysical dependence relation—grounding—with a unified set of formal features backs metaphysical explanation.¹ Others argue that grounding just is, rather than backs, metaphysical explanation.² And some opponents of grounding argue that no single metaphysical dependence relation can back metaphysical explanation and instead deploy a formally and substantively diverse set of metaphysical dependence relations.³ These theorists take the notion of metaphysical explanation to be in good standing, even as they reject the notion of grounding (considered as a unique relation governed by a unified set of formal features). For the purposes of this chapter, I will bracket this last position. Also, for the sake of convenience, I will suppose that to ground is to metaphysically explain. If grounding backs metaphysical explanation instead, the discussion and arguments in this essay will still hold *mutatis mutandis*.

The notion of a metaphysical explanation makes more transparent the relationship between grounding and sufficient reason. Sufficient reason—at least as discussed by Leibniz—is partly a cognitive notion: it is agents who give sufficient reasons. Metaphysical explanation likewise has cognitive connotations: it is agents who provide metaphysical explanations. But while the term ‘explanation’ has cognitive connotations, suggesting that there could be no explanation if there were no agents doing any explaining, ‘metaphysical explanation’ is a term of art.⁴ It transcends particular human interests, perspectives, or capacities and applies to what an omniscient subject—someone who knows all the facts—would be in a position to explain. Construed in this way, metaphysical explanation seems continuous with Leibniz’s conception of a ‘sufficient reason’. Leibniz says, for example:

So far we have just spoken as simple physicists; now we must rise to metaphysics, by making use of the great principle, little used, commonly, that nothing takes place without sufficient reason, that is, that nothing happens without it being possible for someone who knows enough things to give a reason sufficient to determine why it is so and not otherwise.

(Leibniz 1989: 209)

Leibniz’s statement of the PSR implies that a “sufficient reason”—like a metaphysical explanation—is a reason that “someone who knows enough things” could in principle provide. Spinoza too seems to be committed to the claim that a ‘perfect intellect’ could explain everything (see Lin 2017).

But what is metaphysical explanation? I earlier glossed metaphysical explanation as the kind of explanation that tells us what makes it the case that a fact obtains. Can more be said? Many philosophers think that there is a sharp distinction between metaphysical and causal explanation, and some even characterize metaphysical explanation as ‘non-causal explanation’.⁵ Others treat metaphysical explanation and causal explanation as closely related.⁶ The precise relationship between metaphysical and causal explanation is relevant to the question of whether the notion of sufficient reason endorsed by Spinoza and Leibniz is continuous with metaphysical explanation, for both Spinoza and Leibniz often formulate their version of the PSR in terms of causation. If it turns out that sufficient reasons can only be causal reasons and that causal reasons are not metaphysical (in our sense), then a PSR formulated in terms of metaphysical explanation would not have much in common with versions of the PSR endorsed by the early modern rationalists. I will discuss the question of whether sufficient reasons are ever metaphysical or causal in more detail in the next two sections.

Finally, metaphysical explanation and grounding are generally, though not universally, taken to be irreflexive, asymmetric, transitive, and necessitating.⁷ In what follows, I will discuss whether sufficient reason shares these features. Henceforth, unless otherwise noted, I will use ‘explanation’ and ‘explains’ to mean *metaphysical explanation* and *metaphysically explains*, respectively. Also, unless otherwise noted, I will use ‘explains’ or ‘grounds’ to mean *fully explains* or *fully grounds*.

Following notation introduced by Rosen (2010), I will refer to facts by putting a sentence between square brackets. For example, [Snow is white] is the fact that snow is white. If the fact that p grounds the fact that q, I will then write ‘[p] grounds [q]’. If grounding just is metaphysical explanation (rather than a worldly relation that backs metaphysical explanation), then ‘[p] grounds [q]’ can be taken to be shorthand for something like ‘a subject who knows all the facts can metaphysically explain [q] by reference to [p]’.

§2. Grounding in Spinoza

The PSR is not explicitly stated in the axioms of Spinoza’s *Ethics*. Nevertheless, Spinoza’s commitment to the PSR is arguably implicit in the statement of his axioms (see Lin 2017). When Spinoza explicitly formulates the PSR, he does so in terms of ‘cause’ or ‘reason’:

For each thing there must be assigned a cause, *or* reason, both for its existence and for its nonexistence.

(*Ethics Ip11d2*)⁸

The disjunctive ‘cause or reason’ is puzzling. Are ‘cause’ and ‘reason’ equivalent in meaning for Spinoza? Or are they distinct (perhaps overlapping) notions? And how closely is either notion related to grounding or metaphysical explanation? I will address these questions momentarily. But on the face of it, there are at least three apparent barriers to construing Spinoza’s PSR in terms of ground and metaphysical explanation. First, if causes are distinct in kind from ‘metaphysical’ reasons or explanations, then Spinoza’s PSR is not continuous with a PSR formulated in terms of ground, for it is, at least in part, formulated in terms of causation. Second, Spinoza’s PSR is explicitly about the explanation of *things* rather than facts, which suggests that it has narrower scope than a PSR that concerns facts. It suggests, for example, that while each thing (e.g., a red ball) has an explanation, facts concerning the ways those things are (e.g., the fact that the ball is red) do not. Third, Spinoza is seemingly committed to the claim that God explains himself, thereby violating the irreflexivity of explanation. Insofar as metaphysical explanation

and grounding are taken to be irreflexive, it would seem that Spinoza's PSR does not concern metaphysical explanation or grounding.

But there are reasons to believe that even though Spinoza's formulation of the PSR uses 'cause', this does not pick out our contemporary conception of causation. First, for Spinoza, causation involves a conceptual connection (see *Ethics* Iax4, Ip6c2). For all x and y , x causes y if and only if there is a conceptual connection between the concept of x and the concept of y (see Della Rocca 2008; Lin 2017). A causal fact then is then a conceptual fact for Spinoza. Moreover, for Spinoza, a conceptual connection is explanatory (cf. *Ethics* Iax5, IIp5, IIp7s). Thus, effects are conceived or understood through their causes, i.e., a conceptual grasp of an effect requires a conceptual grasp of its cause (cf. *Ethics* Id3). Insofar as contemporary accounts of causation do not take causal relations and conceptual relations to be coextensive, it would seem that Spinoza's account of causation does not bear a close similarity to contemporary accounts of causation. But is causation (in Spinoza's sense) explanatory in the grounding sense? A ground tells us what makes it the case that a fact obtains. And so, because effects are understood through their causes, a causal (and thus conceptual) connection is, for Spinoza, a paradigm case of grounding.

Second, Spinoza's PSR applies to entities that are not apt for standing in causal relations (at least given our contemporary understanding of causation). For instance, Spinoza claims that if a triangle exists, there must be a cause or reason why it exists (*Ethics* Ip11d2). While triangles are plausibly not apt for standing in causal relations, nothing precludes them from being metaphysically explained or grounded (the same point holds if we restrict ourselves to facts about triangles rather than to triangles themselves). Thus, even if reasons are not wholly distinct from causes for Spinoza, and so there is a 'cause' for why a triangle exists, it would seem that Spinoza's understanding of causation is unlike our contemporary notion of causation.

Finally, on Spinoza's view, and unlike on most contemporary views of causation, a total cause necessitates its effects:

From a given determinate cause the effect follows necessarily; and conversely, if there is no determinate cause, it is impossible for an effect to follow.

(*Ethics* Ia3)

Thus causation, for Spinoza, is governed by a necessitation principle. According to the necessitation principle that governs grounding, if a fact grounds another, then necessarily, if the first fact obtains, the second also obtains.⁹ Most philosophers accept that in cases of grounding, the *explanans* can consist in many facts, and some (e.g., Dasgupta 2014) argue that even the *explanandum* can be plural. For simplicity of exposition, my formulation of the necessitation principle takes both the *explanans* and the *explanandum* to consist in single facts. If causation, for Spinoza, is explanatory in the metaphysical sense and governed by a necessitation principle, then what Spinoza means by causation is close (if not identical) to the contemporary notion of grounding. When in §4 I discuss the role necessitation plays in the entailment from the PSR to necessitarianism (the view that all facts are necessary), I will also discuss why a commitment to necessitation is widely taken to be a core formal feature of grounding.

Does Spinoza's PSR have a narrower scope than a PSR that says that every *fact* rather than every *thing* has a sufficient reason? There is reason to believe that Spinoza's PSR extends to facts. First, Spinoza doesn't talk about the explanation of things *simpliciter*, but the explanation for a thing's existence or nonexistence. In addition, there are readings of Spinoza that distinguish facts and things, and yet which plausibly entail that a PSR that applies to things collapses into a PSR that applies to facts. On Michael Della Rocca's (2015) reading of Spinoza, for instance, that there is an unexplained fact entails that there is an unexplained state of substance. If a state of

substance is considered to be a thing on Spinoza's view, then every time we have an unexplained fact, we also have an unexplained thing. Hence, if there is no unexplained thing (as per a PSR that applies to things), there can be no unexplained fact.

Finally, it is not obvious that Spinoza is committed to a violation of irreflexivity. At *Ethics* Id1, Spinoza construes a "cause of itself" as "that whose essence involves existence or that whose nature cannot be conceived except as existing." On at least one understanding, this suggests that God's existence is explained by God's essence. Put in terms of facts, the claim seems to be that the fact that God exists is explained by the fact that it is essential to God that he exists. Absent further assumptions—such as the assumption that God's essence is identical to God—this does not constitute a violation of irreflexivity.¹⁰ It is worth noting that some philosophers reject the claim that explanation is irreflexive. However, even if there are certain exceptions to irreflexivity, it is far from clear that God should be an exception to the rule. Declaring God to be the exception by fiat seems ad hoc and in tension with a thoroughgoing commitment to the PSR.

What might explain the fact that it is essential to God that he exists? It may be that for Spinoza this fact is not apt for explanation. At *Ethics* Ip7, Spinoza says: "It pertains to the nature of a substance to exist". Della Rocca (2008: 49) argues that Spinoza's claim here is that "each substance is such that its existence somehow follows from its very concept or nature". If we then further ask why substances are such that their existence follows from their very concept or nature, the answer might just be: 'that's just what a substance is'. It is unclear, given his commitment to the PSR, whether Spinoza is permitted to simply assume that there is such an existence-involving concept or nature. But for our purposes, it is not at all obvious God's self-causation entails a violation of the irreflexivity of explanation.¹¹ Hence, once again—at least as far as the formal features of grounding are concerned—what Spinoza means by 'sufficient reason' does not obviously come apart from grounding.

§3. Grounding in Leibniz

According to Leibniz's PSR, "we can find no true or existent fact, no true assertion, without there being a sufficient reason why it is thus and not otherwise" (Leibniz 1989: 217). What is a 'sufficient reason' for Leibniz? A hint of the notion of 'sufficient reason' at work in Leibniz's PSR comes from an argument Leibniz presents in *Primary Truths*. According to this argument, the PSR is entailed by Leibniz's conceptual containment theory of truth. And indeed, Leibniz says the following in his July 1686 letter to Arnault:

[I]n every true affirmative proposition, whether necessary or contingent, universal or particular, the notion of the predicate is in some way included in that of the subject. *Predicatum inest subjecto*, otherwise I do not know what truth is . . . For *there must always be some foundation for the connection between the terms of a proposition, and this must be found in their concepts*. This is my great principle, with which I believe all philosophers should agree, and one of whose corollaries is the commonly held axiom that nothing happens without a reason which can always be given, why the thing has happened as it did rather than in another way, even though this reason often inclines without necessitating.

(Leibniz 1970: 337)

On the conceptual containment theory of truth, what makes each true proposition true is the fact that the proposition's predicate is contained in its subject. This containment of predicate in subject is explicit in the case of identities (such as 'A is A') but needs to be shown through the

analysis of concepts in the case of nonidentities (such as ‘A is B’). The sufficient reason for each proposition is an *a priori* demonstration that reduces the proposition to an identity, and it consists in substituting the terms of a proposition with definitions until we (or God) arrive at an identity.

This notion of a ‘sufficient reason’ is not causal. It instead bears close similarity to the notion of grounding. While most philosophers would reject Leibniz’s conceptual containment theory of truth, it is nevertheless Leibniz’s answer to the question ‘what makes it the case that any given proposition is true?’ And Leibniz never seems to have given up his conceptual containment theory of truth. Leibniz here takes a ‘sufficient reason’ to be a truth-making reason. This suggests that Leibniz’s notion of a sufficient reason is closely related to—though perhaps not identical to—the contemporary notion of grounding.

Leibniz also takes God to be “ultimate reason for things”. In discussing the sufficient reason for the existence of the universe (which, for Leibniz, consists in a series of contingent things), Leibniz writes:

Thus *the sufficient reason*, which needs no other reason, must be outside this series of contingent things, and must be found in a substance which is its cause, and which is a necessary being, carrying the reason of its existence with itself. Otherwise, we would not yet have a sufficient reason where one could end the series. And this ultimate reason for things is called *God*.

(Leibniz 1989: 210)

The passage implies that God (or more carefully, perhaps, the fact that God chooses to bring about our world) is the sufficient reason for the existence of contingent things. But bringing about, like causation, is a relation of production rather than constitution. Indeed, Leibniz says that the sufficient reason for the existence of the universe “must be found in a substance which is its cause”. It is not clear whether the sense in which God causes or brings about the existence of the universe can be taken to be grounding. If one, like Fine (2012a), thinks that in cases of grounding the *explanans* is constitutive of the *explanandum* and that the *explanandum* consists in nothing more than the *explanans*, then it would be implausible to think that the existence of the universe is grounded in the fact that God brings it about, for the existence of the universe, for Leibniz, does not *consist in* the fact that God brings it about. For Leibniz, the created is wholly distinct from the creator and does not consist in any fact about the creator. But if the grounding relation can also be productive—rather than merely constitutive—then perhaps the notion can accommodate the relation that God stands in to the existence of the universe on Leibniz’s view.

Finally, it is unclear that Leibniz is committed to a violation of irreflexivity. The above passage says that the necessary being (i.e., God), carries “the reason of its existence with itself”. Like in Spinoza’s case, this claim may be construed as follows: God’s existence is explained by God’s essence, which is contained within God. Leibniz is thus not obviously committed to a violation of irreflexivity. Insofar as the irreflexivity of ground is a core formal feature of grounding, it is thus not obvious that what Leibniz means by ‘sufficient reason’ ever fails to be irreflexive.

§4. The PSR and Necessitarianism

According to a longstanding objection to the PSR, the principle entails necessitarianism: the view that the world could not have been otherwise, i.e., that all facts are necessary. This consequence is generally, if not universally, taken to be unacceptable.¹² My central goal in this section is to examine the argument from the PSR to necessitarianism and assess whether a PSR

formulated in terms of ground also entails necessitarianism. As we shall see, it is not clear that the argument goes through when the PSR is formulated in terms of ground.

In the present discussion, I will focus on a version of the argument that has been put forward by Jonathan Bennett and Peter van Inwagen.¹³ That argument runs as follows. Take the conjunction C of all contingent facts. On the assumption that the PSR is true, C has an explanation. So suppose that some fact [p] explains C. Now [p] is either contingent or necessary. If [p] is contingent, it is itself a conjunct in C. But [p] cannot explain a fact of which it is itself a conjunct (since this would violate the plausible requirement that explanation is noncircular). So [p] has to be necessary. But given necessitation, no contingent fact can be fully explained by necessary facts. C is then either unexplained or it is not contingent. On the assumption that the PSR is true, C cannot be unexplained. It follows that C is not contingent. If C is not contingent, then all facts are necessary. Yet if all facts are necessary, the world cannot be otherwise.

There are reasons to be skeptical of the argument. Levey (2016), for example, argues that the concept *contingent truth* is indefinitely extensible, and so the idea of “all contingent truths” is incoherent. But there are seemingly ways to resist the argument even by simply appealing to principles that govern metaphysical explanation (or, more specifically, ground). Schnieder and Steinberg (2015) argue that the argument relies on a problematic application of irreflexivity. Another reply made by some proponents of grounding (e.g., Schnieder and Steinberg 2015; Dasgupta 2016), is to insist that the conjunction of contingent facts is simply grounded in its conjuncts, taken together. However, McDaniel (2019) points out that if we admit plural grounding—the view that a plurality of facts can collectively ground a plurality of facts—then the Bennett–van Inwagen argument is immune to this last reply (though see a reply to McDaniel in Werner forthcoming).

It is important to note that simply rejecting necessitation as a principle that governs metaphysical explanation—a view defended independently by Leuenberger (2014) and Skiles (2015)—would not be enough to avoid the conclusion of the Bennett–van Inwagen argument. This is because the falsity of necessitation as a principle that governs metaphysical explanation is compatible with the truth of the claim that no contingent fact can be fully grounded in necessary facts. By contrast, Pruss (2006) and Amijee (forthcoming) each argue against the claim that no contingent fact can be fully grounded in necessary facts. Pruss assumes that God has the freedom to do otherwise and so concludes that some putative necessary truths about God could explain all contingent facts without necessitating them. By contrast, Amijee (forthcoming) exploits a parallel between time and modality to show that we should do away with a commitment to the relevant claim as a principle that governs metaphysical explanation.

Is the early modern rationalist argument for necessitarianism susceptible to the ground-theoretic responses? I will show that it falls victim to only the replies made by Pruss (2006) and Amijee (forthcoming). Lin (2012: 419–420) reconstructs Spinoza’s argument for the claim that rationalism leads to necessitarianism as follows:

1. Whatever exists (other than God) is a mode of God. (by Ip15)
2. God exists necessarily. (by Ip11)
3. The existence of the modes follows from the divine nature. (by Ip16)
4. The effects produced by the modes follow from the divine nature. (by Ip26)
5. Whatever follows from something necessary is itself necessary. (suppressed premise)
6. Therefore, there is nothing contingent.

Pruss and Amijee deny that whatever follows from something necessary is itself necessary. Denying this claim requires denying necessitation. Yet it might be difficult to see how 5 (and thus

necessitation) could be rejected. If, like Fine (2012b: 1), one thinks that grounding is akin to logical consequence, then necessitation appears hard to deny. Some (like Rosen 2010: 118) take necessitation to be a feature that differentiates grounding from causal determination. But neither view is compulsory for a grounding theorist: grounding need not be modelled on logical consequence, and features other than necessitation might differentiate grounding from other forms of determination and explanation. Finally, denying 5 does not entail a wholesale denial of necessitation, for a restricted necessitation principle (i.e., one that applies only when the *explanans* is contingent) can still hold.

None of the other canvassed replies to the Bennett–van Inwagen argument would block the early modern rationalist argument. No premise relies on a conjunction of contingent facts. Hence, there is no way for a proponent of grounding to reply by insisting that the conjunction of all contingent truths is grounded in its conjuncts, taken collectively. Nor does the argument rely on a totality of contingent truths or things. And the third premise blocks the reply according to which each contingent truth (or thing) is explained by another contingent truth (or thing) in an infinite series: the premise seems to require that the existence of *each* mode follows from divine nature. The third premise also *a fortiori* blocks the claim that the sufficient reason for the big conjunction can be found in its conjuncts, taken collectively, since no contingent thing in the series of contingent things can be a sufficient reason for the existence of a contingent thing. A similar line of argument can be found in Leibniz, who claims that a contingent thing can never be a “complete explanation” [*ratio*] for the existence of another contingent thing (Leibniz 1989: 149). This final constraint on a sufficient reason—that no contingent thing can be a sufficient reason for the existence of another contingent thing—marks an important distinction between grounding and the early modern notion of a sufficient reason. It would take us too far afield to explore why Leibniz and Spinoza adopted this constraint on sufficient reason and whether a similar constraint might ultimately also apply to grounding.

§5. Contemporary Treatments of the PSR

There are few contemporary defenders of the PSR. This group includes Della Rocca (2010, forthcoming), Pruss (2006), and Amijee (2017). Dasgupta (2016) also develops (though does not endorse) a version of the PSR. Of these contemporary defenses, Dasgupta (2016), Amijee (2017), and Della Rocca (forthcoming) explicitly discuss a PSR formulated in terms of either ground or metaphysical explanation. The versions of the PSR defended by Pruss (2006) and Della Rocca (2010) do not explicitly appeal to grounding or metaphysical explanation but can nevertheless be understood in grounding terms.

A central question I wish to address in the present section is this: does a PSR formulated in terms of ground preserve the strength of the early modern rationalist’s PSR? At least one way of measuring the strength of a PSR is by its consequences. And as mentioned at the beginning of this chapter, the PSR has traditionally been taken to have various formidable consequences. Another way to measure the strength of a PSR is by whether it trivializes intuitively substantive questions. In what follows, I will use these two metrics to assess the strength of contemporary versions of the PSR.

The version of the PSR endorsed by Dasgupta (2016) purports to avoid at least some of the formidable consequences that the PSR is traditionally taken to entail. Because grounding sequences, on Dasgupta’s view, terminate in ‘autonomous’ facts—facts that are not apt for grounding—Dasgupta’s version of the PSR does not entail the existence of God. No fact needs to be ultimately grounded by a fact about the existence of God. Nevertheless, the paradigmatic example of an autonomous fact on Dasgupta’s view is an essentialist fact: a fact of the form [it

is essential to x that p]. And such facts are generally taken to be necessary. Hence, given necessitation (which Dasgupta endorses), and despite his protestations that a kind of contingency is nevertheless preserved, Dasgupta's version of the PSR entails necessitarianism.

Moreover, Dasgupta's ground-theoretic PSR—along with some of the other versions of the PSR formulated in terms of ground—seems to trivialize certain substantive questions that fall under the scope of the traditional rationalist PSR. Consider, for example, the question *why is there something rather than nothing?* On the standard grounding framework, that there is anything is an existentially quantified fact explained by its instances, i.e., facts of the form x exists.¹⁴ But surely, this is not the kind of answer that we are after when we ask *why is there something rather than nothing?* This kind of concern suggests that perhaps the early modern rationalists were not construing sufficient reason in the way we construe ground. Perhaps sufficient reason is more akin to a causal explanation: it seems misguided to answer the question 'what caused there to be something?' by saying 'the table', where the table is one of the existing things. Alternatively, perhaps the fact that the standard view of how existentially quantified facts are grounded makes certain substantive questions trivial suggests that we ought to revise our grounding framework (see Melamedoff 2018 for an argument against the standard view).

Unlike Dasgupta (2016), who develops a view on which a ground-theoretic PSR might be true yet does not argue for it, Della Rocca (2010) provides an argument for the PSR. Della Rocca observes that there are uncontroversial examples of facts for which it is appropriate to demand an explanation. He argues that we frequently appeal to 'explicability arguments'—arguments for the nonobtaining of certain states of affairs on the basis that such states of affairs would be inexplicable. For instance, if we place equal weights on either side of a scale, we infer that one side will not hang down (assuming there are no defects in the scale). If one side were to hang down past the other, we would have an inexplicable state of affairs. Likewise, why think that neither lefty nor righty—fission offshoots in Parfit's (1984) thought experiment—are identical to the original person? Because the obtaining of the state of affairs in which the original person was identical to lefty rather than righty (or vice versa) would be inexplicable.

Della Rocca argues that accepting some explicability arguments puts pressure on one to accept an explicability argument concerning existence facts (facts of the form such-and-such state of affairs exists or obtains). But if all existence facts demand explanation, the PSR is true. Since explicability arguments seem appropriate in many cases, those who deny the PSR must provide a principled reason for why they are not appropriate in the case of existence facts. Della Rocca suggests that the challenge to provide such a principled reason cannot be met by opponents of the PSR. This sort of argument aims to shift the burden of proof from proponents of the PSR over to those who reject it.

It is not obvious which, if any, of the PSR's usual formidable consequences are shared by Della Rocca's PSR (though Della Rocca 2010 seems willing to endorse necessitarianism). And how would a world in which Della Rocca's PSR is true be structured? Would all facts be ultimately explained by God, or by some other self-explaining entity, or by autonomous facts (as per Dasgupta), or would we have to embrace an infinitely descending explanatory regress? Nonetheless, the world as conceived by Della Rocca (2010) certainly has explanatory structure. His recent work defends a more radical position. Della Rocca (forthcoming) argues that a particular kind of Bradleyan explanatory demand with respect to the obtaining of relations entails a radical monism according to which there are no distinctions whatsoever. There are thus neither many things nor one thing—there is simply being. Such a view appears to entail that the world lacks structure, thus rendering moot the question of how a world in which the PSR is true would be structured. Della Rocca further argues that the explanatory demand with respect to the obtaining of relations—and the failure to satisfy it—also commits us to an unrestricted PSR.

Amijee (2017) provides an argument for a *commitment* to the PSR—formulated in terms of metaphysical explanation—rather than for the principle’s truth. Any argument for the truth of the PSR is also an argument for a commitment to it, on the assumption that we ought to be committed to what is true. However, an argument for a commitment to the PSR need not be an argument for its truth. Amijee shows that a familiar species of inquiry—‘structural inquiry’—requires that those who participate in the practice seek metaphysical explanations for facts (where the domain of potential inquiry includes every fact). She then shows that participating in structural inquiry commits one to the PSR in two different ways: (1) it leads to the result that every fact is such that one is committed to its having a metaphysical explanation, and (2) it leads to the result that one is committed to the claim that every fact has a metaphysical explanation. Finally, she shows that we ought to participate in structural inquiry and so ought to be committed to the PSR.

Amijee’s argument for a commitment to the PSR leaves open what explanatory structure a world in which the PSR is true might have. Indeed, in Amijee (2017), she argues that the PSR can be satisfied by a world containing non-terminating explanatory sequences.¹⁵ Moreover, it does not follow from Amijee’s argument for a commitment to the PSR that we ought to be committed to the principle holding in worlds other than our own. In other words, our commitment to the PSR could be a commitment to a contingent principle.

I finally come to the extended treatment of the PSR by Pruss (2006). Pruss argues for a PSR restricted to contingent truths. Pruss also takes his PSR to have some controversial consequences, the most significant of which is the existence of a necessary being. He claims that a circularity worry arises on views that insist that every fact (or entity) in an infinitely long sequence can be explained by the next fact (or entity) in the sequence. In discussing an infinite explanatory sequence concerning chickens and eggs, Pruss (2006: 44) writes:

[I]f we accept infinitely regressive explanations, then we should be willing to say that the existence and activity of the members of the set of eggs are explained by the existence and activity of the members of the set of chickens while the existence and activity of the chickens are explained by that of the eggs. This is circular and clearly fails to answer the question why there are any chickens and eggs at all.

There is much that I find problematic about Pruss’s argument. First, it is not sets (of chickens or eggs) that are the proper *explanans* in an infinite explanatory sequence that consists of chickens and eggs: it is particular chickens and particular eggs. Second, to ask ‘why are there any chickens or eggs at all?’ is to change the question. The question we started with asked what explains the existence of each member of an infinite explanatory sequence and *not* why there are any members in the sequence at all. As noted, on the standard grounding framework, the latter question is answered simply by appealing to particular existing members of the sequence.

I have sketched some major points of overlap and divergence amongst the contemporary treatments of the PSR by Della Rocca, Amijee, Pruss, and Dasgupta. Much more could be written about the details of the various views and arguments, both individually and when taken together. Yet I hope to have said enough to bring out the extent to which the contemporary treatments of the PSR often depart substantially from the rationalist accounts developed by Spinoza and Leibniz.

§6. Explanation and Intelligibility

In this final section, I will explore the connection between intelligibility and explanation. For Spinoza, sufficient reason is intimately connected with intelligibility and understanding. As

Della Rocca (2008: 5) writes, for Spinoza, “to conceive of a thing is to explain it”. Also, as mentioned earlier, for Spinoza effects are understood or conceived through their causes. This suggests that for Spinoza, grasp of the *explanans* suffices for grasp of the *explanandum*. Put differently (and perhaps more strongly), the *explanandum* can be deduced *a priori* from the *explanans*. This type of view is perhaps not entirely alien in the traditional grounding framework. As mentioned, Fine (2012b: 1), for example, treats ground as a relation akin to logical consequence. If ground is like logical consequence, then it is not implausible to think that the grounded can be deduced *a priori* from the ground. However, most discussions of metaphysical explanation stop short of a commitment to the *a priori* deducibility of the *explanandum* from the *explanans* (though the *explanans* must, in some sense, illuminate why the *explanandum* obtains).

Does taking seriously the link between explanation and intelligibility (as Spinoza does) affect contemporary theorizing about grounding and metaphysical explanation? I will suggest that how closely we relate explanation to intelligibility inevitably affects our views about what must figure in the *explanans* for a given thing or fact.

Della Rocca (forthcoming) argues that it is in the nature of relations to be grounded in one or more of their relata. He further claims that because relations are by their nature grounded in one or more of their relata, they depend not only on their relata but also on the grounding relation that obtains between the relation and the relevant relata. While Della Rocca does not draw an explicit connection to Spinoza here, the intuitive plausibility of the view he puts forward stems from construing grounding (and metaphysical explanation) in the way that Spinoza thought of sufficient reason—as a reason that makes its *explanandum* intelligible. Compare: in order to grasp what makes it the case that *p*, it would not be enough to simply grasp its ground. One would need to grasp the ground as a ground for the fact that *p*. This suggests that in order to grasp what makes it the case that *p*, one must not only grasp its ground but also grasp the relationship between the ground and the grounded (i.e., the fact that *p*). Construing grounding as very tightly connected to intelligibility thus informs views about what ought to figure in a full ground for a fact.

Likewise, if we eschew the Spinozist link between the ground and grounded and settle for something weaker, we in turn end up with fewer constraints on what must figure in a full ground. For instance, Amijee (forthcoming) argues that the fact it is raining at 3 p.m. on July 31, 2019, can fully ground the fact that it is raining now. Absent further premises, there is no way to deduce the *explanandum* from the *explanans* in this case, though the *explanans* nevertheless illuminates why it is raining now (i.e., it tell us what makes it the case that it is raining now).

§7. Conclusion

I have shown that Spinoza’s and Leibniz’s notions of sufficient reason are reasonably continuous with the contemporary notions of grounding and metaphysical explanation, and that thinking about the PSR within the grounding framework can help to clarify (and perhaps avoid) some of the unattractive consequences traditionally attributed to the PSR. I also assessed, in a preliminary fashion, the extent to which contemporary treatments of the PSR preserve the strength of the rationalist PSR. Finally, I sketched some of the potential significance of the rationalist connection between grounding and intelligibility for contemporary work on grounding.¹⁶

Related Topics

See chapters from this handbook on: Explanation [Chapter 8]; Necessity [Chapter 10]; Structure [Chapter 27]; Fundamentality [Chapter 23]; Infinite Descent [Chapter 16].

Notes

- 1 Cf. Schaffer (2012, 2016) and Audi (2012).
- 2 Cf. Fine (2001), Litland (2013), and Dasgupta (2014). Wilson (2016) argues that proponents of such views are guilty of conflating metaphysical explanation—a partly epistemic notion—with metaphysical dependence.
- 3 Cf. Wilson (2014) and Koslicki (2015). Wilson argues against both the posited formal features of grounding and the explanatory utility of positing a single relation that backs metaphysical explanation.
- 4 See Martin Glazier's chapter 'Explanation' [Chapter 8] in this volume.
- 5 Cf. Audi (2012), and Correia and Schnieder (2012).
- 6 Schaffer (2016) argues for a close analogy between grounding and causation, Wilson (2018) treats grounding as a type of causation, and Bennett (2017) holds that causation is a kind of building relation.
- 7 See Naomi Thompson's chapter 'Strict Partial Order' [Chapter 17] and Alexander Skiles's chapter 'Necessity' [Chapter 10] in this volume.
- 8 All references to Spinoza's *Ethics* are from *A Spinoza Reader: The Ethics and Other Works*. (E. Curley, Ed. and Trans.). Princeton University Press. The first numeral in references to Spinoza's *Ethics* refers to parts. The letter 'p' refers to proposition, 'c' refers to corollary, 'd' refers to demonstration, and 's' refers to scholium.
- 9 In some discussions of grounding, a commitment to this principle has been referred to as a commitment to 'necessitarianism'. I here refer to the principle as a 'necessitation principle' and use the label 'necessitarianism' to refer to the view that all facts are necessary, i.e., the world could not have been otherwise. In the context of discussions of the PSR, the term 'necessitarianism' has more commonly been used to refer to the latter view.
- 10 At *Ethics* Ip20, Spinoza says that God's existence and God's essence are "one and the same". However, see Silverman (2017) for an argument for the claim that the "one and the same" relation in Spinoza should not be understood as one of identity. See also Marshall (2009).
- 11 In the *Treatise on the Emendation of the Intellect*, Spinoza says, "[T]hat Thought is also called true which involves objectively the essence of some principle that does not have a cause, and is known through itself and in itself" (Spinoza 1985: §70). If the "principle" here refers to God, then Spinoza is saying that God lacks a cause. This appears to be a violation of the PSR, but perhaps Spinoza's claim here is restricted to efficient causes (God lacks an efficient cause, but has a formal cause, i.e., his essence).
- 12 See, for instance, van Inwagen (2009: 149–150). As van Inwagen (2009: 150) states, "a careful examination of the Principle shows that it has a consequence most people would have a very hard time accepting: that all true propositions are necessarily true." Philosophers who accept this formidable consequence include Della Rocca (2010), though in more recent work, Della Rocca argues against all distinctions (including the distinction between the contingent and the necessary).
- 13 Compare Bennett (1984: 115), van Inwagen (2009: 104–107), and Della Rocca (2010: 9n13) for versions of this argument. See Koons (1997) for a formulation of the argument in terms of a mereological aggregate instead of a conjunction.
- 14 It is worth noting that according to Dasgupta (2009), there are no irreducible individual facts, and so general facts obtain unreduced and without any instances witnessing them.
- 15 See also Tuomas Tahko's essay 'Structure' [Chapter27], Ricki Bliss's essay 'Fundamentality' [Chapter23], and Scott Dixon's essay 'Infinite Descent' [Chapter16] in this volume.
- 16 For incredibly useful feedback and discussion, many thanks to Michael J. Raven, Michael Della Rocca, Dominic Alford-Duguid, and the audience at the May 2018 workshop on grounding at the University of Hamburg.

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5

BOLZANO

Stefan Roski

This chapter provides an overview of Bernard Bolzano's views about grounding. On Bolzano's account, grounding is an objective priority relation among true propositions that has certain explanatory features. The chapter briefly highlights historical influences on Bolzano's account of grounding and subsequently provides an overview of the most important aspects of it. As we shall see, Bolzano's account resembles current accounts of metaphysical grounding in many respects and can thus easily be related to many positions in the current debate. This is going to be a main focus of this chapter. Apart from that, we shall investigate some Bolzanian ideas about grounding that differ from the current orthodoxy but may constitute interesting additions, challenges or inspirations for those working in the current debate.

1. Introduction

Due to the unfortunate circumstances of Bolzano's life (he was evicted from his chair at the Charles University in Prague because of his progressive political views), his work has received little attention by his contemporaries and successors in philosophy.¹ In his most creative period, Bolzano was barred from publishing due to a publication ban and was not allowed to teach. He tried to make the best of this situation and devoted most of his time to research. The result is a wealth of material, impressive in both scope and depth. It includes penetrating discussions of topics in the philosophy of logic and language, mathematics, metaphysics and the philosophy of religion, but also in ethics, social and political philosophy and aesthetics. Moreover, it also includes several substantial contributions to mathematics and physics proper, due to some of which (e.g., the Bolzano–Weierstrass Theorem) Bolzano's name was, for a long time, better known to mathematicians than to philosophers.²

The most important narrowly philosophical work Bolzano published in his lifetime is the *Theory of Science* (*Wissenschaftslehre*;³ henceforth *WL*). The *WL* is a four-volume treatise devoted to logic, where logic is broadly conceived and includes parts of epistemology and the philosophy of science. The *WL* has been praised for its antipsychologistic account of propositions and its highly original, rigorous definition of logical consequence that anticipates the definition Tarski gave about 100 years later. It also contains Bolzano's most comprehensive articulation of his account of grounding—an account that has been called “a genuine milestone in the history of grounding” (Correia & Schnieder 2012: 12). Indeed, what Bolzano presents us with in

the *WL* is arguably the first proposal of a genuine *theory* of grounding that we find in history. While notions of ground had been widely employed and discussed by predecessors and contemporaries of Bolzano, his treatment remains outstanding due to its comprehensive scope and highly abstract nature, which resembles current accounts of metaphysical grounding in many crucial respects. Notably, Bolzano considered the notion of grounding to be neither particularly controversial nor unheard-of. Rather, he believed that we either implicitly or explicitly employ the notion all the time in everyday life as well as in science and philosophy (cf. Bolzano 1978: 10–11). Before we will discuss the account, we will thus first take a look at two important sources Bolzano took inspiration from when developing his account.

2. Influences

Bolzano only rarely claims that he presents his readers with so-far unprecedented ideas. Rather, he frequently tries to argue that he merely makes extant ideas more precise or puts them in a more systematic fashion. Accordingly, in the opening part of the *WL*'s grounding sections (*WL* II §§198–222), he proclaims that he has “been strengthened in the view that there is a special relation of grounding between truths by noticing that some of the most penetrating thinkers have been of the same opinion” (*WL* II §198: 341). Two notable thinkers he mentions in this context are Aristotle and Leibniz, each of whom can be associated with two broad historical traditions that influenced Bolzano's views on grounding: a tradition centering around an Aristotle-inspired conception of axiomatic science and a tradition of rationalist philosophers that were engaged in the debate about the Principle of Sufficient Reason (*PSR*). Taking a brief look at what Bolzano took from these traditions will help us clarify the theoretical role Bolzano assigned to grounding and highlight some of its central features.

Aristotle's *Posterior Analytics* famously contains an influential account of science and scientific knowledge (see Chapters 1 and 2). One key ingredient of this account is the idea that a science can be characterized as a collection of truths about a certain subject matter that comes with a certain objective order. The axioms of a given science are the most fundamental truths with respect to this order, prior to any of the science's theorems. Associated with the order comes a special sort of proof, i.e., a sort of proof that is *explanatory* in that it derives its theorem from *reasons why* (*aitiai*) the theorem holds, which are prior to it (cf. Chapter 2). Bolzano took it that when philosophers were appealing to the idea of an order among truths and the associated idea of an explanatory proof, they often had something close to his notion of grounding in mind. On his view, in an ideal Aristotelian science, the axioms are the most fundamental truths in the order of grounding, and explanatory proofs are proofs that derive their theorems from their grounds.⁴ Bolzano was especially interested in the notion of an explanatory proof due to his work on the foundations of mathematics. He complained that the state of mathematics of his time fell short of the Aristotelian ideal. Many common proofs for various theorems in geometry and analysis did not reveal *why* those theorems hold. Much of his work in mathematics proper accordingly consisted in attempts to remedy this defect.⁵ His account of grounding thereby provides a philosophical underpinning of his mathematical practice.

Another important motivation for Bolzano to develop his account was the debate around the *PSR* in rationalism. Due to the influence of Leibniz, the *PSR* was a widely accepted doctrine by the time Bolzano was writing, often appearing as one of the basic ‘laws of thought’ in logic textbooks (cf. Chapter 4 on the *PSR*). In Bolzano's estimation, however, the debate around the *PSR* was beset with confusion, not least because various different notions of *reason*, *cause* or *ground* were not properly kept apart when formulating and arguing for or against the principle. He noticed that at least some rationalists took the notion of a reason pertinent to the

formulation of the *PSR* to be a notion of *objective* (rather than epistemic) *priority* among truths (*WL* I §45.5, II §198: 341ff). Bolzano now took it that *this* notion of a reason is best construed as his notion of ground. As we shall see (Section 4), he then set out to propose analyses for other notions of reason that were employed by rationalists, such as causal and evidential reasons, in terms of grounding. Importantly, Bolzano was convinced that if the *PSR* is formulated in terms of grounding, it turns out to be false, since he believed that there are truths that lack any grounds. Against the background of his theory of grounding, he was, moreover, able to support this belief by rigorous arguments. We will investigate one of these arguments in Section 4.

The fact that Bolzano considered his account of grounding as an attempt of making precise what philosophers belonging to venerable traditions had accepted all along might be of more than mere historical interest. It might provide some evidence against insinuations of current grounding skeptics to the effect that grounding is merely an artificial postulate by 21st-century metaphysicians (cf. Chapters 11 and 12 on skepticism about ground). If that were the case, the fact that Bolzano's account of grounding is strikingly close to current accounts of grounding would appear to be an extraordinary historical coincidence. Such coincidences should, however, not be accepted as genuine if alternative hypotheses are available. And one plausible view of the matter indeed seems to be that current grounding-theorists are at bottom interested in the same notion as Bolzano and his predecessors. Knowingly or unknowingly, they have renewed interest in a topic that has belonged to Western philosophy virtually since its inception in ancient Greece.

3. The Theory

Preliminary Remarks. It is a commonplace in the present debate that the notion of grounding resists attempts at reductive analyses. Bolzano would largely agree with this. Although he occasionally toyed with the idea to define grounding in terms of a suitably restricted notion of entailment (more on this in Section 3), his official position is that grounding is a primitive notion that cannot be reductively analyzed (cf. *WL* II §202). In order to clarify the notion, he uses three complementary strategies that are also employed in the current debate (cf. Chapter 12):

- (i) he provides paradigmatic examples for cases of grounding,
- (ii) he relates grounding to neighboring notions such as, e.g., causation, and
- (iii) he introduces several general principles that grounding satisfies.

Most of this chapter will be concerned with the third strategy, but it will be helpful to have a brief look at the first two to begin with.

In line with a broad consensus in the current debate, Bolzano holds that the sentential connective 'because' can be used to express grounding-claims. The following two examples (loosely adopted from *WL* II §§199, 205) are cases in point:

- (1) Socrates is a Greek philosopher partially because he is Greek.
- (2) That Socrates is pale is true because Socrates is pale.

Bolzano does not, however, take because-claims to be the most perspicuous linguistic device for grounding talk. He considers grounding to be a genuine *relation* that is accordingly best expressed by using a relational predicate. What is conveyed by (1) and (2) is thus better expressed as follows:

- (3) That Socrates is Greek partially grounds that Socrates is a Greek philosopher.
- (4) That Socrates is pale fully grounds that it is true that Socrates is pale.

In the current debate between *predictionalists* and *operationalists* about grounding (on which see Introduction), Bolzano would thus side with the *predictionalists*.

As already mentioned, the *relata* of Bolzanian grounding are all and only *true propositions* or collections⁶ thereof (*WL II* §§203, 208). Bolzanian propositions are abstract objects that have a certain structure, i.e., are composed of certain subpropositional parts in a certain order. They are thus in many ways comparable to Fregean thoughts, although they are more finely individuated than the latter (cf. Künne 1997). The proposition

- (5) that two plus two equals four

is, for instance, different both from the proposition

- (6) that it is true that two plus two equals four,

and from the proposition

- (7) that it is not the case the two plus two is not equal to four.

Bolzano would take each of the latter to be more complex and to involve different constituents than (5). Indeed, he considers (6) and (7) to be grounded in (5). In the current debate between adherents of *worldly* and *representational* conceptions of grounding (Chapter 15), Bolzano would thus side with the *representationalists*.

There is a further reason why because-claims are imperfect vehicles to state that grounding obtains: 'because' can also be used to convey the obtaining of relations other than grounding. *First*, it can be used to state that the relation of *causation* (Chapter 20) obtains (as in 'The window broke because Bart threw a rock at it'); but causation is not to be confused with grounding, according to Bolzano. While grounding exclusively relates true propositions (or collections thereof), and thus certain *abstract* objects, causation exclusively relates *concrete* objects such as states or events (*WL II* §§168, 201). *Second*, because-claims can also be used to express the evidential relation of being a *reason to believe* that a certain proposition is true (as in: 'He must love him very much, because he brought him flowers'). This relation, too, is not to be confused with grounding according to Bolzano. While a reason to believe a given truth *A* may in some cases coincide with one of *A*'s grounds, it may also be a proposition not related to *A* by grounding at all (*WL II* §162, 198).

Outline of the Theory and Basic Distinctions. Although Bolzano did not consider the theory he outlines in the *WL* to be more than a first approximation (cf. *WL II* §195), it is of a *form* that sets it apart from earlier treatments of similar notions. We may indeed think of it as an informal axiomatic theory that operates with a basic undefined notion, characterized by several basic principles from which various corollaries are derived.⁷ In what follows, I will give a condensed overview of the most important aspects of this theory. I will put an emphasis on claims that can be used to compare it to current accounts about grounding.

To begin with, we will have to introduce a number of distinctions. As examples (3) and (4) show, Bolzano makes a distinction between full and partial grounds. While each conjunct of a given conjunctive truth *C* taken separately constitutes a partial ground of *C*, their collection constitutes *C*'s full ground. Next to the partial/full-distinction, there is also a distinction between mediate and immediate grounds. A truth *A* that grounds a conjunct of a given conjunctive truth *C*, for instance, will be a *mediate* ground of *C*. These distinctions will be familiar from most current accounts of grounding (see Introduction). It should be noted, however, that Bolzano applies the partial/full distinction *only* to immediate grounds. Mediate

grounds of a truth are always single truths connected to it via successive steps of partial or full grounding.

How are the notions of partial, full, immediate and mediate ground related? Bolzano appears to consider the notion of a full immediate ground as basic and defines the remaining notions in terms of it.⁸ Partial immediate grounds of a given truth *A* are defined as truths that are parts of a full immediate ground of *A*. As mentioned, the mediate grounds of a truth *A* are all truths that can be reached via successive steps of partial or full immediate grounding, including truths that belong to the full immediate ground of *A* as a limiting case (*WL* II §§216–7).

After these preliminaries, we can now finally approach Bolzano's theory. As it is presented in the *WL*, the theory can be neatly divided into two parts. Adopting (and slightly extending) a distinction from Kit Fine (2012), we may call these parts Bolzano's *pure* and *impure* theory of grounding, respectively. The pure theory of grounding deals with properties of grounding that are independent of the internal structure or shape of its relata. The impure theory, on the other hand, deals with properties of grounding that are dependent on the internal structure or shape of the relata, such as their degree of complexity or the kinds of concepts they contain, etc.

The Pure Theory of Grounding. The core principles of Bolzano's pure theory of grounding correspond to a number of ideas about grounding that are widely endorsed in the current debate; in particular, the ideas that grounding is factive, asymmetric, transitive and nonmonotonic (cf. Chapter 17).⁹ Essentially, Bolzano agrees with all of these ideas. In particular, he accepts the following three principles:

- (Factivity) If *A* mediately grounds *C*, then both *A* and *C* are true. (*WL* II §203)
- (Asymmetry) If *A* mediately grounds *C*, then *C* does not mediately ground *A*. (*WL* II §218)
- (Transitivity) If *A* mediately grounds *B* and *B* mediately grounds *C*, then *A* mediately grounds *C*. (*WL* II §217)

Since partial immediate grounding is a limiting case of mediate grounding, the principles (Factivity) and (Asymmetry) readily entail corresponding principles about partial and full immediate grounding (which Bolzano accepts independently anyway). The same is not true for (Transitivity), however. Perhaps needless to say, neither partial nor full *immediate* grounding are transitive relations (*WL* II §213).

Turning to the idea that grounding is nonmonotonic, things become slightly more complicated. That grounding is nonmonotonic means that adding truths to a given full ground Γ of some truth *C* can result in a collection of truths that is not a full ground of *C*.¹⁰ In the current debate, this claim is often motivated by invoking intuitions about *relevance*. Grounds are to be wholly *relevant* to what they ground, and there is no guarantee that adding arbitrary truths to a given full ground preserves relevance (cf. Fine 2012: 56). It is at least arguable that Bolzano would have agreed to this intuition, and he certainly accepts the nonmonotonicity of full immediate grounding. However, he does not rely on intuitions about relevance to *motivate* the latter. Rather, that full immediate grounding is nonmonotonic follows from a stronger principle that he endorses, namely:

- (Antimonotonicity) There are no collections of truths Γ , Δ and truths *C* such that Γ fully grounds *C*, Γ is proper part of Δ and Δ grounds *C*. (*WL* II §207)

Adding truths to the full immediate ground of a given truth will *never* yield another full immediate ground of that truth. Bolzano's motivation for this latter principle leads us to a more

controversial part of his account of grounding. In *WL* II §207, Bolzano presents a brief argument for (Antimonotonicity) that appears to rely on the following principle:

(Uniqueness) If Γ is a full immediate ground of C , then there is no collection of truths different from Γ that is a full immediate ground of C .

He seems to rely on this principle at various other places as well (see *WL* III §312: 230f, and *WL* IV §528: 266; cf. Roski 2017: Ch. 3.4.2). We may think of (Uniqueness) as capturing the idea that there is no ground-theoretic overdetermination: each grounded truth uniquely determines one, and only one, full immediate ground. This is a strong claim that will likely not be accepted by current grounding theorists (though see Poggiolini 2016). It is, for instance, incompatible with the widely held view that given any two true propositions A and B , each provides a full immediate ground of their disjunction $A \vee B$. Notably, also Bolzano has reasons to reject (Uniqueness). For it seems that some principles from his *impure* theory of grounding are incompatible with the principle (see endnote 11). And indeed, while he seems to *rely* on (Uniqueness) in the course of various arguments, what he officially *states* in the grounding sections of the *WL* is a much weaker principle. Call the collection of all truths that are fully and immediately grounded by the truths Γ , Γ 's *complete immediate consequence*. What Bolzano officially claims, then, is the following:

(Uniqueness_{Weak}) If Δ is the complete immediate consequence of Γ , there is no collection Γ' different from Γ such that Δ is the complete immediate consequence of Γ' . (*WL* II §206)

The collection of *all truths* grounded by a given full ground determines the latter uniquely. This principle allows for *individual truths* to be overdetermined by multiple full immediate grounds and is thus consistent with now-standard ideas about the immediate grounds of disjunctions. (Which is not to say that Bolzano would have accepted these ideas.)

The Impure Theory of Grounding. We now turn to Bolzano's impure theory of grounding. The first aspect of this part of Bolzano's theory we will investigate concerns his views on the interaction between grounding and broadly logical notions, including the concept of truth. Bolzano's ideas here are again remarkably close to those endorsed in the current debate. We will then take a look at another, somewhat more problematic aspect of his impure theory of grounding for which this is not true. This part concerns his views about the interaction between grounding and a certain form of entailment.

One of the most widely discussed topics in the current grounding debate concerns principles about the grounds of truths that have truth-functional connectives or quantifiers as their main operator. A widely held intuition in this connection is that grounding proceeds from less to more complex truths. Truths that possess truth-functional or quantificational complexity are grounded in their simpler constituents. For instance, conjunctive truths are fully and immediately grounded in the collection of their conjuncts; true existential quantifications are fully and immediately grounded in their true instances, etc. (See e.g., Batchelor 2010; Correia 2014; Fine 2012: 58–74; Schnieder 2011; and cf. Chapter 32.) Again, it is reasonable to assume that Bolzano would agree with these ideas to a certain extent. However, since his theory is couched in a traditional, broadly Aristotelian, logical framework, it is not entirely straightforward to assess his views. Bolzano works with a conception of logical form in which truth-functional connectives do not play a central role (cf. Morscher 2018: §3.4). Presumably because of this fact, he does not explicitly formulate principles about the grounds of truth-functionally complex propositions.

That being said, it is striking that Bolzano accepts a number of principles that seem to be motivated by the same kind of intuition that underlies present work on the logic of ground.

His main idea seems to be that complex truths involving broadly logical notions are always grounded in simpler truths that lack those notions. He discusses, for instance, principles about the grounds of truths that contain complex concepts that are composed by certain conjunctive and disjunctive concept-connectives:¹¹

- (Pred-Con) Truths of the form ‘Whatever has *a* has *b* and *c*’ (where ‘*a*’, ‘*b*’, and ‘*c*’ are place-holders for property-designators) are fully and immediately grounded in a collection of two truths of the form ‘Whatever has *a* has *b*’ and ‘Whatever has *a* has *c*’.
- (Sub-Dis) Truths of the form ‘Whatever has *a* or *b* has *c*’ are fully and immediately grounded in a collection of two truths of the form ‘Whatever has *a* has *c*’ and ‘Whatever has *b* has *c*’.

Moreover, Bolzano seems to assume a principle about the grounds of truths that involve iterations of concepts expressing property-negations (*WL* I §45: 206):

- (Pred-Neg) Truths of the form ‘Whatever has *a* has not-not-*b*’ are fully and immediately grounded in truths of the form ‘Whatever has *a* has *b*’.

The intuition that complex truths which are built up by means of broadly logical concepts are grounded in simpler truths that lack those notions also seems to underlie a further principle of Bolzano’s impure theory. He claims that truths that contain the concept of truth as their main predicate are grounded in the propositions this predicate is applied to. In particular, he endorses the following claim (which echoes a famous idea of Aristotle about truth; cf. *Categories* 5: 4^b 8–10 and 14^b 18–22; *Metaphysics* Θ 10: 1051^b 6–9):

- (Aristotle’s Insight) For every truth *A*, *A* is the full immediate ground of the truth that *A* is true. (*WL* II §205: 357)

Two remarks on this principle are in order. *First*, Bolzano also accepts a certain generalization of the principle. Not only does every individual truth *A* fully ground the truth that *A* is true. Also every collection of truths *A*, *B*, . . . fully grounds the truth that each of *A*, *B*, . . . is true. (*WL* II §205: 358) *Second*, due to his acceptance of (Aristotle’s Insight), there is a sense in which Bolzano accepts the intuition that truth is grounded in being, which is at the root of the debate on truth making (cf. Chapters 28 and 36). It should be stressed, however, that Bolzano does not assume that propositions are made true by worldly entities (Tatzel 2002: 11–12). He adheres to the idea that truth is grounded in being only to the extent that truth-ascribing propositions are never fundamental but always grounded in the propositions they ascribe truth to.

Although he does not put it in those terms, Bolzano’s endorsement of principles such as (Aristotle’s Insight) and (Pred-Neg) shows that he considers grounding to be hyperintensional: while, for instance, any truth *A* is necessarily equivalent to the truth that *A* is true, they are not interchangeable in the context of ‘. . . fully immediately grounds . . .’, as this would violate the asymmetry of grounding.

Turning to truths that correspond to quantified statements, matters become less straightforward. On current conceptions, such truths are usually taken to be grounded in their instances (e.g., truths of the form ‘There is an *F*’ are grounded in truths of the form ‘*a* is *F*’). While Bolzano has a comparatively advanced view about logical form and truth-conditions of quantificational propositions, he does not suggest any systematic principles about the way they are grounded. Moreover, some

examples he discusses even seem to be in conflict with current views. Most strikingly, he suggests that the existentially quantified truth *that there is something* is an example of an ungrounded truth, which is incompatible with the idea that every existential quantification is grounded in its obtaining instances (cf. Roski & Schnieder 2019: §4). Regarding universally quantified truths, we do not find any explicit discussion of systematic principles governing their grounds in Bolzano's writings.

We finally turn to a somewhat more contentious aspect of Bolzano's impure theory of grounding. This part of the theory deals with various principles that pertain to the interaction between grounding and the consequence relation of deducibility (cf. *WL* II §221.3–7). (For present purposes, we may think of deducibility as a kind of entailment that does not validate inferences from contradictory premises.¹²) The principles in question are restricted to a certain subclass of all truths, so called *conceptual truths*. Conceptual truths are truths that do not contain any constituent that directly (nondescriptively) refers to a concrete object and are thus either concerned with the abstract realm or else generalize about the domain of concrete objects in one way or another. Typical examples of such truths are mathematical or logical laws as well as metaphysical principles—truths that were traditionally conceived to be knowable *a priori* and susceptible to axiomatic theorizing.¹³

The general intuition underlying Bolzano's principles about the interaction between grounding and deducibility among conceptual truths can be put as follows: in the realm of conceptual truths, grounding generally goes along with a certain form of *deductive economy*. Slightly more precisely, we can put the two central claims as follows:

(Full Ground Economy) If Γ is a full immediate ground of C , then Γ constitutes a *minimal premise set*¹⁴ for C . (*WL* II §221.5)

(Partial Ground Economy) If A is a partial immediate ground for C , then A exhibits a minimal degree of complexity among all equivalent (i.e., mutually deducible) premises. (*WL* II §221.4)

These principles can be considered as motivating certain minimality requirements for ground-revealing *proofs*—principles that Bolzano perceived to be operative in the mathematical practice of his time. A ground-revealing proof for a conceptual truth will neither make use of more premises than strictly necessary nor make use of any premise that is unnecessarily complex.¹⁵

Spelling out these ideas in detail is an intricate matter that would take more space than this chapter can offer, so I have to point the interested reader to pertinent literature (Roski & Rumberg 2016; Roski 2017: Chapter 4.3). It should be mentioned, however, that the principles give rise to certain tensions in Bolzano's account (see *op. cit.*). Recall, for instance, that Bolzano holds that every arbitrary collection of truths Γ is the full immediate ground of the truth that each proposition in Γ is true. But since Γ may be any arbitrary collection of truths, it may well contain deductively redundant propositions such as logical truths. However, no collection of premises that contains deductively redundant truths can be a minimal premise set for any proposition—contrary to (Full Ground Economy).

In addition to the two claims mentioned, Bolzano also endorses a certain generalization of (Full Ground Economy). Again put somewhat roughly, he claims that the order that grounding induces on the class of conceptual truths as a whole is such that the collection of ungrounded conceptual truths constitutes a minimal deductive base for all remaining truths (*WL* II §221.3). This idea seems obviously related to the unification theory of explanation. Spelling it out in detail unfortunately again exceeds the scope of this chapter. (See Roski 2019 for a discussion.)

While Bolzano's principles about the interaction between grounding and deducibility reveal certain tensions in his overall account, the ideas that motivate them are essential to his views on grounding. This is evident from Bolzano's own estimation of his account of grounding in a late text that summarizes his central doctrines and highlights the principles in question as particularly important (Bolzano 1841: 67–68). Apart from that, Bolzano even toyed with the idea that the principles could be turned into necessary and sufficient conditions for grounding to obtain and perhaps might even form the basis of a proper definition of the concept of grounding (see *WL* II §221.7 and the note to §221).¹⁶

4. Further Topics

Monism about Grounds. We have seen that Bolzano takes 'because' to sometimes introduce the relation of grounding, sometimes the relation of causation and sometimes that of being an evidential reason for. However, on Bolzano's account, these different uses are intimately related. This is because he takes the notions that are expressed by causal and evidential uses of 'because' (and cognate phrases such as 'is a ground of') to be analyzable in terms of grounding. Key to his analyses is his account of causation. The main idea of this account is that there is a certain parallelism between the realm of causes and effects and the realm of grounds and consequences. Causal priority among concrete objects such as states of events mirrors priority among truths that ascribe the property of being *actual* (i.e., roughly, existing spatiotemporally) to the respective objects. More precisely, Bolzano proposes the following definition:

(Df. Causation) A state or event *c* causes a state or event *e* iff the truth that *c* is actual grounds the truth that *e* is actual. (*WL* II §§168, 201; cf. Schnieder 2014 for discussion.)

Reasons to believe are then introduced as a special type of causes, namely causes for certain beliefs, and reasons for action are also classified as a subclass of causes. (See *WL* II §313 for reasons to believe; Bolzano 1994: I, 42, 1838: 176ff for reasons for actions.) Bolzano thus defends a certain form of *monism* about notions of ground and reason. There is only one fundamental notion of ground that gives rise to a variety of cognate notions (cf. Roski & Schnieder 2016). While the actual analyses Bolzano presented will undoubtedly appear controversial to contemporary readers, Bolzano's idea to find unity in an apparent diversity of notions of ground or reason is an idea that deserves to be revisited (cf. Berker 2018). Indeed, the view that there is a close connection between causation and grounding has received a certain revival in recent years in the works of Schaffer (2015) and Wilson (2018) (cf. Chapter 20).

Fundamentality. In the current debate on grounding, the view that there are ungrounded truths is widely accepted (cf. Chapters 16, 23 and 27). Bolzano would welcome this agreement. He was convinced of existence of ungrounded truths throughout his philosophical development. That is, he accepted the following claim:

(Fundamentality) For some truth *C*, there is no truth that grounds *C*. (*WL* II §214)

This claim merely entails that there is at least one ungrounded truth. It does not entail that each grounded truth is ultimately dependent on ungrounded ones, i.e., that there is a fundamental

level in the realm of truths. Yet Bolzano seems to hold this stronger claim as well (as his reasoning in *WL* §215 shows). That is, he seems to hold:

(Fundamental Level) There is a collection of truths Φ such that (i) every truth in Φ is ungrounded and (ii) every truth that is not in Φ is mediately grounded in truths in Φ .

Bolzano had a remarkably clear insight about what these two claims entail and what they do *not* entail. To give just one example: he was aware of the fact and came up with various examples to show that both claims are compatible with the existence of infinitely descending grounding-chains. (Cf. Roski & Schnieder 2019: §3, and cf. Chapter 16.)

Notably, Bolzano did not merely *assume* (Fundamentality) and (Fundamental Level). Rather, he tried to establish them by argument. I will confine myself here to just a sketch of the most promising argumentative strategy he proposes for a restricted version of (Fundamental Level). (For a comprehensive discussion of this and other arguments for the existence of ungrounded truths in Bolzano, see Roski & Schnieder 2019.)

We have seen that Bolzano believes that many cases of grounding go along with a reduction of propositional complexity. Truths involving broadly logical concepts are, for instance, grounded in truths that do not contain those concepts. Now, if *every* case of grounding would go along with such a reduction of propositional complexity, the existence of ungrounded truths could be readily established. (Or rather, it would follow from the uncontroversial assumption that there is at least one finitely complex truth.) But of course, the view that *every* case of grounding goes along with a reduction in propositional complexity seems as much in need of argument as the view that there are fundamental truths. What Bolzano saw, however, was that even if not every case of grounding goes along with a reduction of propositional complexity, the existence of ungrounded truths can also be established if grounding merely satisfies a complexity constraint relative to a certain subclass of all truths.

The pertinent subclass is, in Bolzano's view, the class of conceptual truths that we have mentioned earlier in Section 3 (p. 83 above). These truths are subject to three general principles that, against the background of the pure theory of grounding, entail the existence of ungrounded truths. The first principle concerns the building blocks of conceptual truths. Recall that Bolzanian propositions are complex entities that are composed of subpropositional constituents. Some of these constituents are complex themselves but Bolzano holds that they are ultimately composed of certain atomic elements. With respect to conceptual truths, he holds, moreover, that there are only finitely many such atomic constituents:

(Finitude) There is a finite set of atomic constituents from which each conceptual truth is ultimately composed.

If one were to devise a language in which each atomic constituent of a conceptual truth gets assigned exactly one atomic expression, it would be a language with a finite vocabulary. Bolzano measures the *degree of complexity* of a proposition by the number of occurrences of atomic constituents from which it is composed. (Bolzano assumes that the same atomic constituent can occur multiple times in a given proposition—just as a term can occur multiple times in a given sentence.) (Finitude) then entails that there are only finitely many conceptual truths of any given finite degree of complexity. To see this, think again in terms of a language: if there are only finitely many atomic expressions, there cannot be infinitely many different complex expressions that have a given finite maximum length.

Two further premises needed for the argument pertain to the way conceptual truths are grounded. The first says that the grounds of a given conceptual truth will always consist of conceptual truths:

(Closure) If x is a conceptual truth, and y mediately grounds x , then y is a conceptual truth.
(*WL II §221.1*)

Bolzano's thought here is that, for instance, general mathematical truths will never be grounded in truths concerned with any individual concrete object in particular (cf. Textor 2013 for discussion).

The final principle Bolzano needs for his argument is a certain complexity constraint. While grounding might not always go along with a reduction in propositional complexity, at least in the realm of conceptual truths, grounds will never be *more* complex than the respective grounded truths. In other words:¹⁷

(Complexity) If x is a conceptual truth, and y mediately grounds x , then y is at most as complex as x . (*WL II §221.2*)

Given the principles of Bolzano's pure theory of grounding that we have discussed earlier, these three claims entail that there are ungrounded truths. For, recall that (Finitude) entails that there are but finitely many conceptual truths of any finite degree of complexity. (Closure) and (Complexity) entail that the mediate grounds of any conceptual truth C will be conceptual truths themselves and never be of a higher degree of complexity than C . But that means that the set of mediate grounds of any conceptual truth of a finite degree of complexity will be a finite set. Suppose for reductio that there are no fundamental truths. In this case, each truth would rest on an infinite set of mediate grounds, which can't be the case given what we've established earlier. So there are fundamental truths, and, what's more, there is a fundamental level of truths wherein each conceptual truth of finite complexity is ultimately grounded.

While this argument relies on Bolzano's notion of a conceptual truth and the three aforementioned principles governing this notion, the argumentative *strategy* Bolzano pursues is not tied to this category in particular. For principles similar to (Finitude), (Closure) and (Complexity) that allow one to carry out a Bolzano-style argument for ungrounded truths may well hold for other classes of truths. To identify such classes seems to be a worthwhile project also against the background of current assumptions about grounding.

Related Topics

Chapter 1 Ancient (Corkum)
Chapter 2: Demonstration (Malink)
Chapter 4: Principle of Sufficient Reason (Amijee)
Chapter 11: Skeptical Doubts About Ground (Koslicki)
Chapter 12: Anti-Skeptical Rejoinders (deRosset)
Chapter 15: Granularity (Correia)
Chapter 16: Infinite Descent (Dixon)
Chapter 17: Strict Partial Order (Thompson)
Chapter 20: Cause (Wang)

Chapter 23: Fundamentality (Bliss)
Chapter 27: Structure (Tahko)
Chapter 28: Truthmaking (Trogdon)
Chapter 32: Logic (McSweeney)
Chapter 36: Semantics (Fine)

Acknowledgments

For many helpful suggestions, I am grateful to Kit Fine, Stephan Krämer, Benjamin Schnieder, Mike Raven and members of the audience at the Conference on the *Routledge Handbook on Metaphysical Grounding*, Hamburg, 2018. Work on this chapter has been supported by the DFG Project *Relevance* (Grant KR 4516/2–1).

Notes

- 1 One exception: Brentano and his circle of followers as well as Husserl were aware of Bolzano's work. On Bolzano's life and the reception of his works, see (Künne 2008: Chs. 1, 2 and 9) and (Morscher 2018: §§1,13).
- 2 For an accessible overview of Bolzano's mathematical work, including its relation to his philosophical work, see (Rusnock 2000). A recent overview of Bolzano's philosophy is (Rusnock & Sebestik 2019). A classic text is (Berg 1962).
- 3 (Bolzano 1985ff/1837). References will be to the original pagination of the text (in the format: *WL* <volume> §<section>: <page>), which is indicated in the margins of the recent English translation of the *WL* (Rusnock & George 2014).
- 4 This is slightly simplifying Bolzano's more nuanced position; see *WL* IV §§393–4, 483 and cf. (Roski 2014: Chapter 6). Whether Bolzano is correct in his interpretation of followers of the Aristotelian conception of science is a controversial question that will not be assessed here (see Betti & de Jong 2010 for a discussion of the conception in general).
- 5 The most famous work in this connection is his *Purely Analytic Proof* of the Intermediate Value Theorem in (Bolzano 1817). See (Rusnock 2000) for an accessible discussion.
- 6 For the purposes of this chapter, a *collection* of propositions can be understood as a mereological compound that has those propositions as atomic parts; see (Schnieder 2017) for more on Bolzanian collections.
- 7 Notable proposals for rigorous axiomatic reconstructions of Bolzano theory include (Berg 1962: Ch. VI), (Morscher 2016), (Casari 2016: Ch. 7) and (Tatzel 2002). More accessible informal discussions of Bolzano's theory can be found in (Rumberg 2013), (Roski & Rumberg 2016) and (Schnieder 2014). The aforementioned paper by Tatzel also provides a good informal overview of the account next to the axiomatic reconstruction. A book-long treatment is (Roski 2017). (Roski & Schnieder forthcoming) contains translations of Bolzano's most important texts on grounding as well as a number of original essays on the topic.
- 8 Bolzano's text is not completely unequivocal on what notion he takes as basic. The reconstructions in (Berg 1962: Ch. VI), (Morscher 2016) and (Casari 2016: Ch. 7) follow the present approach. Tatzel's (2002) reconstruction takes the relation between a full immediate ground and the collection of all truths immediately grounded by it (its full immediate consequences) as basic.
- 9 While each of these ideas has been problematized, those who argue, e.g., against the asymmetry of partial grounding usually picture what they argue against as 'the orthodoxy'. Note that Bolzano also discusses a nonfactive notion of grounding; see *WL* II §203: 353ff and cf. (Fine forthcoming) for discussion.
- 10 Since current theories of grounding do not generally include specific principles about *immediate* grounding, they usually contain more general nonmonotonicity principles that entail the nonmonotonicity of immediate grounding.
- 11 See *WL* II §§199, 221.7, 227.4 and §227.7. The following three principles involve slight idealizations/simplifications of what Bolzano explicitly states. With respect to the first two principles he requires,

- for instance, that the involved property designators actually pick out different properties. Notice that (Pred-Con) and (Sub-Dis) allow for violations of the aforementioned principle (Uniqueness): A truth of the form ‘Whatever has *a* or *b* has *c* and *d*’ is fully and immediately grounded, on the one hand, in ‘Whatever has *a* or *b* has *c*’ and ‘Whatever has *a* or *b* has *d*’ and, on the other hand, in ‘Whatever has *a* has *c* and *d*’ and ‘Whatever has *b* has *c* and *d*’ (see Rumberg 2013: 451–2).
- 12 See (Siebel 2002) for a concise overview and cf. (Rusnock & Burke 2010). Bolzano appears to assume that grounded truths are in general deducible from their full grounds. He mentions, however, one interesting exception to this rule that concerns the grounds of normative truths; see *WL* II §200. For discussions of the example, see (Tatzel 2003), (Betti 2010) and (Roski 2017: Ch. 4.1).
 - 13 See *WL* II §133. Conceptual truths contrast with *intuitional* truths, which contain at least one constituent that directly (nondescriptively) refers to a concrete object. Paradigmatic examples for such truths are many truths that are expressed by sentences that contain indexical terms (such as ‘This building is tall’). For a comprehensive study of the conceptual/intuitional-distinction in Bolzano, see (Textor 1996). (Textor 2013) provides a concise, accessible overview.
 - 14 That a premise set is minimal means, roughly, that there is no alternative set with *fewer* premises. A fully precise statement of the principle would require several qualifications of the pertinent notion of minimality. For instance, since every truth *A* is deducible from itself, there is a premise set for every truth that is very often smaller than *A*’s full ground, namely the set containing *A* itself. Bolzano can deal with problems like this, but going into these complications would take us too far astray.
 - 15 For a thorough discussion of Bolzano’s account of ground-revealing proofs and its relation to current work in proof theory, see (Rumberg 2013).
 - 16 On Bolzano’s attempt to articulate sufficient conditions for grounding, see (Roski 2017: Ch. 4.5) and (Roski & Rumberg 2016); on his proposal for a definition of grounding, see (Betti 2010) and (Roski 2017: Ch. 4.6).
 - 17 See (Fine forthcoming) for a recent discussion of this claim.

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6

AUSTRO-GERMAN PHENOMENOLOGISTS

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§1 Introduction. Essence, Foundations, Grounds, and Modality

Brentano's heirs, in particular Husserl and Meinong, as well as their students and many philosophers influenced by them, rely heavily on grounding (*begründen*), founding (*fundieren*), and related ties such as dependence (*Abhängigkeit*) and existential relativity. They thought that certain truths or facts ground or explain other facts or truths and that some objects depend on or are founded on other objects, as when a melody depends on certain tones. Their interest in foundation and dependence seems to derive directly from Brentano.¹ The interest in grounding, on the other hand, may well have been mediated by Husserl's acquaintance with Bolzano's pioneering account of what he called the relation of *Abfolge* or grounding.²

In this section, we look at how Brentano's heirs understood grounding and foundation, in particular their relation to essentialism and modality, and say something about the variety of the claims they put forward that employ grounding and foundation. In the following sections, we look in more detail at a handful of philosophical claims formulated in terms of foundation (§2) and grounding (§§3–5).

The ties of foundation and grounding were often taken to be themselves grounded or rooted in essences.³ Essentiality and necessity, however, were not identified. As Reinach puts it, “universality and necessity can both be traced back one step further . . . in every case they are to be found if and only if the predicate is grounded in the essence of the subject” (Reinach 1989a: 70; English tr.: 165). The most important type of necessity, which Husserl calls “objective or ideal necessity”, is law-like and “grounded in essences” (Husserl LU III §7; cf. Mulligan 2004). Such “essential” or “eidetic necessity” is then opposed to “empirical necessity” and “causal necessity”, as “essential or ideal laws” are opposed to “contingent” “empirical laws” (LU III §7; Reinach 1989a). Some noncausal necessities and possibilities are grounded in the essence of objects *tout court*: objects, Husserl says, “can stand to one another in the relation of parts and wholes” and this is “a type of relation which is grounded in the idea of an object” (Husserl LU III §1). Some in the essences of concepts. Others in the essence of certain concepts. Others again in the essence of certain objects, such as colour, emotion, negation, or truth. Finally, they often think of the most important types of grounding and foundation as involving law-like, objective necessity and so as grounded in essence. As we shall see (§4), “grounds” and “founds” are not always distinguished. This may not be an accident. In German “*Grund*” may be used to refer to

the antecedent of a conditional *and* to that which grounds or explains something else. The use of “founds” as an alternative to “grounds” is an indication that a grounding claim is being made. So, too, is the use of “because”.

In the light of the recent interest in analytic philosophy in grounding and dependence, it is worth noting Husserl’s appeal to grounding in his 1891 criticisms of Frege and Russell’s mixed reaction to Meinong’s 1899 (1971: 377–480) philosophy of higher-order objects and the objects they depend on. In his discussion of what has been called Hume’s principle,

there is a one–one correlation between the F’s and the G’s iff the number of F’s is the number of G’s

Husserl suggests that there is such a correlation *because* there is such an identity. “Do we”, he asks skeptically, “call a set of nuts lying before us ‘four’ because it belongs to a certain class of infinitely many sets that can be mutually put into a one–one correlation?”⁴

Russell’s 1904 review of Meinong’s 1899 philosophy of higher-order objects expounds it as follows:

Among objects, there are some that have an intrinsic lack of independence; thus diversity, for example, can only be thought of in relation to differing terms. Such objects are based on others as indispensable presuppositions: Meinong calls them “objects of higher order”, and the presupposed objects he calls inferiora, in respect to which they (the objects of higher order) are superiora. An object which can have an inferior must have one; but an object which can have a superior need not have one . . . Not all objects of higher order are relations: . . . A melody, again, is such; and so is a red square, being compounded of a shape and a colour . . . The above instances make it fairly, plain what class of objects Meinong has in view . . . They have, he says . . ., an internal lack of independence in their nature; they are built on other objects as indispensable presuppositions . . .

(Russell 1904: 207)

He then objects to Meinong’s account of dependence:

there are certain difficulties about this description. In the first place, it is based upon logical priority: the inferiora are in some way prior to their superiora. Now logical priority is a very obscure notion; and so far as can be seen at present, it is one which a careful discussion tends to destroy. For it depends upon the assumption that one true proposition may be implied by another true proposition,—and not the other by the one; whereas, according to symbolic logic, there is a mutual implication of any two true propositions. The appearance of one-sided implication in such cases arises, it would seem, from an unconscious substitution of formal for material implication. Thus it would result that the subsistence or being of a whole cannot presuppose that of its parts in any sense in which that of the parts does not presuppose that of the whole.

(Russell 1904: 207–8)

A further objection stems from Russell’s rejection of the objective necessity Meinong relies on in many of his claims. As Russell notes, “[t]hroughout Meinong’s work, in many crucial points, use is made of the notion of necessity; and some of his most important arguments fail if necessity

is not admitted” (Russell 1904: 208). Russell, however, is impressed by some of Meinong’s examples and qualifies his first objection:

It must be admitted, however, that one-sided inferences can practically be made in many cases, and that consequently some relation or relations other than that considered by symbolic logic must be involved when we infer. One such relation is that with which Meinong is concerned, the relation of the simple to the complex: the simple is prior to the complex in the sense that we can infer it from the complex, whereas the converse inference, even when it is valid, can only be seen to be valid, as a rule, if the conclusion is already known to be true.

(Russell 1904: 208)

The type of priority Meinong has in mind, Russell goes on to argue, is in fact “epistemological priority”. Husserl’s 1900–01 philosophy of foundation is more carefully worked out than Meinong’s.⁵ In §2, we shall look at one application of this philosophy.

When one of Brentano’s heirs makes a claim of the form *The fact that q is grounded in the fact that p in virtue of the essence of γ* the specification of essence is often a reference to something about the fact that q rather than the fact that p. “It is”, as Fine says, “the fact to be grounded that points to its grounds and not the grounds that point to what they may ground” (Fine 2012: 76). The account of the way the truths of truth bearers are grounded in facts set out in §3 conforms to this pattern: that every true judgement has a ground which makes it true is, Pfänder argues, itself grounded in the *essence of judgement* and in the *essence of truth*. But Reinach, for example, claims that if the obligation to *F* is grounded in the promise to *F*, then that this is the case is grounded in the essence of promissings. His example belongs to a large family⁶ in which there is a correlation between the relation of grounding and a non-causal relation of generation (*erzeugen*) between episodes: the event of promising makes an obligation begin to hold.

A common view is that only facts can stand in the grounding relation.⁷ Sometimes, as in Husserl, these are taken to be true truth bearers and sometimes as something more worldly: “it is not existing objects but obtaining states of affairs”, Reinach says, “which stand in the relation of being grounded”, a relation which is “quite different from” the just-mentioned relation of generation, although he calls both of them examples of “univocal determination” (Reinach 1989b: 156). But Meinong, for example, says that although he had long thought that only objectives (states of affairs) can be grounds, this view is in fact “artificial”: some grounds are objectives, others are objects: “the objective that red is not green is not based on any objective”, its grounds are the two objects, red and green (Meinong 1968: 583–4).

In the Austro-German tradition, there is no full account of the philosophy of grounding. But in Reinach, some important elements of such a philosophy are to be found. Thus in the course of arguing that property cannot consist in but rather grounds certain rights, he asserts: “If something is, as a matter of essential lawfulness, grounded in something else, the former cannot be composed of the latter (Reinach 1989b: 194–5). The grounding claim discussed in §4 takes very seriously Reinach’s precept.

Reinach also warns against two mistakes which may be made when a grounding claim is formulated. The first is to

substitute for the object in the essence of which a predication is grounded a more comprehensive object in which the predication has been realised at sometime and somewhere . . .

The second mistake is to confuse “conceptual inconsistency” and “essential inconsistency”, a confusion only encouraged by the expression “conceptual impossibility” (Reinach 1989b: 203).

The distinction already mentioned between necessities, for example necessary possibilities, which are grounded in essences of different kinds, on the one hand, and inessential, natural necessity, on the other hand, is at the heart of Kit Fine’s influential philosophy of modality and essence. But Fine allows also for a second kind of inessential necessitation, *normative necessitation*.⁸ Like Moore in 1922, he thinks that the exemplification of natural properties by an object could not simply necessitate the exemplification by that object of a normative or value property (Fine 2005). In 1915, Meinong’s student, Witasek, had argued likewise that it cannot follow from, say, the colour and spatial properties of an ornament that it is beautiful. The relation between the beauty of the ornament and its natural properties is not, he argues, like that between a higher-order object such as the difference between red and green, on the one hand, and the two colours, on the other hand. This higher-order object is “connected with its inferiora by a priori necessity, whereas such a priori necessity is completely lacking in the case of the relation of beauty to its bearer” (Witasek 1915: 181). If we consider two colours,

it is grounded with necessity in the essence of the two colours and of similarity (difference) whether the two colours are or are not similar. But it is quite otherwise in the case of beauty . . . [I]t does not follow necessarily and a priori from the essence of two [contiguous] colours and of beauty that the contiguous colours must be beautiful.

(Witasek 1915: 182)

Nevertheless, “because the object has such and such properties, it is beautiful” (Witasek 1915: 185). Its beauty depends not merely generically on some natural properties but specifically on certain natural properties of the object. Witasek is therefore a friend of what has been called normative grounding if not of normative necessitation.

Husserl, on the other hand, not only endorses Hume’s principle that the exemplification of a non-normative property cannot simply necessitate the exemplification of a value-property but argues that the exemplification of non-normative, for example natural, properties *may* necessitate the exemplification of value in virtue of the nature of value (Husserl 1988: 70–101). Husserl thus allows for only two types of necessity, essential and inessential, and what has been called normative necessity turns out, on his view, to be a *sui generis* kind of essential necessity.

One of the most interesting examples of the grounding of a normative fact on a non-normative fact in Austro-German philosophy is the view set out in the *Prolegomena* to Husserl’s 1900–01 *Logical Investigations*: logical norms are grounded in logical truths. If one ought not to judge that (p & not- p), this is because it is not the case that (p & not- p),⁹ a view not unlike that of Frege, which was to be rejected by many 20th-century philosophies of logic. In §5 we look at one version of a type of view very popular in the writings of Husserl and Meinong and their followers: deontic norms are grounded in values.

In 1927, Heidegger put an end to the hegemony of essence in phenomenology with his suggestion that essence is not ontologically fundamental. In §5, we look briefly at this episode.

§2 Unity and Composition

The philosophies of mind of Brentano’s heirs are often formulated in terms of relations of dependence or foundation between mental acts or their features. Emotions, Husserl says, are one-sidedly founded on judgments, perceivings, remembering, or expecting. Mental modes and token-contents are mutually dependent. But perhaps the most important application Husserl

makes of the relation of foundation is in his account of the unity of individuals, of the unity of physical things and of mental or psychological things. “A physical thing, such as a house or a tree”, he says, is an individual, that is to say, a temporal and empirical object and, “like all such objects, it has phenomenologically no other unity than that given to it by the unification of attributes (*Beschaffenheiten*), a unity which is grounded in their own contents”. Physical things are independent (*selbständig*) objects and “the unity of independent objects comes about through foundation”, since “whatever truly unifies are the relations (*Verhältnisse*) of foundation” (Husserl 1984a: III §22 A 286).

Similarly, in 1900–01, Husserl endorsed a conception of the self which he attributes to Hume, who famously wrote that it is “the composition of” perceptions “which forms the self” (Hume *Treatise* Appendix). Husserl says he can find no “primitive self” which “underlies” and so unifies experience, no “peculiar self hovering over manifold experiences”. The self is rather “simply identical with their unity of connection”, a unity which is described as phenomenological. This “empirical self”, like empirical, physical things, is a temporal object. The “forms of connection” between the experiences making up a self “are grounded in the nature of the contents” of these experiences and in “the laws to which they are subject” (Husserl 1984a: V §4 A 272).

What is meant by talking of the *phenomenological* unity which is a house or a self? Strictly speaking, the account so far given of the unity of a physical thing is an account of the unity of a *concretum*:

To the unity of a thing there belongs more than one isolated concretum; there belongs to the unity of a thing (ideally speaking) an infinite multiplicity of temporally successive (in the sense of change and endurance) *concreta* which flow into one another continuously and which are of one and the same form, a multiplicity . . . spanned by the *unity of causality*.

(Husserl 1984a: III, §12 A, 261)

What holds of physical things and *concreta* holds too, Husserl thinks, of the “ego as an enduring object” and the “momentary phenomenological ego” (Husserl 1984a: V, §4 A, 332):

Just as the external thing is not the isolated complex of features (*Merkmalcomplexion*) at a moment, but . . . a unity which persists through the multiplicity of actual and possible changes, . . . so too, the ego is first constituted as a subsisting object in the unity which spans all actual and possible changes of the complex of experiences (*Erlebniscomplexion*). And this unity is no longer phenomenological unity but lies in causal law-likeness.

(Husserl 1984a: V, §4 A, 332)

Thus the “phenomenological” unity at a time or perhaps over a short period of a thing or a self, the sort of unity which can be given in experience, derives from relations of foundation which are grounded in essences. But the causal laws required for the unity over time of things and selves are not so grounded. Nevertheless, at one point, Husserl speculates that in “order to get to a thing from a *concretum* a generalisation of the concepts of independence and dependence which transfers these to the domain of succession and causality must be possible.” (Husserl 1984a: III, §12 A, 251).

But Husserl soon abjured the Humean conception of the self in favour of a transcendental ego which resembles in some respects Wittgenstein’s “philosophical self” or “metaphysical subject” (Wittgenstein 1977: 5.641) and Thomas Nagel’s “higher self” (Nagel 1986: 62)¹⁰.

§3 Truth Versus Correctness: Their Bearers, Essence, and Grounds

What are the bearers of truth? Beliefs, judgments, and other mental acts, assertions, or more or less abstract entities such as propositions, statements, and judgements? What are the consequences of answers to these questions for the view that the truth of some or all truths has a ground? Husserl in 1900–01 and then Pfänder argue that truth-bearers are abstract entities. Husserl initially called them propositions. Pfänder and later Husserl call them judgements. In 1921, as we shall see, Pfänder defends a version of truth-maker (more exactly: true-making) maximalism about true judgements: the truth of every true judgement has a ground, which makes it true. To think that judgments or beliefs are truth-bearers, Husserl argues, is to overlook the difference, pointed out by Bolzano, between correctness and truth: judgments or beliefs are correct or incorrect; propositions are true or false. The correctness of judgments and beliefs, as we shall see, has a ground, according to Husserl.

Pfänder's 1921 (231–43, 75–89) account of truth, its grounds, grounding and essence seems to be the first systematic account of these matters. It is a factualist, essentialist version of truth-maker maximalism: every truth has a ground, an obtaining state of affairs or fact, which is a ground of its truth and makes it true, a principle Pfänder calls the *logical* principle of sufficient ground (reason, *Grund*), henceforth: PSG.¹¹ What, Pfänder asks, makes *this* principle true? His answer is that its “own sufficient ground lies . . . in the *essence of judgement* and in the *essence of truth*” (Pfänder 1921: 232). Let us look at one or two of the details.

Pfänder's truth-bearers, like those of Wittgenstein in the same year, “project” (*entwerfen*, Pfänder 2000: 36; Wittgenstein 1977: 2.0212, 3.11, 3.12); they project states of affairs (*ibid.*). Pfänder's truth-bearers (unlike Wittgenstein's candidates) are not mental or linguistic items but non-temporal entities, judgements. What exactly does his logical principle of sufficient reason, his truth-maker principle, assert?

The genuine sense of the PSG is that it specifies in a general way what a judgement requires in order for its claim to truth to be not mere pretence but a satisfied claim. The principle therefore says

Every judgement, in order to be really true, stands necessarily in need of a sufficient ground.

By the *ground* of a judgement is to be understood what can support the assertoric content of the judgement. This ground is “*sufficient*” if it alone suffices to support the complete assertoric content of the judgement, if nothing else is required to *make the judgement completely true*. (Pfänder 2000: 227)

In virtue of what is the principle logical? If there is a *logical* PSG it must

first, deal with purely logical objects; it must, secondly, assert something which is purely logical about these objects; and it must, thirdly, base what it asserts on the specifically logical essence of the objects it is about. Of the different logical objects, only judgements, not concepts nor inferences (*Schlüsse*) are suitable subjects for the [logical] PSG. This principle must therefore assert something about judgements.¹²

Pfänder's claim that true *judgements* are made true should be sharply distinguished from the claim by Husserl and Marty that if a *judging* or belief that *p* is correct, there is something which makes it correct. According to Husserl, the concept of the correctness of judging presupposes the concept of

truth. There is a measure or standard of correctness, something on which correctness is based, namely truth. According to Marty, the way the world is grounds the correctness of judgments. According to Husserl, as we have noted, truth is not correctness. Why not? First, strictly speaking, mental acts and states can be correct but not true, and only propositions can be true. Secondly, correctness, unlike truth, is a normative property. Finally, the concept of truth, unlike that of correctness, is simple.

An interesting generalization of these views is the claim that not only judgments but also emotions and preferences have correctness makers or grounds. In 1908, Marty argues that there is something “independent of the subjective phenomena of loving and hating . . . which grounds this correctness” (Marty 1908: 370). Husserl, too, is of this opinion. The question about the “reasonableness” of some valuing (emoting), is first of all the question “whether it is correct, whether the meant value is really a value” (Husserl 2004: 117) and, secondly, whether it “is not merely correct but completely grounded” (Husserl 1988: 240–4; cf. 278ff.). Thus if the preference for *x* over *y* is correct, this is because *x* is better than *y*.

The endorsement of correctness-makers for judgments and emotions by Husserl and Marty and of truth-makers for truth-bearers by Pfänder represents a sharp break with Brentano’s mature views. According to Brentano’s follower Kraus, Brentano’s peculiar merit consists in the way he brings out and analyzes the *sense* of what is called the objectivity or absoluteness of values (Kraus 1937: 205):

We call an emotion, a feeling or a willing correct, right, or justified, not *because* it agrees with an axiological state of affairs or some eternal values, but *because* a valuing of the object called “valuable” or “good” which announces itself as justified, a valuing characterised as correct, could not possibly value it in any other way.

(Kraus 1934: 76, *emphases added*)

Similarly,

We do not call a judgment correct *because* it is guided by or directed towards some things or states of affairs or more of the same, nor *because* the latter are guided by our thinking, but rather *because* an insightful and evident judgment about the object of the correct judgment or the judgment called true could not possibly have another quality, i.e., could not possibly contradict it.

(Kraus 1934: 76, *emphases added*)

In these two ways, Kraus thinks, Brentano succeeds in avoiding the pitfalls of an all-too-naïve realism and Kantianism.

§4 Values Ground Oughts

The writings of Brentano’s heirs abound in attempts to distinguish different types of normative phenomena—goods and evils, virtues and vices, values, oughts, rights, rightness and correctness—and to defend claims about what grounds what within the normative family. Thus Reinach argues that:

there are “norms” which are founded in the ethical rightness of states of affairs. Because something is ethically right, *it ought to be*, and, if certain further conditions are satisfied, *I ought to do it*.

(Reinach 1989b: 241)

A favourite thesis is that Kant is wrong to claim that the concept of good and evil

must not be determined before the moral law (although it looks as though the former must be the ground of the latter) but only, as here, according to the latter and through the latter.

(Kant *KpV* I: 1, 2)

Rather, good and evil are prior to moral oughts, as are the respective concepts. Kant's view, as he notes, is revisionary: it *looks as though* the moral law must be grounded on the concept of good and evil. Brentano's heirs, on the other hand, trust the appearances. More generally, they argue that moral and also non-moral oughts are grounded in goodness, badness, and betterness, in value, disvalue, and comparative value. Thus Scheler, following Reinach's already-mentioned precept (§1) that grounding claims are not reductive, first rejects any identification of ought and value or any reduction of one to the other:

it *cannot* be said that so-called value-judgments express an "ought-connection," an ought-to-be, instead of an ontological connection, or that "good" and "evil" represent only different kinds of this "ought" . . . The meaning of sentences like "This picture is beautiful" or "This man is good" is not all that this picture or this man "ought" to be something. It is beautiful; he is good . . . Such judgments simply report a *factual state of affairs* . . .

(Scheler 1966: 193, translated)

The non-reducibility (*Nichtzurückföhrbarkeit*) of value-judgements to ought-judgements

is shown by the simple fact that the domain of value-judgements is much wider than that of the ought. We can make value-statements in cases where it would make no *sense* to say that the bearers "ought" to be this or that. This is the case with all aesthetic predicates of objects of nature. Even in the moral sphere, the ought is restricted to single acts of *doing* a deed . . . In the case of deeds, one can of course ask whether a proposition like "This deed is good" simply means that "this deed ought to be done" . . . This, however, is not the case when "good" is attributed to a man, i.e., a *person* in his *being*. Any ethics based on an "ought" must, as an ethics of the ought, err with respect to the true *value of the person* and indeed must *exclude* it; such an ethics can validly retain the person only as an X of possible *doings* that ought to be done. The truth of the matter lies in the opposite direction

(Scheler 1966: 194)

Values "do not *consist* in the ought-to-be-ness of something". If it were otherwise, "there would have to be a whole infinity of different modes and kinds of ought, as many as there are different value-contents" (Scheler 1966: 194).

Just as values are neither identical with nor reducible to oughts, so too, an ought does not "consist" (*bestehen*) in "a relation between value and reality". Rather, an "ought is based (*aufbaue*, built) as a matter of essence on a relation between value and reality". An apparently equivalent "principle" is "that every ought is founded (*fundiert*) in a value (and not vice versa)"; here we have an example where "founds" is used to mean "grounds".

One of Scheler's arguments in favour of this principle—or as he also puts it, the “derivative and limited nature” of ought (*seine abgeleitete und beschränkte Natur*)—is that whenever one speaks of an ought, a grasp (*Erfassen*) of a value must have occurred.

Whenever we say that something ought to happen or ought to be, a *relation* between a positive value and a possible real bearer of this value, such as a thing, an event, etc., is also grasped (*miterfasst*). That a deed “ought” to be presupposes that the “ought” is grasped in the intention of the value of the deed.

(Scheler 1966: 194)

In §3, we noted a version of Brentano's view (due to Kraus), according to which if something exemplifies value, this is because a pro-attitude towards that object is knowable as correct. If correctness is a normative property, then this type of view is incompatible with the view that oughts are grounded in values and turns out to be a relative of the claim by Kant at the beginning of this section. In each case, value is grounded in something deontic.

§5 Does Essence Ground Modes of Being?

Modes (ways, kinds) of being (of existing) are old denizens of the Western ontological zoo. Being, it was thought, splits up into kinds: there is the mode of being peculiar to natural objects, natural being, that peculiar to mental representations, appearances, ideas—intelligible or intentional being. More recently, abstract objects such as numbers and geometrical figures have been said to enjoy their own kind of being, ideal being or subsistence. Van Inwagen (2014: ch. 3) has long highlighted the fact that modes of being have not been very popular in analytic philosophy, where Frege's account of existence as a second-order property has been popular, yet are to be found everywhere in non-analytic 20th-century philosophy. But in a recent book, McDaniel (2017) has provided a spirited defence of modes of being, thus confirming once again the claim that there is no thesis some analytic philosopher is not prepared to defend. The most striking feature of 20th-century defences of modes of being, nearly all of which are due to heirs of Brentano, is that they are committed to many more modes of being than the tradition. The multiplication of modes of being is usually traced back to Heidegger and thence to Sartre, Gadamer, and Foucault. In fact, within Austro-German philosophy, it goes back to Scheler, and the grandfather of 20th-century multiplication of ways of existence seems to be William James (cf. Mulligan 2019).

Thus Heidegger in 1927 allows for the modes of being (*Seinsarten*) of living creatures—being alive; being present-to-hand, *vorhanden*; the mode of being of tools—being ready-to-hand, *zuhanden*; the mode of being of *Dasein*, *Existenz*. Understanding, choosing, and asking are, it is announced, modes of being (*Seinsmodi*) of questioners. And authenticity and inauthenticity are also modes of being of *Dasein*. Scheler's earlier and equally long list of modes or forms of being includes being alive, being real, being ideal, being a person, being valuable, and being possible.

If something has some mode of being, it does so in virtue of its nature or essence. This is a thesis accepted by Meinong, whose term for the mode of being of temporal entities is “existence”:

What exists (*existiert*) must exist at a certain time; existence is attached to time. So little is that the case with obtaining (*Bestand*) that the sort of thing that by nature can only

obtain (*bestehen*) and cannot exist—something ideal, in other words—tolerates no time determinations whatsoever.

(Meinong 1983: 59—tr. modified; Meinong 1977: 75)

It is also accepted by the early phenomenologists. “A mode of being”, Scheler says, “is determined by essence (Scheler 1995: 296; cf. 285). But in 1927, Heidegger famously rejected the claim that essence determines modes of being in the particular case of man or *Dasein*:

The “essence” of this being [*Dasein*] lies in its to-be (*Zu-sein*). Its what-being (*essentia*), must, to the extent that one can speak of it at all, be understood in terms of its being (*existentia*).

(Heidegger 1972: §9)

The mode of being of *Dasein*, the view seems to be, is not grounded in the essence of *Dasein*. On this matter, Heidegger found an influential follower in Sartre.

But by no means all heirs of Brentano were prepared to endorse modes of being. Early (1894) Husserl (1979: 326) rejected them and endorsed Bolzano’s view of being, according to which to be is for a concept to have objectuality, a relative of Frege’s later account. Between 1892 and 1916, Marty waged a vigorous campaign against modes of being employing Brentano’s account of being or existence, according to which to be is to be knowable as correctly acceptable (Textor 2017; Kriegel 2018: ch. 5). Later Husserl, on the other hand, seems to have thought he could combine modes of being and Bolzano’s account of being¹³.

Heidegger’s rejection in 1927 of the then dominant phenomenological view that essence is the fundamental ontological notion in favour of modes of being was combined with the claim that ontology’s subject matter is modes of being. Ontology, as Husserl understood it, formal (mereology, the theory of quantities, of relations and connections, of states of affairs, etc.) and material (the ontology of emotions, of social objects), as well as what Meinong had called the theory of objects, disappeared from view. Their problems and solutions were not to reappear until the end of the 20th century.¹⁴

Notes

- 1 But Brentano may have prompted Husserl’s interest in distinguishing different because. He argues that Aristotle refers to types of cause which do not correspond to his taxonomy of four causes (Brentano 1978: 40–2).
- 2 See Roski this volume [Chapter 5].
- 3 See Zylstra this volume [Chapter 22].
- 4 Husserl 1970: 116, cf. 97. Husserl’s interest in grounding is already apparent in Husserl 1887: 26–7, 44, 56.
- 5 On this philosophy, cf. Smith & Mulligan 1982; Simons 1982; Fine 1995; Correia 2004, 2005; Casari 2007.
- 6 See Massin 2015 for these examples and discussion.
- 7 See Introduction this volume.
- 8 See Richardson this volume [Chapter 13] and Leary this volume [Chapter 34].
- 9 On this view and the relation between Bolzano’s understanding and uses of grounding (*Abfolge*) and Husserl, cf. Mulligan 2020.
- 10 On Husserl’s account of the unity of substances and some related theories, cf. Simons 1994; on Husserl’s early and later accounts of the unity of the self, cf. Baldwin 2013. Husserl’s early remarks about the succession of *concreta* parallel remarks by Brentano and were to be followed by the investigations by

- Ehrenfels and Lewin (Lewin 1922; cf. Smith & Mulligan 1982: 69–74) of different types of genidentity and formulations by Scheler and Lewin (Lewin 1922: 13) of the claim that temporal order is grounded in causal order.
- 11 See Trogdon this volume [Chapter 28] for more on truth-making, Fine this volume [Chapter 36] for more on truth-maker semantics, and Amijee this volume [Chapter 4] for the Principle of Sufficient Reason.
 - 12 Pfänder 2000: 225. On Pfänder's view, its relations to Leibniz and to McTaggart's contemporaneous 1921 view of truth-makers, cf. Mulligan 2009.
 - 13 On modes of being in Austro-German philosophy, cf. Mulligan 2019.
 - 14 Thanks to Fabrice Correia and the Editor for their help.

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PART II

Explanation and Determination



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INTRODUCTION

Michael J. Raven

As inquirers, we seek explanations. We observe seasonal changes in climate. We want an explanation why, and we look to science to deliver. It turns out that the digit 7 occurs exactly 50,000 times in the list of whole numbers from 1 to 99,999.¹ We want an explanation why, and we look to mathematics to deliver. We may believe it is morally wrong to kill one to save five (or, conversely, we may believe it is morally right to kill one to save five). We want an explanation why and look to ethics to deliver. The explanations science, mathematics, ethics, and other domains provide may differ in various important ways. But all these domains of inquiry are united in their search for explanations.

Inquiry into ground is no exception. Questions of ground ask for explanations. But ground is supposed to be distinctive. What makes it distinctive is its involvement with a distinctively metaphysical kind of determinative explanation. Its being distinctively metaphysical is supposed to distinguish it from explanations from other domains, such as science, mathematics, and ethics. Its being determinative is supposed to distinguish it from other forms of explanation, such as causal or probabilistic explanations, in which what is explained needn't constitutively depend on what explains it. But what exactly is this notion of metaphysically determinative explanation?

This broad question invites a host of other questions focused, in one way or another, on the explanatory and determinative aspects of ground. Given that ground is a kind of constitutive dependence, how does it relate to other kinds of dependence? Are explanations of ground the same as ground itself, or are they instead backed by relations of ground? Do explanations of ground themselves have explanations? Do grounds determine what they ground by necessitating them? Should we be skeptical that ground lives up the explanatory ambitions foisted upon it? Are there different varieties of ground?

The essays in Part II focus on these and related questions about the explanatory and determinative aspects of ground.

7. Benjamin Schnieder's "Dependence" discusses notions of existential and ontological dependence, with an eye on how they are connected to the notion of ground. In particular, it is shown how the notion of ground can be employed in a definition of existential dependence.
8. Martin Glazier's "Explanation" focuses on ground's relationship to explanation. Some believe that ground just is a kind of explanation, whereas others believe that ground is a

kind of determination that in some way backs explanations of ground. Glazier's essay surveys these and other potential links between ground and explanations of ground.

9. Jon Erling Litland's "Meta-Ground" is concerned with the question: If A grounds B, what grounds the fact that A grounds B? His chapter begins by discussing what turns on this question. The chapter then compares the two main existing solutions to the problem of meta-ground. It ends by discussing how the problem of meta-ground is connected to other issues in the theory of ground and what are the main issues for future research.
10. Alexander Skiles's "Necessity" surveys two ongoing disputes about the modal import of ground. Is a fact necessitated by its grounds (the necessitarianism vs. contingentism dispute)? And is ground an internal relation (the internalism vs. externalism dispute)? Skiles discusses a number of ways in which these disputes interact with debates over the nature of ground itself, as well as how they interact with debates in other areas of philosophy where claims about what grounds what play a central role.
11. Kathrin Koslicki's "Skeptical Doubts" discusses different types of skeptical doubts one may harbor concerning the notion of ground: "old-school" ground skeptics question the intelligibility or coherence of the ground idiom, while "second-generation" skeptics grant the intelligibility or coherence of the ground idiom but question its theoretical usefulness. "Second-generation" ground skeptics are often particularly concerned with the apparent lack of unity exhibited by the ground idiom or the apparent failure of this idiom to do its intended work of being able to capture or illuminate the notions of relative and/or absolute fundamentality.
12. Louis deRosset's "Anti-Skeptical Rejoinders" responds to global skeptics about ground who argue that the attempt to develop a theory of ground is generally and in principle defective, and that the attempts to apply such a theory will be fruitless. This chapter reviews both the reasons that seem to favor global skepticism and the responses to those reasons by defenders of ground.
13. Kevin Richardson's "Varieties" surveys ground pluralism, the view that there are multiple kinds of ground. The chapter is centered around four questions. Why should one be a pluralist rather than a monist? What are the varieties of ground? What is the sense (if any) in which ground is unified? And what is the meaning of 'grounds'? Each question is answered from the perspective of a ground pluralist.

Note

- 1 This example is from Lange (2019).

Reference

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7

DEPENDENCE

Benjamin Schnieder

§1. Introduction

Not all entities are born equal. Some entities exist only derivatively: they depend for their existence on other, more fundamental entities that make the former exist. To study such existential dependencies is an important task of ontology.

Similarly, not all truths and facts are born equal. Some truths are derivative: they are true because of other, more fundamental truths that make the former true; these latter truths can be called the *grounds* of the former, and the relation holding between grounds and what they ground can be called grounding. Equally, some facts are derivative: they obtain because of other, more fundamental facts, i.e., their *grounds*.

As illustrated by these paragraphs, characterizations of existential dependence and of grounding can be phrased in similar terms. Dependent entities are often called *derivative* and are said to *owe* their existence to other entities that *make* them exist, just as grounded facts are often said to be *derivative* and to *owe* their obtaining to other facts that *make* them obtain. That observation motivates the question how exactly the notion of grounding relates to notions of existential dependence (and also other sorts of dependency; more on that in what follows). This is the main concern of this handbook entry.

As to the structure of what follows: In §2, notions of dependence are characterized in a general way. In §3, notions of existential and ontological dependence are introduced. In §4, proposals about how existential dependence relates to grounding are discussed.

Note that until further notice, grounding will be understood as a productive priority relation between facts: a ground is a fact that brings about a grounded fact (that is to say: the relation is productive), where the grounded fact is less fundamental than the ground (that is to say: grounding is a priority relation). Grounding can be expressed in terms of the predicate ‘*x* grounds *y*’ or in terms of the sentential connective ‘because’, so that ‘*P* because *Q*’ corresponds to ‘the fact that *Q* grounds the fact that *P*’.

§2. Dependence in General

Existential dependence is a particular kind of dependence. Before we take a closer look at what is distinctive of this kind, it should be instructive to take a look at the nature of dependencies

in general. A central observation is that dependence relations are like curries: they vary in flavor and in strength.

The flavor of a dependence relation. Dependence is a relation, and at first glance, it may appear to be a dyadic, i.e., a two-place, relation: one entity depends on another entity (for want of a better term, let us call this latter entity the *dependee*). However, a closer look suggests that what underlies this talk is better conceived of as a four-place relation. The reason is that dependence seems to be relative to certain conditions: an entity depends, for being in a certain condition, on another entity being in a certain condition. An animal, for instance, may depend

- (i) for its happiness,
- (ii) for its health, or
- (iii) for its life

on certain chemical substances

- (i) being contained in its food,
- (ii) being contained in the air it breathes, or
- (iii) being synthesized by the animal.

The conditions to which a particular dependence relation is relativized can be said to determine its *flavor*.

To have a flavor and thus to be relativized to some conditions seems essential to dependence. Arguably, there is no such thing as “bare” dependence that is not, at bottom level, relativized to conditions. This is not to say, however, that there are no two-place notions of dependence; it is only to say that they should be understood or defined in terms of four-place dependence relations. Where we talk about the dependence of an entity on another without specifying the relevant conditions, we often take them to be understood from context. Also, in certain cases, we may implicitly quantify over conditions of a certain sort, so that we say that an entity depends on another entity’s being in some condition of the sort in question. Sometimes we may even quantify over conditions in general and consider an entity as dependent on another entity being in *some* condition or *other*.

The strength of a dependence relation. From the flavor of a dependence relation we can distinguish its *strength*, i.e., the sort of tie that holds its relata together.

Ordinary talk about dependence seems to allow for a range of variable strengths: Sometimes we use dependence talk for contingent, breakable ties—fortunately, emotional dependence and financial dependence hold only contingently. In particular, we sometimes employ dependence talk to report contingent regularities, which may be regularities that hold with exceptions, as when we talk about statistical dependence. Sometimes we restrict dependence talk to cases of stronger, modal ties that bind with some sort of necessity. Sometimes we may restrict dependence talk to productive ties, where it is required that the dependent condition is brought about by the dependee.

Exemplification conditions. The overall nature of a dependence relation is fixed by the combination of its flavor and strength. In particular, the combination of strength and flavor determines the exemplification conditions of a given dependence relation, i.e., the conditions under which the relation holds between certain entities. We can state the exemplification conditions of a particular dependence relation in the following form:

$$x \text{ depends (for being } F) \text{ on } y\text{'s being } G \quad \text{iff} \quad x \text{ is } F \otimes y \text{ is } G$$

The symbol “ $\otimes$ ” is used as a placeholder for some sentential connective; differences in strength then correspond to different connectives. For instance, different sorts of conditionals (“if—then . . .” connections) may be in play. The simplest option would be to use an “only if”, so that the pertinent exemplification condition could semi-formally be stated as follows:

$$x \text{ is } F \rightarrow y \text{ is } G$$

(The arrow “ $\rightarrow$ ” can be read as “if—then . . .”, which is logically equivalent to “—only if . . .”.)

But apart from such a material conditional, stronger conditionals may be involved, for instance a strict conditional (expressible by “necessarily, if—then . . .”), or alternatively a grounding connective such as “because.”

We may also operate with more complex connectives, as for instance a counterfactual with embedded negations, so that we get the following exemplification conditions: if y were not G , x would not be F . In formula, using “ $\Box \rightarrow$ ” as the symbol for the counterfactual connective:

$$\neg Gy \Box \rightarrow \neg Fx.$$

Another example of a more complex operation would be a grounding link conditionalized on the dependent entity’s being in the pertinent condition. Then we can get, for instance, the following exemplification conditions: if x is F , then x is F because y is G ; in other words: if x is F , then the fact x is F is (partially) grounded in the fact that y is G . In formula, using “ $<$ ” for partial grounding:

$$Fx \rightarrow (Gy < Fx).$$

We will take a closer look at some such options later, in course of the discussion of notions of existential dependence that differ in strength.

Independence. Notions of dependence bring along corresponding notions of independence: Given a particular understanding of dependence, an entity can be said to be independent of another if it does not depend on it in the pertinent sense of “depend”; and it can be said to be independent simpliciter if it does not depend on any other entity (in the pertinent sense of “depend”).

Individual versus generic dependence. To conclude this discussion of dependence in general, a distinction between two sorts of dependence deserves to be mentioned that transcends differences in strength and/or flavor. Sometimes we say that an entity depends, for being in a given condition, on *entities* (plural) of a certain sort, as in “a forest depends on trees” or “a child depends on parents”. That way of talking can mean one of two things: it can be understood *distributively*, as a quantification over or plural reference to entities on each of which the dependent entity depends. For instance, if Anna is said to depend on her parents, this can be understood so that Anna depends on each of her parents, i.e., on her mother and also on her father. But on a different reading, one can simultaneously say that an entity depends on entities (plural) of a certain sort while denying that there is any particular entity of the sort on which the dependent entity depends. Thus we may say that a forest depends on trees while denying that there is any particular tree on which the forest depends. So we can use the idiom “ x depends on F s” to report on two different sorts of dependency, which sometimes go by the labels *individual* and *generic* dependence: Individual dependence is dependence on a particular entity. Generic dependence is dependence on a certain genus being instantiated (rather than dependence on any particular specimen of the genus). The distinction is parallel to Quine’s famous distinction between two readings of “John wants a sloop”: There may be a particular sloop John wants, or he may just

want to be relieved of slooplessness. Similarly, that x depends on F s may mean that there are some particular F s on which x depends or that x depends on not being F -less.

Now that the contrast between generic and individual dependence relations has been introduced, let us briefly return to the distinctions between different flavors and different strengths of dependence relations. The formulations used to introduce these distinctions presupposed that we are talking about individual dependence relations. Nevertheless, it is easy to see that generic dependence relations equally differ in flavor and in strength; in fact, it seems that to every individual dependence relation with a particular flavor and strength, there is a corresponding generic dependence relation of the same nature.

In what follows, this chapter will concentrate on individual rather than generic dependence unless noted otherwise.

§3. Existential and Ontological Dependence

Existential dependence. By *existential* dependence, philosophers mean dependence with a particular flavor: Some entities depend *for their existence* on others. So the attribute “existential” is used to indicate the condition with respect to which the dependent entities depend on others. Sometimes, however, it is meant to do a double duty and additionally indicate the relevant condition on the other side of the affair; what we are after then are entities that depend, for their existence, on the existence of other entities.

By way of a terminological note, existential dependence is sometimes also called ontological dependence, which is fitting inasmuch as the term “ontological” can be used to mean as much as “existential”. But the term “ontological” is somewhat flexible.¹ Literally it means about as much as *pertaining to the doctrine of being*, and by “being” one can but need not understand existence; one can also understand something like the identity, nature, and/or essence of a thing. So insofar as entities may also depend for their identity, nature and/or for their essence on others, the term “ontological dependence” can denote dependencies of a different flavor than an existential one. While this entry focusses on existential dependence, one other understanding of ontological dependence will be addressed later.

Relevance and examples of existential dependence. Notions of existential dependence have always figured in ontology, since ontology is—at least in part—an investigation into the nature and extension of existence. We already find such a notion in Aristotle’s *Categories* (1949, ch. 12), where he says that an entity x can be called prior to an entity y if the existence of x is nonreciprocally entailed by the existence of y (so that y cannot exist unless x exists, while x can exist without y). Traditionally, ideas about existential dependence and independence played a particularly important role for how to frame the ontological category of *substance*. Thus, Descartes held that “By *substance* we can understand nothing other than a thing which exists in such a way as to depend on no other thing for its existence” (Descartes 1644).

Philosophers have discussed a range of entities that they took to depend, for their existence, on other entities; here is a list of some typical examples:

<i>existentially dependent entity</i>	<i>the dependee</i>
Jeanne’s courage (generally: a mode/particularized quality)	Jeanne (generally: the bearer of the mode)
the set containing Jean-Luc and François (generally: a set of entities)	Jean-Luc and François (generally: the members of the set)

<i>existentially dependent entity</i>	<i>the dependee</i>
the hole in my shoe (generally: a hole)	my shoe (generally: the object perforated by the hole)
Caesar's death (generally: an event)	Caesar (generally: the participant(s) of the event)
the fictional captain Ahab (generally: a fictional character)	Herman Melville (generally: the author of the story in which the character was introduced)

It should be emphasized that these examples are meant to be cases of *individual* dependence. The idea is that, for instance, a hole not only existentially depends on there being some perforated object or other; it depends on the *particular* object it resides in. Philosophers have also discussed examples of entities that, for their existence, generically depend on entities of other sorts (though not individually). On some conceptions of properties, for instance, the existence of a property presupposes that some thing or other actually has the property; this would render properties generically dependent on objects exemplifying them.

Let us pause for a quick cautionary note: Some statements can look like claims of generic existential dependence while their look is actually misleading.² One may, e.g., say that debtors depend for their existence on creditors, meaning that there cannot be debtors if there are no creditors, and at the same time deny that any single entity that is a debtor depends for its existence on creditors, since no entity that is a debtor is so by necessity. So whereas the original claim at first may seem to express that a debtor *existentially* depends on creditors (taken generically), this misconstrues the claim. The dependence relation involved, born by a debtor to creditors, is not existential but of a different flavor: here one entity depends, *for being a debtor*, on some other entity's being a creditor.

Returning to the list of existentially dependent objects, it should also be emphasized that although the examples are quite often thought to exemplify existential dependence, it is of course not written in stone that they actually do. Some philosophers may reject a particular case simply for the reason that they do not countenance the allegedly dependent entity in their ontology at all (they may, e.g., hold that holes are not entities in the first place). And in some cases, there are philosophical disputes about the nature of the entities in question that bear on the question of existential dependencies; philosophers who think that modes can change their bearers may well deny that a mode existentially depends on its particular bearer. And someone who thinks that events are the fundamental entities in space-time while ordinary objects are actually but a series of events may well deny that an event existentially depends on its participant(s). Moreover, the examples are stated very generally, so that each instance of a certain category (e.g., *holes*, *events*, etc.) depends upon a correlated object. But while it is neat for an ontology if a whole category exhibits a dependency in this way, in some cases there are philosophers who accept that there are dependencies of the sort in question but deny that they are characteristic of the whole category. Take the case of modes: The sharpness of a knife seems to be nothing but the sharpness of its blade. If that is so, there is one sharpness mode with two (related) bearers, the blade and the knife. But since you can destroy the knife by removing its blade while the sharpness of the blade survives, the sharpness mode does not seem to existentially depend on the blade, i.e., one of its bearers. Similar examples can be given for a range of modes but also, for instance, for holes. Claims about the dependence of modes on their bearers (or holes on objects they reside in) may therefore seem to be in need of some restriction.³

Varieties of existential dependence. Now let us take a closer look at what exactly can be meant by “existential dependence”. While we already know the term denotes a dependence relation of a certain flavor, it was pointed out earlier that the flavor of a dependence relation does not settle its whole nature. In fact, we can have a plethora of existential dependencies varying in strength. For ontological enquiries, philosophers usually focus on dependencies whose ties bind at least with the force of metaphysical necessity.

The simplest forms of such existential dependencies are defined solely in modal terms (on modality, see also Chapter 24 of this volume). Thus, that an entity existentially depends on another may be taken to mean that it can only exist if the other exists. Semi-formally, using the box “ $\Box$ ” to express necessity and “ $E!$ ” as an existence predicate, we can define this notion as follows:⁴

$$\text{Modal-Dep}_{E-E} \quad x \text{ existentially depends on } y \quad \leftrightarrow_{\text{def.}} \quad \Box (E!x \rightarrow E!y)^5$$

A variation on this notion would make x ’s existence not dependent on the existence of another entity but on some other condition; for instance, a particular change—a datable event—may depend on another entity’s first having one property, then another. Semi-formally, such a notion can be rendered as follows:

$$\text{Modal-Dep}_{E-F} \quad x \text{ existentially depends on } y \quad \leftrightarrow_{\text{def.}} \quad \Box (E!x \rightarrow Fy)$$

(Modal-Dep_{E-E} can now be seen as a specific instance of the more general form Modal-Dep_{E-F}.)

An alternative modal notion of existential dependence can be defined in terms of a counterfactual conditional. Thus, we may say that an entity depends on another one if the former would have not existed, if the latter had not:

$$\text{Counter-Dep}_{E-E} \quad x \text{ existentially depends on } y \quad \leftrightarrow_{\text{def.}} \quad \neg E!y \Box \rightarrow \neg E!x$$

The modal notions of dependence can be further modified, e.g., by taking the temporal dimension into account, so that one can distinguish between *historical* dependence (x can only exist at time t if y existed at some time t^* up to t) and *permanent* dependence (x can only exist at a time t if y exists at t as well). But in what follows, such further differentiations will be set aside.

Purely modal notions of existential dependence are interesting in their own right and can be fruitfully studied in the context of a modal framework such as possible worlds semantics. But modal notions of dependence are not apt for all purposes. What they do not seem to capture are *productive* sorts of dependence, on which a dependent entity *owes* its existence to the entity it depends on. For the existence of an entity can be modally tied to that of another entity even if the latter does nothing to bring about the existence of the former. For instance, one can easily see that on Modal-Dep_{E-E} every entity counts as dependent on every necessary existent. So given that the number zero exists necessarily, Socrates existentially depends on it in this sense. But it seems clear that he does not owe his existence to the number zero; in fact, the number zero has nothing substantially to do with Socrates.

Relatedly, modal accounts of dependence do not secure that an entity on which another entity depends is more *fundamental* than it or that its existence is *prior* to it. For relative fundamentality and priority are asymmetrical relations; but in a modal sense of dependence, two entities can mutually depend on each other. To use a famous example from Kit Fine (1995: 271), the existence of Socrates and that of his singleton (the set of which he

is the sole member) stand and fall together across the spectrum of possible worlds. So they exemplify a case of mutual existential dependence of modal strength. But one can agree that in the explicated understanding, the two entities depend on one another, and still maintain that there seems to be yet another sense of “depend” in which the singleton existentially depends upon its member in a way in which the member does not depend upon the set. For the existence of the member seems to be the more fundamental fact; it seems prior to it in the order of existence.

A productive sort of existential dependence, on which a dependent entity is less fundamental and posterior to the entity on which it depends, can be defined in terms of grounding (it is assumed here that grounding is asymmetrical; on the issue see also Chapter 17 of this volume). An entity can be said to depend existentially on another entity if its existence is grounded in the existence of the other entity. (This idea can be traced back to Aristotle’s *Categories*, ch. 12. Among things that “reciprocate as to the implication of existence,” Aristotle points out, one can still be called prior to another if the former is the cause or ground—*aitia*—of the latter. On Aristotle’s views concerning grounding and related notions; see also Chapter 1 of this volume.) Using “ $\triangleleft$ ” to express grounding, we have:

$$\text{Ground-Dep}_{E-E} \quad x \text{ existentially depends on } y \quad \leftrightarrow_{\text{def.}} \quad E!y \triangleleft E!x^6$$

The definition allows for different readings, since we can distinguish between different notions of grounding, for instance between full and partial grounding. For ease of presentation, such distinctions will be set aside in what follows so that formulations with “grounding” are in general meant to allow for different specifications; this is why the symbol “ $\triangleleft$ ” is used and not one of the standard symbols for a specific notion of grounding. Note, however, that if one employed the notion of *weak* grounding (Fine 2012: 51ff.), the defined dependency relations would not be asymmetrical; while such notions may deserve to be explored, in the current context, I presuppose that we are working with an asymmetrical notion of grounding, so that the resulting notions of dependence inherit this structural feature (which was one of the motivations to look at these accounts in the first place).

Several variations on the basic ground-theoretical definition are possible. In particular, given that ontologists are usually interested in ties that bind with some necessary force, we may require that the grounding link must hold if the depending entity exists at all:⁷

$$\Box \text{Ground-Dep}_{E-E} \quad x \text{ existentially depends on } y \quad \leftrightarrow_{\text{def.}} \quad \Box (E!x \rightarrow (E!y \triangleleft E!x)).$$

We can also allow for different conditions as grounds of the existence of the dependent entity. We may, for instance, say that an entity owes its existence to another entity if there is *some* fact about the dependee that grounds the existence of the dependent entity. This idea would correspond to the following notion of existential dependence:

$$\text{Ground-Dep}_{E-F} \quad x \text{ existentially depends on } y \quad \leftrightarrow_{\text{def.}} \quad \exists F (Fy \triangleleft E!x).$$

This definition makes use of quantification into predicative position: $\exists F(Fx)$. It is a matter of controversy how to interpret this form of quantification; the issue must be left open here (though for present purposes, a nonsubstitutional and nonobjectual interpretation would be preferable).⁸

We may of course also consider a modally strengthened version of the last definition. As in the case discussed earlier, we may conditionalize on the existence of the dependent entity and

introduce a necessity operator. Because of the quantifier “ $\exists F$ ” involved in our definition, the newly introduced necessity operator now can have different scope, so that we can choose among the following two definitions:⁹

$$\begin{aligned} \text{Ground-Dep}_{E-F} \quad x \text{ existentially depends on } y &\leftrightarrow_{\text{def.}} \exists F \Box (E!x \rightarrow (Fy \triangleleft E!x)) \\ \text{Ground-Dep}_{E-F} \quad x \text{ existentially depends on } y &\leftrightarrow_{\text{def.}} \Box \exists F (E!x \rightarrow (Fy \triangleleft E!x)) \end{aligned}$$

(It is up for debate whether the difference in scope makes a substantial difference or whether the two definitions turn out to be extensionally equivalents.)

Since the present notion of dependence is of particular interest for theories of grounding, let us contemplate it a little longer. The notion captures the idea of an entity owing its existence to another entity in that the existence of the former is brought about by the latter entity, where it is left open how exactly the other entity achieves this, e.g., by its existence or by some other feature or activity. To see the relevance of letting the conditions on the side of the dependee vary like this, let us return to the examples of dependencies given earlier. Under certain assumptions, they are all covered by the present notion of dependence:

- a mode (e.g., Jeanne’s courage) depends on its bearer, since the mode can only exist if its existence is grounded in the bearer’s being a certain way (e.g., courageous)
- a set depends on its members, since the set can only exist if its existence is grounded in the existence of the members;
- a hole depends on the object it resides in, because the hole can only exist if its existence is grounded in the object’s being perforated.
- an event (e.g., Caesar’s death) depends on its participants(s), since the event can only exist if its existence is grounded in the participant(s)’s behaving in a particular way (e.g., dying).
- a fictional character depends on its author, since the character can only exist if its existence is grounded in the author authoring the fiction.

But note that whether or not such examples indeed qualify is open for debate; the notion of dependence alone does not settle the issue, since further assumptions about the categories involved in the cases (e.g., the category of events) will weigh on the stance one takes towards them.

Ontological though not existential dependence. Finally, we turn to a third way of determining the strength of an existential dependence relation, namely by employing the notion of essentiality. An entity may be said to existentially depend on another one if it is essential to the dependent entity that it only exists if the dependee exists; or alternatively: only if the dependee is a certain way. Now some philosophers understand essence as a modal notion so that a property is essential to an entity if the entity cannot exist unless it has the property. The sketched notions of existential dependence using an essential link would then come down to modal notions of existential dependence. But a number of philosophers follow Fine (1994) in resisting the modal analysis of essence and instead considering it to be a primitive notion. In this case, notions of dependence with an essential tie come apart from modally defined notions. If we use “ $\Box_x p$ ” to express that it is essential to x that p , we can give the following definitions of the two notions of existential dependence formulated a few lines ago:¹⁰

$$\begin{aligned} \text{Ess-Dep}_{E-E} \quad x \text{ existentially depends on } y &\leftrightarrow_{\text{def.}} \Box_x (E!x \rightarrow E!y) \\ \text{Ess-Dep}_{E-F} \quad x \text{ existentially depends on } y &\leftrightarrow_{\text{def.}} \Box_x (E!x \rightarrow Fy) \end{aligned}$$

We may also take the condition on the side of the dependee to be variable, so that an entity depends on another if its existence is essentially tied to the dependee being in *some condition or other*.

$$\text{Ess-Dep}_{\text{E-F}} \quad x \text{ existentially depends on } y \quad \leftrightarrow_{\text{def.}} \quad \exists F \Box_x (E!x \rightarrow Fy)$$

Note that these definitions still yield notions of *existential* dependence, since they make the dependent entity depend for its existence on another entity; existence determines the flavor and essence the strength of the defined relations.

However, Fine suggested an alternative notion of ontological dependence based on essence in which existence does not play any role at all: we may say that an entity depends, *for what it is* (for its nature/identity), on another entity whenever the latter entity is involved the essence of the dependent entity.¹¹ Formally, we can capture this idea by quantification into predicative position (though Fine himself does not do this): The dependent entity depends on a dependee if there is *some way* such that it is essential for the dependent entity that the dependee is that way. In formula:¹²

$$\text{Ont-Dep}_{\text{pure}} \quad x \text{ ontologically depends on } y \quad \leftrightarrow_{\text{def.}} \quad \exists F \Box_x Fy$$

(Note that stated like this, the definition requires what Fine calls a notion of *constitutive* essence; on the issue and on a variation of the definition, see Fine 1995: 276ff.)

Lowe proposed a very similar definition of ontological dependence.¹³ Basically, on his definition, an entity is dependent on a dependee if it is essentially a *function* of the dependee:

$$\text{Ont-Dep}_{\text{E-Function}} \quad x \text{ ontologically depends on } y \quad \leftrightarrow_{\text{def.}} \quad \exists \phi \Box_x (x = \phi(y))$$

One can regard this definition as a restricted variant of Fine's. For the formula in the scope of the essence operator can be parsed as the application of the complex predicate " $x = \phi(\dots)$ " to the variable " y ".

Lowe's actual definition moreover introduces a necessity operator, governing the whole definiens:¹⁴

$$\text{Ont-Dep}_{\text{E-Function}} \quad x \text{ ontologically depends on } y \quad \leftrightarrow_{\text{def.}} \quad \Box \exists \phi \Box_x (x = \phi(y))$$

§4. Grounding and Existential Dependence

This final section will take a closer look at possible connections between existential dependence and grounding. Three possible connections will be discussed: definitional links, grounding links, and identity links.

Definitional Links. We have seen one such connection, which is definitional in nature: there is a family of existential dependencies whose strength is defined in terms of grounding links. These definitions have several theoretical virtues: They capture the important idea of dependencies in which the dependent entity owes its existence to the dependee. They are in harmony with the general observation that notions of dependence vary along the dimensions of flavor (here: existential) and strength (here: a productive tie spelled out in terms of grounding). Finally, they embed notions of existential dependence in the overall theory of grounding, allowing us

to study them systematically with the means provided by that framework (such as the available systems of the logic of grounding).

Objections to the definition can be radical or moderate: A radical objection holds that the definition fails to deliver a coherent concept. This could be because by employing grounding talk, the definition already invokes an incoherent concept (or perhaps even a pseudo-concept, because grounding talk fails to properly express any concept), which some of radical grounding sceptics would hold (on grounding skepticism, see Chapters 11 and 12 of this volume). This objection can be set aside here because it targets the notion of grounding in general rather than its particular use in the definition of existential dependence. A different worry concedes that grounding vocabulary is intelligible but holds that it generates opaque contexts barred to quantification.¹⁵ This would render the ground-theoretical definitions of dependence incoherent, since they indeed involve quantification into a grounding context. Albeit not quite as general as the first worry, this objection still goes well beyond the topic of this handbook chapter; it has to be addressed in discussions about the logic and semantics of grounding vocabulary (on the logic of grounding, see Chapter 14 of this volume).

A more moderate objection would admit that the definition yields a coherent concept but add that

- (i) this concept is not properly expressed by the term “existential dependence”: Whatever this is, the objection goes, it is not existential dependence as commonly understood; and/or
- (ii) hold that the obtained concept is theoretically uninteresting and devoid of useful applications.¹⁶

In connection with the first of these objections, someone may more generally object to the plethora of different dependence definitions considered earlier, and hold that instead of such a variety, there is only one unique notion of dependence (or perhaps a small family, all based on the same core concept, which is not grounding).¹⁷ However, inasmuch as the definitions are meant to spell out what “existential dependence” can mean in accordance with how “depend” and “dependence” are used in common English, instead of just introducing a philosophical term of the art, a pluralistic view that allows for a range of different concepts indeed seems advisable in light of the highly flexible use of those terms.

A general point to make in light of the second moderate objection is that its force seems to depend heavily on the theoretical interests of the objector. Certainly one can fruitfully debate whether the ground-theoretical notions serve particular, well-defined purposes; but here is not the place to enter into such a debate. Clearly the ground-theoretical notions of dependence are worthwhile for anybody generally interested in the theory of grounding. And inasmuch as one does not endorse a flat ontology on which no entities owe their existence to others, the definitions are not empty but will be applicable to particular cases; and then it would certainly seem to be a significant ontological question which entities exist because of which other entities and also whether there exist entities that do not exist because of any other entities.

Let us consider a final, more specific criticism of the ground-theoretical notions, which can be called the *PRIORITY OBJECTION*. It was first voiced by Francesco Orilia, when he wrote that it has been

proposed that x 's ontological dependence on y can be analyzed as the fact that x 's existence is ontologically explained by the existence of y . This seems to lead to the conclusion that x ontologically depends on y because the proposition that x exists is explained by the proposition that y exists and thus perhaps puts the cart before the

horse. For one might rather think that it is the other way around, namely, that the proposition that x exists is explained by the proposition that y exists, because x ontologically depends on y .

(Orilia 2009: 334, fn. 3)¹⁸

Note that instead of “ontological explanation”, one can read “grounding” (otherwise, the objection would not be relevant to the present topic). So Orilia thinks that the ground-theoretical definitions set the wrong priorities: existential dependence should ground existential grounding claims, not the other way around. There are different things to respond to this. First, “dependence” is a highly flexible term allowing for relations of different strength; but not every legitimate understanding of “dependence” is capable of underwriting grounding claims. So if it is claimed that existential dependence should ground grounding claims, one better be able to single out the relevant understanding of “dependence”. The ground-theoretical definition gives us an understanding on which existential dependence is indeed closely related to grounding; but if the definition is rejected because of the PRIORITY OBJECTION, one would like to see an alternative way of carving out the pertinent understanding of “dependence” that can support the claim that dependence grounds grounding. Second, Orilia explicitly targets the simple ground-theoretical definition on which x depends on y iff x exists because y exists (i.e., Ground-Dep_{E-E} from Section 3). But can the objection be applied to the other ground-theoretical definitions, in particular to the one on which x depends on y iff y is something such that x exists because y is it; in formula: $\exists F (x \text{ exists because } Fy)$? (Which is the definition favored in Correia 2005; Schnieder 2006.) The definiens of this definition is an existentially quantified statement. As such, it will be immediately grounded in its true instances, as for example: x exists because y exists; or: x exists because y authored a certain fiction; or: x exists because y is courageous, etc. So if the definiens were also grounded in the existential dependence of x , one would ordinarily expect this to be a case of mediate grounding, running through the instances of the definiens. But those instances have variable predicates on the side of the grounds (“exists”, “authored a certain fiction”, etc.). Now it seems rather mysterious how a uniform sort of ground, namely that x existentially depends on y , should ground grounding statements that vary in this way.¹⁹ It seems fair to say that as it stands, the PRIORITY OBJECTION at least leaves a number of open questions that should be addressed by someone who endorses it.

Identity link. Let us take a look at another possible link between grounding and existential dependence. Could it be that grounding *just is* existential dependence, under a different linguistic guise? More specifically, could it be that

GROUNDING=DEPENDENCE A fact x grounds a fact y iff y existentially depends on x (where for facts, to exist is to obtain)?

Such a view is often associated with Jonathan Schaffer.²⁰ But we need not discuss here whether this is Schaffer’s considered view. The view itself deserves our attention.

As it stands, the view clearly requires supplementation before it can be properly evaluated. In particular, there are quite diverse conceptions of the nature of facts that the ontological market has on offer. However, the nature of facts is highly relevant to understand whether some facts existentially depend on others and, if so, on which.

Be this as it may, a standard example of grounding can already cast serious doubts on the viability of the view. Consider the following fact:

SOME Someone is a philosopher.

This is an existentially quantified fact, and such facts are usually taken to be grounded in their true instances. In the case at hand, one such instance would be the fact:

VAN Quine is a philosopher.

So VAN should ground SOME. But whatever the exact nature of facts may be, this latter fact does not existentially depend on VAN on any notion of dependence that has at least modal force. For SOME could obtain without VAN obtaining; after all, worlds in which Quine never studied philosophy may still contain countless other philosophers.²¹

Not-quite identity link. The prospects of identifying grounding with existential dependence between facts seem dim. However, there is a somewhat related proposal which may look more promising (Zylstra 2018; see also Section 2.2 of Chapter 22 in this volume). Consider the following notion of generic dependence that is existential in flavor and essential in strength: Let R be some relation. A fact f depends on facts being R -related to f iff it is essential to f that it obtains only if some obtaining facts are R -related to it. Semi-formally, we may state the notion as follows:

Fact f generically existentially depends on facts that are R -related to $f \leftrightarrow_{\text{def.}} \Box_f (f \text{ obtains} \rightarrow \exists f^* (f^* \text{ obtains} \ \& \ f^* \text{ is } R\text{-related to } f))$

Now consider some obtaining fact f that generically depends on facts being related to it one or more relations $R_1, R_2, \dots$. We know that for each of the relations $R_1, R_2, \dots$, there must be some obtaining facts related to f by that relation (otherwise, f would not obtain). Let us say that all these facts form the *actual basis* of f . The proposal now is that grounding can be defined as follows:

Fact f is grounded in fact $f^* \leftrightarrow_{\text{def.}} f^*$ belongs to the actual basis of f

In other words: f is grounded in f^* iff there is some relation R such that f generically depends on facts being R -related to it, and f^* is an obtaining fact thus related to f . (Note that the given presentation slightly streamlined Zylstra's proposal but so as to preserve its core.)

The proposal seems to deliver the goods in many standard cases. Consider, for illustration, the common claim that a disjunctive fact is grounded in its obtaining disjunct(s). For Zylstra's proposal to yield this result, we have to make the plausible assumption that if f is a disjunctive fact, then it is part of f 's essence that if f obtains, so does some fact which is a disjunct of it. Hence, a disjunctive fact generically depends upon facts that stand to it in the relation of being one of its disjuncts. Now if a disjunctive fact f obtains, at least one of its disjuncts obtains as well and hence belongs to the actual basis of f . This in turn validates the claim that the disjunctive fact is grounded in its obtaining disjunct(s).

It should be noted that grounding is defined not quite *as* a form of existential dependence on the current proposal but still as something very closely related to it. The view clearly rests on some contentious assumptions about facts and their essence (some of which are addressed in Zylstra 2018); but it does not seem easily defeated by standard assumptions on grounding.

Related Topics

Ch. 1 "Ancient" (Phil Corkum)

Ch. 11 "Skeptical Doubts" (Kathrin Koslicki)

- Ch. 12 “Anti-Skeptical Rejoinders” (Louis deRosset)
 Ch. 14 “Logics” (Francesca Poggiolesi)
 Ch. 17 “Strict Partial Order” (Naomi Thompson)
 Ch. 22 “Essence” (Justin Zylstra)

Notes

- 1 “Ontology” is a philosophical term of the art; apparently it was coined in a book from 1606 written by a hardly known philosopher called Jakob Lorhard (see Lamanna 2014: 146).
- 2 As was pointed out in Simons 1987: 296f.
- 3 The sharpness example is from Levinson 1980; see Schnieder 2004 for discussion.
- 4 For a seminal study of modal notions of existential dependence, see Simons 1987: ch. 8. For an application of modal notions of dependence in the ontology of fictional objects, see Thomasson 1999: ch. 2.
- 5 A definition of a corresponding notion of generic dependence would be:

$$x \text{ existentially depends on } Fs \leftrightarrow_{\text{def.}} \Box (E!x \rightarrow \exists y Fy)$$
- 6 A definition of a corresponding notion of generic dependence would be:

$$x \text{ existentially depends on } Fs \leftrightarrow_{\text{def.}} E!x \rightarrow \exists y (Fy \ \& \ (E!y \rightarrow Fy))$$
- 7 A similar proposal (formulated with the connective “only because”) was made, though not endorsed, by Lowe (1998: 145).
- 8 For such an account of nonnominal quantification, see Rayo & Yablo 2001.
- 9 Such definitions were proposed in Correia 2005 (narrow scope of “ $\exists F$ ”) and Schnieder 2006 (wide scope of “ $\exists F$ ”).
- 10 The first of these definitions is stated in Fine 1995: 273. A definition of a corresponding notion of generic dependence would be:

$$x \text{ existentially depends on } Fs \leftrightarrow_{\text{def.}} \Box_x (E!x \rightarrow \exists y Fy)$$
- 11 Fine 1995: 274f.
- 12 A corresponding notion of generic dependence is arguably less straightforward to state as in the other cases; see Fine 1995: 288f.
- 13 Lowe 1998: 149.
- 14 For some discussion of Lowe’s proposal, see Correia 2005: 46ff.; Lowe 2013.
- 15 The worry is indicated, though not endorsed, in Lowe 1998: 146.
- 16 For the second assumption, see Koslicki 2015.
- 17 If I understand her correctly, Barnes 2018: 53 makes such assumption. She also ascribes it some other authors, including the author of Schnieder 2006; at least the latter ascription seems wrong, see Schnieder 2006: 401.
- 18 The objection can also be found in Koslicki 2013: 47.
- 19 This second response is formulated in more detail in Schnieder 2017.
- 20 Schaffer 2009, 2010 contain passages that at least suggest this view. See Schnieder 2017 for some more specific references and some discussion.
- 21 For further discussion, see Schnieder 2017, and also Rydén 2018.

Further Readings

For further survey articles on ontological see Correia 2008; Tahko & Lowe 2016.

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8

EXPLANATION

Martin Glazier

Metaphysical ground, like other topics in philosophy, is the subject of intense disagreement. What is it? What principles govern it? How can we know anything about it? Controversy surrounds these and other questions about ground. But if there is one *uncontroversial* claim in this area, it is that ground is deeply linked with a certain form of explanation, what we will call *grounding explanation*. This link and this form of explanation are the subject of this chapter.

For its enthusiasts, grounding explanation is both ubiquitous in ordinary life and central to many of philosophy's biggest questions. In ordinary life, we offer grounding explanations of why this chess player is victorious (because her opponent's king is in checkmate), of why this coffee is hot (because its molecules have a high mean kinetic energy), or of why these two nations are at war (because they are bombing each other). And in philosophy, grounding explanation has been thought to be what is at issue when we ask whether this or that phenomenon 'reduces' to something else. Kit Fine, for instance, writes:

What is it to explain the appearance of a world with minds in terms of a mindless world or the appearance of a world with value in terms of a purely naturalistic world? My own view is that what is required is that we somehow *ground* all of the facts which appear to presuppose the reality of the mental or of value in terms of facts which do not presuppose their reality.¹

This chapter surveys the philosophical literature on grounding explanation and its connection to metaphysical ground. I begin by discussing explanation in general (§1) before turning to grounding explanation in particular (§2). I then take up the question of whether and how this form of explanation relates to reality (§3). I turn finally to ground (§4).

1. Explanation

In daily life, as well as in more theoretical pursuits, we seek and offer explanations of facts. Here are some examples, or potential examples, of explanation.

- (1) The window shattered because Suzy threw a rock at it.
- (2) Jones went to the post office in order to mail a package.

- (3) Since Meghan Markle is married to Prince Harry and Harry is the queen's grandson, Markle is a member of the royal family.
- (4) Billy's taunt hurt Jimmy's feelings and that's why it was wrong.
- (5) What explains why nothing has ever traveled faster than light is that the laws of nature prohibit that.
- (6) The reason why Anaxagoras failed in his attempt to square the circle is that it is impossible to do that.
- (7) Arthritis is a disease of the joints and so no one can have arthritis in the thigh.

We should distinguish the *act* of explaining something from the *explanation* that such an act aims to convey.² For example, I might perform an act of explaining by saying to my friend, 'The window shattered because Suzy threw a rock at it.' The aim of this act is to convey to my friend a certain explanation—namely, that the window's shattering is accounted for by Suzy's throwing a rock at it. My main concern in this chapter will be explanations, not acts of explaining.

Which act conveys, or best conveys, a given explanation might be thought to be somehow relative. Suppose, for example, that someone asks me why I was late to the NYU colloquium. With some questioners, it might be enough to answer that I took the express train. But if my questioner is unfamiliar with New York's subway system, this answer is liable to be puzzling: how could I be late on account of taking the *express* train? To remove this puzzlement, I will need to explain that although the express train is generally quicker than the local, it does not stop near NYU's philosophy department, and so I had to walk several blocks. One might take this to show that which act best conveys the explanation of my tardiness is relative to what my interlocutor knows and perhaps relative in other ways as well.

But even if this is right, it does not show that the explanations *conveyed* by such acts are relative. Even if which act best conveys an explanation depends on what certain people know or expect or find intelligible or care about, that does not show that what explains what depends on these things. Explanation *might* turn out to be a relative matter, but it might also be perfectly objective.³

We began this section with a list of examples. These examples show that the category of explanation is quite diverse. Some explanations, for instance, concern what causes what, while others do not.⁴ But there are nonetheless certain features common to all explanations. Every explanation, for instance, has both a fact that is explained, called the explanandum, as well as one or more facts that do the explaining, called the explanans. Thus in the first example on the list, the explanandum is the fact that the window shattered, and the explanans is the fact that Suzy threw a rock at it. (Technically, the plural of 'explanans' is 'explanantia', but I will not use it, and at any rate I will mostly confine myself to cases in which the explanans consists of only one fact.)

Our concern in this section has been the explanation of one fact in terms of some other facts. There may be forms of explanation that involve things other than facts. Consider, for instance, Socrates' singleton set. There is a sense in which we may explain what this object is by saying that it is the set whose sole member is Socrates. One might view this as an explanation not of a fact but of an object.⁵ In this chapter, however, we will set such 'nonfactual' forms of explanation aside.

2. Grounding Explanation

The focus of this chapter is a particular form of explanation that we will call *grounding explanation*. Some potential examples of grounding explanation were given in the chapter's introduction; here are a few more.

- (1) What makes it the case that the Golden Gate Bridge is red is that it is this particular vermillion shade.

- (2) Bowser is big and Bowser is bad, and that's why Bowser is big-and-bad.
- (3) Since Socrates is a philosopher, there is someone who is a philosopher.
- (4) Because snow is white, it is true that snow is white.
- (5) The reason why this figure is a regular pentagon is that it has five sides of equal length and five angles of equal measure.
- (6) There is a table here in virtue of there being wood arranged here in a certain 'tablewise' way.
- (7) There is a hole in this piece of paper because it is perforated.

Like the category of explanation in general, the category of grounding explanation displays considerable diversity. There are explanations which are logical in character, explanations concerning the relationship between determinates and determinables, explanations involving parts and wholes, explanations involving the dependence of what is true on what the world is like, and still others besides.

Despite this diversity, however, philosophers have tended to think there are certain claims that hold of grounding explanation in general. Some of these they have taken to follow simply from the fact that grounding explanation is a form of explanation. For example, explanation of *any* form is irreflexive, antisymmetric, and transitive, and so grounding explanation must be irreflexive, antisymmetric, and transitive as well.⁶ And in a grounding explanation, as in any explanation, the explanans must be relevant to the explanandum. This entails, among other things, that if *A* has a grounding explanation in terms of *B*, then *A* will not in general have a grounding explanation in terms of *B* together with some arbitrary additional fact.

If these claims hold of grounding explanation in general, then this category has at least a certain 'formal' kind of unity. But philosophers have taken the category of grounding explanation to be unified in other ways as well.⁷ For example, many have thought that grounding explanation can be characterized in general as that form of explanation in which we explain some fact by citing other facts that 'constitute' it.⁸ The fact that the Golden Gate Bridge is red, for instance, is constituted by the fact that it is vermilion. Or again, perhaps the existence of this table is constituted by the tablewise arrangement of certain pieces of wood.

But even if this constitutive characterization is correct, is it illuminating? Do we have a good grasp of the notion of constitution?

It is at least clear that constitution is different from causation. The constitutive relationship between the Golden Gate Bridge's being vermilion and its being red is different from the causal relationship between Suzy's throw and the window's shattering. But beyond this contrast with causation, how is the notion of constitution to be understood?

One option is to understand it as simply a form of identity. On this understanding, the constitutive characterization of grounding explanation requires the explanans and explanandum of such an explanation to be identical. However, most philosophers have thought that these must be distinct, and so they have tried to spell out the constitutive characterization in other ways.⁹ Fine, for instance, gives the following gloss:

[I]t is natural in such cases to say that the explanans or explanantia are constitutive of the explanandum, or that the explanandum's holding consists in nothing more than the obtaining of the explanans or explanantia.¹⁰

And in a similar vein, Rosen writes:

Metaphysical grounding is often distinguished from other explanatory relations by the metaphorical thought that the grounded fact is 'nothing over and above' the facts that ground it; that it is not really a 'further fact' or an 'addition to reality'.¹¹

Fine's and Rosen's glosses suggest a broadly 'lightweight' conception of constitution. Although the explanans of a grounding explanation is not strictly identical to the explanandum, the difference between them is somehow insubstantial.

But other philosophers have glossed the notion differently. According to Schaffer, for instance, claims of grounding explanation 'have the feel of concerning the constitutive generation of a dependent outcome'.¹² Does Schaffer's talk of 'generation' mean he rejects the idea that the explanandum of a grounding explanation is no addition to reality? Is his conception of constitution more 'heavyweight' than that of Fine and Rosen? It is not clear. Whether these different glosses reflect genuinely different conceptions of grounding explanation, and if so, whether this difference can be stated precisely, remains a matter for further research.

Philosophers have offered other characterizations of grounding explanation that do not involve the notion of constitution, but these are more controversial. For example, it is sometimes suggested that a grounding explanation is one in which the explanans and explanandum are connected as a matter of metaphysical necessity: necessarily, if the explanans obtains, so does the explanandum. But this is not universally accepted.¹³ And there may be other forms of explanation, such as mathematical explanation, that also obey this condition.¹⁴ Or again, it is sometimes suggested that a grounding explanation is one that is not diachronic. But mathematical explanation is nondiachronic as well, and it may even be that certain grounding explanations *are* diachronic.¹⁵

One also sees grounding explanation characterized simply as 'metaphysical' explanation. Sometimes this is just a matter of terminology: some authors use the phrase 'metaphysical explanation' to mean grounding explanation.¹⁶ But in some cases it may be more than merely terminological, since it may stem from an assumption that grounding explanation is the only distinctively metaphysical form of explanation. This assumption is controversial. One might think, for instance, that if something is a certain way by its very nature, then this explains why it is that way,¹⁷ and this might be thought to be a metaphysical explanation that is not a grounding explanation.¹⁸ Or one might think that if something *must* be true, then this explains why it *is* true,¹⁹ and perhaps this too is a metaphysical explanation that is not a grounding explanation.

3. The Question of Realism

Given any form of explanation, whether it is grounding explanation or some other form, we may ask how it relates to reality. To see what I mean by this, let us look at a passage from Carl Hempel, one of the most important theorists of *scientific* explanation. Describing his influential deductive–nomological account of such explanation, Hempel wrote:

[A] D–N explanation answers the question 'Why did the explanandum–phenomenon occur?' by showing that the phenomenon resulted from certain particular circumstances, specified in $C_1, C_2, \dots, C_p$, in accordance with the laws $L_1, L_2, \dots, L_r$. By pointing this out, the argument shows that, given the particular circumstances and the laws in question, the occurrence of the phenomenon *was to be expected*.²⁰

There are in fact two opposing views of explanation suggested by this rich passage. The first is realism.²¹ On this view, explanation is 'backed' by real or worldly determination. Something in the world makes something else exist or happen (or one thing 'results' from another, as Hempel puts it), and this determination or making underwrites an explanation of whatever it is that is determined or made. For example, if the shattering of this window is causally explained by

Suzy's throwing a rock at it, a realist about causal explanation will take this explanation to be backed by some form of real determination, perhaps the fact that Suzy's throw *made* the window shatter.

On the antirealist view, by contrast, explanation is not backed by determination. It is not possible to say much more than this about antirealism in general, for there are many different versions of the view with little to unite them apart from their rejection of realism. On one version of antirealism, suggested by the passage from Hempel, explanation is simply a matter of expectability. One fact explains another just in case, given the first, the second is to be expected. Another version of antirealism is 'unificationist': *A* explains *B* just in case there is a deductive argument from *A* to *B* that fits into a pattern of arguments that best unifies or systematizes the facts.²² A third version is 'pragmatic', rendering explanation a contextually sensitive matter.²³ The commitment common to these and other antirealist views is simply that explanation is not backed by real determination. (The antirealist may allow, of course, that the explanans and explanandum of an explanation are themselves perfectly real.)

Just as there are realist and antirealist views of scientific explanation, so there are realist and antirealist views of grounding explanation. But in the case of grounding explanation, antirealist views have so far seen little discussion. On one such view, grounding explanations hold in virtue of psychological dispositions.²⁴ (In calling this view antirealist, I do not mean to suggest that psychological facts are unreal. The view is antirealist because psychological facts, even if they are real, are not standardly taken to constitute a form of determination.) Another version of antirealism is unificationist.²⁵ And it is perhaps also possible to develop a pragmatic view of grounding explanation. What other antirealist views there might be is a matter for further research.

In light of the relatively undeveloped state of antirealism, let us turn now to realism. On one popular realist view, grounding explanations are backed by determination relations of a certain sort, what might be termed relations of constitutive generation or production.²⁶

But what do these generation relations *relate*? Defenders of this relationalist form of realism differ on this question. Some, for instance, take generation relations to hold between properties: vermilionness generates redness. Others take the relations to hold between facts: the fact that snow is white generates the fact that it is *true* that snow is white. Still others think that generation relations can hold between entities of any category—between one object and another object or between a fact and a property and so on. It might be thought, for instance, that Socrates generates his singleton set.²⁷

These generation relations, according to the relationalist realist, back grounding explanations. But this view raises two questions. First, what *is* backing? Second, which generation relations back which grounding explanations? (Analogous questions apply to the other forms of realism we will discuss, though we will not address them.)

Let us take the first question first. One possibility is to understand backing itself in terms of explanation. For a grounding explanation to be backed by generation relations, on this view, is simply for it to be explained in some way by those relations.²⁸ Another possibility is to take backing to be a matter of information content. On this view, a grounding explanation is backed by generation relations in the sense that it contains information about generation relations.²⁹ Yet a third possibility is to understand backing in terms of mereological or part-whole relations. Suppose one takes an explanation to be a complex whose parts are an explanans, an explanandum, and some generation relations that serve to connect the two. Given this view of explanation, one might then say that an explanation is backed by those generation relations that are parts of it.

What about the second question? Which generation relations back which grounding explanations? The answer to this question depends in part on what these relations relate. Do they

relate facts? Properties? Entities of arbitrary category? If the relata are facts, then a simple answer to our second question is available: the grounding explanation of one fact *A* in terms of another fact *B* is always backed by *A*'s being generated by *B*, and conversely, *A*'s being generated by *B* always backs a grounding explanation of *A* in terms of *B*.³⁰ (Matters are less clear if the relata are something other than facts. If Socrates generates singleton Socrates, for example, exactly which grounding explanations does this back?)

The simple answer faces an objection from a case due to Jonathan Schaffer. To understand the objection, notice that the generation relation, whatever it is, is naturally taken to be transitive. If the fact that *A* generates the fact that *B*, and the fact that *B* in turn generates the fact that *C*, then it is natural to think that the fact that *A* will generate the fact that *C*. And grounding explanation, according to the simple answer, corresponds one to one with generation. So since the latter is transitive, the former must be too.

Yet Schaffer presents a case in which grounding explanation might be thought *not* to be transitive. The case concerns a sphere with a small dent. That the sphere has this particular dent, he suggests, in part explains why it has the exact overall shape that it does—namely, dented here and spherical everywhere else. And that it has that exact overall shape in turn explains why it is more or less spherical. Yet it may seem that the sphere's having this particular dent does not explain, even in part, why it is more or less spherical. For as Schaffer observes, 'the presence of the dent makes no difference to the more-or-less sphericity of the thing'.³¹

One response on behalf of the simple answer is this. Distinguish an immediate or direct form of generation, which is not transitive, from a mediate or indirect form, which is.³² Just as one gets a relation of ancestorship by chaining together parent-child relations, so one gets a relation of mediate generation by chaining together relations of immediate generation. Now say that what backs grounding explanation is *immediate* generation. This would accommodate Schaffer's case, for now neither grounding explanation *nor* generation is transitive.

But this response goes too far. Even if grounding explanation is not transitive in general, there are surely many cases in which chaining occurs. For example, because the Golden Gate Bridge's being vermilion explains its being red, and because its being red in turn explains its being colored, it follows that its being vermilion explains its being colored. But there is no reason to think that the bridge's being vermilion immediately (as opposed to mediately) generates its being colored. This response to the simple answer, then, has difficulty accommodating such cases.

Should this answer therefore be abandoned? Stephan Krämer and Stefan Roski have recently argued that it should.³³ On their view, in order for a fact *A* to explain another fact *B*, *A* must not only generate *B* but must also *make a difference* to whether *B* obtains. This difference-making requirement means that grounding explanation need not correspond one to one with generation, and so even if the latter is transitive, the former need not be. For Krämer and Roski, Schaffer's dented sphere is a case of generation without difference making—and so of generation without explanation.

There may, of course, be competitors to the simple answer other than the kind of difference-making answer that is suggested by Krämer and Roski's argument. In fact, these answers may not even be in competition. One might admit multiple notions of grounding explanation, each of which obeys a different answer to the question of which generation relations back which explanations. Perhaps one notion of explanation obeys the simple answer, while another obeys the difference-making answer, so that there is no need to choose between them.

Let us now turn to a second form of grounding-explanatory realism, which we will call 'nomicist'. For the nomicist realist, grounding explanations are backed by laws of metaphysics.³⁴ Laws of *nature* are familiar: it is a law of nature, for example, that like charges repel. But some

philosophers have also recognized the existence of laws of *metaphysics*. One might take it to be a law of metaphysics, for instance, that everything vermilion is red. Suppose now that one explains why the Golden Gate Bridge is red by saying that it is vermilion. The nomicist realist can take this explanation to be backed by the law that everything vermilion is red.

Views of laws of *nature* are standardly classed as Humean or anti-Humean. On Humean views, the laws are mere summaries of independently existing regularities in nature. On anti-Humean views, by contrast, the laws play an active role in determining the course of events or in making events happen. Maudlin describes this sort of nomic determination this way: 'The universe started out in some particular initial state. The laws of temporal evolution operate, whether deterministically or stochastically, from that initial state to generate or produce later states'.³⁵

The distinction between Humean and anti-Humean views of laws of nature is familiar. Far less familiar is an analogous distinction between Humean and anti-Humean views of laws of *metaphysics*.³⁶ On Humean views, laws of metaphysics are mere summaries of independently existing regularities in the facts, such as the correlation between vermilionness and redness. On anti-Humean views, by contrast, the laws play an active role in determining the facts or in making the facts obtain.

An anti-Humean view appears inevitable once one adopts the nomicist form of realism. A realist, after all, is someone who takes grounding explanations to be backed by real determination. But Humean laws do not determine the facts; they merely summarize them. Only anti-Humean laws provide the kind of determination or 'making' that realist explanation requires.

Turning now to a third realist view, the *essentialist* realist takes grounding explanation to be backed by the essences or natures of things.³⁷ Suppose, for example, that one explains why singleton Socrates exists by pointing to the existence of Socrates himself. One might then take this explanation to be backed by the fact that singleton Socrates is essentially the set whose sole member is Socrates.

But will this kind of essentialist backing be available in every case? A number of philosophers have explored views on which the answer is 'yes'. On these views, whenever there is a grounding explanation of *A* in terms of *B*, there will be some items involved in *A* or *B* (or both) whose essences require there to be some kind of link between *A* and *B*.³⁸ A defender of the essentialist form of realism might take grounding explanations to be backed by these links.

The most developed version of essentialist realism, due to Kelly Trogdon, is inspired by the mechanistic view of causal explanation.³⁹ That view takes causal explanation to be backed by causal mechanisms, understood as complex systems or as physical processes. In an analogous way, Trogdon takes grounding explanation to be backed by what he calls *grounding mechanisms*: relations whose essences require certain determination relations to obtain. Set formation, the relation holding between a set and its members, provides a good illustration. For Trogdon, it is essential to this relation that, if some objects exist, then their existence determines or makes it the case that a *set* exists that contains exactly those objects. Set formation, then, turns out to be a grounding mechanism. As such, it backs various grounding explanations, such as the explanation of the existence of singleton Socrates in terms of the existence of Socrates himself.

We have now considered three broad kinds of grounding-explanatory realism: relationalism, nomicism, and essentialism. We should bear in mind that these may not be exclusive, for some may turn out to reduce to or be otherwise compatible with others. Nor are they necessarily exhaustive, for there are further realist views that we have not yet considered.

One such view is primitivist. The realist views considered earlier are all committed to the claim that what backs a given explanation is something other than that very explanation. But this might not be true. Perhaps grounding explanation need not be backed by some distinct, underlying form of determination. Perhaps it is *itself* a form of determination. (Objection: this

is no longer realism, since by definition realism requires grounding explanations to be backed by real determination. Reply on behalf of the primitivist realist: grounding explanations can be backed by themselves.)⁴⁰

It may even be possible to hold a 'semiprimitivist' form of realism along the following lines. Some, even most, grounding explanations are backed by a distinct form of determination: generation relations, laws of metaphysics, or whatever you like. But certain grounding explanations are, in primitivist realist fashion, backed by themselves.

One potential motivation for semiprimitivist realism arises from logical grounding explanations. One can give a grounding explanation of why Bowser is big and bad by saying that he is big and he is bad. But some philosophers have thought it impossible to understand this case, and other cases of logical grounding explanation, on the nonprimitivist realist model. Even if one is willing to say that *many* grounding explanations are backed by a distinct, underlying form of determination, how could this be true of the logical cases? What could possibly underlie them? To such philosophers, these cases have seemed to involve utterly basic explanatory relationships. These considerations might move one to adopt a version of semiprimitivist realism. Thus one might hold that while nonlogical grounding explanations are backed by a distinct form of determination, logical grounding explanations are backed by themselves.⁴¹

4. Grounding Explanation and Ground

Grounding explanation, at least by that name, has seen far less discussion than something called 'ground'. But what is ground, and what does it have to do with grounding explanation?

These questions are ambiguous, because philosophers have used the term 'ground' in two different ways. Some have used it simply to mean grounding explanation.⁴² On this usage, to say that *A* grounds *B* is simply to say that *A* provides a grounding explanation of *B*. But other philosophers have used 'ground' in a different way, to mean a relation of constitutive determination or generation or production.⁴³

In the first, explanatory sense of ground, the relationship between ground and grounding explanation is simply identity. But matters are less simple if ground is understood in the second, determinative sense. What one says about the relationship between ground in this sense and grounding explanation will depend in part on one's views on grounding-explanatory realism.

Consider first the relationalist realist. Ground as a determination relation finds a natural place in her view of grounding explanation. For since the relationalist realist takes grounding explanation to be backed by determination relations, it will be natural for her to take these relations to be nothing other than relations of ground. As Schaffer, himself a relationalist realist, writes:

One should distinguish the worldly relation of grounding from the metaphysical explanations between facts that it backs, just as one should distinguish the worldly relation of causation from the causal explanations between facts that it backs.⁴⁴

For the relationalist realist, then, it is natural to take the relationship between ground (in the determinative sense) and grounding explanation to be one of backing: the former backs the latter.

Consider next the primitivist realist. Ground as a determination relation also fits naturally with her view, though in a different way than with that of the relationalist realist. For the primitivist realist, grounding explanation itself is a form of determination. And so for her (and her alone) it will be natural to take the two uses of 'ground' to have one and the same referent. There is the explanatory use, which refers to grounding explanation. And there is the determinative use, which, for the primitivist realist, will also refer to grounding explanation. For her, then, the

relationship between grounding explanation and ground, in either the explanatory or determinative sense, is one of identity.

Ground in the determinative sense perhaps fits less straightforwardly with other forms of realism, though the combination can usually be made to work. Consider Trogdon's mechanistic version of essentialist realism, for instance. For him, grounding explanations 'are representations of grounding relations [i.e., relations of ground in the determinative sense] as being instances of grounding mechanisms'.⁴⁵ His view of grounding explanation therefore preserves a place for ground as a determination relation.

Finally, what about the antirealist? For her, grounding explanation requires no form of real determination to back it. And so it will be natural for her to deny that there is any such thing as ground in the determinative sense. For her, ground in this sense simply does not exist, and so there is no relationship at all between it and grounding explanation.^{46,47}

Notes

- 1 Fine (2012, 41).
- 2 This distinction is familiar from the literature on explanation in the philosophy of science, e.g., Bromberger (1965, 104) and Strevens (2008, 6–7). Although that literature has focused largely on causal explanation, many of the considerations raised there apply equally to grounding explanation. This literature, I believe, constitutes an invaluable resource for philosophers of ground that has only just begun to be exploited.
- 3 Thompson (2016) and Dasgupta (2017) consider views on which grounding explanation possesses a kind of relativity.
- 4 Some explanations, then, are *noncausal*. It is true that some philosophers, such as Lewis (1986), have held that the explanation of events is always causal. But this thesis is not so popular today, and even Lewis did not extend the thesis to explanation in general.
- 5 Compare Fine (2015, 297).
- 6 These properties are defined as follows. Irreflexivity: nothing has a grounding explanation in terms of itself. Antisymmetry: if *A* has a grounding explanation in terms of *B*, then *B* has no grounding explanation in terms of *A*. Transitivity: if *A* has a grounding explanation in terms of *B*, and *B* has a grounding explanation in terms of *C*, then *A* has a grounding explanation in terms of *C*. Irreflexivity has been challenged by Jenkins (2011) and Wilson (2014), asymmetry by Wilson (2014), and transitivity by Schaffer (2012). I discuss Schaffer's challenge further in what follows; for discussion of related issues, see Naomi Thompson's contribution to this handbook [Chapter 17].
- 7 Not all philosophers have taken grounding explanation to be unified; skeptics about unity include Wilson (2014, 2016) and Koslicki (2015). See Koslicki's and deRosset's contributions to this handbook [Chapters 11 and 12] for further discussion of this issue.
- 8 Dasgupta (2017), for instance, offers such a characterization. I hasten to add that some philosophers, such as Audi (2012, 709), disagree with the constitutive characterization.
- 9 Jenkins (2011) and Wilson (2014) have suggested that there may be certain grounding explanations in which the explanans and explanandum are identical. But even if this is so, there will presumably be other grounding explanations in which the explanans and explanandum are distinct. For example, the fact that the Golden Gate Bridge is red can be given a grounding explanation in terms of the distinct fact that it is vermilion. Thus if grounding explanation is to be characterized in terms of constitution, the latter cannot be understood as a form of identity.
- 10 Fine (2012, 39).
- 11 Rosen (2017, 280).
- 12 Schaffer (2017, 303).
- 13 Leuenberger (2014) and Skiles (2015) are notable opponents of this thesis. For further discussion, see Skiles's contribution to this handbook [Chapter 10].
- 14 Steiner (1978) is a classic discussion of mathematical explanation.
- 15 Rosen (2017, 280), for instance, considers the fact that a given utterance of 'arthritis' means *arthritis*, and he suggests that the grounding explanation of this fact will appeal to facts about the behavior of speakers at various times.
- 16 I myself did this in Glazier (2016).

- 17 Kment (2014, 163).
- 18 Glazier (2017).
- 19 Thus van Inwagen (1996, 95) writes, ‘If showing that it is impossible for a certain state of affairs to obtain doesn’t count as answering the question why that state of affairs does not obtain, I don’t know what would count’.
- 20 Hempel (1965, 337).
- 21 This explanatory notion of realism is discussed by Kim (1988, 1994), Ruben (1990, ch. 7), and Taylor (2017), among others. It should be distinguished (despite this section’s winking title) from the metaphysical notion of realism discussed in Fine (2001), although the two are not wholly unrelated. For further discussion of metaphysical realism, see Olla Solomyak’s contribution to this handbook [Chapter 26].
- 22 Kitcher (1981) gives an influential unificationist account of scientific explanation.
- 23 Van Fraassen (1980) and Achinstein (1984) develop pragmatic views of scientific explanation, though their focus is perhaps on acts of explaining as opposed to explanations themselves.
- 24 Miller and Norton (2017, n. 8).
- 25 Dasgupta (2017, 80–81); Kovacs (2017, 2942).
- 26 Versions of this view are discussed in Audi (2012), deRosset (2013), Schaffer (2016), Dasgupta (2017, 80–81), and Kovacs (2017, 2935–2946). Such determination relations are often called relations of ground, but I wish to leave discussion of ground and of the term ‘ground’ until the final section.
- 27 On a related realist view, the facts that back grounding explanations are properly stated by means of a sentential operator rather than (as we have done) by means of a predicate. The distinction between these two forms of realism is related to the distinction between the operational and predicational views of ground. See the introduction to this handbook for further discussion.
- 28 Compare Kovacs (2017, 2934).
- 29 This view requires an explanation to be something that can contain information—i.e., something like a proposition. Lewis (1986) defends a propositional view of causal explanation.
- 30 This answer must be generalized to accommodate cases in which one fact is explained in terms of a plurality of facts. I will suppress this and related complications.
- 31 Schaffer (2012, 127). Schaffer’s response to the problem is to take both generation and grounding explanation to be contrastive. So understood, he argues, both can be seen to be transitive after all. I will not discuss this response here.
- 32 For further discussion of the mediate/immediate distinction, see the introduction to this handbook.
- 33 Krämer and Roski (2017). Their view builds on Strevens’s (2008) account of scientific explanation.
- 34 Laws of metaphysics have been discussed by Rosen (2006), Sider (2011, 274–280), Wilsch (2015a, 2015b), Glazier (2016), and Schaffer (2017), among others. See Wilsch’s contribution to this handbook [Chapter 30] for further discussion.
- 35 Maudlin (2007, 174).
- 36 It is briefly discussed in Wilsch (2015b, 3306–3307).
- 37 Much of the contemporary discussion of essence stems from Fine (1994).
- 38 Rosen (2010) and Fine (2012) develop views of this kind; see also Dasgupta (2014b).
- 39 Trogdon (2018).
- 40 Schaffer (2016, 84) offers a brief argument against primitivist realism.
- 41 Compare Kovacs (2017, 2942). For discussion of related issues, see Michaela McSweeney’s contribution to this handbook [Chapter 32].
- 42 Examples include Litland (2013), Dasgupta (2014a), and Wilsch (2015a).
- 43 One might also use ‘ground’ to mean a determinative *operation*. However, I am not aware of any philosopher who has used the term this way.
- 44 Schaffer (2012, 124); see also Audi (2012), deRosset (2013), and Trogdon (2013).
- 45 Trogdon (2018, 1290).
- 46 The philosophical terrain charted in this section has been partitioned differently by some other authors. Raven (2015) distinguishes ‘unionist’ views, on which ground and grounding explanation are identical, from ‘separatist’ views, on which they are distinct. But from the present perspective, this distinction is ambiguous owing to the ambiguity of the term ‘ground’. The relationalist realist, for instance, will be a unionist if ground is understood in the explanatory sense but a separatist if it is understood in the determinative sense. However, we should not go as far as Dasgupta (2017, n. 8) in characterizing Raven’s distinction as largely verbal. For if ground is understood in the determinative sense, then the

debate between unionists and separatists will turn largely on the *substantive* question whether grounding explanation itself is a form of determination.

- 47 I would like to thank Kit Fine, Daniel Fogal, Mike Raven, and Olle Risberg for their written comments on earlier drafts of this chapter. I am also indebted to Kevin Richardson, Erica Shumener, and the participants at a workshop on metaphysical grounding at the University of Hamburg. I am grateful for the support of the Deutsche Forschungsgemeinschaft and of the Program of Postdoctoral Fellowships at the Universidad Nacional Autónoma de México.

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9

META-GROUND

Jon Erling Litland

Suppose the facts Γ ground the fact φ . Then it is a fact *that* Γ grounds φ . And we may ask what grounds *it*. What is the answer? And what turns on this? This is the question of *meta-ground*, *grounding ground*, or *iterated ground*.

The goal of this chapter is to introduce the reader to the state of the debate about meta-ground and to indicate some areas for future research. Even though the problem of meta-ground is a fairly small piece of the larger literature on ground, it is impossible to cover everything. Since I want to indicate what I take to be the most interesting areas for future research, regrettably some subtleties in the existing views had to be suppressed.¹

Background and Notation

Let us use $<$ as a sentential operator for strict, full, factive, mediate ground. (We will introduce some more grounding operators later.) We thus opt for operationalism over predicationism about ground; but most of the discussion can be transposed to the predicationist framework. (For more on this distinction, see the editor's introduction.) This operator has the following grammar: when Γ are some sentences and φ a sentence, then $\Gamma < \varphi$ is a sentence. When Γ contains just a single sentence, we may pronounce $\Gamma < \varphi$ as " φ because Γ ". Officially, the question of grounding ground is then: what are the Δ such that it is because it is the case that Δ that φ because Γ ? For ease of expression, we will, however, often talk about grounding as a relation between *facts*, and then the question of grounding ground is just what are the grounds for the grounding facts.

The problem of meta-ground is thus *not* the problem of grounding the grounding operation (or relation) itself. For the operationalist and predicationist alike, this question is senseless: an operator (or relation) like $<$ is not a sentence, and so an expression like $\Gamma < <$ is simply ungrammatical.² Let us now consider what turns on the question of meta-ground. We focus on the Collapse Problem and the Problem of Recombinability.

The Collapse Problem

Could we say that grounding facts are *ungrounded*? What we can call the Collapse Problem has persuaded many otherwise. (This terminology comes from (deRosset 2013: 6–8.) Bennett

(2011), Sider (2012: 143–45), Dasgupta (2014: 560–62), and Litland (2017a: 283–85) have made similar arguments.) Let us say that a fact is *fundamental* if it is ungrounded. (For further discussion of fundamentality see Bliss’s essay “Fundamentality” [Chapter 23] and Solomyak’s essay “Realism” [Chapter 26].) We may also wonder what it is for an *object* to be fundamental. For now, we work with the following definition: an object b is *fundamental*_o iff b is a constituent of a fundamental fact. (Our definition of fundamentality requires that there are ungrounded facts. For some ways of defining fundamentality without this assumption, see Fine (2010b, 2001) and Raven (2016).)

The Collapse Problem is that if grounding facts are ungrounded, all objects b are fundamental_o. Consider some fact involving b , e.g., the fact that b exists (call it Eb). Either Eb is ungrounded or it isn’t. If it is, b is fundamental_o. If it isn’t, there is some such that $\Gamma < Eb$. But since the fact that $\Gamma < Eb$ is ungrounded, b is fundamental_o after all.

If the grounding facts are ungrounded, it is a fact that all objects are fundamental *in the above sense*: but is this a problem? For some it clearly is. Sider’s principle of Purity says, in our terminology, that fundamental facts should only contain fundamental objects. For Sider, the Collapse is indeed a serious problem. But the principle of Purity has its home in Sider’s theory of “joint-carving” expressions, and it is not obvious that a grounding theorist should accept Purity. (For more on Sider’s views, see Tahko’s essay “Structure” [Chapter 27].)

Say that an object is fundamental_e if there is nothing that grounds its *existence*. Nothing in the collapse argument shows that all objects are fundamental_e; and maybe it is enough if there are objects that are not fundamental_e even though all objects are fundamental_o.

The defender of the Collapse argument can respond that there is nothing wrong with the definition of fundamental_e; and they can even grant that fundamentality_e is an interesting notion. But that does not mean that it is not a problem if every object turns out to be fundamental_o.

One often sees claims like that the grounded “consists in” the grounds, or that the grounded is “nothing over and above” the grounds (Fine 2012: 39, 2001: 15–16). This talk is related to another metaphor. In making the ungrounded facts obtain, God makes all the other facts obtain. How serious the Collapse Problem is depends on how seriously we are to take the talk of the grounded being nothing over and above the grounds.

For suppose that the existence of electron e is ungrounded. Then the existence of $\{e\}$ is grounded in the existence of e , and so $\{e\}$ is not fundamental_e. But suppose the fact $Ee < E\{e\}$ is ungrounded. Then, when creating the world, in order to make $\{e\}$ exist, it would not be enough for God to make e exist; God also would have to make it the case that $Ee < E\{e\}$. Thus the existence of $\{e\}$ *would* be something over and above the existence of e . It is (non)-fundamentality_o and not (non)-fundamentality_e that is required to draw this distinction.

The Argument From Free Recombination

Another reason to hold that grounding facts should be grounded is the thought that fundamental facts are freely modally recombinable—an idea memorably expressed in the *Tractatus*: “Any one can either be the case or not be the case, and everything else remain the same” (Wittgenstein 1921: § 1.21).

The idea that the fundamental facts are freely recombinable is quite commonly accepted in the grounding literature. (See e.g., (Bennett 2017: 190–92; Cameron 2010; Schaffer 2010: 40; Ismael and Schaffer 2016; for some criticisms of free recombination, see Wang 2016; Wilson 2010).) If the fundamental facts are freely recombinable, then the grounding facts cannot be ungrounded. For suppose that ϕ is fundamental and that $\phi < \psi$, for some ψ . Ground is factive

in the sense that, necessarily, if $\varphi < \psi$ then φ . But this means that we cannot have both $\varphi < \psi$ and $\neg\varphi$, contradicting that $\varphi < \psi$ and φ are freely recombinable. It follows from this that $\varphi < \psi$ has to be grounded.

Loss (2015) builds on this observation to argue that if every fact is grounded in some ungrounded facts and φ is fundamental, then $\varphi < \psi$ is at least partially grounded in φ . Here's a rough outline of the proof. (For a careful presentation, I refer the reader to Loss's paper.) In addition to the factivity of ground, Loss's argument requires that the grounds necessitate the grounded in following sense: (This principle is not uncontroversial—for more, see Skiles's essay "Necessity" [Chapter 10].)

(Necessitation) If $\Gamma < \varphi$ then $\Box(\bigwedge\Gamma \rightarrow \varphi)$.³

Let Γ be some ungrounded facts such that $\Gamma < (\varphi < \psi)$. By (Necessitation) we get that $\Box(\bigwedge\Gamma \rightarrow \varphi < \psi)$. By the factivity of ground we have $\Box(\varphi < \psi \rightarrow \varphi)$. It follows (by uncontroversial modal principles) that $\Box(\bigwedge\Gamma \rightarrow \varphi)$. Thus Γ and φ are *not* freely recombinable. Since φ and Γ are fundamental, however, the only way this could happen is if φ is one of the Γ . That means that φ partially grounds $\varphi < \psi$.

Loss's argument only works for ungrounded φ , but if the solution to the problem of meta-ground has to be *uniform*, this is some evidence that, in general, Γ will be a partial ground for $\Gamma < \varphi$.

Regress?

Could every fact about ground be grounded? Suppose that

$$(1) \quad \Gamma_0 < \varphi$$

Then if every grounding fact is grounded there is Γ_1 such that

$$(2) \quad \Gamma_1 < (\Gamma_0 < \varphi)$$

and there is Γ_2 such that

$$(3) \quad \Gamma_2 < (\Gamma_1 < (\Gamma_0 < \varphi))$$

and so on.

This gives us infinitely many distinct facts. (At least, it does this if we can assume that what it is for $\Gamma_1 < (\Gamma_0 < \varphi)$ to be the case is distinct from what it is for $\Gamma_0 < \varphi$ to be the case.⁴ This is not entirely obvious; for more on such issues of "fineness of grain", see Correia's essay "Granularity" [Chapter 15].) But does this give us an infinite regress?

It does not. What may be worrying about a regress of ground is that we have a fact φ_0 that cannot obtain except on account of its ground φ_1 . φ_1 in turn, cannot obtain, except on account of its ground φ_2 , and so on. In that case, there is something to the thought that "being would be infinitely deferred, never achieved" (Schaffer 2010: 62). Whether this thought is right is far from clear. (See further Dixon's essay "Infinite Descent" [Chapter 16] and Bliss's Essay "Fundamentality" [Chapter 23].) Fortunately, we do not have to worry about this.

That every grounding fact is grounded does not lead to a regress in this sense. For suppose $\Gamma_1 < (\Gamma_0 < \varphi)$; then what Γ_1 grounds is not Γ_0 — Γ_0 may be a collection of ungrounded facts for all we know—but rather the distinct fact that Γ_0 grounds φ . Thus there is no regress.

Restricting the Relata?

Could the problem of meta-ground be ill posed? One could hold that grounding is a relation between *facts*, where not every meaningful sentence corresponds to a fact. (Such a view is held by Audi (2012), though not in connection with the problem of meta-ground.) And one could hold that a true statement like Γ grounds ϕ does not correspond to a fact. In that case, the problem of meta-ground does not arise.

Such a view is intriguing, but since such a view has not been fully developed, I am, apart from the next comment, going to set it aside for the remainder of this chapter. One worry about this strategy is that even if one takes *ground* to be a relation holding between facts in this demanding sense, it seems that we can make sense of a notion of ground that is not so demanding. For cannot we, for any true statement ϕ , make sense of the question “why ϕ ?” where the answer to the question would be that in virtue of which it is the case that ϕ ?

Desiderata on a Solution

What are the desiderata for a solution? If we are moved by the Collapse Problem and want the ungrounded facts to be freely recombinable, we should at least require that our account of grounding ground solve the Collapse Problem and allow for free recombination. We should also require that the account of meta-ground respect any clear cases. Are there such cases? Clark (2018: 39) has offered the following:

- If $\phi < \psi$ and $\psi < \theta$, then $(\phi < \psi, \psi < \theta) < (\phi < \theta)$ (This case is also offered by Dasgupta (2014: 566n18).)
- If $\phi < \theta$ and $\psi < \theta$, then $((\phi < \theta), (\psi < \theta)) < (\phi, \psi < \theta)$ ⁵

Especially the first principle has a plausible ring to it: “Why does ϕ ground $(\phi \vee \psi) \vee \theta$? Because ϕ grounds $\phi \vee \psi$ and $\phi \vee \psi$ grounds $(\phi \vee \psi) \vee \theta$.”

As we will see, the existing accounts of meta-ground do not respect these intuitive judgments. This does not show that these accounts are wrong, but their defenders do have to explain away these intuitions.

Proposed Solutions

Assuming that grounding facts need to be grounded, what solutions have been proposed? The solutions come in two broad types. Proponents of *systematic* solutions hold that there is a schema such that the grounds for grounding facts fit the schema. Those who defend unsystematic accounts reject just that. Until very recently, most of the work has focused on systematic accounts, so those will be our focus.

Upwards Anti-Primitivism

Most who think that grounding facts are grounded hold that the grounds partially ground that the grounds ground the grounded. (As noted, if we accept free recombination and the grounds are fundamental, Loss (2015) shows that this is forced on us.) Litland (2017a), deRosset (2013), and Bennett (2011) take this one step further and argue that if $\Gamma < \phi$, then $\Gamma < (\Gamma < \phi)$. In fact, it is natural to take them to hold that Γ is the sole *immediate* full ground for $(\Gamma < \phi)$, though only Litland (2018) explicitly commits to this. This view goes by various names—“The Straightforward

Account” (Litland 2017a), “Simple Reductionism” (Dasgupta 2014)—but here we follow Bennett (2017) and call it Upwards Anti-Primitivism.

Upwards Anti-Primitivism solves the Collapse and Free Recombinability problems. Since, according to the Upwards Anti-Primitivist, every grounding fact is grounded, there is no reason to think that the grounding facts are freely recombinable. The Collapse is averted too. For if Fb is grounded in some fact q in which b does not occur, then $q < Fb$ is just grounded in q —a fact in which b does not occur.

Bennett’s Upwards Anti-Primitivism goes beyond merely saying that if $\Gamma < \phi$, then $\Gamma < (\Gamma < \phi)$. Bennett holds that the grounds are responsible for the grounded and the relation between the grounds and the grounded in a very strong sense. To see what she has in mind, suppose that *physicalism* is true; and suppose that being in a given physical state grounds that one desires coffee. Bennett asks, “[i]n virtue of what does it do so? Well, . . . in virtue of *itself*. Part of what it is to be that complex physical fact is to be a realization of a coffee desire” (Bennett 2017: 196).

How are we supposed to understand the locution “part of what it is to be”? One natural view takes this to be an *essentialist* claim. (For more about the various notions of essence, see Zylstra’s essay “Essence” [Chapter 22].) Let us use Υ_p for the sentential operator “it’s true in virtue of what it is for it to be the case that p that . . .”. Then it seems that Bennett commits herself to the claim:

$$(\text{Upwards Essence}) \quad p < q \rightarrow \Upsilon_p(p \rightarrow p < q)$$

But (Upwards Essence) seems to have counterexample. For instance, the existence of Socrates grounds the existence of $\{\text{Socrates}\}$. But it is not plausible that it is essential to the existence of *Socrates* that if he exists then his existence grounds the existence of $\{\text{Socrates}\}$. (The existence of Socrates “knows nothing” of sets, as the saying goes.)

There are also some moderately strong theoretical reasons for rejecting (Upwards Essence). Suppose, for the sake of illustration, that an action’s being good is grounded in its maximizing utility. Then Bennett is committed to holding that it is in the nature of utility maximization that anything that maximizes utility is good. But this prevents us from adopting the following necessary condition on something’s being nonnormative. (For more on this type of constraint, see Rosen (2017).)

(Nonnormativity) If the property P is wholly nonnormative, then no normative feature figures (nontrivially) in its essence.

It is not entirely clear if Bennett means to commit to (Upwards Essence); but if not, then what does “part of what it is to be” mean? Let us now go on to discuss versions of Upwards Anti-Primitivist that do not accept (Upwards Essence).

Grounding and Explanatory Arguments

The proposals of Litland (2017a) and deRosset (2013) both turn on a connection between ground and explanatory arguments. (Others have also noted the connection with explanatory arguments. Wilsch (2015a, 2015b) tries to account for grounding in terms of a deductive nomological model of explanation, and Kovacs (Forthcoming(a)) develops a “unificationist” account of metaphysical explanation.) Let me sketch Litland’s proposal.

Litland makes use of the notion of non-factive ground. Whereas if ϕ factively grounds ψ , both ϕ and ψ have to be the case, ϕ can nonfactively ground ψ even though neither ϕ nor ψ

are the case. We do, however, require that if ϕ nonfactively grounds ψ and ϕ is the case, then ψ is the case. (For more on the distinction between factive and nonfactive ground, see Fine (2012: 48–50).) Let us use $\Rightarrow$ as an operator for nonfactive ground.

Litland's idea is that for Γ to nonfactively ground ϕ is for there to be an explanatory argument $\mathcal{E}$ from premises exactly Γ to conclusion ϕ . He proposes the following introduction rule for the non-factive grounding operator $\Rightarrow$:

$$\frac{\frac{\overline{\Gamma}^1}{\mathcal{E}} \phi}{\Gamma \quad \phi} 1$$

Figure 9.1

In this rule, we require that $\mathcal{E}$ be an explanatory argument. The superscript “1” above Γ means that we assume Γ as premises. The “1” next to the horizontal line above $\Gamma \Rightarrow \phi$ means that in passing to the conclusion $\Gamma \Rightarrow \phi$ the premises Γ are discharged. Litland requires that in arguments of this form, *all* the premises are discharged: the conclusion $\Gamma \Rightarrow \phi$ thus depends on no assumptions.

Litland then holds that this argument *itself* is explanatory. That means that we can apply the $\Rightarrow$ -I rule *again* as follows.

$$\frac{\frac{\frac{\overline{\Gamma}^1}{\mathcal{E}} \phi}{\Gamma \quad \phi} 1}{\Rightarrow (\Gamma \quad \phi)}$$

Figure 9.2

What does this mean? Fine (2012: 47–48) distinguished between a truth's being *ungrounded* on the one hand and having the *empty ground* or being *zero-grounded* on the other. Put in terms of explanatory arguments, this distinction is not mysterious. An ungrounded fact is one that is the conclusion of no explanatory argument; a zero-grounded fact is one that is the conclusion of an explanatory argument with no (undischarged) premises.

For Litland, all true nonfactive grounding claims are thus zero-grounded. A factive grounding claim $\Gamma < \phi$ is taken to be grounded in Γ together with $\Gamma \Rightarrow \phi$. In this way, we ensure the characteristic Upwards Anti-Primitivist thesis that $\Gamma < \phi$ is grounded in Γ .

Unlike Litland, deRosset's version of Upwards Anti-Primitivism does not make use of the notion of nonfactive ground. For deRosset, the factive grounding operator $<$ would be directly governed by the rule:

$$\frac{\Gamma \quad \mathcal{E} \quad \phi}{\Gamma \quad \phi}$$

Figure 9.3

This rule differs from Litland's rule for $\Rightarrow$ in that the premises Γ are *not* discharged. By taking arguments of this form to be themselves explanatory, deRosset too can derive the characteristic Upwards Anti-Primitivist thesis that $\Gamma < \varphi$ is grounded in Γ . The difference from Litland's view is that $\Gamma < \varphi$ is not partly grounded in $\Gamma \Rightarrow \varphi$. (For a detailed account of a logic of iterated ground based on these ideas, see Litland (2018).)

Before we discuss how this version of Upwards Anti-Primitivism relates to (Upwards Essence), let us consider some objections to Upwards Anti-Primitivism.

Objections to Upwards Anti-Primitivism

Clark (2018) objects to Upwards Anti-Primitivism by appealing to the intuitive desiderata mentioned earlier. (His objections also apply to the connectionist views discussed in what follows.) One of the plausible claims he mentioned is that if $A < B$ and $B < C$ then $(A < B), (B < C) < (A < C)$. According to the Upwards Anti-Primitivist, however, the ground for $A < C$ is A . Some, like Litland, hold that the *only* (nonempty) immediate ground for $A < C$ is A ; such Upwards Anti-Primitivists are forced to reject Clark's intuitive claim.

One can argue against the intuitive claims as follows: " $A < B$ and $B < C$ does not ground $A < C$; rather, when $A < B$ and $B < C$ are both the case this ensures that the connection between A and C is such that $A < C$. But this does not mean that the grounds for $A < C$ itself are $A < B$ and $B < C$." The Upwards Anti-Primitivist can try to explain away the pull of Clark's principle by noting that one often will mention grounding facts in explaining what are the grounds of some other facts; but one should not infer from this that the grounding facts themselves are part of the grounds for the fact to be explained. For instance, one may wonder why it is the case that $PV(QVR)$? It seems fine to answer: "Because Q and Q grounds QVR and QVR grounds $PV(QVR)$." But few would hold that QVR 's grounding $PV(QVR)$ is a ground for $PV(QVR)$.

Those who accept the connection between ground and explanatory arguments have a further response. For such philosophers can hold that we only introduce grounding *operators* to have a way of speaking, in the object-language, about the existence of explanatory arguments. And there are many ways of introducing such operators. Instead of proceeding in Litland's way, one could introduce an operator $\multimap$ such that $\Gamma \multimap \varphi$ was true only when there was a one-step argument from Γ to φ . (This would correspond to *immediate* ground.) And instead of giving $\Rightarrow$ the above introduction rule, one could say that $P \Rightarrow Q$ can be explanatorily inferred from $P \multimap R_0, R_0 \multimap R_1, \dots, R_n \multimap Q$. Proceeding in this way, one gets Clark's proposal.

More concessively, an Upwards Anti-Primitivist can accept Clark's intuitive principles and allow that $A < C$ can have other immediate grounds than A . In response to this, Clark points out that one is then forced to admit that grounding facts are overdetermined. But while overdetermination may be problematic in general, the present case seems fairly benign. For suppose $A < C$ is grounded in $(A < B), (B < C)$ as well as in A . $(A < B), (B < C)$ are not *independent* grounds for $A < C$. For A grounds $A < B$ by Upwards Anti-Primitivism, and since B grounds $B < C$ by Upwards Anti-Primitivism, it follows by the transitivity of ground that A grounds $B < C$.

Against Litland's version of Upwards Anti-Primitivism, one can object that it is too deflationary. Grounding, some may object, is an objective relationship turning only on how things stand in the world. It has nothing to do with anything as linguistic as an *argument*. Moreover, whether one finds an argument to be explanatory depends on all sorts of pragmatic and subjective features that are irrelevant to ground.

On the other hand, some may like this deflationary feature because it allows us to give an "expressivist" account of grounding. For we can say that an explanatory argument with conclusion φ is one in the giving of which one expresses one's attitude that this settles the question

“why $\varnothing$?”. (For two (very different) antirealist views about grounding itself, see Dasgupta (2017) and Miller and Norton (2017), and for more about explanation, see Glazier’s essay “Explanation” [Chapter 8].)

The most common objection to Upwards Anti-Primitivism is the following. Both $p \vee p$ and $\neg p$ are grounded in p . But according to the Upwards Anti-Primitivist the ground for both $p < p \vee p$ and $p < \neg p$ is simply p . Dasgupta objects: “the grounds are surely different and involve something about disjunction in the first case and negation in the second. It is because of the way disjunction works that p is a sufficient explanation of why $p \vee q$, while it is because of how negation works that p is a sufficient explanation of why $\neg p$ ” (Dasgupta 2014: 573).

An initial response is to agree that it is because of how disjunction works that p grounds $p \vee p$, whereas it is because of how negation works that p grounds $\neg p$. What the Upwards Anti-Primitivist will object to is that this means that there has to be a difference in the *grounds* for $p < \neg p$ and $p < p \vee p$. Most grounding theorists are happy with the idea that distinct facts can have exactly the same grounds— $\neg p$ and $p \vee p$, arguably being an example (though see Correia’s essay “Granularity” [Chapter 15])—so why should not $p < p \vee p$ and $p < \neg p$ itself be an example? (For more on this line of argument see Bennett (2017: 201–2) and Dasgupta (2019).)

One can expand on this response by holding that $p \vee p$ and $\neg p$ are grounded in p in different *ways* (Litland 2017a). But the difference in the ways in which $p \vee p$ and $\neg p$ are grounded does not mean that the grounding facts $p < p \vee p$ and $p < \neg p$ have to have different grounds. If one takes this line, one owes an account of what ways of ground are. Krämer (2018) has developed a theory of ways of ground that deals with conjunction, disjunction, and negation, but there is no theory yet that deals with grounding facts, let alone a general theory of ways of ground. This is an area in which more work is required.

Finally, the Upwards Anti-Primitivist can use essentialist resources to account for the difference in how disjunction and negation works. We can illustrate the idea with an example. Suppose that $p < p \vee p$. The grounding connection between p and $p \vee p$ is not just a “one-off”; the same relationship obtains between any disjunction and its disjuncts (though see Krämer’s essay “Puzzles” [Chapter 18]). In particular, we have the following generalization (we here quantify directly into sentence position): $\forall p \forall q ((p \rightarrow p < p \vee q) \wedge (q \rightarrow q < p \vee q))$. (This is an instance of the *Formality* principle of (Rosen 2010: 131).) But this is no accidental generalization: it is essential to disjunction that this generalization holds. That is, we have

$$(EV) \quad \Box \forall p \forall q ((p \rightarrow p < p \vee q) \wedge (q \rightarrow q < p \vee q))$$

(This is an instance of the *Mediation* principle of Rosen (2010: 131); see also Audi (2012).)

Consider now $p < \neg p$. This, too, is not a one-off. It is an instance of the generalization $\forall q (q \rightarrow q < \neg q)$. That generalization is not accidental either, and we have

$$(E_-) \quad \Box \neg \forall q (q \rightarrow q < \neg q)$$

The Upwards Anti-Primitivist can now meet Dasgupta’s challenge by holding that (EV) is distinct from (E₋).

It is important to see that accepting (EV) (or (E₋)) by itself does not provide an answer to the problem of meta-ground. (This was in effect pointed out by Audi (2012).) This can be obscured by the gloss often given to essentialist claims. It is common to pronounce $\forall \varnothing \varnothing$ as “it is true in virtue of the nature of disjunction that $\varnothing$ ”. But this “in virtue of” should not be taken to indicate

ground. (For one thing, it has the wrong grammatical form: it is a relation between a sentential operator—disjunction—and a proposition, but ground is a relation between propositions.) In particular, accepting (E_-) is—contra (Bennett 2017: 205–6)—compatible with Upwards Anti-Primitivism.

But even though (E_-) is compatible with Upwards Anti-Primitivism, claims like (E_-) play a central role in the main competitor to Upwards Anti-Primitivism.⁶

Connectivism and Trialism

The main systematic alternative to Upwards Anti-Primitivism is *Connectivism*. Consider again a case of disjunctive grounding: $p < p \vee q$. There is a type of “general connection” between disjuncts and disjunctions. What the connectivist proposes is that the fact that $p < p \vee q$ is grounded in p together with this general connection. What is this general connection? One possibility is that it is a fact about essence, for instance $\Box \forall p \forall q ((p \rightarrow p < p \vee q) \wedge (q \rightarrow q < p \vee q))$. Another possibility is that it is simply a *necessity* like: *it is necessary that, for all p and all q if p then $p \vee q$* . Or it could be a conceptual truth like: *it is a conceptual truth that for all p and for all q , if p then $p \vee q$* . Or it could be a fact about metaphysical law, like: *it is a metaphysical law that for all p and for all q , if p then $p \vee q$* .

Let us work with the essentialist view—Essentialist Connectivism. The essentialist connectivist holds that (EV) together with p grounds $p < p \vee q$ and that (E_-) together with p grounds $p < \neg \neg p$. Since (EV) and (E_-) presumably are different facts, this means that $p < p \vee p$ and $p < \neg \neg p$ have different grounds—just as Dasgupta wants.

This just deals with two examples. The essentialist view, in general, works by holding

- (i) that any grounding claim $\Gamma < \phi$ is an instance of a generalization Φ ,
- (ii) there are some constituents C of Γ, ϕ such that $\Upsilon_C \Phi$, and
- (iii) $\Gamma, c\Phi$ ground that $\Gamma < \phi$.

(This formulation is rough in various ways, but it will work for our purposes; for more careful formulations, I refer the reader to (Fine 2012; Rosen 2010; Dasgupta 2014).)

However, as far as the Collapse Problem is concerned, this only pushes the problem back a bit. The question becomes what grounds facts like (EV). It is at this point that Dasgupta makes his most distinctive move. Dasgupta holds that the essentialist facts that ground grounding facts are ungrounded. (He thus opts for the *brute essentialist* version of *brute connectivism*. Other versions of brute connectivism are *brute necessitarianism*, *brute conceptualism*, and *brute nomicism* (Dasgupta 2014: 568).) This might not be so problematic in the case of (EV), but consider a grounding fact like:

(Bachelor) Bob is a man, Bob is unmarried < Bob is a bachelor

According to the brute essentialist, (Bachelor) is partially grounded in the following essentialist fact E_{Bachelor} :

$$\text{Bachelor} \quad \forall x (\text{Man}(x) \wedge \text{Unmarried}(x) \rightarrow (\text{Man}(x), \text{Unmarried}(x) < \text{Bachelor}(x)))$$

For Dasgupta, it is quite plausible that E_{Bachelor} is ungrounded. But that means that the *property* of being a bachelor is fundamental. That is an unwelcome result. (Though it is not as bad as it could be—at least we do not have to hold that *Bob* is fundamental!)

Dasgupta's solution is to adopt a version of *trialism*. (The terminology is from deRosset (2013).) The trialist idea, in general, is that we should not count an object as fundamental just because it figures in an ungrounded fact; it has to figure in a *special* sort of ungrounded fact. We are of course free to draw a distinction within the ungrounded facts; the challenge for the trialist is to show that it corresponds to an important distinction. Dasgupta (2014: 575) draws the distinction in terms of a distinction between *substantive* and *autonomous* facts. Substantive ungrounded facts are apt for explanation, they just happen not to have one. Autonomous facts, on the other hand, are not even apt for explanation.

How can we motivate the distinction between substantive ungrounded facts and merely autonomous ones? Dasgupta motivates it in two ways. One is by analogy with causal explanation. Some facts are apt for causal explanation but do not have any. (An example might be the initial configuration of the universe.) Some facts—mathematical ones—do not have causal explanations, but nor are they apt to have them. Since Dasgupta equates grounding with metaphysical explanation, his autonomous facts are those facts that are not apt for metaphysical explanation.

It is not clear that this way of motivating the distinction works. Facts are apt for *causal* explanation if they concern matters in time and space: this accounts for why mathematical facts are not apt for causal explanation. But a fact seems apt to be grounded if, well, it is a *fact*. It is not clear that the analogy suffices to motivate Dasgupta's distinction. (For more on this criticism, see Glazier (2017) and Raven (Forthcoming).)

The second analogy likens autonomous facts to (stipulative) *definitions*. One can ask how to prove a *theorem*, and one can ask how to prove an *axiom* from other axioms. But it makes no sense to ask how to prove a definition. For instance, set-inclusion $\subseteq$ is typically defined in terms of membership as follows $A \subseteq B \leftrightarrow^{\text{df}} \forall x (x \in A \rightarrow x \in B)$. Dasgupta thinks we would make a mistake if we tried to prove this definition. Dasgupta suggests that autonomous facts stand to ground as definitions stand to proof.

Sider (Forthcoming), Glazier (2017), and Raven (Forthcoming) criticize the analogy on the grounds that we cannot get definitions *both* to correspond to facts *and* to be unprovable. They point out that there are different things we can mean by "definition." First, we can mean a statement in the *meta-language*—the sentence $A \subseteq B$ is an abbreviation for the sentence $\forall x (x \in A \rightarrow x \in B)$. In this case, it is clear why the definition cannot be proved, but this is just because the definition is not a statement in the *object-language*. Secondly, we can take definitions to be *acts* of stipulation. But then it is even more clear that definitions do not correspond to statements. Finally, we could take the definition to be the object language statement $A \subseteq B \leftrightarrow \forall x (x \in A \rightarrow x \in B)$, which is an abbreviation of the logical truth $(\forall x (x \in A \rightarrow x \in B) \leftrightarrow \forall x (x \in A \leftrightarrow x \in B))$. The problem now is that *this* statement *is* provable.

A distinction that is related to the autonomous/substantive distinction is drawn by Fine (2005). Fine distinguishes between *worldly* facts that turn on how the circumstances turn out, and *transcendental* facts that are true irrespective of the circumstances. For Fine, a fact of the form $p \vee \neg p$ (where p is contingent) is worldly even though it is necessarily true. The reason for this is that *how* the fact obtains varies depending on whether it is p or rather $\neg p$ that is the case. The transcendental facts, on the other hand, do not turn on anything that varies from world to world. Fine says that "we might think of the transcendental facts as constituting the invariable framework within which the variation takes place . . . these are the facts that provide the framework in which God makes the decisions that he does" (Fine 2005: 325–26). (Compare the "scaffolding" metaphor (Dasgupta 2014: 562).) If the worldly/transcendental distinction can be made out, it would be natural to say that it is only the objects that figure in ungrounded worldly facts that are really fundamental.

Even though it is hard to make out the autonomous/substantial distinction, the Upwards Anti-primitivist should not be too comfortable in pointing out the problems caused by holding that E_{Bachelor} is ungrounded. For whether or not E_{Bachelor} plays a role in *grounding* grounding facts, it is at least still a fact. And if it is ungrounded, we face the Collapse Problem. (What the grounds for essentialist or necessary truths are is one of the more pressing problems in the theory of ground.)

Schaffer (2017a, 2017b) develops a brute nomicist version of trialism. For Schaffer, a meta-physical explanation has three parts. There are the *grounds*, there is the *grounded*, and then there is the *link*; the principle describing how the grounds ground the grounded. Like Dasgupta, Schaffer takes the linking principles (at least the most basic ones) to be ungrounded. And, like Dasgupta, he thinks that even though the linking principles are ungrounded, this does not mean that we should count the items occurring in them as fundamental.

But the way Schaffer thinks about this is quite different from the way Dasgupta (and Fine) think about it. We have already seen the slogan: “If A grounds B , then God only had to bring about A in order to bring about B .” Unlike most authors in the grounding literature, Schaffer thinks this slogan is, strictly speaking, false. For God *had* to do something else, namely, put the linking principles in place! This, of course, is exactly what Dasgupta and Fine denies it even makes sense for God to do.

I think Schaffer is best seen as accepting the conclusion of the Collapse argument: if an object is fundamental if it is a constituent of a fact that God had to bring about, then Schaffer accepts that all objects are fundamental in that sense. What Schaffer (2017a: 19–21) goes on to argue is that we cannot hope to do better than this and that this is enough for grounding to do the work philosophers have wanted it to do.

One might worry that on Schaffer’s view, the grounds would not necessitate the grounded. For if God had to do more than just make it the case that ϕ in order to bring it about that ψ , how could it be necessary that if ϕ then ψ ? But for Schaffer ϕ necessitates ψ iff ψ is the case in any situation in which ϕ is the case—as long as that situation respects the actual linking principles (Schaffer 2017b: 15–16).

Unsystematic Accounts

Given the problems faced by both the Upwards Anti-Primitivist and the Connectionist, Sider (Forthcoming) has recently proposed that we reject their common presupposition: that there is a uniform account of what grounds grounding facts. To motivate this, consider physicalism—understood as the view that all mental facts are grounded in physical facts. In accepting physicalism, one just commits oneself to there being *some* account of how the mental is grounded in the physical; one does not commit oneself to there being a uniform schema that specifies the physical grounds for any given mental fact. So, Sider wonders, why should it be different with grounding facts?

What we should do instead is to list some types of facts amongst which the grounds for grounding facts can be found and then show how we can make this work in a range of cases. Given its proudly piecemeal nature, I cannot here hope to do justice to Sider’s view, but let me illustrate how it avoids the Collapse Problem.

Sider argues that we can find the grounds for grounding facts amongst

- (i) patterns in what actually happens
- (ii) modal facts

- (iii) facts about the form or constituents of the grounding fact in question
- (iv) metalinguistic facts
- (v) facts about fundamentality.

Consider the fact that Trump won the election grounds that Trump won the election or Trump is bankrupt (symbolically: $Wt < Wt \vee Bt$). Suppose we want to ground this grounding fact in a pattern in what actually happens. The relevant pattern is that anyone who wins the election either wins the election or is bankrupt: $\forall x (Wx \rightarrow Wx \vee Bx)$. So maybe what grounds that $Wt < Wt \vee Bt$ is Wt together with the generalization $\forall x (Wx \rightarrow Wx \vee Bx)$.

The properties of winning the election and being bankrupt are, intuitively, not fundamental; to avoid the Collapse Problem, the generalization $\forall x (Wx \rightarrow Wx \vee Bx)$ has to have grounds that not involve neither winning nor bankruptcy. Either we can find such grounds or not; if we cannot, then the Collapse Problem arises already for generalizations and has nothing in particular to do with meta-ground, or else we can solve it, and meta-ground does not create trouble in this case. (A similar strategy works if we take the grounds for $\forall (x \ W \ x \rightarrow W \vee \ x \ \ B \ \ x)$ to be a *modal* generalization like $\forall (x \ W \ x \rightarrow W \vee \ x \ \ B \ \ x)$.)

Connections and Open Issues

I will end by discussing some connections to other issues in the theory of ground.

Grounding Negations

If one is moved by the Collapse Problem then showing that all grounding facts are themselves grounded is just the beginning. For take a false statement about ground—e.g., that the vase’s being cobalt grounds that Trump is president. Since this grounding statement is false, its negation is true. What grounds it? If the vase is not cobalt, then one could perhaps say that it is the vase’s *not* being cobalt that grounds the negation of the grounding claim. But what if the vase *is* cobalt? One possibility is to say that in this case, the grounding claim is zero-grounded (Litland 2017a: 306); another possibility is to say that it is the vase’s being cobalt that does the job. But nobody has developed a systematic account of these issues.

Accounting for the Grounding Operation Itself

While it does not make sense to ask what *grounds* the grounding operation, there are good questions in the vicinity—and those questions may be more important than the question of meta-ground. Let me mention a few:

The grounding operation could, in some sense, be “constructed” or “built” from other operations. (The relevant notions of building or construction would not be ground.) To illustrate some options: $<$ is a sentential operator. In higher-order logic, it makes perfect sense to have higher-order *relations* between sentential operators. One important higher-order relation is identity. In a higher-order setting, we can then ask whether grounding is identical to some operation that is otherwise given.

In a higher-order framework, we can also form the claim that $<$ exists— $E(<)$. (Here E is a higher-order existence predicate.) And it does make sense to ask whether $E(<)$ is grounded or ungrounded. It is not obvious that holding that $E(<)$ is ungrounded gives rise to Collapse worries, but more work must be done here. Schaffer (2017a) has recently argued that some

metaphysical laws must be ungrounded. (On these arguments, see further Wilsch’s essay “Laws of Metaphysics” [Chapter 30]). Similar arguments could perhaps be used to motivate that $E(<)$ is ungrounded.

More generally, there could be *dependence* relations between sentential operators. (For more on dependence and its connection with essence see Schnieder’s essay “Dependence” [Chapter 7] and Zylstra’s essay “Essence” [Chapter 22]. For instance, the complex operator O that takes any three propositions p, q, r to the proposition $p \wedge (q \vee r)$ plausibly depends on the operations $\wedge$ and $\vee$. In a higher-order language, we can then ask whether $<$ depends on other operations or whether it is independent. An interesting view is that the grounding operation depends on various “small-g” grounding relations (in the sense of (Wilson 2014). For more on “small-g” grounding, see Koslicki’s “Skeptical Doubts” [Chapter 11], “deRosset’s “Anti-Skeptical Rejoinders” [Chapter 12], and Richardson’s “Varieties” [Chapter 13].)

Internality

Most authors writing on ground have accepted that ground is an internal relation in the following sense:

$$(\text{Internality}) \quad \Gamma < \phi \rightarrow \Box (\Gamma \wedge \phi \rightarrow \Gamma < \phi)$$

All of deRosset (2013: 14–15), Litland (2017a) and Dasgupta (2014: 569) and Bennett (2011) accept—though Bennett (2017) has her doubts—that if $\Gamma < \phi$, then it is necessary that if Γ is the case then ϕ is the case. And given their views on grounding ground this commits them to (Internality).

But (Internality) contradicts standard principles in the logic of ground (Litland 2015). (For more on this and other problems with (Internality) see Skiles’s essay “Necessity” [Chapter 10] and Krämer’s essay “Puzzles” [Chapter 18].) So either there is something wrong with the existing proposals about grounding ground, or some assumptions from the logic of ground have to be given up.

Puzzles of Ground

The existing proposals about what grounds ground have—reasonably enough—abstracted from the problems posed by the puzzles of ground. But the puzzles of ground interact with the problem of grounding ground. The existing solutions to puzzles (Litland 2015; Fine 2010a; Krämer 2013; Woods 2018) have all been worked out for languages without grounding operators in the object language. Once we introduce grounding operators into the object language, the constructions have to be redone, taking into account one’s favorite proposal about grounding ground. This is not a trivial exercise, since principles about grounding ground can generate new puzzles of ground (Litland 2017b; Korbmacher 2015).

Grounding Pluralism

Fine (2012) suggests that there are three different notions of ground—normative, natural, and metaphysical ground. (For more on such “grounding pluralism” see Richardson’s “Varieties” [Chapter 13]. For criticism of pluralism, see Berker (2018).) All the discussions of the problem of meta-ground have focused on metaphysical ground. Once grounding pluralism is on the table, the question of grounding ground splits into several different question, for instance: what are the

normative grounds for facts about normative ground? Particularly interesting is the question: what metaphysically grounds facts about normative ground?⁷

It is in connection with *metaphysical* ground that the slogan: “All God had to do was to make sure the grounds obtain” is plausible. If normative ground is distinct from metaphysical ground, then it does appear plausible to hold that God indeed had to do more than to make the action be a promise-keeping, in order to make the action good. God also had to put in place the linking principle that connects promise-keeping with goodness. For those who accept the distinction between normative/natural and metaphysical ground, Schaffer’s line on fundamentality may be called for—at least for normative and natural ground.

Related Topics

The following chapters in the handbook are related to the issues discussed in the present chapter:

Chapter 7: Dependence
Chapter 8: Explanation
Chapter 10: Necessity
Chapter 15: Granularity
Chapter 16: Infinite Descent
Chapter 18: Puzzles
Chapter 22: Essence
Chapter 23: Fundamentality
Chapter 26: Realism
Chapter 27: Structure
Chapter 30: Laws of Metaphysics

Notes

- 1 I especially regret not being able to discuss (Kovacs Forthcoming(b)) and how the problem of meta-ground is affected by whether one takes grounding to be metaphysical explanation or rather to “back” metaphysical explanation.
- 2 I should note that Schaffer (2009: 375–76) allows grounding to be cross-categorical; for him, the question “what grounds the grounding relation” is sensible. We will not have anything to say about positions like Schaffer’s. There are, however, good questions in the vicinity of the question “what grounds the grounding operation”—we return to those later.
- 3 Here “ $\bigwedge$ ” is the conjunction of all the sentences in Γ , and “ Υ ” is the necessity operator.
- 4 Here “what it is for . . . to be the case is for . . . to be the case” is meant as a higher-order identity relation. For more on such higher-order identities, see (Dorr 2016).
- 5 This is an instance of the principle of Amalgamation; see Poggioli’s essay “Logics” [Chapter 14] and (Fine 2012).
- 6 Further objections against Upwards Anti-Primitivism are given by (Carnino 2016) and Wallner (Forthcoming); considerations of space did not allow me to discuss them here.
- 7 As Thompson (2019) points out, a similar issue arises for Bennett’s pluralism about building.

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10

NECESSITY

Alexander Skiles

Introduction

Grounding is widely thought to bear some interesting relationship to another notion of perennial metaphysical interest: *necessity*. Yet it is controversial what that relationship is. Much of the controversy surrounds two disputes over the modal import of **grounding facts** (i.e., facts about which particular facts grounds which).

Consider first **the necessitarianism versus contingentism dispute**. Using notation from Rosen (2010), say that a plurality of (one or more) facts, Γ , **necessitates** the fact that p , i.e., $[p]$, iff the following conditional necessarily holds: if the facts in Γ hold, so does $[p]$. (Thus [it is cold] necessitates [it is either rainy or cold] but not [it is both rainy and cold]). Then one may ask: is it necessarily the case that if Γ grounds $[p]$, then Γ also necessitates $[p]$? In other words, is **necessitarianism** true? Or is it false, in which case **contingentism** is true? (Those who endorse **grounding pluralism**—roughly, the view that there are multiple types of grounding, and these are in some metaphysically interesting sense on equal footing—may read this characterization, and others to follow, as schematic for whichever grounding relations they countenance. Thus, those who believe that some but not all types of grounds necessitate count as contingentists for the purposes of this chapter. I provide some examples later; see Richardson [Chapter 13] in this volume for further discussion of grounding pluralism.)

Consider next the **internalism versus externalism dispute**. Say that a relation a plurality of facts, Γ , bear to a fact $[p]$ is **internal** to Γ and $[p]$ iff the following conditional necessarily holds: if the facts in Γ and $[p]$ all hold, then Γ bears that relation to $[p]$. (Thus it is internal to the facts [it is cold] and [it is either rainy or cold] that the first is a disjunct of the second, while it is not internal to them that they appear as examples in this chapter. I should stress that this definition of ‘internal relation’ is stipulative, meant only to apply to relations between facts, and not to be confused with other definitions of ‘internal relation’ in the literature, e.g., those given by Armstrong (1989: 43) and Lewis (1986: 62).) Then one may ask: is it necessarily the case that if Γ grounds $[p]$, then the grounding relation between Γ and $[p]$ is also internal to them? In other words, is **internalism** true? Or is it false, in which case **externalism** is true?

Both disputes have seen a flurry of attention recently, for two reasons. First, they interact with numerous important **first-order issues regarding grounding**; these concern specific claims about which types of facts ground (or do not ground) which and appear across the philosophical

landscape. Second, they interact with numerous important **second-order issues regarding grounding**; these concern specific claims about the nature, varieties, logical regimentation, semantics, epistemology, and scope of grounding itself and have advanced to the forefront of contemporary metaphysics (cf. Skiles (2015: 748–9) for a partial list of both sorts of issues). The goal of the present chapter is to survey where the two disputes currently stand and highlight some ways in which they interact with first-order and second-order issues about grounding.

To make this goal manageable, though, I must narrow the chapter's scope in a few respects.

First, to frame the disputes as crisply as possible and reflect the terms in which they are usually conducted, I made several nontrivial assumptions about both grounding and necessity. Rather than catalogue them all here, I will bring them into focus if and when they bear on the two disputes. That said, it is worth mentioning at the onset that by “necessity”, I mean *metaphysical* necessity rather than other varieties the notion might come in (see Kment (2017) for an overview of the varieties of necessity). Some necessitarians believe that there are multiple varieties of grounding, each entailing a distinctive variety of necessity: metaphysical grounding entails corresponding *metaphysical* necessities, natural grounding *natural* necessities, normative grounding *normative* necessities, and (perhaps) so on (see e.g., (Berker 2018: 746–60; Fine 2012: 37–40; Litland 2018; Richardson [Chapter 13] this volume; Rosen 2017: 137–8) for discussion of views in the vicinity). Yet since this is a handbook on *metaphysical* grounding, I will focus on how metaphysical grounding relates to metaphysical necessity.

Second, another dispute regarding how grounding relates to necessity is the perennial first-order dispute regarding what, if anything, grounds facts about which facts hold with necessity. In addition to the voluminous literature on this issue in discussions of systematic accounts of modality (see van Cleve (2018) for a recent overview), a related issue is whether there is any in-principle obstacle to a necessarily obtaining fact being grounded in contingently obtaining ones (see e.g., Blackburn (1993), Cameron (2010), Hale (2002), Hanks (2008), Lange (2008), Morato (2014), Schnieder (2011: 457–8), and Wildman (2017, 2019)). Although I mention a few ways in which these matters interact with the two disputes at issue in this chapter, a comprehensive survey goes beyond what I can carry out here.

Third, one might expect a chapter on the relationship between grounding and necessity to also discuss the relationship between grounding and **supervenience**, where, roughly, $[p]$ supervenes on the facts in Γ iff $[p]$ cannot vary with respect to whether it holds unless one or more facts in Γ vary with respect to whether they hold. Although I briefly discuss certain issues about the relationship between grounding and supervenience that bear on the two disputes, otherwise I will not investigate them here, as they are discussed in detail by Kovacs (Chapter 24, this volume). I will also set aside other modal relationships in the vicinity of necessitation one might reasonably ask how grounding relates to—for instance, if Γ grounds $[p]$, whether it follows that $[p]$ **counterfactually depends** on Γ (i.e., if at least some of the facts in Γ had *not* held, then neither would have $[p]$). Bennett (2017: 47–8) presents three putative counterexamples, one that directly targets this principle (cf. Krämer and Roski (2017: 1199–200) for discussion of others), as well as two others that indirectly undermine it given the widely held belief that in cases of mental-physical realization and material composition, corresponding grounding facts hold. For accounts of grounding that link it to other, more sophisticated notions of counterfactual dependence, see, e.g., Krämer and Roski (2017), Schaffer (2016), and A. Wilson (2018a, 2018b).

Caveats and clarifications aside, here is the plan. In §1, I discuss the main arguments in the literature for necessitarianism and consider responses to them. In §2, contingentism is subjected to the same treatment. In §3, I survey the internalism versus externalism dispute. (Since the necessitarianism vs. contingentism dispute has been explored in more depth than the internalism

vs. externalism dispute, and because many of the maneuvers available within the first dispute have analogues in the second, I will focus more attention on the former.)

I. Necessitarianism

There is what seems to be a general consensus that not every instance of necessitation is an instance of grounding. The counterexamples are many and well known and can be found as far back as Aristotle (*Categories* 14b: 15–22) and Bolzano (see e.g., 1837/2014: §162), so I shall not rehearse them here. In contrast, most participants in both the first-order and second-order literature on grounding believe the converse and thus endorse necessitarianism. Necessitarianism is no doubt rightly characterized as “the default view among proponents of grounding”, as Bliss and Trogdon (2016: §5) do. In the contemporary literature, Audi (2012a, 2012b), S. Barker (2012), J. Barker (2019), Bennett (2011), Bernstein (2016), Bhogal (2017), Cameron (2018), Chilovi and Pavlakos (2019), Correia (2005, 2018), Dasgupta (2014a, 2014b, 2016), deRosset (2010, 2013a, 2013b), Dixon (2018), Fine (2012), Glazier (2017), Guigon (2018), Horvath (2018), Kroedel and Schulz (2016), Litland (2017), Loss (2016, 2017), Lovett (2019), McDaniel (2019), O’Conaill (2017), Peterson (2016), Rosen (2010), Roski (2017), Russell (2016), Saenz (2015), Trogdon (2013), Wang (2016), Wilsch (2015), and Witmer et al. (2005) are among those that either endorse, argue for, or defend necessitarianism outright, commit to principles that clearly entail it, or else take necessitarianism on board as a *prima facie* plausible although perhaps defeasible constraint on theorizing about grounding. Others, e.g., Kovacs (2019) and Norton and Miller (2019), do so for the notion of *metaphysical explanation* rather than (explicitly) that of grounding, although the relationship between the two is a complicated matter; see Glazier (Chapter 8 this volume) for further discussion. I lack the expertise to provide even an incomplete list from the historical literature, but I suspect one would find many examples here too (e.g., Stang (2019), fn. 22 provides textual evidence that Kant was a necessitarian).

But why should one believe necessitarianism?

One common thought is that necessitarianism is in some way encoded into the very idioms used to communicate what grounding is in the first place. Grounding is commonly taken to be—as a rough first gloss—a type of *determination* relation, at least in the sense that a fact is *made the case by*, and *holds in virtue of*, its grounds. But if that is so, how could a fact fail to be necessitated by its grounds? As Shamik Dasgupta puts the point, one would think that what grounds the fact that a conference occurs are certain activities of its participants; yet “if it were possible” for the latter facts to hold without the former, then “presumably their acting like that was not the full story of what *makes* the event count as a conference after all” (2016: 393 (his emphasis)). In a similar vein, Tobias Wilsch claims that “the key motivation” for necessitarianism is “conceptual”, insofar as “[t]he concept of ground and necessitation overlap on the intuitive notion of ‘metaphysical sufficiency’: grounds are metaphysically sufficient conditions for what they ground and metaphysical sufficiency in turn suffices for necessitation” (2015: 3295).

Variations on this theme recur throughout the second-order literature on grounding (see e.g. Fine 2012: 37–40) and neighboring notions such as *truthmaking*, as well as in the first-order literature both in and outside metaphysics. For examples from the truthmaking literature, see Armstrong (1997: 116), Cameron (2018: 340), and Correia (2005: 61); see Trogdon (Chapter 28 this volume) for a general overview of the relationship between grounding and truthmaking. Moreover, the thought has also been used to support what in §2 I call *circumstantialist* versions of contingentism (Bennett 2017: 50).

Yet this line of thought seems inconclusive for at least two reasons.

First, the modal import of these idioms used to pretheoretically describe what grounding is are at least as opaque as the modal import of grounding itself. Take Wilsch's claim that grounds are 'metaphysically sufficient conditions' for what they ground. It is clear what it is for Γ to be a plain old sufficient condition for $[p]$ (i.e., if the facts in Γ hold the $[p]$ does too) and what it for this to be a metaphysical necessity. But if *that* is how 'metaphysically sufficient condition' is to be understood, Wilsch's claim would provide no *motivation* for necessitarianism; it would be the assertion that the view is true, and this is precisely the issue. But if it is *not* understood in this manner, why believe that metaphysically sufficient conditions necessitate? It is difficult at best to see what motivation there could be for necessitarianism about *metaphysical sufficiency* that could not serve equally as a direct motivation for necessitarianism about *grounding* itself, thus obviating the need to appeal to a conceptual link between them. Similar comments can—and have been—made about 'determination', 'making the case', 'in virtue of', and other (allegedly) necessitation-implying idioms commonly used to pretheoretically characterize grounding (see Leuenberger (2014: 152–3), Skiles (2015: 736–7), Trogon (2013: 467), and Wasserman (2017: 74–5) for variations on this criticism and others).

Second, it is not uncommon in metaphysics to learn that our concept of some phenomenon must be revised somewhat due to countervailing intuitive and/or theoretical pressure (cf. Plunkett and Cappelen forthcoming for a general overview of the topic of 'conceptual engineering'). It may well be that part of our concept of being a person is that they are three-dimensional objects, for instance, or part of our concept of acting freely that it requires the possibility that one could have done otherwise. But (as is well known) there is serious discussion in metaphysics over whether these alleged conceptual truths are nonnegotiable. Similarly, even if our *pretheoretic* concept of grounding entails that it is a necessitation relation, grounding *itself* may not perfectly comport with that concept, whereas a *more refined* concept sensitive to broader theorizing about grounding may reflect what it is like more accurately (cf. Bliss (2018: 74), who raises a similar point against those who appeal to what is allegedly "true by definition" in order to argue that grounding has certain structural features).

Both comments extend to attempts to motivate necessitarianism that introduce grounding by saying that a fact is 'nothing over and above', 'no addition to reality' beyond, 'constitutively' explained by, or 'consists in' its grounds, in ways that matter when evaluating the overall economy of a theory that countenances such a grounding relationship (cf. Fine (2012: 6–7)). What is more, such attempts face a further problem: 'nothing over and above' and related idioms are legendarily slippery. If no fact can even partially ground itself—as is widely (although not universally) presumed—then no fact can literally *be* its grounds, individually or taken as a collective.¹ But how could a fact *not* be 'an addition to reality' over and above its grounds if it is distinct from each and from all of them? For this reason, many in the grounding literature reject that grounding implies nothing-over-and-aboveness. One example is Audi (2012b: 708–9). In contrast, Berker (2018: fn. 1) argues that grounding does not imply nothing-over-and-aboveness by appealing to alleged differences in their structural features, while Melnyk (2016) criticizes various ways of motivating this implication in the context of formulating physicalism.

Those who do claim that grounding implies nothing-over-and-aboveness face the challenge of explicating nothing-over-and-aboveness in a manner consistent with the irreflexivity of grounding—or, if not that, at least accounting for how grounding can accomplish the theoretical tasks that nothing-over-and-aboveness is supposed to accomplish. However, a number of the extant attempts in this direction are either silent about the modal import of grounding (e.g., deRosset (2017)), are meant to characterize merely some varieties of grounding (e.g., Goff (2017: 42–8)), or are even specifically designed to allow for grounding without necessitation (e.g., Skiles (2015: 740–1)).

Rather than appealing to how we pretheoretically describe what grounding is, another way to argue for necessitarianism is to appeal to a specific *theory* of grounding (here I use ‘theory’ in a liberal way that includes nonreductive approaches.) To illustrate how such a theory might lead one to necessitarianism, suppose a fact that consists in the instantiation of a determinable property, e.g., [the ball is red], is grounded in a fact that consists in the instantiation of one of its determinates, e.g., [the ball is scarlet].

- According to **nomological** theories, every instance of grounding is subsumed by a general ‘law of metaphysics’ that, in this case, presumably has as its content the following universal fact: for *any* *x whatsoever*, if [*x* is scarlet] holds, then so does [*x* is red]. If—as is widely presumed—such a law of metaphysics holds with necessity if it holds at all, and if—as is also widely presumed—that a law of metaphysics necessitates the universal fact it embeds, then it follows that [the ball is scarlet] necessitates [the ball is red]. Proponents of nomological theories who extend this reasoning to all instances of grounding must also accept necessitarianism. A variation on this reasoning can be found in Wilsch (2015: 3299). For more on the notion of metaphysical lawhood and its potential connections to grounding, see Wilsch (Chapter 30 this volume).
- According to **essentialist** theories, every instance of grounding is accompanied by a fact about ‘what it is’ for the derivative fact and/or its constituents to be the very things that they are ‘by their very nature’, understood in a way that is more fine-grained than what can be captured with standard modal notions (Fine (1994, 1995) are highly influential discussions of the relevant notion of essence). The content of this fact about what is essential to what may be the same universal fact as considered earlier, or it could simply be an *instance* of it—one might simply hold, for instance, that it is part of the essence of [the ball is red] that if an obtaining fact satisfies a certain condition—one that [the ball is scarlet] in fact satisfies—then [the ball is red] holds as well. Proponents of essentialist theories who hold that such an essentialist fact hold with necessity, that it necessitates the fact it embeds, and that all instances of grounding fit this mold thus must also accept necessitarianism. Variations on this reasoning can be found in Audi (2012a: 697) and Trogon (2013). And although they do not explicitly endorse this reasoning, Correia (2013) and Rosen (2010) endorse principles that clearly commit them to the premises underlying it. Lovett (2019: 4), using an account of grounding in terms of ‘generalized identity’ due to Correia and Skiles (2019), presents an argument for necessitarianism that counts as an essence-based one if Correia and Skiles are correct that every fact about generalized identity is a fact about essence (*ibid.*: 648–54). For more on the notion of essence and its potential connections to grounding, see Zylstra (Chapter 22 this volume).
- According to **explanationist** theories, every instance of grounding is accompanied by a fact about how the derivative fact is ‘metaphysically explicable’ in terms of its grounds and thus holds ‘because’ of them in some distinctively noncausal sense. Concrete proposals differ in their details, although a seemingly plausible constraint for any explanation to be good is that it not be subject to what deRosset (2010: 79) calls “confounding cases”, which include possible situations in which the explanans holds but not the explanandum.² Proponents of explanationist theories who place such a constraint on (the relevant type of) metaphysical explanation in particular are thus committed to necessitarianism: to say that the metaphysical explanation of [the ball is red] solely in terms of [the ball is scarlet] has no confounding cases is just to say that it is impossible for the latter fact to hold without the former. A variation on this reasoning can be found in deRosset (2010), although it is not entirely clear to me whether the necessitarian arguments based on ‘determination’, ‘makes it the case’,

etc. intuitions discussed earlier are, at their core, really intuitions about the modal import of the metaphysical explanations often communicated with these idioms, in which cases the authors mentioned should be included here as well. For more on the notion of metaphysical explanation and its potential connections to grounding, see Glazier (Chapter 8 this volume).

- According to **necessitation** theories, the claim that grounds necessitate is built in right from the start. For instance, Guigon (2018) develops a sophisticated counterpart-theoretic semantics for grounding statements with this feature, such that “[the ball is scarlet] necessitates [the ball is red]” always states a truth in contexts of utterance in which their grounding relationship is salient due to how the contexts select counterpart relations between facts. For another instance, Miller and Norton (2017) offer an antirealist approach to grounding, one that takes grounding-relevant observations—or perhaps even grounding itself (Norton and Miller 2019: fn. 4)—to be the product of how our evolved cognitive mechanisms happen to processes certain apparent nonsymmetric instances of necessitation, such as those holding between [the ball is scarlet] and [the ball is red]. For another account that analyzes grounding partially in terms of necessitation, see Correia (2018).

If, in the final analysis, a theory that implies necessitarianism performs better than its contingentist or contingentist-neutral rivals, then that fact would seem to provide at least some defeasible reason to accept the view. Of course, whether this is indeed the case remains hotly contested. Be that as it may, the contingentist may also question whether these theories *do*, in fact, entail necessitarianism. For instance, here are two obstacles for the entailment from nomological and essentialist accounts. Related obstacles can be raised for the entailment from explanationist and necessitation accounts, as well as others that target specific versions of one or more of these styles of argument, but I lack the space to explore them here. (For instance, (Richardson forthcoming; Skiles 2015: 741–5; Skow 2016: 110–11; Wasserman 2017: 71 all challenge explanationist arguments for necessitarianism by challenging deRosset’s claim that good explanations lack confounding cases.)

Consider putative cases in which an accidental generalization of the form $[\forall x(Fx \rightarrow Gx)]$ is grounded in yet not necessitated by its instances $[Ga], [Gb], \dots$, where $a, b, \dots$ are all the F s there are (cf. Skiles 2015; see McSweeney, Chapter 32 this volume, for further discussion of this case and the grounding of logically complex facts more generally). According to nomological accounts, this instance of grounding must be subsumed by a law of metaphysics. Now, laws of *nature* can display a quite complicated logical form: they can have *ceteris paribus* clauses, or govern the chances of certain types of events occurring, or have only a limited domain of application, and so on. Similarly, it is hard to see why being a *law* would prevent a law of *metaphysics* from having something like the following conditional form:

(LC) It is a law of metaphysics that: for any $x_1, x_2, \dots$ that are all the F s there are, if $[Gx_1], [Gx_2], \dots$ hold, then $[\forall x(Fx \rightarrow Gx)]$ holds as well.

In contrast, essentialist accounts instead subsume instances of grounding under truths about the essence of the derivative fact in question and/or its constituents. But here too, it is unclear why an essentialist truth could not have the logical form of something like (EC):

(EC) It is part of the essence of $[\forall x(Fx \rightarrow Gx)]$ that: for any $x_1, x_2, \dots$, that are all the F s there are, if the facts $[Gx_1], [Gx_2], \dots$ hold, then $[\forall x(Fx \rightarrow Gx)]$ holds.

Even if (LC) and (EC) hold necessarily, neither of them entails that $[Ga]$, $[Gb]$, . . . necessitate $[\forall x(Fx \rightarrow Gx)]$, since neither entails that $a, b, \dots$ are in fact all the F s there are, much less that there is no additional F that is not a G . Yet still, even if the nomological or essentialist account is true, $[Ga]$, $[Gb]$, . . . would still meet the conditions for grounding $[\forall x(Fx \rightarrow Gx)]$, so long as they are all the F s, and even though they do not necessitate it. But it is hard to see why (LC) could not be a candidate law of metaphysics, or (EC) an essentialist truth, that is without some independent reason to believe necessitarianism. Therefore, one major obstacle to using these accounts to argue for necessitarianism is to explain why they are not, but without appealing to necessitarianism.

To be clear, I am *not* arguing that there are *no* nomological or essentialist accounts that entail necessitarianism. For instance, the nomological account given by Wilsch (2015), which takes a fact and its grounds to correspond to the conclusion and premises of a deductively valid argument—entails necessitarianism, given that deductive entailment presumably holds (at least) with metaphysical necessity. Rather, I am pointing out that accepting one or the other *need not* entail necessitarianism, as contingentist alternatives exist that do not seem obviously problematic. Similarly, see Zylstra (2018) for an essentialist account of grounding that, he argues, does not entail necessitarianism.

A second obstacle is to explain why one should believe that the laws of metaphysics or essentialist truths—whatever form they take—are necessarily true. Some have argued that there is little good reason to believe this. This line has been pushed specifically with respect to metaphysical lawhood by Balaguer (2018: 3989) and with respect to essence by Leech (2018); for broader defense of the contingency of ‘metaphysical principles’ of these and related sorts, see e.g., Cameron (2007), Miller (2009, forthcoming), and Williams (2006).

Moreover, one might argue against their being necessarily true by appealing to certain, although admittedly controversial, claims about the metaphysics of facts. As an illustration, suppose that (LC⁻) and (EC⁻) actually hold, which have the same form as (LC) and (EC) without the underlined restrictions. Suppose further that a fact of the form $[\forall x(Fx \rightarrow Gx)]$ exists only if it obtains. And suppose further still that an essentialist truth or law of metaphysics can possibly be false if something it has as a constituent can fail to exist. Under these suppositions, (LC⁻) and (EC⁻) are actually true yet possibly false: namely, when $[\forall x(Fx \rightarrow Gx)]$ fails to hold, as both (LC⁻) and (EC⁻) include it as a constituent. Yet a situation in which (LC) and (EC) could have been false is one in which $[Ga]$, $[Gb]$, . . . hold when a, b , are not all of the F s, and so one in which $[\forall x(Fx \rightarrow Gx)]$ does not obtain, and thus one in which $[Ga]$, $[Gb]$, . . . do not ground it. A separate argument, then, is needed to show that this is impossible, one that does not appeal to the fact that (LC⁻) and (EC⁻) actually hold.

Rounding out this (brief, open-ended) survey of the case for necessitarianism, certain other arguments instead turn on certain ground-theoretic matters that do not fit neatly under any of the previous headings. Moreover, they concern principles that are in a sense independent of the nature of grounding itself. For instance, Roberto Loss has recently given two arguments that might be thought to put some pressure on the contingentist. One is that contingentism cannot be consistently combined with the view that grounding is transitive and noncontrastive, the rejection of priority monism (here, the thesis that the only fundamental fact about the existence of a concrete thing is one about the existence of the mereological fusion of every concrete thing), and a ban on fundamental general and negative facts (Loss 2017). The other is that when necessitarianism is combined with a plausible principle about the modal recombining of contingent fundamental facts, one can prove a partial answer to the question of what if anything grounds facts about the grounding facts themselves (Loss 2016).

II. Contingentism

Although still a minority position, interest in and endorsement of contingentism about grounding has increased in recent years. Sympathetic treatments of contingentism or outright endorsements that assume there to be one type of grounding can be found in Bricker (2006), Chudnoff (2013), Dancy (2004), Emery (2019), Leuenberger (2014), Schaffer (2010), Schnieder (2006), Skiles (2015), Wasserman (2017), and Zangwill (2008), while authors sympathetic to grounding pluralism that take a contingentist stance toward some types of grounding include Bennett (2017: 54–5), Cohen (2018), Rettler (2017: 634), and Richardson (forthcoming). For closely related discussion that is sympathetic to contingentism about truthmaking, see, e.g., Barceló Aspeitia (2015), Briggs (2012), Heil (2003), Mellor (2003), Parsons (1999), and Stenwall (2016).

But why should one believe contingentism?

One motivation for contingentism is a *general* desire to avoid necessary connections of the sort that necessitarianism countenances. For instance, Naomi Thompson gives voice to this type of worry when she writes that

primitive grounding relations are ‘spooky’ in much the same way as primitive causal relations are often considered spooky. For those who take grounds to necessitate what they ground, the analogy is particularly strong; grounding relations are *necessary connections* in nature, and to the extent that we are suspicious of such connections, we should be suspicious of the grounding relation so understood.

(Thompson 2018: 32–3)

This suspicion may come from any of a number of quarters, including a general skepticism about *de re* modality (e.g., Quine 1953) or the thought that “the Humean prohibition of necessary connections . . . gives us our best handle on the question of what possibilities there are”, as Lewis (2001: 611) puts it when discussing similar suspicions with truthmaking. For an expression of the ‘brute necessity’ worry in the context of arguing for a conventionalist view of derivative ontology, see Heller (2008: especially 85–6).

There are numerous replies at the necessitarian’s disposal, however.

First, the motivations behind skepticism about necessary connections of the relevant sort have been subject to withering criticism in recent decades; for instance, see Whittle (2009) and Wilson (2010), and see van Cleve (2018) for an overview of issues surrounding the possibility of brute necessities.

Second, there is strong intuitive pressure to allow that at least *some* grounds necessitate. For instance, how could [the ball is scarlet] hold without [the ball is red] holding as well? A form of contingentism forced to reject such innocuous instances of grounding seems too extreme a form of contingentism to accept.

Third, some substantive accounts of grounding attempt to explain why grounds necessitate or at least show why it is no mystery that grounding has the modal import it does. Authors who take on this project explicitly include Correia (2018), Guigon (2018), and Wilsch (2015).

And fourth, in presentations of the Humean prohibition of necessary connections, nearly always a dispensation is granted to those holding between ‘wholly distinct entities’ having no constituents in common. Yet given this dispensation, it is at best unclear whether grounding facts violate the Humean prohibition, properly understood. For instance, if one holds a ‘Russellian’ conception of the facts—one according to which they have objects, properties, and other entities as constituents—then overlap is arguably the rule rather than the exception: [the ball is scarlet] and [the ball is red] share the ball as a constituent. Authors who endorse such a

conception of facts include Rosen (2010) and Audi (2012a, 2012b). For further discussion of grounding and the ‘wholly distinct entities’ restriction, see Ismael and Schaffer (2016: 8–9) and Schaffer (2016: 76).

Another way to motivate contingentism is to instead develop counterexamples to necessitarianism. The following is a partial list of alleged (although not always endorsed) cases of grounding without necessitation, listed by the derivative facts in question (first in order) and their alleged grounds (second in order):

- General facts of various sorts, such as the fact that every gold sphere on Earth’s surface is less than a kilometer in diameter; and the particular facts that are their instances (e.g., Bricker 2006; Barceló Aspeitia 2015; Skiles 2015; B. McDaniel 2017)
- Normative facts, such as a person’s having a moral reason for or against performing an action; and the nonnormative features of the relevant situation (e.g., Dancy 2004: ch. 3; Zangwill 2008)
- A proposition’s being known; and the fact that it is truly, justifiably believed (e.g., Chudnoff 2013; Zangwill 2018)
- Where lies the sharp cutoff in a sorites series for a vague expression, such as “is bald”; and more fundamental facts, such as the exact number of hairs the members of the series have (e.g., Wasserman 2017)
- The existence and features of a composite of material objects; and facts solely about the objects that compose it (e.g., deRosset 2013b; Skiles 2015)
- Facts about the social world, such as the fact that a certain piece of paper is a United States dollar; and facts about the behavior, intentions, knowledge, etc. of relevant communities (e.g., Mikkola 2015: 791)
- Facts about phenomenal consciousness, such as the fact that one is having an experience as of a red surface; and underlying neurophysiological facts (e.g., Leuenberger 2014: 157–61)
- Facts about the dispositions of a thing, such as its fragility; and facts about its microstructure, perhaps along with the relevant laws of nature (e.g., Leuenberger 2014: 161–5)
- Facts about a sequence of indeterministically linked events, or the outcomes of indeterministic processes; and the indeterministic laws that govern them, perhaps together with the inputs of the relevant indeterministic processes (e.g., Taylor 2018: 212–5; Emery 2019)
- Certain facts that hold with necessity (or that hold permanently); and certain other facts that hold only contingently (or that hold only temporarily; e.g., Amijee forthcoming)

In principle, the contingentist might argue from these specific cases in one of two ways. One way is to argue *directly* that this or that case is an instance of grounding without necessitation. Another way is to argue *indirectly*. For instance, one might argue that although no single counterexample is thoroughly decisive—there are, after all, a number of ways for the necessitarian to dig in their heels with respect to any one of them—together they have evidential force, as contingentism may (one might argue) offer a simple, unified explanation for why threats to necessitarianism reappear across a broad, diverse range of philosophical subject matter, which the necessitarian must respond to in a more piecemeal fashion.

Broadly speaking, there are two broad ways for the necessitarian to respond. First, one way to respond is to argue that Γ is *not even a partial ground* of $[p]$. Call these **displacement strategies**. For instance, those who deny standard versions of physicalism about phenomenal consciousness—either because they take facts about it to be either *ungrounded* or grounded solely in *nonphysical* facts of some sort, or *naturally* grounded in the physical facts in some way that is mediated by physicalistically unacceptable bridge laws—would of course reject

Leuenberger's claim that phenomenal facts are grounded but not necessitated by physical facts (as he is well aware—see 2014: 158, 161). Another instance of this strategy is to argue that although Γ does not ground $[p]$ alone, Γ 'irreducibly plurally' grounds (and necessitates) a collection of facts that include $[p]$ and argue that this is good enough for demoting $[p]$'s fundamentality status (cf. Dasgupta 2014a). Although they may prove useful for dealing with some of the putative counterexamples to necessitarianism, displacement strategies appear to be ill-equipped to deal with others. It seems dubious, for instance, to reply to Chudnoff and Zangwill's counterexamples by claiming that particular facts about knowledge are not even partially grounded in facts about true justified belief, or only grounded in them en masse.

A second way to respond to a putative counterexample is to instead argue that Γ is a *merely partial* ground of $[p]$. Call these **supplemental strategies** and the envisaged facts required to bridge the link between grounding and necessitation **grounding supplements**. For instance, in reply to Chudnoff and Zangwill's counterexamples, one might claim that the envisaged supplements are facts specifying that certain 'anti-Gettier' conditions are satisfied, such that in these circumstances, true justified belief necessitates knowledge. However, contingentists have replied to supplemental strategies by arguing that the proposed grounding supplements run contrary to certain antecedently plausible principles about which facts are capable of grounding which. For instance, Chudnoff anticipates the supplemental strategy envisaged above by arguing that (i) partial grounds of a fact must play a role in explaining why it holds and (ii) as such, they must be "natural enough" in the sense of Lewis (1983) and Sider (2011) to explain the relevant facts about knowledge; and yet (iii) such anti-Gettier conditions are notoriously disjunctive and gerrymandered and thus unsuited to this task.

Setting aside their dispute with necessitarians, there is a further internal dispute among contingentists between those who endorse **circumstantialism** and those who do not (I borrow this terminology from Bennett (2017: 53)). According to circumstantialists, if Γ grounds $[p]$, then there is some nontrivially specifiable collection of facts, C , which I will call **circumstantial supplements**, such that the facts in Γ and C together necessitate $[p]$. Thus, for instance, the circumstantialist may hold that even if Chudnoff is correct that anti-Gettier conditions cannot serve as a *grounding* supplement in a case of knowledge, if taken as *circumstantial* supplements, they need not satisfy the more stringent conditions that partial grounds do and (for all that has been said) can be specified in some sufficiently illuminating way. Although circumstantialism has emerged in the recent literature as a way to *deflate* the necessitarianism versus contingentism dispute, at least in the form in which it is typically conducted, contingentists might find circumstantialism independently attractive. Sympathetic discussions of circumstantialism can be found in, e.g., Bennett (2017: 49–55), Donaldson (2017: 789–90), and Cohen (2018).

However, circumstantialists face three main challenges.

First, if Γ grounds $[p]$, then there are of course many *trivial* circumstantial supplements: $[p]$ itself, the fact that Γ grounds $[p]$ (supposing that grounding is factive), the fact that the proposition $\langle [p] \text{ holds} \rangle$ is true, and so on. Supplementing Γ with a collection including any of these facts would obviously guarantee that $[p]$ holds too; but if that is all that circumstantialism requires, the view would be uninteresting. So, then, are there any *nontrivial* conditions on circumstantial supplements that exclude such facts? It is not clear that this is so. One approach is to require that circumstantial supplements must also at least partially ground not $[p]$, but rather the fact $[\Gamma \text{ grounds } [p]]$ —see Skow (2016: 110–11) for relevant discussion. Another approach is to hold that circumstantial supplements must, following Epstein (2015: 69–87), partially explain $[p]$ by *anchoring* it, which he argues is a form of metaphysical explanation but not an instance of grounding. Still another approach is to take the circumstantial supplements to serve as 'background conditions' for, or to 'enable', Γ 's grounding of $[p]$ in some other way, but the success

of this strategy would of course require much more detail about what these notions amount to. And yet another approach still is suggested by Bennett, who adds the condition that if Γ grounds $[p]$ in some yet-to-be-specified circumstances, C , then C includes neither $[p]$ nor anything that fully grounds $[p]$ (2017: 52). However, further conditions would appear to be needed to avoid trivialization. Adding a fact like $[p \vee (q \wedge \neg q)]$ to C , for instance, seems to trivialize circumstantialism, yet most grounding theorists would deny that $[p \vee (q \wedge \neg q)]$ is either identical to or fully grounds $[p]$. One can exclude this fact by adding to Bennett's proposal that C also not include any fact that requires $[p]$ as an at least partial ground, but whether this modification is enough to handle other potential threats is an open question.

A second challenge is that several of the putative cases of grounding without necessitation mentioned before appear to be counterexamples to circumstantialism too. For instance, Wasserman (2017) argues for contingentism by arguing that there are possible scenarios exactly similar with respect to the number and distribution of hairs on heads, *as well as every other fact*, save for which people are bald near the 'borderline' of this property's extension. Nonetheless, Wasserman argues, facts about who is and is not bald are grounded in the number and distribution of hairs on heads. Yet if this is a counterexample to contingentism, it is also a counterexample to circumstantialism, as there is no difference between these scenarios—and thus no difference in the circumstances that hold in these scenarios—other than the differences in the relevant facts about baldness.

Third and finally—to wrap up our brief tour of contingentism—the circumstantialist must also face the question of whether the very distinction between grounds versus circumstances is in some sense an 'objective' one, as opposed to a distinction between, e.g., which types of facts are taken as salient for certain explanatory purposes, and if it is not, whether that impugns the metaphysical significance of theorizing about grounding. One author who worries about the objectivity of this distinction is Trogdon (2013: 478), and two others who address versions of this worry as they pertain to the grounding of reasons include Bader (2016) and Zangwill (2008: 114–17).

III. Internalism Versus Externalism

Compared to that between necessitarians and contingentists, the dispute between internalists and externalists has received far less explicit attention. Even so, roughly contemporaneous authors who have commented on it have in some cases asserted the widespread acceptance of internalism (e.g., Bernstein 2016; Litland 2015) *and* in some cases denied it (e.g., Bhogal 2017). Be that as it may, it can certainly be said that many in the current literature *do* in fact endorse internalism or else endorse principles that appear to commit them to internalism. Authors clearly sympathetic to internalism—in the sense of 'internal' at issue in this chapter—include Audi (2012b), Bennett (2011), Correia (2005: 61), and Rosen (2010), while some describe grounding as internal but in a sense closer to how Armstrong and Lewis use 'internal' (e.g., Schaffer 2016: 94–5). Litland (2015: fn. 1) lists a number of authors as committed to internalism yet who do not explicitly endorse it, although presumably he does so due to their background views about the relationship that grounding bears to essence (Dasgupta 2014b, 2016; Trogdon 2013), explanation (deRosset 2010, 2013a), and weak grounding (Fine 2012). Similarly, Lovett (2019) argues that internalism is a commitment of the view developed by Correia and Skiles (2019).

But why should one believe internalism?

It should be noted, to begin with, that necessitarianism does not *entail* internalism—since necessitation is insufficient for grounding, it is logically possible for Γ to necessitate $[p]$ without grounding it—one cannot immediately infer the latter from the former. Nonetheless, certain

lines of argument thought to support necessitarianism might also be thought to support internalism once suitably modified. For instance, a number of authors sympathetic to nomological and essentialist accounts of grounding have employed reasoning such as that canvassed earlier in order to motivate *both* necessitarianism and internalism. See, for example, Audi (2012b), Rosen (2010), and Wilsch (2015). For another instance, one might argue for internalism *by way of* necessitarianism, by arguing that if Γ grounds $[p]$, then Γ also grounds the fact that Γ grounds $[p]$; hence if grounds necessitate, then the facts in Γ (and a fortiori the facts in Γ and $[p]$) cannot all hold unless Γ grounds $[p]$. See, for example, Bennett (2011) and deRosset (2013a), and for more discussion of issues pertaining to such principles of ‘meta-ground’ and their relationship to internalism, see Litland (Chapter 9 this volume).

But as above, so below—such arguments for internalism face similar challenges as those that beset their necessitarianism counterparts. For instance, the externalist might claim that even if grounding facts imply corresponding laws of metaphysics or essentialist truths, these may—for all that has been said—be like (LC) and (EC) in that they only concern the relationships that the facts in Γ bear to $[p]$ *in certain external circumstances*, remaining silent when they do not. For another instance, the externalist might claim that if Γ grounds $[p]$, then it is only Γ *together with the fact that certain obtaining external circumstances hold* that grounds the fact that Γ grounds $[p]$. In both cases, the internalist would then have to explain why these externalist-friendly proposals are independently problematic or argue that the facts in Γ and $[p]$ could not have all held except in those particular circumstances.

Why might one instead believe externalism?

It should be noted to begin with that just as necessitarianism does not entail internalism, neither does contingentism entail externalism—even if Γ fails to necessitate $[p]$, it may still be that Γ must ground $[p]$ whenever the facts in Γ and $[p]$ all obtain—and so one cannot immediately infer the latter from the former. Nonetheless, certain lines of argument thought to support contingentism might also be thought to support externalism, once suitably modified; for an example, see Leuenberger (2014: 165–6). Indeed, there seems to be a general template for converting counterexamples to necessitarianism into counterexamples to internalism, regardless of the details of the case. Here is the idea (cf. Skiles 2012: 118–9). Suppose that $[p]$ grounds but does not necessitate a certain fact $[q]$. Let $[r]$ be any arbitrarily chosen fact that is not modally or ground-theoretically connected to either $[p]$ or $[q]$. Now consider the fact $[(p \wedge q) \vee r]$. Given widely held principles governing the grounding of logically complex facts (cf. Fine 2012), in a possible situation where $[p]$ grounds $[q]$, we have it that $[p]$ alone grounds $[(p \wedge q) \vee r]$. Yet given our starting assumptions, there is also a possible situation in which $[p]$ holds but not $[q]$, yet nonetheless $[r]$ holds, and thus one in which $[p]$ and $[(p \wedge q) \vee r]$ both hold even though the former does not ground the latter.

Finally—to wrap up—there are methods of arguing for externalism that make no appeal to contingentist-supporting considerations and indeed are compatible with its denial. One example is provided by Litland (2015), who presents a sophisticated counterexample to internalism involving contingent self-reference, one that otherwise only assumes certain widely held principles about the logic of grounding (e.g., regarding how disjunctive facts and facts about which sentences are true get grounded). Another example is a putative instance of what A. Wilson (2018b) calls **grounding early preemption**, named after a structurally similar phenomenon discussed in the causation literature (cf. Bernstein 2016: 28). Consider the following five facts:

- $[p_1]$ = the fact that Bad Bill unlawfully physically assaulted someone
- $[p_2]$ = the fact that Bad Bill should be found guilty of assault
- $[q_1]$ = the fact that Bad Bill unlawfully and premeditatedly killed someone

$[q_2]$ = the fact that Bad Bill should be found guilty of murder

$[r]$ = the fact that Bad Bill committed a crime

According to Wilson, the legal circumstances in Bad Bill's jurisdiction could be such that if $[p_1]$ holds but not $[q_1]$, then $[p_1]$ would ground $[r]$ —but only by way of $[p_1]$'s grounding $[p_2]$. Yet the legal circumstances could also be such that if someone should be found guilty of murder, then someone cannot and thus should not be found guilty of assault. So in this jurisdiction, if $[p_1]$ holds but $[q_1]$ holds too, then $[q_1]$'s grounding of $[r]$ by way of $[q_1]$'s grounding $[q_2]$ 'pre-empt's' $[p_1]$'s grounding of $[r]$, as $[q_2]$'s holding is incompatible with $[p_1]$'s grounding $[p_2]$. Yet even if this is a potential counterexample to internalism, it need not be a counterexample to necessitarianism: even though $[p_1]$'s grounding of $[r]$ is preempted, it is still a case in which $[p_1]$ and $[r]$ both obtain, and thus no threat has yet been raised to $[p_1]$'s necessitating $[r]$.³

Related Topics

Glazier, Martin (2019), "Explanation", Chapter 8 this volume.

Kovacs, David (2019), "Modality", Chapter 24 this volume.

Litland, Jon (2019), "Meta-Ground", Chapter 9 this volume.

McSweeney, Michaela (2019), "Logic", Chapter 32 this volume.

Richardson, Kevin (2019), "Varieties", Chapter 13 this volume.

Trogon, Kelly (2019), "Truthmaking", Chapter 28 this volume.

Wilsch, Tobias (2019), "Laws of Metaphysics", Chapter 30 this volume.

Zylstra, Justin (2019), "Essence", Chapter 22 this volume.

Notes

- 1 Note here and elsewhere that I am speaking of what Fine (2012) calls *strict* grounding as opposed to *weak* grounding, the latter which—according to Fine (2012), Correia and Skiles (2019), and Lovett (2019)—is supposed to subsume instances of identity.
- 2 Strictly speaking, in this context, deRosset is endorsing a view stronger than necessitarianism that—for our purposes, as deRosset's own formulation differs somewhat—can be roughly construed as the claim that facts of the same 'form' ought to be grounded in the same sort of way. Somewhat more precisely, let x be any object and let Γ be any collection of facts, and suppose that Γ grounds some fact of the form $[\dots x \dots]$, one which may or may not have x as a constituent. If so, then the following conditional necessarily holds: for any object y , if the facts in Γ all hold, then so does $[\dots y \dots]$, i.e., a fact of the same form as $[\dots x \dots]$ in which y is a constituent (where y may or may not be x). See Amijee (forthcoming), Audi (2012b), Fine (2012), Moran (2018), Rosen (2010), and Wilsch (2015) for further discussion of this and other principles in the conceptual neighborhood. Given space limitations, I leave it for the reader to investigate how these 'formality' principles stand or fall given the sorts of issues that arise in the two disputes at issue in this entry.
- 3 Many thanks to Mike Raven for inviting me to participate in this volume, as well as for his kind patience. Thanks as well to Fatema Amijee, Nathaniel Baron-Schmitt, Louis deRosset, Aaron Griffith, David Kovacs, Kelly Trogon, and Nathan Wildman for helpful discussion on the topics of this chapter. Research for this entry was conducted while funded by grants from the Bank of Sweden Tercentenary Foundation and the Swiss National Science Foundation, and I am grateful for their generous support.

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11

SKEPTICAL DOUBTS

Kathrin Koslicki

1. Grounding Enthusiasm, Grounding Skepticism, and the Central Question

Grounding enthusiasts (also known as “groundhogs”, see Fine 2020) tell us that something is to be gained by recognizing that certain factual connections and/or certain nonfactual connections are all grounding connections. Often, phenomena such as those in (1) are cited as putative examples of *factual* grounding connections, while phenomena such as those in (2) are cited as putative examples of *nonfactual* grounding connections:¹

- (1) a. Moral/Natural: The fact that an act is a telling of a lie grounds the fact that the act is morally wrong.
- b. Truthmaking: The truth of the proposition that snow is white is grounded in the existence of the state of affairs, snow’s being white.
- c. Logical Cases: The fact that the ball is red grounds the fact that the ball is red or round.
- d. Determinate/Determinable: The fact that the ball is crimson grounds the fact that the ball is red.
- e. Genus/Species: The fact that this geometrical figure is a square is grounded in the fact that this geometrical figure is an equilateral rectangle.
- (2) a. Sets/Members: The singleton set containing Socrates is grounded in its sole member, Socrates.
- b. Holes/Hosts: The holes in a piece of Swiss cheese are grounded in the piece of Swiss cheese in which they reside.
- c. Abundant/Sparse Properties: The abundant property, *grueness*, is grounded in some combination of sparse properties.
- d. Boundaries: The boundary around a football field is grounded in the football field it delimits.
- e. Tropes: A rose’s redness trope is grounded in the rose in which it is present.

Although the factual and nonfactual connections cited in (1) and (2) are here expressed using the grounding idiom, this formulation in itself should by no means be regarded as an endorsement of or as straightforward evidence in favor of grounding enthusiasm. These same connections can

also be expressed using other connectives, such as “in virtue of”, “because”, “is explained by”, “is made the case by”, or “depends on”, instead of “is grounded in”. Regardless of the terminology we use to state the connections at issue, the grounding enthusiast’s claim that these (and/or other) examples are in fact best viewed as grounding connections is a substantive claim and one that is contested by those who dispute the theoretical utility of the grounding idiom. (More in what follows on the method of giving examples of alleged grounding connections in support of grounding enthusiasm.)

What exactly are we supposed to learn from the grounding enthusiast’s alleged insight that factual and/or nonfactual connections such as those cited in (1) and (2) are all grounding connections? In what follows, I shall refer to this as “the Central Question”. *Grounding skeptics* (also known as “ground busters”, see Fine 2020), depending on the version of grounding skepticism they endorse, respond to the Central Question as follows: either (i) we learn nothing at all from the grounding enthusiast’s alleged insight; or (ii) what we learn from the grounding enthusiast’s alleged insight can be better stated in terms that do not appeal to the grounding idiom. In either case, so the grounding skeptic reasons, the grounding idiom lacks theoretical utility, and we therefore might as well continue to go about the business of trying to clarify the nature of the factual and/or nonfactual connections at issue without appeal to the grounding idiom.²

2. Varieties of Grounding Skepticism

We can distinguish between different varieties of grounding skepticism. Chronologically speaking, the first wave of grounding skepticism, or what I shall call “old-school” skepticism, primarily calls into question the overall coherence or intelligibility of the grounding idiom. Since a notion can presumably only be theoretically useful if it is intelligible and coherent, this form of skepticism, if successful, would undermine the theoretical utility of the grounding idiom by questioning one of its presuppositions. The next wave of grounding skepticism, or what I shall call “second-generation” skepticism, focuses not so much on the overall coherence or intelligibility of the grounding idiom but instead challenges the theoretical utility of the grounding idiom in other ways. In particular, “second-generation” skeptics raise doubts concerning (i) the alleged unifying power of the grounding idiom; (ii) the alleged power of the grounding idiom to capture and elucidate the distinction between the fundamental and the nonfundamental; and (iii) the alleged metaphysical (as opposed to mind-dependent, epistemic or psychological) utility of grounding claims.³

2.1 “Old-School” Grounding Skepticism

“Old-school” skepticism finds its expression, for example, in Daly 2012 and Hofweber 2009 (see also Clark and Liggins 2012; Kovacs 2017; Rayo 2013, 2014.) According to Chris Daly, grounding theorists can defend the intelligibility of the grounding idiom by opting for one of the following two strategies: they can either take the grounding idiom to be defined in terms whose intelligibility can be taken for granted; or they can take the grounding idiom to be primitive but convey its meaning in other ways. Primitivists like Paul Audi, Gideon Rosen, and Jonathan Schaffer (Audi 2012a, 2012b; Rosen 2010; Schaffer 2009) opt for the latter strategy and try to convey the meaning of the grounding idiom by (a) specifying its logical properties, (b) citing its connections with other terms, and (c) giving examples. Daly argues that the cumulative effect of (a) through (c) nevertheless misses its target. For, first, the logical properties ascribed to the grounding idiom (e.g., factivity, referential transparency, well-foundedness, irreflexivity, asymmetry, transitivity, nonmonotonicity, necessitation, etc.) are not sufficiently fine-grained to identify

the grounding idiom uniquely. Secondly, the other terms with which the grounding idiom might be connected either cannot be taken to be antecedently any more well understood than the grounding idiom itself (e.g., “fundamentality”; “degrees of reality”; etc.); or, if these terms can plausibly be taken to be antecedently more well understood than the grounding idiom (e.g., “reduction”; “explanation”; etc.), then their connection to the grounding idiom is in turn debatable. Thirdly, any putative examples of the grounding idiom, insofar as they are construed as applications of an idiom whose intelligibility is in question, will not be any more comprehensible to the “old-school” skeptic than the idiom they are meant to illustrate. If, however, the examples in question can be construed in other terms whose intelligibility is not in doubt (e.g., supervenience, identity), then they of course no longer serve to illustrate the grounding idiom. Thus, either way, the “old-school” skeptic will fail to be swayed by the primitivist’s attempt to defend the intelligibility of his idiom by citing purported examples of grounding connections. In sum, while Daly’s skeptical arguments do not (and are not meant to) establish that the case for adding the grounding idiom to our language is completely hopeless, he does take himself to have shown that the primitivist’s defense of the intelligibility of the grounding idiom requires more than the considerations cited in (a) through (c). We will see that primitivists have, in their more recent work, heeded Daly’s advice and adduced additional machinery in response to the skeptic’s objections.

Thomas Hofweber is similarly concerned about the intelligibility of the grounding idiom, especially if the grounding idiom is to be accepted as primitive. In Hofweber’s view, the grounding idiom, as it is employed for example in Fine 2001 and Schaffer 2009, is illustrative of a style of metaphysics he calls “esoteric”, as opposed to “egalitarian”, metaphysics. Egalitarian metaphysics, according to Hofweber, conceives of the discipline of metaphysics as aiming to answer questions (e.g., the question of whether change is possible) that are formulated using ordinary terms (e.g., “change”), accessible to nonmetaphysicians as well. Esoteric metaphysics, in contrast, conceives of the discipline of metaphysics as aiming to answer questions (e.g., the question of what grounds what) that are formulated using “distinctively metaphysical terms” (e.g., “ground”), i.e., terms that are either not employed outside of metaphysics at all or at least not in the sense in which they are intended to be used by the esoteric metaphysician (Hofweber 2009: 267). If a “distinctively metaphysical term” is taken as primitive, then there must be some way to convey its meaning to “outsiders”; otherwise, so Hofweber reasons, metaphysics becomes an objectionably exclusive endeavor into which only those who already profess to be converts can gain entry.

Like Daly, Hofweber is also skeptical that primitivists can avail themselves of the method of giving examples to allow “outsiders” to catch on to their “distinctively metaphysical” use of the grounding idiom. When they do so, Hofweber argues, primitivists tend to be guilty of a “bait-and-switch” tactic (Hofweber 2009: 268): the examples they give, insofar as they are intelligible to “outsiders” at all, make use of perfectly ordinary notions of priority instead of illustrating the “distinctively metaphysical” notion of priority that is supposed to be associated with the primitivist’s use of the grounding idiom. To illustrate, in Hofweber’s view, the relation between a true disjunction and its true disjuncts cited earlier in (1.c) is perfectly intelligible to “outsiders”, but it does not illustrate a “distinctively metaphysical” notion of priority; rather, it should be understood in terms of logical entailment (viz., the truth of the disjunct logically entails the truth of the disjunction, while the converse is not the case). Thus, in each case, in which a purported example is supposed to illustrate the “distinctively metaphysical” priority relation associated with the grounding idiom, Hofweber’s strategy is to offer some alternative construal in terms of a different notion of priority he takes to be perfectly accessible to the “outsider” (e.g., causal, conceptual, logical, or counterfactual priority). If no such reconstrual is possible, then Hofweber

would presumably conclude that no sense at all can be made of the purported application of the grounding idiom. At the end of the day, then, primitivists, according to Hofweber, cannot succeed in bringing “outsiders” into the fold by citing alleged instances of their “distinctively metaphysical” grounding idiom, and those who are concerned about the unintelligibility of a “distinctively metaphysical” notion of priority will continue to harbor their “old-school” skeptical doubts, even after having encountered a list of alleged instances of the grounding idiom.

2.2 “Second-Generation” Grounding Skepticism (I): Unity

As described in the previous section, “old-school” skeptics like Chris Daly and Thomas Hofweber challenge the theoretical utility of the grounding idiom by denying one of its presuppositions, viz., that the grounding idiom is at least coherent and intelligible. “Second-generation” skeptics, in contrast, include theorists who object to the grounding idiom for other reasons, even while being willing to grant that the types of connections at play in (1) and (2) cannot be understood in terms “old-school” skeptics might find acceptable (e.g., identity, modality, parthood, supervenience, realization, causation, counterfactual dependence, logical entailment, and the like). In this section, we will consider one such source of skeptical doubts that has recently gained traction among “second-generation” skeptics, viz., skeptical doubts concerning the alleged power of the grounding idiom to *unify* what at first sight appears to be a heterogeneous collection of factual and/or nonfactual connections, such as those cited in (1) and (2).

Grounding enthusiasts like Paul Audi, Gideon Rosen, and Jonathan Schaffer have responded to the Central Question, in part, by citing the alleged unifying power as one of the main theoretical virtues they want to claim for their primitive grounding idiom (Audi 2012b: 689; Rosen 2010: 114; Schaffer 2009: 376–377). But the hypothesis that all alleged grounding connections present us with a unified phenomenon (or what I shall call the “unity hypothesis”) is open to several different interpretations of varying strengths, and grounding enthusiasts therefore would do well to clarify and provide arguments for the particular version of this hypothesis they wish to put forward (Koslicki 2015a: 314ff). In its strongest form, the unity hypothesis holds that all alleged grounding connections exhibit a single *species* of relation or operation. A second, somewhat weaker interpretation posits that distinct specific grounding connections at least form a single *generic* kind. A third, yet weaker, reading requires only that distinct connections which are collected together under the single rubric of grounding at least *objectively resemble* each other in certain theoretically significant ways. M. Cameron 2014 expands the range of options further by suggesting that Aristotle’s notion of *focal meaning* (“*pros hen*” or “core-dependent homonymy”) might provide grounding enthusiasts with an additional intermediate option between the second “single genus” reading and the third “objective resemblance” reading of the unity hypothesis. According to this reading, distinct uses of the grounding idiom are neither straightforwardly equivocal nor straightforwardly univocal; rather, they can all be traced back to a single, yet-to-be-specified central use of the grounding idiom, just as everything that has being, in Aristotle’s view, in some way depends on the primary substances. (See also Rettler 2017 for a defense of the view that, while grounding itself is monistic, the term, “grounding” refers to different relations in different contexts.)

In response to invitations from “second generation” skeptics like Koslicki 2015a and J. Wilson 2014 to disambiguate and defend their invocation of the unity hypothesis, grounding enthusiasts have appealed to at least the following three types of considerations: (i) the apparent heterogeneity of grounding connections should be traced not to differences among the types of connections at issue but rather to differences among the types of entities that figure in these alleged grounding connections (Audi 2012b; Raven 2017); (ii) the apparent preservation of formal

properties across different uses of the grounding idiom indicates that a single generic type of grounding connection is operative in these various contexts (Berker 2017); and (iii) the fact that all alleged grounding connections can be described using a single formalism, viz., that utilized by structural equations models of causation, supports the hypothesis that all grounding connections belong to a single species (A. Wilson 2017) or at least a single genus (Schaffer 2016a, 2016b). (See also Kovacs 2017, who—although himself a grounding skeptic—borrows from the literature on explanation in the philosophy of science and considers various potential unification strategies while ultimately remaining doubtful about their effectiveness.)

2.2.1 *The Source of Heterogeneity*

“Second-generation” skeptics have noted that there are certain obvious apparent differences between cases of alleged grounding connections, such as those cited in (1) and (2) (Koslicki 2015a: 317ff; J. Wilson 2014: 568ff). For example, in the determinate/determinable case illustrated in (1.d), if the ball’s being red (all over) is grounded in the ball’s being crimson (all over), then that same determinable fact cannot also at the same time be grounded in the ball’s being maroon (all over), since a ball cannot simultaneously be both crimson (all over) and maroon (all over). In contrast, in the logical case illustrated in (1.c), for example, over-determination of the grounded fact by its alleged ground is permissible: a disjunctive fact, such as the ball’s being red or round, may obtain because one of its constituent facts obtains (e.g., the ball’s being red), or because its other constituent fact obtains (i.e., the ball’s being round), or because both of its constituent facts obtain (viz., the ball’s being red and the ball’s being round). Some such differentiating feature or combination thereof can be found by means of which each of the cases illustrated in (1) and (2) can be distinguished from all the other cases. At best, then, given the apparent differences between alleged grounding connections, “second-generation” skeptics conclude that the unity hypothesis still stands in further need of justification. At worst, however, if grounding enthusiasts are unable to provide plausible arguments in favor of their preferred reading of the unity hypothesis, then some form of pluralism concerning the types of connections at play in such cases as (1) and (2) seems warranted.

Grounding enthusiasts like Paul Audi and Michael J. Raven have countered that the apparent heterogeneity visible in the phenomena collected together under the rubric of grounding should be blamed on the heterogeneity of the entities that figure in these connections and not on the types of connections themselves (Audi 2012b: 689; Raven 2016: 633–634). By way of analogy, Raven observes that, in the case of logical entailment, for example, we are not inclined to infer from the heterogeneity of cases to which logical entailment can be applied that a multiplicity of logical entailment relations must therefore be stipulated. To illustrate, when we existentially generalize from an instance, the premise of our inference is more specific than the conclusion; in contrast, when we infer a particular instance from a universal generalization, the premise of our inference is less specific than its conclusion. Nevertheless, despite this difference in specificity, we hold that the logical entailment relation which applies in both cases is the same.

At this point in the dialectic, the danger, from the point of view of the “second-generation” skeptic, is that, when we are asked to abstract away from the details of specific applications of the grounding idiom to particular cases, we thereby also in effect lose sight of what is metaphysically most interesting and informative about the nature of the connections at issue. To illustrate, as Koslicki 2015a and J. Wilson 2014 have pointed out, when we are presented for example with a bare factual grounding claim of the form, [p] grounds [q], we are given very little guidance as to how to answer philosophically significant more specific questions concerning the relationship between [p] and [q] or their constituents. The following list of such questions which may or may

not be left open, depending on the particular conception of grounding at issue, is not by any means intended to be exhaustive (see J. Wilson 2014 and Koslicki 2015a for additional examples, the latter of which also includes questions regarding nonfactual connections that are seemingly left unanswered by their paraphrases in terms of the grounding idiom):

- (3) a. Is [p] identical with [q]?
- b. Are [p] or [q] real or unreal? And if so, in what sense?
- c. Is [q] reducible to [p]? And, if so, in what sense?
- d. Is the connection between p and q normatively, nomologically, metaphysically, or logically necessary?
- e. Can a special kind of causal efficacy be ascribed to [q] or its constituents?
- f. Do [p] or some of its constituents functionally realize [q] or some of its constituents?
- g. Do [p] and [q], or some of their constituents, stand in the determinate/determinable relation?
- h. Do [p] and [q], or some of their constituents, stand in the part/whole relation?
- i. Is there a set-theoretic relationship between the constituents of [p] and [q]?
- j. Do the constituents of [p] figure in a real definition of the constituents of [q]?
- ...

Unless or until grounding enthusiasts can deliver answers to these and other specific questions concerning the nature of the connections at issue, “second-generation” skeptics are likely to feel that very little philosophical progress has been made toward illuminating the nature of the connections at issue by subsuming them under the rubric of grounding. We already knew to begin with that some interesting explanatory or dependence relation or other holds in cases to which the grounding idiom is applied; but relabeling these connections as grounding connections does not help us understand any more deeply than we already did at the outset what kind of explanatory or dependence relation is operative in each case. In fact, from the perspective of those who believe that phenomena such as those cited in (1) and (2) illustrate a plurality of distinct dependence relations, subsuming this multiplicity under a single label, if anything, represents a dialectical step backward, since it suggests the presence of a unitary connection where in reality there is none.

2.2.2 The Preservation of Formal Properties

Berker 2017 argues in favor of the generic unity of grounding by directing our attention to the fact that certain formal properties that are commonly ascribed to the grounding idiom (e.g., asymmetry and transitivity) appear to be preserved across different (e.g., normative and metaphysical) contexts. Berker develops several specific cases that are intended to bring out that the transitivity and asymmetry principles for alleged “mixed” normative/metaphysical cases of grounding are just as plausible as the corresponding transitivity and asymmetry principles for “pure” cases that purportedly concern only normative or metaphysical grounding connections. The best explanation for this apparent preservation of formal properties, in Berker’s view, is that a single generic grounding connection is at play in these different contexts.

Although Berker’s argument is most directly addressed to a view he calls “moderate grounding pluralism”, a modified version of essentially the same argument is also intended to target a type of view he labels “extreme grounding pluralism”. “*Moderate grounding pluralists*”, e.g., Fine 2012, are not grounding skeptics; rather, these theorists firmly believe in the theoretical utility of the grounding idiom but accept multiple distinct types of grounding connections,

e.g., metaphysical grounding, normative grounding, and natural grounding (see Richardson 2020). These distinct types of grounding connections, according to the moderate pluralist, are not reducible to or definable in terms of a single basic type of grounding connection. (See also Mikkola 2015 for an argument to the effect that, if the grounding idiom is to apply to social phenomena, then we seem to be pushed toward accepting at least a form of moderate grounding pluralism; we will return to the issue of how or whether the grounding idiom is applicable to the domain of social ontology in Section 2.3.2.) By contrast, “*extreme grounding pluralists*” are grounding skeptics, viz., theorists belonging to the “second generation” of grounding skepticism (e.g., Koslicki 2015a, 2016; J. Wilson 2014, 2016): extreme grounding pluralists do call into question the theoretical utility of the grounding idiom and hold instead that a plurality of irreducible dependence relations is needed to characterize phenomena such as those cited in (1) and (2).

To illustrate, consider a normative context, such as that cited in (1.a), in which a natural fact (e.g., the fact that an act is a telling of a lie) allegedly normatively grounds a moral fact (e.g., the fact that the act in question is morally wrong). Now consider a logical context, such as that cited in (1.c), in which a disjunctive fact (e.g., the fact that the act under consideration is either morally wrong or not morally wrong) is allegedly grounded in one of its constituent facts (viz., the fact that the act in question is morally wrong). Following Fine 2012, such a case is to be viewed as an instance of metaphysical grounding, mediated here by the essence of disjunction. According to Berker’s mixed transitivity principle, we are now permitted to conclude that the natural fact in question (viz., the fact that the act under consideration is a telling of a lie) grounds the disjunctive fact (viz., the fact that the act in question is either morally wrong or not morally wrong) by way of grounding one of its constituent facts (viz., the fact that the act in question is morally wrong). The best explanation for the intuitive plausibility of this inference, and hence for the intuitive plausibility of the mixed transitivity principle which licenses this inference, in Berker’s view, is that the type of grounding connection that is operative in both contexts is a single generic notion of grounding. A similar argument, according to Berker, can be used to establish a corresponding result using a mixed asymmetry principle.

Litland 2018 has replied to Berker’s arguments on behalf of the moderate grounding pluralist. And although a response from the point of view of the extreme grounding pluralist has yet to be developed, these theorists may well want to mirror and adapt for their own purposes some of the moves Litland makes in defense of moderate grounding pluralism. In the case of intuitively plausible inferences that appear to be licensed by “mixed” transitivity principles, for example, pluralists of any stripe are likely to invoke separate explanatory connections for distinct types of contexts (e.g., one type of explanatory connection that accounts for the relation between natural and moral facts and a different type of explanatory connection that accounts for the relation between disjunctive facts and their constituents). In addition, extreme grounding pluralists will want to supplement their account with some sort of justification for why chaining together distinct explanatory connections is, under certain circumstances, permissible and yields a further “non-rigged-up” explanation (Berker 2017: 23–24).

2.2.3 *Positing a Single Formalism*

In defense of the hypothesis that grounding presents us with a unified phenomenon, some grounding enthusiasts have proposed to take seriously the idea that grounding either is or is at least analogous to a kind of causation, viz., *metaphysical* causation (see Wang 2020). (In what follows, I continue to use the terms “grounding” and “causation”, respectively, for what these theorists have in mind when they speak of “metaphysical causation”, on the one hand,

and “nonmetaphysical” or “nomological causation”, on the other hand.) Jonathan Schaffer, for example, defends the weaker view, viz., that grounding is merely analogous to causation (Schaffer 2016a, 2016b), while Alastair Wilson adheres to the stronger view, viz., that grounding literally is a species of the genus, causation (A. Wilson 2017). More specifically, Schaffer and A. Wilson argue that both grounding and causation are best approached through the lens of a single formalism, viz., that utilized by SEM (or “structural equation model”) frameworks of causation. Defenders of this line of reasoning take the fact that a single formalism can be used to characterize both cases to indicate that grounding is at least as unified as causation. (For further endorsements of the idea that grounding and causation are in some fashion closely related, see also Bennett 2011: 93–94, 2017; Fine 2012: 40; Schaffer 2012: 122; Sider 2011: 145.)

The legitimacy of the SEM-style argument in favor of the unity hypothesis can be and has been challenged on multiple grounds. For one thing, the assumption that the SEM framework in fact provides a workable approach to causation is highly controversial (J. Wilson 2016). Secondly, even if the plausibility of SEM approaches to causation is granted for the sake of the argument, it is debatable whether the SEM framework transfers as smoothly from the case of causation to the case of grounding as Schaffer and A. Wilson would have us believe (Koslicki 2016; J. Wilson 2016). Thirdly, the purported similarities between causation and grounding can also be disputed independently of the particulars associated with the SEM framework (Bernstein 2016; Shaheen 2017). Given these challenges, then, it is fair to say that, SEM-style arguments notwithstanding, skeptical doubts concerning the unity hypothesis, at this point, are still alive and well.

2.3 “Second-Generation” Grounding Skepticism (II): Fundamentality

In Section 2.2, we considered reasons for and against thinking that the theoretical utility of the grounding idiom derives, at least in part, from its alleged power to unify what might otherwise strike us as a heterogeneous collection of phenomena. In addition to this first type of response, an extremely widespread second response to the Central Question defends the theoretical utility of the grounding idiom by appeal to its alleged power to capture and/or elucidate the distinction between the fundamental and the nonfundamental (see Bliss 2020). Thus, following this second style of response to the Central Question, what we learn from being told that a fact, [p], is grounded in a fact, [q], is at least in part that [p] is less fundamental than or posterior to [q]; that [q] is more fundamental than or prior to [p]; that [p]’s obtaining is not a brute, unexplained phenomenon, but rather that [p]’s obtaining is to be explained by appeal to the obtaining of some other fact, namely [q]. (An analogous strategy is supposed to apply in nonfactual, non-propositional cases as well; but see Koslicki 2015a for a discussion of some of the difficulties in finding a ground-theoretic way of expressing relative fundamentality relationships among nonfactual, nonpropositional entities.)

Although some version of the second style of response to the Central Question seems to be almost universally accepted by grounding enthusiasts, different theorists choose different vehicles to give voice to the general sentiment underlying this second response. The following partial list illustrates (in no particular order) some of the variations on the general theme just cited that are represented in the literature on grounding: among other things, the grounding idiom is said to be needed, for example, in order to express the layered hierarchical structure of reality (deRosset 2013); to fix the direction of relative fundamentality or priority (R. Cameron 2016; Raven 2017); to analyze claims concerning ontological dependence and/or the substance/nonsubstance distinction (Correia 2005, 2008; Rydéhn 2018; Schaffer 2009; Schnieder 2006, 2020); to formulate a hyperintensional theory of intrinsicness (Bader 2013; Rosen 2010); to articulate a sensible form of nominalism about properties (Melia 2005); to encapsulate the

idea that some truths are metaphysically “light-weight” (deRosset 2017: ms.); or to make room for a suitable notion of ineliminability (Raven 2016, 2017). Moreover, without their idiom, so grounding enthusiasts have argued, it is impossible to understand certain philosophical positions or substantive disputes in philosophy properly, e.g.: physicalism (Schaffer 2003, 2009); the realism/antirealism debate (Fine 2001); the relation between normative ethics, meta-ethics, and metaphysics (Berker 2017); disagreements in feminist metaphysics or social ontology more generally (Mikkola 2015; Schaffer 2017; Sider 2017); the mind/body problem or the debate between substantialists and relationalists about space (Dasgupta 2014, 2017); as well as the idea of sparseness or “nothing-over-and-aboveness” (Bryant 2020; R. Cameron 2008; Leary 2020; Passinsky 2020; Ney 2020; Schaffer 2007, 2009, 2010; Solomyak 2020).

In addition, in the minds of some grounding enthusiasts (e.g., R. Cameron 2016; Schaffer 2016b), a theory that incorporates the grounding idiom has greater expressive powers than a comparable theory that helps itself only to a primitive notion of absolute fundamentality but lacks the grounding idiom (J. Wilson 2014, 2016). According to the former type of theory, so the reasoning goes, the absolutely fundamental can simply be defined in terms of the relatively fundamental, viz., as that which, while being itself ungrounded, serves as a ground for other derivative phenomena. But the reverse, so these theorists argue, is not the case, since there appears to be no similar straightforward mechanism by means of which the relatively fundamental can be defined in terms of the absolutely fundamental without implicitly appealing to (some terminological variant of) the grounding idiom.

2.3.1 Formal Properties of the Grounding Idiom

Grounding connections are often assumed to be factive, referentially transparent, well founded, irreflexive, asymmetric, transitive, nonmonotonic, and metaphysically necessary. Overall, given these alleged formal properties, grounding enthusiasts have taken their idiom to be suitable for the formulation of a hierarchically structured picture of reality, eventually bottoming out in an absolutely fundamental layer that in turn fixes and metaphysically explains everything else. But various arguments have been mounted in the literature that call into question some of these alleged formal properties commonly associated with the grounding idiom. (In some cases, these arguments are formulated in terms of ontological dependence rather than grounding; however, insofar as grounding enthusiasts are committed to the idea that statements concerning ontological dependence can be understood as grounding claims, the objections in question will also affect the grounding idiom.) Barnes 2018, for example, proposes that dependence relations should not be assumed to be asymmetric and that this provides a reason for distinguishing dependence from ground, assuming that ground must be asymmetric. Thompson 2016a argues that grounding should be taken to be nonsymmetric rather than asymmetric (see also Thompson 2020). Jenkins 2011 raises objections against the irreflexivity of dependence. Bliss 2014, R. Cameron 2008, Lubrano 2018, Rosen 2010, and others critically discuss the assumption that grounding is well founded (see also Dixon 2020). Schaffer 2012 regards explanatory claims as contrastive, which calls into question the transitivity of grounding. Rodriguez-Pereyra 2015 puts forward reasons for thinking that grounding is neither irreflexive nor asymmetric nor transitive. Skiles 2015 has questioned necessitarianism and holds instead that grounding connections may be contingent (see also Skiles 2020). On the whole, if these sorts of arguments succeed and we have reason to doubt that the grounding idiom really has the formal properties that are commonly ascribed to it, then we may in turn wonder whether this notion can in fact be used to accomplish the theoretical tasks for which it is intended. In particular, unless the grounding idiom has the requisite formal properties, it cannot perform the purported theoretical job of

imposing a hierarchical order connecting the (absolutely and/or relatively) fundamental to the (absolutely and/or relatively) nonfundamental.

2.3.2 Potentially Problematic Applications of the Grounding Idiom

According to the second style of response to the Central Question, the grounding idiom is said to furnish us with the preferred concept by means of which to express the connection between the (absolutely and/or relatively) fundamental and the (absolutely and/or relatively) nonfundamental. Recently, however, some critics have questioned whether the grounding idiom can really be helpfully applied to nonfundamental phenomena within the social domain (e.g., Barnes 2014, 2017; Mikkola 2015, 2017). Barnes 2014, for example, writes that ground-theoretic approaches to (meta-)metaphysics tend to privilege the absolutely fundamental and characterize existence questions as trivial, thereby apparently ruling out the prospect for feminist metaphysics as a sub-discipline of metaphysics. Such approaches, according to Barnes, make it difficult to appreciate the substantivity of certain disputes among feminist philosophers, e.g., concerning the nature or existence of gender, since these disputes concern nonfundamental rather than fundamental phenomena; moreover, existence questions, in these contexts, are often not regarded as trivial.

In response, grounding enthusiasts (e.g., Raven 2017; Schaffer 2017; Sider 2017) have stressed that ground-theoretic approaches have the resources to capture the importance of derivative phenomena. Raven 2017, for example, replies to Barnes's concerns by arguing that the mere acceptance of ground does not require "fetishizing the fundamental"; rather, so Raven remarks, metaphysics is concerned no less with the nonfundamental than with the fundamental: "Much of our interest in the fundamental is to use it to account for the nonfundamental" (Raven 2017: 644). In light of criticisms mounted by Mikkola 2015, which challenge the local applicability of ground to value-laden inquiry, Raven notes that ground and value-laden inquiry can coherently combine (Raven 2017: 645–649). (See also Passinsky 2020.)

2.3.3 Alternative Approaches

Alternatives to ground-theoretic approaches have been proposed in the literature. For example, Lipman 2016 argues that, when it comes to a characterization of antirealism, simple error-theoretic approaches are preferable to ground-theoretic approaches. In general, so Lipman writes, metaphysics must be about what does and does not obtain and should concern itself with questions of fundamentality only when commonsense or scientific verdicts are lacking or deliver conflicting results (Lipman 2016: 604).

In my own work, I have argued that, in order to arrive at a sufficiently fine-grained approach to relative fundamentality for nonfactual, nonpropositional phenomena, we must recognize multiple distinct dimensions along which an entity or type of entity may be classified as more or less deserving of substance status (Koslicki 2012, 2013a, 2013b, 2015a, 2015b, 2018). Currently available idioms of grounding do not reflect, for example, distinctions among the following dimensions of relative fundamentality and nonfundamentality: (i) whether an entity is essentially *abstracted* from something more complex; (ii) whether an entity is essentially *constructed* out of other entities; (iii) whether an entity is *natural* or *artificial* (e.g., essentially the result of a creative act involving an intentional agent); or (iv) whether an entity is more or less *unified*. However, just as one and the same thing can be both good in some respects (e.g., dancing) and bad in others (e.g., playing basketball), so one and the same entity, according to the multidimensional conception, can also be both more fundamental than another in certain respects (e.g., its degree of unity) and less fundamental than another in other respects (e.g., its naturalness or artificiality).

It is only once we make room for multiple dimensions of fundamentality and nonfundamentality, so I argue, that we can do justice to the data that presents itself to the ontologist and meta-ontologist.

A different kind of pluralism has been proposed by J. Wilson 2014, 2016. According to J. Wilson, there is no need to posit a grounding idiom (or, in her terms, “big ‘G’ Grounding”), since the work for which this notion is intended is in fact already performed by a multiplicity of what she calls “small ‘g’ grounding relations” (e.g., parthood, composition, realization, constitution, set membership, and the like). In order to account for the directionality of relative fundamentality relations, J. Wilson appeals in addition to a primitive notion of absolute fundamentality and to further “suppositions/facts about the natures of the non-fundamenta and how (via one or other small-g relation) the non-fundamenta stand to one another” (J. Wilson 2016: 195).

J. Wilson and I thus agree that grounding is too coarse-grained to perform the metaphysical work for which it is intended and that a pluralism of some kind is called for to carry out these responsibilities properly. But we arrive at these conclusions in different ways and differ with respect to certain crucial commitments we take on board along the way. For one thing, I do not follow J. Wilson in regarding the relations she calls “small ‘g’ grounding relations” themselves as ontological dependence relations. Rather, my own position is that these relations *induce* different varieties of ontological dependence in different circumstances and in different respects. Thus, in certain cases and in certain respects, the parts composing a whole may depend on the whole in question; and in certain cases and in certain respects, a whole may also depend on its parts. But we lose the ability to make these distinctions if we simply view parthood itself as a relation of ontological dependence. Secondly and relatedly, I depart from the earlier position of J. Wilson 2014 that absolute fundamentality together with what she calls “small ‘g’ grounding relations” gives us the apparatus sufficient to capture the *directionality* of relative fundamentality. Even if we assume an absolutely fundamental level as fixed, such relations as parthood can still induce ontological dependence relations going in both directions, both towards the absolutely fundamental and away from the absolutely fundamental. In order to capture the directionality of relative fundamentality, it is thus necessary to make room for the different varieties of ontological dependence, in addition to what J. Wilson calls “small ‘g’ grounding relations”. Thirdly, I am not convinced that it is legitimate, from the perspective of a metaphysician, to assume an absolutely fundamental level as fixed; rather, like Rosen 2010 and others, I advocate maintaining a neutral stance concerning the question of whether relations of relative fundamentality turn out to be well founded. (See Schaffer 2016b and J. Wilson 2016 for a recent exchange in which J. Wilson addresses some of these concerns.)

2.4 “Second-Generation” Grounding Skepticism (III): Deflating the Grounding Idiom

Up to this point, we have considered two types of responses to the Central Question: the first stresses the alleged power of the grounding idiom to unify an apparently heterogeneous collection of phenomena; the second defends the theoretical utility of the grounding idiom by citing its alleged power to capture and elucidate the distinction between the fundamental and the nonfundamental. In addition to these two types of responses, I want to end by mentioning at least briefly a third possible response to the Central Question. This third reply proceeds by recommending a properly deflated conception of the grounding idiom. In response to the Central Question, deflationists may allow that something is to be gained by classifying certain connections as grounding connections; however, the lessons we learn, so deflationists will stress, are more modest and less metaphysical than the initial (and continued) hype connected with the

grounding idiom may have led us to believe (Dasgupta 2017; Maurin 2019; Miller and Norton 2017; Thompson 2016b).

Along these lines, Dasgupta 2017, for example, argues that the grounding idiom, as it is utilized by certain grounding enthusiasts and grounding skeptics alike, has been improperly inflated, with the result that the theoretical usefulness of this notion has thereby been obscured. Rhetoric aside, the grounding idiom, in Dasgupta's view, should be understood as merely playing the role of "limning many issues of intellectual interest" (Dasgupta 2017: 76–77). Once our expectations of what the grounding idiom can deliver have been appropriately adjusted, Dasgupta concludes that the concerns of grounding skeptics can be put to rest and grounding enthusiasts can still accomplish their goals with a deflated conception of ground. Despite his assurances to the contrary, however, grounding skeptics may well feel vindicated by what remains of the grounding idiom after its ambitions have been toned down in the way Dasgupta indicates.

A number of critics (e.g., Maurin 2019; Miller and Norton 2017; Thompson 2016b) also play into the hands of the grounding skeptic by objecting that the grounding idiom has been confused with a metaphysical notion, when it in fact represents a psychological, mind-dependent, and/or epistemically imbued explanatory mechanism. Thompson 2016b, for example, presses grounding enthusiasts to spell out in more detail exactly how they see their grounding idiom as being connected to metaphysical explanation and finds that these theorists are then pushed toward a pragmatic and agent-centered conception of explanation, thus threatening the purported mind-independence of grounding connections. Maurin 2019 similarly arrives at the skeptical conclusion that more work is required on the part of the grounding enthusiast in order to substantiate the purported connection between grounding and metaphysical explanation. Miller and Norton 2017 propose to make sense of our grounding-related judgments by way of a psychological explanation and without positing a grounding idiom; instead, they appeal to cognitive mechanisms that have evolved to detect correlations and causal connections, together with the modal apparatus of supervenience, necessitation, and entailment.

3. Conclusion

In this chapter, we reviewed several varieties of grounding skepticism as well as responses that have been proposed by grounding enthusiasts to considerations raised by grounding skeptics. Grounding skeptics, as we conceived of them here, are theorists who belong to one of the following two schools of thought. "Old-school" grounding skeptics doubt the theoretical utility of the grounding idiom by denying one of its presuppositions, viz., that this notion is at least intelligible or coherent. "Second-generation" grounding skeptics call into question the theoretical utility of the grounding idiom for other reasons; their skeptical doubts tend to focus on one of the following three purported theoretical virtues grounding enthusiasts ascribe to their idiom: (i) its alleged power to unify an apparently heterogeneous collection of phenomena; (ii) its alleged power to capture and/or elucidate the distinction between the fundamental and the non-fundamental; or (iii) the alleged metaphysical (as opposed to mind-dependent, epistemic, or psychological) utility of grounding claims. As we have noted, grounding enthusiasts have already formulated responses to many of the objections described in this chapter. At this point, however, it is fair to say that the state of the literature is still evolving, and no conclusive judgment can therefore be reached as of yet as to whether grounding enthusiasts or grounding skeptics have gained the upper hand in these debates. In the meantime, though, grounding skeptics continue to maintain that the classification of factual and/or nonfactual connections, such as those cited in (1) and (2), under the rubric of grounding does not really help us illuminate the nature of the

connections at issue; instead, from the view of the grounding skeptic, we are better off studying these various connections separately and in their own right.

Related Topics

- Bliss, Ricki (2020) “Fundamentality,” Chapter 23 this volume.
Bryant, Amanda (2020) “Physicalism,” Chapter 35 this volume.
Dixon, Scott (2020), “Infinite Descent,” Chapter 16 this volume.
Fine, Kit (2020) “The Essential Glossary of Ground,” Chapter 38 this volume.
Leary, Stephanie (2020) “Normativity,” Chapter 34 this volume.
Ney, Alyssa (2020) “Mind,” Chapter 33 this volume.
Passinsky, Asya (2020) “Social Entities,” Chapter 37 this volume.
Richardson, Kevin (2020) “Varieties,” Chapter 13 this volume.
Schnieder, Benjamin (2020) “Dependence,” Chapter 7 this volume.
Skiles, Alexander (2020) “Necessity,” Chapter 10 this volume.
Solomyak, Olla (2020) “Realism,” Chapter 26 this volume.
Thompson, Naomi (2020) “Strict Partial Order,” Chapter 17 this volume.
Wang, Jennifer (2020) “Cause,” Chapter 20 this volume.

Notes

- 1 I use the terms “factual” and “nonfactual” here to draw a distinction between the types of entities that can allegedly figure in grounding connections (i.e., whether these entities are facts or propositions or whether they are, rather, nonfactual or nonpropositional entities, e.g., objects, properties, relations, and the like). The contrast in question is therefore not to be confused with the distinction between “factive” and “nonfactive” expressions, e.g., as it pertains to the question of whether grounding claims should be regarded as factive or nonfactive.
- 2 What is important, here, is of course the nature of the *connection* at issue, not the expression(s) we use to single out these connections. Thus, if grounding enthusiasts, for example, stopped using the term, “grounding” and related expressions (e.g., “ground”, “ground-theoretic”, etc.) but continued to view the connections at issue as grounding-connections (under a different label), then grounding enthusiasts and grounding skeptics would continue to disagree with one another but use a different vocabulary to formulate their dispute.
- 3 We should note here that, since different grounding enthusiasts accept different claims, not every grounding enthusiast will recognize their own position in connection with every objection I discuss in this chapter. Thus, the skeptical doubts raised in what follows are merely meant to survey the kinds of objections that have been mounted against some version or other of grounding enthusiasm defended in the literature. Given space constraints, I unfortunately cannot always differentiate between the different versions of grounding enthusiasm that are or are not affected by a particular criticism that is being considered. Moreover, since this chapter is intended to give a survey of skeptical doubts, I am often merely reporting what has been said in the literature rather than expressing my own agreement with a particular point. In the case of some of the objections I discuss in what follows, grounding enthusiasts can no doubt formulate plausible responses (or have already done so). Readers are referred to the references I give if they wish to investigate how these debates have played out in the literature in more detail.

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ANTI-SKEPTICAL REJOINDERS

Louis deRosset

The recent groundswell of interest in the theory of grounding has been met in some quarters with a skeptical reaction. There are two kinds of skepticism that should be distinguished. Relatively local skeptical doubts concern the appropriateness of applying grounding to this or that particular theoretical purpose. Consider, for instance, Bader's proposal to use the notion of ground to characterize the distinction between a reason for doing something and conditions that enable that consideration to be a reason (Bader 2016). One might marshal ethical considerations against this particular application without casting doubt on the utility of ground more generally for doing other theoretical work. The kind of skepticism that is our focus here is more ambitious. *Global skepticism* about ground is the view that the attempt to develop a theory of ground is generally and in principle defective, and attempts to apply such a theory will be fruitless. There are several global skeptics in the literature (Daly 2012), (Hofweber 2009: 269–72), (Koslicki 2015), (Thompson 2016a), (Turner 2016), (Wilson 2014). Are their doubts warranted? In this chapter, I review both the reasons that seem to favor global skepticism and the responses to those reasons by defenders of ground. I suspect that the number of published global skeptics is a tiny fraction of the total population of global skeptics. So rather than merely offering a piecemeal enumeration of skeptical arguments and responses in the literature, I will attempt to fit them into a more general scheme.

1. Varieties of Skepticism

Claims of ground are canonically expressed by sentences of the form

- (1) ϕ in virtue of the following facts: that ψ_0 , that $\psi_1, \dots$ (Raven, THIS VOLUME: Introduction).

There are, then, three varieties of global skepticism.

HARD ELIMINATIVISM No claim of the form (1) is true;

SOFT ELIMINATIVISM (Even) if some instances of (1) are true, these sentences are generally unfit for theorizing because the notion of grounding which they deploy is junk.

REVOLUTIONARY REDUCTIONISM (Even) if some instances of (1) are true and fit for theorizing, the theoretical work they do can be done better by sentences that do not employ any notion of grounding.

It is easy to see that this taxonomy is exhaustive, in the sense that global skepticism requires the truth of at least one of the three claims. For, if each is false, then there are true claims of ground that are useful in theorizing, and the theoretical work that they do is done no better by any sentences that deploy notions other than grounding. In other words, if each of *HARD ELIMINATIVISM*, *SOFT ELIMINATIVISM*, and *REVOLUTIONARY REDUCTIONISM* are false, then the project of theorizing by appeal to truths of the form (1) is not generally and in principle defective, and there is no reason to doubt that the theory can be applied fruitfully. A given consideration favors global skepticism, then, only if it provides reason to think that one or more of the three skeptical claims is true.

Theorists of ground have marshalled three sources of insight into the nature of ground. These sources of insight appear to establish a presumption against global skepticism. The first source of insight concerns ongoing investigations into questions of ground. Most areas of philosophical inquiry regularly truck in the idea. Ethicists wonder, for instance, what made Oswald's assassination of Kennedy morally wrong. Opinions on this question differ. Kantian ethicists propose one sort of answer, utilitarians another, and other theorists still others. Similarly, a central question in epistemology concerns what it is in virtue of which I am justified in believing that I am not dreaming. In metaphysics, we sometimes wonder what it is in virtue of which things have the modal or temporal properties that they do. These examples could be multiplied.¹

Investigations into what grounds what are not limited to philosophy. Organic chemists, for instance, can tell us what makes ethyl alcohol miscible in water. Physical chemists can tell us, in a general way, what makes the fluid in one system hotter than the fluid in another. I once heard a theoretical physicist characterize her research on the radio as an investigation into what makes gravity so weak. As the chemical examples demonstrate, some investigations into questions of ground have already met with success. See (Raven, *THIS VOLUME*: Introduction) for more examples and discussion.

A second source of insight concerns claims of ground that are easy to evaluate. The weather today is sunny. So it's either sunny or snowy. It is overwhelmingly plausible to think that these two circumstances are related: the weather is either sunny or snowy in virtue of being sunny. It is overwhelmingly implausible to think that it is sunny in virtue of being either sunny or snowy. Aristotle plausibly held that the statement that there are human beings is true in virtue of there being human beings (*Categories*: 14b14–22). It is even more plausible to deny the converse grounding claim, that there are humans in virtue of the truth of the statement. So we have a large class of grounding claims and denials of grounding claims that enjoy a high degree of plausibility.²

The third source of insight is that we are familiar in practice with the constraints governing such explanations, since we liberally indulge in constructing and criticizing them, at least in our professional lives. I trust, for instance, that you are intimately familiar with at least some of the metaphysical, ethical, epistemological, and scientific investigations I have mentioned. You have participated in some of these debates. You have proposed or objected to arguments for or against the grounding claims that characterize them. We will encounter particular examples of such constraints when we discuss skeptical arguments in detail in what follows.

So we have a large class of ongoing inquiries into what grounds what. Some of these inquiries have met with success, yielding well-established conclusions about what grounds what. We have a large class of paradigmatic truths and paradigmatic falsehoods concerning what grounds what. And we fluently participate in the practice of proposing and criticizing grounding claims, giving us intimate familiarity with the constraints that govern their assessment. These facts establish a defeasible presumption in favor of the theoretical utility of at least some grounding claims. Global skeptics claim that the inquiries are misbegotten or mischaracterized, the successes

illusory or mischaracterized, the paradigms chimerical or mischaracterized, and the practices ill founded. We will need arguments to establish the skeptic's conclusions.

2. Old-School Skepticism

Some global skeptics have been happy to supply arguments. These arguments rarely explicitly connect the dots between the considerations advanced and the three varieties of skepticism we have specified. In this chapter, I will borrow Koslicki's taxonomy (THIS VOLUME: Chapter 11) of global skepticism, which distinguishes between *old-school skepticism*, where it is not at all difficult to see how the considerations in question yield global skepticism, and *second-generation skepticism*, where the skeptical implications are somewhat harder to discern. Let's start with old-school skeptical challenges. We will discuss in particular charges that claims of ground are unintelligible or impossible to assess. These challenges are generally held by enthusiasts of ground to be met by one or more of our three sources of insight.

2.1. Unintelligible?

A skeptic might contend that the notion of ground is unintelligible (Daly 2012). If so, then clearly hard eliminativism is correct, since claims that are unintelligible are not also true. Daly has explored this form of skepticism. Following Nelson Goodman (Goodman 1954; quoted at (Daly 2012: 89–90)), Daly's skeptic contends that, though we have no criterion for a notion's intelligibility that we may apply to the notion of ground, we do not need one. We have the capacity to discern intelligible notions, and we should take seriously judgments of unintelligibility that issue from its exercise. Daly's skeptic judges that the notion of ground is unintelligible.³

The charge of unintelligibility is difficult to square with our three sources of insight. There are cases in which it is very easy to assess claims about what grounds what. The claim that it is not the case that it is sunny in virtue of being either sunny or snowy is highly plausible. If the notion of ground were unintelligible, it should not be so. The claim that it is either sunny or snowy in virtue of being sunny is also highly plausible. If the notion of ground were unintelligible, it should not be so. The fact that some inquiries into what grounds what have been successful also entails that the notion of ground is intelligible. Chemical investigations have discovered that the fact that one fluid has a higher temperature than another is grounded in the facts concerning the kinetic energies of their constituent molecules. If the notion of ground were unintelligible, this success would not have occurred.

In light of these facts, what are we to make of Daly's skeptic's judgment of unintelligibility? I do not doubt that many philosophers judge that they do not understand the grounding talk that is filling the journals these days. And, I think, those judgments are to be taken very seriously indeed. But, grounding enthusiasts will deny that those judgments, taken with full seriousness, merit a skeptical conclusion. First, there is a difference between judging that I do not understand something and judging that nobody does or could understand it. I, personally, do not understand

(2) every elliptical equation can be correlated with a modular form.

This is the Taniyama-Shimura conjecture, proved by Andrew Wiles in the course of his proof of Fermat's Last Theorem. My incomprehension does not show that Wiles's proof is unintelligible. It shows that the limits of intelligibility extend far beyond my own capacities. Still, Daly suggests that the skeptic does make a further judgment, to the effect that grounding locutions

are unintelligible, and we ought to take that further judgment seriously. This brings us to the second point: Daly's argument requires us to weigh the skeptic's unsupported judgment of unintelligibility against chemists', epistemologists', etc. (perhaps unsupported) judgments that they do understand grounding locutions. Grounding enthusiasts contend that the balance of considerations favors intelligibility. So we should take very seriously Daly's skeptic's unsupported judgment of unintelligibility. But we should also take seriously other theorists' (unsupported) judgments of intelligibility and even more seriously their judgment that they themselves understand the grounding claims they are considering. This is not to deny that people may be confused without realizing it. Instead, it is to suggest that we should seek to understand them unless and until independent evidence of unintelligibility emerges.

2.2. Impossible to Assess?

A second kind of old-school skepticism accedes to the intelligibility in principle of grounding claims of the form (1). Nevertheless, it holds that such claims are impossible to assess, for one reason or another.⁴ Perhaps the notion of ground is simply too unclear for serious theorizing. One might have the same attitude toward grounding locutions, for instance, as one does toward Dr. Martin Luther King, Jr.'s famous claim that the moral arc of the universe is long but that it bends towards justice. We might reasonably think that this claim is true so far as it goes but that the notion of the moral arc of the universe is too unclear to have a place in moral theory. Alternatively, one might think grounding claims are clear enough but that dispositive evidence concerning their truth is inaccessible to us. So, like claims about exactly how many human beings were born in the year 17 C.E., the truth about what grounds what will forever elude us.

However such a view might be motivated, it immediately encounters difficulty accommodating the three sources of insight highlighted by grounding enthusiasts. As we have noted, there are claims about what grounds what that are easy to assess. As we have also noted, even where the questions are more interesting and the assessment is more difficult, some of our investigations into grounds have yielded firm, affirmative verdicts. Finally, even where our investigations have not yielded firm results, our ongoing, robust practice of assessing grounding claims imposes constraints. Consider, for instance the utilitarian claim

(3) Oswald's assassination of Kennedy was wrong in virtue of failing to maximize utility.

Critics have objected that there are cases in which actions that fail to maximize utility are not morally wrong. It has been held, for instance, that framing and punishing an innocent person to stop a crime wave might fail to maximize utility without being wrong (Carritt 1950). It is not clear that these objections succeed. The utilitarian may dispute the premise, arguing, for instance, that failing to frame the innocent in such circumstances is wrong after all. The utilitarian may amend the original explanatory proposal, arguing, for instance, that the assassination is wrong in virtue of being prohibited by a rule whose adoption would maximize utility.⁵ But this much is clear: the utilitarian may *not* blithely accept the premise and stick with her explanatory proposal. This is because her original grounding claim commits her to there being no possible cases in which an act fails to maximize utility but is not morally wrong. These commitments facilitate the assessment of grounding claims in certain circumstances. In some cases, as in the dispute over utilitarianism, it is difficult to ascertain whether we are in those circumstances. But whatever difficulties we may encounter are not due to any difficulty in principle in assessing grounding claims.

2.3. Have We Mischaracterized the Cases?

Global skeptics are, of course, aware of the pervasive phenomena that provide the grounding enthusiasts' three sources of insight. But, the skeptics contend, the cases in question are not best characterized as involving ground. Skeptics have contended that grounding enthusiasts are correct to think that these are sensible inquiries being pursued by philosophers and scientists, we have made progress on some of them, and there are genuine insights expressed by certain claims that encapsulate the progress. But, skeptics contend, none of the questions that are the subject of the investigations and none of the answers we have found involve the notion of grounding. Grounding enthusiasts have simply mischaracterized the cases.

This rejoinder would be unreasonable if there were no plausible alternative characterization of the cases, on which they involve some phenomenon other than ground. Global skeptics have suggested alternative construals. So, for instance, Daly (2012: 96, 98) has suggested that the cases in question all involve identity, supervenience, or entailment by laws of nature. The idea is that the supposed examples of grounding are just inaptly expressed examples of claims of supervenience, identity, or entailment by laws of nature.

To illustrate each of the strategies, consider the physical chemist's claim that what makes one fluid hotter than another is the kinetic energies of their molecules. Daly's skeptic might suggest that this is an inapt or unclear way of saying something true: that the former's being hotter than the latter is identical to the molecules of the former having a higher mean kinetic energy than the molecules of the latter. This doesn't quite handle the case, since the chemist tells us that the molecules' having a certain *specific* mean kinetic energy is what makes the first fluid hotter than the other. The claim is that the fluid in system A is hotter than the fluid in system B in virtue of either some fact expressible by a long conjunction of the form

- (4) System *A* contains exactly particles $a_0, a_1, \dots, a_i \wedge a_0$ has a kinetic energy of n_0 joules $\wedge a_1$ has a kinetic energy of n_1 joules $\wedge \dots \wedge$ system *B* contains exactly particles $b_0, b_1, \dots, b_j \wedge b_0$ has a kinetic energy of m_0 joules $\wedge b_1$ has a kinetic energy of m_1 joules, . .

or (collectively) the facts expressed by its conjuncts. It would be implausible to suggest that the fluids' having just those molecules with just those energies is identical to *A*'s being hotter. In response, the skeptic might suggest that the clearest way to state the claim in question is that the *hotter than* relation among fluids supervenes on the facts concerning the kinetic energies of particles together with the facts concerning which molecules are in each fluid. Again, this doesn't quite handle the case, since the supervenience claim is general, and the claim to which the physical chemist alludes concerns the specific kinetic energies of specific molecules. So the skeptic might finally suggest here that it is a law of mathematics, and so, presumably, a law of nature, that any two systems *A* and *B* whose ingredient molecules have just those kinetic energies are such that the mean kinetic energy of the molecules in *A*'s fluid is greater than that of the molecules in *B*'s.

A persistent theme of the literature on ground is that such proposed surrogates for grounding cannot plausibly characterize these cases; see (Kovacs THIS VOLUME: Chapter 24) for discussion. As we have seen, in many—perhaps all—of the cases, identity claims will not capture the force of the erstwhile grounding claims. This leaves supervenience and laws of nature as skeptical-friendly alternatives to accommodating the paradigms. Both of the alternatives share a defect: they are not plausible surrogates for grounding because they are insensitive to the relative fundamentality ordering—the order of dependence and determination—implicit in such cases (Berker 2017), (deRosset 2011), (Fine 2012a: 38–9), (Rosen 2010), (Schaffer 2009), (Trogon 2009), (Wilson 2014: §IV).

Let's consider laws of nature first. It is a law of logic, and so, presumably, a law of nature, that if the desk at which I write is both brown and wooden, then it is brown. And yet it would be utterly implausible to summarize this insight by saying that the desk's being both brown and wooden makes it brown or that it is brown in virtue of the fact that it is both brown and wooden. So it is implausible to think that the grain of truth in what we are trying inartfully to express when we use grounding locutions is expressed instead by the claim that there is a law of nature to the effect that, if the alleged grounds obtain, then the alleged grounded circumstance also obtains. In particular, physical chemists are not plausibly characterized as suggesting that the relation between the relative temperature of A and B and the particular kinetic energies of their particles is relevantly similar to the relation between the color of the desk and the fact that it both has that color and is wooden. Identity obviously also won't serve as a proxy for grounding in the chemical case. That leaves supervenience as the last refuge offered by Daly as a surrogate for the notion of ground. And arguments against the idea that supervenience can serve in place of grounding are common (Berker 2017), (deRosset 2011), (Fine 2012a: 38–9), (Rosen 2010), (Schaffer 2009), (Trogdon 2009), (Wilson 2014: §IV); see (Kovacs THIS VOLUME: Chapter 24) for discussion.

Hofweber (2009: 268–70) suggests *conceptual*, *causal*, or *counterfactual priority* as surrogates. It is not at all obvious that the particular kinetic energies of the molecules in the case at hand are *counterfactually* prior to A's being hotter than B, since they could very easily have been slightly different in a way that does not affect A and B's relative temperature. Similarly, it is highly dubious that their kinetic energies are *causally* prior to the relative temperatures of A and B, and, moreover, many cases of causal priority are not plausibly mischaracterized as cases of ground; see (Wang THIS VOLUME: Chapter 20) for discussion of the relations between ground and causation. That leaves *conceptual* priority. Again, the proposed surrogate is often present in cases that aren't cases of apparent ground. Any instance of $(\phi V(\psi \wedge \chi))$ contains ψ as a constituent, and thus, presumably, ψ is conceptually prior to $(\phi V(\psi \wedge \chi))$. If ϕ is itself a conceptual truth, then $(\phi V(\psi \wedge \chi))$ is even conceptually entailed by ψ . But, for many choices of ϕ , ψ , and χ , in a case in which ψ is true but χ is not, ψ does not appear to ground $(\phi V(\psi \wedge \chi))$.

Grounding enthusiasts contend that some of the proposed surrogates that we have discussed *undergenerate*, by failing to apply in cases in which ground appears to be at issue. So, for instance, identity clearly does not cover all of the cases. They also contend that the other proposed surrogates we have discussed *overgenerate*, by falsely predicting the appearance of ground in cases in which there is no such appearance. Naturally, different surrogates might be proposed, and variations on the proposals we have briefly consider might be advanced. Grounding enthusiasts doubt that the results will plausibly characterize the sorts of cases that provide their putative sources of insight into grounding: the proposed surrogates will either undergenerate or overgenerate.

3. Second-Generation Skepticism

Old-School skeptics propose views that are difficult to reconcile with the three alleged sources of insight on which grounding enthusiasts draw. Second-generation skeptics, instead, propose views that accommodate the sources of insight. So they avoid the difficulties faced by old-school skeptics. But, grounding enthusiasts contend, they encounter other difficulties. Among the principal difficulties second-generation skeptics' arguments face is the very thing that distinguishes them from old-school skeptics: it is hard to see how the claims at issue, even if true, engender global skepticism. In particular, it is hard to see how the claims in question support hard eliminativism, soft eliminativism, or revolutionary reductionism. We will discuss two varieties of second-generation skepticism. One is based on a claim that there is no unified kind of

dependence and determination that falls under the notion of grounding, and the other is based on the idea that grounding claims on their own do not give us theoretically useful information.

3.1. *Disunified?*

One form of second-generation skepticism appeals to the idea that the phenomena at issue in cases in which grounding claims are true are very disparate.⁶ Because the phenomena are so disparate, the skeptic argues, theorizing about them under a single label is bound to end in failure. Sometimes the complaint is couched in terms of the metaphor of *fineness of grain* (Koslicki 2015), (Wilson 2014): it is alleged that grounding is not fine-grained enough to appropriately characterize the immense variety of situations in which grounding claims are true. Thus, the skeptic argues, other notions are needed to take proper account of the variety we find; see (Richardson THIS VOLUME: Chapter 13) for a discussion of this variety.⁷

This form of skepticism faces three challenges from enthusiasts of ground. The first challenge concerns whether the diversity to which the skeptic appeals indicates that ground itself is disunified in any interesting way. Some enthusiasts of ground agree that we find a wide diversity of phenomena in cases in which grounding claims are true but argue that the diversity at issue is due to diversity in the things connected by ground rather than to diversity in the mode of connection (Raven 2017). To illustrate, no one thinks it problematic or indicative of global skepticism that some grounded facts involve the Pope and others do not. There are certainly two kinds of cases, the Pope-involving and non-Pope-involving cases. The two kinds of cases are disparate, but the differences trace to differences in the grounded facts rather than the mode of ground. We simply have two independent dimensions of variation: there are congeries of facts that involve the Pope and congeries that do not; and there are congeries of facts connected by ground and congeries that are not. Grounding enthusiasts suggest that the diversity highlighted by skeptics has a similar character. Thus, the alleged disunity of the phenomena is not good reason on its own to conclude that the grounding connections themselves are disunified in a way that precludes sensibly theorizing in terms of the notion of ground.

Jason Turner, for instance, has worried that putative instances of ground differ importantly (Turner 2016). In some cases, Turner notes, grounding is accompanied by *counterfactual determination*. A circumstance ϕ counterfactually determines a circumstance ψ iff ϕ and ψ both obtain, but if ϕ had not obtained, neither would ψ . For convenience, let's abbreviate the 'would' counterfactual connective using ' $\Box \rightarrow$ ', so that 'it is sunny $\Box \rightarrow$ it is not raining' abbreviates the counterfactual 'if it had been sunny, then it would not have been raining.' It is commonplace to claim

(5) {Socrates} exists in virtue of the fact that Socrates exists.

Assuming this grounding claim is true, it is accompanied by counterfactual determination:

(6) $\neg$ Socrates exists $\Box \rightarrow \neg$ {Socrates} exists.

The closest worlds in which Socrates fails to exist are also worlds in which his singleton set fails to exist. On the other hand, the truth of

(7) It is either sunny or cloudy in virtue of the fact that it is sunny

is not accompanied by counterfactual determination, since

(8) $\neg$ it is sunny $\Box \rightarrow \neg$ (it is either sunny or cloudy)

is not true. The closest worlds in which it isn't sunny are ones in which it is cloudy instead.⁸

The first challenge, in this case, requires one who would infer the disunity of ground from this sort of diversity to say why we shouldn't infer instead that counterfactual determination is simply an independent dimension of variation, like involving the Pope. Turner has offered a case of ground with counterfactual determination and a case of ground without counterfactual determination. Cases in which there is neither a connection of ground nor counterfactual determination are very common. My child's height is neither grounded in nor counterfactually determined by the price of tea in China. And it is fairly easy to come up with cases of counterfactual determination that are not also cases of ground. Consider the claim that I wore blue jeans today. This claim is in fact true but could very easily have been false, since I might easily have worn shorts instead. Now consider some 'stickier' truth that could not so easily have been false. For concreteness, let 'P' abbreviate the truth that there are fewer than seven refrigerator-sized perpetual motion machines packed into your kitchen. Then the grounding claim

(9) I wore blue jeans in virtue of the fact that (I wore blue jeans $\wedge$ P)

is clearly false. However, the conjunction does counterfactually determine its left-hand conjunct:

(10) $\neg$ (I wore blue jeans $\wedge$ P) $\Box \rightarrow \neg$ I wore blue jeans.

If the conjunction had been false, it would have been its left-hand conjunct that was false, since the right-hand conjunct could not so easily have failed. The closest worlds in which the conjunction fails are not ones in which you have to pick your way past perpetual motion machines while puttering around your kitchen. So counterfactual determination of the sort to which Turner appeals seems like a completely independent phenomenon from grounding. There is diversity among these cases, but there is no reason to infer from that diversity that the notion of ground itself bifurcates in any interesting way.

The second challenge that this form of skepticism faces is to counter arguments by some enthusiasts of ground that the cases of ground exhibit striking commonalities that indicate the presence of a common form of dependence and determination. Schaffer, for instance, has argued that the truth of grounding claims always indicates some explanatory relation between the grounds and what they ground.⁹ Moreover, Schaffer claims, ground can be regimented using the apparatus of structural equation models (Schaffer 2016); See (Wang, *THIS VOLUME*: chapter 20). Similarly, grounding enthusiasts contend that grounding claims indicate relations of relative fundamentality, unlike causation, counterfactual dependence, or other relations of dependence and determination (Berker 2017), (Schaffer 2016), (Raven 2017), (Rosen 2010), (deRosset 2013); see (Bliss *THIS VOLUME*: Chapter 23). Similarly, enthusiasts of ground contend that it has certain formal features, including (a generalization to the multigrade case of) transitivity and asymmetry (Berker 2017), (Fine 2012a), (Litland 2015), (Raven 2017), (Rosen 2010); and have developed logics to characterize these formal features (Fine 2012a, 2012b), (Correia 2010; Correia 2014), (Litland 2015), (deRosset 2015), (Schnieder 2011), (Poggioli *THIS VOLUME*: Chapter 14); see (Jenkins 2011), (Thompson 2016b), and (Wilson 2014) for dissenting views.

Finally, Berker has contended that grounding claims chain with one another to yield claims of dependence and determination that are clearly akin to grounding; but in cases in which

relations are relevantly diverse, we cannot chain instances together to yield anything like the sort of relation with which we started. So, for instance, from

- (11) A acted wrongly in telling him in virtue of the fact that A could have maximized utility by not telling him.
- (12) Either A acted wrongly in telling him or A acted in a way she believed to be wrong in virtue of the fact that A acted wrongly in telling him.

we may infer that there is ‘some non-rigged-up sense’ in which A’s acting either wrongly or in a way she believed to be wrong is grounded in A’s failing to maximize utility. Consider, by contrast, a clear case of pluralism: there is the relation *being bigger in geographic area than* and the relation *being bigger in population than*. Each is colloquially expressed by the comparative predicate “is bigger than.” But chaining instances does not yield any instance of any relation of comparative size. So for instance, from

- (13) New York City is bigger in population than Montana
- (14) Wyoming is bigger in size than New York City

we may not infer that there is any ‘non-rigged-up sense’ in which Wyoming is bigger than Montana (Berker 2017: §5.1).¹⁰

It would seem that, if these enthusiasts of ground are correct about all of these respects of unity, developing and applying a theory of ground is a perfectly sound project. We might expect that theorizing to consist in part in regimenting grounding claims using one of the indicated formalisms, describing its formal features, and tracing its implications for explanation and relative fundamentality. Thus, this second battery of considerations is not just a challenge to the kind of skepticism we are presently discussing but also part of an attempted affirmative case for the utility of theorizing about ground. It is unsurprising, then, that skeptics about ground doubt that cases of ground are genuinely unified in the ways we have indicated and that much of the skeptics’ criticisms have been aimed at these claims of unity.¹¹

The third challenge faced by the proposal to base skepticism about ground on the claim that the phenomena at issue are disparate is that it is not at all clear that disunity of the relevant sort provides any reason for skepticism. One kind of *grounding pluralism* holds that there are a number of different varieties of ground, none defined in terms of the others; see (Richardson THIS VOLUME: Chapter 13). This is not a reason to think that theorizing about ground is a generally defective enterprise. In fact, it seems most plausibly construed as a contribution to the theory of ground. Similarly, there are hundreds of thousands of kinds of beetles, exhibiting massive diversity of form and function. It does not follow that theorizing about beetles in general is defective. Such theorizing depends, of course, on the fact that there is a common taxonomic category, *Coleoptera*, to which these organisms all belong and that there are substantial commonalities among them that do not extend to other creatures, in causal history if nothing else.¹²

Suppose, however, that the kinds of ground are even more diverse than the kinds of beetles, exhibiting little in the way of commonalities. Suppose, for instance, that it were discovered that there are many kinds of ground, as disparate from one another as the two relations we colloquially mark using ‘bigger than’. For the reasons highlighted by Berker (2017), this would be a very surprising discovery about ground. Our theorizing, naturally, would have to mark the distinction between the many kinds of ground, and we might have to be careful to avoid, among other pitfalls, chaining instances of the disparate varieties to yield nonsense relevantly analogous

to the conclusion that Wyoming is ‘bigger than’ Montana. This is pretty close to the worst-case scenario for the disunity of ground, but it is no reason to object in advance to the possibility of developing a perhaps pluralistic theory of ground. Indeed, very prominent enthusiasts of ground appear to have theories of just this sort.¹³ So there is no quick inference from the truth of the skeptic’s premise that the phenomena we mark using grounding locutions are disparate to global skepticism.

3.2. *Uninformative?*

Another form of second-generation skepticism about ground contends that grounding claims on their own provide little or nothing in the way of information about matters of antecedent interest.¹⁴ So Koslicki, for instance, argues that claims of ground are not *specific* enough to provide us with the details we need to understand a relation of dependence and determination in particular situations. There are significant differences, Koslicki notes, between the way in which the existence and features of sets depend on and are determined by the existence and features of their members (on a Finean theory) and the way in which the existence and features of a hylomorphic compound depends on the existence and features of its matter and form (on an Aristotelian theory; Koslicki 2015: esp. 340–1). These differences matter for theorizing, Koslicki argues: simply lumping both cases together as cases of grounding without delving into the details is theoretically unhelpful.

Similarly, Wilson contends that claims of ground on their own do not provide “even basic illumination about [n]or allow even basic assessment of claims of metaphysical dependence” in particular cases (Wilson 2014: 544–5). Wilson identifies three types of questions that, she says, have to be answered to provide illumination or allow assessment of claims of dependence. Consider a grounding claim relevant to the mind–body problem:

(17) Joe is amused in virtue of the fact that Joe is in neural state N.

Suppose for the sake of illustration that (17) is true. Wilson argues that (17), by itself, does not entail an answer to any of the following questions:

EXISTENCE Does the fact that Joe is amused exist?

REDUCTION Is the fact that Joe is amused reducible to the fact that Joe is in neural state N?

EFFICACY Does the fact that Joe is amused have any causal powers that the fact that Joe is in neural state N does not have?

And, she says, these questions need to be answered to assess and gain illumination about the claim that Joe’s amusement is dependent on and determined by his neural state.¹⁵

Why think claims of ground by themselves entail no answer to either Wilson’s three questions or the matters of detail to which Koslicki alludes? To fix ideas, let’s focus on entailment on the sort treated in the first-order logic one learns as an undergraduate. Call this sort of entailment *narrowly logical entailment*. Then there is no narrowly logical entailment between the truth of (17) and answers to any of Wilson’s three questions. Similarly, to be told that {Joe} exists in virtue of the fact that Joe exists narrowly logically entails nothing about whether {Joe} is a compound of Joe and an Aristotelian form. So (17) lacks the sort of narrowly logical entailments that would answer the questions that Wilson and Koslicki think are crucial. As a result, philosophers may, without any formal contradiction or rank incoherence, hold (17) without taking any position

on existence, reduction, or efficacy. Moreover, Wilson argues, philosophers working on ground could easily take opposite sides of these questions while affirming (17) (Wilson 2014: 545ff.).¹⁶

Skepticism on the basis of the charge that grounding claims are uninformative faces two related challenges. The first is that it is not clear how un informativity of the relevant sort provides any reason for global skepticism. Assuming that the claim that (17) is uninformative on its own is consistent with its being true, evidently the charge provides no reason for hard eliminativism. The idea must be, then, that the charge of un informativity supports either soft eliminativism or revolutionary reductionism. But it is not yet clear why it should do so.

The skeptic's premise, recall, is that the truth of (17) does not narrowly logically entail answers to the sorts of questions that need to be answered for us to understand or assess the claim that Joe's amusement depends on and is determined by his neural state. As a result, philosophers may, without any formal contradiction or rank incoherence, hold (17) without taking any position on those questions. The narrowly logical facts are adequately clear, as is the *prima facie* coherence of each of the positions to which Wilson draws our attention.

But these facts seem to be the basis for an invitation to further develop the theory of grounding, tracing its links to existence, reduction, and causal powers rather than for a charge that the very project of developing such a theory is misbegotten. No doubt the evidence adduced for the theory will draw on considerations that go beyond logic, narrowly construed. This is a very common phenomenon. Compare the theory of ground with the theory of electromagnetism. The claim, for instance, that a given body has unit negative charge does not narrowly logically entail much of physical interest. To get interesting consequences from an attribution of unit negative charge in a particular case, we need to appeal to claims about charge that go far beyond logic. These claims are the target of theorizing and draw connections between charge and work, energy, force, mass, voltage, etc. Similarly, the profound confusion in the middle of the 20th century about the nature and features of what we nowadays call "metaphysical necessity" was no reason to think that theorizing about it would be fruitless. There were skeptics, famously including Quine, but they had other grounds for their skepticism (Quine 1943, 1947, 1966); see (Kaplan 1986) for discussion. Now, there is no guarantee *ab initio* that the theory of ground will end up succeeding nearly as well as the theory of electromagnetism or (even) the theory of modality. But, it seems, the only way to find out is to develop the theory and see how it goes. So the un informativity of grounding claims seems on its own to be no basis for global skepticism.

The second challenge is that similar considerations appear to apply to almost any phenomenon of interest. To illustrate, consider Wilson's position. As we have seen, Wilson contends that grounding claims are uninformative and that the theoretical work supposedly done by grounding is better done by a battery of other relations, which she calls "small-g" grounding relations (Wilson 2014: 539).¹⁷ One of these relations is mereological composition. Suppose we are given a particular base-ball *b*, mereologically composed of certain material proper parts $p_0, p_1, \dots, p_n$, and consider the claim

(18) *b* is mereologically composed of $p_0, p_1, \dots, p_n$.

This claim on its own does not narrowly logically entail answers to questions analogous to EXISTENCE, REDUCTION, or EFFICACY.¹⁸ As a result, philosophers may, without any formal contradiction or rank incoherence, accept (18) while taking any position on each of the analogues of EXISTENCE, REDUCTION, or EFFICACY. Moreover, this problem is not peculiar to ground and mereological composition. It will plausibly generalize to almost any notion of interest. Philosophers are resourceful, and *prima facie* coherence is fairly easy to come by.

So if un informativity is taken to support some form of global skepticism about grounding, then the analogous phenomenon with respect to almost every other phenomenon should be

taken to support the analogous skeptical position about those phenomena. As Wilson's example demonstrates, global skeptics do not typically take their skepticism to be quite that global. More to the point, grounding enthusiasm is on firm ground if there is no more (and no less) reason for skepticism about ground as there is for skepticism about metaphysical modality, mereological composition, or electrical charge.

4. Conclusion

We have discussed only a few of the many potential arguments for global skepticism. Still, the discussion reveals two general lessons for the contemporary state of the debate between enthusiasts of ground and global skeptics. First, old-school skepticism looks to grounding enthusiasts as if it runs afoul of the three sources of insight into the nature of grounding. Those include the fact that ground figures in a large number of inquiries in a wide variety of areas, some of which have been successful; that there are paradigmatic cases in which it is easy to assess claims of ground; and that even in harder cases where success has eluded us, we are familiar in practice with the constraints governing the truth conditions for claims of ground. Indeed, any form of global skepticism must ultimately find a way to reconcile itself with these three sources of insight, since they establish a presumption against each of the three forms of skepticism we have identified.

Second-generation global skeptics at least defer the challenge of reconciling themselves to these three sources of insight. The cost of accommodation, however, is that the link between the considerations the skeptics advance as premises and global skepticism is more difficult to discern. The second lesson is that these considerations, including the alleged disunity of grounding phenomena and the alleged uninformativity of grounding claims, look to enthusiasts of ground more like invitations to theorizing than objections.

Related Topics

Raven "Introduction", Glazier "Explanation" [Chapter 8], Koslicki "Skeptical Doubts" [Chapter 11], Richardson "Varieties" [Chapter 13], Poggiolini "Logics" [Chapter 14], Wang "Cause" [Chapter 20], Bliss "Fundamentality" [Chapter 23], Kovacs "Modality" [Chapter 24], and McSweeney "Logic" [Chapter 32].

Notes

- 1 The fact that these ongoing philosophical investigations provide paradigms of the deployment of the notion of ground is noted by (Rosen 2010), (Fine 2012a), (Raven 2012), and (deRosset 2013). See also (Raven, *THIS VOLUME*: Introduction) and the essays in §5 of this volume.
- 2 For more examples, see (Fine 2012a: §1.7); see also (McSweeney, *THIS VOLUME*: Chapter 34) for doubts about some of these cases.
- 3 Hofweber (2009: 268–70) suggests that, though there are all sorts of relations other than ground that induce priority orders, there is no notion of ground for any claim of the form (1) to deploy (Hofweber 2009: 271). Attempts to suggest that there is such a notion, but that it is "esoteric" in the sense that it is available only to those with training in metaphysics, are bound to fail. This is a form of hard eliminativism that is closely related to Daly's, though Hofweber, unlike Daly, appeals to the empirical fact (if it is a fact) that the history of esoteric metaphysical research is littered with failure.
- 4 This is one of the grounds for skepticism suggested by Wilson (2014: 540). She links this criticism to the further claim that grounding claims are uninformative, taken on their own. We will discuss that further claim in §3.2.
- 5 See the discussion in (Smart 1973).
- 6 Koslicki (2015), Turner (2016), and Wilson (2014) suggest skeptical conclusions on this basis; the claim of disunity is disputed by Berker (2017) and Schaffer (2016).

- 7 This last claim is related to the charge that grounding claims on their own are uninformative, which we will discuss in what follows.
- 8 To be clear, Turner does not explicitly avow global skepticism on the basis of this sort of diversity. He does offer a few locally skeptical considerations concerning certain applications of grounding, though it is not clear what connection those considerations have to the diversity he highlights.
- 9 See (Schaffer 2016). The claim that ground bears an intimate relation to explanation is a core and commonplace commitment in the literature; see (Glazier, THIS VOLUME: Chapter 8) for discussion.
- 10 This argument runs into trouble because claims of causal dependence and determination appear to chain with claims of ground to yield some claim of dependence and determination that is clearly akin to ground. So, for instance, from (15) My painting efforts (causally) made the table black (16) The table's being black makes it (by grounding) either black or blue, we may infer that there is "some non-rigged-up sense" in which my painting efforts made the table either black or blue. But enthusiasts of ground typically want to distinguish ground from causation. This point is inspired by remarks by (Schaffer 2016: §4.3) and (Kim 1993).
- 11 Essentially all published skeptics urge this point; see especially (Koslicki 2015) and (Wilson 2014).
- 12 Koslicki (2015) argues against the idea that the disparate varieties of ground she discusses are all species of a common genus. Her argument assumes (though, as I read her, merely for the purposes of argument against (Rosen 2010)) a broadly Aristotelian conception of the relationship between a genus and its species, on which there must be a *real definition* of the species in terms of the genus and some differentia. It is not clear that this Aristotelian conception applies to the actual variety of beetles.
- 13 See (Richardson THIS VOLUME: Chapter 13). Nonskeptical pluralists will hold that we have to be careful about drawing modal conclusions from claims of ground, since there are different kinds of necessity entailed by the different kinds of grounding claims.
- 14 This is a key point in Koslicki's consideration of what she terms the *Central Question*: "What exactly are we supposed to learn from the grounding enthusiast's alleged insight that the . . . connections [cited in a diverse range of cases] are all grounding connections?" (THIS VOLUME: Chapter 11)
- 15 Wilson thinks the answers to the questions of the existence of psychological states, their distinctness from physical states, and their distinctive causal efficacy are crucial to the assessment of competing views. But she presents this as merely her own view of the dialectical situation, and she sensibly suggests that other reasonable views about the basis for assessment of grounding claims and associated views are possible (Wilson 2014: 549). She identifies these three issues as the basis of assessment, then, solely for the sake of illustration. I am happy to play along.
- 16 On EXISTENCE, though the most natural position, perhaps, is to think that (17) entails that Joe's amusement exists, Wilson (2014: 545–6) reads Fine and Rosen as denying the entailment. For what it's worth, I interpret both authors differently from Wilson. On REDUCTION, Wilson (2014: 545, 574) cites (Rosen 2010: 124) as contending that reduction (of a certain sort) is not only consistent with but entails grounding, while many theorists hold that, by the irreflexivity of grounding together with a conception of reduction that requires identification of facts, grounding is incompatible with reduction. Again, it seems to me that there are subtle features of Rosen's view that undercut the appearance of disagreement here. The case of EFFICACY is the most straightforward, since grounding applies both in cases like (17), in which the idea of causal efficacy seems to get purchase, and also in cases in, e.g., ethics or metaphysics of modality, in which the facts in question appear not to have causal powers. Grounding claims seem to say nothing on their own about what causes what.
- 17 One wrinkle is that the work is done by the battery of other relations together with a notion of fundamentality. So the theoretical work that is supposedly done by, say, the claim that a certain baseball *b* exists in virtue of the existence and features of its material proper parts is better done by the claims (i) that *b* has certain material proper parts; and (ii) that those parts are fundamental with respect to *b*.
- 18 On EXISTENCE, though the most natural position, perhaps, is to think that (18) entails that *b* exists, van Inwagen (1990) can be read as denying the entailment, at least so long as (18) is uttered "outside the ontology room." On REDUCTION, the most plausible view is that *b* is not identical to its material proper parts, but there is an interesting and important view, a version of *Composition as Identity*, that defends the identification; see (Wallace 2011) and the essays in (Cotnoir and Baxter 2014). The case of EFFICACY is the most straightforward, since composition claims seem to say nothing on their own about what causes what.

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13

VARIETIES

Kevin Richardson

1. Introduction

Philosophers often make claims like:

- (1) An act is morally right *in virtue of* the fact that it maximizes happiness.
- (2) *What it is* for x to be water is for x to be composed of H_2O .
- (3) The proposition p is true *because* the fact that p obtains.

These claims differ in two ways. *First*: they have different subject matters; (1) is about moral acts, while (2) is about natural kinds. *Second*: they use different locutions to express explanatory dependence; (1) uses “in virtue of”, while (3) uses “because.” Despite these differences, claims (1)–(3) are unified *qua* expressions of grounding dependence.

If the grounding theorist is right, (1) through (3) can be given more perspicuous formulations.

- (1)’ The fact that an act is right is *grounded* in the fact that it maximizes happiness.
- (2)’ The fact that x is water is *grounded* in the fact that x is composed of H_2O .
- (3)’ The fact that p is true is *grounded* in the fact that p obtains.

Instead of using a variety of locutions, we use a single technical term—“grounds”—that we take to express metaphysical dependence. Instead of vaguely gesturing toward some kind of dependence or other, we refer to a specific kind of dependence whose properties we characterize in detail. We have gained clarity by *lumping together* certain dependencies under the umbrella category of grounding. Could we gain further clarity by *splitting up* this umbrella category into different subcategories? The *grounding monist* says: no (Audi 2012: 688; Rosen 2010: 114; Schaffer 2009: 376–377). The *grounding pluralist* says: yes (Richardson 2018, forthcoming; Bennett 2017; Cameron 2015; Fine 2012; Griffith 2014, 2018; Litland 2013; Rettler 2017).

Pluralists think there are varieties of grounding. Claims like (1)’ through (3)’ are not fully perspicuous with respect to what grounds what. Instead of saying that π grounds λ , we should say: π grounds₁ λ , or π grounds₂ λ , etc. Each kind of grounding will have different properties, so it is important to clarify which kind we are talking about.

In this chapter, I describe the state of the art for pluralist theories of grounding. Every pluralist must answer four questions:

- Why should one be a pluralist rather than a monist? (§2)
- What are the varieties of grounding? (§3)
- What is the sense (if any) in which grounding is unified? (§4)
- What is the meaning of “grounds”? (§5)

In what follows, I give various representative pluralist answers to these questions.

Notice that I have omitted the question: what is the difference between monism and pluralism? Here is a simple answer: pluralism says there are multiple kinds of grounding; monism says there is exactly one kind of grounding. Of course, this simple answer is too simple. It overlooks the importantly different ways in which grounding can be pluralistic (or monistic); one person’s pluralism can be another person’s monism, and vice versa. I discuss these senses, but I do not structure my discussion around the conflict with monism. Rather, I focus on explicating views that have—for some reason or other—been deemed pluralist.

For ease of exposition, I make three assumptions: (i) there are grounding relations; (ii) grounding relations relate facts; (iii) grounding relations covary with metaphysical explanations; more precisely: π grounds λ if and only if π metaphysically explains λ . (See the INTRODUCTION and EXPLANATION [Chapter 8] of this handbook for explanation and justifications of these assumptions.) I will flag cases where these assumptions make a significant difference to the views discussed.

2. Why Should One Be a Pluralist Rather Than a Monist?

The main reason to be a pluralist, as opposed to a monist, is that pluralism gives us more fine-grained (i.e., more specific) kinds of dependence. Monism, pluralists argue, gives us an overly coarse-grained kind of dependence.

Grounding skeptics insist on this last point. Wilson (2014: 549) says:

[Suppose] someone claims that the mental is Grounded in the physical. Am I in position to know whether I should agree with them? Not at all . . . Absent further information about the specific grounding relation(s) supposed to be at issue, I am stuck: I am not in position to assess, much less endorse, the claim that the mental is Grounded in—is metaphysically dependent on, nothing over and above—the physical.

Similarly, Koslicki (2015: 340) writes:

[When] presented with a grounding claim of the form, “[p] grounds [q]”, we are left in the dark with respect to many other questions which ideally should be resolved by a sufficiently fine-grained approach to relative fundamentality.

Suppose we know that the mental properties are grounded in the physical properties. Wilson and Koslicki think this information is only informative if we know *additional* facts about the explanatory relationship between the mental and physical properties.

Do physical properties *cause* mental properties? Are mental properties *composed of* physical properties? Do physical properties *realize* mental properties? These questions are all questions

about explanatory dependence, but none of these questions are resolved by the claim that the mental is grounded in the physical.

But if metaphysical grounding is supposed to clarify metaphysical dependence, how could it leave so many of these questions open? Grounding tells us that one thing metaphysically depends on another, but it tells us nothing about *how* or *why* one depends on the other. To answer those specific questions about dependence, we ought to focus on fine-grained dependence relations like causation, composition, realization, etc.

Traditionally, this line of reasoning has been thought to support skepticism about grounding *simpliciter*. The idea is that *every* conception of grounding will be one where grounding is insufficiently fine-grained. Consequently, grounding should be rejected entirely. (For more detailed presentations of this argument, see Wilson (2014), Koslicki (2015), and the SKEPTICAL DOUBTS chapter [Chapter 11] of this volume.)

There is another conclusion we might draw from these arguments, however. Instead of taking them to imply grounding skepticism, we could take them to support grounding pluralism. If the problem is that grounding is insufficiently fine-grained, we might solve the problem by positing fine-grained grounding relations. (Another possibility is to be a pluralist about *metaphysical building* relations, where grounding is one of many such relations. See Bennett (2017) for this proposal.)

Suppose that realization, composition, causation, and other dependencies are varieties of grounding. In that case, grounding claims give answers to *exactly* the kinds of questions that grounding skeptics believe ought to be answered. Moreover, grounding relations are fine-grained in the way that skeptics believe they should be.

The resulting view is not a skeptical one. I am not a skeptic if I believe (a) there are different species of animals and (b) it is usually more helpful to refer to the distinct species of animals rather than the genus *animal*. Similarly, one is not a grounding skeptic just because one believes (a) that there are different kinds of grounding and (b) that it is often more helpful to refer to the specific kinds of grounding than the umbrella category.

Can this proposal dispel every kind of grounding skepticism? Probably not. You might argue—as Wilson and Koslicki do—that the umbrella category of grounding is too heterogeneous. To answer this objection, the pluralist must give an account of what unifies the grounding relations, and such an account will vary from theory to theory.

Independently of whether pluralism converts skeptics, it does address certain worries about the fine-grainedness of grounding. We do not have to be like the poor biologist who cannot distinguish between species of animals. We can embrace the rich diversity of grounding relations. (See the ANTI-SKEPTICAL REJOINDERS chapter [Chapter 12] of this volume for this type of response.)

3. What Are the Varieties of Grounding?

One reason to be a pluralist is that we need more fine-grained varieties of grounding. But what are those varieties, exactly?

Imagine constructing a list of grounding relations. Which relations are on the list? How do we determine which relations get to be on the list? To answer the first question, I will start by answering the second. I outline the three most common ways to individuate—or single out—the varieties of grounding. These ways of individuating grounding relations correspond to different lists of grounding relations.

Disclaimer. I will ignore grounding relations that can be trivially defined. For example, you could say that π *cupcake-grounds* ν if and only if (i) π grounds ν and (ii) π and ν are facts about

cupcakes. If cupcake-grounding is a grounding relation, it is a trivial one. It is a trivial kind of grounding in the same way that *red-headed human* is a trivial kind of human. I only survey varieties of grounding that are (in some intuitive sense) nontrivial. I assume that several distinctions between grounds—like partial/full, weak/strict, immediate/mediate, rigid/nonrigid—feature trivial kinds of grounding. (Though even if some of these kinds of grounding were nontrivial, I would not have the space to discuss them. See Makin (2017) for the distinction between rigid and nonrigid grounding. See Fine (2012) for the other distinctions.)

3.1. *Small-g Pluralism*

Small-g pluralism is the view that the grounding relations are just those relations that are of distinctive interest to metaphysicians. Metaphysicians are in the business of explaining general features of reality. The various grounding relations play a role in such explanations, and that is what makes them part of a unified class.

This account is inspired by a critic of grounding, Jessica Wilson. After criticizing the informativeness of a monistic grounding relation (or big-G Grounding, as she calls it), she proposes that metaphysicians should appeal to more familiar (and more fine-grained) kinds of dependence relations.

Wilson (2014: 539) calls these relations *small-g* grounding relations, and the set of such relations includes

type identity, token-but-not-type identity, functional realization, the classical mereological part-whole relation, the causal composition relation, the set membership relation, the proper subset relation, and the determinable-determinate relation, among others.

Wilson presents these relations for two reasons.

First: metaphysicians have been discussing the small-g relations for decades prior to the contemporary emphasis on metaphysical grounding. They are relations that are clearly of interest to metaphysicians. This gives evidence for—or perhaps even constitutively determines—the fact that these relations are metaphysical dependence relations.

Second: these more specific relations will answer the important questions about dependence that big-G grounding fails to answer. Small-g grounding, not big-G grounding, gives a more perspicuous representation of the informal dependence claims we make.

Some grounding pluralists accept the idea that many of the dependence relations that Wilson discusses are grounding relations; they reject, however, Wilson's skeptical gloss on grounding (Bennett 2017; Griffith 2014, 2018; Rettler 2017).

Instead of seeing the small-g relations as *competing with* the notion of big-G grounding, small-g pluralists see these relations as *extensions of* big-G grounding. Perhaps big-G grounding is a genus of which the small-g grounding relations are species. In that case, it would be wrong to see the small-g relations as competing with big-G grounding; that would be like seeing the species *human* as competing with the genus *animal*.

Once one accepts the general thesis, one needs a canonical list of small-g grounding relations. Below is a list of possible candidates for small-g relation-hood, along with the persons nominating them for candidacy.

- Realization (Bennett 2017; Griffith 2018)
- Mereological composition (Wilson 2014; Bennett 2017)

- Ontological dependence (Rettler 2017)
- Truthmaking (Bennett 2017; Rettler 2017; Griffith 2014)
- Reduction (Rettler 2017)
- Emergence (Bennett 2017)

This is an abridged list. The current list has the shape it does because

(a) it describes, by my lights, the most plausible instances of small-g grounding, (b) it represents the diversity of relations that metaphysicians have proposed for candidacy, and (c) it captures how much overlap exists between the lists of small-g grounding theorists.

This pluralist theory has the advantage of familiarity. The various small-g relations have already been extensively studied by philosophers, so we have some grip on the properties of small-g relations.

One big question about small-g pluralism concerns what (if anything) unifies the class of small-g grounding relations. My definition of small-g pluralism tells us that the small-g relations are united in virtue of being the relations that metaphysicians are interested in, but you may ask, “Why are metaphysicians interested in *those* relations?” From this perspective, the list of relations may seem arbitrary. The apparent disunity of small-g relations is not a problem for Wilson, who is a skeptic of grounding, but it may be a problem for small-g pluralists. (See Berker (2017) for an extensive critique of small-g pluralism.) The small-g pluralist has to explain away or justify the apparent disunity of the class of small-g grounding relations.

3.2. Subject Matter Pluralism

Subject matter pluralism is the view that the varieties of grounding are individuated by the varieties of subject matters involved. To explain this idea, it is best to start with an analogous idea concerning modality.

Philosophers and linguists note that modality has different “flavors”. Consider the following sentences.

- (4) Humans cannot fly.
- (5) Red balls must be colored.
- (6) Everyone must be treated with respect.

In each case, something is said about what must or must not be the case. However, the flavor of “must” varies.

For example, (4) is naturally necessary; it is necessary given the laws of nature of this world. Nonetheless, we can imagine a possible world where the laws of nature are different, where in that world, humans *can* fly. So (4) is naturally necessary but not metaphysically necessary. (5) is metaphysically necessary; red is a color, so every red object is a colored object. Finally, (6) is a normative necessity. From an ethical perspective, everyone must be treated with respect. But as an empirical fact, some people are not treated with respect. (6) is neither naturally nor metaphysically necessary.

Different flavors of modality correspond to different kinds of subject matters. Natural modality concerns natural (or physical) subject matters, normative modality concerns normative subject matters, and so on. These different kinds of modality have different properties.

You might think something similar holds in the case of grounding. Specifically, Fine (2012) thinks there are three basic kinds of grounding—metaphysical, natural, and normative—where those kinds of grounding correspond to parallel kinds of modality. (For more on normative grounding, see the *NORMATIVITY* chapter [Chapter 34] of this handbook.)

Consider the following.

- (7) The fact that Gwen's act is right is grounded in the fact that Gwen kept her promise.
- (8) The mental facts are grounded in the physical facts.
- (9) The fact that x is an individual is grounded in the fact that x is a bundle of properties.

On Fine's view, each statement of ground (7) through (9) might be thought to be associated with a distinctive modal statement.

- (10) It is *normatively* necessary that: if Gwen's act fulfills a promise, then Gwen's act is right.
- (11) It is *naturally* necessary that: if certain physical facts obtain, then certain mental facts obtain.
- (12) It is *metaphysically* necessary that: if x is a bundle of properties, then x is an individual object.

Fine conceives of the different kinds of grounding as differing in strength, where "statements of metaphysical ground are the strictest form of in-virtue-of claim" (2012: 38).

Fine thinks there are only three kinds of grounding, but you might think there are more. Alternatively, you may think there are different kinds.

For example, sometimes people speak of *conceptual* (or *representational*) grounding. Conceptual grounding concerns grounding between concepts or representations as opposed to nonrepresentational objects, properties, or facts.

There are apparent cases of conceptual grounding without metaphysical grounding. Read $\langle p \rangle$ as: the proposition that p .

- (13) $\langle p \rangle$ grounds $\langle p \wedge p \rangle$.

(13) tells us that a proposition p grounds a conjunction with itself. This seems true in some sense. Conceptually, there may be an explanatory asymmetry between $\langle p \rangle$ and $\langle p \wedge p \rangle$. However, as far as the world is concerned, $\langle p \rangle$ and $\langle p \wedge p \rangle$ are identical; they represent the same states of affairs, after all. So there is no nonconceptual grounding relation between them (assuming that grounding is irreflexive).

There might also be cases of metaphysical grounding without conceptual grounding. As Chalmers (2012: 453) points out: "a claim about a table might be metaphysically grounded by microphysical truths about charge, spin, and the like, but it is not plausibly conceptually grounded in those truths." The concept *table* does not have all of the relevant truths about the microphysical structures of tables packed into it. If it did, we could do physics from the armchair.

Or so one might argue. Unsurprisingly, philosophers disagree about the exact properties of conceptual grounding and its relation to worldly grounding. For more discussion, see Correia and Skiles (2019), Smithson (forthcoming), and the GRANULARITY chapter [Chapter 15] of this handbook.

If conceptual grounding is another variety of grounding, it may have an associated form of necessity: conceptual necessity.

- (14) Tables are composed of atoms arranged table-wise.
- (15) Bachelors are unmarried males.
 - a. It is conceptually necessary that: if x is a bachelor, x is an unmarried male.

(14) is metaphysically necessary but not conceptually necessary. (15) is metaphysically *and* conceptually necessary, where the relevant conceptual necessity is (15-a).

There may be further kinds of grounding. Epistemic grounding, social grounding, mathematical grounding—for each ontological category, there might be a kind of grounding associated with it.

Another possibility is that the Finean idea—that the varieties of grounding are associated with varieties of necessity—is false even though subject matter pluralism is true. For example, normative grounding might have a special normative force, but this force may not correspond to a distinctive kind of necessity. In general, you might accept grounding pluralism without accepting modal pluralism.

3.3. *Explanation-Theoretic Pluralism*

Explanation-theoretic pluralism is the view that the varieties of grounding are individuated by the varieties of metaphysical (or grounding) explanation. To explain this view, I will briefly describe a problem that it has been thought to resolve.

Consider the following principle.

TRANSITIVITY: If ϕ partially grounds ψ , and ψ partially grounds χ , then ϕ partially grounds χ .

Grounding theorists debate about whether this principle holds. The friends of nontransitivity offer putative counterexamples to TRANSITIVITY. Those who accept TRANSITIVITY try to explain away those examples. (For more details, see the STRICT PARTIAL ORDER chapter [Chapter 17] of this handbook.)

For example, Schaffer (2012) asks us to imagine a sphere O with a maximally determinate shape S , where O has a small dent D in it. Then the following grounding claims seem plausible. (Read $[q] < [p]$ as: the fact that p partially grounds the fact that q .)

DENT-TO-SHAPE: $[O \text{ has shape } S] < [O \text{ has dent } D]$

SHAPE-TO-SPHERE: $[O \text{ is nearly spherical}] < [O \text{ has shape } S]$

DENT-TO-SHAPE is plausible because the dent in the sphere is partially responsible for its shape. SHAPE-TO-SPHERE is plausible because the shape of O contributes to its being nearly spherical. But if TRANSITIVITY holds, then so does the following.

DENT-TO-SPHERE: $[O \text{ is nearly spherical}] < [O \text{ has dent } D]$

Intuitively, DENT-TO-SPHERE is implausible. The near-sphericity of O doesn't need the dent; even stronger: the near-sphericity of O exists *in spite* of the dent, not because of it. The dent doesn't make a metaphysical contribution to the near-sphericity of O .

So it seems that DENT-TO-SHAPE and SHAPE-TO-SPHERE are true, but DENT-TO-SPHERE is false. If this is so, we have a counterexample to TRANSITIVITY.

The friends of transitivity are not convinced. They think the dent does make a contribution to the near-sphericity of O . The dent tells us the *way in which* O is nearly spherical. The near-sphericity of O is surely grounded in the way O is nearly spherical.

This is where the debate begins. The standard approach in the literature is to pick a camp—either you think grounding is transitive or nontransitive—and then argue about what these kinds of examples show. The presupposition of the debate is that there is a single kind of grounding being discussed and that this kind of grounding will be either transitive or nontransitive.

But what if this presupposition is false? What if there are multiple grounding relations at issue here? In that case, it is possible that one kind of grounding is transitive while another kind is nontransitive. This proposal does not neatly fit into the standard ways of thinking of grounding—as either uniformly transitive or uniformly nontransitive. Rather, it constitutes (what I call) a Third Way position.

Litland (2013) and Richardson (2018) defend a Third Way position. They argue that there are different kinds of grounding corresponding to different kinds of metaphysical explanation. And depending on what kind of metaphysical explanation we have in mind, we should think grounding is either transitive or nontransitive.

Grounding is nontransitive if we think grounding corresponds to why-explanation. The fact that *O* has a dent doesn't explain *why* *O* is nearly spherical. Call this kind of grounding *why-grounding* ($\prec_{why}$).

Nonetheless, the dent does tell us *how* (or the way in which) *O* is nearly spherical. The object is nearly spherical, in part, by having a certain dent. Grounding is transitive, then, if we think grounding corresponds to how-explanation. Call this kind of grounding *how-grounding* ($\prec_{how}$).

The revised account of the examples may look like this. (Read $[q]$

$\prec_{why+how}$ $[p]$ as: $[q] \prec_{why} [p]$ and $[q] \prec_{how} [p]$.)

DENT-TO-SHAPE_{*w+h*}: $[O \text{ has shape } S] \prec_{why+how} [O \text{ has dent } D]$

SHAPE-TO-SPHERE_{*w+h*}: $[O \text{ is nearly-spherical}] \prec_{why+how} [O \text{ has shape } S]$

DENT-TO-SPHERE_{*h*}: $[O \text{ is nearly-spherical}] \prec_{how} [O \text{ has dent } D]$

This proposal preserves the transitivity of how-grounding but not the transitivity of why-grounding. The moral of the story is that the putative counterexamples to TRANSITIVITY reveal the existence of two kinds of grounding. Once we distinguish between these kinds, we can dissolve the traditional debate about whether grounding is transitive.

Litland (2013) proposes how- and why-grounding mainly as a way to make sense of the debate about the transitivity of grounding. Richardson (2018) gives a more detailed account of the two kinds of grounding and argues explicitly for grounding pluralism.

In both cases, the view sketched is an *explanation-theoretic* pluralist theory, a theory where the kinds of grounding are individuated by the kinds of metaphysical (or grounding) explanation. The focus on explanation stems from a more general view about the relationship between grounding and explanation.

In general, we motivate the existence of grounding by appealing to its explanatory character. Metaphysical explanation is seen as a guide to grounding and vice versa. As an extension of this strategy, we might motivate the existence of multiple kinds of grounding by appealing to the multiple kinds of metaphysical explanation it provides. (See the EXPLANATION chapter [Chapter 8] for more considerations on grounding explanation.)

Here, I have identified how-explanation and why-explanation as two possible kinds of metaphysical explanation, but there may be other kinds of explanation. Richardson (forthcoming) argues that there is a kind of grounding corresponding to “what-it-is” explanations.

For example, sometimes metaphysical explanations come in the form of specifying *what it is* for things to be case. What is it for something to be water? For it to be H₂O. What is it for someone to be a bachelor? For someone to be married. This kind of what-it-is explanation has been studied under various names—real definition, the just-is relation, generic identity—but it may turn out to be a kind of grounding explanation (Correia 2017; Dorr 2016, 2005; Linnebo 2014; Rayo 2013; Rosen 2015; Correia and Skiles 2019). Given the explanation-theoretic view,

it follows that metaphysical what-it-is explanation corresponds to a form of grounding. Richardson (forthcoming) calls this variety of grounding *what-grounding*.

Another possible kind of explanation is *difference-making* explanation. The intuitive idea behind the dented sphere example is that the dent does not make a positive difference to the near-sphericity of *O*; worse: it makes a negative difference. Krämer and Roski (2017) give a precise formulation of difference-making using a set of counterfactuals. They then argue that good grounding explanations involve (what they call) difference-making grounds. Richardson (forthcoming) identifies difference-making grounding with why-grounding.

Explanation-theoretic pluralism is especially attractive to *unionists*, who identify grounding with metaphysical explanation (Dasgupta 2017; Litland 2013; Miller and Norton 2017; Richardson 2018, forthcoming; Thompson 2016, 2018). If grounding *just is* metaphysical explanation, then it follows that the varieties of metaphysical explanation are varieties of grounding.

For *separatists*, who reject a tight link between grounding and metaphysical explanation, explanation-theoretic pluralism is especially *unattractive* (Audi 2012; Maurin 2018; Schaffer 2012, 2016; Trogon 2013). One might agree that there are different kinds of metaphysical explanation without thinking there are different kinds of grounding. (See Krämer and Roski (2017) and Schaffer (2012) for views like this.)

4. What Is the Sense (If Any) in Which Grounding Is Unified?

When considering varieties of grounding, we often encounter two different questions about the unity of grounding.

The first question is: what do the varieties of grounding have in common? This question can only be answered by appealing to what those varieties are, exactly. For example, if the small-g pluralist is correct, then the varieties of grounding have the common feature of being of interest to metaphysicians. This answer may or may not be satisfactory, but it counts as a possible answer to this unity question.

Another question is: in what sense (if any) is grounding unified? This is not a question about what the different kinds of grounding have in common. Rather, it is a question about the kind of unification involved; it is a question about how the umbrella notion of grounding—call it *generic grounding*—relates to the notions that fall under it.

There are several options here.

- **NOMINALISM:** There are multiple basic grounding relations that merely resemble one another.
- **FUNCTIONALISM:** To be a grounding relation is to play a certain functional role. Or: grounding is a functional kind.
- **DEFINITION-THEORETIC:** There is a fundamental grounding relation that defines each nonfundamental grounding relation. Or the varieties of grounding define the generic kind of grounding.
- **DETERMINABLE-DETERMINATE:** Grounding is a determinable that has specific ways of grounding as determinates.
- **GENUS-SPECIES:** Grounding is a genus of which its varieties are species.

I will discuss the first three options, since they are the most commonly discussed.

4.1. Nominalism

Grounding pluralism is often associated with the idea that grounding is *disunified*. The most obvious sense in which grounding could be disunified is if there were no generic grounding

relation; rather, there are only multiple, equally basic varieties of grounding relations. I call this view *nominalism*.

For example, a nominalist must say there is why-grounding and how-grounding but no such thing as grounding *simpliciter*. There is no generic grounding relation. Each grounding relation is equally basic, and there is no common kind of grounding—generic grounding—that they fall under. (For the view that there is a generic kind of grounding defined by specific kinds of grounding, see §4.3.)

Nominalism abandons the project of unifying the varieties of grounding, and you might think this is a positive consequence of the view. Suppose you are a small-g pluralist who thinks grounding is more disunified than not. Instead of trying to unify the small-g relations, you might be better off defending their disunity. On this view, the grounding relations seem disunified because they are.

One possible problem with nominalism is that it makes it difficult to distinguish between the nominalist and the skeptic. The skeptic says, “There is no relation of grounding, only a bunch of different relations that vaguely resemble one another”. But this seems to be exactly what the nominalist thinks! So either (a) purported skeptics are actually pluralists or (b) purported pluralists are actually skeptics. Either conclusion will be surprising.

Another view—closely related to nominalism—is that generic grounding exists but it is defined in terms of the specific kinds of grounding. So generic grounding might exist but how- and why-grounding are more fundamental; generic grounding is derivative upon how- and why-grounding. I discuss this kind of view in §4.3.

Nominalism is not a view widely held (if at all) by grounding pluralists, but it is certainly a view casually attributed to pluralists. Pluralism is often thought to imply disunity of some sort, and nominalism clearly captures the idea that grounding is disunified.

4.2. Functionalism

Functionalism is the view that grounding relations are unified by the *functional role* of grounding, where a functional role is akin to a job description. To understand the general idea of functionalism, let us first consider more familiar functionalist accounts.

What are bridges? They are made of different materials and they often take very different forms. Nonetheless, bridges have their functional role in common. Bridges are walkable, they surmount obstacles, they help you get from one location to another, etc. *Bridge* is a functional kind. Bridges are realizers of the functional property of being a bridge.

In philosophy, functionalism is most common in the philosophy of mind. To give an account of mental states, you might define mental states via the role they play. For example, a mental state *P* is a pain state just in case being in *P* tends to cause its experiencer to think something is wrong with their body. Pain is defined by the pain-role. Pain states are *realizers* of the pain-role.

In both cases, we specify constraints that tell us when something counts as an instance of the relevant kind. For grounding functionalism, we need to specify the grounding role. Rettler (2017: 13–14) writes:

As a first pass, the job description for the grounding role is something like: relates the fundamental to the non-fundamental, relates the relatively more fundamental to the relatively less fundamental, lays out the structure of the world, says which things depend on which other things, explains why something exists, and explains why some thing has a property.

For Rettler, the relations that satisfy this functional role include ontological dependence, truth-making, reductive analysis, and metaphysical explanation. There are doubtless more grounding relations that satisfy this functional role.

Different pluralists may disagree about the functional role of grounding. In any case, functionalism tells us that grounding is unified in the sense that the grounding relations realize a common functional role. Grounding—or the property of being a grounding relation—is the functional kind, while the realizers of this kind are grounding relations.

This view provides a particularly natural way of making sense of small-g pluralism. Big-G Grounding is the kind; small-g grounding relations are the realizers. To respond to the objections of disunity, then, the small-g pluralist may only need to specify a functional definition of grounding. This is still a challenging task, but it is a more manageable one, given functionalist resources.

If one thinks there are no grounding relations, functionalism might be better formulated in terms of concepts. The general concept of grounding would be a functional concept. This concept either (a) refers to relations that realize the concept—but are not, strictly speaking, grounding relations—or (b) the concept is realized by different concepts of grounding. The details are tricky, but the point is that functionalism can accommodate different ways of conceiving of grounding.

4.3. *Definition-Theoretic*

On the *definition-theoretic approach*, we unify grounding relations by either (a) defining the grounding relations in terms of generic grounding or (b) defining generic grounding in terms of the varieties of grounding.

What is a *definition*? A definition could be some kind of grounding relation, although this would immediately raise questions about the grounding of grounding. For the sake of simplicity, I will assume that definitions correspond to other notions of metaphysical analysis (like reduction, perhaps). We just need some notion of defining one thing in terms of another.

Given a notion of definition, we are now in position to consider the first possibility—that the various grounding relations can be nontrivially defined in terms of a single grounding relation. To make this proposal concrete, let us consider a version of subject-matter pluralism. On the Finean view, there are three kinds of grounding: natural, normative, and metaphysical.

Suppose we think there is a generic grounding relation. How do we define the specific groundings relations in terms of generic grounding? Here is one proposal.

ϕ metaphysically grounds ψ just in case (a) ϕ generically grounds ψ and (b) it's metaphysically necessary that: if ϕ , then ψ .

To define the other varieties of grounding, we simply substitute in the various types of grounding and necessity. Call this the *top-down* approach to unifying grounding; we use generic grounding to define its varieties.

Fine (2012: 39) briefly considers and rejects this proposal. He proposes a counterexample. Consider the following facts.

(16) A specific act x maximizes happiness.

(17) x is right.

(18) x is either right or not right.

Fact (18) is generically grounded in (17), which you may think is normatively grounded in (16). Given the transitivity of generic grounding, (18) is generically grounded in (16). Notice that it's metaphysically necessary that: if (16), then (18). (Why? Because (18) is itself metaphysically necessary.) Given the presence of generic grounding and metaphysical necessity, it follows from our reductive definition that (16) metaphysically grounds (18). However, you might not think this is a genuine case of metaphysical grounding.

The intuition is that the varieties of grounding are distinguished by more than the presence of different kinds of necessity. Rather, there is something different about the flavor or force of grounding itself that changes. One way to accommodate this sense is to build it into the definition of metaphysical grounding. For example:

ϕ metaphysically grounds ψ just in case ϕ generically grounds ψ with metaphysical force.

In this case, you have a definition of metaphysical grounding parting in terms of generic grounding, but the definition is less trivial than the previous one. The notion of *metaphysical force* may be primitive, and even if it is not, it will require substantive theorizing to unpack it.

So far, I have considered definitions of the kinds of grounding in terms of generic grounding. An alternative definition-theoretic approach would be to define generic grounding in terms of its kinds.

Consider the following definition.

What it is for ϕ to generically ground ψ is for either ϕ to normatively ground ψ , ϕ to naturally ground ψ , or ϕ to metaphysically ground ψ .

On this view, the varieties of grounding are fundamental. Generic grounding is derived from its varieties. This view is attractive if you think there can be no definition of the varieties of grounding, yet you still want to speak of a generic notion. Call it the *bottom-up* approach to unifying grounding; we are defining generic grounding in terms of its varieties.

Berker (2017) argues that the bottom-up approach to defining grounding cannot succeed. Suppose metaphysical and normative grounding are both transitive. (Assume these claims are of partial ground.)

TRANSITIVITY_{M+N}: If ϕ metaphysically/normatively grounds ψ , and ψ metaphysically/normatively grounds χ , then ϕ metaphysically/normatively grounds χ .

It is plausible that mixed versions of transitivity hold, linking metaphysical and normative grounding. For example:

TRANSITIVITY_{N→M}: If ϕ normatively grounds ψ , and ψ metaphysically grounds χ , then ϕ (in some nontrivial sense) grounds χ .

But if the varieties of grounding are fundamentally distinct, why does TRANSITIVITY_{N→M} (and other mixed logical principles) hold? This result is too much of a miracle. The best explanation of these principles is that there is a single generic grounding relation by which the other grounding relations are defined. Or so the argument goes. (See Litland (2018) for a pluralist response to Berker (2017).)

5. What Is the Meaning of “Grounds”?

If there is only one kind of grounding, the meaning of “grounds” is straightforward. “Grounds” means grounds. If there are multiple grounding relations, it is unclear what to say about the contents of our grounding claims. While the semantics of “grounds” does not matter to the varieties of grounding themselves, it does matter to grounding theorists who want to be clear about their claims. I cannot give a comprehensive survey of the various semantic options for pluralists, but I will correct a common confusion about the semantic consequences of grounding pluralism.

Sometimes people suggest that pluralism implies that grounding-talk is *ambiguous*. The paradigm of semantic ambiguity is the word “bank.” The word refers to rivers and financial institutions. The word has at least two meanings, and those meanings are unrelated to one another. So if grounding-talk is ambiguous, the word “grounds” refers to disparate relations. This view is sometimes expressed as the idea that “grounds” is not univocal (i.e., it is equivocal). (See Trogon (2013), Correia and Schnieder (2012: 30), and Tahko (2013).)

The ambiguity model will not work for most grounding pluralists, however. This is because most grounding pluralists think the different utterances of “grounds” have something in common—namely, grounding. Again, If grounding is unified and grounding-talk refers to multiple kinds of grounding, this cannot be a case of ambiguity.

Ambiguity is sometimes mistaken for context sensitivity. Consider modal language. “Must” can have a metaphysical or ethical import, depending on the context. The context sensitivity of modal language pairs well with modal pluralism. Similarly, the context sensitivity of grounding-talk pairs well with grounding pluralism. Suppose someone says (19).

(19) The physical facts ground the mental facts.

- (a) The physical facts naturally ground the mental facts.
- (b) The physical facts metaphysically ground the mental facts.

Depending on the context, the meaning of (19) will be either (19-a) or (19-b). Let *contextualism* be the view that grounding-talk is context-sensitive in this way. (See Cameron (2015) for an Aristotelian contextualist theory.)

So grounding pluralists are not forced into thinking grounding-talk is ambiguous; they could be contextualists about grounding-talk. More generally, pluralists are not forced into *any* particular semantic theory. You could think that grounding-talk isn’t context-sensitive; rather, it either (a) refers to a generic notion of grounding, or (b) quantifies over the behavior of some (or all) grounding relations. There are subtle versions of each of these semantic theories.

Ultimately, “grounds” is a relatively new technical term, and it is unclear how to evaluate the semantics of such terms. Instead of determining how we *actually* use the term, it may be more useful to determine how we *ought* to use the term. The important semantic project may be more prescriptive than descriptive.

6. Conclusion

I have surveyed the current state of grounding pluralism. My discussion has centered around four questions about grounding pluralism, but there are other important questions about pluralism.

What are the logical relations between the various grounding relations? Are there specific applications of grounding pluralism to first-order metaphysical debates? If grounding relations do not covary with grounding explanations, what might grounding pluralism look like? Are there other viable semantic theories for pluralism?

I leave these questions for future research. The purpose of the current chapter has been to survey the varieties of grounding. The hope is that knowledge of these varieties will sharpen our debates about what grounds what. As Fine (2002: 281) puts it:

Philosophers like to think of themselves as having found the key to the universe. But where there are many locks, it should be recognized that we may have need of many keys.

Related Topics

- Explanation [Chapter 8]
- Skeptical Doubts [Chapter 11]
- Granularity [Chapter 15]
- Strict Partial Order [Chapter 17]
- Normativity [Chapter 34]

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PART III

Logic and Structure



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INTRODUCTION

Michael J. Raven

In the literature on ground, characterizations of it often begin with statements of its various logical or structural features. There are several important rationales for this.

One rationale concerns the explanatory or determinative aspects of ground. Many of ground's logical or structural features are often assumed to derive directly from its explanatory or determinative aspects. Investigating these features is therefore not an idle pursuit of formal technicalities. It is, instead, an attempt to characterize what ground is at its very core. To illustrate, if explanations cannot be circular and if ground is or provides a kind of explanation, then one would expect ground itself to be acyclic. Or, for another illustration, if explanations must come to an end and if ground is or provides a kind of explanation, then one would expect there to be no infinitely descending chains of ground. There are other illustrations too of the interplay between the logical or structural features of ground and its explanatory character.

A second rationale concerns the logical coherence of ground. It was discovered in the early 20th century that the so-called naïve conception of set was paradoxical. This had profound implications for logic, mathematics, and various ambitious philosophical projects concerning their foundations. One lesson many learned from this harrowing episode is that seemingly innocuous notions may lead to logical incoherence. One might then wonder whether ground is such a notion. As it turns out, there are various puzzles of ground that threaten to lead to paradox. Ultimately, we should like to know how to react to these threats. But perhaps we may resist alarmist reactions, at least for now. The liar paradox threatens the logical coherence of truth. The sorites paradox threatens the logical coherence of vague terms, which likely includes most of the terms we ordinarily use. But few have thought that the correct response to these paradoxes was the alarmist reaction of dispensing with truth or vague terms in general. Perhaps we may also resist such an alarmist reaction when faced with the puzzles of ground.

The chapters in **Part III** focus on aspects of ground's logic or structure, especially those that concern its explanatory or determinative aspects, or threaten to lead to paradox.

14. Francesca Poggiolesi's "Logics" explores in a detailed yet accessible way the different logics of ground that have been proposed in recent years. Logically, the notion of ground can be formalized in three different ways: as an operator, as a predicate, or as a meta-linguistic relation. The structure of the chapter follows this tripartite distinction and aims at evaluating the virtues of each approach.

15. Fabrice Correia's "Granularity" focuses on the question of how fine-grained ground is. He puts forward a precise formulation of the question and explores the possible answers by means of notions that he defines for the purpose at hand and of general principles framed in terms of these notions. Along the way, Correia clarifies the important distinction between worldly and representational conceptions of ground and shows which role it plays in the granularity issue.
16. T. Scott Dixon's "Infinite Descent" focuses on two questions about the infinite descent of ground: what it is and whether it is possible. Dixon notes that infinite descent is possible in both infinitist and foundationalist metaphysics, given a plausible characterization of metaphysical foundationalism. He subsequently turns the discussion toward the question of whether nonfoundationalist infinite descent is possible. He then surveys a number of arguments, some in favor of the possibility of nonfoundationalist infinite descent and some against it.
17. Naomi Thompson's "Strict Partial Order" explores aspects of the logic of ground. A strict partial order is transitive, irreflexive, and asymmetric. Whilst it is generally assumed that ground exhibits these properties, this has consequences for other aspects of the study of ground, and so each assumption has been challenged in the literature. The present essay critically discusses those challenges, and explores reasons for thinking that ground forms a strict partial order in the first place.
18. Stephan Krämer's "Puzzles" discusses a range of paradoxical arguments, leading from apparently plausible, broadly logical assumptions about ground to unacceptable conclusions. It begins by describing the different kinds of arguments of this sort that have been identified in the literature, and then surveys the solutions to these puzzles that have so far been proposed, indicating their respective advantages as well as the main challenges they face.

14

LOGICS

Francesca Poggiolesi

Introduction

The founding fathers of modern logic, illustrious scholars such as Frege, Hilbert, or Russell, used this discipline to answer mathematically foundational issues; however, starting from the 1960s, logic has also been fruitfully used to deal with key philosophical notions, such as necessity, knowledge, time, probability, vagueness, and many others. But what exactly does it mean to study a philosophical notion from a logical point of view? It standardly means to use a predicate or a sentential operator to denote the philosophical concept under investigation and then to formulate and discuss axioms¹ that adequately characterize the predicate or the operator. For the sake of clarity, let us consider the example of knowledge, which is commonly formalized via the operator K (e.g., see Hendricks and Symons (2015)). Amongst the axioms that describe the operator K , one of the most famous is the axiom T , which is written as $K(A) \rightarrow A$, where A is a formula, and can be read as “if something is known ($K(A)$), then it is true (A)”. The axiom T does not tell us what someone knows, but individuates a distinctive feature of knowledge, namely that it is factive: if someone knows something, then that something is true.

Axioms characterizing a certain philosophical notion are put together to constitute *formal systems*, and each formal system can be looked at from the two following perspectives:

- from a syntactic point of view, one analyzes which theorems are *provable* in a formal system $\mathbf{S}$. We write $\mathbf{S} \vdash A$ to denote that the formula A is provable or derivable in, or is a theorem of, the formal system $\mathbf{S}$.
- from a semantic point of view, one analyzes which sentences are valid (i.e., *true* in every situation, or more technically in every *model*) in the semantics corresponding to the system $\mathbf{S}$. We write $\mathbf{S} \models A$ to denote that A is valid in the semantics corresponding to the system $\mathbf{S}$.

The famous theorems of validity and completeness establish an equivalence between the semantic level and the syntactic level: formulas that are valid (true in every model) in the semantics corresponding to the system $\mathbf{S}$ are all and only the theorems provable in $\mathbf{S}$.

Amongst the several philosophical notions that have been logically taken into account in the last 50 years, two play a prominent role: the notions of *provability* and *truth*. The peculiar logical role of these notions is clearly indicated by the fact that both have been formalized into two

quite different ways. On the one hand, they have been treated just as the other philosophical notions, namely they have been formalized via predicates or sentential operators. This kind of treatment has produced impressive results: to mention a few, the formalization of truth via the predicate *T* has given rise to the wide literature concerning paradoxes of truth (e.g., see Gupta (1982) and Kripke (1975)); the formalization of provability via the predicate *Bew* (from the German *Beweisbar* for *provable*) has given rise to Gödel's incompleteness theorems (1986, 1990), while the formalization of provability via a sentential operator has given rise to the modal logic of provability (Boolos (2010)).

But truth and provability have also been treated as meta-linguistic symbols: as we have said, the metalinguistic symbol $\vdash$ —denotes provability in a formal system **S** and as such it has been widely studied by the cornerstone works of Frege, Russel, Hilbert (see van Heijenoort (1967)) and Gentzen (1969); the symbol $\models$, meaning truth in a certain semantic model, represents the basis of the notion of validity and has been central since the work of Tarski (1936). The importance of the treatment of truth and provability as metalinguistic symbols can hardly be overestimated: in this respect truth and provability are the two milestone concepts on which logic itself lies and the two main tests to which any logical formal system is confronted.

Let us finally come to the notion of *grounding*, which is the central notion of this chapter and is standardly described as a relation amongst truths (or facts) that is noncausal and explanatory in nature. The concept of grounding has been long neglected or forgotten in the history of logic (exceptions can be found in this book, as in Chapter 5 by Roski). This fact is all the more astonishing once we realize that grounding seems to cover the same special logical role as the kindred notions of truth and provability: like its cousin concepts, grounding can be fruitfully formalized into two different ways, namely as (i) a predicate or sentential operator or as (ii) a metalinguistic relation. This double formalization, which testifies to the importance of grounding as a logical notion, will structure this chapter. The next section will be dedicated to studies of grounding under the perspective of (i), which is the most developed in the contemporary literature; while Section 3 will focus on approaches adopting perspective (ii).

In order to slightly constrain the scope of this survey, let us emphasize that here we will mean by 'grounding' a relation from many to one, which cannot be iterated,² and embraces a representational conception of grounding (see also Chapter 15 of this book by Correia).³ Most of the work developed in the formalization of grounding concerns such a notion. Moreover, we will mainly focus on the propositional logic of grounding, leaving aside the links between grounding and quantifiers: this choice is motivated by the fact that the links between quantifiers and grounding is already extensively treated by McSweeney in Chapter 32.

Finally, let us emphasize that in the last decade, the literature on the logic of grounding has blossomed: there is a wide range of studies that vary in their objects, conceptions, and features. In order to clarify the presentation and classification of all these studies, we first introduce some distinctions relative to the notion of grounding.

As underlined by Fine (2012b) and Correia (2014), grounding is thought to come in various types; amongst those, there are *metaphysical* grounding, *logical* grounding, *conceptual* grounding, and *natural* grounding. For any type of grounding, three further distinctions apply: the one between full, complete, and partial grounding, the one between immediate and mediate grounding, and the one between strict and weak grounding.

According to Fine (2012b), *A* is a partial ground of *C* if *A* on its own or together with some other truths is a ground of *C*. Thus, given that *A* and *B* are the full ground of *AAB*, each of *A* and *B* will be a partial ground of *AAB*. The notion of full ground is never explicitly defined, but it is suggested that *A* is a full ground of *C* if the truth of *A* is sufficient to guarantee the truth of *C*. Bolzano's distinction (cf. Chapter 5 by Roski) between complete and partial ground is slightly

different. Following the analysis of Sebestik (1992) and Tatzel (2002), for Bolzano the multiset (a multiset is a set where the number of occurrences of the same formula counts) of all, and only, those truths each of which contributes to ground the truth C is a complete ground of C . On the other hand, each of the truths that compose the complete ground of C , as well as each strict submultiset of them, is said to be a partial ground of C . It seems that Bolzano and Fine are aiming at similar, if not the same, distinction, although the way they draw the line between the two concepts is different. For Fine, a partial ground of a truth C can also be a full ground of C , while for Bolzano, this can never be the case, since partial and complete grounds are two disjoint concepts.

Let us now move to the distinction immediate and mediate. If we could describe this distinction with proof-theoretic terminology, we would say that immediate grounding corresponds to a single (irreflexive) proof-step, while mediate grounding corresponds to a sequence of several steps of immediate grounding. In other terms, while immediate grounding is a relation that does not seem to be further reducible, mediate grounding is definable as the transitive closure of immediate grounding. As for the last distinction, strict grounding does not allow a truth to ground itself (it is irreflexive), while grounding in the weak sense allows, in fact requires, a truth to ground itself (it is reflexive).

2. Grounding as a Predicate or as a Sentential Operator

In this section, we introduce those logics of grounding that formalize grounding either as a predicate or as a sentential operator. The pros and cons of each approach are quite straightforward: the predicate approach provides a richer and stronger framework, but it is ontologically and formally more demanding; the operator approach is notable for its simplicity but might be more limited in its expressive power.

2.1. Grounding as a Sentential Operator

The first and most comprehensive contemporary logical study of the notion of grounding has been developed by Fine (2012a, 2012b) in two cornerstone papers in which he formalizes the notion of metaphysical grounding in most of its variations, namely full and partial and weak and strict. In what follows, we will take Fine's logic as the reference logic for grounding as a sentential operator and as the benchmark against which we present other approaches.

2.1.1 Fine's Logic

Language. Fine adds to the first-order classical language four symbols denoting the four notions of weak full, weak partial, strict full, and strict partial grounding. Accordingly, four novel types of logical formulas can be constructed in this language. Assuming that $M, N, \dots$ denote sets of formulas and $A, B, C, \dots$ formulas, we have:

- $M \leq B$, M is a weak full ground for B ,
- $A \preceq B$, A is a weak partial ground for B ,
- $M < B$, M is a strict full ground for B ,
- $A \prec B$, A is a strict partial ground for B .⁴

Syntax. In this given language, Fine formulates axioms and rules that describe the four notions just introduced. These axioms and rules are divided between *pure* and *impure*.⁵ pure rules describe the structural features of the several notions of grounding, while the impure rules describe the

relationships between grounding and the other logical connectives and quantifiers. Let us start describing the former rules.

Pure (or structural) rules. The pure rules for grounding can be divided into two sets: those describing the relations between the four symbols introduced (they are called *Subsumption rules*), and those describing the properties that each symbol enjoys. As for the former, these outline the following relations:

- from strict grounding claims, one can infer corresponding (full or partial) weak grounding claims;
- from full grounding claims, one can infer corresponding (strict or weak) partial grounding claims;
- from weak full grounding claims, one can infer strict full grounding claims as long as each of the grounds is a strict partial ground of the conclusion.

As for the structural rules introducing properties, we have that:

- weak full grounding enjoys identity and cut, namely (respectively):

$$\frac{}{A \leq A} \quad \frac{M_1 \leq A_1, M_2 \leq A_2, \dots, A_1, A_2, \dots \leq C}{M_1, M_2, \dots \leq C}$$

- strict partial grounding enjoys noncircularity, namely:

$$\frac{A < A}{\perp}$$

- weak partial grounding enjoys transitivity, which is also enjoyed by the following two combinations

$$\frac{A \leq B, B < C}{A < C} \quad \frac{A < B, B \leq C}{A < C}$$

Given the presence of the transitivity and cut rules, we assume Fine is dealing with mediate notions of grounding, although this is not specified.

Impure (logical) axioms and rules. The impure rules for grounding can be divided into *introduction* and *elimination* rules. The introduction rules are:⁶

- *Conjunction:* $A, B < A \wedge B$.
- *Disjunction:* $A < A \vee B; B < A \vee B; A, B < A \vee B$.
- *Double negation:* $A < \neg \neg A$.
- *Negative conjunction:* $\neg A < \neg(A \wedge B); \neg B < \neg(A \wedge B); \neg A, \neg B < \neg(A \wedge B)$.
- *Negative disjunction:* $\neg A, \neg B < \neg(A \vee B)$.
- *Universal quantifier:* $P(a_1), P(a_2), \dots < \forall x P x$
- *Existential quantifier:* $P(a), E(a) < \exists x P x$, where $E(a)$ expresses the fact that a exists.
- *Negative universal quantifier:* $\neg P(a_1), E(a) < \neg \forall x P x$
- *Negative existential quantifiers:* $\neg P(a_1), \neg P(a_2), \dots, T(a_1, a_2, \dots) < \neg \exists x P x$, where $T(a_1, a_2, \dots)$ stands for the totality claim, i.e., $a_1, a_2, \dots$ are all of the individuals that there are.

The elimination rules are:

- *Conjunction*: from $M < A \wedge B$, one can infer $M \leq \{A, B\}$.
- *Disjunction*: from $M < A \vee B$, one can infer $M \leq A; M \leq B; M \leq \{A, B\}$, where the semi-colon stands for a disjunctive form.
- *Double negation*: from $M < \neg \neg A$, one can infer $M \leq A$.
- *Negative conjunction*: from $M < \neg(A \wedge B)$, one can infer $M \leq \neg A; M \leq \neg B; M \leq \{\neg A, \neg B\}$.
- *Negative disjunction*: from $M < \neg(A \vee B)$, one can infer $M \leq \{\neg A, \neg B\}$.
- *Universal quantifier*: from $M < \forall x P x$, one can infer $M \leq \{T(a_1, a_2, \dots), P(a_1), P(a_2), \dots\}$.
- *Existential quantifier*: from $M < \exists x P x$, one can infer $M \leq \{T(a_1, a_2, \dots), P(a_{11}), P(a_{12}), \dots\}; M \leq \{T(a_1, a_2, \dots), P(a_{21}), P(a_{22}), \dots\}; \dots$ where the $a_{k1}, a_{k2}, \dots$ run through all of the nonempty subsets of the a for which $P(a)$ is true.
- *Negative universal quantifier*: from $M < \neg \forall x P x$, one can infer $M \leq \{T(a_1, a_2, \dots), \neg P(a_{11}), \neg P(a_{12}), \dots\}; M \leq \{T(a_1, a_2, \dots), \neg P(a_{21}), \neg P(a_{22}), \dots\}; \dots$
- *Negative existential quantifiers*: from $M < \neg \exists x P x$, one can infer $M \leq \{T(a_1, a_2, \dots), \neg P(a_1), \neg P(a_2), \dots\}$.

Let us briefly comment on the rules just introduced. As for the *Subsumption* rules, these seem natural and straightforward. As for the structural rules expressing properties enjoyed by the grounding relation, let us underline that according to the orthodox view (see Raven (2013)) (mediate) grounding is an irreflexive, asymmetric, and transitive relation. Whilst transitivity and asymmetry have been widely discussed (see Chapter 17 by Thompson for further details on these discussions), irreflexivity does not seem to have been questioned; on the contrary it has led to some interesting puzzles of grounding (Fine 2010; Krämer 2013) which are reminiscent of puzzles of truth (cf. Chapter 18 by Krämer). Thus it is quite bizarre to find a weak notion of grounding in Fine's logic. On reflection, one realizes that Fine needs such a notion in order to formulate its elimination rules. Let us note that Fine's system does not contain the weakening rule:⁷ we take this to be a nice feature of the system, since all grounds must be relevant to the conclusion, and weakening typically breaks down relevance. Finally, we remark that from the structural rules enumerated, one can easily derive the rule of *Amalgamation* that, according to Fine (2012b), corresponds to a plausible feature of strict full grounding:

$$\frac{M_1 < C, M_2 < C, \dots M_n < C}{M_1, M_2, \dots M_n < C}$$

Chapter 32 by McSweeney contains an extensive discussion of the introduction rules, where the problem of the overdetermination of disjunction, as well as the several features of quantifiers, are taken into account. Let us then dwell on a point that, although important, is often neglected in the current literature. Any discipline, from mathematics to biology, from metaphysics to moral philosophy, seems to have its own internal, explicit or implicit, criteria to test whether a result has been properly carried out. Logic is no exception in this sense, and it enjoys several criteria like the subformula property, the harmony property, or separability property⁸ (e.g., see Poggiolesi (2010a) for an extensive list and discussion). These criteria are meant to ensure a formal as well as a conceptual adequacy (of the rules) of a formal system. Logics of grounding—insofar as they are logics—should be tested on the background of these criteria, but unfortunately this analysis is lacking in the current literature (the only exception

is Poggiolesi (2016b)). This is a regrettable gap. For example, despite their great virtues, Fine's introduction and elimination rules do not seem to enjoy many of the proof-theoretic properties of rules.

Finally, let us comment on the representational approach to grounding, which is present in many logics of grounding discussed in this chapter. According to this perspective, grounding should be sensitive to how facts are represented. However, not every linguistic difference should be relevant to ground-theoretic status. As an example, even for a representationalist, it should be plausible to hold that $A \wedge B$ is ground-theoretically equivalent to $B \wedge A$. As we will see later, while both Correia (2017) and Poggiolesi (2018) propose a general criterion to account for that intuition, in Fine's system, there is nothing of this sort: the logic does not include or rely on any proper theory of ground-theoretic equivalence.⁹ As a result, it cannot be proved in Fine's logic that $A \wedge B$ and $B \wedge A$ have the same grounds, for example.

Semantics. Fine's semantics for its logical system of grounding is a truth-making semantics, based on the idea that a *fact* f makes true a certain sentence A (a wide discussion of truth-making semantics is developed in Chapter 36 by Fine). According to it, a true sentence A can be considered as the set of facts that make it true. In this sense, there is a strong resemblance between facts and possible worlds of Kripke semantics for modal logic (e.g., see Poggiolesi (2010b)): in Kripke's semantics, a true sentence A can also be considered as the set of possible worlds where A is true. Nevertheless, there are three main differences between worlds and facts. The first is that while an accessibility relation holds among worlds of Kripke semantics, there is no accessibility relation among facts. Secondly, worlds are complete, i.e., they settle the truth-value of every proposition, while facts are partial, i.e., they can be considered as parts of possible worlds. Finally facts, differently from worlds, can be fused: given the fact f that Mary is tall and the fact g that Mary is thin, we can have the fused fact $f.g$ that Mary is tall and thin. The set of facts that verifies the sentence A is closed under fusion and is denoted by $|A|$.

We now have all the necessary notions to describe the four symbols for grounding introduced earlier from a semantics point of view.

- M is a weak full ground for B , $M \leq B$ is true, if, and only if, $f_1.f_2 \dots f_n$ verifies B (i.e., $f_1.f_2 \dots f_n$ is a member of the verification set $|B|$), whenever f_1 verifies A_1 , $\dots$, and f_n verifies A_n , for $M = \{A_1, \dots, A_n\}$ (i.e., whenever each of $f_1.f_2 \dots f_n$ is a member of the respective verification-sets of $|A_1|, \dots, |A_n|$).
- A is a weak partial ground for B , $A \leq B$ is true, if, and only if, the set M , composed by A plus other sentences, is a weak full ground of B .
- M is a strict full ground for B , $M < B$ is true, if, and only if, $M \leq B$ is true, and B is not a weak partial ground for any of $A_1, \dots, A_n$ composing M .
- A is a strict partial ground for B , $A < B$ is true, if, and only if, $A \leq B$ is true and $B \leq A$ is not true.

Fine (2012a) shows validity and completeness between the structural rules of his system and the semantics just introduced. As Fine (2012b: 73) says, "this provides some kind of vindication both for the system and for the semantics". However, although Fine attempts to extend this semantics to connectives and quantifiers and provides attractive definitions, semantics cannot be shown to be sound with respect to the logical part of his system (this is partly due to the fact that the semantics cannot discriminate among the semantic values of, for example, A and $AV A$). This gap leaves open the question of the adequacy of Fine's rules.

2.1.2 Other Logics Treating Grounding as a Sentential Operator

Schnieder (2011). The main goal of Schnieder’s approach is to give a rigorous formalization of the sentential expression ‘because’; as a result, he formulates a logic which, following our distinctions, could be read as a logic capturing the notion of strict partial mediate metaphysical grounding.

Let us briefly illustrate Schnieder’s approach. In order to construct a logical system, Schnieder firstly extends the first-order language with the connective ‘because’, and secondly, he adds to the classical natural deduction calculus rules describing the new connective. These rules can be divided into structural and logical rules.

Structural rules. Since Schnieder introduces only one new connective, he does not need to formulate Subsumption rules. On the other hand, he formulates rules that describe the properties of the connective ‘because’: these are factivity, asymmetry, transitivity. As an example of these rules, here are the factivity rules:

$$\frac{A \text{ because } B}{A} \qquad \frac{A \text{ because } B}{B}$$

Logical rules. As for the logical rules, Schnieder only has introduction rules; Schnieder’s introduction rules express the same relations conveyed by Fine’s introduction rules, although in a quite different form (as pointed out before, Correia (2017) correctly notes that Fine’s rules could be rewritten as Schnieder’s ones). For the sake of illustration, here are the rules concerning the grounds of a conjunction (with $i = \{1,2\}$):

$$\frac{A_1, A_2}{A_1 \wedge A_2 \text{ because } A_i}$$

Or these are the rules concerning the grounds of a disjunction (with $i = \{1,2\}$):

$$\frac{A_i}{A_1 \vee A_2 \text{ because } A_i}$$

Schnieder’s system is proved to conservatively extend classical natural deduction. Moreover, Schnieder extends his approach to the quantifiers and underlies that other variations of his system could be obtained by substituting classical logic as the basic framework with other logics.

deRosset (2015). In his 2015 paper, deRosset proposes a new semantics for a subsystem of Fine’s pure logic of grounding. His motivation for constructing such a semantics, which is an alternative to Fine’s one, is philosophical. deRosset seems to be dissatisfied with Fine’s semantics for the following three reasons: (i) the semantic characterization of grounding in terms of truth-making does not seem to capture the metaphysical idea that animates grounding-enthusiasts. (ii) The rule of amalgamation, which is validated by Fine’s semantics, might be questioned. (iii) Fine’s semantics rules out a view according to which facts are “sparse”, namely there are truths that does not relate to any fact. By working with hypergraphs, assuming as primitive the relation of immediate grounding, and taking into account all subsets of the powerset of the domain of facts, deRosset avoids problems (i) through (iii).

Correia (2017). In his 2017 paper, Correia proposes an impure logic of metaphysical grounding, where grounding is treated in its several variations (full, strict and weak, immediate and mediate). Since he deals with the same objects as Fine's logic, let us summarize the main differences between the two. Like Fine, Correia considers both strict and weak grounding; but whilst Fine assumes weak grounding to be more fundamental, Correia does the opposite. Secondly, while Fine also covers partial grounding, Correia only analyzes full grounding. In Fine's logic, there is no treatment of ground-theoretic equivalence; in Correia's logic, the issue is fully treated within the introduction of the symbol $\approx$, which is described by structural rules as well as by introduction and elimination rules that define its interaction with the other connectives. *Syntactically*, Correia's logic shares the same impure grounding rules as Fine's. As for the pure rules, Correia needs to add to Fine's rules those that regulate links between the notion of grounding and the notion of factual equivalence. Also there is a rule taking into account the complexity ingredient (note that an analogous rule can be found in Poggioli (2018)) and rules describing excluded middle and noncontradiction with grounding formulas. As for this latter rule, *Semantically*, Correia's approach relies on three main notions: the notion of structure, the notion of proposition, and the notion of complexity. Each sentence of the language is interpreted into a proposition of a structure via a function. Propositions are ordered from atomic to more complex ones.

2.2. Grounding as a Predicate

The literature on the formalization of grounding as a predicate, although important, is not as widespread and rich as the one on grounding as a sentential operator. Indeed, with the exception of Rosen (2010), where some axioms on grounding as a predicate are suggested, the only papers where the issue is fully developed are Korbmacher (2018a, 2018b). We dedicate the rest of the section to a brief description of his approach.

Korbmacher (2018a, 2018b). The main goal of Korbmacher's approach is to reproduce Schnieder's logic with a predicate for grounding rather than an operator.

Languages. Let L be the language of Peano Arithmetic; let L_T be the language L extended with the truth predicate T ; finally, let $L_{TR}^{<=}$ be the language that extends L_{TR} by the addition of the binary grounding predicate $<=$, which allows us to construct formulas of the form $x <= y$ to be read as x is a strict, partial, mediate grounding of y .

Syntax. Let **PA** be the axiomatic system of Peano arithmetic based on the language L ; **PAT** is the system that extends **PA** with the missing instances of the induction scheme over L_{TR} ; finally **PAG** is the axiomatic system that extends **PAT** with all the missing instances of the induction scheme over $L_{TR}^{<=}$. The predication theory of ground **PG** consists of the axioms of **PAG** plus other axioms describing the logical as well as structural features of grounding (and truth). As for what we might call 'pure axioms of grounding', they describe the following features: irreflexivity, transitivity, and factivity. Just to give an idea of the form of such axioms, irreflexivity is conveyed in the following way: $\forall x \neg(x <= x)$. As for the impure axioms of grounding, these are divided into *upward-directed axioms* and *downward-directed axioms*. Each impure rule of Schnieder's system corresponds to two axioms: one that says that if the grounds are true, then the grounds ground the conclusion, while the other says that if the conclusion is true, then the grounds ground the true conclusion. As an example, consider again conjunction rules in Schnieder's system:

$$\frac{A_1, A_2}{A_1 \wedge A_2 \text{ because } A_i}$$

(with $i = \{1, 2\}$). These rules correspond to the following two axioms:
 $\forall x \forall y (T_i(x) \wedge T_i(y) \rightarrow (x \leq x \wedge y) \wedge (y \leq x \wedge y))$ upward-directed axiom
 $\forall x \forall y (T_i(x \wedge y) \rightarrow (x \leq x \wedge y) \wedge (y \leq x \wedge y))$ downward-directed axiom

The system **PG** is consistent, but it is not a conservative extension of **PA**; it is valid, though not complete, with respect to Schnieder's logic of grounding via a certain translation.

Semantics. Starting from a standard model of the positive theory of truth (see Halbach (2011)), Korbmacher constructs a standard model for the predicational theory of grounding. The model is obtained by the construction of grounding trees, which were firstly introduced by Correia (2014). As Korbmacher underlies, the semantics he constructs strongly relies on the fact that the starting model is standard and would break down if applied to a non-standard model.

The predicational theory of grounding put forward by Korbmacher is proof-theoretically stronger than the operator approach (at least that of Schnieder (2011)). Moreover, it provides a unique framework where grounding and truth co-exist, making it easy to analyze their relations: in this framework, it is possible to formulate an answer to the famous puzzles of grounding put forward by Fine (2010) and Krämer (2013), as well as to study and analyze the ground-theoretic paradoxes of self-reference. Korbmacher suggests several ways to extend his logical system of grounding, namely via the notion of fundamentality or via a theory of definitions.

3. Grounding as a Meta-Linguistic Relation

Grounding as a meta-linguistic relation can be naturally introduced via the notion of proof. At least since Aristotle (1984), two kinds of proof have been distinguished in accordance with the goals of proof (see also Chapter 2 by Malink). On the one hand, there are *proofs-that*, namely proofs by means of which we prove *that* something is true, and on the other hand, we have *proofs-why*, namely proofs by means of which we prove *why* something is true. The distinction is easy to grasp from some real-world examples. From the fact that temperatures are higher in summer than in winter, we can prove that (well-functioning) thermometers are higher in summer than in winter; and conversely from the fact that thermometers are higher in summer than in winter, we can prove that temperatures are higher in summer than in winter. However, we only have a proof-why in the first case: the fact about temperatures explains why—or is part of a proof why—thermometers are higher in the summer. By contrast, the fact about thermometers does not contribute to a proof why the temperatures are higher in the summer: it is not the case that thermometers being higher in summer than in winter explains why temperatures are higher in summer than in winter.

Note that the dichotomy between proofs-that and proofs-why naturally maps onto another central dichotomy for this chapter, namely that between the relation of inference and the relation of grounding. Indeed, as a proof-that is about drawing inferences—in proving that something is true, a proof-that also shows from what that something can be inferred—in an analogous way, proofs-why bare open the links relating a conclusion with its grounds: in showing why something is true, a proof-why also reveals the grounds. Note that an analogous idea, although expressed in different terms, can be found in deRosset (2013).

The dichotomy proofs-that and proof-why has been regarded with a certain scepticism in contemporary logic. Whilst we have seen that the distinction seems very natural informally, certain doubts might arise about its applicability in formal contexts. In particular, while proofs-why have mainly been ignored in 20th century logic, proofs-that have been rigorously formalized thanks to the work of Frege, Russell, Hilbert (see van Heijenoort (1967)), and, perhaps most of all, Gentzen (1969). In this literature, the metalinguistic relation of derivability, which has been mentioned in

the Introduction and is denoted by the symbol $\vdash$, seems to fruitfully capture the logical aspects of proofs-that. This relation is standardly defined by rules, which are divided into introduction and elimination rules. It also finds its linguistic counterpart (at least in certain logics) via a result known as the *deduction theorem*, in the (standard) material implication. This formalization, which is at the same time simple and elegant, has given rise to impressive results, such as cut-elimination, decidability, complexity, interpolation theorems, and to the whole field known today as *proof theory*.

Given the close links between proofs-that and proofs-why and the great developments of the study on proofs-that relying on this (metalinguistic) formalization, some grounding theorists have developed an analogous formalism to capture proofs-why. Therefore, as the logical aspects of proofs-that have been formally captured by the metalinguistic symbol $\vdash$, new metalinguistic symbols have been created for formally capturing the logical aspects of the notion of proof-why. Note that as the formalization of proofs-that naturally leads to a formalization of the notion of inference given their close relation, in the same way, the formalization of a proof-why is also naturally a formalization of the notion of grounding.

As we mentioned at the beginning of this chapter, logic mainly relies on the two key notions of truth (validity) and provability, formalized as metalinguistic symbols. Formalizing (proofs-why and thus) grounding as a metalinguistic symbol puts it on the same level as truth and provability, with consequences for the very basis of logic. As far as we understand, such was the revolutionary role attributed to grounding by the great philosopher and mathematician Bernard Bolzano, who attempted to logically capture the grounding relation (and more generally the notion of proofs-why) through the notion of exact derivability. We refer to Chapter 4 by Roski for a detailed description of Bolzano's work.

3.1. Logics Treating Grounding as a Metalinguistic Relation

In the contemporary literature, there are three works where grounding is associated to a metalinguistic relation: (i) Batchelor (2010), (ii) Correia (2014), and (iii) Poggiolesi (2016a, 2018). In (i), Batchelor introduces the idea of proofs of a special kind that provide the grounds of what is proved but does not systematically develop a logic around this intuition. In (ii) and (iii), logics of grounding as metalinguistic relation are analyzed; however, while in Poggiolesi, this type of formalization is explicitly defended, Correia does not seem to explicitly defend this position.¹⁰ For this reason, we start by presenting (iii) before moving on to (ii).

Poggiolesi (2016a, 2018). Poggiolesi's work is strongly inspired and influenced by Bolzano's insights; according to this perspective, it focuses on the notion of complete and immediate strict logical grounding, since, according to Bolzano, this is the most fundamental type of grounding and thus the appropriate starting point for analyzing the notion. Just as the notion of proof-that is formalized by the metalinguistic relation of derivation and denoted by the symbol $\vdash$, the notion of proof-why (and thus of grounding) is formalized by the metalinguistic relation of *formal explanation* that is denoted by the symbol $\sim$. In accordance, (standard) inferences rules are supplemented with grounding rules, which express an authentic dependence between premises and conclusion and have the following form:

$$\begin{array}{c} M \\ \sim \sim \sim \\ C \end{array}$$

to be read as: the multiset M is the full and immediate strict formal explanation (or ground) of C .

Ground-theoretic equivalence. In Poggiolesi’s account, formulas that are equivalent by associativity and commutativity of conjunction and disjunction have the same grounds. This is rendered in a simple way. Bold letters, such as **A**, **B**, **C**, . . . denote sets of formulas containing all formulas that are ground-theoretic equivalent. Grounding rules contain bold letters. So, for example, the grounding rule for conjunction has the following form

$$\begin{array}{c} A, B \\ \sim \sim \\ \mathbf{A \wedge B} \end{array}$$

which means that not only do the formulas A and B formally explain (or ground) the formula $A \wedge B$, but they also formally explain (or ground) all formulas that are equivalent to $A \wedge B$ by associativity and commutativity of conjunction and disjunction.

Grounding rules for disjunction and negation. In Poggiolesi’s logic, the grounding rules for conjunction, whilst having a specific form due to the use of the metalinguistic symbol (and the account of ground-theoretic equivalence), convey a principle analogous to all the other logics of grounding. Grounding rules for disjunction and negation in Poggiolesi’s logic are more peculiar and worth a brief explanation. The rules for disjunction contain a new element called *robust condition*. Consider the example of the disjunction $A \vee B$, in the situation where the formula A is true. In this case, A is certainly a ground for $A \vee B$; but in order for A to be the complete ground of $A \vee B$,¹¹ it is necessary to specify that B is false, i.e., it is necessary to say that B does not play any role in grounding $A \vee B$ and thus that A does the job on its own. In other words, it is the falsity of B that ensures that or is a (robust) condition for A to be the *complete* ground of $A \vee B$. Thus, A is the complete and immediate formal ground of $A \vee B$ under the robust condition that the negation of B is true. The use of robust conditions solves the problem (as McSweeney puts it; see Chapter 32) of overdetermination of disjunction and is rendered by means of square brackets, as in the following examples of grounding rules for disjunction:

$$\begin{array}{ccc} p, q & [\neg q] p & [\neg p] q \\ \sim \sim & \sim \sim \sim & \sim \sim \sim \\ p \vee q & p \vee q & p \vee q \end{array}$$

As concerns the negation connective, first note that Poggiolesi’s logic is the only one to have an unique rule for negation. This rule validates the grounding principles for negation analogous¹² to the negation axioms in Fine’s logic, but it also shows what unites them. The formulation of this rule is made possible by the use of the two metalinguistic relations of derivability and formal explanation. The rule formalizes the following grounding analysis of the negation connective. Consider the question: what is the (complete and immediate) ground of a negative truth, such as “the ball is not cubic”? A natural answer is that the ground is “the ball is spherical”: after all, “the ball is spherical” is true, and “the ball is spherical” and “the ball is cubic” are contradictory—one can derive a contradiction from them—so it seems to be the reason why, or the grounds of or to provide a formal explanation why “the ball is not cubic”. In formal terms, the idea here is that the ground for the truth of a formula of the form $\neg A$ is a formula B (satisfying a certain complexity constraint) such that from A and B , a contradiction is derivable.

Justification of grounding rules. Let us underline that almost all work on the logic of grounding does not treat the issue of *motivating* or *arguing* for certain grounding rules (or axioms) rather

than others (this is also emphasized by McSweeney in Chapter 32). Poggioli is an exception to this trend: all the rules of her logic of grounding are motivated by a rigorous conception of grounding developed in Poggioli (2016a), according to which grounding is definable in terms of (positive and negative) derivability and complexity.¹³

Introduction and elimination rules for the connective ‘because’. Just as the metalinguistic symbol of derivation is reflected in the (formal) language via material implication (the deduction theorem says that $A \vdash C$ if, and only if, $\vdash A \rightarrow C$), in an analogous way, formal explanations are reflected in the language via the ‘because’ connective, denoted by the symbol ω . The new connective ω is treated like the other classical connectives: by means of introduction and elimination rules that only deal with it (and do not specify the relation between ω and the other logical connectives) and enjoy several other proof-theoretic properties. To give the reader an idea of the introduction and elimination rules for the connective ω , let us describe the introduction rule. This rule mimics the rule that introduces material implication, but at the grounding level. Indeed, as this latter says that if there exists a derivation from a formula A to a formula B , one can infer the formula $A \rightarrow B$ (and ‘discharge’ or remove the A); analogously, the rule that introduces the connective ω roughly says that if there exists a formal explanation from A to B , then one can infer $A \omega B$ (and then discharge the A , as well as the whole formal explanation).

Correia 2014. In his 2014 paper, Correia treats the notion of logical grounding, strict and weak, from a syntactic as well as from a semantic perspective. Both in the syntactic and in the semantic approaches, the central notion is that of tree: a mediate grounding relation between a set of formulas M and a formula A is a tree, where complexity is either stable or it increases. In the syntactic approach, each step of the tree is determined by an introduction grounding rule, while in the semantic approach, each step of the tree is determined by a transition from truth to truth. A set of formulas M strictly grounds a formula A , in symbols $M \omega A$, if, and only if, there is a tree for A from M . A set of formulas M weakly grounds a formula A , in symbols $M _ \omega A$, if, and only if, either A belongs to M or there is a subset N of M , such that $N \omega A$.

Syntax. As for the structural rules, strict grounding enjoys irreflexivity and cut, whilst weak grounding enjoys reflexivity, cut, and weakening. Since, as we have said in Section 2.1, grounds must be relevant to the conclusion and weakening typically leads to violations of relevance, the notion of grounding in Correia’s logic may exhibit such violation. As for the introduction rules that define a grounding tree, these convey the same principles of Fine’s logic and are expressed by means of the standard inferential line, namely they have the form:

$$\frac{M}{B}$$

to be read as M grounds B . Finally, let us also note that, unlike Correia (2010, 2017), in Correia (2014), there is no treatment of grounding-theoretic equivalence.

Semantics. The two metalinguistic relations of strict ω , as well as $_ \omega$ weak grounding, are semantically defined by trees (as in the approach proposed by Korbmacher). The links between these consequence relations and other more famous notions of consequence relations are investigated: it is shown that strict grounding is included in weak grounding, which is in its turn included in first-degree entailment consequence relation, which is a sub-consequence of the classical logical consequence.

	<i>Strict/ Weak</i>	<i>Full (or Complete)/ Partial</i>	<i>Immediate/ Mediate</i>	<i>Metaphysical/ Logical</i>	<i>Syntactic/ Semantic</i>	<i>Operator/ Predicate/ Metalinguistic relation</i>
Correia 2014	S & W	F	M	L	Sy & Se	MR
Correia 2017	S & W	P	M	M	Sy & Se	O
Fine (2012a, 2012b)	S & W	F & P	M	M	Sy & Se	O
Korbmacher (2018a, 2018b)	S	P	M	M	Sy & Se	P
Poggiolesi (2016a, 2018)	S	C	I	L	Sy	MR
Schnieder (2011)	S	P	M	M	Sy	O
deRosset(2015)	S & W	F & P	M	M	Se	O

Summary of all the logics of grounding introduced, where the letters of each column corresponds to the initial of the words occupying the first case of the column itself.

Related Topics

- Chapter 2: Aristotelian Demonstration, reductio and demonstration
- Chapter 5: Bolzano, ground in the work of Bolzano
- Chapter 15: Granularity, the granularity of ground
- Chapter 18: Puzzles, the logical puzzles of ground
- Chapter 32: Logic, ground's application to logic
- Chapter 36: Semantics, ground's application to truth-maker semantics

Notes

- 1 Although it is more common to firstly describe a novel operator or predicate via axioms, it is also possible to do that via rules.
- 2 Litland (2018) gives a comprehensive study of the logic of iterated grounding.
- 3 For the difference between factualist and representational conception of grounding, see Correia and Schnieder (2012). The only logic of grounding embracing a factualist conception is Correia (2010). In his work, Correia has developed his logic both at the syntactic and at the semantic level, enriching it by a treatment of ground-theoretic equivalence. However, Krämer and Roski (2015) have shown that Correia's logic involves theorems concerning grounding, which turn out to be opposite to our intuitions.
- 4 In the literature, these formulas are usually called *sequents*. We use the term 'formula' to prevent confusion with the notion of sequent in logic, which denotes a metalinguistic object (e.g., see Poggiolesi (2010a)).
- 5 The distinction between pure and impure rules corresponds to the standard distinction in logic between structural and logical rules (e.g., see Poggiolesi (2010b)).
- 6 The main difference between an axiom and a rule is that while an axiom does not depend on anything, in a rule, the conclusion depends on its premises. Although Fine uses the word 'rule', the formulas listed are axioms rather than rules since they do not depend on anything. Note that Correia (2017) has reformulated them as rules; for example, he substitutes $A, B < A \wedge B$ with *from* A, B , *infer* $A, B < A \wedge B$. We prefer to be faithful to Fine's presentation.
- 7 The weakening rule is the structural rule that allows one to add to the premises of a derivation a new formula; it represents the formal counterpart of the monotonicity property. Hence in a calculus where there is no weakening rule (nor explicit nor derivable nor admissible), the derivability relation is monotone or relevant.

- 8 A calculus is said to satisfy the *subformula property* if, and only if, any theorem of the calculus can be proved by means of a derivation that does not contain any redundancy, i.e., no formula is introduced in the derivation if it is not essential to prove the theorem itself. Corresponding introduction and elimination rules are said to be *harmonious* if, and only if, they satisfy a certain balance amongst them, where the idea of ‘balance’ has been formally captured in several different ways; see for example Francez and Dyckhoff (2012). Finally, introduction rules are said to be *separated* if, and only if, they introduce only one new symbol at a time.
- 9 A recent study of this issue is Krämer (2019).
- 10 In other words, whilst one interpretation of Correia (2014) is in metalinguistic terms, the author does not seem to rule out an alternative interpretation where grounding remains in the language.
- 11 As opposed to a full ground; see Introduction.
- 12 Note that according to Poggiolesi’s logic and in contrast with Fine’s, the grounds of formulas of the form $\neg(A * B)$, where $*$ is either a conjunction or disjunction, vary depending on the form of A and B . If A and B are formulas that begin with a negation, e.g., they are of the form $\neg p$, $\neg q$, respectively, then the grounds have the form p , q . Otherwise, the grounds will have the form $\neg A$, $\neg B$. This difference is required to respect a complexity measure. It is also proof-theoretically adequate, as shown in Poggiolesi (2016b).
- 13 Poggiolesi’s original definition of complete and immediate grounding is formulated for the standard connectives of classical logic. It has been extended in many ways: to the framework of relevant logic (Poggiolesi 2020a), to other less standard connectives of classical logic (Poggiolesi and Francez 2020a) and to the notion of partial (immediate and mediate) and complete and mediate grounding (Poggiolesi 2020b).

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15

GRANULARITY

Fabrice Correia

Grounding is a hyperintensional notion: necessarily equivalent sentences need not be equivalent from a ground-theoretic perspective. How fine-grained, exactly, is grounding? There is a striking lack of consensus on this question. In this chapter, I try to systematize and review the main options that have been put forward in the literature. For reasons that have to do with both naturalness and convenience, I for the most part take the question to be about what is sometimes called, following Kit Fine's (2012a) terminology, *strict full grounding*, and I take for granted a conception of grounding as a *relation* that is *many-to-one* and *non-factive*.¹ I discuss the consequences of making alternative assumptions only in the very last section.²

1. Grounding Relata, Facts and Sentences

One of the important questions to be addressed in a theory of grounding qua relation is that of the nature of its relata.³ Are they states of affairs? Propositions? Entities of yet another kind? No specific answer will be presupposed before we proceed. For lack of a better alternative, I will use the term “fact” as a neutral label to cover the relevant category of entities, whatever it may be.⁴ Thus, every grounding relatum will by stipulation be a fact. Those facts that are not grounding relata I will call “ghostly”. (For an illustration of how a fact might be ghostly, take the facts to be propositions, and think of the view that some propositions are not truth-apt, coupled with the view that the grounding relata must be truth-apt.)

The standard designators for facts are nominalized truth-evaluable sentences. If the language from which the sentences are taken is (plain or regimented) English, the typical designators are of types ‘the proposition that *S*’, ‘the state of affairs that *S*’ and ‘the fact that *S*’, where *S* is a sentence of English.⁵ In line with my use of the term “fact” as a neutral label, I will often use the form “the fact that *p*” when neutrality matters.

If the designator ‘the fact that *S*’ denotes something, I will say that the sentence *S* *corresponds* to the fact that is denoted. It is natural in the present context to assume that every sentence, whatever the language, corresponds to *at most* one fact. I will not address the question of exactly which sentences should be taken to correspond to facts, nor will I address the inverse question of which facts have sentences corresponding to them—although both questions are certainly of interest. It will be convenient to work with a function [*·*] from sentences of any languages to

facts which takes each sentence that corresponds to a fact to the fact it corresponds to (so that e.g., [“Socrates is wise”] is the fact that Socrates is wise).⁶

Given one or more background languages that supply truth-evaluable sentences, there are thus three classes of entities to distinguish: the class **F** of all the facts, the class **R** of all the grounding relata—these two classes are language-independent—and the class **C** of all the facts to which sentences of the language(s) correspond. Both **R** and **C** are subsets of **F**, and as I previously stressed, there may be ghostly facts, i.e., **R** may be a proper subset of **F**. Likewise, **C** may be a proper subset of **F**. On any reasonable choice of language(s), **R** and **C** will overlap. It will be important to keep these considerations in mind throughout the chapter.

2. What Is the Question?

The granularity question for grounding is, roughly, the question of which linguistic distinctions grounding is sensitive to. Let me use “<” for “ground(s)”.⁷ The question can be made more precise by invoking the notion of ground-theoretic equivalence, which is defined as follows:

GROUND-THEORETIC EQUIVALENCE: Two facts x and y are *ground-theoretically equivalent* iff they play the same ground-theoretic role, i.e., iff the following two conditions are satisfied:

- (a) Given any plurality of facts Z : $Z < x$ iff $Z < y$;
- (b) Given any plurality of facts Z and fact z : $Z, x < z$ iff $Z, y < z$.⁸

(This relation is indeed by definition an equivalence relation.) Note that two facts trivially fail to be ground-theoretically equivalent if one but not the other is ghostly and that two facts are trivially ground-theoretically equivalent if both are ghostly. The granularity question can then be formulated as follows:

GRANULARITY QUESTION: For which distinct sentences S and T that correspond to grounding relata is it the case that $[S]$ and $[T]$ are ground-theoretically equivalent?

Three points are worth stressing here. First, the question is *not*: For which distinct sentences S and T that correspond to *facts* is it the case that $[S]$ and $[T]$ are ground-theoretically equivalent? Given previous remarks, the question is indeed easily settled when at least one of S and T corresponds to a ghostly fact; the interesting cases involve sentences that correspond to facts that are ground-theoretically tied to other facts. Second, the granularity question presupposes as given one or several background languages that provide the sentences it quantifies over. When people discuss the fineness of grain of grounding, they normally do not bother to specify which languages are under focus, but in principle this should be done. (I will myself for the most part remain silent on the issue.) Third, ground-theoretic equivalence is a relation between facts, not between sentences (unless, of course, facts are sentences).

3. Ground-Theoretic Equivalence and Identity

By Leibniz’s Law, for all facts x and y such that $x = y$, x and y are ground-theoretically equivalent. Should the converse be also accepted? Given the way ghostly facts behave vis-à-vis ground-theoretic equivalence, the interesting question is whether the following general principle should be taken to hold:

IDENTITY: Given any grounding relata x and y , if x and y are ground-theoretically equivalent, then $x = y$.

If IDENTITY holds, then the formulation of the granularity question can be significantly simplified:

GRANULARITY QUESTION, SIMPLIFIED: For which distinct sentences S and T that correspond to grounding relata is it the case that $[S]$ and $[T]$ are identical?

Interestingly, explicit reference to ground-theoretic equivalence is here completely absent. By contrast, if IDENTITY does not hold, then appeal to ground-theoretic equivalence remains mandatory.

The question whether IDENTITY should be accepted is substantive. The facts are theoretical posits, entities whose nature is at least in part given by the theoretical role they are intended to play. If this role is stipulated to be that of being the relata of the grounding relation, as this is the case in Correia (2017a: 515), then there is considerable pressure to answer positively: for what could then possibly be the motivation for distinguishing between facts that are indistinguishable from a ground-theoretic perspective? But if the facts' defining theoretical role is taken to be another one, for instance if they are taken to be Fregean thoughts individuated by some cognitive role, then there can be room for a negative answer.

Important for the discussion to come is the characterization of the facts in terms of a relation between sentences (be they of the same language or not) I call "descriptive equivalence". We use sentences to describe the world, to say how it is. Some linguistic differences between such sentences make no difference for that purpose. Thus, for instance, the sentences "It is rainy and windy" and "It is windy and rainy" describe the world as being exactly the same way despite their syntactic differences; and the same is true—or so one may want to argue—of the sentences "Lake Geneva is filled with water" and "Lake Geneva is filled with a substance composed of H_2O molecules". When this is the case, the sentences are said to be descriptively equivalent. Formally, I adopt the following definition:

DESCRIPTIVE EQUIVALENCE: Sentences S and T are *descriptively equivalent* iff they describe the world as being the same way.

Descriptive equivalence is a relation between sentences. But it is tempting to postulate the existence of a category K of extra-linguistic entities whose role is to "back" this relation, so that two sentences are descriptively equivalent just in case they express the same entity of kind K .⁹ The *facts* of Correia (2010) are precisely a category of entities that can be characterized in this way,¹⁰ and the "worldly" conception of grounding that I put forward there—to be accurate: the worldly conception that takes grounding to be a relation rather than a notion expressed by a sentential operator—takes the grounding relata to be members of that category. We thus have here a particular conception of the facts that does not stipulatively associate them with the theoretical role of being relata of the grounding relation.

Say that fact x is a *proximal ground* for fact y iff (i) x grounds y , and (ii) for every fact z that grounds y , either $z = x$ or z grounds x . A proximal ground is, as it were, a ground through which all other grounds (if any) must pass. Granted that there is a relation of immediate grounding,¹¹ proximal grounds should presumably be counted as immediate grounds, but the converse is not true: plausibly, the fact that $1 = 1$ and the fact that $2 = 2$ are each immediate grounds for the fact that $(1 = 1 \text{ or } 2 = 2)$, but neither is a proximal ground for the latter. The view that every grounding relatum is a proximal ground fails on some conceptions of grounding, e.g., on the approach put forward in Fine (2012b) given certain natural assumptions.¹² But it holds on some other conceptions. Thus, the theory of ground-theoretic structures developed in Correia (2017a: §3) posits the existence of a negation operation $-$ on facts (on the intended understanding of the operation, the negation of the fact that p is the fact that it is not the case that p), and has it that

every fact x is a proximal ground for the fact $\neg\neg x$ (Facts 3.2.4 and 3.2.5, p. 521). The same is true in the theory put forward in Krämer (2018), where the operation of negation is defined rather than taken as a primitive (Theorem 6, p. 812, and Definition 2, p. 809).

The concept of a proximal ground is relevant for the question whether IDENTITY holds, thanks to the following interesting result, which can be established on the orthodox assumption that grounding is irreflexive:¹³

PROXIMAL: If every grounding relatum is a proximal ground for some fact, then IDENTITY holds.

Proof. Let x and y be grounding relata that are ground-theoretically equivalent. If we assume that every grounding relatum is a proximal ground for some fact, we have some y^+ such that (i) y grounds y^+ , and (ii) for every fact z that grounds y^+ , either $z = y$ or z grounds y . Since x and y are ground-theoretically equivalent, we then have by (i) that x grounds y^+ , and so by (ii) that either $x = y$ or x grounds y . Yet x cannot ground y , for if it did, then—again since x and y are ground-theoretically equivalent— y would ground itself, which is impossible. Hence, $x = y$.

According to the theory of ground-theoretic structures developed in Correia (2017a), grounding is irreflexive, and PROXIMAL accordingly allows one to infer that on that theory, IDENTITY holds. The theory developed in Krämer (2018) admits of several versions, some forcing grounding to be irreflexive (these seem to be favored by Krämer), some not. PROXIMAL accordingly allows one to infer that on the former versions of the theory, IDENTITY holds. Note that PROXIMAL only gives a *sufficient* condition for IDENTITY to hold. There are theories—for instance, given certain assumptions, the theory put forward in Fine (2012b)—according to which grounding is irreflexive, IDENTITY holds, but some grounding relata fail to be proximal grounds.¹⁴

4. Worldly versus Representational

We just saw that on the views put forward in Correia (2017a) and Krämer (2018), every fact x grounds the “doubly negated” fact $\neg\neg x$. Given that grounding is not reflexive, which is granted on both views, at least some fact x fails to be ground-theoretically equivalent to $\neg\neg x$. By contrast, on the view developed in Correia (2010), as well as on the semantic theory advocated in Fine (2012a) (which extends the theory of Fine (2012b) to take care of the truth-functional connectives), there cannot be such ground-theoretic connections. Indeed, on both views, grounding is irreflexive, and negation “cancels out” itself: the fact that $\neg\neg p$ *just is* the fact that p .

These different takes regarding double negation can serve to illustrate the distinction between *worldly* and *representational* conceptions of grounding.¹⁵ On the last pair of views mentioned, the difference between “Socrates is wise” and “It is not the case that it is not the case that Socrates is wise” is merely representational and hence of no ground-theoretic significance. These views endorse a worldly conception of grounding. By contrast, on the first pair of views, the representational difference matters. These views construe grounding as representational.

How exactly is the worldly versus representational distinction to be characterized? A neat way of doing it involves the concepts of ground-theoretic equivalence and descriptive equivalence introduced earlier (Sections 2 and 3):

WORLDLY VS REPRESENTATIONAL: Grounding is *representational* iff it is not worldly. Grounding is *worldly* iff for all sentences S and T (of any languages) that correspond to grounding relata, if S and T are descriptively equivalent, then $[S]$ and $[T]$ are ground-theoretically equivalent.¹⁶

The worldly versus representational distinction manifests itself in a particularly clear way in cases involving logically equivalent sentences. The case of double negation can be redescribed using the vocabulary involved in the definition of “worldly”. It is extremely plausible to hold that given any sentence S that describes the world as being a certain way, S and the corresponding doubly negated sentence $\neg\neg S$ are descriptively equivalent.¹⁷ Taking this for granted, on a conception of grounding as both worldly and irreflexive, there cannot be such a sentence S such that S corresponds to a fact x and $\neg\neg S$ to a fact y with x grounding y . By contrast, agreeing that double negation leaves invariant the descriptive content of sentences does not ipso facto prevent friends of a representational conception of grounding from holding that the fact corresponding to some sentence S grounds the fact corresponding to $\neg\neg S$, even if they assume that grounding is irreflexive. Other cases of this sort will be discussed in what follows.

The worldly versus representational distinction also manifests itself in nonlogical cases, of course. Take again the sentences “Lake Geneva is filled with water” and “Lake Geneva is filled with a substance composed of H_2O molecules”. They are descriptively equivalent, or so it is plausible to claim. Suppose both correspond to a fact, say x in the case of the first sentence and y in the case of the second. If grounding is both worldly and irreflexive, x cannot ground y ; if grounding is representational, there may well be such a ground-theoretic connection, even if grounding is irreflexive. Other nonlogical examples are easy to concoct.

It is perhaps tempting to think that the worldly versus representational distinction can be equivalently characterized solely in terms of the nature of the grounding relata by saying that grounding is representational if its relata are representations and worldly if they are not representations.¹⁸ This alternative characterization has its virtues, but it can be disputed.

The second half of the alternative characterization, namely the view that if the grounding relata are not representations, then grounding is worldly, would seem to be hardly objectionable. Rejecting the view would require holding that there are at least two sentences S and T that describe the world as being the very same way and yet correspond to distinct grounding relata. There does not seem to be anything properly *incoherent* in doing so while holding that the grounding relata are not representations, but it is hard to see which conception of the facts could possibly underpin such a combination of views.

The first half of the alternative account seems to be more contentious. Consider a view that combines the following two claims: (i) the grounding relata are Fregean thoughts and (ii) for all descriptively equivalent sentences S and T , there are thoughts x and y such that S corresponds to x , T corresponds to y , and x and y are ground-theoretically equivalent. This view does not seem to be incoherent, and yet it takes the grounding relata to be representations and grounding to be worldly.

The point can be strengthened. Define the property of “being worldly” as applied to the category of facts as follows:

WORLDLINESS FOR FACTS: The category of facts is *worldly* iff for all sentences S and T (of any languages) that correspond to facts, if S and T are descriptively equivalent, then $[S]$ and $[T]$ are identical.

If the category of facts is worldly, then grounding is worldly. Now the view that the category of facts is worldly would seem to be compatible with the view that the facts are representations. Of course, the view described earlier is one on which the category of facts is *not* worldly, since, say, the sentences “Lake Geneva is filled with water” and “Lake Geneva is filled with a substance composed of H_2O molecules” correspond to distinct Fregean thoughts that arguably enter into ground-theoretic ties and yet are descriptively equivalent. But Fregean thoughts are particularly

fine-grained, and there is nothing incoherent in the view that there are categories of representations that are significantly less fine-grained. In particular, it does not seem to be incoherent to hold that the facts are representations that constitute a category that is worldly in the sense introduced. In the *Tractatus*, thoughts (Gedanken) are representations, and—or so is it plausible to claim given the textual evidence—no two distinct thoughts can represent the same state of affairs (Sachverhalt). If we agree that sentences *S* and *T* are descriptively equivalent iff *S* and *T* correspond to thoughts that represent the same state of affairs, then the Tractarian view of thoughts and states of affairs entails that the category of thoughts is worldly.

These objections to the view that grounding is worldly only if its relata are not representations have some force, but I think something can still be said in favor of the view. The grounding relata are entities that are canonically picked out by (plain or nominalized) sentences. Now the representations that we *normally* take to be picked out by sentences constitute a category that is not worldly. In addition, once it is granted that the grounding relata are representations of that sort, the view that grounding is not worldly is overwhelmingly *plausible*. Take again the sentences “Lake Geneva is filled with water” and “Lake Geneva is filled with a substance composed of H₂O molecules”, assume they express Fregean thoughts *x* and *y*, respectively, and also assume that grounding relates such thoughts. Then it is hard to resist the view that *x* and *y* together ground the thought *z* expressed by the conjunction “Lake Geneva is filled with water & Lake Geneva is filled with a substance composed of H₂O molecules” (I make essentially this point in (Correia 2010: 258–259)). Given that the two conjuncts are descriptively equivalent, the conjunction is arguably itself descriptively equivalent to them. Assuming that *z* cannot help ground itself, it follows that *z* is not ground-theoretically equivalent to any of *x* and *y*, and hence that grounding is not worldly.

To summarize, even though the alternative characterization of the worldly versus representational distinction cannot, strictly speaking, replace the original, it nevertheless seems to be a good approximation, since it is plausible on the assumption that the representations it appeals to are those we standardly take truth-evaluable sentences to pick out.

The worldly versus representational distinction is directly relevant to the granularity question as formulated in Section 2. The view that grounding is worldly yields a straightforward *partial* answer to the question: where *S* and *T* are distinct sentences that correspond to grounding relata, [*S*] and [*T*] are ground-theoretically equivalent *if* *S* and *T* are descriptively equivalent. The answer is partial because the converse might fail, i.e., nothing rules out cases in which there are two sentences that correspond to ground-theoretically equivalent grounding relata even though the sentences are not descriptively equivalent. Such cases will be ruled out if IDENTITY holds and the category of facts is “reverse-worldly”, this being defined as follows:

REVERSE-WORLDLINESS FOR FACTS: The category of facts is *reverse-worldly* iff for all sentences *S* and *T* (of any languages) that correspond to facts, *S* and *T* are descriptively equivalent if [*S*] and [*T*] are identical.

If *S* and *T* are distinct sentences that correspond to grounding relata, the connections between their being descriptively equivalent, the corresponding facts’ being identical and these facts’ being ground-theoretically equivalent is depicted in Figure 15.1:

Thus, to partially summarize, if grounding is worldly, IDENTITY holds and the category of facts is reverse-worldly, then there is a full answer to the granularity question framed in terms of the concept of descriptive equivalence: two sentences *S* and *T* satisfy the relevant condition *iff* they are descriptively equivalent. If IDENTITY fails or the category of facts is not reverse-worldly, we have a partial answer to the question: two sentences *S* and *T* satisfy the relevant condition *if*

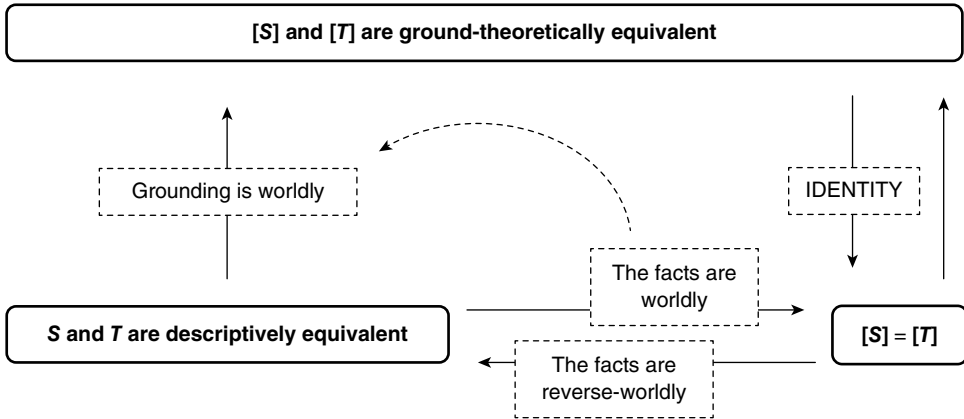


Figure 15.1

they are descriptively equivalent. In both cases, the concept of descriptive equivalence plays an important role. By contrast, the concept of descriptive equivalence is of little help to answer the granularity question if grounding is taken to be representational.

Given these clarifications, where should we place the existing views about grounding on the worldly/representational map? The question is not easy to answer, since the question of when two sentences are descriptively equivalent is itself likely to generate controversies. Let me here offer a *personal* answer relative to a range of (what I take to be) representative views. Importantly, here (as before) I focus on conceptions of grounding as a relation as opposed to a notion expressed by a sentential operator, and I leave the discussion of the latter conception to the last section of this chapter.

Worldly Conceptions

As I emphasized in the previous section, Correia (2010) construes the facts as nonlinguistic correlates of descriptive equivalence. On that view, the category of facts is both worldly and reverse-worldly, and since it is worldly, grounding itself is worldly.

Assuming the semantic approach put forward in Fine (2012a), it is natural to take two sentences to be descriptively equivalent just in case they are verified by the very same states, i.e., just in case they are associated with the same “verification set” (see Correia 2016). If we identify the facts with the verification sets, which is also natural, we again end up with a view according to which the facts are both worldly and reverse-worldly and grounding is consequently worldly.

A further illustration of a worldly conception of grounding can be found in the work of Paul Audi (2012a, 2012b). Audi takes the grounding relation to be states of affairs, and he has a fairly narrow conception of what states of affairs there are—on his view, logically complex sentences, except perhaps special cases, fail to correspond to states of affairs. Yet Audi’s states of affairs are arguably both worldly and reverse-worldly, and accordingly, Audi arguably defends a conception of grounding as worldly.

Representational Conceptions

As I previously argued, Correia (2017a) and Krämer (2018) both encode representational conceptions of grounding because of the way they treat double negation. Further illustrations

involving truth-functional operators can be given. Thus, for instance, while it can be argued that for any truth-evaluable sentence S , S itself, $\lceil S \wedge S \rceil$ and $\lceil S \vee S \rceil$ are all descriptively equivalent, the corresponding facts are not taken to be ground-theoretically equivalent, because on these views, grounding is not reflexive and $[S]$ grounds both $\lceil S \wedge S \rceil$ and $\lceil S \vee S \rceil$ for any S that corresponds to a fact.

Gideon Rosen's (2010) conception of grounding is also clearly representational. On Rosen's view, grounding is irreflexive (2010: 116), and his theory validates the principle that any $[S]$ grounds both $\lceil S \wedge S \rceil$ and $\lceil S \vee S \rceil$ provided that $[S]$ holds (p. 117; this latter qualification is present because Rosen takes grounding to be factive). A further reason why grounding is representational on Rosen's view stems from Rosen's "Grounding-Reduction Link", the principle that when its being the case that p reduces to its being the case that q , the fact that p is grounded in the fact that q (122 ff). If for example we grant, with Rosen, that John's being a bachelor reduces to his being an unmarried male, the Link has it that the fact that John is a bachelor is grounded in the fact that John is an unmarried male. Yet the sentences "John is a bachelor" and "John is an unmarried male" are arguably descriptively equivalent.

5. Fact-Identity, Ground-Theoretic Equivalence and Truth-Functions

Sentences of type " $[S] = [T]$ " play a central role in theorizing about the granularity of grounding (have again a brief look at Figure 15.1). There currently are several well-worked-out conceptions of when sentences of that type are true solely in virtue of the truth-functional logical form of the component sentences S and T . I here present five of them, each in the form of a logical system that delivers (only) such identities as theorems.¹⁹ Following the notation used in the relevant papers, the identities are formalized as sentences of the form " $A \approx B$ ".²⁰

Each system is characterized by a set of axioms and a set of rules. I run through each system in turn.

System $\mathbf{C}_{2017}$ is defined by the following axioms and rules:

- A1. $A \wedge B \approx B \wedge A$
- A2. $A \vee B \approx B \vee A$
- A3. $A \approx A$
- R1. $A \approx B/B \approx A$
- R2. $A \approx B, B \approx C/A \approx C$
- R3. $A \approx B/A \wedge C \approx B \wedge C$
- R4. $A \approx B/A \vee C \approx B \vee C$
- R5. $A \approx B/\neg A \approx \neg B$

The system, introduced in Krämer *forthcoming*, is extracted from the general postulates for propositional equivalence introduced in Correia (2017a).

System $\mathbf{K}_{\text{int}}$ is defined like $\mathbf{C}_{2017}$, but with R5 replaced by

- R6. $A \approx B/\neg\neg A \approx \neg\neg B$

and the list of axioms augmented with the De Morgan equivalences

- A4. $\neg(A \wedge B) \approx \neg A \vee \neg B$
- A5. $\neg(A \vee B) \approx \neg A \wedge \neg B$

This is the “intermediate system” introduced in Krämer *forthcoming*.

System $\mathbf{K}_{\text{ext}}$ is defined like $\mathbf{K}_{\text{int}}$ but with the extra axioms

$$\text{A6. } A \wedge A \approx A \vee A$$

$$\text{A7. } A \vee A \approx \neg\neg A$$

and the extra rules

$$\text{R7. } A \leq B, A \leq C / A \leq B \wedge C$$

$$\text{R8. } A \leq B / A \leq B \vee C$$

where $A \leq B$ is short for $A \vee B \approx B \vee B$. This is the “extensional system” introduced in Krämer *forthcoming*.

The axiomatic basis of system $\mathbf{C}_{2016}$ is like that of $\mathbf{K}_{\text{int}}$, minus A3 and R6, plus the extra axioms

$$\text{A8. } A \approx \neg\neg A$$

$$\text{A9. } A \approx A \wedge A$$

$$\text{A10. } A \approx A \vee A$$

$$\text{A11. } A \wedge (B \wedge C) \approx (A \wedge B) \wedge C$$

$$\text{A12. } A \vee (B \vee C) \approx (A \vee B) \vee C$$

$$\text{A13. } A \wedge (B \vee C) \approx (A \wedge B) \vee (A \wedge C)$$

It is introduced in Correia (2016). Note that A3 and R6 have been removed from the axiomatic basis but are still provable in the system.

Finally, the axiomatic basis of system $\mathbf{C}_{2010}$ is the same as that of $\mathbf{C}_{2016}$ plus the extra distributivity axiom

$$\text{A14. } A \vee (B \wedge C) \approx (A \vee B) \wedge (A \vee C)$$

$\mathbf{C}_{2010}$ is nothing but R. B. Angell’s (1989) system AC for analytic equivalence but in other axiomatic clothes; the axiomatization of the system I just gave is taken from Fine (2016). This system corresponds to the logic of factual equivalence put forward in Correia (2010). The reason why I moved from the 2010 conception of factual equivalence to the 2016 conception is that A14 leads to trouble once grounding is understood along the lines put forward in the 2010 paper. For more details, see (Correia 2016); see also (Krämer & Roski 2015), where the issue has been independently noticed.

The five systems are related as shown in Figure 15.2, where each arrow represents strict inclusion.²¹

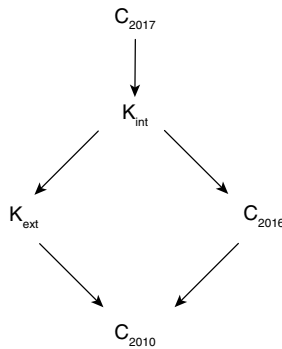


Figure 15.2

To complete this diagram, it should be noted that neither of $\mathbf{K}_{\text{ext}}$ and $\mathbf{C}_{2016}$ is properly included in the other.

As should be clear from the previous section, I take it that any logic suited for a worldly conception of the category of facts should have A8, A9 and A10 as theorems, because I believe that any sentence that describes the world as being a certain way is descriptively equivalent to its double negation, its self-conjunction and its self-disjunction. This eliminates $\mathbf{K}_{\text{ext}}$, and hence $\mathbf{K}_{\text{int}}$ and $\mathbf{C}_{2017}$. Since $\mathbf{C}_{2016}$ is contained in $\mathbf{C}_{2010}$, if the former is suited for a worldly conception of the facts, so must be the latter: the move from $\mathbf{C}_{2016}$ to $\mathbf{C}_{2010}$ consists in adding an extra principle of identity, and one cannot get a violation of worldliness that way. The extra axiom, namely A14, is independently problematic (see my comments earlier).

The five logics discussed have been construed as logics of fact-identity. Can they also be understood as logics of ground-theoretic equivalence? This crucially turns on whether IDENTITY (see Section 3) holds. In the theory of ground-theoretic structures of Correia 2017a and the theories of grounding put forward in Krämer (2018, *forthcoming*), IDENTITY is validated (see Section 3). Hence, given these theories, $\mathbf{K}_{\text{ext}}$, $\mathbf{K}_{\text{int}}$ and $\mathbf{C}_{2017}$ must be taken to be logics of ground-theoretic equivalence. (The reasons put forward above for taking these logics to be unsuited for a worldly conception of *the facts* are accordingly reasons to take them to be unsuited for a worldly conception of *grounding*.) The truthmaker semantical treatment of factual equivalence put forward in Correia (2016) goes hand in hand with the semantic approach to grounding advocated in Fine (2012b) and further developed in Fine (2012a).²² As we saw (see again Section 3), IDENTITY holds on Fine's approach given certain assumptions. On this account, $\mathbf{C}_{2016}$ must also be seen as a logic of ground-theoretic equivalence. I do not know whether IDENTITY holds on the approach put forward in Correia (2010).

6. Must a Choice Be Made Between a Worldly and a Representational Conception?

Should we prefer a worldly conception over a representational conception? Two main arguments have been put forward in favor the first option. One argument is that we should endorse a conception of grounding as worldly because grounding is supposed to carve reality at the joints, and grounding can do that only if it is worldly (Correia 2010: 258–259). The other argument is that representational conceptions of grounding, or at least the most plausible ones, lead to logical puzzles, while a worldly conception is naturally immune from them (see Krämer's chapter [Chapter 18] in this volume).²³

A possible reply to the first argument is pluralist in nature: even if we need to recognize a notion of grounding that is joint-carving, this does not preclude the view that there is in addition a notion of grounding which is not joint-carving but which is nevertheless of significance for metaphysics.²⁴ It would not do here to retort that only joint-carving notions can be of significance for metaphysics. The notion of *being more joint-carving than*, i.e., of *carving reality at the joints in a better way than*, for instance, surely belongs to the possible tools of metaphysical inquiry, but it is not itself joint-carving: the Fregean thought that Lake Geneva is filled with a substance composed of H_2O molecules is more joint-carving than the Fregean thought that Lake Geneva is filled with water, say, but the two thoughts describe the world as being exactly the same way.

A natural reaction to the second argument also appeals to grounding pluralism. Even though there may be a worldly notion of grounding that escapes the puzzles, there may also be one or more representational notions that are philosophically significant but are directly threatened by them (see Krämer's chapter [Chapter 18] in this volume).

Assume that there is room for both worldly and representational notions of grounding. Surely then, the two notions, or the two families of notions, are intimately connected. But how? A very natural suggestion is that to each worldly notion, there corresponds a representational notion defined in terms of it by “inserting” representational elements in the relation. Suppose given a worldly relation of grounding that relates, say, states of affairs. The idea is to define a relation between representational entities, say propositions, via the following biconditional:

$$\pi_1, \pi_2, \dots \text{ground}_R \rho \text{ iff the state of affairs that } \pi_1 \text{ is true, the state of affairs that } \pi_2 \text{ is true, } \dots \\ \text{ground}_W \text{ the state of affairs that } \rho \text{ is true}$$

(where “ π_1 ”, “ π_2 ”, ... and “ ρ ” range over propositions). This account looks promising if we focus on the interaction between grounding and the truth-functions: even though, for instance, the state of affairs that p cannot possibly ground_W the state of affairs that p and p (because we do not have two states of affairs here but only one), the view that the proposition that p grounds_R the proposition that p and p is very plausible, at least on some standard conceptions of propositions.

As far as I know, this view about the connection between worldly and representational grounding has never been discussed in the literature, but it certainly deserves scrutiny. More generally, the question of how worldly and representational conceptions of grounding are related to one another deserves serious investigation.²⁵

7. Dropping the Initial Assumptions

Throughout the chapter, I focused on strict full grounding understood as a nonfactive, many-to-one relation. What would have changed if another conception of grounding had been assumed instead?

Focusing on *partial* rather than full grounding would have made little sense. Ground-theoretic equivalence defined in terms of partial grounding is weaker than the concept defined in Section 2, and accordingly any answer to the granularity question framed in terms of the latter yields an answer to the granularity question framed in terms of the former. Yet the converse may fail, and therefore the wise choice is to work with a full notion.

Weak grounding is a more basic concept on some approaches to grounding. It is so, for instance, on the worldly approaches of Correia (2010) and Fine (2012a, 2012b). In both cases indeed, strict grounding can be defined in terms of its weak counterpart by saying that facts X strictly ground fact x iff (i) X weakly ground x and (ii) x does not help weakly ground any of X .²⁶ On these approaches, it would therefore be natural to study the granularity question by first focusing on the weak notion. On other approaches, by contrast, strict grounding is the more basic notion. This is the case of the representational views put forward in Correia (2017a) and Krämer (2018). These two views indeed define in the same way a weak notion in terms of the basic, strict notion.

Assuming that grounding is *many-many* rather than *many-one* would have forced us to define ground-theoretic equivalence in a way that takes the many-many character of grounding into account. The details are obvious, and the shape of the discussion would have been essentially the same.

By contrast, assuming that grounding is *factive* would have had an important consequence. Under the proposed definition of ground-theoretic equivalence, two facts that fail to hold

(i.e., obtain or be true) are ground-theoretically equivalent if grounding is factive. Yet we want to say, for instance, that the fact that Socrates was a British philosopher and the fact that there is no pope in 2018 are *not* ground-theoretically equivalent. The obvious suggestion to avoid this problem is to “modalize” the definition, by saying that two facts are equivalent in the new sense iff necessarily, they are equivalent in the old sense if they both hold. On this account, facts that cannot possibly hold automatically count as ground-theoretically equivalent, and hence the account potentially diverges from the account based on nonfactive grounding on such facts: while on the former account, the fact that $2 \neq 2$ and the fact that $\text{water} = \text{CO}_2$ must be taken to be ground-theoretically equivalent, on the latter account, they need not be. This is an important difference, but if we agree to understand the phrase “grounding relatum” throughout the previous discussion as meaning the same as “possible grounding relatum”, then this difference will not matter much. After all, the granularity question is about the sentences that correspond to *grounding relata*, not about unrestrictedly all the sentences that correspond to facts.

The move from a predicate conception of grounding to an *operator* conception is certainly the most substantive of all. Let us still use “ $<$ ” for grounding, but now as a sentential operator: “ $p, q, \dots < r$ ” is now short for, say, “Its being the case that p , that $q, \dots$ makes it the case that r ” or “Its being true that p , that $q, \dots$ makes it true that r ”. The move requires appeal to both singular and plural quantification into sentential position. Once these resources are in place, a binary operator for ground-theoretic equivalence and a unary operator for being a grounding relatum can be defined in the obvious way. Corresponding to the binary predications of type “ S corresponds to x ”, we need to introduce hybrid sentences of type “ S signifies that p ” or something similar. A new binary operator “ $\approx$ ” is also needed to express “fact-equivalence”: “ $p \approx q$ ”, which can be read “For it to be the case that p is for it to be the case that q ” or “For it to be true that p is for it to be true that q ”, says what we would want to say in the predicate setting using “the fact that $p = \text{the fact that } q$ ”.²⁷ With these linguistic resources in place, the previous discussion would go in much the same way, with the relevant adjustments of course (for instance, the phrasing “the category of facts is worldly” should be abandoned and replaced by “ $\approx$ is worldly” or something equivalent).

Here is a brief classification of operator views about grounding relative to the worldly versus representational distinction.

Worldly Conceptions

The “official” logic of grounding developed in Correia (2010) is formulated in an object language where grounding is expressed by means of a sentential operator: the grounding relation that is also countenanced in the paper, which I have often referred to, is there used to provide a semantics for the object language. The operator conception of grounding that is put forward in the paper is of course also worldly. Correia and Skiles (2019) draws on Correia (2010) but diverges from it among other things insofar as it (i) defends the view that grounding is to be understood in terms of factual equivalence and (ii) advocates a logic of factual equivalence that is weaker than the logic put forward in Correia (2010) (see Section 5).

Similar remarks hold of Fine (2012a, 2012b): the official language is an “operator language”, and it is only in the semantics that grounding is treated as a relation. The corresponding conception of grounding is of course also worldly. Importantly, these remarks concern the whole, proof-theoretic and semantic approach of Fine (2012b), but *only* the semantic approach of Fine

(2012a) (see what follows). Litland (2016) develops Fine's worldly approach for many-many grounding.

Representational Conceptions

The "official" logic of representational grounding developed in Correia (2017a) assumes that grounding is expressed by means of an operator, while the associated semantics is given in terms of ground-theoretic structures in which, as we saw, grounding is treated as a relation.

The proof-theoretic account of grounding in Fine (2012a), contrary to the semantic account, treats grounding as representational. The system put forward by Fine in that paper is similar to the system I put forward in Correia (2017a).

Other instances of an operational-cum-representational conception of grounding can be found in Correia (*forthcoming*), Fine (2010), Litland (2017, 2018a, 2018b) and Schnieder (2010).²⁸

Related Topics

The chapters of the handbook that are most relevantly connected to this entry are the following: LOGIC [Chapter 32], LOGICS [Chapter 14], NORMATIVITY [Chapter 34], PUZZLES [Chapter 18], SEMANTICS [Chapter 36], VARIETIES [Chapter 13].

Notes

- 1 Fine opposes strict to weak and full to partial (Fine 2012a: 50–54). When I talk of "grounding understood as a relation", I mean grounding understood as a notion expressed by means of a relational predicate, as opposed to a notion expressed by means of a sentential operator (on the distinction, see, e.g., Correia 2010: 253–254). While most take grounding to be many-to-one, some think it can be many-to-many (Dasgupta 2014; Litland 2016). On factive versus nonfactive grounding, see Fine 2012a: 48–50, and Correia 2014: 35–36. On all these distinctions, see also the introduction to this handbook.
- 2 I should emphasize that I focus on *metaphysical* grounding, as opposed to, e.g., normative or logical grounding. The question of granularity is of course relevant to all notions of grounding. Correia 2014 and Poggiolesi 2016, 2018 contain detailed material that can be used to discuss the granularity of logical grounding. See the VARIETIES chapter [Chapter 13] of this handbook for a discussion of the varieties of grounding and the NORMATIVITY chapter [Chapter 34] on normative grounding.
- 3 To avoid any misunderstanding, let me stress that by "relatum of grounding", I will throughout mean something that *actually* helps ground something or is *actually* grounded by something.
- 4 The label "fact" is not ideal given my assumption that grounding is not factive. On that assumption indeed, if facts are taken to be propositions, then it should be admitted that some of them *fail to be true*, and if they are taken to be states of affairs, it should be admitted that some of them *fail to obtain*. This is of course at odds with the standard ways of understanding the term "fact".
- 5 The corner quotes are used for Quinean quasi-quotation.
- 6 There is thus a slight departure from standard notational conventions here, which have it that [Socrates is wise] (notice the absence of quotation marks) is the fact that Socrates is wise.
- 7 The symbol is usually used as a sentential operator expressing strict full grounding rather than as a predicate, but no confusion should arise from my nonstandard use.
- 8 On the intended understanding of "plurality", there is an empty plurality. I will not discuss the question whether the empty plurality can be a ground, i.e., to use an expression introduced in Fine (2012a: 47–48), whether there can be "zero-grounding".
- 9 Meanings are sometimes introduced as entities whose role is to "back" the relation of synonymy in exactly this sense.

- 10 Like here, “fact” in Correia 2010 is used in such a way that some facts fail to obtain.
- 11 See Fine 2012a: 50–51, and also the introduction to this handbook, on the distinction between immediate and mediate grounding.
- 12 On Fine’s approach, each natural model includes a maximal fact, the set of all the states of the model. Such a fact cannot ground anything, and given the natural assumption that the model has at least two states, that fact is not ghostly.
- 13 The proof is adapted from Krämer 2018, where Stephan Krämer invokes the view that every fact grounds its “double negation” (see what follows) to show that IDENTITY holds in his theory given the assumption of irreflexivity. The proof actually establishes a stronger result than PROXIMAL, namely: if every fact is a proximal ground, then given any facts x and y , if x and y ground the same facts, then $x = y$. IDENTITY and the stronger proposition that facts that ground the same facts must be identical are established in a similar way in Correia 2017a: 526.
- 14 I have a nontrivial proof that IDENTITY holds in Fine’s (2012b) theory, which makes use of the assumption that every state that has proper parts is a fusion of some of its proper parts.
- 15 I introduced the distinction in Correia 2010. I there use the label “conceptual” rather than “representational”, and I explicitly apply the labels “worldly” and “conceptual” not to conceptions of grounding but to conceptions of facts and what I call “factual equivalence”. Yet the worldly versus representational distinction that I try to clarify in what follows is at the core of the paper. The distinction is also discussed in Krämer and Roski 2015, Fine 2017 and Correia and Skiles 2019, and it is implicit in Correia and Schnieder 2012, §3.3.
- 16 Given that the granularity question is explicitly a question about sentences and the facts they correspond to, the proposed characterization is both natural and convenient. An alternative, “nonlinguistic” characterization is possible, though. Say that two facts x and y are *metaphysically equivalent* iff to say that x holds and to say that y holds is to describe the world as being the same way. The suggestion is to say that grounding is worldly iff for all grounding relata x and y , if x and y are metaphysically equivalent, then x and y are ground-theoretically equivalent (and that grounding is representational iff it is not worldly, as before). Throughout this chapter, appeal to descriptive equivalence could be systematically replaced by appeal to metaphysical equivalence, *mutatis mutandis*.
- 17 See the excellent argument in Dorr 2016: 62–63.
- 18 The distinction as applied to grounding understood as a relation is cashed out in this way in Krämer and Roski, p. 60, and Correia 2017b: 58.
- 19 This section heavily relies on Krämer *forthcoming*, which develops the views of Krämer 2018.
- 20 To be completely accurate, I should stress that in each of these systems, “ $\approx$ ” is understood as a sentential operator. However, in each case, there is a corresponding semantics that assigns, or allows one to assign, semantic values to truth-functional sentences in such a way that “ $A \approx B$ ” is taken to hold just in case the semantic value of “ A ” is the same as the semantic value of “ B ”. It is accordingly admissible to interpret the systems in the way I suggest.
- 21 The only fact depicted in the figure that is not trivial is that every theorem of C_{2017} is a theorem of K_{int} . This is established in Krämer *forthcoming*.
- 22 I also introduce another semantic approach, which is equally good for the purpose of characterizing C_{2016} , but which is not directly connected to Fine’s approach to grounding. On grounding and truth-maker semantics, see the SEMANTICS chapter [Chapter 36] of this handbook.
- 23 Krämer (2013) focuses on a family of similar puzzles first discussed in Fine 2010. The presentation of these puzzles assumes an operator view of grounding, but the puzzles can also be formulated if we assume a predicate view. Peter Fritz (*ms*) discusses a puzzle of a different kind, based on a version of the Russell–Myhill paradox. The puzzle crucially appeals to the notion of immediate grounding and to principles involving this notion, in particular to principles governing its interaction with conjunction and disjunction. On the worldly approaches of Correia 2010 and Fine 2012a, 2012b, it is possible to define a notion of immediate grounding in terms of the usual notion, but it can be shown that on *any* reasonable way of doing it, all the principles put forward by Fritz fail.
- 24 I myself came to see this soon after the publication of Correia 2010 (see Correia 2011). Something like this form of pluralism is discussed in the VARIETIES chapter [Chapter 13] of this handbook. Fine (personal communication) is sympathetic to such a pluralism, and insists on the importance of the notion of representational grounding as a tool for identifying where disagreement on ground-theoretic matters may lie.

- 25 I have recently put forward a view about the connection between worldly and representational grounding of a quite different sort (Correia 2017b: §8). I do not have the space to present it here.
- 26 In Correia 2010, X weakly ground x iff the conjunction of X is a disjunctive part of x . (To be fully accurate, since grounding is there understood as factive, a further clause to the effect that X hold should be added.) On Fine's (2012a) account, disjunctive parthood can also be defined on the facts, and the biconditional connecting weak grounding and disjunctive parthood also holds. The two views are thus very similar, as Fine himself recently emphasized (Fine 2017: 685–687).
- 27 Such a notion is invoked in the context of theorizing about grounding, e.g., in Correia 2010, 2016, 2017a and Correia and Skiles 2019, but also in other contexts: see for instance Rayo 2013 and Dorr 2016.
- 28 I am grateful to Kit Fine, Peter Fritz, Stephan Krämer, Kevin Mulligan and Michael J. Raven for useful feedback. This work was supported by the Swiss National Science Foundation Consolidator project BSCG10_157792.

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16

INFINITE DESCENT

T. Scott Dixon

1. Introduction

Once one accepts that certain things metaphysically depend upon or are metaphysically explained by other things, it is natural to begin to wonder whether these chains of dependence or explanation must come to an end. This chapter surveys the work that has been done on this issue—the issue of grounding and infinite descent. I frame the discussion around two questions:

Question 1. What is infinite descent of ground?

Question 2. Is infinite descent of ground possible?

In addressing the second question, I will consider a number of arguments that have been made for and against the possibility of infinite descent of ground. When relevant, I connect the discussion to two important views about the way reality can be structured by grounding: metaphysical foundationalism and metaphysical infinitism. A third such view, metaphysical coherentism, countenances cyclic grounding chains. Due to limitations on space, I will not discuss this view in what follows, though I will have cause to discuss cyclic chains. For further discussion of coherentism, see “Strict Partial Order” (Chapter 17 in this volume).

Before I begin, I must introduce certain definitions and principles that will make it possible to state the central questions, and the answers to them, more precisely. I take full grounding as primitive, at least provisionally, and assume that it is a relation that holds between a single entity on the one hand and one or more entities on the other. I express full grounding claims as instances of the schema

x is fully grounded by Γ ,

where ‘ x ’, ‘ y ’, and ‘ z ’ are singular variables and ‘ T ’ and ‘ Δ ’ are plural ones.¹ Each ranges over the entities in the domain of the grounding relation, whatever they may be. To simplify the formulation of various claims, I assume a plural variable can be assigned to no objects. While many authors restrict the grounding relation to facts, others (e.g., Schaffer 2009: 375–76) are more liberal about the sorts of things that may stand in the grounding relation. I frame the following discussion in a way that is neutral between the two sorts of view. I define partial grounding in the standard way.

Partial Grounding. x is *partially grounded* by $y =_{df}$ there are Γ such that y is among Γ and x is fully grounded by Γ .²

I'll say that x is *properly* partially grounded by y if and only if x is partially, but not fully, grounded by y .

Full grounding is generally thought to be governed minimally by the following two axioms.

Full Irreflexivity. x is not fully grounded by x, Γ .

Full Cut. If x is fully grounded by y, Γ and y is fully grounded by Δ , then x is fully grounded by Γ, Δ .³

These two axioms together entail that full and partial grounding are asymmetric, and that partial grounding is irreflexive.

Full Asymmetry. If x is fully grounded by y, Γ , then y is not fully grounded by x, Δ .

Partial Asymmetry. If x is partially grounded by y , then y is not partially grounded by x .

Partial Irreflexivity. Nothing is partially grounded by itself.

And Full Cut on its own entails the following more familiar transitivity principles usually thought to govern full and partial grounding.

Full Transitivity. If x is fully grounded by y and y is fully grounded by z , then x is fully grounded by z .

Partial Transitivity. If x is partially grounded by y and y is partially grounded by z , then x is partially grounded by z .

Thus these axioms ensure that partial grounding is a strict partial order—i.e., an irreflexive and transitive (and thus asymmetric) relation.

It will be helpful to have available the notion of a grounding structure.

Grounding Structures. Γ form a *grounding structure* $=_{df}$ there are x and y among Γ such that x is partially grounded by y .

It will also be helpful to have available the notion of a grounding chain.

Grounding Chains. Γ form a *grounding chain* $=_{df}$ (i) Γ form a grounding structure and (ii) for every x and y among Γ , either x is partially grounded by y , y is partially grounded by x , or $x = y$.

With these definitions and principles in hand, we are ready to address the two questions, beginning with Question 1: What is infinite descent?

2. What Is Infinite Descent of Ground?

What is infinite descent of ground? The most straightforward, and, I think, the best answer to this question is that it is something that is exhibited by any grounding structure which contains a grounding chain that has no end, i.e., that is such that one can move along it from node to node in a “downward”, or *groundward*, direction (i.e., in the direction of a ground) forever. In other words, infinite descent of ground is what is exhibited by any grounding structure which

contains an *infinitely descending grounding chain*. But what is an infinitely descending grounding chain? The characterization I just provided is rather imprecise. Here is a more precise one:

Infinitely Descending Grounding Chains. Γ form an *infinitely descending grounding chain* $\stackrel{\text{df}}{=}$ (i) Γ form a grounding chain and (ii) for every x among Γ , there exists a y among Γ such that x is partially grounded by y (cf. Rabin and Rabern 2016: 371).

Any node you pick in an infinitely descending grounding chain will have a ground in the groundward direction in the chain, as with C_1 in Figure 16.1. For the purposes of the following discussion, I assume that this definition captures the notion of an infinitely descending grounding chain and thus the notion of ground-theoretic infinite descent more generally.

It is an interesting question whether a cyclic chain would count as exhibiting infinite descent. One way to allow for the possibility of cyclic grounding chains is by denying Partial Transitivity, since one can avoid violations of Partial Irreflexivity so long as nothing immediately grounds itself. But the definition of grounding chains I gave in the previous section presupposes the truth of Partial Transitivity. It would not count pairwise distinct entities a_0 , a_1 , and a_2 as forming a grounding chain if a_0 is grounded by a_1 and a_1 is grounded by a_2 but no other grounding claims hold amongst them. This is because a_0 and a_2 are distinct and yet neither is grounded by the other. Thus one must modify the definition of grounding chains to work in contexts in which Partial Transitivity is not assumed to be true. The most straightforward way to do so is as follows.

Grounding Chains (Revised). Γ form a *grounding chain* $\stackrel{\text{df}}{=}$ (i) Γ form a grounding structure and (ii) for every x and y among Γ , either x is partially grounded⁺ by y , y is partially grounded⁺ by x , or $x = y$,

where partial grounding⁺ is the transitive closure of partial grounding on its domain, which can be defined as follows. (Let S be the domain.)

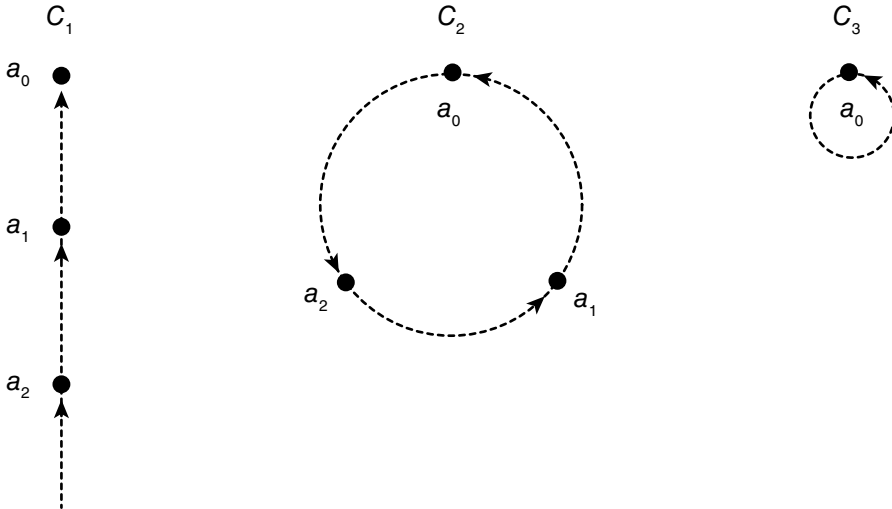


Figure 16.1 Nodes represent the relata of grounding. A directed dashed line running to a node x from a node y indicates that x is partially grounded by y . The only other grounding claims that are assumed to hold are those entailed by Partial Transitivity.

Transitive Closure of Partial Grounding. The transitive closure of partial grounding on a set S is the smallest relation R such that, for any x, y , and z in S , (i) if x is partially grounded by y , then $x R y$, and (ii) if $x R y$ and $y R z$ then $x R z$.

Given this revised definition of chains, cyclic grounding chains can be defined as follows.

Cyclic Grounding Chains. Γ form a *cyclic grounding chain* =_{df} (i) Γ form a grounding chain and (ii) the transitive closure of partial grounding on $\{\Gamma\}$ is reflexive (i.e., everything in $\{\Gamma\}$ is partially grounded⁺ by itself),

where $\{\Gamma\}$ is the set of exactly those things among Γ .

As it happens, the definition of infinitely descending chains I gave at the beginning of this section counts cyclic chains as such, given this definition of cyclic chains (and the revised definition of grounding chains). For any noncyclic infinite descending chain, any node in it will have a (partial) ground distinct from itself in the chain (as with C_1). For any cyclic chain, any node in it will also have at least one ground in the chain. But whether that ground is distinct from that node will depend on how grounding is axiomatized. Anyone who countenances cyclic chains will deny that partial grounding is a strict partial order, since its being so rules out such chains. But one can allow for cyclic chains by rejecting either Partial Transitivity or Partial Irreflexivity. If partial grounding is antitransitive on C_2 (i.e., it is not the case that, for any x, y , and z in C_2 , if x is partially grounded by y and y is partially grounded by z , then x is partially grounded by z), then every ground of every node in C_2 will be distinct from that node. This case presupposes the falsity of Partial Transitivity. If partial grounding is transitive on C_2 , however, each of its nodes will be one of its own grounds. And, of course, if the chain consists of just a single node (as with C_3), that node will be one of its own grounds. These latter two cases presuppose the falsity of Partial Irreflexivity.

Allowing cyclic chains to count as infinitely descending ones does not do injustice to the notion of infinite descent, as one might expect. There is a perfectly natural sense in which a cycle of ground involves infinite descent. Indeed, the ways I've informally characterized infinite descent, at the beginning of this chapter and at the beginning of this section, already count cyclic chains as doing so. Cyclic chains do not come to an end. Moreover, one can move groundward from node to node in a cyclic chain forever. Ricki Bliss and Graham Priest (2017: 67, 2018: 9) seem sympathetic to this more liberal sense of the terms “infinite descent” and “infinitely descending grounding chains”. Interesting as it is, I will not investigate the possibility of cyclic infinite descent in what follows, as space permits me only to focus on the possibility of the noncyclic variety. I will henceforth assume that grounding is axiomatized as I have axiomatized it in Section 1, so that partial grounding is a strict partial order. For further discussion of the possibility of cyclic chains, see “Strict Partial Order” (Chapter 17 in this volume).

3. Is Infinite Descent of Ground Possible?

I now move on to Question 2. Is infinite descent of ground possible? Given the definition, this question can be understood as: Are infinitely descending grounding chains possible? Surprisingly, proponents of the two major views about how reality is structured by grounding, according to which grounding is a strict partial order—metaphysical foundationalism and metaphysical infinitism—both seem to answer yes to this question. But a closer investigation of these views reveals a nearby question, on the answer to which there is the expected disagreement between them. In order to explain these two points, I will first characterize these two views about

how reality is structured by grounding, beginning with an informal statement of metaphysical foundationalism.

Metaphysical Foundationalism (Preliminary). There must be a bottom layer of fundamental entities, which ground all other entities.⁴

One typical way to define fundamentality is as follows.

Fundamentality. x is *fundamental* =_{df} nothing partially grounds x .⁵

Something is nonfundamental just in case it is not fundamental (i.e., it is grounded by something). With fundamentality characterized in this way, foundationalism has seen a number of advocates. Jacek Brzozowski (2008), Ross Cameron (2008), Kit Fine (2010: 105), Jonathan Schaffer (2010: 37, 62, 2016: 95), and Karen Bennett (2011) have all endorsed or defended the view. But the view is endemic outside the grounding literature as well. Many a complaint about a view resulting in a vicious regress can plausibly be traced to a tacit foundationalist picture of the world.

This characterization of foundationalism is admittedly imprecise. The view is most often stated by adding an axiom to the standard axiomatization for grounding, which I provided in Section 1. While this axiom has come to be referred to as a “well-foundedness axiom for grounding”, and to the condition it imposes as “well-foundedness”, the most plausible formulation of it is not an instance of the standard mathematical definition of a well-founded relation. For this reason, it is perhaps better to refer to the axiom and to the condition it imposes differently. In what follows, I refer to it as a “well-groundedness axiom for grounding” and to the condition it imposes as “well-groundedness”. As noted by Scott Dixon (2016) and Gabriel Rabin and Brian Rabern (2016), there are several potential candidates one might adopt as one’s well-groundedness axiom for grounding. Four of the most plausible candidates are:

No Infinitely Descending Chains. There are no infinitely descending chains.

No Maximal Infinitely Descending Chains. There are no maximal infinitely descending chains.

No Weakly Maximal Infinitely Descending Chains. There are no weakly maximal infinitely descending chains.

Full Foundations. For every nonfundamental entity x , there are some fundamental entities Γ (where there is at least one x among Γ) such that x is fully grounded by Γ .

The notions of maximal chains and weakly maximal chains can be defined as follows.

Maximal Grounding Chains. Γ form a *maximal grounding chain* =_{df} (i) Γ form a grounding chain and (ii) nothing partially grounds every x among Γ .

Weakly Maximal Grounding Chains. Γ form a *weakly maximal grounding chain* =_{df} (i) Γ form a grounding chain and (ii) there are no Δ that fully ground every x among any final segment of Γ .

And the notion of a final segment of a chain can, in turn, be defined as follows.

Final Segments. Δ form a final segment of Γ =_{df} (i) Γ form a grounding chain, (ii) there is an x among Δ , (iii) Δ are among Γ , and (iv) every x among Γ , but not among Δ , is partially grounded by every y among Δ .

A final segment of the infinitely descending chain C_1 , depicted in Figure 16.1, for example, will be any chain formed by $a_n, a_{n+1}, a_{n+2}, \dots$ for any $n \geq 0$.

The notions of maximal chains and weakly maximal chains are not entirely straightforward, so I'll illustrate them by making reference to the grounding structures depicted in Figure 16.2.⁶ In S_1 , the chain $a_0, a_1, \dots, b$ is maximal, since there is nothing that partially grounds each of the things in it. It is also weakly maximal, since there is nothing that *fully* grounds each of the things in any final segment of it. (If a chain is maximal, then it is weakly maximal, since, if there is nothing that partially grounds everything in a chain, then there are no things that fully ground everything in any final segment of it.) The chain consisting of just the a s is neither maximal nor weakly maximal, since there is something (namely, b) that fully (and so partially) grounds each of the a s. Similarly, in S_2 , the chain $a_0, a_1, \dots, b$ is maximal (and weakly maximal), again, since there is nothing that partially grounds each of the things in it. And the chain consisting of just the a s isn't maximal, since b partially grounds everything in it. But it *is* weakly maximal, since no things *fully* ground everything in any final segment of it. In S_3 , the chain consisting of the a s is maximal, since there isn't anything that partially grounds each of the things in it (and so it is weakly maximal also).

Of the four most plausible candidate well-groundedness axioms mentioned above, Dixon and Rabin and Rabern note that Full Foundations (or an equivalent of it) is the most plausible. They provide considerations in support of it, in the form of grounding structures which intuitively should be considered acceptable by foundationalist lights but which are ruled out by some of the other contenders, and structures which intuitively should *not* be acceptable by foundationalist lights but are allowed by some of them. S_1 and S_3 , for example, should be considered acceptable by foundationalist lights. Following Schaffer (2010: 37), Dixon (2016: 447) notes that one of the motivations for foundationalism is to validate the idea that the derivative (the nonfundamental) must have its source in, or acquire its being from, the nonderivative (the fundamental). In S_1 , there is something nonderivative, namely b , which fully grounds every derivative thing in it. b might be God, who has decided to create an infinitely descending chain of dependent entities. Each thing in that chain would be dependent on God as well as on those

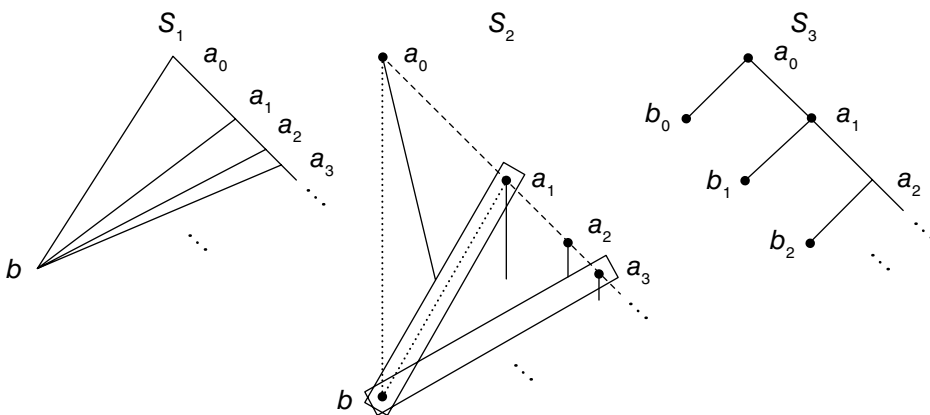


Figure 16.2 A solid/dashed/dotted line running in a downward direction from a node x to another node y indicates that x is fully/partially/properly partially grounded by y . A solid line connecting a node x to a solid box enclosing nodes $y_1, y_2, \dots$ indicates that x is fully grounded by $y_1, y_2, \dots$. The only other grounding claims that are assumed to hold are those entailed by Full Cut and Full and Partial Transitivity.

things groundward relative to it in the chain.⁷ In S_3 , while there is no *single* nonderivative thing which fully grounds every derivative thing, each derivative thing in it is fully grounded by *some* nonderivative thing or other (a_i is fully grounded by, for example, b_i for each $i \in \{0, 1, \dots\}$). An infinitely disjunctive fact, $[p_0 \vee (p_1 \vee (p_2 \vee \dots))]$, each disjunct of which is true and expresses a fundamental fact, might be seen as an example of this sort of structure. Each disjunctive subformula, $[p_i \vee (p_{i+1} \vee (p_{i+2} \vee \dots))]$, would be fully grounded by the fundamental $[p_i]$, while also being fully grounded by the nonfundamental $[p_{i+1} \vee (p_{i+2} \vee (p_{i+3} \vee \dots))]$.⁸ Neither is true of S_2 . While each derivative thing is *partially* grounded by b , not all (indeed, none) is *fully* grounded by it or by any other nonderivative thing. It is fully grounded only by b together with something derivative (a_i is fully grounded by, for example, b, a_{i+1} for each $i \in \{0, 1, \dots\}$).⁹

Of the contenders, Full Foundations is unique in that it does not make the wrong determination about any of these grounding structures. No Infinitely Descending Chains incorrectly rules out S_1 and S_3 . (It is this candidate which comes closest to an instance of the standard mathematical definition of a well-founded relation.) No Maximal Infinitely Descending Chains incorrectly allows for S_2 and incorrectly rules out S_3 . And No Weakly Maximal Infinitely Descending Chains incorrectly rules out S_3 . Only Full Foundations allows for S_1 and S_3 but rules out S_2 . Dixon and Rabin and Rabern conclude on this basis that Full Foundations is the correct well-groundedness axiom. Henceforth, when I use the phrase “well grounded”, I have Full Foundations in mind.

If Dixon and Rabin and Rabern are right, metaphysical foundationalism can be more precisely characterized as follows.

Metaphysical Foundationalism (Final). For every nonfundamental entity x , there are some fundamental entities Γ (where there is at least one x among Γ) such that x is fully grounded by Γ .¹⁰

(Presumably, many foundationalists will take this thesis to hold necessarily.) However, as Dixon and Rabin and Rabern note, and as should be clear from the preceding discussion, there are infinitely descending grounding chains that satisfy Full Foundations, i.e., every (nonfundamental) entity in them is fully grounded by some fundamental entities or others. S_1 and S_3 both contain examples of such chains. So it would appear that infinitely descending grounding chains are possible *even by foundationalist lights*.

But what of infinitely descending chains that are *not* well grounded? That is, what about infinitely descending chains which contain entities which are *not* fully grounded by any fundamental entities? Are they possible? This is the more philosophically interesting question, and the question about which there is the expected disagreement between metaphysical foundationalism and infinitism. The foundationalist says no, while the infinitist says yes. For the purposes of this discussion, metaphysical infinitism can be formulated as follows.

Minimal Metaphysical Infinitism. There is at least one infinitely descending grounding chain, at least one entity in which fails to be fully grounded by any fundamental entity or entities.

Metaphysical infinitism is often stated in much stronger terms. Perhaps most notably, Matteo Morganti’s (2014, 2015) argument for metaphysical infinitism (to be discussed in more detail later) can plausibly be construed as supporting the idea that *every* entity in the domain of grounding sits atop an infinitely descending grounding chain, and perhaps even that this

is *necessarily* so, despite the fact that Morganti himself (2014: 233) characterizes the view very minimally, as I have done above. (Naomi Thompson (2016: 41) calls the former view “strong infinitism” and calls minimal infinitism “weak infinitism”.) In any case, however the infinitist characterizes her view, any plausible statement of it will at least entail the minimal version of it stated above. And that is enough to set it apart from foundationalism, as any view that entails that thesis allows for the possibility of non–well–grounded infinitely descending grounding chains.

It would appear, then, that the interesting question is not whether infinite descent *simpliciter* is possible but whether *non–well–grounded* infinite descent is possible. Proponents of foundationalism and infinitism both answer yes to the former question. It is the latter question which sets them apart, as foundationalists answer no while infinitists answer yes.

Question 2 (Revised). Is *non–well–grounded* infinite descent of ground possible?

We turn our attention to this question. I will not take a stand on the answer to this question in this chapter. Instead, I will survey a number of arguments for answering this question no (Section 4) and yes (Section 5).

4. Non–Well–Grounded Infinite Descent Is Not Possible

Many have expressed support for the idea that grounding is well grounded, and hence that non–well–grounded infinite descent is not possible. Perhaps a greater number still have assumed its truth, for example, as a premise, sometimes explicit, sometimes implicit, in a regress argument against some view or other. See, for example, Brzozowski (2008: 199–201), Fine (2010: 105), Schaffer (2010: 61–65), Bennett (2011: 30), and Dixon (2018: §5). In short, the view that non–well–grounded infinite descent of ground is not possible, often under the guise of foundationalism, has, deserved or not, enjoyed a profusion of adherents compared to its rivals. Perhaps for this very reason, arguments for foundationalism have been difficult to come by. The few that there are come in the form of arguments for foundationalism. In this section, I will look at three of the most prominent recent discussions in favor of foundationalism, which, among all others, most deserve to be called “arguments”.

Perhaps the best known and most familiar of these are *inheritance arguments*. This sort of argument is deployed by Schaffer (2010: 37, 62, 2016: 95), who traces it back to Leibniz and Aristotle. (See **Part I** of this volume for more on grounding in the history of philosophy.) A version of the argument can be put as follows. Reality (or being or existence) is a quality that every entity in the domain of grounding must have in full measure, and it is a quality that can be transmitted from one such entity to another only by grounding. In particular, such an entity *x* has the relevant quality in full measure if and only if either (i) *x* is fundamental or (ii) *x* is fully grounded by some entities, each of which has that quality in full measure. Were every nonfundamental entity *x* not fully grounded by some fundamental entities or others, then *x* could not have the quality in full measure. After all, as Schaffer says,

One cannot be rich merely by having a limitless sequence of debtors, each borrowing from the one before. There must actually be a *source* of money somewhere. Likewise something cannot be real merely by having a limitless sequence of ancestors, each claiming reality from its parents. There must actually be a source of reality somewhere. Just as wealth endlessly borrowed is never achieved, so reality endlessly dependent is never realized.

(Schaffer 2016: 95, *ital. orig.*)

But every nonfundamental entity *must* have the quality in full measure (since *every* entity in the domain of grounding must). So every nonfundamental entity must be fully grounded by some fundamental entities or others. Of course, independent reasons need to be given for Full Irreflexivity and Full Cut for a full defense of foundationalism as it is usually formulated. But the inheritance argument does a lot of heavy lifting for the view.

Another important argument for foundationalism is due to Cameron (2008). He argues for it on pragmatic grounds. Other things being equal, he says, we should avoid theories which posit infinitely descending chains of explanatory relations such as grounding if we have an alternative theory available which provides a unified explanation of the relevant phenomena. This effectively puts foundationalism at an advantage over infinitism in any context where such a theory exists. Consider some set of phenomena, P , consisting of some facts which we wish to explain. Suppose there are two theories, T_1 and T_2 , such that T_1 provides a full explanation of each element of P by a single fundamental entity f , while T_2 does so by an infinitely descending chain of facts $f_1, f_2, \dots$. Many will agree that T_1 is preferable to T_2 . And Cameron's criterion yields this result—a result that is friendly to foundationalism. Further, Cameron's criterion may, or a slightly stronger version of it would, yield a foundationalism-friendly result even when there are a countably infinite number of facts in P , $g_1, g_2, \dots$, and a theory, T_3 , which explains each element of P with another element of P , so that the elements of P form an infinitely descending grounding chain. Depending on how heavily we prefer unified explanations to nonunified ones, we can employ a criterion like Cameron's that will justify our choosing T_1 over T_3 , even though it posits a fact that T_3 does not. This would be because it posits a *unified* explanation of the phenomena we would like to explain, whereas T_3 does not.

A final argument for foundationalism that is worth noting is due to Bliss (2019). Once each dependent (nonfundamental) entity that exists has been provided with a full explanation, one might think that there is still something that needs to be explained. Such a situation is, after all, compatible with a model in which Full Foundations is false. Suppose, for example, that every entity is fully grounded by some entities or others, but they are not all fully grounded by some *fundamental* entities or others. (Perhaps all that exists is an infinitely descending chain of full grounds.) After considering and rejecting several possibilities, Bliss concludes that what still needs to be explained is why there are any dependent entities *whatsoever*, i.e., why there are some dependent entities rather than none. This, Bliss argues, can only be explained by positing independent (fundamental) entities. This is due, according to Bliss, to the *externality assumption*: “no dependent entity can explain why there are any dependent entities whatsoever” (*Ibid.*: 11). Moreover, Bliss thinks that this assumption is defensible, since she takes it to be plausible that one cannot explain why there are any members of a substantial kind *whatsoever* by invoking only members of that kind, even if that explanation goes on forever—a view inspired by Stephen Maitzen (2013: 260).

To be sure, these arguments for foundationalism can be criticized. See Bliss (2013: 407–8), Bohn (2018: 170), and Trogon (2018) for objections to the inheritance argument. And while this is not an objection to Cameron's theory-choice argument *per se*, several have noted that it isn't strong enough to ensure that foundationalism holds necessarily (see Cameron 2008: 12–13; Morganti 2014: 240; Bliss 2018: 76; Trogon 2018: 182). Moreover, it isn't even strong enough to ensure that foundationalism should be preferred to infinitism when the only foundationalist alternative to infinitism posits nonunified explanations of the relevant phenomena. I know of no one who has yet taken issue with Bliss's argument, but one might wonder whether dependent entities form a substantial kind. In any case, whatever one might think of these three arguments, they constitute the most compelling strategies that have been deployed in recent years in attempts to show that there is no non–well-grounded infinite descent. But what of considerations *in favor of* non–well-grounded infinite descent? This is the question to which I turn next.

5. Non–Well-Founded Infinite Descent Is Possible

Arguments for non–well-grounded infinite descent of ground are a bit more numerous. They often come in the form of defenses of metaphysical infinitism. But even these arguments are rarely direct arguments for the *necessary* or even *actual* existence of non–well-grounded infinitely descending chains. As I suggested, foundationalism has enjoyed a dialectical advantage in the debate, and so arguments for alternative views are often intended merely to defend an alleged theretofore unappreciated viability or possibility of non–well-grounded infinitely descending grounding chains.

Schaffer (2003) and Tuomas Tahko (2014) argue that infinitely descending chains are possible if the descent is “boring” or “repetitive”, i.e., if the structure repeats itself in a certain sense.

There is no *novelty* in the structure after a certain point. . . . The boring part of the structure that repeats infinitely could be of any length, as long as it starts anew eventually.

(Tahko 2014: 261, *ital. orig.*)

Schaffer and Tahko speak of the boring part of a boring infinitely descending structure as a fundamental supervenience base. Once one has determined the laws of just one of the segments which repeat in this base, there is no more interesting scientific work to be done (see Schaffer 2003: 512).¹¹ Schaffer thinks that there is something to be said for boring infinite descent. He argues that a picture on which everything supervenes on a boring infinitely descending supervenience base is better off, evidentially, than foundationalism (*Ibid.*: 510–11). The latter is, he says, committed to a particle-based theory of fundamental entities. But the claim that everything ultimately decomposes to particles, Schaffer thinks, is an article of faith, not supported by empirical evidence (*Ibid.*: 504–05). The former, on the other hand, merely postulates a fundamental supervenience base containing infinite descending chains of dependence. So it is compatible with, for example, a pure field theory, which postulates fundamental entities which are infinitely divisible.¹² Tahko (2014: §4) adds to Schaffer’s case for the possibility of boring infinite descent, citing a model of the structure of quarks and leptons which is consistent with the relevant observational data.

Another important argument in favor of non–well-grounded infinite descent is due to Morganti (2014, 2015). He thinks that grounding may be understood in a way analogous to the way Peter Klein (e.g., 2007) conceives of justification. Reality (or being or existence) isn’t *transferred* from grounds to grounded. Instead, it *emerges* from one (or more) infinite chains which descend from a given entity. Morganti thinks that this move is, in the case of a metaphysical dependence relation like grounding, at least as plausible as if not more so than it is in the case of an epistemic dependence relation like justification. This is because metaphysical infinitism appears to be immune from the objection that haunts the epistemological infinitist: that an epistemic agent, with limited cognitive resources, could never hold in her mind all the beliefs (an infinite number of them!) which, according to the account, allegedly justify a given belief only together. The relata of the metaphysical dependence relationships Morganti has in mind (presumably facts, or something like facts) exist whether or not there is anyone around who is capable of representing them. As I mentioned at the end of Section 3, while Morganti’s goal is merely to argue for the *possibility* of infinite descent, his argument can be naturally construed as supporting the idea that, *necessarily*, *every* entity in the domain of grounding sits atop an infinitely descending grounding chain. This is because the emergence model of reality can naturally be taken to be exclusive: reality can *only* emerge; it *cannot* be transferred in full

measure. If so, then something can have reality in full measure *only* if it sits atop an infinitely descending grounding chain. And given that everything *must* have reality in full measure, this would mean the emergence view holds necessarily. See Lubrano (2018) for a recent challenge to emergence-based infinitism.

Michael J. Raven (2016) develops a view according to which an entity may be fundamental without being foundational. Note, however, that he uses these terms differently than I have been using them. For Raven, an entity is foundational just in case it is fundamental, in the way I defined fundamentality (i.e., as something that is ungrounded). A fundamental entity, for him, however, is an entity that is *ineliminable*, in the sense that “reality cannot be completely described without it” (*Ibid.*: 609). To get the idea across, Raven provides the following examples of an (comparatively plausibly) ineliminable fact and an eliminable fact, respectively.

- (E) Electron e^- is negatively charged.
- (F) FEMA is a federal agency.

He thinks that one can also evaluate the *constituents* of facts for eliminability. Of the constituents of (E) and (F), e^- and the property *being negatively charged* are more plausibly ineliminable than FEMA and the property *being a federal agency*. What is important in connection with the present discussion is that Raven’s view allows one to make out a conception of fundamentality even if every grounding chain is not well grounded. The materiality of a gunky object, *a*, for example, can be understood as fundamental, even though every grounding chain which the fact that *a* is gunky sits atop is infinitely descending and not well grounded. (A gunky object is one which has a proper part and is such that each of its proper parts has a proper part.) Its materiality is grounded in the materiality of its left and right parts, the materiality of each of which is grounded in *its* left and right parts, and so on. But materiality is ineliminable from these series of explanations, and thus it is fundamental on Raven’s account. So, while Raven does not directly argue for the possibility of non–well-grounded infinitely descending chains, he removes an obstacle one might envisage for infinitist views according to which every grounding chain is not well grounded: they preclude the existence of fundamental entities, and one might have independent reason to think that such entities exist.¹³

Two final arguments for non–well-grounded infinite descent are due to Einar Duenger Bohn (2018). He argues first that the metaphysical possibility of gunky and junky objects, along with the truth of at least one of the following two theses, entails that non–well-grounded infinitely descending chains are possible.

- Parts Ground Wholes.** Necessarily, for any *x* and *y*, if *x* is a proper part of *y*, then *y* is partially grounded by *x*.
- Wholes Ground Parts.** Necessarily, for any *x* and *y*, if *x* is a proper part of *y*, then *x* is partially grounded by *y*.

(A junky object is one that is a proper part of something and is such that everything it is a proper part of is a proper part of something.) Bohn then provides some considerations to the effect that the actual world might be both gunky and junky (what he calls “hunky”), in which case metaphysical foundationalism would *actually* be false. Bohn deploys a second argument against any view that posits any fundamental facts (e.g., foundationalism) by arguing that it cannot answer the question of why they exist, and (he presumes) there must be such a reason. This presumption is given by what he calls “the metaphysical principle of sufficient reason”:

Metaphysical Principle of Sufficient Reason (MPSR). Every fact has a metaphysical explanation.

Bohn thinks that even if the MPSR isn't true, it is better to avoid violating it without good reason. Since views about the metaphysical structure of reality exist which satisfy the principle, he thinks that, all else being equal between them and foundationalism, they should be preferred to foundationalism. For more on grounding and the principle of sufficient reason, see "Principle of Sufficient Reason" (Chapter 4 in this volume).

6. Concluding Remarks

Infinite descent of ground is what is exhibited by any infinitely descending chains of ground. The question of whether infinite descent, so understood, is possible has an easy answer. Assuming that metaphysical foundationalism is best captured by the addition of the most plausible well-groundedness axiom for grounding, Full Foundations, to the standard axiomatization for grounding consisting of Full Irreflexivity and Full Cut, that answer is yes, whether one is a metaphysical foundationalist or infinitist. But answering the nearby question of whether *non-well-grounded* infinite descent is possible is not nearly so easy. I surveyed some of the most compelling considerations for answering this question yes and no. I leave it to the reader to decide which is the best way to answer it.

Related Topics

See "Strict Partial Order" (Chapter 17 this volume) for more on cyclic chains and coherentism. See "Fundamentality" (Chapter 23) for more on arguments for and against the existence of infinitely descending chains of ground (in the form of arguments for and against fundamentality). For more on the connection between grounding and the principle of sufficient reason, see "Principle of Sufficient Reason" (Chapter 4).

Notes

- 1 I therefore take the view that the fundamental notion of grounding is expressed by a predicate rather than an operator. Despite this, what is said in what follows can be recast in operationalist terms. For further discussion of the predicate-operator debate, see the Introduction to **Part III** and "Granularity" (Chapter. 15) in this volume. Also, I have in mind Fine's (2012a, 2012b) notion of strict ground, as opposed to weak ground. It is the former of which, after all, that is supposed to capture the explanatory notion at issue.
- 2 See Rosen 2010: 115; Audi 2012: 698; Fine 2012a: 50; Raven 2013: 194 for endorsements of this definition. I should note that despite its wide-ranging acceptance, this definition is not completely free from controversy. Some have argued that a partial ground of something need not be among some things which fully ground it. See, for example, Leuenberger 2019. I ignore this issue in what follows.
- 3 This principle is not quite strong enough to do everything a cut principle for full grounding should do, though it is sufficient for my purpose here. See deRosset 2014: 720 for a completely adequate principle, which is derivable in Fine's (2012a, 2012b) system as well.
- 4 For other informal characterizations of foundationalism, see, for example, Schaffer 2010: 37; Bliss 2014: 245, 2018: 76, 2019: 359; Raven 2015: 323; Brzozowski 2016: 59; Thompson 2016: 40; Barnes 2018: 65; Morganti 2018: 257; Rabin 2018: 38; Tahko 2018: 240.
- 5 Not everyone accepts this definition of fundamentality, but it will serve my purpose for the time being. I will have cause to discuss an alternative conception of fundamentality in what follows. I also ignore the distinction between being ungrounded and being zero-grounded, which could complicate a

- definition of fundamentality. See Fine 2012a: 47–48. See “Fundamentality” (Chapter. 23) in this volume for further discussion of how to conceive of fundamentality.
- 6 A chain is maximal iff it has no (partial) lower bound and is weakly maximal iff it has no full lower bound. This is the terminology used by Rabin and Rabern (2016: 372).
 - 7 For other potential examples of this sort of structure, see Bliss 2013: 416, 2014: fn. 9; Dixon 2016: §4, Rabin and Rabern 2016: §4.1.
 - 8 This example is from Dixon 2016: §7; Rabin and Rabern 2016: 355–56. While Litland (2016: §5) raises concerns about its plausibility, it is one of the most straightforward ways at least to give the reader a sense of the sort of examples that might exhibit this structure. For other examples of this sort of structure, see Litland 2016; Dixon 2016: §7.
 - 9 See Dixon 2016: §6; Rabin and Rabern 2016: 372 for further discussion of this sort of structure.
 - 10 While many foundationalists will presumably endorse Full Irreflexivity and Full Cut in addition, it is not obvious that foundationalism requires the truth of these two axioms. Views which allow for cyclic grounding chains, as long as every nonfundamental entity is fully grounded by a plurality of fundamental entities, and views in which Full Cut is false, but in which every nonfundamental entity is related to a plurality of fundamental entities by the closure of full grounding under Full Cut on its domain, may well count as foundationalist. For the sake of brevity, I set such views aside here and focus only on views which validate both axioms. Analogous remarks apply to my formulation of metaphysical infinitism.
 - 11 A Pascalian worlds-within-worlds situation might be an example of this, according to which each particle in the universe comprises a microuniverse, each particle of which comprises yet another microuniverse, and so on (see Schaffer 2003: 505). It might be enough that each embedded universe has the same natural laws as each of the others. Even if the worlds differ radically otherwise (i.e., different events occur in them), this may be enough to constitute boring infinite descent. Such a world may be the setting of the adventure described in Hasse 1936.
 - 12 Fine (2009: 174–75) discusses a similar example of boring infinite descent, exemplified by the Thalean view that everything is composed of water conjoined with the Aristotelian view that water is infinitely divisible. See also Fine 2001: 27.
 - 13 Raven’s characterization of fundamentality as ineliminability can be operative in, for example, Rosen’s (2018: 157) characterization of metaethical nonnaturalism, which appears to allow for infinitely descending chains of normative facts. Raven can regard normativity as fundamental—a natural, perhaps even essential, thesis for a nonnaturalist to endorse—even if there are no ungrounded normative facts.

Further Reading

- Dixon, T. S. (2016) “What Is the Well-Foundedness of Grounding?” *Mind* 125, 439–68.
(For a comprehensive look at the most plausible well-groundedness axioms, and extensive argumentation for Full Foundations.)
- Litland, J. E. (2016) “An Infinitely Descending Chain of Ground without a Lower Bound,” *Philosophical Studies* 173, 1361–69.
(For an example of a chain like S_3 (Figure 16.2) that, Litland argues, is more plausible than those discussed in Dixon 2016 and Rabin and Rabern 2016.)
- Rabin, G. O. and B. Rabern (2016) “Well-Founding Grounding Grounding,” *Journal of Philosophical Logic* 45, 349–79.
(For a comprehensive look at the most plausible well-groundedness axioms, and the connection between well-groundedness and the meta-grounding debate.)

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STRICT PARTIAL ORDER

Naomi Thompson

Introduction

This chapter explores aspects of the logic of ground. A strict partial order is transitive, irreflexive, and asymmetric. Whilst it is generally assumed that grounding exhibits these properties, this has consequences for other aspects of the study of ground, and so each assumption has been challenged in the grounding literature. The present chapter critically discusses those challenges and explores reasons for thinking that grounding forms a strict partial order in the first place.

According to the orthodox conception of grounding, grounding imposes a *strict partial order* on the entities in its domain: it is transitive, irreflexive, and asymmetric. Grounding is *transitive* if when A grounds B and B grounds C, A grounds C. More formally, transitivity =_{df} $\forall x \forall y \forall z [(x \prec y \wedge y \prec z) \rightarrow x \prec z]$ (where $x \prec y$ is to be read as ‘x partially grounds y’). Grounding is *irreflexive* if nothing grounds itself: irreflexivity =_{df} $\neg \exists x [x \prec x]$. It is *asymmetric* if when A grounds B, B doesn’t also ground A. Asymmetry =_{df} $\forall x \forall y [x \prec y \rightarrow \neg(y \prec x)]$. Each of these three principles is taken to hold for full as well as for partial ground, where A is a partial ground for C if A on its own or with some other grounds is a full ground for C (see Fine 2012: 50). Together, the principles ensure that grounding is capable of ordering reality, relating the facts or entities in its domain in linear, nonrepeating chains.

It is perhaps most natural in this context to think about grounding as a relation which forms a strict partial order on facts, but we might also think in terms of entities of various other ontological categories. I’ll use both formulations here. Those commentators who take grounding to be a sentential connective can take grounding to form a strict partial ordering on sentences (and hence on the reality represented by those sentences) without taking grounding itself to be a relation, because sentences of ground are transitive, irreflexive, and asymmetric.¹

Other relations that form a strict partial order include *being older than* and *being an ancestor of*. Just as it is not the case that everybody is related to everybody else via the *being older than* relation (some people are the same age) the order that grounding relations form is partial rather than total: not everything appears on the same grounding chain. Perhaps (the existence of) my kitchen table is grounded in (the existence of) the atoms that compose that table, and perhaps (the existence of) your sofa is grounded in (the existence of) the atoms that compose your sofa. It is very unlikely that any grounding relations obtain between the atoms that compose my table and those that compose your sofa, and so grounding forms a merely partial order.

All three of the features that render grounding a strict partial order have been challenged in the grounding literature. In some cases, these challenges have been mounted in an attempt to modify or to reform the orthodox conception of grounding. In others, they have been used to challenge the very notion of ground itself, or one or more of its key roles. In the next section, I'll outline why proponents of grounding generally take grounding relations to be transitive, irreflexive, and asymmetric. I'll then discuss arguments for and alleged counterexamples to grounding exhibiting each of those three properties. Note that the relevant properties are not independent of one another; asymmetry entails irreflexivity, and when transitivity is assumed, irreflexivity implies asymmetry.

1. Why Think Grounding Relations Form a Partial Order?

The point is often made in the literature that grounding is an explanatory relation (see Chapter 8 on "Explanation"). It is sometimes claimed that this explanatory aspect of ground imposes on the notion a distinctive logic, which includes transitivity, irreflexivity, and asymmetry (see Raven 2015: 327). If *unionism* is true and so grounding *just is* an explanatory relation, then we should expect grounding to have the features that explanatory relations have. If *separatism* is true and so grounding relations back a certain kind of explanation, then we should expect the formal features of explanation and grounding (largely) to be shared, since the features of the relevant kind of explanation would be inherited from those of grounding. The relevant kind of explanation here is *metaphysical explanation*. If separatism is true and metaphysical explanations track grounding relations, the inheritance runs from grounding to explanation rather than from explanation to grounding.² Those features of explanation then cannot serve as justification for grounding being thought to have the formal features that it does, on pain of circularity. Note too that were we to claim that the inheritance ran in the other direction, we would need some independent specification of the formal features of *metaphysical explanation*, which threatens to be too similar to the notion of grounding to be illuminating (see Daly 2012: 91).

It is plausible that explanation in general is transitive, irreflexive, and asymmetric. It seems to be important that explanations can chain—that they can build on one another. We don't expect to be able to explain a fact or event in terms of that very same fact or event, and circular explanations are *prima facie* bad explanations. If these are features of all notions of explanation, then they are features of metaphysical explanation. If grounding just is metaphysical explanation, then grounding relations must be transitive, irreflexive, and asymmetric.

Why think that explanations in general exhibit those features? One reason is that causal explanations seem to exhibit them,³ and causal explanation is our explanatory paradigm. But a plausible way to account for the fact that causal explanations seem to exhibit these features is that causal *relations* are generally considered to be transitive, irreflexive, and asymmetric.⁴ Unless we think of grounding as a kind of 'metaphysical causation' (Wilson 2017; Schaffer 2016), it is a mistake to infer from our paradigmatic causation-based understanding of explanation that grounding is transitive, irreflexive and asymmetric. Grounding is usually explicitly taken to be a *noncausal* dependence relation, and so we should not simply expect its formal features to be shared with those of causal explanation. We might, however, argue that grounding and causation are similar enough to share logical properties (and so to be entitled to try to explain why grounding is a strict partial order in terms of general features of explanation, which causal explanation exemplifies), but this is a substantive thesis.

A different way we might account for the formal features of explanation is via the role that explanation plays in our epistemic lives. Transitivity is a seemingly important feature of explanations, allowing us to modify the kind of true explanation we give depending on the

context. Whilst it is widely held to be obvious that circular explanations are bad explanations, it is much less obvious precisely why this is thought to be the case. (One candidate reason is that causal explanation is our explanatory paradigm and causal loops generate paradoxes, but this we discussed earlier.) A different suggestion is that circular explanations are objectionably trivial or uninformative; they fail to contribute any new explanatory information. A ban on circular explanation can be used to justify both the asymmetry of explanation and its irreflexivity, since we can think of self-explanation as an extremely tight explanatory circle.

The friend of grounding is likely to want to resist accounting for the formal features of grounding in terms of explanation when the formal features of explanation are themselves accounted for in epistemic terms such as these. Grounding is generally supposed to be a thoroughly metaphysical relation, not the sort of thing whose formal features might derive from the importance of respecting agent-dependent notions such as informativeness or understanding. But there is more to be said here to justify the claim that this is indeed the situation, as there is scope to understand both explanation and notions such as informativeness, as well as the connection between explanation and understanding, in terms that are not obviously agent dependent. This way of understanding metaphysical explanation at least is likely to appeal to the friend of grounding.

Thus far we have assumed that explanations do indeed exhibit the properties of a strict partial order, but there are reasons to resist this assumption. *Holistic explanations* do not have an asymmetric structure. Instead, holistic explanations are built out of mutually supporting pieces of information such that there are no ‘unexplained explainers’. Each part of a system of explanation is explained by the other parts of the system. The model is much like a coherentist’s model of justified belief, where justification is a feature of being part of a web of mutually supporting beliefs. If explanations can be holistic, neither asymmetry nor (assuming transitivity holds) irreflexivity are features of all explanations. I’ve said nothing here to motivate thinking that we should countenance holistic models of explanation,⁵ but at the very least it is plausible that we shouldn’t dismiss such explanatory structures without further argument (Barnes 2018: 67).⁶

Though there is undoubtedly a close connection between grounding and (metaphysical) explanation, we do not have reason to take the logical properties of grounding to be entirely settled by reflections on the properties of explanation. Reflecting on the explanatoriness of particular grounding claims can be a useful guide to our discussion of grounding, but we should be cautious about taking such reflections to be conclusive evidence.

We touched earlier on a different kind of motivation for thinking that grounding relations form a strict partial order, which does not have anything to do with the alleged explanatory character of grounding. One of the primary purposes of engaging in grounding-talk is to offer a characterization of reality’s structure; grounding relates the more fundamental to the less fundamental in a hierarchical structure. According to the orthodox, foundationalist conception of reality’s structure (see Chapters 23 on “Fundamentality” and 27 on “Structure”) that structure terminates in some fundamental facts or entities. According to the infinitist conception (see Chapter 16 on “Infinite Descent”) the structure continues infinitely. In both cases the chains formed by grounding relations are linear and non-repeating because grounding relations are taken to be transitive, irreflexive, and asymmetric.

To deny that grounding forms a strict partial order is to suggest a radically different conception of reality’s structure, at least for those who think of grounding as the relation that characterizes that structure.⁷ In the penultimate section we’ll touch on how such structures might look, but for now it is enough to note that if reality has a foundationalist or an infinitist structure characterized in terms of grounding, then grounding must form a strict partial order.

2. Is Grounding Transitive?

Transitivity is an assumption of almost all of the grounding literature, and the assumption of transitivity is generally made without further discussion. Inferences that rely on the transitivity of grounding generally seem valid, as in the following case:

- (1) Biological facts are grounded in chemical facts;
- (2) Chemical facts are grounded in physical facts;
- (3) Therefore, biological facts are grounded in physical facts

Notwithstanding the plausibility and prevalence of the assumption that grounding is transitive, Schaffer (2012) develops some alleged counterexamples to the transitivity of grounding. One example (see also Chapter 18 of this volume) involves a slightly dented sphere which is more or less spherical, and has maximally determinate shape *S*. (Note that Schaffer is concerned here with partial grounding):

- (a) The fact that the thing has the dent it does grounds the fact that the thing has shape *S*.⁸
- (b) The fact that the thing has shape *S* grounds the fact that it is more-or-less spherical
- (c) The fact that the thing has the dent it does grounds the fact that it is more or less spherical

Schaffer takes (a) to hold because were it not for the dent the thing's shape would have been different. *Having shape S* is a determinate of the determinable *being more or less spherical*, and determinables are fully grounded in their determinates. This accounts for the truth of (b). Given transitivity, we should conclude that (c), but Schaffer claims that (c) is implausible, because if anything, the dent detracts from, rather than explains, the more or less sphericity of the thing (Schaffer 2012: 126–127).

Schaffer takes this and two further alleged counterexamples to transitivity to motivate adopting a *contrastive* treatment of grounding. Roughly, this involves taking grounding statements to have the form: the fact that *A* rather than *A** grounds the fact that *B* rather than *B**. In this form, Schaffer argues, a version of transitivity for grounding can be preserved. From the previous contrastive statement and the fact that *B* rather than *B** grounds the fact that *C* rather than *C**, we can infer via transitivity that the fact that *A* rather than *A** grounds the fact that *C* rather than *C**. This amounts to taking transitivity to hold between *differences*, and Schaffer (2012: 132) takes irreflexivity and asymmetry to hold in the same way, thereby preserving a strict partial ordering structure over these differences.

Schaffer's alleged counterexamples to the transitivity of grounding have been challenged. Litland (2013: 22–23) argues that if we replace Schaffer's merely partial grounding claims with full grounding claims, the allegedly counterintuitive conclusion no longer seems problematic. Replacing (a) with full grounding claim (a') gives us: the fact that the thing has the dent it does and elsewhere is shaped like *S'* fully grounds the thing having shape *S*. (b) is already a full grounding claim, and (c') becomes: the fact that the thing has the dent it does and elsewhere is shaped like *S'* fully grounds the thing's being more or less spherical. There is seemingly nothing implausible about (c'), and since (c) clearly follows from it via the definition of partial grounding, (c) too must be unproblematic.

The same point is brought out independently and in a different way by Raven (2013: 199), who argues that the fact that a different dent would also suffice for the more or less sphericity of the thing doesn't preclude *this* dent from accounting for the more or less sphericity on *this* occasion. Both Litland and Raven provide responses to all three of Schaffer's alleged counterexamples, strengthening the case for thinking that grounding is transitive.

Rodriguez-Pereyra (2015: 521–522) also objects to Schaffer’s counterexamples, rejecting (in the case we’ve been discussing) the intuition that the thing wouldn’t have the shape it has were it not to have the dent it does. But once we flesh out the grounding claim as in (a’), that claim seems much harder to resist, giving us the reason to reject the counterexample explained earlier. Rodriguez-Pereyra goes on to provide his own alleged counterexample to the transitivity of grounding, which requires that we take truthmaking to be a grounding relation. He reasons as follows (2013: 523), where I have modified the example slightly to present it explicitly in terms of grounding:

- (d) The fact that grass is green grounds the truth of <grass is green>⁹
- (e) <grass is green> grounds the truth of <there are propositions>
- (f) The fact that grass is green grounds the truth of <there are propositions>

According to Rodriguez-Pereyra, by transitivity, (f) follows from (d) and (e), but (f) is false. The ground for <there are propositions> must itself be a proposition, since existential generalizations are grounded in their instances.¹⁰ However, this alleged counterexample can be resisted. What is grounded in (d) is the *truth* of <grass is green>, but what plays the grounding role in (e) is <grass is green> itself. We should then replace (e) with (e’): the truth of <grass is green> grounds the truth of <there are propositions>, but (e’) is false for the same reason Rodriguez-Pereyra takes (f) to be false.

Let’s discuss one final proposed counterexample presented by Tahko (2013: 336). Tahko’s example concerns what he calls ‘truth-grounding’, which we might think captures the idea of truth-making as grounding (that truth is grounded in being) but does so by avoiding a controversial assumption made by Rodriguez-Pereyra; that grounding obtains cross-categorically. Here is Tahko’s example:

- (g) The fact that a particular bottle of beer, *b*, has a stable macrophysical structure partially grounds the fact that *b* exists
- (h) The fact that the Pauli Exclusion Principle holds partially grounds the fact that *b* has a stable macrophysical structure
- (i) The fact that the Pauli Exclusion Principle holds partially grounds the fact that *b* exists

Given transitivity, (i) follows from (g) and (h). The Pauli Exclusion Principle says that no two identical fermions can be in the same quantum state. It explains the properties of atoms and their classification in the periodic table, some properties of complex molecules, and ultimately the stability of matter (Eisert 2013). The reason Tahko takes this to be a failure of transitivity is that the obtaining of a fundamental law governing the existence of all macrophysical objects whatsoever is not explanatory with respect to the existence of any particular object; it is far too general (Tahko 2013: 337).

There are two ways we might object to Tahko’s example. The first involves questioning the alleged implausibility of (i). It is a matter for debate whether general principles ought to be included amongst the grounds of facts or whether they are merely ‘enabling conditions’ (a similar debate plays out in the literature on causation). If they are to be included amongst the grounds, then perhaps we have no failure of transitivity here. If not, we should also question the appearance of the principle in (g). Suppose we think general principles can (sometimes) appear among the grounds of facts. Does the appearance of a general principle here render (i) false? Tahko appeals to a principle of relevance for truth grounding: ‘if the fact that *q* is not relevant to the truth of proposition *p*, then it is not the case that *q* helps ground the truth of *p*’ (2013:

335). But the principle is controversial. Suppose the truth of a disjunction is made true by each of its disjuncts. Then we might say that the truth of one disjunct renders the truth of the other irrelevant, but it is still the case that the truth of each disjunct grounds that of the disjunction (see Raven (2013: 198)).

Moreover, it seems quite plausible that the principle *is* relevant in grounding the fact that *b* exists, if only in much the same way as the Big Bang is relevant to my typing this chapter. One might insist (with Tahko) on a stricter reading of the relevance constraint, but this leads to the second way we might object to the counterexample. That *b* has a stable macrophysical structure seems itself only fairly distantly relevant to the existence of *b*. Grounding is often introduced with examples such as ‘the fact that beer bottle *b* exists is fully grounded in the fact that the atoms that compose *b* exist’, with no mention of stable macrophysical structure.¹¹

None of the alleged counterexamples we have considered constitute a persuasive case for giving up on the transitivity of grounding. This is perhaps no great surprise: it is extremely important to our ability to generate a comprehensive worldview that explanations chain. A structure that failed to obey transitivity might be extremely fragmented, depending on whether there are failures of the transitivity of full as well as partial grounding. Even if we were to find some cases of failure of transitivity compelling, the friend of grounding might insist that the cause of the apparent failure has to do with our cognitive or epistemic inclinations; that what we find illuminating or take to be understood doesn’t chain, and that this should not be taken to imply any failure of transitivity for grounding itself.¹² Nevertheless, we will not (unless stated) assume transitivity in what follows, and so will treat irreflexivity and asymmetry separately.

3 Is Grounding Irreflexive?

The intuitive motivation for thinking that grounding is irreflexive is that nothing can account for its own existence; nothing is self-explanatory. In rejecting the irreflexivity of grounding, we seemingly reject the idea that grounding orders the world; a fact that grounded itself would occupy two levels of the fundamentality hierarchy at once, and this is seemingly impossible—nothing can be more fundamental than itself. But that on its own is a bad argument for irreflexivity. One should not stipulate that grounding must be irreflexive because it forms a strict partial order and use that to conclude that grounding is irreflexive (Bliss 2018: 80). Instead, we must look for independent motivation for the irreflexivity of grounding and consider the efficacy of purported counterexamples.

At this stage, it will be helpful to introduce a distinction between *mediate* and *immediate* grounding. Mediate grounding is the transitive closure of immediate grounding. The fact that London is a city is immediately grounded in the fact that London is in the UK and the fact that it has been granted city status by a British monarch. (It is not, contrary to popular belief, grounded in the fact that London has a cathedral.) The fact that London is a city is ultimately grounded (let’s suppose) in facts about fundamental particles, but this kind of grounding is mediate (it is mediated by many further grounding steps). Though (as we saw earlier) grounding is plausibly considered transitive, immediate grounding is not. If *A* immediately grounds *B* and *B* immediately grounds *C*, it is not the case that *A* immediately grounds *C*.

Imagine a small circle of grounding such that *A* immediately grounds *B*, *B* immediately grounds *C*, and *C* immediately grounds *A*. Here we have no violation of irreflexivity, because nothing immediately grounds itself. But when we switch our attention to the more familiar notion of mediate grounding, we are faced with such a violation. Assuming that grounding is transitive, each of the entities in our small circle of ground (mediately) grounds itself. (Because of

the way in which each entity grounds the others, we also have a violation of asymmetry—more on that in the next section). Violations of irreflexivity might come in this form, as well as the more direct form of something immediately grounding itself.

Let's start by considering a direct case of putative reflexive grounding suggested by Fine (2010).¹³ Fine takes the case to be a puzzle of ground, as the conclusion is 'an absurdity, given that nothing can hold in virtue of, or partly in virtue of, itself' (Fine 2010: 98). Take the fact that everything exists. The fact that everything exists is itself amongst the existing things, and so is one of the things that partially grounds the fact that everything exists. The fact that everything exists partially grounds the fact that everything exists, in a direct violation of irreflexivity. This is one of a number of puzzles of ground generated in part by taking grounding to be irreflexive. One potential solution to such puzzles is simply to reject the assumption that grounding is irreflexive. (Of course, this is not the only option. For a detailed discussion of this and other puzzles of ground, see Chapter 18 of this volume.)

A different potential counterexample is discussed in Jenkins (2011).¹⁴ A popular physicalist position in the philosophy of mind identifies particular types of mental states with particular types of brain states. Despite this identification, it seems as though we can explain some subject S's being in pain in terms of her being in brain state B but that the explanation doesn't run in the other direction. If, as is common, we cash out this explanatory claim in terms of grounding, it follows that S's being in brain state B grounds S's being in pain. But if S's being in pain is identical to her being in brain state B, it follows that S's being in pain grounds S's being in pain, which is a clear violation of irreflexivity.

There are a couple of ways we might resist this potential counterexample to irreflexivity. First, we might insist on restricting the relata of the grounding relation to facts, in which case the argument would run as follows:

- (i) The fact that S is in pain is identical to the fact that S is in brain state B
- (ii) The fact that S is in brain state B grounds the fact that S is in pain
- (iii) The fact that S is in pain grounds the fact that S is in pain

Given irreflexivity, (iii) is unacceptable. But we are not forced to accept it. On a coarse-grained conception of facts whereby facts are identified with worldly entities, (i) is true. However, anybody with this conception of facts will be motivated to reject (ii) on the very grounds that the relevant facts are identical, and so no grounding relation holds between them. Accepting (ii) indicates a fine-grained conception of fact-individuation which is sensitive to how worldly items are conceptualized, but such a conception of facts precludes accepting (i); the relevant facts are different (for more on this distinction, see Chapter 15 of this volume on "Granularity").

A second way to resist the counterexample doesn't require thinking of grounding as a relation between facts. A thoroughgoing identity theorist might reject the claim that mental states are grounded in brain states as failing to pay proper attention to the identity thesis. Mental states *just are* brain states, and so there aren't two distinct things for grounding relations to obtain between (this amounts to a denial of (ii)).¹⁵

A systematic investigation of the claim that grounding is irreflexive is made by Bliss (2018). One of the arguments she considers against irreflexive grounding is that it amounts to a kind of 'bootstrapping'; anything that is self-grounded would have to bootstrap itself into being. A way to precisify what might be wrong with this bootstrapping is to think about what kind of fact or entity a self-grounding fact or entity would have to be. There is historical precedence for self-grounding facts in Leibniz's conception of God, where God's existence is taken to be self-explanatory because it is God's essence to exist, and from God's essential

existence follows God's necessity and a multitude of divine properties. It might seem then that self-grounding facts have the potential to commit us to a number of other, metaphysically weighty theses.

But as Bliss (2018: 84) points out, this doesn't seem like a case of strict self-grounding. Rather, it is a case where the fact that God exists is explained by the fact that God has this particular divine essence, and these two facts are distinct. Moreover, there is no obvious link between a fact's being self-explanatory and its being necessary and so no reason to think that self-grounding facts require this particular sort of metaphysical baggage.

A different reason to insist that grounding is irreflexive has to do with the apparent failure of reflexive explanations: explanations of the form 'A because A' are generally considered trivial and uninformative and hence useless. Bliss (2018: 87) suggests that to the contrary, we might take instances of reflexive metaphysical explanation to indicate that we have discovered a self-dependent fact or entity or that we have reached a breakdown in our explanatory progress. This might be because we are ignorant of any further path to progress, in which case one might think that there is some potential (irreflexive) explanation available. But it might be because we have arrived at a fact not apt to be explained.

The idea that something might 'bootstrap itself into being' is undoubtedly strange, though we might wonder whether it is any stranger than the idea that some things simply exist or are the case, as on the foundationalist picture (Bliss 2018: 76). Irreflexivity for grounding can certainly be maintained, though in order to deal with the puzzles of ground it helps generate, some other desirable feature of grounding might have to be given up.

4. Is Grounding Asymmetric?

To deny that grounding is asymmetric is to accept that grounding relations can relate facts or entities in a circle. As with irreflexivity, there are two different ways in which asymmetry might be violated. A violation of asymmetry for immediate grounding generates a tight circle, but assuming grounding is transitive, there might be large networks of grounding relations such that a given fact or entity appears only distantly in its own grounding history.

Whilst it might seem obvious that grounding must be asymmetric (just as it might seem obvious that no good explanation can be circular) some purported counterexamples should give us pause for thought. Take two propositions, P_1 and P_2 . P_1 is the proposition $\langle P_2 \text{ is true} \rangle$ and P_2 is the proposition $\langle P_1 \text{ is true} \rangle$. If we assume that both propositions are true, the fact that P_1 is true is grounded in the fact that P_2 is true, and the fact that P_2 is true is grounded in the fact that P_1 is true (Thompson 2016: 46–47). This is an immediate violation of the asymmetry of grounding. A similarly direct violation of asymmetry seemingly occurs when we consider the poles of a magnet: the existence of the north pole of a magnet is plausibly grounded in the existence of the south pole of the magnet and vice versa (Bliss 2014: 248).

A third example is borrowed from Fine (2001: 11), who uses it to make a different point. Facts about the values of mass, density, and volume of a homogenous fluid are related such that any two may be said to ground the third. Assuming that grounding is transitive, this constitutes a violation of both asymmetry and irreflexivity.

Finally, some structuralist approaches to mathematics and science can be interpreted as requiring large networks of interdependent grounding-related entities. A structuralist approach to mathematics says that:

The number 2 is no more and no less than the second position in the natural number structure; and 6 is the sixth position. Neither of them has any independence from the

structure in which they are positions, and as positions in this structure, neither number is independent of the other.

(Shapiro 2000: 258)

As before, there are ways to resist these alleged counterexamples. We might deny that facts about the truth of propositions are grounded in the way required for the first example. In the third, we might question whether it is really appropriate to describe the situation as one involving grounding as opposed to (e.g.) supervenience or whether facts about some two of the values are in fact prior, but we are ignorant of which they are. Structuralist approaches to mathematics are controversial and are perhaps better understood as requiring mutual ontological dependence than interdependent grounding. Similarly, one might object that the second example deals with ontological or essential dependence rather than with grounding.¹⁶

Unlike arguments against transitivity and (to some extent) irreflexivity, arguments for and against asymmetry often proceed by examination of the kinds of grounding structures that might result from denying that grounding is asymmetric. To deny that grounding is asymmetric is to make room for the possibilities of grounding loops and interdependent networks. If we assume that fundamentality is to be characterized in terms of grounding, then such structures are *anti-foundationalist*, because they do not require that anything be absolutely fundamental (see “Fundamentality” [Chapter 23 of this volume] and “Infinite Descent” [Chapter 16 of this volume]). Arguments against foundationalism motivate alternative accounts of reality’s structure, including infinitism and the *metaphysical coherentism* that is a consequence of rejecting asymmetry.¹⁷

To deny that grounding is asymmetric is not to insist that grounding is symmetric: that whenever A grounds B, B also grounds A. We can conceive of grounding structures where some parts of the structure are interdependent networks, and other parts are chains that end in loops of various sizes (the smallest of which are instances of reflexive grounding). There might be many chains that end with a fundamental fact or entity, and yet more that continue infinitely.

Bliss (2014) defends metaphysical coherentism against the objection that grounding loops entail vicious infinite regresses. If A grounds B, B grounds C, and C grounds A, there are a finite number of grounding relations and relata of those relations, and each entity in the loop is grounded. Having moved around the loop once, there is nowhere else to go. No infinite regress is generated. What *is* perhaps generated is an infinite number of inferences; we can continue to move around the circle as many times as we like, and there is perhaps no natural stopping point. But we generally have no cause to do so. If we do choose to make multiple inferences, whether a vicious regress is generated or not depends on our explanatory project (Bliss 2014: 253). Depending on the view we adopt of the connection between explanation and grounding, we might insist that any inferential regress should not cause problems for our theory of grounding.

A related argument against metaphysical coherentism says that whilst in the small loop described earlier each individual fact or entity is explained, there is no explanation for the system as a whole. A metaphor might be helpful here. The foundationalist can think of fundamental facts or entities as generators of a kind of ‘reality fluid’, which flows up via grounding relations (‘pipes’) from the fundamental to the derivative. On the coherentist view, we can explain how the fluid moves around the pipes but not how it got to be in the system in the first place.

In response, the coherentist can push the foundationalist to expand on the metaphor of fluid-generation. The fundamental facts or entities in the foundationalist’s system are brute, in the sense that they don’t admit of any kind of explanation. This, the coherentist might insist, is no more of an explanation for the system as a whole than they themselves are able to offer. Moreover, whilst the coherentist has an explanation for each fact or entity in the system, by hypothesis, the foundationalist takes some facts or entities to be unexplained.

Closely related to the coherentist's account of metaphysical structure are holistic accounts of explanation. As mentioned, holistic explanations are those where propositions that are part of the system are explained by the system as a whole. This kind of explanation is unpopular because such explanations exhibit a kind of circularity. Barnes (2018: 66) argues that objecting to circular *arguments* is no reason to object to symmetric dependence and the circular explanations it gives rise to: circular arguments are valid but don't play the epistemic role we expect (they fail to provide additional reason to believe the conclusion), but *metaphysical* explanation is generally explicitly taken to be nonepistemic.

One final argument against violations of grounding asymmetry is that one main aim of engaging in grounding talk is to describe how the fundamental is related to the derivative, and whatever relations take us 'down to the bedrock' must be asymmetric in order to do their job (Barnes 2018: 54). This might be viewed as another version of the bad argument for irreflexivity we discussed at the beginning of Section 3; we shouldn't assume that grounding relations form a strict partial order and then use that to argue that grounding is asymmetric. However, it is important to note that if we think that the relation 'is more fundamental than' forms a strict partial order but that grounding doesn't, we can no longer express claims about fundamentality in terms of grounding.¹⁸ This is a fairly severe cost, as being able to express such claims was an important part of the motivation for introducing the idioms of grounding into our metaphysical vocabulary.

5. Concluding Remarks

As we have seen at various points, that grounding is often thought to form a strict partial order is one of the components used to generate a number of puzzles of ground. One way to diffuse some of the puzzles is thus to give up on one or more of transitivity, irreflexivity, and asymmetry for grounding. One of the things to weigh up in considering those puzzles then is the relative cost of giving up on grounding being a strict partial order. The puzzles and some potential solutions are discussed in detail in the next chapter of this volume (Chapter 18).

There is perhaps little we can do to conclusively establish whether grounding forms a strict partial order. Alleged counterexamples are always subject to attack, especially when we are willing not to hold fixed too many other properties of grounding. Direct arguments that grounding must form a strict partial order have intuitive appeal, but struggle to free themselves from the suggestion that they beg the question. A large part of the problem here is that grounding is (at least to some extent) a philosopher's term of art, introduced to the contemporary discussion to play a particular role. If we find a way conclusively to settle what that role is, whether or not grounding forms a strict partial order will likely fall out as a consequence. Without a consensus on precisely what notion we are talking about, or how to identify and regiment it, whether or not grounding forms a strict partial order will remain a matter for debate.¹⁹

Related Topics

"Fundamentality" [Chapter 23]; "Puzzles" [Chapter 18]; "Infinite Descent" [Chapter 16]

Notes

- 1 The choice between predicationist and operationalist formulations of grounding is discussed in the introduction to this volume.

- 2 Unless the separatist holds that grounding relations hold in virtue of the holding of metaphysical explanations, and so inherit their formal features from those of metaphysical explanation. I am not aware of anybody defending such a view.
- 3 For a discussion of the relation between grounding and causation, see Chapter 20 of this volume.
- 4 These features have sometimes been challenged in the literature, and if the challenges are sustainable, then we have less reason to think of even causal relations and the explanations that correspond to them as exhibiting the features of a strict partial order.
- 5 See Peacocke (1979) for a thorough defense of a particular version of explanatory holism.
- 6 Note though that Barnes thinks that grounding and cognate notions of priority couldn't be symmetric if they are to relate the fundamental to the derivative.
- 7 Whilst this is the most common contemporary view, some have argued otherwise. Bennett (2017) argues that fundamentality is to be characterized in terms of *building*, where grounding is merely one amongst a family of building relations. Sider (2011) takes his notion of *structure* to be primitive and takes what is fundamental to be that which is perfectly structural. Others (e.g., Barnes 2012; Wilson 2014, 2018) take a notion of fundamentality itself to be primitive and do not appeal to any further notions in characterizing reality's structure.
- 8 Schaffer speaks of *S* being dented rather than it having the particular dent it does, but the latter is required to ground *S* having the *particular* shape it does (see Litland 2013: 21 f.n. 11).
- 9 <grass is green> is to be read as 'the proposition that grass is green'.
- 10 See Chapter 18 (this volume) on puzzles of ground.
- 11 That *b* has a stable macrophysical structure is of course required for *b* to exist, and so there is *some* sense in which the fact that *b* has a stable macrophysical structure grounds the fact that *b* exists, but this is the same sense in which the Pauli Exclusion Principle is relevant to *b*'s existence.
- 12 Of course, the matter is not settled unless we have a clear conception of the relationship between grounding and metaphysical explanation—see Chapter 8 of this volume.
- 13 Krämer (2013) presents an even simpler version of this puzzle that makes use of sentential quantification rather than quantification over facts.
- 14 Jenkins' discussion explicitly concerns metaphysical dependence, but it is common and in keeping with the spirit of Jenkins's paper to cash out such metaphysical dependence claims in terms of grounding.
- 15 Raven (2013: 197) suggests that the notion that Jenkins is interested in is not in fact *grounding* but something closely related such as ontological dependence, where we can say that *y* ontologically depends on *x* iff were *y* to exist, *x* would exist. Since everything trivially ontologically depends on itself, mental states might both depend on and be identical with brain states. However, on this interpretation, we should be happy to accept '*S*'s pain depends on *S*'s pain', which we are assuming sounds bad enough to generate the puzzle.
- 16 For more on the relationship between grounding and dependence, see Chapter 7 of this volume.
- 17 Thompson (2016) calls the view *metaphysical interdependence*.
- 18 At least, not in the straightforward way previously assumed. There might be reasons to take every member of a grounding loop to be equifundamental, or to think in terms of relative rather than absolute fundamentality (see Chapter 23 of this volume on "Fundamentality" and Chapters 5 and 6 of Bennett 2017).
- 19 Thanks to Darragh Byrne, Kit Fine, and Mike Raven for helpful comments and discussion.

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18

PUZZLES

Stephan Krämer

Ground seems to satisfy certain *structural* principles, like irreflexivity, asymmetry, and transitivity (cf. Chapter 17). Ground also seems to satisfy certain principles concerning its interaction with *logical concepts* like disjunction and existential quantification, such as the principle that a true existential quantification is grounded by its true instances (cf. Chapter 14). Finally, ground seems to satisfy some principles linking it to concepts like that of a fact, proposition, or truth, which might be called broadly logical concepts. An example is the principle that if the proposition that *P* is true, then it is true because *P*. (As is common in the literature, I sometimes use “because” to convey metaphysical grounding.) We are faced with a *logical puzzle of ground* when the combination of a number of principles of these sorts, given standard logical inference rules, yields an absurd conclusion.

Kit Fine, in his 2010 paper “Some Puzzles of Ground”, first showed that there are logical puzzles of ground. To give you a taste, here is a sketch of one. It is very plausible that some proposition is true. If so, it seems we may infer that the proposition *that some proposition is true* is true. By the principle stated above, this proposition is true *because* some proposition is true. But by the principle that an existential quantification is grounded by any true instance, we *also* have that some proposition is true *because* (among other things) the proposition *that some proposition is true* is true. Combining these two results, we obtain a violation of the asymmetry of ground.

This chapter reviews the variety of logical puzzles of ground that have been identified in the literature, describes the solutions that have been proposed, and indicates what the main challenges are that these solutions face. I begin by introducing relevant notation as well as the key concepts and principles that will subsequently be used in formulating the puzzles before turning to the puzzles themselves. In principle, there is a huge number of different derivations of contradictions from the relevant principles about ground. Many of them are essentially alike, so that any reasonable solution to one will immediately provide a solution to the other. Some of them exhibit more substantial differences, however, and I shall try to describe all the substantially different types of puzzles. I then briefly discuss what desiderata we might impose on adequate solutions to the puzzles before I finally turn to the solutions themselves. Many of these, once developed in detail, involve a fair bit of formal machinery. I shall mainly attempt to convey the basic philosophical ideas underlying and motivating the technical work; readers keen on the details will have to consult the primary texts.

Background

To state the logical puzzles of ground clearly and concisely, we employ the usual symbolism and write “ $P < Q$ ” to say that P partially grounds Q . Note that “ $<$ ” combines with *sentences*. In this way, the notation fits with the *operator* view of grounding rather than the predicate view (see the editor’s introduction for an explanation of the distinction). However, both the puzzles and the solutions to be discussed could be transposed, with minor adjustments, to the predicational setting.

The logical puzzles of ground all invoke certain principles of the so-called *Pure Logic of Ground* (Fine 2012b), comprising purely structural rules for ground. The most important ones are that partial ground is irreflexive, asymmetric, and transitive. We shall formulate these principles in the form of inference rules:

<i>Irreflexivity</i>	$A < A$	/ $\perp$
<i>Asymmetry</i>	$A < B, B < A$	/ $\perp$
<i>Transitivity</i>	$A < B, B < C$	/ $A < C$

(Inference rules are stated in the form “Premise, Premise, . . . / Conclusion”, allowing the empty list of premises as a special case. “ $\perp$ ” stands for absurdity.) *Irreflexivity* follows from *Asymmetry*, and *Asymmetry* is derivable given *Irreflexivity* and *Transitivity*, so either of *Irreflexivity* and *Asymmetry* could be dropped as a basic rule. However, since some puzzles require only *Irreflexivity* and some only *Asymmetry*, it is best to state them separately.

Next, several principles from the *Impure Logic of Ground*, linking ground to the logical connectives and quantifiers, play an essential role in the puzzles:

$\exists$ -Grounding	$B(a)$	/ $B(a) < \exists x B(x)$
$\forall$ -Grounding	$\forall x B(x)$	/ $B(a) < \forall x B(x)$
$\vee$ -Grounding	A	/ $A < A \vee B$

Most of the puzzles also invoke some means of talking about either sentences or things suitably related to sentences, such as propositions or facts. Our focus here will be mainly on puzzles that talk about propositions rather than sentences or facts; where the differences between these options matter, this will be made explicit. For brevity, instances of “ $\langle P \rangle$ ” are used as shorthand for corresponding instances of “the proposition that P ”, and instances of “ Tp ” for instances of “ p is true”. We make use of three rules concerning, respectively, the existence of propositions, the truth of a proposition, and the grounds of a proposition’s being true:

<i>P-Existence</i>		/ $\exists x x = \langle A \rangle$
<i>T-Introduction</i>	$p = \langle A \rangle, A$	/ Tp
<i>T-Grounding</i>	$Tp, p = \langle A \rangle$	/ $A < Tp$

Finally, we need the standard logical rule of $\exists$ -Elimination, which says that a conclusion C may be derived from $\exists x B(x)$ and some auxiliary assumptions $A_1, \dots, A_n$ if it may be derived from $B(a)$ and $A_1, \dots, A_n$, provided that a does not occur in any of $\exists x B(x)$ and $A_1, \dots, A_n$.

The Puzzles

All of the puzzles to be discussed essentially involve an instance of what may be called *circular aboutness*. A simple example of this is a proposition that *quantifies* over a range of things including

that proposition itself. Thus, the proposition that there are propositions exhibits circular aboutness (short: is circular). Another class of examples involves *singular self-reference*. Suppose my first utterance this morning was: “The first proposition I express today is interesting”. Then the proposition I expressed in making this utterance is also circular. (In principle, the circles of aboutness can be longer, when one proposition is about another, which is in turn about the first, but we shall mainly be concerned with the simple cases.)

In all of the puzzles, the existence of a circular *aboutness* relationship is then exploited to derive, using the principles introduced earlier, the existence of a circular *grounding* relationship. This yields a violation of the principles of *Irreflexivity*, *Transitivity*, and/or *Asymmetry*, which jointly exclude the possibility of a grounding circle. In many cases, the circles are indeed so small as to include only two members, so that *Asymmetry* is directly violated without the help of *Transitivity*. As we shall see, there is even an example of a one-element circle of ground, directly violating *Irreflexivity*.

We begin with a detailed presentation of the puzzle mentioned in the introduction to this article, centering on the proposition that some proposition is true. Both this puzzle and the next are adapted with minor modifications from (Fine 2010: §5). The arguments will be presented in a Lemmon-style notation for derivations in which each line represents one step in the derivation, consisting in, first, the (possibly empty) list of assumptions upon which the derived sentence depends, second, the number of the line, third, the derived sentence, and fourth, the rule in whose application the step in question consists.

	(1)	$\exists y \gamma = \langle \exists x Tx \rangle$	P-Existence
2	(2)	$\exists x Tx$	Assumption
3	(3)	$p = \langle \exists x Tx \rangle$	Assumption, for $\exists$ -Elimination
2, 3	(4)	Tp	3, 2 T-Introduction
2, 3	(5)	$\exists x Tx < Tp$	3, 4 T-Grounding
2, 3	(6)	$Tp < \exists x Tx$	4 $\exists$ -Grounding
2, 3	(7)	$\perp$	5, 6 Asymmetry
2	(8)	$\perp$	1, 3, 7 $\exists$ -Elimination

Two comments. First, beyond the rules introduced in the previous section, the derivation depends solely on the highly plausible assumption (2) that there is at least one truth. In many contexts, even that assumption can be dispensed with, since most logical systems allow the premise-free derivation of some statement, from which (2) may then be derived using P-Existence and T-Introduction. Second, the proposition $p = \langle \exists x Tx \rangle$ whose existence is established in the first step of the derivation is circular, for p is itself within the range of the quantifier $\exists x$ used in expressing p . If $\exists x$ was restricted so that p lies outside its range, we would not be justified in applying $\exists$ -Grounding to obtain $Tp < \exists x Tx$.

A contradiction can also be derived for suitable *universally* rather than existentially quantified circular propositions:

	(1)	$\exists y \gamma = \langle \forall x (Tx \vee \neg Tx) \rangle$	P-Existence
2	(2)	$\forall x (Tx \vee \neg Tx)$	Assumption
3	(3)	$p = \langle \forall x (Tx \vee \neg Tx) \rangle$	Assumption, for $\exists$ -Elimination
2, 3	(4)	Tp	3, 2 T-Introduction
2, 3	(5)	$\forall x (Tx \vee \neg Tx) < Tp$	3, 4 T-Grounding
2, 3	(6)	$Tp < Tp \vee \neg Tp$	4 $\vee$ -Grounding
2	(7)	$Tp \vee \neg Tp < \forall x (Tx \vee \neg Tx)$	2 $\forall$ -Grounding

2, 3	(8)	$Tp < \forall x (Tx \vee \neg Tx)$	6, 7 <i>Transitivity</i>
2, 3	(9)	$\perp$	5, 8 <i>Asymmetry</i>
2	(10)	$\perp$	1, 3, 9 $\exists$ - <i>Elimination</i>

Beyond the rules of the previous section, this derivation of a contradiction depends only on the assumption that everything is either true or not true. Again, in many logics, we can derive this claim. The first step in the derivation consists in using *P-Existence* to establish the existence of a circular proposition, in this case $\langle \forall x (Tx \vee \neg Tx) \rangle$. Similarly as in the first puzzle, it is essential that the proposition be construed as itself within the range of the quantifier used in expressing it; otherwise the application of $\forall$ -*Grounding* would not be justified. And just as before, the contradiction is obtained by establishing a circle of ground. Only in this case, the circle is slightly longer than before. To establish its individual links, we need both $\vee$ -*Grounding* and $\forall$ -*Grounding*, whereas previously we only needed $\exists$ -*Grounding*, and to obtain a violation of *Asymmetry*, we first need to apply *Transitivity*.

Let me now state a puzzle that invokes a proposition employing singular reference to itself, adapted from Fine (2010: 117n15):

	(1)	$\exists y y = \langle 0 = 0 \vee Ty \rangle$	<i>P-Existence</i>
2	(2)	$p = \langle 0 = 0 \vee Tp \rangle$	Assumption, for $\exists$ - <i>Elimination</i>
3	(3)	$0 = 0 \vee Tp$	Assumption
2, 3	(4)	Tp	2, 3 <i>T-Introduction</i>
2, 3	(5)	$Tp < 0 = 0 \vee Tp$	4 $\vee$ - <i>Grounding</i>
2, 3	(6)	$0 = 0 \vee Tp < Tp$	4, 2 <i>T-Grounding</i>
2, 3	(7)	$\perp$	5, 6 <i>Asymmetry</i>
3	(8)	$\perp$	1, 2, 7 $\exists$ - <i>Elimination</i>

Beyond the rules of the previous section, this derivation of a contradiction depends only on the classical logical truth that $0 = 0 \vee Tp$. The crucial first step uses *P-Existence* to establish the existence of a proposition that “says of itself” that either it is true or $0 = 0$. As in the first case, the circle of ground that we establish has only two members, so no appeal to *Transitivity* is needed to establish the contradiction. (For further discussion of this puzzle, see (Litland 2015: 488f).)

One can construct variations of these puzzles that do not require *P-Existence*, *T-Introduction*, and *T-Grounding*. First, as already mentioned, in place of propositions, one may appeal to sentences or facts. Talk of facts, sentences, or propositions serves a similar purpose in the arguments: it is used to transform a sentence like “ $\exists x Tx$ ” into a singular term “ $\langle \exists x Tx \rangle$ ”, the denotation of which may then be assumed to be itself within the range of a quantifier in the sentence—or, in the case of self-reference, referred to by another singular term in the sentence.

Second, that same effect can also be achieved by more direct means, through the use of *higher-order* quantifiers—specifically, quantifiers binding variables in the position of *sentences* or *predicates*. Once such quantifiers are in play, it is plausible to suppose that they satisfy analogous grounding principles to the first-order ones. In particular, given the principle $\exists$ -*Grounding* that every true instance of a first-order existential quantification grounds that quantification, it is natural also to accept its sentential and predicational higher-order analogues

$\exists$ - <i>Grounding_s</i>	$B(A)$	$/ B(A) < \exists P B(P)$
$\exists$ - <i>Grounding_p</i>	$B(F)$	$/ B(F) < \exists X B(X)$

according to which every true instance of a higher-order existential quantification grounds that quantification. Using $\exists$ -Grounding_S, a particularly efficient construction of a logical puzzle of ground becomes possible (Krämer 2013):

1	(1)	$\exists P P$	Assumption
1	(2)	$\exists P P < \exists P P$	1 $\exists$ -Grounding _S
1	(3)	$\perp$	2 Irreflexivity

The circle of aboutness here results directly from the fact that “ $\exists P P$ ” is seen as a *true instance of itself*, justifying the application $\exists$ -Grounding_S. (A hybrid of this argument and its earlier Finean counterpart is given in (Correia 2014: §6), using both quantification into sentence position and a version of T-Grounding.)

To construct a puzzle using $\exists$ -Grounding_p, we require two additional principles that govern the abstraction of a predicate using the λ operator. Intuitively, “ λ ” may be read as “is such that”, so that “ $\lambda x(Fx)a$ ” for example means: *a* is such that it is *F*. The principles we require are roughly analogous to T-Introduction and T-Grounding:

λ -Introduction	$B(a)$	/ $\lambda x(B(x))a$
λ -Grounding	$\lambda x(B(x))a$	/ $B(a) < \lambda x(B(x))a$

Using these two principles in conjunction with $\exists$ -Grounding_p, we can derive a contradiction as follows (cf. Donaldson 2017: 794):

1	(1)	$\exists X Xa$	Assumption
1	(2)	$\lambda x(\exists X Xx)a$	1 λ -Introduction
1	(3)	$\exists X Xa < \lambda x(\exists X Xx)a$	2 λ -Grounding
1	(3)	$\lambda x(\exists X Xx)a < \exists X Xa$	2 $\exists$ -Grounding _p

A circle of aboutness here results from the fact that in taking $\lambda x(\exists X Xx)a$ to be an instance of $\exists X Xa$, we are taking the existential quantifier to include in its range the very property ascribed to *a* in $\exists X Xa$. For both types of higher-order quantification, one can also construct versions of the puzzles using universal rather than existential quantification.

It is possible to formulate a large number of further variations on these puzzles. Most of them depend on essentially the same assumptions and therefore introduce nothing interestingly new. One way to try to obtain interestingly new puzzles is to look for alternative interpretations of the predicate “*T*” for which at least the specific applications of T-Grounding and T-Introduction in the given puzzles may be argued to yield true results. For example, it appears that when “*T*” is read as “Socrates knows” in the very first puzzle, we obtain another compelling version. (“Knows” should here be read so that knowing a proposition *p* is knowing that *P*, when $p = \langle P \rangle$.) For the assumption that $\exists x Tx$ then amounts to the very plausible claim that Socrates knows something. T-Grounding now expresses the plausible principle that whenever Socrates knows that *P*, then *P* < Socrates knows that *P*. The rule of T-Introduction is of course invalid under this reading, since Socrates doesn’t know everything. But all that is required for the argument is the claim that Socrates knows that Socrates knows something, which is independently plausible. By $\exists$ -Grounding, we then obtain that (Socrates knows that Socrates knows something) < Socrates knows something, and by T-Grounding, that Socrates knows something < (Socrates knows that Socrates knows something). So once more, we have a violation of *Asymmetry*.^{1,2}

Desiderata for a Solution

Having considered the range of logical puzzles of ground, it is worth considering in general terms what we should expect of a satisfactory solution to the puzzles. Although this may not be wholly uncontroversial, I suspect that most philosophers would be happy to accept, at least under some reasonable interpretation, the following four desiderata.

Generality. A solution should cover all the puzzles of ground, not just a few of them. Of course, partial solutions may still be of interest, but their plausibility ultimately depends on the possibility of extending them in such a way as to provide a general solution.

Uniformity. A solution should be appropriately uniform across all cases. This is harder to make precise than the previous desideratum, but there are some clear cases of appropriately uniform (would-be) solutions and objectionably disuniform ones. For instance, a general ban on circular aboutness would constitute a uniform solution. A solution that rejects the relevant application of *P-Existence* in the first puzzle but only the application of *Transitivity* in the second puzzle, on the other hand, would appear objectionably disuniform. It would be overly simplistic to demand that a uniform solution reject a single principle invoked in all puzzles. But it may perhaps be plausible to demand, when different premises are rejected, that there be a unified explanation both of their falsity and of their initial appeal.

Proportionality. Given that all the assumptions required to generate the puzzles seemed initially plausible, there is no way around the fact that something that looks intuitive will have to go. But a solution should at least preserve as much of our intuitive views as possible: if one solution retains more of our intuitive judgements than another, then this constitutes a comparative advantage of that solution. We might add that other things being equal, preference should be given to retaining those principles that are most central to our overall philosophical and scientific theorizing.

Independent Motivation. A solution should provide independent motivation for rejecting the assumptions that the solution gives up. Thus, if the rejection of a particular principle is proposed, then a reason should be offered for giving up that principle independently of its role in the puzzles of ground.

Proposed Solutions

We turn finally to the question of how the logical puzzles of ground may be resolved. It is impossible within the confines of this survey to do justice to the various ingenious proposals that have been made on how best to do this. Instead, I shall focus on trying to explain the basic philosophical ideas underlying these various proposals and to say which of our desiderata would seem to be the most challenging ones to meet when developing these ideas.

Banning Circular Aboutness. As already emphasized, all of the puzzles can be described as exploiting some form of circular aboutness to derive an instance of circular grounding. Similar forms of circular aboutness lie at the heart of the more familiar *Liar* paradox of truth: a proposition that says of itself that it is not true seems to be both true and not true. In light of this, it is at first glance quite tempting to conclude that the solution to the puzzles must lie in the rejection as incoherent of such circular aboutness. One would then reject the applications of *P-Existence* in the first three puzzles, and in the higher-order versions, deny that $\exists P P$ and $\lambda x(\exists X Xx)a$ are genuine instances of $\exists P P$ and $\exists X Xa$, respectively.

At second glance, however, this seems to be an overreaction. For many examples of circular aboutness seem entirely harmless, such as the proposition that all the propositions I express today are interesting. This may be true or, more likely, false, but it does not seem paradoxical. The situation is even clearer with respect to sentences. Thus, consider

(X) The unique sentence on this page labelled (X) contains more than five words.

which is both self-referential and unproblematically true.

Typing. A natural reaction to this difficulty is to try to further specify conditions under which circular aboutness creates problems. Now, in each of the puzzles that apply *P-Existence*, the sentence expressing the problematic proposition p is not only *about* itself but contains an application of the *truth-predicate* to (a range of objects including) itself. What is illegitimate, one might therefore suggest, is not circular aboutness *per se* but application of a truth-predicate to a proposition which is itself expressed using that same truth-predicate. This is the conclusion that (Tarski 1944) drew from the Liar paradox (though for sentences, not propositions).

It would seem implausible, though, to conclude that one simply cannot coherently ascribe truth or untruth to propositions that are themselves about the truth of some proposition. Instead, one may advocate a *typed* conception of truth, on which there are many truth-predicates $T_1, T_2, \dots$ of different types or levels $1, 2, \dots$. The application of a truth-predicate T_m to a proposition p is then allowed only if p is expressed using only truth-predicates of levels $n < m$. In the context of the theory of ground, a version of this solution is presented in (Korbmacher 2017); (Donaldson 2017: 795f) describes a way in which it may be transposed to the predicational higher-order setting (the additional extension to sentential higher-order quantification is unproblematic).

The main difficulty facing this kind of approach—sometimes called the *predicative* approach in the literature—is that it seems to impose excessively severe and counterintuitive constraints on what may be said and thereby violates the desideratum of *Proportionality*. Suppose, for example, that I say to you that all of the claims you’re making today are true, and you say that some claim I’m making today is false. Then there is no way to assign levels to the truth-predicates we use under which we both succeed in generalizing over the entirety of the other’s claims today. But this seems implausible. Moreover, our claims need not be paradoxical at all. For instance, if you make some other false claim, then my claim seems simply false and your claim true. (However, if all our other claims today are true, then the utterances described do give rise to a Liar-like paradox. This illustrates the striking fact that empirical matters can help determine if a self-application of truth is paradoxical.)

Building on Kripke’s Constructions. Considerations like these led Saul Kripke, in his ground-breaking “Outline of a Theory of Truth” (1975), to develop a more sophisticated approach that allows for some self-applications of truth-predicates to receive a determinate truth-value without giving rise to paradox. Building on Kripke’s methods, (Fine 2010) has described (though not advocated) solutions to the puzzles of ground that likewise permit circular aboutness while avoiding in a principled way the corresponding circularities of ground; parts of (Litland 2015) are devoted to extending and improving on these solutions.

Informally put, the key idea of the Kripkean approach is to classify a sentence A containing T as true (false) iff A ($\neg A$) can be established through a certain step-wise procedure, *starting from sentences not containing T* . As long as neither A nor $\neg A$ has been established in this way—and in some cases, in particular when A is paradoxical, this never happens— A counts as neither true nor false. If A is of the form $T(\text{“}B\text{”})$, then A ($\neg A$) may be established if B ($\neg B$) has been established. There are then a number of possible views on how A may be established when it is of another form, such as $T(\text{“}B\text{”}) \vee C$, say. It would take us too far afield to discuss all, or indeed any, of them in detail. What follows is a sketch of what happens to (sentential versions of) our examples under perhaps the most promising, “Strong Kleene” approach. Consider the sentence “ $\exists s T(s)$ ”—“there is a true sentence”. Despite the fact that it self-applies T , it may be established via the following steps: $0 = 0 \Rightarrow T(\text{“}0 = 0\text{”}) \Rightarrow \exists s T(s)$. Similarly, the sentence $s = \text{“}0 = 0 \vee T(s)\text{”}$, although it self-applies T , can easily be established: $0 = 0 \Rightarrow 0 = 0 \vee T(s)$.

How does all this lead to a solution to the puzzles of ground? Fine's basic idea is that we may take this step-wise construction to indicate how truths involving T are *grounded* by truths obtained at a previous stage in the construction. Roughly, the idea is that a truth A involving T is grounded by some truth B iff there is an appropriate way of establishing A that proceeds via B . The step-wise derivations would be taken to show, for example, that $T("0 = 0") < \exists s Ts$ and that $0 = 0 < (0 = 0 \vee Ts)$. Any putative circles of ground can in this way be avoided. For example, $\exists s T(s)$ cannot be established on the basis of $T(" \exists s T(s) ")$. For the only way the latter could be established is if we *already had* established $\exists s T(s)$. And likewise $0 = 0 \vee T(s)$ cannot be established on the basis of $T(s) = T("0 = 0 \vee T(s) ")$, for the only way to establish this would be if we *already had* established $0 = 0 \vee T(s)$.

The biggest challenge for this kind of approach, it might be thought, is the joint satisfaction of *Proportionality* and *Generality*. Some versions of the approach quite drastically weaken classical logic (the "Weak Kleene" approach in Fine), and some others yield a drastic revision of the impure logic of ground (the "Supervaluational" approach). The "Strong Kleene" approach lies between these extremes, weakening both classical logic and the impure logic of ground, but less drastically so. For example, $\exists$ -Grounding and $\vee$ -Grounding are slightly weakened so as to yield only that *some* true instance must ground a true existential generalization, and *some* true disjunct must ground a true disjunction, not that any such instance or disjunct does this. While these revisions may be tolerable, it is not easy to see how the approach may be extended to the fact-based or the higher-order versions of the puzzle at similarly limited cost. As Fine points out, in application to a fact-based version of the puzzles, we are led to deny the apparent truism that everything exists. Similarly, an extension to the higher-order setting may prevent us from endorsing any universal quantification $\forall P B(P)$.

Allowing Circularity of Ground. A different option is to take the puzzles to show that there can be circles of ground after all. This option is favoured by (Correia 2014: 54f), (Rodríguez-Pereyra 2015), and (Woods 2018). This solution seems adequately *general* and *uniform*. It is less clear that there is *independent motivation* for allowing circles of ground, but a number of authors have argued for cases of symmetric or reflexive ground for independent reasons (see Chapter 17). The main issue, though, would seem to be *Proportionality*, since a mere rejection of *Irreflexivity* and *Asymmetry* would surely violate this constraint. For even if the puzzles have convinced us that circles of ground can sometimes arise, it would seem that this can happen only in special circumstances. Rather than merely being rejected, the principles should then be weakened so as to apply only in "normal" circumstances. The challenge is then to say which circumstances are special and which are normal.

(Woods 2018) addresses this task, maintaining that violations of *Irreflexivity* arise only in cases of *vacuous* grounding: cases in which the particular content of the truth that does the grounding does not matter, so that any other truth might replace it, and we should still have a true grounding claim. This claim, Woods argues, also helps to dissolve the intuitive discomfort with accepting instances of self-grounding. For once we have recognized that in certain cases, all we need to ground a given truth is some truth, *no matter which*, then we should not be surprised that even the truth to be grounded itself does the job. In addition, we seem to keep a fairly strong version of *Irreflexivity*. For in many cases, it will be relatively straightforward to establish that a putative instance of self-grounding would be nonvacuous and thus may still be ruled out by the envisaged weakening of *Irreflexivity*.

The main challenge, perhaps, for this approach is to show that in *all* versions of the puzzles, the putative violations of *Irreflexivity* are instances of vacuous grounding. This is easy for some of our examples, like $\exists P P < \exists P P$, but less so for others. As it stands, the solution does

not straightforwardly apply to the predicational higher-order case establishing $\exists X Xa < \exists X Xa$, though perhaps it can be suitably extended. After all, there does seem to be some sort of vacuity involved here, too. Roughly, all it takes to ground $\exists X Xa$ is *any* truth *about a*—no wonder, then, that $\exists X Xa$ itself does the trick, too. Whether the strategy works in all cases is an open question.

Replacing Strict by Weak Grounding Principles. Fine (e.g., 2012a: 51ff, 2012b: 1–4) makes a distinction between the familiar, *strict* variety of grounding and a somewhat less familiar, *weak* variety. Every case of strict grounding is also taken to be a case of weak grounding but not the other way round. For unlike strict grounding, weak grounding is not supposed to be asymmetric or irreflexive. Indeed, it is taken to be *reflexive*, so that every truth weakly grounds itself. We may continue to focus on the case of partial grounding. In that case, strict and weak grounding may be taken to be related in a particularly simple way in that a case of weak partial grounding ($\leq$) is always either strict or mutual, so $A < B$ if and only if ($A \leq B$ and not $B \leq A$). With the notion of weak grounding at hand, another approach to the puzzles, defended in (Lovett forthcoming), is to weaken the relevant *impure* grounding principles—specifically, T-, $\exists$ -, $\forall$ -, and V-Grounding—by replacing $<$ in the conclusion by $\leq$. This blocks the derivation of an absurdity in the very last step, since $\leq$ satisfies neither *Irreflexivity* nor *Asymmetry*.

This weakening of the impure rules can be motivated on independent if not uncontentious grounds. (Correia 2010) has argued that grounding should be considered a *worldly* phenomenon and that there is no plausible worldly difference to be made between, for example, a truth A and its self-disjunction $A \vee A$. If so, A and $A \vee A$ should be interchangeable in the context of ground. (For more on the worldliness or otherwise of grounding, see Chapter 15.) Indeed, this is borne out under Fine’s truthmaker semantics for ground (2012a, 2017). But then V-Grounding immediately clashes with Irreflexivity, since from $A < A \vee A$ we may, by interchange of A and $A \vee A$, obtain $A < A$. On this view, it is therefore natural to weaken V-Grounding in the way proposed and to permit the inference to $A < A \vee B$ only given the additional assumption that $\neg(A \vee B \leq A)$. That assumption is always false in the applications made in the puzzles of the V-Grounding rule. Similar considerations apply with respect to $\exists$ - and $\forall$ -Grounding and their higher-order counterparts. Finally, the proposed weakening of the T-Grounding rule may be motivated on the basis of a deflationary conception of truth, on which, again, there is no worldly difference between a truth A and the truth $T\langle A \rangle$.

This approach appears to provide both a *general* and a *uniform* solution: it offers a common motivation for weakening the various rules, namely that their plausibility depends on an excessively fine-grained conception of ground. Moreover, since this general thought puts pressure on the rules under attack irrespective of the puzzles of ground, the approach also seems to satisfy *Independent Motivation*. But it might be argued that the approach does not do as well with respect to *Proportionality*. For note that it is not enough to claim that there is *an* interesting, worldly notion of grounding on which the impure rules must be weakened, and so the puzzles are avoided. The claim must be that there is *no* coherent notion of grounding, worldly or otherwise, that satisfies all these principles. And it has seemed very plausible to many—including, indeed, Correia in later work (e.g., 2011, 2014)—that although there may be this worldly notion of ground, there is *also* another, representational notion which does satisfy the original, “strict” impure rules. Once this is granted, we still require a story about how to avoid the puzzles for this other notion of ground.

Alleging Equivocation on “Grounds”. Another strategy for responding to the puzzles is to argue that they commit a fallacy of equivocation. The idea would be that “grounding”, as currently used by philosophers, is ambiguous between a number of interpretations and that there is no

single interpretation of the term under which enough of our principles of grounding turn out true that we can generate one of the puzzles. A response of this sort is defended in (Peels 2013).

Adapted to our setting, Peels' proposal may be stated as follows: *T-Grounding* and $\exists$ -*Grounding* require different interpretations of ' $\prec$ ' to come out valid. In order that $\exists$ -*Grounding* be valid, ' $\prec$ ' must be given an interpretation under which grounding is a matter of one fact's *being an instance of* another. Under that interpretation, however, *T-Grounding* is not valid, since the fact that A is not an instance of the fact that $T\langle A \rangle$. In order for *T-Grounding* to be valid, ' $\prec$ ' must be given an interpretation under which grounding is a matter of one fact's *being ontologically more basic than and comprised by* another. Under that interpretation, however, $\exists$ -*Grounding* is not valid, since the fact that $T\langle \exists x Tx \rangle$ is not ontologically more basic than the fact that $\exists x Tx$ —indeed, it seems less basic—even though it is an instance of the latter. Finally, *Asymmetry* and *Transitivity* hold only under a uniform interpretation of ' $\prec$ ', and they do not hold under the (uniform) interpretation of ' $\prec$ ' as expressing the disjunction of the two forms of grounding just distinguished.

Prima facie, the proposal scores well with respect to *Proportionality*, in that it retains a version of each of the initially plausible-seeming principles invoked in the puzzles, though only in a restricted form. There may also be *Independent Motivation* for recognizing a multiplicity of notions of grounding. For claims like this, albeit often in a grounding-critical or -skeptical context, have been defended by several authors for reasons independent of the puzzles (cf. esp. (Koslicki 2015), (Wilson 2014), and Chapters 11–13). The main challenge would seem to be the desideratum of *Generality* in combination with *Uniformity*: it is unclear if the proposal can be extended, in an appropriately uniform way, to all the puzzles.

It seems straightforward to extend it to puzzles using $\forall$ -*Grounding* instead of $\exists$ -*Grounding*, since it may again be argued that this principle targets an *instance-directed* interpretation of " $\prec$ ". With some more work, it may perhaps also be extended to puzzles only invoking $\forall$ -*Grounding*, since the relation between a disjunction and its disjuncts seems very similar to the relation between an existential quantification and its instances and arguably dissimilar to that between the fact that $T\langle A \rangle$ and the fact that A. In particular, it may be argued that in the puzzle cases the disjunct–disjunction relationship fails to be a relationship from the ontologically more basic to the less basic. For these are cases in which the disjunct is Tp , where p is the very proposition expressed by the entire disjunction $Tp \vee 0 = 0$. It is harder to see, however, how the proposal could be extended to the puzzles involving higher-order quantifications. Here the sentential version appears particularly troublesome, since it invokes only one grounding principle, the apparently instance-directed $\exists$ -*Grounding*.

Maintaining the Metaphysical Transparency of Truth and Higher-Order Quantification. It is a familiar idea in the theory of truth that the notion of truth exhibits a distinctive kind of *transparency*. The basic thought is that applying "it is true that" to a sentence seems to have only a minimal effect on the content expressed. What is said by an instance of "It is true that P ", it seems, is hardly more than what is said by the corresponding instance of " P ". One might be tempted to infer from this that instances of " P " and " $T\langle P \rangle$ " are ground-theoretically equivalent: they ground and are grounded by exactly the same things. This would lead one to reject *T-Grounding*, but it would not avoid the puzzles, since from $\exists$ -*Grounding*, we still obtain that $T\langle \exists x Tx \rangle \prec \exists x Tx$. Since it is being maintained that instances of " P " and " $T\langle P \rangle$ " ground the same things, we may still infer that $\exists x Tx \prec \exists x Tx$, in violation of *Irreflexivity*.

However, there is an alternative way of understanding the transparency of truth that does block the derivation of the puzzle (deRosset manuscript). The idea is that truth is *metaphysically* transparent in the sense that truth-ascriptions play no robust (*metaphysically*) *explanatory* role: all of their explanatory power is merely inherited from the proposition to which truth is being

ascribed. In particular, the fact that $\langle P \rangle$ is true *does not ground anything*—for to ground something is to have a robust explanatory role. In explanatory contexts, that fact merely serves as a *placeholder* for its own ground, the fact that P , and thereby is able to explain whatever may be explained by its being the case that P .

It is clear how the proposal avoids those puzzles of ground which make use of truth-ascriptions. In order to yield a fully *general* and *uniform* solution, it needs to be supplemented by analogous claims of metaphysical transparency for any alternative devices that can be used in place of the truth-predicate to generate a puzzle. For instance, it would also have to be claimed that predications involving λ -terms and/or higher-order quantifications do not ground anything but can only derivatively play an explanatory role by standing in for their instances or λ -free counterparts. *Prima facie*, it is particularly difficult to see how the proposal could be extended in a uniform way to the knowledge-versions of the puzzle, like the one about Socrates' knowing that Socrates knows something. The overall plausibility of the proposal and the extent to which it satisfies *Independent Motivation* seems to depend in large part on the plausibility of claiming that truth-ascriptions, higher-order quantifications, and so on literally do not ground anything and on the extent to which this idea serves to capture, in an independently attractive way, the intuition that truth is somehow transparent.³

Related Topics

Logics [Chapter 14]

Granularity [Chapter 15]

Strict Partial Order [Chapter 17]

Semantics [Chapter 36]

Notes

- 1 (Whitcomb 2012) uses a similar style of reasoning to derive a contradiction not from the assumption that Socrates knows something but from the assumption that there is an omniscient being and proposes to blame the contradiction on the latter assumption, concluding that an all-knowing God could not exist. (Rasmussen, Cullison, and Howard-Snyder 2013) and (Peels 2013) respond.
- 2 Two more puzzles concerning ground should be mentioned only to be set aside, since they are of a very different kind to those described earlier and require separate discussion. First, it is possible to construct puzzles invoking ground that are more similar to the classical Liar paradox. Famously, given a self-referential "Liar"-proposition p such that $p = \langle \neg Tp \rangle$, we obtain a contradiction by observing that the assumption that p is true leads to the conclusion that p is not true, which in turn leads to the conclusion that p is true. But as shown by (Korbmacher 2015), given common assumptions about ground, we can easily construct a sentence equivalent to " $\neg Tp$ ". For example, given the principles that grounding is *factive* and that any truth helps ground its disjunction with any other proposition, the sentence " $\neg(p < (p \vee 0 = 1))$ " is equivalent to " $\neg Tp$ ". As a result, the assumption that there exists a proposition $p = \langle \neg(p < (p \vee 0 = 1)) \rangle$ yields a contradiction in much the same way as the assumption that the Liar-proposition exists. Second, (Fritz manuscript) shows that plausible and widely accepted assumptions about ground seem to conflict with a version of Cantor's Theorem by requiring the existence of a distinct proposition for each plurality of propositions. Although some solutions to this problem may also provide the required materials for a solution to the above puzzles and vice versa (such as Fritz's own proposed solution), the puzzle is too different and too complex to be discussed as part of this entry. See Chapter 15 for some more discussion of Fritz's puzzle.
- 3 For helpful comments, feedback, and advice, I would like to thank Fabrice Correia, Louis deRosset, Kit Fine, Jon Erling Litland, Adam Lovett, and Mike Raven.

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PART IV

Connections



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INTRODUCTION

Michael J. Raven

One way of approaching what ground is proceeds *from above*. The idea is to triangulate what ground is from its theoretical role and its connections to other theoretical notions. An approach from above often begins with a rough idea of what ground's theoretical role is supposed to be. We may then rely on this to guide our investigations of how ground connects to other notions. Charting these connections promises to give us a clearer vision of what ground is and what it is not.

But which theoretical role for ground should guide these investigations? Although the matter is not entirely free from controversy, one particular role is often cited. It is that ground helps stratify reality into levels. Facts or phenomena belonging to a higher level hold in virtue of or are grounded in facts or phenomena from a lower level. (One may or may not add that some level is ungrounded.) This general appeal to stratification is often combined with particular views about what the levels are and how they are arranged in a hierarchy. For example, a certain physicalist picture might take the sociological to be grounded in the psychological, the psychological in the biochemical, the biochemical in the chemical, the chemical in the physical, with nothing else grounding the physical. By contrast, a certain idealist picture might take all nonmental phenomena to be ultimately grounded in the mental. As different as these pictures may be, they have in common the abstract idea that reality is stratified by connections of ground. Ground may serve other theoretical roles too. But this abstract idea, or something near enough, is often taken to be among its main theoretical roles. We may use it to guide our investigations into how ground connects to other notions.

But which notions should be investigated for their connections to ground? The possibilities are endless in principle. This scuttles any reasonable hope of a comprehensive investigation here. The relevant possibilities, however, are far fewer in practice. In the literature, one finds at least some provisional consensus on which notions are most worthy of investigation. These include the notions of analyticity, cause, emergence, essence, fundamentality, modality, ontology, realism, structure, and truth-making.

The essays in Part IV survey potential connections between ground and these related notions.

19. Tom Donaldson's "Analyticity" explores ground's connections to analyticity. The logical empiricists (and many later philosophers influenced by them) have claimed that some statements are analytic: true by virtue of the meanings of their constituent words. To use a

- familiar example, it is claimed that to explain the truth of ‘All bachelors are unmarried’ it suffices to observe that bachelor means *unmarried man*. Donaldson revisits this long-standing debate, using the notion of ground.
20. Jennifer Wang’s “Cause” discusses the claim that ground and causation present a unified phenomenon. Proponents of the ground-causation analogy have proposed that they can be modelled using the same formal framework, perhaps because they are two species of the same genus. In any case, there appear to be significant similarities between the two notions—for instance, both ground and causation back explanations, are generative, and involve laws. But there appear to be significant differences as well, as critics have pointed out.
 21. Stephan Leuenberger’s “Emergence” offers a brief sketch of theories of emergence and notes that the literature contains little explicit discussion of the relationship between the two relations. The chapter then tries to identify a number of ways in which recent theorizing about ground can provide resources for theorists of emergence.
 22. Justin Zylstra’s “Essence” considers various proposals from the literature on how ground and essence are related. These include reductive proposals: reductions of essence to ground, reductions of ground to essence, and reductions of both to a third party. They also include proposed entailments: entailments from essence to ground and entailments from ground to essence. It is also considered how essence and ground have been jointly applied toward various theoretical ends.
 23. Ricki Bliss’s “Fundamentality” discusses the notion of fundamentality as it is commonly articulated in the ground framework. She briefly considers the notion of relative fundamentality before setting it aside and focusing on the arguably more central notion of absolute fundamentality. She discusses the ways in which fundamentality has been both captured and characterized in the literature; as well as the various arguments philosophers have offered in defense of the view. She concludes by addressing the question of whether the ground framework is actually required for articulating an account of fundamentality.
 24. David Mark Kovacs’s “Modality” surveys the connection between ground and modality, in particular supervenience. Supervenience is necessary co-variation: the A-facts supervene on the B-facts iff there can be no difference in the A-facts without some difference in the B-facts. The survey explores three possible connections between ground and supervenience: (1) supervenience can be analyzed in terms of ground, (2) grounded facts supervene on their grounds, and (3) ground and supervenience overlap in their theoretical roles.
 25. Noël Saenz’s “Ontology” focuses on three ways ground and ontology are said to relate. One way involves ground’s ability to provide a safe and sane way of admitting certain kinds of things in our theories. Another way involves ground’s ability to show how we should measure ontological simplicity. And a third way involves ground’s ability to restrict what things or kinds of things can depend on other things or kinds.
 26. Olla Solomyak’s “Realism” explores the ways in which questions of ground are related to questions of realism. Questions of these two kinds appear to be tightly connected, but there has not been consensus on precisely how these connections should be understood. Solomyak’s chapter spells out several ways in which we might understand the relationship between questions of ground and questions of realism and thereby clarifies the role for questions of ground in metaphysical inquiry more generally.
 27. Tuomas Tahko’s “Structure” explores ground’s connections to structure (joint-carving, naturalness). The notion of structure is often invoked in connection to ground, because ground is understood to impose constraints on the “structure of reality”. Specifically, ground is thought to impose a hierarchical structure. There is another, technical sense of structure,

sometimes captured with reference to the notion of “joint-carving”. This notion concerns the most fundamental distinctions that we may use to categorize reality. Both of these senses of structure as well as their potential connections are discussed.

28. Kelly Trogon’s “Truthmaking” discusses how ground and truthmaking might be related. Trogon focuses on two ground-theoretic accounts of truthmaking in particular. According to the first proposal, truthmaking is a special case of ground—some facts make true a proposition when the former ground a certain alethic fact corresponding to the latter. The second proposal imposes an additional constraint, namely that all truthmakers are fundamental in the sense that they are ungrounded. The discussion is framed in terms of the ideas that truthmaking is theoretically useful in that it helps us adjudicate between rival philosophical theories.

ANALYTICITY

*Tom Donaldson***1. A Plea for Ideological Toleration**

Much has been written about grounding. Much has been written about analyticity. Very little has been written about grounding *and* analyticity. Why?

I suspect that part of the answer is that philosophers who work with the notion of ground (groundnuts?) think of it as a *replacement* for the older notion of analyticity. Grounding is the smartphone; analyticity is the telegraph. Think of the question of how ‘physicalism’ should be defined in the philosophy of mind. It was once common to define ‘physicalism’ as the doctrine that every psychological sentence is synonymous with some sentence in physical language (e.g., Carnap 1959). Today, metaphysicians are more likely to define “physicalism” as the thesis that every fact is grounded by some physical facts (Bryant 2020; Dasgupta 2015).

While I agree that the ground-theoretic definition of ‘physicalism’ is preferable to Carnap’s definition, I don’t think we should jump to the conclusion that the notion of analyticity has *no* role to play in philosophy today. I think, for example, that there are good epistemological reasons to think that the truths of elementary arithmetic are analytic.¹ And this is of course consistent with the point that ‘physicalism’ is properly defined in ground-theoretic terms.

Perhaps another reason why philosophers who study ground are wary of the notion of analyticity is that they often maintain that facts are highly structured and fine-grained. For example, Gideon Rosen writes:

I shall suppose that facts are structured entities built up from worldly items—objects, relations, connectives, quantifiers, etc.—in roughly the sense in which sentences are built up from words. . . . Facts are individuated by their constituents and the manner of their composition. This yields a very fine-grained notion. If p and q are distinct propositions, then the fact that $p \vee \sim p$ is distinct from the fact that $q \vee \sim q$. And this is as it should be. The fact that $p \vee \sim p$ might obtain in virtue of the fact that p . But p cannot possibly ground the fact that $q \vee \sim q$, except in special cases.²

This clashes with the claim—common among theorists of analyticity—that a fact is like a block of jelly which can be ‘carved up’ into constituents in different ways.³ (An appealing claim, by the way, to those of us who have a taste for desert landscapes.)

Does this imply that theorists of ground and theorists of analyticity cannot work together? I think not, though we *do* have an indication that if philosophers from the two factions are to work together, some concessions may have to be made, on one side or the other.

In this chapter, I will begin to explore the question of what happens if we take seriously *both* the notion of analyticity *and* the notion of ground. I will do this by revisiting the old idea that analytic truths are “true in virtue of meaning”. One warning is necessary before we start. I will indulge in the convenient but questionable practice of assuming that ground is a relation among *facts*. Nothing of consequence hangs on this, however, so readers who are wary of this way of thinking about grounding can make suitable adjustments as they read. See the Introduction to this volume for discussion.

2. Ayer’s Thesis

A.J. Ayer famously claimed that when a sentence is analytically true, it is true “solely in virtue of the meaning of its constituent symbols” (Ayer 1946, pg. 16). Let’s give this thesis a name:

Ayer’s Thesis

When a sentence is true, its truth can be explained wholly by appeal to semantic facts.

Even those who are skeptical of the analytic/synthetic distinction should admit that Ayer’s Thesis has some *prima facie* appeal. Suppose, for example, that we are asked to explain why “Every moon is a satellite” is true. We will naturally reply simply that “moon” just means *natural satellite of a planet*, and so *of course* “Every moon is a satellite” is true. And if we are asked for more details, it is hard to understand why more details would be needed.

So, as I say, Ayer’s Thesis has some *prima facie* appeal. And yet powerful objections have been raised. One of these objections is that the phrase “in virtue of” is obscure or that it is unclear what species of explanation is at issue. The theorist of ground will of course be tempted to give “in virtue of” a ground-theoretic interpretation.

Ayer’s Thesis then becomes:

Ayer’s Thesis (Ground-Theoretic Version)

When a sentence *S* is analytic, the fact that *S* is true is fully grounded by semantic facts.

In what follows, I will present some objections to this ground-theoretic version of Ayer’s Thesis and discuss how a proponent of the thesis should respond.

I begin the discussion in the next section by discussing one particularly prominent objection to Ayer’s Thesis—what Jared Warren calls the “master argument” (Warren 2016). Because the master argument is both complex and important, I break down my discussion into two parts. In Section 3, I introduce the master argument and consider some replies. In Section 4, I reconsider the master argument ground-theoretically. In Sections 5 and 6, I consider putative counterexamples to Ayer’s Thesis.

2. The Master Argument⁴

Consider the German sentence “Schnee ist weiß.” This sentence is true, one might say, because it means that snow is white, and because snow is indeed white. Hence, to fully explain the truth of the sentence, one would have to explain two things: why it means that snow is white and why

snow is indeed white. Similarly, the sentence “Gras ist grün” is true because it means that grass is green and because grass is, in fact, green.

We might put the point, more generally, as follows:

Suppose that a sentence *S* expresses a true proposition *p*. Then a complete explanation of the truth of *S* must contain (a) an explanation of why *S* expresses the proposition *p* and (b) an explanation of the truth of *p*.

This formulation, however, is apt to lead us into difficult debates about the nature of propositions, so I suggest an alternative, schematic formulation. In what follows, “*S*” is to be replaced by a term for a sentence and “*S*^T” with the translation of this sentence into English:⁵

Suppose that *S* is true. Then a complete explanation of the truth of *S* must contain (a) an explanation of why *S* means that *S*^T and (b) an explanation of why *S*^T.

Boghossian (1996) calls this a “truism”—let’s adopt the label.

Let’s apply the Truism in the case of a paradigmatic analytic truth, “Every moon is a satellite.” The proponent of Ayer’s Thesis will insist that the truth of the sentence can be explained wholly by appeal to semantic facts. However, the objector will insist that to explain the truth of the sentence fully, one must explain the following two things:

- (a) The sentence means that every moon is a satellite.
- (b) Every moon is a satellite.

But, the argument goes on, it is simply not plausible that (b) is explained purely by semantic facts. The movements of the heavenly bodies are entirely independent of sublunary facts about language. It is crazy to think (it is argued) that the movements of the Moon or Io or Ganymede are explained by facts about the way we talk, write, or think.

In summary, the master argument is this. It is claimed that the Truism and Ayer’s Thesis together lead to an Absurd Conclusion. It is then argued that we must reject Ayer’s Thesis in order to avoid the Absurd Conclusion.

Ayer’s Thesis

When a sentence is true, its truth can be explained wholly by appeal to semantic facts.

The Truism

Suppose that *S* is true. Then a complete explanation of the truth of *S* must contain (a) an explanation of why *S* means that *S*^T and (b) an explanation of why *S*^T.

The Absurd Conclusion

One can explain why every moon is a satellite wholly by appeal to semantic facts.

How might the proponent of Ayer’s Thesis reply?

Well, they might argue that the so-called Absurd Conclusion doesn’t deserve its name. While it might seem obvious at first glance that the Absurd Conclusion is untenable, on reflection, it is not straightforward to articulate a compelling argument against the claim. Some have attempted to show that the Absurd Conclusion is indeed absurd by arguing that it has obviously false *modal* consequences. For example, it might be suggested that the Absurd Conclusion implies that if our linguistic practices had been different, moons would not have been satellites.⁶ Brett Topey

(2019), building on earlier work by Wright (1985), has argued that the Absurd Conclusion does *not* have such obviously silly modal consequences.

If the proponent of Ayer's Thesis does not wish to accept the Absurd Conclusion, they may decide instead to deny the so-called Truism. This response has a more radical and a more moderate form. According to the more radical form of the response, the "Truism" is a core commitment of a *representationalist* meta-semantic theory, which should be rejected in favor of an *anti-representationalist* or *use-based* theory.⁷ Since groundnuts are unlikely to be sympathetic to this response, I will not discuss it. Instead, I will focus on a more moderate view, according to which the Truism is correct in the usual run of things, but there are exceptions. Allow me to explain.

Suppose that someone asks you *why* Hesperus is Phosphorous. You might in response review the evidence that we have for the claim that Hesperus and Phosphorous are identical. Alternatively, you might reply by talking about how the names "Hesperus" and "Phosphorous" were introduced, explaining how these two different names became attached to the same planet. But suppose your interlocutor insists that these are misinterpretations of the question, saying:

You misunderstand me. I am fully acquainted with all of the evidence, and I am totally convinced that Hesperus is indeed Phosphorous. And I understand how this planet came to have two different names. My question is not meta-semantic or epistemological. Rather, my question is metaphysical: I want you to explain why the planet Hesperus is identical with the planet Phosphorous.⁸

You may feel at this point that there is something defective about this question. There is simply nothing to be explained here—there is no "Why?" about it. Agustín Rayo puts the point in the following way:

Suppose it is agreed on all sides that Hesperus (and Phosphorus) exist. Someone says: "I can see as clearly as can be that Hesperus is Phosphorus; what I want to understand is *why*." It is not just that one wouldn't know how to comply with such a request—one finds oneself unable to make sense of it. The natural reaction is to either find a charitable reinterpretation of the question ("Why does one planet play both the morning-star and the evening-star roles?") or reject it altogether ("What do you mean *why*? Hesperus *just is* Phosphorus").⁹

Rayo introduces a useful term for such cases: he says that a sentence ϕ is "why-closed" if and only if "one is unable to make sense of the question 'Why is it the case that ϕ ?'". In our example, the why-closed sentence is "Hesperus is Phosphorous" (or, if you prefer, "If Hesperus exists, then it is Phosphorous").

Now, why-closed sentences are counterexamples to the so-called Truism. The Truism implies that to explain the truth of the sentence "Hesperus is Phosphorous," one must explain the following two facts:

- (a) "Hesperus is Phosphorous" means that Hesperus is Phosphorous.
- (b) Hesperus is Phosphorous.

However, in this case, we can't make sense of the demand for an explanation of (b)—there is simply nothing to explain here. Thus, in this case, an explanation of (a) is sufficient to explain the truth of the sentence "Hesperus is Phosphorous". So the Truism fails in this case.

There are certain logical truths which are plausibly why-closed. If you ask an ethicist to explain why it is wrong to lie, they are likely to have an answer ready. However, if you ask an ethicist to explain the truth of the conditional, *if it is wrong to lie, then it is wrong to lie*, the likely response is a puzzled squint. Similarly, if you ask a philosopher of mind to explain why stones aren't capable of having beliefs, you will likely receive an articulate and thoughtful answer. If you ask why *if a thing is conscious, it's conscious*, you will be ushered out of her office.

The point bears emphasis. It's plausible that to explain $(p \rightarrow q)$, it is sufficient to provide what must be *added* to an explanation of p to produce what is also an explanation of q . For example, suppose we have two balls, one on the left and one on the right, both of which are red. Consider the fact that *if the left-hand ball is red, both balls are red*; plausibly, we can explain the truth of this conditional by pointing out that the right-hand ball is red.¹⁰ With this thought in mind, consider $(p \rightarrow p)$. What needs to be added to an explanation of p to produce an explanation of p is: *precisely nothing!* Hence, instances of the schema $(p \rightarrow p)$ are why-closed. A similar argument can be used to motivate the claim that $(p \rightarrow (q \rightarrow (p \wedge q)))$ and $((p \wedge q) \rightarrow p)$ are similarly why-closed.¹¹

Back to analyticity. It is plausible, I think, that analytic sentences such as "Every moon is a satellite" are why-closed in Rayo's sense. For suppose we are asked to explain why every moon is a satellite. We are likely to reply that "moon" just means *natural satellite of a planet*, and so *of course* every moon is a satellite. But suppose our interlocutor goes on:

You misunderstand me. I didn't ask for an explanation of the truth of the English sentence "Every moon is a satellite", I asked you to explain an astronomical fact—viz. that every moon is a satellite. And you can't explain this astronomical fact by reference to facts about the English language: the movements of the heavenly bodies are independent of facts about how we talk. So, again, why is it the case that every moon is a satellite?

No doubt a well-trained metaphysician will be able to dream up some answer to this question. One might suggest that the fact that every moon is a satellite is explained by facts about Titan, Ganymede, Io *etc.* Or one might suggest that the fact is explained by facts about the essences of the properties involved. But it is hard to escape the thought that this is to give our interlocutor's bizarre question more respect than it deserves. If one understands what "Every moon is a satellite" means, one ought to be able to see that no explanation is needed of the fact that every moon is a satellite. There is no "Why?" about it.

In summary, we can reply to the master argument by insisting on two points. First, why-closed sentences are exceptions to the Truism. Second, analytic sentences are why-closed. I should stress that this reply to the master argument is not new: it has appeared several times in the literature, variously clothed. It is implicit in Rayo (though Rayo prefers to avoid the word "analytic"). It is prominent in the literature on truthmaker theory—see for example the work of Jamin Asay and Amie Thomasson.¹² As far as I know, however, it has not yet been given a ground-theoretic treatment: this is my task in Section 3.

3. The Master Argument, in Ground-Theoretic Terms

Let's take a look at the Master Argument again, this time taking a ground-theoretic approach. Some notation will be helpful.¹³ I will represent facts by putting the corresponding sentences in square brackets. For example, [Snow is white] is the fact that snow is white.

Let S_0 be the sentence "Every moon is a satellite." Giving the Truism a ground-theoretic interpretation, we claim that $[S_0 \text{ is true}]$ is fully grounded by $[S_0 \text{ means that every moon is$

a satellite] and [Every moon is a satellite]. Now $[S_0]$ means that every moon is a satellite] is a semantic fact. But what about [Every moon is a satellite]? What grounds *this* fact? Some will suggest that it is a universal generalization, and it is therefore grounded by its instances—including facts about Titan, Ganymede, and Io. Perhaps others will suggest that it is grounded by facts about the essences of the properties involved. No doubt other answers are possible. But what may seem clear is that [Every moon is a satellite] is grounded by nonlinguistic facts. It is, after all, a fact of astronomy, not a fact about language. But then we seem to be stuck with the conclusion that $[S_0 \text{ is true}]$ is *not* fully grounded by linguistic facts after all.

How might we reply?

We could, of course, simply accept the conclusion that [Every moon is a satellite] is grounded by semantic facts. But I think that this is a nonstarter. There might be something to be said for the view that semantic facts explain [Every moon is a satellite] *in some sense of 'explain'*; however, the claim that semantic facts *ground* [Every moon is a satellite] is beyond far-fetched. To claim that astronomical facts are *grounded* by facts about language would be to endorse a bizarre kind of linguistic idealism.

And so, I suggest, proponents of the ground-theoretic version of Ayer's Thesis should instead insist that "Every moon is a satellite" is why-closed, and hence it is a mistake to ask for a ground for the fact it expresses. This fact, we may say, simply doesn't require explanation.

But this reply is puzzling. If [Every moon is a satellite] is not grounded, it would seem to follow that [Every moon is a satellite] is a *fundamental* fact. And this is extremely implausible, if only because moons are not fundamental objects.¹⁴

This brings us, I think, to the crux of the issue. If we are to sustain the ground-theoretic version of Ayer's Thesis, I suggest, we must maintain that facts such as [Every moon is a satellite] are in some sense trivial or empty, that they "don't require explanation"—but it is not easy to make sense of this claim within the ground-theoretic framework.

At this point, intransigent ground-theorists may think that Ayer's Thesis must be rejected. For my part, I think that this would be too hasty. The claim that there are why-closed sentences in nonfundamental vocabulary is widely shared and highly intuitive. If this point cannot be accommodated within the theory of ground, many will see this as a defect of groundology. Theorists of ground would do well to explore the idea that there are non-fundamental facts which "do not require explanation."

My own suggestion is this: the proponent of Ayer's Thesis should say that [Every moon is a satellite] is *zero-grounded*. The notion of zero-grounding is due to Kit Fine (2012), and perhaps deserves some explanation. Some facts have a ground that consists of three facts; for example:

$[A \wedge B \wedge C]$ is grounded by $[A]$, $[B]$, and $[C]$.

We might say that $[A \wedge B \wedge C]$ is 'three-grounded'. Some facts have a ground that consists of two facts; these are 'two-grounded'. For example:

$[A \wedge B]$ is grounded by $[A]$ and $[B]$.

Some facts are one-grounded; for example:

$[\exists x \text{ Dog}(x)]$ is grounded by $[\text{Dog}(\text{Fido})]$

Now, Fine claims that some facts have a ground that consists of *no facts at all*. Such facts are said to be 'zero-grounded'. Here is one of Fine's examples. Suppose that our language contains a

conjunction operator $\wedge$ which operates on any number of sentences $A, B, C, \dots$. We might then maintain that a conjunction is always grounded by its conjuncts, thus:

$[\wedge(A, B, C, \dots)]$ is fully grounded by $[A], [B], [C], \dots$

Now consider the empty conjunction, $T := \wedge()$. T (like all conjunctions) is grounded by its conjuncts—though in this case there are none of them. So T is zero-grounded.

Perhaps another example will be helpful—this time, the example is mine, not Fine’s. It is very plausible that the existence of a set is grounded by the existence of its members. For example, $[\{\text{Athos, Porthos, Aramis}\} \text{ exists}]$ is fully grounded by $[\text{Athos exists}], [\text{Porthos exists}],$ and $[\text{Aramis exists}]$. One might maintain, as a limiting case of this principle, that $[\emptyset \text{ exists}]$ is zero-grounded.

It is important to distinguish zero-grounded (or trivial) facts from ungrounded (or fundamental) ones. As Fine puts it:

There is a . . . distinction to be drawn between being zero-grounded and being ungrounded. In the one case, the truth in question simply disappears from the world, so to speak. . . . But in the case of an ungrounded truth . . . the truth is not even generated and simply stays in place.¹⁵

Now, back to those moons. If it is granted that $[\text{Every moon is a satellite}]$ is zero-grounded and that $[S_0 \text{ means that every moon is a satellite}]$ and $[\text{Every moon is a satellite}]$ fully ground $[S_0 \text{ is true}]$, it will follow by the cut principle¹⁶ that $[S_0 \text{ expresses } p_0]$ fully grounds $[S_0 \text{ is true}]$ —thus, $[S_0 \text{ is true}]$ is fully grounded by a semantic fact, just as the ground-theoretic version of Ayer’s Thesis requires.

More generally, I suggest, the proponent of Ayer’s Thesis should say that the facts expressed by analytic sentences are zero-grounded. This is, I suggest, the best way to incorporate into the theory of ground the claim that analytic sentences are why-closed. In Sections 4 and 5, we’ll consider putative counterexamples to this position, due to Timothy Williamson and Louis deRosset.

4. Analyticity and the Logic of Ground

Timothy Williamson has offered¹⁷ the following sentence as a problem case for Ayer’s Thesis:

(B) Either Barbara is a lawyer or Barbara is not a lawyer.

Suppose that Barbara is indeed a lawyer. Now it may seem obvious that (B) is an analytic truth. So if my discussion so far has been on track, the proponent of the ground-theoretic version of Ayer’s Thesis will say that the fact that (B) expresses is zero-grounded. However, this conflicts with the standard account of the logic of disjunction, according to which this fact is *not* zero-grounded; it is, rather, grounded by $[\text{Barbara is a lawyer}]$.¹⁸

I see two possible replies to this complaint.

First, one might doggedly insist that the fact that (B) expresses *is* zero-grounded, and that our theories of the logic of ground must be modified to accommodate this point. This is perhaps not out of the question. Groundology is a young science, and it’s not implausible that some of our views about the logic of ground need revision. What’s more, theorists of ground often present their account of the logic of disjunction with very little argument, as McSweeney (2020, Chapter 32) emphasizes. However, to accept that the fact that (B) expresses is zero-grounded would

be a significant concession for the theorists of ground. If you flick through this book, you will see that much in the theory of ground is controversial—but there is very little controversy on the question of how disjunctive facts are grounded.

Perhaps, then, it is the theorists of analyticity who should make a concession. Perhaps the proper conclusion is that (B) is not analytic, in the requisite sense,¹⁹ even though it is a classical logical truth. In support of the move, we may point out that there have been *independent* arguments for the claim that instances of the law of excluded middle are not analytic.²⁰

Having denied that (B) is analytic, we may reasonably be asked to give some positive characterization of the class of analytic truths. I will return to this issue in Section 6.

5. deRosset's Objection

Now for another alleged counterexample to Ayer's Thesis. Louis deRosset²¹ defines "grassgreen" in the following way:

Suppose we stipulate that "grassgreen" is to be a predicate that expresses the property *being green* if, as a matter of fact, grass is green, and *not being green* otherwise.

Now, consider this sentence, (G):

(G) Grass is grassgreen.

Now we can show that (G) is true, using the following simple argument:

Suppose that grass is in fact green. Then "grassgreen" expresses the property *being green* and so "Grass is grassgreen" is true. Suppose on the other hand that grass is not green. Then "grassgreen" expresses the property *not being green* and so "Grass is grassgreen" is again true. So, either way, "Grass is grassgreen" is true.

Thus, we can see that (G) is true, just by reflecting on the meanings of the words that comprise the sentence. Thus, (G) is analytic.

Now, as we all know, grass is in fact green, and so "grassgreen" expresses the property *being green*. Thus, (G) expresses the fact [Grass is green]. But this is obviously a substantive, contingent, botanical fact. It is not zero-grounded. Thus, we seem to have a counterexample to the suggestion that analytic sentences are why-closed and express zero-grounded facts.

It seems to me very clear that the fact expressed by deRosset's sentence is not zero-grounded. Thus, if we are to rescue Ayer's Thesis, we must somehow restrict Ayer's Thesis or exclude deRosset's sentence from the class of analytic sentences.

In a discussion of this issue, David Chalmers defends a relative of Ayer's Thesis by restricting it to so-called super-rigid sentences.²² If we follow him, our thesis would become:

Ayer's Thesis (Chalmersian Version)

When a super-rigid sentence *S* is analytic, the fact that *S* is true is fully grounded by semantic facts.

Chalmers's definition of "super-rigid" is complex, but what's important here is that every super-rigid sentence is comprised entirely of "epistemically rigid" words, where a term is "epistemically rigid" just in case one can know *a priori* what it refers to. Now "grassgreen" is not

super-rigid because one cannot know *a priori* whether it refers to the property *being green* or the property *not being green*.²³ Thus, (G) is not a counterexample to the new Chalmersian version of Ayer's Thesis.

This maneuver may save Ayer's Thesis—but at a significant cost. As Chalmers recognizes,²⁴ natural-kind terms such as "swan" are not super-rigid. Thus, paradigmatic analytic sentences such as "Every cygnet is a juvenile swan" and "Every ewe is a female sheep" are not covered by the Chalmersian version of Ayer's Thesis.

My view, then, is that the proponent of Ayer's Thesis should respond to de Rosset's example by arguing that "Grass is grassgreen" is not analytic in the requisite.

6. A Proposal

In Sections 4 and 5, we looked at two putative counterexamples to Ayer's Thesis:

- (B) Either Barbara is a lawyer or Barbara is not a lawyer.
- (G) Grass is grassgreen.

In both cases, I recommended that the proponent of Ayer's Thesis should reply that the given sentence is not analytic in the relevant sense. But this leaves them with the task of providing some *positive* characterization of the class of analytic truths.

It will help to describe the task slightly differently. Our goal is to draw a line around a class of truths, to be called 'analytic'. The class should be not so large as to introduce counterexamples to Ayer's Thesis. At the same time, it should not be so small as to strip Ayer's Thesis of theoretical interest. In this section, I will present my own response to this problem—but I hope that others will do better in the future.

In Section 4, I introduced Kit Fine's sentence "T", the empty conjunction, which expresses a zero-grounded fact. Using this sentence as a starting point, we can introduce a number of schemas all instances of which will be zero-grounded:

$$\begin{aligned} &T \vee p \\ &\neg\neg(T \vee p) \\ &\neg(\neg T \wedge p) \end{aligned}$$

I would now like to conjecture that certain schemas involving conditionals have the same property, including perhaps these:

$$\begin{aligned} &p \rightarrow p \\ &p \rightarrow (q \rightarrow (p \wedge q)) \\ &\forall x (p \rightarrow p) \end{aligned}$$

Now if this conjecture is true, then the following sentences express zero-grounded facts:

- If Edith is an adult female sheep, Edith is an adult female sheep.
- If Hugo is a male, then if Hugo is a sibling then Hugo is a male sibling.
- Every juvenile swan is a juvenile swan.

Now I suppose that if you take a sentence and substitute a phrase in that sentence with a synonym, the resulting sentence has the same meaning as the original and expresses the same fact.²⁵

So if the three sentences just listed indeed express zero-grounded facts, so do these:

If Edith is an adult female sheep, Edith is a ewe.

If Hugo is male, then if Hugo is a sibling then Hugo is a brother.

Every cygnet is a juvenile swan.

This brings me to my proposal. As I say, I conjecture that further research in the logic of ground will deliver a number of schemas, each instance of which expresses a zero-grounded fact. Let's say that a sentence is 'analytic' if it can be obtained from an instance of such a schema by substituting synonyms for synonyms. My proposal is that Ayer's Thesis is true when 'analytic' is interpreted in the manner I have suggested.

7. Conclusion

I hope to have convinced you of two things. First, that Ayer's Thesis, when formulated in a sufficiently restricted form, may be true. Second and more importantly, that theorists of ground and theorists of analyticity need not be locked in a permanent antagonism. Indeed, there is interesting work to be done at the intersection of these two areas of research.

In particular, I would like to mention a few open problems that we've come across:

- What grounds conditional facts?
- Can there be nonfundamental facts which nevertheless "don't require explanation"?
- What is the *best version* of Ayer's Thesis? That is, what is the *strongest* version of the thesis that is not susceptible to counterexample?

Related Topics

- Physicalism [Chapter 35]
- Granularity [Chapter 15]
- Logic [Chapter 32]

Notes

- 1 Perhaps the most detailed defense of this position is that developed by the "neofregeans". Wright 1983 was an influential early presentation of the neofregean position; for more recent work, see Hale and Wright 2001. See also Rayo 2013.
- 2 Rosen 2010, pg. 114–115. It's worth noting that theorists of ground differ to some extent on the question of granularity; for discussion, see Correia (2020, Chapter 15).
- 3 Surely the most influential statement of this idea is in §64 of Frege 1974 (though Frege doesn't mention jelly and focuses on *thoughts* rather than facts). See also the discussion of whether states of affairs are "crystalline" or "plastic" in Macbride 2003. For more recent discussion, see Rayo's (2013) discussion of "metaphysicalism".
- 4 Boghossian 1996 is probably the most well-known statement of the master argument. See also Sider 2011; Williamson 2007; Yablo 1992; Sober 2000. Topey (2019) and Warren (2016) claim to have found versions of the master argument in much earlier sources.
- 5 For simplicity, we'll suppose that S contains no context-sensitive expressions.
- 6 For example, Blackburn (1986) claims that one can't explain a necessary truth by appeal to contingent facts about language, because "the explanation, if good, would undermine the original modal status: if that's all there is to it, then [for example] twice two does not have to be four". Quoted in Topey 2019.
- 7 This approach is recommended in Warren 2016. Price (2013) explains the contrast between "representationalist" and "anti-representationalist" theories of language and defends the latter.

- 8 Alternatively, when asked why Hesperus is Phosphorous, you might attempt to explain the “diagonal proposition” associated with the sentence “Hesperus is identical to Phosphorous” (see Stalnaker 1978). Let us suppose that your interlocutor insists that this too is a misinterpretation.
- 9 Rayo 2013, pg. 54. Papineau (1998) makes similar claims. It is worth noting that Rayo’s claim that “Hesperus is identical to Phosphorous” is why-closed is not controversial. For another take, see Shumenner’s “Identity and Ground,” 2020, and her forthcoming.
- 10 I have taken this example from Øystein Linnebo (manuscript).
- 11 Mathematically inclined readers may at this point ask whether this idea can be implemented in a formal semantics. One approach is to extend Kit Fine’s (2012) exact truthmaker-semantics for ground by adding a conditional. One may say that the following is sufficient for a fact f to verify $(p \rightarrow q)$:

Given any g such that g verifies p , the fusion $f \bullet g$ verifies $(p \wedge q)$.
 We should also add that any falsifier for p will also verify $(p \rightarrow q)$. I leave the details for future work—this is not the place to explore the technicalities.
- 12 In particular, see Asay forthcoming and Thomasson 2007, ch. 2.
- 13 I have taken this notational device from Rosen 2010.
- 14 Perhaps some readers will say that moons *are* fundamental objects. I suggest that such readers come up with their own example of analytic sentence about nonfundamental objects and work with this example instead.
- 15 Fine 2012, pg. 48. For further discussion of zero-grounding, see Litland (2020, Chapter 9).
- 16 See Fine 2012, pg. 56.
- 17 See Williamson 2007, pg. 60. Williamson focuses on a truthmaker-theoretic version of Ayer’s Thesis, but his point applies with equal force to the ground-theoretic version.
- 18 To put the point with a little more precision: the claim that the fact that (B) expresses is zero-grounded conflicts with the elimination rule for disjunction presented in section eight of Fine 2012.
- 19 I add the qualification “in the requisite sense”, because one might argue that the sentence in question is not analytic in the sense relevant to Ayer’s Thesis but *is* analytic in some other sense of that vexed term. For example, one might say that the sentence is “epistemically analytic” but is not “epistemically analytic”—for this distinction, see Boghossian 1996.
- 20 Rumfitt (2012) argues that excluded middle is not justifiable on the basis of the “sense” of the negation and disjunctions signs—which amounts, I think, to saying that it is not analytic. Tennant 1996 is another important article—though Tennant adopts an antirealist stance quite foreign to grounding-theory.
- 21 See deRosset 2015.
- 22 Chalmers 2012, pg. 453. The “relative of Ayer’s Thesis” that Chalmers discusses is the claim that, given true sentences A and B, if A “conceptually grounds” B, then A metaphysically grounds B.
- 23 In more detail: a word is “super-rigid” if it is both “epistemically rigid” and “metaphysically rigid” (Chalmers 2012, pg. 474). A word is “epistemically rigid” if its “referent can be known *a priori*”, or, to put it another way, if it “picks out the same referent in every epistemically possible scenario” (pg. 470). A word is “metaphysically rigid” if, by stipulation, it refers to the same object with respect to every metaphysically possible scenario (Kripke 1980, pg. 21).
- 24 Chalmers 2012, pg. 453. Space does not permit a thorough discussion of the point that natural-kind terms are not epistemically rigid. However, here is a short argument to motivate the point. Consider “dog” and “fox”. You no doubt know that these two words refer to different kinds. However, this is something you know *a posteriori*—no amount of armchair reflection could establish that the terms don’t refer to the same kind. Thus, there are some epistemically possible scenarios in which the two terms refer to the same kind. Thus, at least one of the terms refers to different kinds with respect to different epistemically possible scenarios. And so at least one of the two terms is not epistemically rigid.

The canonical presentation of the point that natural-kind terms are not epistemically rigid is Putnam 1975, though Putnam himself did not use the term ‘epistemically rigid’.
- 25 This assumption is more substantial than it might first seem to be. For discussion, see Schnieder 2010.

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20

CAUSE

Jennifer Wang

1. Introduction

Compare the following claims:

- (1) The window shattered because Suzy threw a rock.
- (2) The shirt is red because it is maroon.
- (3) The singleton set containing Sonia Sotomayor exists because Sonia Sotomayor exists.

Each of (1), (2), and (3) explains one state of affairs in terms of another.¹ But whereas (1) is causal, (2) and (3) apparently involve a noncausal form of explanation, which according to many indicates the presence of a grounding relation. The notion of ground is often introduced as the metaphysical analogue of the notion of cause. For the most part, no more is said about the connection between the two notions, for instance, the extent of the analogy or whether theorizing about one notion might shed light on the other. But in recent literature, some philosophers have developed a sustained analogy between grounding and causation, with the end goal of showing that they present a unified phenomenon. Although there are other questions about the connection between grounding and causation, this chapter centers on the unification claim.²

The unification claim will be made more precise in Section 2. Section 3 details the main alleged similarities between grounding and causation, with special attention to the interventionist and mechanistic accounts. Section 4 details objections to the unification claim, and Section 5 comments on future directions for work. In what follows, I will assume that there are grounding and causal relations—the discussion that follows may be reformulated for those who hold that there are truths about grounding or causation, even if there are no corresponding relations in the world.

2. Some Positions

I follow Sara Bernstein (2016: 22) in distinguishing two theses:

Unity: Grounding and causation are species of the same genus.

Illumination: Issues in grounding can be illuminated by appealing to causation and vice versa.

Note that Bernstein actually characterizes Unity (or “grounding-causation unity,” as she calls it) as the thesis that grounding and causation are merely “nominally distinct,” a phrase apparently borrowed from Schaffer (2016a: 94). But as far as I know, no one defends the thesis that grounding just is nomological causation. Meanwhile, Illumination (or “grounding-causation comparison”) is a methodological claim that admits of degrees.

There is variation in how and whether Unity and Illumination are defended. Alastair Wilson (2016, 2018) argues that grounding just is metaphysical causation and that metaphysical and nomological causation are both species of causation. Jonathan Schaffer (2016a, 2016b) and Karen Bennett (2017) hold that grounding is not a species of a causation but that both are species of directed determination (for Schaffer) or building (for Bennett). These defenders of Unity thus have an obvious path toward Illumination: the notions of ground and cause are mutually illuminating because they share the features unifying the genus. However, all acknowledge that there are significant differences between grounding and causation, just as there are differences between two species of any genus.

Other metaphysicians accept Illumination without accepting Unity. For instance, Kelly Trogdon (2018) does not take a stance on Unity; however, he argues that there are important similarities between causal mechanisms and what he calls grounding mechanisms. Arguably, many defenders of ground are in the same camp, in the sense of accepting some degree of similarity between ground and cause without outright endorsing Unity. Grounding has been introduced as “metaphysical causation” in articles that do not further pursue the analogy, e.g., in Schaffer (2012: 122) and Ted Sider (2011: 145). More commonly, the topic has been introduced using notions such as noncausal determination, explanation, or dependence, e.g., in Paul Audi (2012: 686), Selim Berker (2018: 507), and Michael Clark and David Liggins (2012: 812). However, because such defenders do not draw extended, direct comparisons between grounding and causation, I will focus primarily on the views defended by Wilson, Schaffer, Trogdon, and Bennett.

Finally, Bernstein (2016) has argued that both Unity and Illumination are false. There may be similarities between grounding and causation, but this does not show that they stand in any interesting relation to each other, at least not with the significance that has been ascribed. Kathrin Koslicki (2016) does not explicitly reject either thesis, but she objects to some of Schaffer’s claims about the similarities between grounding and causation via interventionist models. Presumably, these criticisms would also apply to Wilson, who also appeals to interventionist models to make the case for Unity.

3. In Defense of Unity and Illumination

Defenders of Unity and/or Illumination point to a long list of similarities between grounding and causation to justify their claims. In Section 3.1, I catalogue some general similarities between the two notions. Sections 3.2 and 3.3 address specific alleged analogies with structural equation models (as defended by Schaffer and Wilson) and mechanistic models (as defended by Trogdon). Section 3.4 examines Bennett’s views on building relations, including her defense of potentially illuminating analogies between grounding and causation. In each of these sections, some of the differences acknowledged by the defenders of Unity and/or Illumination will be noted.

3.1 General Similarities Between Grounding and Causation

The following list is not intended to be a full account of similarities between grounding and causation but is rather a somewhat miscellaneous list of similarities that have been pointed out

in the literature. Items (i) and (ii) are relatively superficial; items (iii) through (v) may indicate deeper connections.

- (i) Both arguably form partial orders.

Where there is disagreement over such features, it seems that structurally similar objections and replies may be advanced. See Wilson (2018: 727–28) and Schaffer (2016a: 55), as well as Chapter 17 of this volume.

- (ii) The full vs. partial ground distinction is analogous to the sufficient vs. contributory cause distinction.

Just as there can be multiple sufficient causes of a state of affairs, there can be multiple full grounds of a state of affairs.

- (iii) Both can be understood as production-based or dependence-based notions.

Ned Hall (2004) distinguishes between a production-based notion of causation and a dependence-based notion. Causal production may be made more precise, but it concerns the idea of one state of affairs generating or bringing about another. Causal dependence is understood through patterns of counterfactual dependence; in its simplest form, for two states of affairs x and y , x causes y iff y depends upon x iff had x not occurred, y would not have occurred.

Grounding may likewise be understood as production based or dependence based, though the details may differ. Both Schaffer (2016a: 54) and Wilson (2018: 733) draw this distinction in the grounding case, though Wilson identifies grounding with grounding dependence. Trogon (2018: 1292) seems to suggest that grounding is grounding production; Bennett (2017: 68–69) explicitly endorses this, though she also acknowledges that both grounding and causation are reflected in patterns of counterfactual dependence.

- (iv) Both back explanations.

In response to some why-questions, e.g., Why did the window break?, we cite nomological causes. In other cases, e.g., Why does singleton Sonia Sotomayor exist?, the answers appeal to grounds. Just as causal explanations are backed by causal relations in the world, metaphysical explanations are backed by grounding relations in the world. Schaffer (2016a: 84–87) argues that we should not conflate relations in the world with explanations among facts or sentences. In fact, he holds (89) that there is just one kind of explanation, which can be backed by grounding or causation or both. See also Schaffer (2016c: 145–46). Wilson (2018: 728–29), Trogon (2018: 1303), and Bennett (2017: 61) also accept the distinction between grounding or causal relations and explanations. See Chapter 8 of this volume for discussion of what “backing” means in the context of grounding explanation.

- (v) Both are similarly connected to notions such as laws, necessity, and counterfactuals.

Because of the connection to explanation, both grounding and causation thereby stand in similar relations to other notions.

Particular cases of grounding or causation are thought to be supported by more general principles, respectively, metaphysical or nomological laws. This claim is compatible with many different views about the nature of such laws; for the grounding case, see Chapter 31 of this volume.

Both relations are thought to be necessitating: if the grounds exist, then the grounded must as well; likewise, if the cause occurs, then the effect must as well. However, there is controversy in each case. For discussion of grounding necessitation, see Chapter 10 of this volume. For the causal case, see Elizabeth Anscombe (1975).

Finally, as noted in item (iii) of this list, causation is believed to have intimate ties to counterfactual conditionals—counterfactual dependence is one possible mark of a causal connection. Likewise, counterfactual dependence may indicate a grounding connection, though there may be a need to venture into *counterpossible* dependence in the case of grounding (see Section 4.2).

3.2 Structural Equation Models

Schaffer (2016a, 2018) and Wilson (2016, 2018) each independently argue that there is a systematic analogy between ground and cause, as evidenced by the fact that each may be captured using the structural equations models of the interventionist framework. This constitutes their case for both Unity and Illumination, for the structural equation models allegedly provide parallel insights in both cases. Note that they are only considering cases in which there is a determination relation between the causes (grounds) and their effects (grounded), since grounding is typically thought to be necessitating.

The interventionist framework was developed as a theory of causation by Joseph Halpern and Judea Pearl (2005a, 2005b), Christopher Hitchcock (2001), Judea Pearl (2009), and James Woodward (2003), among others. The following is a simplification of the formalism actually used in the causation literature, though it is not much simpler than that described in the grounding literature.

A structural equations model consists of:

- (i) a set of variables and the values they take (specifically, a set of endogenous variables and a set of exogenous variables),
- (ii) a set of structural “equations” (see what follows) exhibiting the dependencies between variables, and
- (iii) an assignment of actual values to variables (specifically, an assignment of values to the exogenous variables).

Consider the simple causal example in which Suzy throws a rock through a window, which then shatters. We can model the connection between the relevant types of events as follows. Let C be the variable that takes the value 1 if the rock is thrown and 0 otherwise, and let E be the variable that takes the value 1 if the window shatters and 0 otherwise. The single structural equation in this case is:

$$E = C$$

Importantly, the symbol “=” is conventionally used to denote not identity but a counterfactual dependence relation indicating how the variable on the left depends upon the variable (or variables) on the right. If the model assigns $C = 0$, then $E = 0$, and if the model assigns $C = 1$, then $E = 1$. This model thus represents the causal dependency of the window’s shattering upon whether or not Suzy throws the rock. Note that throughout, the models strictly speaking concern types rather than tokens of events and causal relations. There have been attempts to characterize token causation via causal models, e.g., in Hitchcock (2001).

A causal model may be visually represented by a directed graph with nodes corresponding to variables.

$$C \longrightarrow E$$

We can determine the truth values of certain counterfactuals by performing interventions on the variables upon which other variables depend in the model. For instance, whatever the actual value of C , we intervene on it by setting $C = 1$. Given the structural equation associated with the model, it follows that $E = 1$. Likewise, an intervention of $C = 0$ yields $E = 0$. From these observations, we may infer some counterfactual truths, namely:

- (4) If Suzy had thrown the ball, the window would have shattered.
- (5) If Suzy had not thrown the ball, the window would not have shattered.

Both Schaffer and Wilson draw an analogy between this simple case of cause and (variants on) the following two cases of ground. The first is the case of a shirt's being maroon grounding its being red. Let C be the variable that takes the value 1 if the shirt is maroon and 0 otherwise, and let E be the variable that takes the value 1 if the shirt is red and 0 otherwise. As in the causal case, the single structural equation is:

$$E = C$$

Likewise, let C take the value 1 if Sotomayor exists and 0 otherwise, and let E take the value 1 if singleton Sotomayor exists and 0 otherwise. This model thus supports the following counterfactuals:

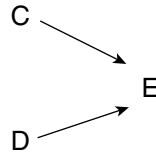
- (6) If Sotomayor had existed, then singleton Sotomayor would have existed.
- (7) If Sotomayor had not existed, then singleton Sotomayor would not have existed.

Schaffer explicitly adopts contrastive theories of both cause and ground (and presumably Wilson would agree). We cannot simply think of Suzy's throwing the rock as the cause of the window's shattering; rather, Suzy's throwing the rock rather than not throwing the rock is the cause of the window's shattering rather than remaining intact. Likewise, the shirt's being maroon rather than its being navy grounds the shirt's being red rather than its being blue. Thus, in the model above, $C = 0$ represents the shirt's being navy, not the shirt's failing to be maroon, and $E = 0$ represents the shirt's being blue, not the shirt's failing to be red. Note that the contrastive treatment leads to the failure of transitivity in both cases (Schaffer 2005; Hitchcock 2001).

These are only the simple cases. One of the motivations for the interventionist account of ground is the ways in which it parallels the interventionist account of cause in more complicated cases. For instance, consider cases of symmetric causal overdetermination, in which two states of affairs obtain, each of which is by itself sufficient for some effect. If Suzy and Billy both throw rocks that hit the window at the exact same time, then the window's shattering is overdetermined. In this case, C and D represent whether Suzy and Billy each throw their rocks, and E represents whether the window breaks:

$$E = \max(C, D)$$

This can be represented graphically as:



For a case of grounding that uses the same structural equation, consider Elena Kagan and her singleton set alongside Sotomayor and hers. C represents whether Sotomayor exists, as before, and D represents whether Kagan exists. Unlike before, E represents whether either singleton Sotomayor or singleton Kagan exists.

Interventionism about causation is thought to be able to handle cases that simple counterfactual theories cannot, such as symmetric overdetermination, omission, and early pre-emption. For examples of omission and early pre-emption in both the causal and grounding cases, see Section 5 of Wilson (2018), though see also Section 4.1. Note, however, that interventionist accounts are still counterfactual theories of causation, since the models support counterfactual reasoning via interventions on variables in the model. Interventionist models thus do not provide reductions of causation or of grounding, and not all interventionist models correspond to genuine cases of grounding or causation.

Wilson also holds the following difference between grounding and causation: causation, unlike grounding, is mediated by the laws of nature. For Wilson, mediating principles determine which structural equations are part of models of genuine cases of causation (or grounding). So in the causal case, the dependencies between the variables of an interventionist model are determined by the laws of nature. They thus differ in their relationships with concrete physical processes; causal dependencies but not grounding dependencies have consequences for how physical events in the world unfold. Wilson (2018: 746) holds that grounding is instead mediated by principles of logic or metaphysics such as the laws of logic, the axioms of set theory, or the laws of mereology. Schaffer (2016a: 57–58) also thinks that grounding stands in a similar relation to laws of metaphysics as causation does to the laws of nature.

3.3 Causal- and Grounding-Mechanical Models

Trogon (2018) has defended Illumination by appealing to causal mechanisms and causal-mechanical explanation. This section follows Trogon's lead in characterizing his view and may not be representative of how others would understand these notions. See Holly Andersen (2014a, 2014b) for different senses of the term "mechanism."

Trogon assumes a systems view of causal mechanism, on which causal-mechanical models describe how the parts of a system interact with each other to produce a phenomenon. This corresponds to a type of scientific explanation, namely, causal-mechanical explanation. For Trogon, explanations are propositions or other abstract structures that accurately and informatively represent the world's structure. Thus causal explanations accurately and informatively represent some aspect of the world's causal structure.

Likewise, claims Trogon, there is a type of metaphysical explanation that appeals to grounding-mechanical models. In the grounding case, mechanisms are instances of metaphysical determination relations (or chains of such relations), where the relata are constituents of the facts that stand in grounding relations. For instance, corresponding to the fact that Sotomayor exists grounds

the fact that her singleton set exists, there is a metaphysical determination relation (namely, set formation) holding between Sotomayor and her singleton set. This is a grounding mechanism.³ On an alternative view, the nomic subsumption view, grounding explanations appeal to laws of metaphysics. The grounding mechanism view has no such requirement on grounding explanation. This parallels the broad division between theories of causal explanation that appeal to laws of nature versus the causal mechanisms views that do not.

As noted, Trogdon defends only Illumination, not Unity. In favor of Illumination, he argues that the notion of grounding-mechanical explanation can do theoretical work in philosophy. Namely, it creates a demand for a specification of the grounding mechanisms in play whenever someone makes a grounding claim. Consider the debate between the priority monist and the priority pluralist about concrete objects. Whereas the priority pluralist recognizes a plurality of fundamental objects, the priority monist holds that the entire concrete cosmos is the sole fundamental object. According to Trogdon, the priority monist must specify how each “subcosmos” state of affairs ultimately depends upon the cosmos. Likewise, the priority pluralist must specify how each state of affairs ultimately depends upon the fundamental concrete objects.

3.3 Building Relations

Bennett’s (2017) approach differs from previous defenders of Unity and/or Illumination in not treating grounding as the most significant metaphysical analogue of causation. Instead, Bennett subsumes both under the class of building relations, which are a family of metaphysical determination relations. These include relations such as composition, constitution, set formation, realization, as well as grounding and causation. The addition of the latter two is unusual in two respects. First, grounding is placed alongside the more familiar relations rather than as the genus to their species or in some unifying role. Second, causation is included among rather than contrasted with such relations.

Bennett recognizes that there are differences between individual building relations, including the nature and number of relata. But they are unified by satisfying jointly necessary and sufficient conditions, namely, being asymmetric, necessitating, and generative, where generativity is, according to Bennett, what licenses locutions such as “in virtue of,” “because,” and “explain.” This does not mean that there is a single building relation of which the other relations are determinates or species or which unifies the other relations. Nonetheless, Bennett holds that they form an interesting and theoretically useful kind.⁴

Bennett explicitly defends Illumination in the course of arguing that causation is a building relation—in fact, her defense of Illumination constitutes her defense of Unity. She compares causation with so-called “vertical” building relations, those that hold synchronically rather than diachronically. Bennett also argues that the allegedly “vertical” noncausal building relations may also hold across time; see her (2017: 83–99). Here are five questions about causation that can also be asked about other building relations, including grounding (and whose answers may illuminate each other):

- (i) Is causation built?
- (ii) Must chains of causes terminate?
- (iii) Does every state of affairs have a cause?
- (iv) Is there causal overdetermination?
- (v) What are causal roles?

Bennett's point isn't that some answer to one of these questions carries over to the grounding case or vice versa but that the analogy may be helpful in exploring answers in each case. Bennett (2017: 78) also suggests adopting the methodological principle that "where and when parallel questions arise about causing and vertical building, the default position should be to adopt parallel answers to them."

4. Objections to Unity and Illumination

The main objections to Unity and Illumination come from two sources: Bernstein's (2016) wholesale critique of both Unity and Illumination and Koslicki's (2016) specific objections to Schaffer's interventionist models for grounding. This section lists some differences between grounding and causation that underlie potential or actual objections to Unity or Illumination.

Note that some differences have been acknowledged by defenders of Unity or Illumination themselves who do not take such differences to undermine either thesis. After all, none of them claim that causation and grounding are exactly analogous. But this might elicit a skeptical reaction: if defenders are not claiming an exact analogy, how should we evaluate Unity and Illumination? What similarities or dissimilarities should make a difference to the truth of these theses? Arguably, these questions have not been satisfactorily addressed in the literature.

4.1 General Differences

- (i) Causal explanation is significantly different than grounding explanation.

Michael J. Raven (2015: 325) argues that it is not obvious how to model grounding explanation on causal explanation. Grounding explanations need not involve "traditional hallmarks" of causal explanation, e.g., transference of power, a nontrivial statistical relationship, or asymmetric counterfactual dependence. And grounding explanations can occur without causal explanations.

- (ii) Causation but not grounding can be indeterministic.

Schaffer (2016a: 95) acknowledges this difference, saying that once the restriction to deterministic causation is lifted, the formalisms for causation and grounding no longer coincide. This and the next difference are among Schaffer's reasons for rejecting Wilson's claim that grounding and nomological causation are both species of a single causal relation (though he does not direct his criticisms at Wilson, since the views were developed independently).

- (iii) Grounding but not causation must be well-founded.

Schaffer holds that grounding chains must terminate (see also Schaffer 2003). But in the causal case, there is no "transference of reality," and thus causal chains need not terminate.

- (iv) Grounding is synchronic whereas causation is diachronic.

This difference is generally accepted. At the same time, Wilson (2018: 729–30) points out that there are potential counterexamples. For instance, quantum theory posits what looks like simultaneous causal action-at-a-distance, and Wilson's being human rather than a swampman may be grounded in his past causal history. According to Wilson, the reason that grounding is generally synchronic while causation is

generally diachronic is that only the latter is mediated by laws of nature, which typically relate states of affairs at different times. This difference underlies other differences between grounding and causation, e.g., that grounding is necessitating, whereas causation can be indeterministic. Bernstein (2016: 24–25) objects that this difference is significant.

There is a related alleged difference between grounding and causation: whereas the latter obtains between wholly distinct states of affairs, the former obtains between merely nonidentical states of affairs. See Gideon Rosen (2010: 118) and Schaffer (2016a: 75–76). Koslicki objects to the weaker constraint of nonidentity in the grounding case; see her (2016: 109–11).

- (v) The comparison between causal production/dependence and grounding production/dependence is inappropriate.

Bernstein objects to the claim in Section 3.1 that there is a distinction between grounding production and grounding dependence that mirrors the distinction in the causal case. She points to Schaffer's claim that whereas states of affairs inherit their "reality" from their grounds, there is no analogous transfer from cause to effect. She adds that there is no analogue in grounding of the view that causation involves a transfer of energy or "mark." Note that Bennett, who identifies grounding with grounding production, acknowledges that there are accounts of causation that involve transmission of a conserved quantity, unlike in the grounding case; see her (2017: 69).

- (vi) There may be important differences between causal omission and preemption on the one hand and grounding omission and preemption on the other.

Causation by omission seems possible—for instance, one's failure to water a plant may cause it to die. However, Bernstein (2016: 26–27) argues that there is no direct analogue in the grounding case, since "non-things" (as she puts it) cannot be grounds. She also argues (2016: 28–30) that causal preemption fails to have a direct grounding analogue. Consider a classic case of preemption, in which Suzy's rock throwing prevents Billy's own rock throwing from shattering the window. Wilson (2018: 24) suggests a grounding analogue: the presence and arrangement of his particles preempts some subset of them constituting a person. But Bernstein points out grounding preemption must involve intimately connected relata at different levels of specificity, whereas causal preemption may involve entirely distinct relata at the same level of specificity. Furthermore, counterfactual reasoning about grounding preemption requires reasoning about the impossible; see the next subsection.

4.2 *Objections to Grounding Interventionism*

The objections in this subsection are framed as objections against the interventionist model of grounding, even though they may have more general upshots.

- (i) Counterfactual reasoning involving grounding requires counterpossibles.

In Wilson's case of grounding preemption in the last subsection, the presence and arrangement of his particles (**Particles**) preempts some subset of them (**Subset**) constituting a person. While **Subset** can exist without **Particles**, **Particles** cannot exist without **Subset**. So reasoning about whether the person would still exist if **Particles** but not

Subset existed requires (nontrivial) counterpossibility, which has been independently controversial; see Francesco Berto and Mark Jago (2018).

Wilson himself acknowledges the need for counterpossibles in his interventionist framework. In the language of this framework, an independent intervention upon **Subset** is not possible, since **Subset** may not exist without **Particles**. Wilson (2016: 719) points out that counterpossibles are required for even simple cases, such as Sotomayor's existence grounding her singleton's existence. His own conclusion is merely that acceptance of grounding is tied to acceptance of counterpossibles.

- (ii) The interventionist framework cannot capture various grounding dependencies.

Consider again the case of a shirt's being maroon grounding its redness. On Schaffer's contrastive interventionist framework, we let C be the variable that takes the value 1 if the shirt is maroon and 0 if it is navy, and let E be the variable that takes the value 1 if the shirt is red and 0 if it is blue. The single structural equation is $E = C$.

Koslicki (2016) objects that there are several dependencies not captured in this model. First, the alternative to the shirt's being maroon (or red) is the shirt's being navy (or blue), when really there are many other options. But a revised model on which $C = 0$ if the shirt is not maroon (or $E = 0$ if the shirt is not red) incorrectly represents the dependencies, since $E = 0$ does not follow from $C = 0$. Second, the model does not capture certain necessities, such as the fact that it is metaphysically necessary that the shirt has some color or other; see Koslicki (2016: 109) for other examples. Koslicki then argues that the shirt case is not really analogous to the simple causal case of the ball and the window; it is more like a case of massive causal preemption, since the shirt's being a determinate color prevents it from having any other determinate color. But massive causal preemption is a problematic case for interventionism about causation. See Schaffer (2016a: 65) for acknowledgment of the complications with preemption cases. See Wilson (2016: 186–91) for further discussion.

5. Implications

If Unity, or at least Illumination, is correct, there are obvious upshots. Arguably, the upshots are most beneficial for defenders of grounding, given that causation is the more familiar and accepted theoretical posit. In fact, Schaffer (2016a: 91) takes his defense of Unity to constitute an argument for grounding monism, the view that there is a single relation of grounding. Grounding skeptics like Jessica Wilson (2014, 2016) and Koslicki (2015) accept only specific determination relations such as set formation and constitution. According to such skeptics, these relations are not instances or species of a more general grounding relation.

Unity is consistent with the view that there are many different grounding relations; however, Schaffer argues that just as the formalism of structural equations models unifies notions of causation, it also unifies notions of grounding. See also Schaffer (2016b: 166–67). Note that Trogon (2018: 1292–93) is explicit that he is not arguing this. On the other hand, if Illumination and thus Unity are false, we cannot use causation as a starting point for theorizing about grounding, except to explain what it is not. While this would not entail grounding skepticism, it would take away an important resource for defenders of grounding.

★ Many thanks to Holly Andersen, Tom Donaldson, and Mike Raven for comments on drafts of this chapter.

Related Topics

- Chapter 8: M. Glazier, “Explanation”
Chapter 10: A. Skiles, “Necessity”
Chapter 11: K. Koslicki, “Skeptical Doubts”
Chapter 13: K. Richardson, “Varieties”
Chapter 31: N. Emery, “Laws of Nature”

Notes

- 1 I am going to use the term “state-of-affairs” as a neutral stand-in (insofar as this is possible) for whatever the relata of grounding or causation are, with some exceptions. For a discussion of the relata of ground, see the introduction of this volume; for the relata of causation, see Schaffer (2016c).
- 2 Historical connections between causation and grounding-like notions deserve to be explored as well. As this task requires historical as well as philosophical expertise, I leave it to the relevant experts. See, for instance, Phil Corkum (2016), Katy Meadows (Forthcoming) and Benjamin Schnieder (2014), as well as Chapter 1 of this volume.
- 3 Trogdon discusses a potential complication, given that he leaves it open whether there are instances of grounding relations that are not grounding mechanisms; see his (2018: 1304–5).
- 4 She in fact holds that there is a family resemblance rather than a natural kind; see Chapter 3 of her book.

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21

EMERGENCE

Stephan Leuenberger

1. A Tale of Two Literatures

Grounding and emergence are two “vertical” relations, linking different levels of reality. Their study belongs to what is sometimes called “inter-level metaphysics”. My brief is to critically survey how these relations have been compared, contrasted, and connected. There is little explicit discussion for me to report on, alas. In light of this, I shall do two things. First, I shall give an introduction to the literature on emergence, pitched at a reader who has some familiarity with work on grounding. (Grounding needs no introduction here, in light of its starring role in the other chapters of this volume.) Second, I shall highlight a few ways in which an appeal to grounding could illuminate the notion of emergence.

At the outset, I should note that the literature on emergence has rather different characteristics from the one on grounding. The latter—if I am allowed to paint with a broad brush—is relatively homogeneous, in the sense that a small number of seminal papers serve as a shared point of reference. (You know which ones I have in mind.) The working assumption is that different authors talk about one and the same relation when they use the term “grounding”. There is thus no urgent need to define the term, which is generally taken as primitive. Disagreements are treated as substantive rather than merely verbal. (However, see Kathrin Koslicki, “Skeptical Doubts”, Chapter 11 in this volume, for a different take on the working assumption.)

Grounding is taken to be topic neutral, and contributors tend to be interested in general and noncontingent claims about it. Relatedly, the literature gives prominence to certain formal questions—concerning the grammatical and logical form of grounding claims and various structural properties of the grounding relation, or alternatively, of a corresponding operator. In so far as its examples are from outside the province of logic, it is largely indifferent to the question whether they are actual or merely possible. (See **Part III** of this volume, “Logic and Structure”.)

The literature on emergence is strikingly different on all those counts. It is dispersed over a number of subdisciplines of philosophy, with different intellectual traditions, notably philosophy of mind, philosophy of science, and to a lesser extent metaphysics. Indeed, collections on emergence (e.g., Bedau and Humphreys (2008); Clayton and Davies (2006)) tend to cast the net even wider and include contributions from various natural and social sciences, as well as speculative proposals from the margins of mainstream science. Accordingly, the literature is highly heterogeneous. This means that whatever generalizations I am about to offer on the literature should be taken with a grain of salt.

It would be bold, if not fanciful, to claim that all contributors to the literature on emergence make claims about one relation and substantively disagree about its features. By and large, authors are aware of this and accordingly try to offer a definition of “emergence”. (A sample will be given later in this chapter.) Primitivism about emergence is not a popular approach.¹ But a definition or analysis is not the be-all and end-all of the efforts of theorists of emergence. Typically, the Holy Grail is the solution of a recognized philosophical mystery, such as that of consciousness or of mental causation. That is, the main desideratum for a definition is that it makes “emergentism” about consciousness, or about the causal powers of the mind, a view that is not just plausible but illuminating and perhaps even exciting. So an example of the application of the concept serves much more than a merely illustrative purpose, and it matters greatly whether the example is drawn from the actual world. In contrast, it is not a high priority to define a topic-neutral concept, or at least one which allows for the formulation of interesting generalizations. Little attention is paid to the question whether emergence is a transitive relation, for example.

2. Emergence and Levels of Reality

As the remarks in the preceding section indicate, a short, precise, and generally acceptable characterization of emergence is not to be had. Accordingly, I shall approach the topic slowly and in a somewhat round-about way.

Emergence, I submit, involves novelty along some dimension. In ordinary discourse, we might talk about the emergence of a certain phenomenon at a certain point in time, or over a period of time—a monopoly in the market for text-processing software, say. The implication is that the phenomenon is novel in the sense that it was not present prior to that time, even though it may well continue to be present later. In general, the dimension along which there is emergence need not be temporal. To allow us to make sense of novelty, it does need to be directed, unlike spatial dimensions.

In the philosophical discussion of emergence, the relevant dimension is given by a stratification of reality into vertically arranged levels or layers. If a level is lower than another one, it counts as more fundamental. A phenomenon is novel at some level if it does not occur at any lower level, even though it may occur at a higher level. The characterization of emergence given in an influential article by David J. Chalmers appeals to such a background stratification into levels:

We can say that a high-level phenomenon is strongly emergent with respect to a low-level domain when the high-level phenomenon arises (in some sense) from the low-level domain, but truths concerning that phenomenon are not deducible even in principle from truths in the low-level domain.

(Chalmers 2006: 244)

There is a great deal packed into this characterization, but for the moment, I just wish to draw attention to the entailed difference in levels between what is emergent and that with respect to which it is emergent. Chalmers, like most other writers on emergence, does not say more about how levels are to be understood. That the world is layered is typically presupposed rather than asserted among theorists of emergence.

By default, the relation generating the levels can be understood to be parthood. Indeed, many authors explicitly define emergence to be a relation holding between properties of a whole and properties of its parts. The stratification of reality into mereological layers that is typically taken for granted in such discussions is nicely articulated in the classic Oppenheim and Putnam (1958).

However, parthood is just one relation that can generate a vertically oriented dimension. Another one, of course, is grounding. Indeed, it is often said that grounding allows us to make sense of the notion of relative fundamentality that is closely associated with the layered conception of the world.

The idiom of “grounding” and cognate terms allows us to ask whether levels that are lower in the mereological ordering are also more fundamental. Popular views such as microphysicalism would appear to be committed to a positive answer. But priority monism, as advocated in Schaffer (2010), holds that the cosmos, the object which includes all others as parts, is the most fundamental, in the sense that all the others depend on it.

As I noted already, the term “emergence” is used in more than one way. On one use, it is tied to a mereological layering of the world. In that sense, Schaffer’s priority monism may be described as a view according to which the cosmos has emergent properties. (Though see Section 4 for a qualification of that remark.) On another use, “emergence” is tied to a layering generated by grounding or dependence. In that sense, “emergent” cannot apply to Schaffer’s cosmos or its properties, since they are not higher-level phenomena. On the contrary, properties of parts might turn out to be emergent, in that sense, from properties of the whole—although that idea has not yet been explored, to the best of my knowledge.²

Grounding and its cognates may thus be of use in the articulation of a layered conception of the world that forms the background of debates about emergence. It must be acknowledged, however, that this is only a promissory note so far. There is no straightforward way to define relative fundamentality, or levels of reality, in terms of grounding, as the discussion in chapter 6 of Bennett (2017) shows. The simplest proposal for explicating that *f* is at a lower level than *g* is that *f* partially grounds *g*. But while this condition is presumably sufficient, it is not necessary: a fact about the quantum properties of a particular electron in Spain is arguably at a lower level than a fact about population density in Scotland, even though it does not ground it.

All may not be lost, though. Many emergence claims do not rely on levels being defined across disconnected chains but only on comparisons of relative fundamentality within a given chain. A sufficient condition for relative fundamentality may be enough for that purpose.

3. Emergence Claims and Their Logical Forms

For now, I postpone the questions whether there can be an informative analysis of emergence and how it might look. In this section, I do not discuss how emergence relates to grounding but rather how methods familiar from the study of grounding might be of use in the study of emergence.

One of the contributions of the literature on grounding to philosophical progress has been the regimentation of discourse involving such terms as “grounds” and “in virtue of”. Since a *Handbook* chapter offers me as good a platform as any to recommend a regimentation of the language of emergence, I shall begin by asking what we should take as the canonical form of emergence claims. This question has already received a sophisticated answer by Hempel and Oppenheim (2008), and I shall present a modified version of their account. For simplicity, I shall assume that the background layering is mereological.

What is perhaps the most familiar use of any word in the family is the one-place predicate “is emergent”. This predicate is informally applied to entities of a variety of ontological categories. In the specialized literature on emergence, it is typically a predicate of properties. It seems to me, though, that something is emergent because it is emergent from

something else and that the locution “is emergent from” makes for a better starting point for an analysis. (I choose the clunkier “is emergent from” over “emerges from” because of its connotations.)

While “is emergent from” suggests a binary predicate, we actually need at least a ternary one, to allow that one and the same property is emergent in one thing but not in the other. My first stab at a canonical form is this:

F is emergent from A in x

Here, F is a property, A a class of properties and relations, and x a particular. If F emerges from A in x, then F can be called an “emergent property” of x; and A can be called the “emergence base” of F in x.

I shall shortly refine this formulation and discuss in more detail under what conditions it is true. But the basic idea is clear, given our assumption that the background layering is mereological: if F emerges from A in x, then (i) x is a complex object which has F, (ii) the parts of x do not have F, but have some properties in A, and (iii) condition (i) and facts involving the parts of x and A stand in a suitable relationship, to be specified.

I called the above a “first stab”. The reason is that in general, a relativization of the “is emergent from” relation to particulars does not seem enough. In my explanation, I said the parts of A have some properties in A. But which parts? All of them? All of its proper parts? It seems that we need to have a specification of what parts count.

Consider the human brain as a candidate bearer of emergent properties such as consciousness, and let the putative emergence base A consist of all intrinsic properties of the relevant parts. The brain is made up of two hemispheres, which, we may suppose, are both intrinsically conscious. Then presumably, consciousness does not count as being emergent from the intrinsic properties of the brain relative to a decomposition into the two hemispheres: it is not a novel feature of the whole brain relative to that decomposition. But the brain has other decompositions: into cells (and intercellular material), into molecules, into atoms, into subatomic particles, etc. Relative to some of them, consciousness may well be emergent.

Hempel and Oppenheim expressed this as a relativization to a parthood relation, or a meaning for the term “part of”:

The volume of a brick wall, for example, may be inferable by addition from the volumes of its parts if the latter are understood to be component bricks, but it is not so inferable from the volumes of the molecular components of the wall. Before we can significantly ask whether a characteristic W of an object w is emergent, we shall therefore have to state the intended meaning of the term “part of”.

Rather than to a meaning of “part of”, I shall take the relation to be relativized to some parts that compose the bearer of the property. (For convenience, I shall say “a class of parts” rather than speaking plurally of “some parts”.) Accordingly, I propose that canonically, emergence claims have four argument places:³

F is emergent from A in x relative to Y

The idea is that among the members of the class Y, there are things that compose x and that are potential bearers of properties in A.⁴

To summarize the necessary conditions for the truth of “F is emergent from A in x relative to Y” mentioned so far:

- x has F.
- No member of Y has F.
- Some members of Y compose x.

Notice that the first two conditions entail that x does not belong to Y—an emergent property of a thing is emergent from facts involving proper parts of that thing, not from facts involving that thing itself.⁵

A property’s being emergent is thus triply relative. We can derelativize in the usual ways: by fixing the value of one of the variables, perhaps as a function of context, and by quantification. We may say that F is emergent from A relative to Y iff for all x that have F, F is emergent from A in x relative to Y. When it comes to the variable A, universal quantification will not do. We will want to exclude properties such as “being a part of something that is F”, which would in some sense trivialize the question what the relationship between the properties of the parts and the properties of the whole is. Once again, this was clearly recognized by Hempel and Oppenheim:

If a characteristic of a whole is counted as emergent simply if its occurrence cannot be inferred from a knowledge of all the properties of its parts, then, as Grelling has pointed out, no whole can have any emergent characteristic. Thus . . . the properties of hydrogen include that of forming, if suitably combined with oxygen, a compound which is liquid, transparent, etc. Hence the liquidity, transparency, etc., of water can be inferred from certain properties of its specific constituents.

One salient choice for A is the class of intrinsic properties and relations. Assuming that everything is composed of mereological atoms—that there is no “gunk”, in other words—the class consisting of all of them is a nonarbitrary choice for our Y.

After a triple derelativization, we can define the monadic “is emergent” in terms of the quaternary predicate:

F is emergent =_{df} for all x, F is emergent from the intrinsic properties and relations relative to the mereological atoms.

In the absence of explicit relativizations, and given the assumption that there are mereological atoms, this often provides a sensible reading of emergence claims.

Grounding claims typically take facts as relata, and we might wonder whether emergence claims can also be cast in such form. Perhaps our form for emergence claims is general enough to yield emergence claims for a singular fact [Fa] as a special case:

[Fa] is emergent from A relative to Y if F is emergent from A in a relative to Y.

It is not clear whether this sufficient condition should also count as necessary: maybe [Fa] can count as emergent not because of any novelty in the constituent property F but because the existence of the particular a is emergent.

If we disregard this worry, we can move the form of emergence claims even closer to that of grounding claims, by taking it to be a relation between one fact and a class (or plurality) of facts.

Let $f(A, Y, x)$ consist of all and only facts of the form $[G(y_1, \dots, y_n)]$, for some G in A and y_i that belong to Y and are part of x . We can then define:

$[Fa]$ is emergent from $f(A, Y, x) =_{\text{df}}$ $[Fa]$ is emergent from A relative to Y .

I do not know whether this can be usefully extended to facts and classes of facts of different forms, e.g., facts involving relations, or quantified facts.

4. Janus-Faced Emergence

So far, I have dodged the most difficult question: What is the required relationship between x 's being F and the distribution of properties and relations from A among the parts of x that are in Y ? I have already suggested that the former needs to be at a higher level than the latter, but that is clearly not a sufficient condition. The short answer to the question what sort of relationship is required beyond that seems to be: a relationship that is intimate but not too intimate.

Emergent properties are Janus faced. On one side, they contrast with "resultant" ones: paradigmatically, mass and charge properties of wholes, obtainable by summation from masses and charges of the parts. If the background layering is mereological, we may also call such properties "microreducible".

On the other side—although this claim may be more controversial—they contrast with properties of wholes that do not depend in any way on properties of the lower level. Given a mereological background layering, we can call them "macrofundamental" properties. Clear examples of the latter are hard to come by. Perhaps they include so-called distributional properties as they are understood by philosophers such as Josh Parsons (2004) and Jonathan Schaffer (2010: 59). (Hence properties of Schaffer's cosmos may not count as emergent even if the layering is mereological, qualifying a remark I made in Section 2.) On their view, a whole may have the property of being polka-dotted, where the instantiation of that property is not the result of it having homogeneous parts with different colors; rather, heterogeneity among its parts is derivative from or a result of the whole having the distributional property.⁶

This Janus-faced nature of emergent properties is already evident in a classic discussion by C.D. Broad (1925). He contrasts the theory of emergence with two other accounts. On the one hand, he distinguishes it from what we might call "natural pietism":

[According to this view, certain] characteristically different ways of behaving may be regarded as absolutely unanalysable facts which do not depend *in any way* on differences of structure or components. . . . This . . . consists in holding that there are certain differences which cannot be explained, even in part, but must simply be swallowed whole with that philosophic jam which Professor Alexander calls "natural piety".

(54–5)

On the other hand, he distinguishes the theory of emergence from "the ideal of pure mechanism", exemplified by gravitating particles as conceived by classical mechanics.

A clear expression of such a double contrast is to be found in a more recent classic discussion, by James van Cleve (1990: 222–223): "Emergence. . . contrasts with two alternatives, a looser alternative on the left and a tighter alternative on the right. On the left, we do not even have dependence or determination; on the right, we have full reducibility".

This suggests two necessary conditions for emergence: it entails the disjunction of dependence and determination, and it entails the lack of full reducibility. It is useful to express this schematically, with relations S and R , where R entails S : if emergence holds, then S holds but R does not.⁷ If this necessary condition is also held to be sufficient, I shall speak of an “ S -without- R ” or “ S -but-not- R ” account of emergence. Accounts of that type are legion in the literature. For example, emergent properties are said to be:

- Nomologically supervening but not metaphysically supervening (van Cleve 1990; Noordhof 2010)
- Metaphysically supervening but not functionally reducible (Kim 2006)
- Metaphysically supervening without there being an explanation for their supervenience (Vision 2011)
- Dependent but fundamental (Barnes 2012)
- Dependent but autonomous (Wilson 2015)

Needless to say, these slogan-characterizations do not capture the details of these respective proposals. But for reasons of space, they will have to do here.

5. Grounding in an S -but-not- R -Account of Emergence

In Section 2, I introduced a quaternary emergence predicate. An “ S -but-not- R ” analysis of this predicate might take the following form:

F is emergent from A in x relative to $Y =_{df}$

- $l_s(F, x)$ stands in S to $r_s(A, x, Y)$, but
- $l_R(F, x)$ does not stand in R to $r_R(A, x, Y)$.

Here, l_s maps to F and x to an entity that is of the right category to fill the “left” argument place of S ; and r_s maps A , x , and Y to an entity (or plurality of entities) that is of the right category to fill the “right” argument place of S ; and likewise for l_R and r_R , mutatis mutandis. So $l_s(F, x)$ and $r_s(A, x, Y)$ might simply be F and A , if S is the relation of supervenience; or they might be $[Fx]$ and $f(x, A, Y)$, if S is the relation of nomic necessitation.

Remaining at the schematic level a bit longer, we now have three relations on the table:

- S , the broader relation, whose holding is a positive condition for emergence.
- R , the narrower relation, whose failing to hold is a negative condition for emergence.
- S -without- R , or emergence.

Given the brief for this chapter, it is appropriate to ask whether grounding might be one of these three relations.

Consider first the hypothesis that grounding is S -without- R , or emergence. For that to fly, we would need to be able to identify suitable relations S and R . Perhaps we might take the logically stronger relation R to be identity (or, more precisely, class membership, or the “is one of” of plural logic).⁸ It is not clear whether we could capture S in any other way rather than disjunctively, as grounding-or-identity. But whether or not grounding can be fitted in such a scheme, the hypothesis must be rejected as implausible. Typically, emergence is taken to rule out not only identity but also a kind of full reducibility that can hold between numerically

distinct properties or facts—the kind of case where one thing is nothing over and above the other. If there are such cases, they will be cases of grounding (Rosen 2010: 122) but not of emergence.

A second hypothesis is that grounding is the positive emergence condition. Before evaluating this, I need to introduce a distinction that is now standardly made in the literature: that between weak and strong emergence. A stock example of weak emergence in the literature is provided by the Game of Life: simple low-level rules give rise to patterns at the higher level, such as “gliders”, which we would not expect based on mere inspection of the rules, in the absence of simulation. More generally, if a phenomenon is weakly emergent, we do not typically gain understanding of it by looking at the properties of the parts of the system which exhibits the phenomenon. As this rough characterization indicates, weak emergence is at bottom an epistemic matter.

Strong emergence, in contrast, would force us to expand our ontology. There are no uncontroversial examples, but there is a tradition that thinks that consciousness is strongly emergent from the microphysical properties of the brain.

With this distinction in hand, let us return to the second hypothesis: that grounding is the positive emergence condition. There is no obvious objection to this thought in the case of weak emergence. To a first approximation, weak emergence might be nontransparent grounding. The lack of transparency would need to be of a particular kind: it should not just be due to the complexity of the fact being grounded. Rather, the grounded fact needs to be simple, no longer containing the grounding facts as constituents. Thus we would have the following formula (if we wish to fit this into our formula, we can read “transparency” as “transparent grounding”):

weak emergence = grounding without transparency

Whether this characterization is better than alternative ones that have been offered of weak emergence is not a question I can take up here. For what it is worth, I think that with weak emergence, the main challenge is to pin down the negative condition (what I have called “transparency” here), not the positive condition.

However, grounding clearly fails to be the positive condition on strong emergence. It is not easy to argue against this by counterexample, that is by exhibiting a case of emergence that is not a case of grounding. The reason is that there are no uncontroversial cases of emergence. Indeed, many philosophers think that the notion is incoherent, and much of the literature is trying to address their worries. As the introduction to a recent collection puts it:

One critical focus [of debates about emergent properties] is on the very intelligibility of such “emergence”: critics of the notion are suspicious that emergent properties inhabit a kind of halfway house that is, by its nature, unstable, given that such properties are characterized by two features that are in tension with one another.

(Macdonald and Macdonald 2010: 1)

It is fair to say, though, that if there are instances of emergence, i.e., of S without R, they will not be instances of grounding.⁹ The relation S will then be distinct from grounding.

The third hypothesis to consider is that grounding is the negative emergence condition, i.e., the relation R. I take it that weak emergence is compatible with grounding. However, the proposal is quite plausible for strong emergence, on the face of it. It suggests the following equation, with S to be filled in:

strong emergence = S without grounding

This would be the most straightforward way of connecting grounding and emergence. Given the connections between grounding and metaphysical supervenience, functional reduction, and fundamentality, it seems to be in the same spirit as some of the other proposals listed above.

Indeed, using grounding rather than metaphysical supervenience in a definition of grounding might have an advantage that will be familiar to devotees of grounding. Jessica Wilson (2005: 438) pointed out that if natural laws are necessary, then “[e]mergent properties will . . . supervene with metaphysical necessity on their physical base properties”. Depending on the kind of necessitarianism at issue, the putatively emergent property may not be grounded on the base property. If so, the grounding-theoretic formulation will not face the problem Wilson raises.

This last point generalizes. A number of authors have appealed to modal conditions when defining emergence (supervenience, necessitation, modal recombability, or the lack thereof). In some cases, these suffer from the sort of problems that have seen modal conditions replaced by ground-theoretic ones in other contexts. They may be hostage to controversial views about what is possible, as the example of the last paragraph shows; or they may fail to make distinctions that are fine-grained enough. Formulating these conditions in terms of grounding promises to solve these problems.

In the earlier schematic proposal, S has not been specified. But we might try to cash it out in ground-theoretic terms too, appealing to the distinction between partial and full grounding:¹⁰

strong emergence = partial grounding without full grounding

As it stands, this proposal is too simple. Extrinsic properties, like being less tall than the Eiffel Tower, should not count as strongly emergent simply because their instantiation is partially but not fully grounded by intrinsic properties of the proper parts of a thing.

Perhaps we could require that strongly emergent properties are such that their instantiation lacks full grounds altogether rather than simply lacking full grounds involving their proper parts. But on the orthodox account, some facts partially ground a fact just in case they are among facts that fully ground that fact. Partial grounds can always be supplemented to form a full ground. On that account, nothing could be strongly emergent according to the earlier formula.

In the literature on grounding, nonorthodox notions of partial ground have been identified. A promising candidate is Fine’s strict partial ground, which can be contrasted with strict full ground to yield the following formula:

strong emergence = strict partial grounding without strict full grounding

A discussion of this notion of strict partial grounding and its potential to help us understand emergence is beyond the scope of this chapter.¹¹

6. Conclusion

While there is little explicit discussion of how grounding and emergence relate, I have identified a number of ways in which the literature on grounding can provide a resource for theorists of emergence.

Use grounding to define levels. The idea of emergence seems to presuppose a background picture of reality as divided into different levels. Standardly, such levels are understood in mereological terms: things at lower levels compose those at higher levels. The relation of grounding can be used to articulate a different segregation into levels, which need not coincide with the mereological one.

Regiment the idiom of emergence, grounding-style. The logical form of various emergence-claims and the logical relations between them are often less than fully transparent. The literature on grounding shows the usefulness of regimentation and provides an exemplar of how to go about it.

Replace modal conditions by ground-theoretic ones. Emergence has often been defined partly in modal terms. Such definitions tend to be vulnerable to the same sort of objections that have plagued modal accounts of other “vertical” relations—objections that have motivated appeals to grounding in other contexts.

Grounding as a positive condition on weak emergence. Definitions of emergence are typically conjunctive, imposing a positive and a negative condition. What these conditions are will depend on whether the target notion is weak emergence or strong emergence. For weak emergence, grounding is a plausible positive condition, with the corresponding negative being epistemic in nature. (This point, as well as the next one, can be seen as illustrations of the foregoing point that modal conditions may be replaced by ground-theoretic ones.)

Grounding as a negative condition on strong emergence. Strong emergence—which is of greater metaphysical interest than weak emergence—is plausibly incompatible with grounding, and the absence of grounding may indeed be its characteristic negative condition.

Conversely, the literature on emergence also provides a great resource for theorists of grounding, especially those interested in adding variety and lifelikeness to their diet of examples.¹²

Related Topics

Chapter 27: Structure

Chapter 23: Fundamentality

Notes

- 1 The multiplicity of uses of “emergence” may still be harmful even once it is generally acknowledged. In a recent book-length study entitled *Emergence*, Paul Humphreys suggests that “in an intellectually ordered world, we would stop using the word ‘emergence’ and use more precisely defined terms . . . Yet the word has irresistible attraction and I do not have the power to ban it . . . I shall retain the term as a convenience” (Humphreys 2006: xviii).
- 2 The interaction between the mereological layers and the dependence layers raises some tricky questions. See Steinberg (2015) for discussion.
- 3 I omit Hempel and Oppenheim’s fifth argument place, a scientific theory.
- 4 Why do I not require Y to be such that x is its fusion? Given the assumption that fusions are unique, the requirement would allow us to simplify our canonical formulation by dropping x, since it could be recovered from Y. The reason is that we need both argument places to be able to express claims such as “For all cells x, x is emergent from A in x relative to the class of molecules”.
- 5 As Michael J. Raven points out, the two conditions entail that the existence of x cannot be emergent from any properties relative to any decomposition, since whatever parts belong to a decomposition also have the property of existence. However, they do allow that certain essential properties of x are emergent—its kind membership, say, or its haecceity.
- 6 Sometimes, “emergent” is arguably used in such a way to include macrofundamental properties, as when David Lewis acknowledges that “there might be emergent natural properties of more-than-point-sized things” (Lewis 1986: x).
- 7 Mnemonically, we can think of S as “supervenience” and R as reducibility, but some accounts of emergence use different relations.
- 8 The hypothesis that grounding is S-but-not-identity has the virtue of accounting for the widely assumed irreflexivity of grounding.
- 9 Compare Karen Bennett’s argument that while grounding is a building relation in her sense, emergence is not because its instantiations fail to have implications about relative fundamentality (2017: 64). Obviously, instantiations of S will not have any implications that instantiations of S-without-R fails to have.

- 10 Wilson (2014: 544) claims that emergence “cannot be characterized by appeal to Grounding alone,” adding in a footnote that “at least two primitively related primitive relations”, which she calls “Partial Grounding” and “Complete Grounding”, would be required.
- 11 In unpublished work, the author objected against defining emergence in terms of full grounding without orthodox partial grounding and suggested the use of strict partial grounding; see Wilson 2018: 506–507, 2019. Related ideas are discussed in Leuenberger (2019), but with a focus on fundamentality rather than emergence.
- 12 Many thanks to the editor as well as the participants at a Hamburg workshop dedicated to this Handbook for their useful comments. This work was supported by the Arts and Humanities Research Council [grant number AH/M009610/1].

Further Reading

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22

ESSENCE

Justin Zylstra

Recent theorizing about essence describes essence as “conceived on the model of definition”, such that “we may define an object, or say what it is” (Fine 1994a: 2). So conceived, something is essential to an item if and only if it is part of what it is to be that item (Correia 2012a; Dasgupta 2016; Fine 1994a, 1995a). For example, it is essential to electrons that their rest mass is strictly positive, since it is part of what it is to be an electron that its rest mass is strictly positive. By contrast, it is not essential to electrons that they were studied by Thomson, since it is not part of what it is to be an electron that electrons were studied by Thomson. After all, electrons may not have been studied by Thomson. Moreover, it is not essential to electrons that they travel at less than speed of light, even though they must travel at less than the speed of light, since it is not part of what it is to be an electron that it travels at less than the speed of light. Rather, electrons travel at less than the speed of light *in virtue of* having a rest mass that is strictly positive, perhaps in addition to certain facts of relativistic kinematics.

Recent theorizing about ground describes ground as a “distinctively metaphysical kind of explanation, in which the explanans and explanandum are connected . . . through some constitutive form of determination” (Fine 2012a: 37). So conceived, $\Phi_1, \Phi_2, \dots$ ground Ψ if and only if Ψ holds *in virtue of* $\Phi_1, \Phi_2, \dots$, where Φ and Ψ , with or without numerical subscripts, are variable sentences. For example, suppose that it is sunny. From this it follows that it is sunny or windy, irrespective of whether or not it is windy. Moreover, it is sunny or windy *in virtue of* its being sunny. In this sense, the presence of sun *determines, or is constitutive of*, the presence of sun or wind. By contrast, suppose that it is sunny and windy, and so it follows that it is sunny. Nevertheless, it is not sunny *in virtue of* its being sunny and windy, since the presence of sun *and* wind does not determine, or is not constitutive of, the presence of sun alone. After all, there is more to the weather if it is sunny *and* windy than if it is sunny alone.¹

When we investigate some phenomenon, it pays to think about how that phenomenon relates to other phenomena. For example, when we investigate metaphysical necessity, it pays to think about how it relates to (e.g.) quantification. In fact, we tend to think that these relations are constitutive of metaphysical necessity. For example, whether the Barcan Formula and its converse are true makes a difference to the nature of metaphysical necessity. The case is similar with respect to essence and ground: it pays to think about how metaphysical definition (i.e., essence) and metaphysical explanation (i.e., ground) are related, by which I mean that it pays to

think about how facts expressed by statements of the form ‘It is essential to x that Φ ’ relate to facts expressed by statements of the form ‘ Ψ in virtue of $\Phi_1, \Phi_2, \dots$ ’, where x is a variable name.

Philosophers have recently explored various interesting relations between metaphysical grounding and essence, not all of which are compatible. For example, it has been argued, among other things, that essence reduces to ground (Gorman 2014; deRosset 2013a); that ground reduces to essence (Correia 2013; Zylstra 2018a); and that essence and ground can jointly characterize physicalism (Dasgupta 2014), moral laws (Rosen 2017), and even metaphysics itself (Fine 2012b, 2017b). The aim of this chapter is to provide an overview of this literature. To achieve this aim in a systematic way, I divide the chapter into four sections. In the first section, I review the literature on essence. In the second section, I outline various reductive proposals: reductions of essence to ground, reductions of ground to essence, and reductions of both to a third party. In the third section, I outline various proposed entailments: entailments from essence to ground and entailments from ground to essence. In the fourth section, I outline how essence and ground have been jointly applied toward various theoretical ends.

1. Conceptions of Essence

The literature on essence does not describe a unique kind of essence. In this section, I outline the different conceptions of essence. According to the syntax I adopt, statements of essence are formed using a binary operator that takes a name and a sentence and gives a sentence. For example, ‘It is essential to Socrates that Socrates is human’ is a statement of essence. An alternative formulation of the syntax of essence takes the essence operator to be a predicate modifier. On this formulation, the essence operator is a unary operator that takes a predicate and gives a predicate. For example, if ‘human’ is a predicate, then ‘essentially human’ is an essentialist predicate, and ‘Socrates is essentially human’ is a statement of essence. Again, I adopt the sentential formulation here. But on the different formulations of the syntax of essence, see Fine (1994b).

1.1. *Constitutive and Consequentialist Essence*

Constitutive essence is linked with definition in the way described at the outset. On this conception of essence, it is essential to x that Φ if and only if it is part of what it is to be x that Φ , where x is a nominal variable and Φ is a sentential variable. Constitutive essence is not closed under logical consequence. For example, although it is constitutively essential to Socrates that Socrates is human, and Socrates’s being human logically implies that Plato is a sophist or not a sophist, it is not constitutively essential to Socrates that Plato is a sophist or not a sophist.

By contrast, consequentialist essence satisfies the following closure condition: if it is essential to x that Φ_1 and it is essential to x that $\Phi_2 \dots$ and Ψ is a logical consequence of $\Phi_1, \Phi_2, \dots$ then it is essential to x that Ψ . For example, although it is not constitutively essential to Socrates that Plato is a sophist or not a sophist, it is consequentially essential to Socrates that Plato is a sophist or not a sophist.

What is the order of definition between constitutive and consequentialist essence? On one view, constitutive essence is taken as primitive and consequentialist essence is defined as the closure of constitutive essence under logical consequence, perhaps for a restricted relation of logical consequence (Correia 2012a; Fine 1995a). On an alternative view, consequentialist essence is taken as primitive and constitutive essence is defined by restriction on consequentialist essence. For example, on one view, something is constitutively essential to some items when it is not consequentially essential to those items in virtue of other things that are consequentially essential to those

items (Fine 2012a). For example, it is consequentially but not constitutively essential to Socrates that Socrates is human or not a philosopher, since it is consequentially essential to Socrates that he is human or not a philosopher in virtue of its being consequentially essential to Socrates that Socrates is human. By contrast, it is constitutively essential to Socrates that Socrates is human, since it is not in virtue of anything consequentially essential to Socrates that it is consequentially essential to Socrates that he is human.

1.2 Objectual, Generic, and Sentential Essence

The essence operator is a binary operator that takes a name and a sentence and gives a sentence. This gives *objectual essence*. But the syntax of essence may be extended in such a way that the operator takes items from syntactical categories other than names (Correia 2006, 2013). On one extension, the operator may take a predicate and a sentence and deliver a sentence. For example, if ‘prime number’ is a predicate, then ‘It is essential to being a prime number that for all x , x is prime if and only if x is a number with no nontrivial divisors’ is a statement of essence. This gives *generic essence* (Correia 2006).

On another extension, the operator may take a sentence and a sentence and deliver a sentence. So extended, the statement ‘It is essential to its being the case that $\sim\Phi$ that Φ if and only if $\sim\Phi$ ’ is a statement of essence. This gives *sentential essence* (Correia 2013). Correia (2006) takes objectual essence as definable in terms of generic essence: for it to be essential to x that Φ just is for it to be essential to being identical with x that Φ . Correia (2013) suggests that sentential essence is primitive.

We can sometimes express the same ideas with different syntactical items. For example, we may express strict implication using a monadic sentential operator and material implication, or we may express strict implication using a binary sentential connective (i.e., the fishhook). As Correia and Skiles (2019) note, we can express the three types of essentialist statements using the variable-binding connective of *generalized identity* (Correia and Skiles 2019; cf. Dorr 2016 and Rayo 2013). We get the following equivalences.

- It is essential to Socrates that Socrates is human iff for something to be identical with Socrates just is for it to be human and identical with Socrates.
- It is essential to being a vixen that all vixens are female foxes and all female foxes are vixens iff for something to be a vixen just is for it to be a female fox.
- It is essential to its being the case that $\sim\Phi$ that Φ is materially equivalent to $\sim\Phi$ iff for it to be the case that $\sim\Phi$ just is for it to be the case that Φ .

Correia and Skiles are noncommittal on whether generalized identity is at bottom essentialist (Correia and Skiles 2019: 13). Insofar as it is, we get an alternative framework for expressing claims of objectual, generic, and sentential essence. This is important, since recent literature surrounding essence and ground is ideologically rich: it contains kinds of essence (Fine 1995a), as well as generalized identity (Correia and Skiles 2019), ground (Fine 2012a), structure (Sider 2011), building relations (Bennett 2011), degrees of being (McDaniel 2013), fundamentality (Fine 2001), etc.

1.3 Immediate and Mediate Essence

Some items have relational essences: for some $x_1, \dots, x_n$, R : it is essential to x_1 that $Rx_1 \dots x_n$, where R is a predicate variable. For example, it is essential to $\{\text{Socrates}\}$ that Socrates is a

member of $\{\text{Socrates}\}$. Moreover, it is essential to Socrates that Socrates is human. On an immediate conception of essence, it is not essential to $\{\text{Socrates}\}$ that Socrates is human.

By contrast, mediate essence is closed under the following inclusion principle: if it is essential to x_1 that $Rx_1 \dots x_n$ and it is essential to x_1 that Φ , then it is essential to x_1 that Φ . For example, it is mediately essential to $\{\text{Socrates}\}$ that Socrates is human, since it is essential to $\{\text{Socrates}\}$ that Socrates is a member of $\{\text{Socrates}\}$ and it is essential to Socrates that Socrates is human (Fine 1994b).

1.4. Individual and Collective Essence

The syntax of essence may be extended to plural names as well. On this conception, the statement ‘It is essential to conjunction and negation that they are truth-functionally complete’ becomes a statement of essence.

Collective essence is intuitive. We make plural predications all the time, some of which do not distribute over the pluralities to which the plural predicates are attributed. If our singular predications divide into essential and accidental, so too should our plural predications. Indeed, there are compelling examples of such attributions. For example, conjunction and negation are truth-functionally complete, in the sense that the set of connectives can be used to write out every truth-table. But neither conjunction nor negation alone is truth-functionally complete. Moreover, it seems to be part of what conjunction and negation are that they form a truth-functionally complete set of connectives. Collective essence is crucial to fulfilling various of the theoretical ambitions undertaken by theorists of essence. For example, it is crucial to the reduction of necessity to essence (Correia 2012; Zylstra 2018b), and it is crucial to the possibility of a reduction of ground to essence (Zylstra 2018a).

2. Reduction

There are various reductive proposals concerning essence and ground in the literature, and they run in different directions. Some take essence to be reducible to ground, while others take ground to be reducible to essence. Still others take both essence and ground to be reducible to something else.

2.1. Reduction of Essence to Ground

Some take essence to reduce to ground. On this view, for something to be essential to an item is for it to be explanatorily basic *for that item* (deRosset 2013a; Gorman 2014). More specifically, for it to be essential to x that Φ is for the proposition that Φ to have x as a constituent and for there be no Ψ such that the proposition that Ψ has x as a constituent and Ψ partially grounds Φ . For example, it is not essential to $\{\text{Socrates}\}$ that it contain Socrates or not, since $\{\text{Socrates}\}$ contains Socrates or not in virtue of containing Socrates. By contrast, it is essential to $\{\text{Socrates}\}$ that it contains Socrates, since $\{\text{Socrates}\}$ has no features in virtue of which it contains Socrates.

Zylstra (2018c) argues that the analysans of this analysis is neither necessary nor sufficient for its analysandum. In the necessity direction, it is essential to Socrates that Socrates is an animal. But Socrates is an animal in virtue of being human. Moreover, it is essential to Socrates that Socrates is human. In the sufficiency direction, suppose that Socrates is located at place P . There does not seem to be any other truths about *Socrates* in virtue of which he is located at P . But it is not essential to Socrates that he is located at P . Gorman (2014) is open to the possibility that some essential truths are contingent. But it is controversial to claim that location is among them.

2.2. Reduction of Ground to Essence

We can assume that the reductions of ground to essence are provided in terms of constitutive essence. What varies is whether they involve objectual, generic, or sentential essence and, in addition, what syntactical items we use to express them.

Correia (2013) proposes a reduction of ground in terms of sentential essence, as well as a plural version of material implication. On this view, for it to be that $\Phi_1, \Phi_2, \dots$ ground Ψ is for it to be that (i) $\Phi_1, \Phi_2, \dots$ and (ii) it is essential to what it is for it to be the case that Ψ that IF $\Phi_1, \Phi_2, \dots$ THEN Ψ , where “IF $\Phi_1, \Phi_2, \dots$ THEN Ψ ” abbreviates “It is true that Φ_1 & It is true that $\Phi_2 \dots$ only if Ψ ”. For example, suppose that it is sunny and windy in virtue of its being sunny and its being windy. Then the analysis says that it is sunny and it is windy, and it is essential to its being the case that it is sunny and it is windy that IF it is sunny, it is windy THEN it is sunny and it is windy. Plural material implication is required to avoid violating the noncircularity of grounding. Whether or not this definition successfully evades worries about noncircularity is a further question.

Correia (2013) makes clear that the proposed account is committed to the claim it is essential to its being the case that something is a philosopher that IF Socrates is a philosopher THEN something is a philosopher. So, according to the view, Socrates is involved in the constitutive essence of what it is to be the case that something is a philosopher. This creates a kind of dependence of the existential fact on Socrates, which may be regarded as problematic (Correia 2013; Fine 2012a). Correia finds the essentialist claim “quite plausible, and [is] therefore happy to hold that there is this dependency” (Correia 2013: 293).

Zylstra (2018a) proposes an analysis of ground in terms of objectual essence modeled on existential dependence. Suppose that f, g , etc. are sorted nominal variables ranging over possible facts; and suppose that F, G , etc. are sorted plural nominal variables ranging over pluralities of possible facts. Zylstra proposes the following reduction: for F to ground f is for it to be the case that for some R : it is essential to f that f exists only if there are some G such that RfG and G exist; and RfF and F exist. For illustration, consider the following example. Suppose that something is a philosopher in virtue of the fact that Socrates is a philosopher. Plausibly, it is essential to the fact that something is a philosopher that it exists only if there are some G such that G exist and witness the fact that something is a philosopher; and the fact that Socrates is a philosopher exist(s) and witness(es) the fact that something is a philosopher. On Zylstra’s reduction of ground, there is no commitment to the dependency mentioned earlier, since nothing making reference to Socrates occurs *within the scope of the essence operator indexed to the grounded fact*.

Against these reductive proposals is the view that “there are two fundamentally different types of explanation. One is of identity, or of what something is [essence]; and the other is of truth, or of how things are [ground]” (Fine 2012a: 79–80). The view is defended in Glazier (2017), where it is argued that explanatory claims such as *Socrates is human, because it is essential to Socrates that Socrates is human*, are not to be interpreted in terms of ground. For discussion and criticism of Glazier (2017), see Zylstra (2018b, 2019). For further discussion of the reduction of ground to essence, see Carnino (2014).

2.3 Reduction to Something Else

An alternative to either reducing essence to ground or ground to essence is to reduce both essence and ground to something else. In the literature, we see two proposals of this sort. Fine (2015) proposes a reduction of essence and ground to constitutively necessary and sufficient conditions. Correia and Skiles (2019) propose a reduction of essence and ground to generalized

identity. Admittedly, my use of the term ‘reduction’ is very liberal in this context, since it is a real question whether the reduction bases are neither essentialist nor ground-theoretic. The authors of the accounts do not claim otherwise. At any rate, they mark significant contributions to the literature that do not necessarily fit the headings of the previous two subsections.

2.3.1. Constitutively Necessary and Sufficient Conditions

If ‘ Φ only if Ψ ’ holds, then Ψ is a necessary condition on Φ . For short, let us say that *the fact that Ψ is materially necessary for the fact that Φ* . For example, consider:

Grass is green only if it is raining.

Suppose that it is raining. Here the fact that it is raining is materially necessary for the fact that grass is green. But this is not *strictly* necessary, since possibly, grass is green and it is not raining. By contrast, consider:

Necessarily, 2 is prime only if 2 is abstract.

Here the fact that 2 is abstract is strictly necessary for the fact that 2 is prime. However, the fact that 2 is abstract is not constitutively necessary for the fact that 2 is prime, since an account of what it is to be prime does not require appeal to the distinction between abstract and concrete entities. Now consider:

2 is prime only if 2 is a number with no nontrivial divisors.

Here there is a constitutive connection. But this does not mean that the fact that 2 is prime constitutively suffices for the fact that 2 is a number with no nontrivial divisors. On the contrary, the fact that 2 is a number with no nontrivial divisors constitutively suffices for its being prime. This is because an account of what it is to be prime involves appeal to being a number with no nontrivial divisors.

Fine’s proposal is to *reduce*, in the liberal sense, (nonfactive) ground to a generic form of constitutively sufficient conditions and to *reduce*, again in the liberal sense, essence to a generic form of constitutively necessary conditions. Then, for example, for it to be essential to Socrates that Socrates is human is for it to be that something’s being human is constitutively necessary for its being identical with Socrates. Moreover, for it to be that Φ , Ψ grounds $\Phi \& \Psi$ is for it to be that Φ and Ψ jointly constitutively suffice for $\Phi \& \Psi$. Finally, generic claims of ground, such as ‘Something is a human or a mountain in virtue of its being human’ becomes ‘Something’s being human constitutively suffices for its being human or a mountain’. On Fine’s view, generic ground is interpreted in terms of the theory arbitrary objects (Fine 2015).

2.3.2. Generalized Identity

The reduction of essence to generalized identity was discussed earlier. In this section, I focus on the reduction of ground to generalized identity. I ignore generic cases of ground.

The reduction can be stated as follows: for it to be the case that $\Phi_1, \Phi_2, \dots$ ground Ψ is for it to be the case that Φ_1 and $\Phi_2 \dots$ and Φ_1 is a disjunctive part of Ψ and Φ_2 is a disjunctive part of Ψ and $\dots$ and Ψ is not a conjunctive part of a disjunctive part of Φ_1 and not a conjunctive part of a disjunctive part of $\Phi_2 \dots$. Both the notion of *disjunctive part* and *conjunctive part* of a

disjunctive part can be defined in terms of generalized identity. Φ is a disjunctive part of Ψ if and only if there is some χ such that for it to be the case that $\Phi \vee \chi$ is for it to be the case that Ψ . Moreover, Φ is a conjunctive part of a disjunctive part of Ψ if and only if for some χ : $\Phi \& \chi$ is a disjunctive part of Ψ .

Fine (2017a) shows that there is a way to understand this analysis in terms of truthmaker semantics. The idea is that $\Phi_1, \Phi_2, \dots$ ground Ψ if and only if (i) every verifier for $\Phi_1 \& \Phi_2 \& \dots$ is a verifier for Ψ and (ii) the fusion of verifiers for Ψ properly contains the fusion of verifiers for Φ_1 and for Φ_2 , etc. This way, Ψ has some verifier that is not part of any verifier contained in anything making up the grounds. Then in cases of ground, it is impossible to conjoin what is grounded with anything the result of which is a disjunctive part of any of the grounds. For a discussion of the differences between Fine's approach and that of Correia and Skiles, see Correia and Skiles (2019: Section 4). For a specification of truthmaker semantics and its connection to ground, see Fine (2012a, 2017a). See also Fine's contribution to the present volume [Chapter 36].

3. Entailments

Even if none of the reductive proposals are successful, there still may be interesting entailments between essence and ground. In this section, I review what philosophers have taken ground to entail about essence and essence about ground.

3.1. Entailments From Essence to Ground

In addition to its definitional role, it is often thought that essence plays an explanatory role: "essence is a definable nature with explanatory power" (Williams and Charles 2013: 125). This is an important idea, since an explanatorily impotent essence is merely honorific: it confers a status on some features or truths without assigning any theoretical duties. Let any statement of the form 'It is essential to x that Φ ' be a *statement of essence* and let ' Φ ' be its *prejacent*. Moreover, let an *essentialist truth* be any true essentialist statement. Insofar as ground is explanatory, claims regarding what essentialist truths ground make up part of what we might call *the explanatory role of essence*.

Rosen (2010) claims that essentialist truths ground their prejacent, a thesis he labels *Essential Grounding*. Rosen supports Essential Grounding with two arguments. (i) Essential Grounding offers non-universal grounds for some universally quantified truths (2010: 119). (ii) Essential Grounding allows for an explanatory argument for particular grounding connections (2010: 130–131). Glazier (2017) claims, contra Rosen, that essentialist truths cannot, in general, ground their prejacent. His support for this claim is (a) that Essential Grounding implies that (e.g.) ungrounded physical facts are in fact grounded (2017: 2875); and (b) that Essential Grounding is inconsistent with some necessary conditions on grounding disjunctions and existentials (2017: 2876–2877). For example, we tend to think that some set of sentences Δ ground $\Phi \vee \Psi$ only if Δ does the grounding work of a disjunct. But if essentialist truths ground their prejacent, and the prejacent is a disjunction, then there is a disjunction that has a ground that does not do the grounding work of a disjunct.

Zylstra (2019) argues that we can use the existing truthmaker semantics for ground, extend it to constitutive essence, and use the resulting semantics, the *essence-making semantics*, to demonstrate the falsity of Essential Grounding. First, $\Phi_1, \Phi_2, \dots$ strictly ground Ψ if and only if (i) every verifier for $\Phi_1 \& \Phi_2 \& \dots$ is a verifier for Ψ and (ii) the fusion of verifiers for Ψ properly contains the fusion of verifiers for Φ_1 and for Φ_2 , etc. Second, a state s verifies an essentialist statement S within the essence-making semantics if and only if (i) s is identical with the fusion of

verifiers for S and the proposition that S , i.e., the set of verifiers for S , belongs to the essence of the item in question. The essence of an item is determined by a function in the essence-making model that takes an item to its essence: the set of propositions that make it what it is (Zylstra 2019: 9–13). The question then arises as to whether *it is essential to x that Φ strictly grounds Φ* . It is demonstrated that this cannot be the case, because condition (ii) on strict ground fails within the essence-making semantics. For a scholarly study of the explanatory role of essence in Aristotle, see Sirkel (2018).

It is common to think that ground satisfies *necessitation*: if Δ ground Φ , then it is necessary that Δ only if Φ , where Δ is the conjunction of sentences belonging to Δ . Friends of necessitation include Dasgupta (2016), deRosset (2013), Fine (2012a), and Trogdon (2013). Foes of necessitation include Leuenberger (2014) and Skiles (2015).

In response to foes of necessitation, O’Conaill (2017) argues that essence can help defend the necessitation of ground. Leuenberger (2014) argues (e.g.) that certain conceivability scenarios falsify the claim that the physical facts entail the phenomenal facts; and yet, he claims, the physical facts ground the phenomenal facts. O’Conaill (2017) argues that the grounding-essence link from Section 3.2 implies that the physical facts entail the phenomenal facts *given the grounding claim*.

Trogdon (2013) also defends the view that essence can help to secure the necessitation of ground. For Trogdon, a *connecting question* is of the form ‘Why should Φ obtain given Δ ?’. A connecting question is *cognitively insignificant* if and only if possibly, the connecting question is nonsubstantive given full knowledge of the essences of the facts involved and their constituents. Trogdon argues that if Δ grounds Φ , then (a) the connecting question is cognitively insignificant, and (b) the connecting question is cognitively significant only if $\Delta \rightarrow \Phi$ is entailed by some essential truths of the facts involved or their constituents. Given the necessity of essence, Trogdon shows that we secure the necessitation of ground.

3.2 What Ground Entails About Essence

Ground is taken to be the strongest form of metaphysical explanation or determination. The strength of this explanatory connection suggests that when (e.g.) the fact that it is sunny grounds the fact that it is sunny or windy, it is not an accident that this explanatory connection holds between the facts involved. In other words, their explanatory connection is in some way *essential*.

Fine (2012a) claims that grounding connections entail general conditions for the truth of what is grounded that are essential to what is grounded. For example, suppose that something is a philosopher in virtue of Socrates being a philosopher. Then it is essential to its being the case that something is a philosopher that, for any x , x is a witness to the existential iff something is a philosopher (Fine 2012a: fn.24).

Rosen (2010) claims that essence underwrites particular grounding connections. Suppose that it is sunny or windy in virtue of its being sunny. What explains this grounding connection? On Rosen’s view, the grounding connection is explained by the fact that every obtaining fact grounds its disjunction with any fact; that this is essential to disjunction; and that the fact that it is sunny obtains. In general, every grounding connection entails some essential connection that underwrites it.

O’Conaill (2017) proposes the following link between grounding and essence: If F fully ground f , then there are essential truths characteristic of the properties instantiated in F or in f which either include or entail that necessarily, F obtain only if f obtains. For example, suppose that the sign is colored in virtue of being red. Then plausibly, it is essential to being colored that for any x : it is necessary that the fact that x is red obtains only if the fact that x is colored obtains.

One difference between O’Conaill’s view from Fine and Rosen is that grounding connections are taken to possibly entail something essential to the *grounding facts*, or the properties involved in them, and not merely the grounded fact.

Dasgupta (2014) claims that particular grounding connections are grounded partially in facts about essence. For example, suppose that it is sunny or windy in virtue of its being sunny. Then this grounding fact obtains in virtue of the fact that it is sunny, in addition to the fact that it is essential to disjunction that, for any Φ , Φ only if Φ grounds $\Phi \vee \Psi$. Dasgupta takes this to be an improvement upon the existing approaches defended in Bennett (2011) and deRosset (2013b), according to which a grounding connection is grounded only in its constituent grounds, for example, $\Phi < (\Phi < (\Phi \vee \Psi))$.

4. Joint Applications

So far, focus has been on the connections between essence and ground. But also of interest are their applications. In this section, I focus on applications that involve both essence and ground, which we may call *joint applications*.

4.1. Expressive Power

Dasgupta (2016) claims that essentialist truths are *autonomous*, where essence is construed as constitutive and immediate. An essentialist truth is autonomous if and only if it is not apt to be grounded. Dasgupta further claims that the principle of sufficient reason can be expressed in terms of grounding and autonomy, at least given the claim that many essentialist truths are autonomous. (Exceptions include, for example, truth-functional compounds of essentialist truths, which are apt to be grounded in familiar truth-functional ways.) On this view, the principle of sufficient reason is stated as follows: every nonautonomous, or substantive, fact, i.e., every fact that is apt to be grounded, is grounded in some congeries of autonomous facts. Dasgupta also thinks these ideas can be used to characterize physicalism as follows: every substantive non-physical fact is grounded in some congeries of physical and autonomous facts. This allows one to maintain physicalism as a grounding thesis while at the same time denying the explanatory claim that physicalism is true only if all nonphysical facts obtain in virtue of some congeries of physical facts. Finally, Dasgupta takes this to generalize to other positions, such as normative naturalism. For more on the connection between ground and the principle of sufficient reason, see Fatema Amijee’s contribution to this volume [Chapter 4].

Rosen (2017) claims that we can use essence and ground to give an account of what it is for something to be a moral law. For example, suppose it is a moral law that killing animals is wrong. What is it for this to be a moral law? According to Rosen, for it to be a moral law that killing animals is wrong is for it to be (i) normatively necessary that killing animals is wrong, and (ii) it lies in the nature of being wrong that if it is normatively necessary that killing animals is wrong, then any action of killing an animal, together with the normative necessity, grounds the fact that the action is wrong.

Leary (2017) claims that essence and ground may be jointly applied to characterize a version of normative nonnaturalism that meets the supervenience challenge. The supervenience challenge is to explain how normative properties supervene on nonnormative properties given the nonnaturalist’s claim that some normative properties are *sui generis*. Leary’s view is that this challenge is met “by claiming that all the normative facts are fully grounded in the natural facts, but [the non-naturalist] explains why certain normative facts are grounded in certain natural facts by appealing to essences” (Leary 2017: 97). For more on the connection between ground, essence,

and metaethics, see Stephanie Leary's contribution to this volume [Chapter 34]. For challenges to the claims of expressive power, see Wilson (2016).

4.2. *Essence, Ground, and Real Definition*

What is it for something to be an electron? What is it for something to be a moral law? There seems to be a reading of these questions according to which the answer is asymmetric. For example, if it is the case that for something to be an electron is for it to be a particle with negative charge, spin half, and a rest mass of 9.1×10^{-31} kg then it is not the case that for something to be a particle with negative charge, spin half, and a rest mass of 9.1×10^{-31} kg is for it to be an electron. These are cases of *real definition*.

Correia (2017) proposes an analysis of real definition in terms of generalized identity and ground: to be F is defined to be G if and only if for x to be F just is for x to be G and being F is generically grounded in being G, where F and G are variable predicates. Rosen (2015) also proposes an analysis of real definition in terms of grounding: to be F is defined to be G if and only if it is essential to being F that everything is either F or G only if its being F is grounded in its being G. The upshot is that essence and ground together can give the asymmetry of real definition that essence alone may struggle to deliver.

4.3. *Metametaphysics, or Metaphysics Itself*

Although some say metaphysics is about what is fundamental (Schaffer 2009; Sider 2011), this view has been forcefully argued to be too restrictive (Barnes 2014). An alternative view is that metaphysics is about the nature and structure of reality. This divides metaphysics into *naïve* and *foundational* projects (Fine 2012a).

Naïve projects concern the essences of things: What is knowledge? What is the Higgs Boson? What is it for an action to be right? What is gender? Questions of naïve metaphysics may have answers even if the phenomena in question are not fundamental. For example, even if facts about gender are grounded in various social facts, and ultimately in physical facts (in addition, perhaps, to autonomous essentialist facts), it may nevertheless be the case that gender *is*, in the sense of essence, a system of embedded hierarchies (Haslanger 2000). On gender and essence, see also Witt (2011).

Foundational projects concern what is fundamental in reality. Grounding is indispensable here, even if fundamentality is not definable in terms of ground (Fine 2017b). For example, we may determine that physicalism is true, even if we do not yet know what the ultimate physical constituents of reality are essentially. The upshot is that both essence and ground are central to metaphysics, at least on the conception of metaphysics as concerned with the nature and structure of reality. For more on ground and fundamentality, see Ricki Bliss's contribution to this volume [Chapter 23]. For more on ground and realism, see Olla Solomyak's contribution to this volume [Chapter 26].

5. Conclusion

The aim of this chapter has been to survey the literature on the connections between essence and metaphysical grounding. This does not mean that there are no conclusions to draw from the land that has been surveyed. We can safely conclude that the literature on the connections between essence and ground is rich, diverse, and growing. This only speaks to the *importance* of

the theoretical questions *What is it?* and *Why is it?* that beat at the heart of philosophy, and that correspond to essence and ground, respectively.²

Notes

- 1 Unless I specify otherwise, all explanatory locutions, including ‘ground’, ‘in virtue of’, and ‘because’, are taken to express connections of metaphysical grounding. In addition, I interchange the terms ‘ground’ and ‘grounding’. For more on the general principles and logic of ground, see the contributions by Fabrice Correia [Chapter 15], Francesca Poggiolesi [Chapter 14], and Kevin Richardson in this volume [Chapter 13].
- 2 Thanks to Louis deRosset, Manuel Lechthaler, Tom Pashby, Mike Raven, and Riin Sirkel for helpful comments.

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23

FUNDAMENTALITY

Ricki Bliss

The world appears to contain many and varied things: minds, asteroids, peonies, flamingos, tubas. It also seems reasonable to believe that it contains facts about minds, asteroids, peonies, flamingos and tubas as well: the fact that my mind exists, the fact that peonies are scented, and so on. Amongst these contents of the world, it appears as though some things depend upon other things. Of course, everyone might agree that storms cause the swooshing of leaves on trees, and in this sense, that swooshings depend upon storms, but one might also think that there is something more to be said here: trees depend on their branches, leaves, xylem and phloem in a way that also appears to be distinctively non-causal. The fact that my mind exists depends on facts about the existence of my body that would, again, appear to be non-causal. The sense of dependence that is at issue here is commonly formulated in the language of grounding.

Eventually, so the story goes, chains of entities ordered by these grounding relations terminate in something fundamental. Reality has an over-arching hierarchical structure, then, where what supports that structure is something basic or fundamental—something below which there is nothing further. This view, or something very much like it, is the view that dominates the contemporary analytic tradition. Let us call this view metaphysical foundationalism. It is a view that we are to understand as pertaining to contingent reality. Although much can be and surely one day will be, said about dependence orderings amongst necessary existents and how those necessary existents find their place in the greater order of things, for now, the literature is very much focused on the shape of contingent reality.

Not everyone agrees on what the relata of these grounding relations are—are they facts? Are they substances? In order to remain as neutral as possible, however, and not be needlessly distracted, I shall speak largely in the language of entities. In places I switch between talk of things, facts and entities, though, on the assumption that the reader understands that I do not care to be making assertions or assumptions about how we ought to be understanding grounding. With this in mind, let us understand metaphysical foundationalism, then, as characterised by the following commitments (Bliss and Priest 2018: 2):

1. The hierarchy thesis: reality is hierarchically structured by grounding relations that are anti-symmetric, transitive, and anti-reflexive.
2. The fundamentality thesis: there is some thing(s) which is fundamental.

3. The contingency thesis: whatever is fundamental is merely contingently existent.
4. The consistency thesis: the dependence structure has consistent structural properties.

Strictly speaking, species of foundationalism require only the commitment to something fundamental in order to be considered as such. That the standard understanding of foundationalism involves the additional aforementioned commitments is important, however, because those commitments interact in significant ways with the fundamentality thesis. In addition to this, the views that are thought to be alternatives to foundationalism differ on a variety of the additional commitments. The metaphysical infinitist denies the fundamentality thesis (Tahko 2014; the metaphysical coherentist denies both the fundamentality and hierarchy theses (Morganti 2018; Thompson 2018); and the metaphysical rationalist denies the contingency thesis (Dasgupta 2016). Metaphysical foundationalism is, in fact, a package of views, and we shall, in this entry, consider it as such.

The notion of fundamentality has, in more recent times, come not only to be associated with the thought that there is something ultimate, basic or at the bottom, but also with the thought that some things are nearer or further from or equidistant with that bottom. Thus, we have at our disposal, in fact, not only a notion of absolute fundamentality but also one of relative fundamentality.

This chapter is focused largely upon absolute fundamentality. Although the notion of relative fundamentality is an important one, it is commonly thought to be captured by the asymmetry, transitivity and irreflexivity of the grounding relation to which it is intimately related. Although, as we will see, this supposition is questionable, our discussion of relative fundamentality will be relegated to a cursory and brief portion of the chapter—§1. More extensive discussions of relative fundamentality can be found in Saenz [Chapter 25], Tahko [Chapter 27] and Thompson [Chapter 17] in this volume. In §2, I move onto a discussion of absolute fundamentality proper. In §2.1, I consider the common ways in which the notion has been captured, and in §2.2, the ways in which it has been characterised. In §3, I present what the most common arguments in defence of the view are. And in §4, I offer a brief discussion of whether we need grounding in order to cash out accounts of fundamentality.

1. Relative Fundamentality

Let us suppose that a human being depends upon their parts; or that facts about the existence of human beings are grounded in facts about the existence of their parts. Let us also suppose that a human being—or facts about the existence thereof—is not the kind of thing that is fit to be absolutely fundamental. We can say, then, that a human being is less fundamental than its parts—because higher up the grounding chain—and that what's absolutely fundamental is more fundamental than the human being. These relations also serve as the converse of one another; where a human being is less fundamental than its parts, the parts are more fundamental than the human being. For good measure, we can also assume that whatever it is that is at the fundamental level is equifundamental with everything else that is down there. The relations of less fundamental than, more fundamental than and equifundamental with are all relations of relative fundamentality.

How can we best capture this notion of relative fundamentality? We can begin by noting that relations of relative fundamentality have at least three important features. The first of these is that the relations of less fundamental than and more fundamental than are directed: nothing can be both more fundamental than and less fundamental than something else relative to an ordering; and nothing can be either more fundamental than or less fundamental than itself. Second, it is as a result of chaining together instances of the more fundamental than and less fundamental

than relations respectively that the hierarchy is generated. And third, relations of relative fundamentality might be thought to have a role to play in allowing us to track the distance that non-fundamental entities find themselves relative to what is absolutely fundamental.

At first blush, it is not at all unreasonable to suppose that the relations of more fundamental than and less fundamental than can be captured in terms of the anti-symmetry, anti-reflexivity and transitivity of the relation, along with appeal to the notion of absolute fundamentality. After all, it seems that it is exactly these features of the relation, in addition to there being some ungrounded things, that deliver the kind of over-arching structural features that the relations of relative fundamentality aim at capturing. But to suppose that relative fundamentality is best captured in this way is too quick. First, the transitivity of the relations has the effect of flattening out the structure—everything is but one step from what is absolutely fundamental. Second, we appear able to understand things as more or less fundamental than other things without believing those things to stand in any relations of ground to one another whatsoever. It seems reasonable to suppose that my father is less fundamental than Jupiter but also that he bears no relation of ground to the great milky orb. Finally, not only may there not well be anything absolutely fundamental in the first place, but we seem perfectly able to understand relative fundamentality without it. In other words, we ought to be able to capture notions of relative fundamentality without relying on that of absolute fundamentality. This is not to suggest that appeal to some notion or other of grounding is hopeless but that whatever that notion is will require refinement.

In light of this strife, how are we to formulate notions of relative fundamentality? In her *Making Things Up*, Bennett offers a proposed multi-clause definition of the more fundamental than relation in terms of her notion of building. I reproduce Bennett's definition here in terms of the language of 'grounding':

(MFT) x is more fundamental_R than y =_{def}

- 1) x is fewer grounding steps away from the fundamental entity(ies) that terminate its unique chain than y is from the fundamental entity(ies) that terminate its unique chain, or
- 2) x at least partially grounds y , or
- 3) x stands in the ancestral of a grounding relation to y , or
- 4) x is absolutely fundamental and y is not, or
- 5) x belongs to some kind K and y belongs to some kind K^* such that
 - i. neither K nor K^* includes both grounded and ungrounded members, and
 - ii. y does not belong to K and x does not belong to K^* s, and
 - iii. K^* s are typically and normally grounded in K s. (Bennett 2017: 164)

This proposed definition is complicated. Presumably, the complexity of the definition is a price worth paying if it allows us to adequately capture the notion under issue. What the problems might be with this definition are beyond the scope of the present work. For now, it is enough to note that a good working account of relative fundamentality is actually harder to come by than the commitment to the hierarchy thesis would otherwise indicate. I leave discussions of relative fundamentality for elsewhere.

2. Absolute Fundamentality

Moving on from the notion of relative fundamentality, let us turn our attention to that of absolute fundamentality. Although notions of relative fundamentality are implicit in much of the

literature, explicit discussions of fundamentality are generally aimed at the notion of absolute fundamentality.

The thought that there is something fundamental is, indeed, a very old one. For Plato, for example, what is absolutely fundamental was the form of the good—with everything else, including all of the other forms, depending upon it. For the Aristotle of the *Categories*, there are primary substances upon which the secondary substances and modes depend. For the Aristotle of the *Metaphysics*, there is prime matter without which there would be nothing else. For the Early Moderns, such as Spinoza and Leibniz (Newlands 2013), God was the ultimate ground. For Heidegger, it is Being (Casati 2018). And for the contemporary analytic metaphysician, it is commonly thought to be whatever our final and best physics will tell us is the case. The notion of fundamentality is by no means the hostage of the Western tradition. For the Buddhist Abhidharmika, the dharmas are the basic units of being (Priest 2018). For the Hindu traditions, there is Brahman. And for Japanese thinkers such as Nishida, there is absolute nothingness. The list goes on.

In the following two subsections, we take a look at how it is that the notion of fundamentality is captured and characterised before turning to a discussion of reasons to believe there is anything fundamental in the first place.

2.1 Capturing Fundamentality

The aforementioned list of noteworthy examples of that which is considered fundamental is varied. It seems reasonable to ask, what, if anything, do God, Absolute Nothingness, Being and the fundamental particles of physics have in common? It turns out, in some important senses, quite a lot. In particular, the fundamental posits of most foundationalist theories are thought to play a particularly important explanatory role and that in order to play this explanatory role, they must themselves be unexplained or independent. In other words, our fundamentalia are very often thought of in terms of the notion of unexplained explainers. In its contemporary guise, the two aspects of an unexplained explainer are captured in terms of notions of independence and completeness.

First, let us consider fundamentality understood in terms of independence. According to the independence criterion of fundamentality, if something is fundamental, then it is independent, where independence is understood in terms of ungroundedness. In other words, if something is fundamental, then it is ungrounded. Note that I have refrained from defining fundamentality in terms of independence; I have avoided using the biconditional—something is fundamental if and only if it is independent. My reason for doing so is that one needs not, indeed, one ought not, commit to the right-hand side of the conditional: if something is independent, then it is fundamental. Consider necessary truths, for example. By the lights of many folks, these truths are ungrounded, and yet we do not want to say they are fundamental. The reason for this is that if grounding involves necessitation and the fundamentalia are necessary existents, then all contingency drops out of the world (Skiles Chapter 10 in this volume). At least this would seem to be the case if all our fundamentalia turn out to be necessary beings. This conclusion is not rejected by all—Spinoza, for example—but it is a bitter pill to swallow, nonetheless (see Dasgupta 2016). Or consider facts about sense datum. One might think that such facts are ungrounded without wanting to think that they are also fundamental.

It seems undeniable that fundamentality understood in terms of a very particular kind of independence—independence understood in terms of grounding—is not only the dominant but also the core conception of fundamentality. In addition to fundamentality understood in terms of independence, however, there is a second conception of fundamentality that is very

often explicitly or implicitly assumed: fundamentality in terms of completeness (see, for example, Sider 2011: 105; Schaffer 2009: 377). Informally, the completeness criterion on fundamentality aims at capturing the idea that whatever is fundamental is supposed to explain or ground everything else. More precisely, the completeness criterion is captured in terms of the notion of unique minimal completeness. A set is complete if and only if its members ground everything else. A set is minimal if and only if no subset of that set is complete. And a set is unique just in case no distinct set is complete. So a unique minimally complete set is a set such that its members ground everything else, no subset of that set grounds everything else, and no set distinct from that set grounds everything else either (see Bennett 2017: §5.3).

There is an additional, noteworthy, account of fundamentality available in the literature: fundamentality formulated in terms of naturalness (see Sider 2011: vii). According to such an account, what it is to be fundamental is to be perfectly natural. I shall remain silent on this approach to fundamentality, as it is not normally formulated in the language of grounding.

What, if any, are the connections between independence and completeness? I am inclined to think they are many, although I am not sure everyone would agree with me. Consider, first, the suggestion that independence and completeness aim at capturing the two sides of the notion of an unexplained (independence) explainer (completeness). Let us call such explainers ultimate explainers and the sense of explanation at issue ultimate explanation. Can something unexplained fail to be an ultimate explainer? Absolutely. Consider, again, facts about numbers. The fact that $2 + 2 = 4$ does not look to explain everything else. Similarly, with facts about sense-datum. Can ultimate explainers fail to be unexplained? I think not. The issues here are, in fact, thick and many. But for illustrative purposes, consider the following. If the collection of things that stands in need of explanation is the collection of grounded things, then anything that is grounded is a member of that collection. But if what is supposed to explain that collection is itself grounded (not independent), then it belongs amongst the collection of things to be explained. On pain of circularity, it would seem that our ultimate explainer(s) must be unexplained (see Bliss 2019).

The relationship between the independence and completeness criteria on fundamentality stands in need of much further exploration. And unfortunately, here is not the place for such a project. Instead, let us now turn our attention to the variety of ways in which the fundamentalia have been characterised in the literature.

3. Characterising the Fundamental

As the aforementioned list of candidate fundamentalia would indicate, that which has been considered to be fundamental has been characterised in a number of different ways. Not only does God seem like a remarkably different kind of thing to Absolute Nothingness—He is perfect whereas Absolute Nothingness is, well, nothing—He is also remarkably different to the fundamental particles of physics—amongst many things, God resides at the top of the Great Chain of Being, whereas fundamental particles reside at the bottom; God is a necessary being, whereas particles are merely contingent. What it is to be fundamental can be captured in a very thin or minimal way—if something is fundamental, then it is independent or ungrounded. Without a lot of additional metaphysical assumptions, the notion of fundamentality, although metaphysically loaded, is not particularly interesting. At least it is not interesting in the sense that a mere definition of fundamentality leaves us with important questions such as ‘what are the fundamentalia like?’ and ‘what do they do?’. In this section, we shall focus primarily on the first question. In §3, we shall have occasion to think some more about answers to the latter question.

Before moving onto consideration of particular accounts of fundamentality in the literature, it is important to begin by briefly discussing the categories from which fundamental entities can be drawn. On views according to which grounding obtains only between facts, candidate fundementalia will be fundamental facts. Although talk in terms of fundamental facts seems natural and acceptable, philosophers have pointed out that there is, at the same time, something odd about such a view. One source of consternation is that such a view seems to restrict us as regards what our broader commitments might be for the metaphysics of facts. In particular, if facts are thought to be structured entities, then even though facts might not be grounded in their constituents, it looks like our fundamental facts are not, then, metaphysical rock bottom. Instead, on such a view, we have our fundamental facts, but what turns out to be absolute rock bottom will be whatever the constituents of (at least some of) those facts are. And this brings us to what has been gestured at as a further problem: facts just don't seem to be the right kind of thing to be considered absolutely rock bottom.

On views according to which relations of ground range across categories, the kinds of entities able to be fundamental also range across categories. Fundamental entities can be facts, propositions, truths, substances, properties or relations. According to Jonathan Schaffer, for example, what is fundamental is the cosmos (Schaffer 2010). According to other views, what is fundamental are simple substances. And on other views still yet, what turns out to be fundamental are certain kinds of properties and relations. Settling any of these matters is not within the remit of this chapter. All that matters for the reader to understand at this stage is that different kinds of things have been considered candidate fundementalia.

By the lights of much of the contemporary analytic tradition, whatever it is that turns out to be fundamental will surely be posited by our final and best physics. In other words, that which is fundamental is physical. To believe something like this picture is what it is to be a physicalist. Now, there are some subtleties here that deserve immediate attention. First, in order to be a physicalist one needs not assume that there is something fundamental. Physicalism is perfectly compatible with the thought that there is nothing fundamental—for the physicalist infinitist, reality admits of ever deeper physical layers. Second, it is an open question for the physicalist whether grounding chains run downwards from wholes to parts, or upwards from parts to wholes. In other words, the physicalist must decide in which direction the priority ordering runs.

To be a physicalist is just to think that reality is first and foremost a physical kind of place. If there are such things as experiences of God, explosions of joy, intelligent thoughts, then they are all, ultimately, explainable in terms of something physical. But in terms of the over-arching order of things, it is open to the physicalist to suppose that what is of absolute ontological priority is at the bottom, so to speak, or at the top (see Bohn 2009). In other words, it is open to the physicalist to be either a priority pluralist or a priority monist. According to priority pluralism, the direction of the ontological priority ordering runs from wholes down to their parts. For the priority pluralist, what is fundamental are the ultimate physical constituents or parts of the cosmos (see Schaffer 2009). For the priority monist, on the other hand, what is fundamental is the ultimate physical whole or the entire cosmos. For the physicalist, then, some important decisions need to be made. Is there something fundamental? And if so, is it at the bottom (priority pluralism), or is it at the top (priority monism)? To the best of my knowledge, a view that we might think of as some kind of priority midlingism—according to which what is ultimately ontologically prior resides at the middle level, perhaps at the level of chairs and planets—is not one that has any purchase in the contemporary literature.

Although physicalism is surely the dominant view, there are, however, some additional and noteworthy views on the table. For substance dualists, what is fundamental will include both

physical and mental substances. On such a view, according to the book of the world, amongst the basic constituents of reality are both mental and physical entities. For the idealist, on the other hand, unlike for the physicalist, for whom reality's basic constituents are physical, and unlike for the substance dualist for whom both physical and mental substances are fundamental, what is fundamental is purely mental. Not a terribly popular view nowadays, but certainly one with an illustrious past.

So far, the views we have described are all views according to which whatever is fundamental is contingent. But perhaps nowhere have accounts of fundamentality been as well developed as in the literature on the existence of a necessary being, namely, God. God, according to the Judeo-Christian tradition, is (i) a necessary being, (ii) either absolutely independent or self-grounded and (iii) the ultimate ground of everything else. Cast in the language of the aforementioned distinction between priority pluralism and priority monism, accounts according to which God exists are accounts that look to be species of priority monism. Of course, matters here are complicated. According to many accounts of God, God creates or actualises essences or some kind of simple substances, and it is from these seeds that the rest of reality derives. By the lights of many accounts of God, then, whilst God is the ultimate explainer and ground of all else, internal to the created world, there are also simple substances from which everything else is said to spring.

Still further accounts of fundamentality can be found in the literature. According to Kit Fine, what is fundamental is that which is real. In particular, what is fundamental is reality as it is in itself—reality as it is intrinsically structured (Fine 2001: 25–26). Although Fine's more general account of the structure of the world avails itself of the notion of grounding, fundamentality is formulated in terms of the notion of the real. Michael J. Raven, on the other hand, argues for a conception of fundamentality understood in terms of ineliminability, where what it is to be ineliminable is to be such that not all facts about that thing are grounded in facts about other things. Particularly striking about Raven's novel account is that it is possible that the world contains fundamental entities without having a foundation. In other words, there may well be entities that are ineliminable (fundamental), but nothing that serves as foundational (Raven 2016).

4. Arguing for Fundamentality

Let us turn now to a consideration of the reasons we have to believe that there is, or ought to be, something fundamental. Somewhat surprisingly, remarks in the literature that one can assume are aimed at serving as justifications for the view are often presented in the form of terse statements, as opposed to arguments. To cite but one example, Jonathan Schaffer claims that 'if there is nothing fundamental, being is infinitely deferred and never achieved' (Schaffer 2010: 62). What this statement is an assumption in a version of a regress argument the conclusion of which is that there must be something fundamental (see Bliss 2019).

Let us look both more closely and more broadly at regress arguments in defense of fundamentality. Regress arguments fit a certain kind of structure. First, these arguments require the inclusion of an assumption that stipulates a goal. Very often, this goal is presented in the form of an explanatory target. We might think, for example, that there is an explanation for why things have being or where being comes from or something of the like. Although this goal needs not be expressed in the language of explanation. One might prefer to suppose that we can or must account for where being comes from, for example. There are a variety of ways in which this target can be expressed. Next, arguments of this form must include an assumption, to the effect that entities of a certain sort cannot do the kind of work that the goal is aiming at. In the argument

to hand, we are supposed to believe that anything non-fundamental cannot ultimately or really explain why things have being, for example (Bliss 2019).

From an assumption about a goal, in combination with an assumption about what cannot arrive us at that goal, we are able to derive the conclusion that there must be something fundamental. Packed into this very simple argument structure are a number of weighty and often overlooked assumptions, amongst which include the threat of looming regress.

To see where the regress rears its ugly head, let us look to the second assumption. Why can't something non-fundamental account for why things have being? Suppose we were to invoke some non-fundamental entity to explain why things have being. But as that entity is dependent, we can ask where its being comes from. The answer here might well be easy. But if what explains that thing is also dependent, then we can ask where its being comes from, and we are off to the races. We can think of this as (i) the never-ending questions regress. Alternatively, suppose we want to know why things have being. And suppose we understand that as asking a question about, say, my cup, our planet, a puppy, etc. But for any non-fundamental thing we can invoke to explain why things have being, we can ask the very same question of it—so where does this thing's being come from? This leaves us with what we can call (ii) the same questions' regress—where, crucially, we are forced to keep asking the same question because we don't appear to answer it in the first place. We might imagine a scenario in which when asked why there is anything whatsoever, the response is 'because my garden shears exist'. One can imagine responding to this in frustration and saying 'Ok, ok, but why do they exist?!' In both cases, it is because the supply of something non-fundamental as a means of reaching our goal generates a regress—and a regress that is thought to be problematic or vicious—that we are supposed to conclude that our second crucial assumption is well motivated (see Bliss 2019).

But what do fundamental and non-fundamental entities have to do with being? Good question. To understand how we are to connect up these notions, let us return to our explanatory target. The explanatory target I have suggested can be extracted from Schaffer's quip stands in need of further refinement. First, if the aim is to explain how things have being, then, strictly speaking, anything with being will fall foul of our objective—for surely its being will invite new or repetitive investigation. If what we wish to explain is how things have being, then, on pain of regress, we ought to conclude that whatever explains that must not have being. Of course, it is open for Schaffer to conclude that what explains being is something without being, and let's just call that 'fundamental', but this is not the view he is aiming for (the cosmos surely has being!). So to rephrase the explanatory target to be more faithful to Schaffer's actual position, we might conclude that what we are, in fact, seeking an explanation for is why grounded things have being, for example. But this is not the only modification to the target that needs to be made.

Consider the fact that you exist. What explains this fact? Of course, the story is going to be complex, but in order to metaphysically explain your existence, we will make appeal to facts about the existence of your parents (assuming essentialism about origins), facts about your parts, and so on. As these facts or entities also appear to be dependent, we can ask similar questions about what metaphysically explains them. And once again, we appear to be off to the races. It is important to note, though, that at each stage of our regress, we do appear to be uncovering successful explanations. I'm sure everyone would agree that your parents at least partially metaphysically explain your existence. Of course, this is not the whole story, but as part of a story, it seems to be acceptable. Recognising that our regress yields an abundance of explanations (indeed, a potentially infinite number of them) is important, for it draws our attention to what cannot be at issue here: that we haven't explained anything. Where *a* is grounded in *b*, *b* in *c*, *c* in *d* and so on *ad infinitum*, although we have an explanation for each element in our chain, what we appear not to have an explanation for is the entire chain taken together. But let's suppose,

now, that that chain is grounded in some first element of a new chain (or all the elements of that chain taken together), so on ad infinitum. Each element is grounded in the element beneath it and each chain in a further chain. Is there anything missing? It seems that there is, for what we have not yet encountered an explanation for is how the whole heaping lot of it got to be in the first place (Bliss 2013).

Understood in this light, a regress argument in defence of fundamentality takes as one of its assumptions that there is something that stands in need of explanation, and this something cannot be explained in terms of any grounded entity or collection of grounded entities. It is the job of whatever it is that is fundamental to explain why any grounded things have being whatsoever, why there are any grounded beings at all or something of the like. A reason to think there must be something fundamental, then, is not that I need explaining or you need explaining but rather that the whole lot of it does!

How, exactly, to make sense of this explanatory target is itself a tricky issue. For present purposes, however, it is enough to understand that the epistemic stability of metaphysical foundationalism would seem to require that the fundamentalia play the role of explaining not this thing here and that thing there but something much more general (see Bliss 2019).

i Arguments From Theoretical Virtue

There are several arguments from theoretical virtue in defence of fundamentality available in the literature. The first of these is offered by Ross Cameron. According to Cameron, although it is not the case that there must be something fundamental, it is contingently true of the actual world that there is something fundamental. Why? Because a theory of reality that posits the existence of something fundamental is more unified than one that does not. To quote Cameron at length,

if there is an infinitely descending chain of ontological dependence, then while everything that needs a metaphysical explanation (a grounding for its existence) has one . . . there is no collection of objects that explains the existence of every dependent *x*. This is a theoretical cost; it would be better to be able to give a common metaphysical explanation for every dependent entity. We can do that only if every dependent entity has its ultimate ontological basis in some collection of independent entities.

(Cameron 2008: 1)

There is a lot packed into these remarks. I don't wish to focus on what Cameron thinks needs explaining here but rather, upon why he thinks something fundamental needs to do the explaining. What Cameron seems to be suggesting is that the fundamentalia lend unity to a theory insofar as they allow us to explain, say, this thing here and that thing over there in terms of the same thing, namely, something fundamental.

Philosophers have pointed out, rightly in my view, that the position as stated, and the notion of unity in operation, are not entirely clear (See Paseau 2010, for example). According to priority monism, there is one fundamental entity, namely, the entire cosmos. The priority monist does, in fact, look to hold a theory according to which this thing here and that thing over there are explained in terms of the same thing, namely, the cosmos. But for the priority pluralist, for whom the fundamental level is populated by possibly infinite fundamental entities, the situation is different. Some fundamental entity *x* in Paris may explain some table in a Parisian bistro, but it would appear to have no explanatory role to play in explaining the furniture in my local diner. Alternatively, we might suppose that it is the fundamentalia taken together that explains all the

grounded x's, but this also seems strange. At best, it seems we are able to capture something of explanatory value in positing a theory that allows us to explain all the grounded x's in terms of something else, namely, something independent. But whilst such a theory might well have its virtues, it is not clear how unity is one of them. At least, it is not clear how such a theory is more unified than a theory without anything fundamental.

Two further versions of arguments from theoretical virtue in defence of fundamentality are available in the literature. Bliss (2019) presents an argument that relies not so much on the thought that foundationalist theories exhibit some virtue or other more so than non-foundationalist theories but rather that fundamental elements must do some theoretical work if we can justify positing them. Her reasons for thinking so are two-fold. First, it is reasonable to suppose that our theoretical posits do something—whatever those posits or the theories that contain them happen to be. Although the theoretical roles played by the posits of a theory may be varied, all theoretical posits have some role to play in the theory. Consider an example: Lewisian concrete possible worlds are admitted into his modal realism because they allow us to account for modality. They are not there because Lewis woke up in a particularly Dadaist frame of mind and decided to throw them in for good measure. No. They are there because they do an enormous amount of work. So too for the foundationalist. If the foundationalist cares to have their position in good epistemic standing—if they care to be able to justify their view—then a very standard way of understanding theory choice and construction would suggest that the fundamentalia must have a theoretical role to play. Second, foundationalist theories involve positing entities that are without explanation—they are independent. This is, in fact, a theoretical cost. In order to justify not only including some theoretical posit or other in our theory but including some theoretical posit that is costly, the fundamentalia need to pull their theoretical weight (see Bliss 2019).

A second and related argument, also presented in Bliss (2019) proceeds as follows. Metaphysical foundationalism posits the existence of a likely enormous number of unexplained entities. This is a theoretical cost. Contrast metaphysical foundationalism with metaphysical infinitism. According to the metaphysical infinitist, there is nothing fundamental; dependence chains ordered by relations of ground are infinitely downwardly descending. Where a is grounded in b, b in c and so on ad infinitum, it would seem to be the case that everything has an explanation, unlike with metaphysical foundationalism. On the face of it, metaphysical infinitism looks to be the more powerful theory. Unless, of course, it's not. What if where a is grounded in b, b in c and so on ad infinitum, there is something left out that stands in need of explanation? If metaphysical foundationalism allows us to explain that thing, then we have reason to believe that metaphysical foundationalism is the more powerful theory. Absent this additional something, the scales look to tip in favour of the infinitist.

Of course, considerations of theoretical virtue are downstream from other basic considerations; considerations such as whether a theory is coherent, for example. But given that many people believe foundationalism to be in otherwise good standing, arguments from theoretical virtue appear to have an important role to play in the literature.

5. Grounding and Fundamentality

Finally, it is worth pausing to address a more basic issue: must we employ the grounding framework as a means of understanding fundamentality in the first place? One reason to think that we must is if what it is to be fundamental just is to be ungrounded. Or perhaps if what it is to be fundamental just is to be ungrounded and to explain everything else. In other words, we might think we need the grounding framework in order to understand fundamentality if that notion simply has no life of its own outside of that framework.

But, of course, a few thousand years of philosophical thought tells us that this simply isn't true. The thought that there is something that is rock bottom that plays an important role in accounting for or explaining everything else would appear to be as old as the tradition itself. The notion of grounding, on the other hand, would appear to be much more modern. Indeed, it is certainly the case that many accounts of fundamentality have been formulated in language other than that of grounding; and that accounts of fundamentality can be understood without any appeal to anything that looks like grounding either.

There is also precedent in the contemporary literature for philosophers believing that we need not invoke the notion of grounding in order to make sense of that of fundamentality. Consider Sider's account in terms of naturalism, where what it is to be fundamental just is to be purely natural (Sider 2011: vii.). Both Kit Fine and Jessica Wilson also take issue with the suggestion. Wilson argues not only that we need not formulate fundamentality in terms of grounding but that we ought not. In particular, she is of the view that to say that something is fundamental is just to confess that there is nothing more to be said about that thing. To cash fundamentality out in terms of grounding, however, is to say something further about that thing, namely, that it is ungrounded (Wilson 2014: 560). Fine, in a similar spirit, says that 'what I would like to suggest . . . is that we reject the idea that the absolute notion of fundamental reality is in need of relational underpinning. The conception of reality that we are after is simply the conception of reality as it is in itself' (Fine 2001: 25).

6. Conclusion

Much about the notion of fundamentality remains to be explored. Do arguments in defense of the view appeal to the principle of sufficient reason? Do they involve contrastive questions? If so, what are they? Must everything in the uniquely minimally complete base be independent? Are formulations of fundamentality not in the language of grounding equally as fruitful? The list goes on. Wherever the literature on fundamentality heads to next, one hopes that it maintains in its sights the extremely rich and developed discussions of the notion throughout the history of various traditions.

Cross-References

Infinite Descent [Chapter 16]; Necessity [Chapter 10]; Ontology [Chapter 25]; Strict Partial Order [Chapter 17]

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24

MODALITY¹

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1. Introduction

How is grounding related to modality? This question is ambiguous, since several phenomena fit under the broad label ‘modality’. First, we could ask whether grounds necessitate what they ground (‘Grounding Necessitarianism’). Second, we could ask whether grounding is an internal relation, i.e., whether in every possible world in which some fact and its grounds obtain they automatically stand in the grounding relation. Third, we could ask how grounding is related to modal notions, in particular supervenience, which used to be assigned similar theoretical roles.

Alex Skiles’s contribution to this volume [Chapter 10] discusses the first and the second of these questions, and in the context of a broader discussion of meta-grounding, Jon Erling Litland [Chapter 9] also touches on the second; the present chapter will entirely focus on the relation between grounding and supervenience. But to fruitfully discuss this issue, we need a minimal regimentation of the target notions. I will treat grounding as a many-one relation between facts. I will use the ‘[‘, ‘]’ notation to form names of facts from sentences; so, ‘[A]’ should be read as ‘the fact that A’. I will also use small Greek letters for variables that range over facts and capital Greek letters for sets of facts (I will often represent the plurality of facts on the grounding side of the relation as a set of facts).² Unless noted otherwise, I will use ‘grounding’ for *full* and *factive* grounding. For the most part, I will bracket two further (otherwise important) questions: whether grounding is itself metaphysical explanation or instead a relation that ‘backs’ such explanations (see Martin Glazier’s entry in this volume [Chapter 8]) and whether there is only one kind of grounding or many (see Kevin Richardson’s entry [Chapter 13]).

Supervenience is necessary co-variation: speaking in general terms, the A-entities supervene on the B-entities iff the A-entities cannot vary without some variation in the B-entities. We can distinguish various types of supervenience depending on the ontological category of the relata, the modal force with which they co-vary, and the scope of the supervenience thesis. To make the discussion manageable, I will narrow down my target notion in each of these three respects. First, although the specialized literature usually focuses on property supervenience (see McLaughlin and Bennett 2018), I will primarily focus on supervenience between facts. Since I already regimented grounding as a relation between facts, this will make it easier to detect interesting connections between grounding and supervenience. By default, I will take the

supervenience base to be a set of facts (an assumption I will revisit in Section 3). Second, since our topic is metaphysical grounding, I will require the modal force of covariation to be at least as strong as metaphysical necessity. Third, I will focus on supervenience theses that assert patterns of co-variation between entire possible worlds ('global supervenience') rather than co-variation within a possible world ('local supervenience').³ Since our main focus is on the modal profile of grounding rather than its uniformity within a possible world, this too is a natural choice. With this minimal regimentation under our belt, I will discuss three questions: (1) can either grounding or supervenience be defined in terms of the other; (2) does either entail the other; (3) do their theoretical roles overlap to some extent?

2. Definition

Few explicitly offered supervenience-based analyses of grounding. Armstrong often treated supervenience as a grounding-like relation (see Armstrong 1997), but he arguably relied on a nonstandard notion of supervenience (see Bricker 2006: 267–68). In recent work, however, Kris McDaniel offers an actual analysis of grounding partly in terms of supervenience (I slightly rephrased his formulation to match our chosen regimentation):

(*McDaniel's Thesis*): $[a \text{ is } F] \text{ grounds } [b \text{ is } G] \text{ iff } G \text{ supervenes on } F \text{ and } F \text{ is more natural than } G$ (cf. McDaniel 2013: 12)

A similar criterion has been proposed by Philipp Bricker, albeit as a merely sufficient condition (here, too, I slightly revised his formulation):

(*Bricker's Thesis*): If a set of facts Γ are fundamental and a fact ϕ supervenes on Γ , then the facts in Γ ground ϕ (cf. Bricker 2006: 272)

The main difference between the two theses (beside Bricker's proposing only a sufficient condition of grounding) is that in the analysans clause McDaniel uses the relative notion of *being more natural than*, whereas Bricker uses the word 'fundamental' for Lewis's (1983) notion of a *perfect naturalness* (originally devised for properties but often extended to facts).

These differences won't matter much here; the two proposals face similar problems. Take the fact [2 is an even number] and some arbitrary fundamental microphysical fact, say, [electron *e* has negative charge]. Intuitively, neither [2 is an even number] nor the property of being an even number is fundamental. On the assumption that the facts of pure mathematics are necessary, [2 is an even number] trivially supervenes on [electron *e* has negative charge] (no two worlds can differ with respect to the former without differing with respect to the latter, since they cannot differ with respect to the former *period*). For similar reasons, the *property* of being an even number also trivially supervenes on the property of having negative charge (as McDaniel's definition requires). Yet it seems implausible that [2 is an even number] is grounded in [e has negative charge].

This example spells trouble for both McDaniel's and Bricker's Thesis. It also exemplifies an important difference between supervenience and grounding, which will come up in the discussion to follow from time to time: grounding is a hyperintensional relation at least in the sense that (unlike supervenience) it doesn't automatically hold between pairwise necessarily coextensional relata (cf. McLaughlin 1995). Now, Bricker's and McDaniel's Theses also appeal to naturalness, and we cannot in advance rule out a reductive definition of grounding in terms of supervenience and some other (hyperintensional) notion. But it's not clear how such a

definition would go, and most contributors to the grounding literature aren't optimistic about the prospects of finding one.

Could we instead define supervenience in terms of grounding? Since supervenience is defined in modal terms, this question boils down to whether we can dispense with primitive modality in favor of grounding-theoretic notions. While many philosophers are sympathetic to the idea that grounding facts are prior to modal facts, few endeavored to offer a straightforward grounding-based *analysis* of modality. What comes closest is a recent proposal by Boris Kment, who offers the following definition of metaphysical necessity:

(*Law Analysis*) A proposition is metaphysically necessary iff it is true throughout the sphere around actuality that contains the worlds that match actuality with respect to the metaphysical laws (2014: 188)

Kment uses the word 'world' neutrally for possible *as well as impossible* worlds. A "sphere around actuality" is a class of worlds each member of which is closer to the actual world than any world not in the class; Kment thinks that the class of worlds that matches ours with respect to the metaphysical laws forms just such a sphere. Third, Kment doesn't define 'metaphysical law'; he accepts the notion as a primitive. If this analysis is correct, we could use it to analyze supervenience by replacing the modal idiom in it with its analysans (see Tobias Wilsch's entry to this volume [Chapter 30] for more on grounding and metaphysical laws).

One might try to turn the Law Analysis into a directly grounding-based one by understanding metaphysical lawhood in terms of grounding. A relatively straightforward way of doing so has been offered by Glazier (2015: 23–26), who understands law statements as generic grounding statements that feature a primitive variable-binding operator. Kment himself spells out the connection between grounding and metaphysical laws differently and maintains that grounds, in conjunction with the metaphysical laws, metaphysically explain what they ground. So there are two kinds of metaphysical explanantia, grounds and metaphysical laws, just like there are two kinds of scientific explanantia, causes and natural laws (Kment 2014: 164).

This means that if we accept both grounding and metaphysical explanation as primitives, we can use them to define metaphysical laws. To be clear, Kment himself offers no such definition; the forthcoming analyses are inspired by Kment's work, but he would likely reject them. Our task would be easier on the (controversial) assumption that no world contains uninstantiated laws: then we could say that in any world w , L is a metaphysical law iff for a set of facts Γ and a fact ψ (each distinct from L), the members of Γ fully ground ψ , L and the members of Γ together fully metaphysically explain ψ , and L is not even a partial ground of ψ . Things are more complicated if we allow for uninstantiated laws (as Kment does); in that case, the following revision may be proposed: for any world w , L is a metaphysical law iff either (i) for some set of facts Γ and some fact ψ (each distinct from L), the members of Γ fully ground ψ , L and the members of Γ together fully metaphysically explain ψ , and L is not even a partial ground of ψ , (ii) or if L is uninstantiated, then for some set of facts Γ and some fact ψ (each distinct from L), if L and the members of Γ together fully metaphysically explained ψ , then the members of Γ would fully ground ψ and L would not even be a partial ground of ψ .

If one of these analyses is on the right track, we can get a definition of metaphysical supervenience ultimately in terms of worlds, grounding, metaphysical explanation, and (if we allow uninstantiated metaphysical laws) counterfactuals. The resulting definition would be a highly complicated one with several subordinate clauses, and I won't attempt to spell it out here. However, one might also wonder whether such an involved analysis would be worth the benefits in the first place. One commonly acknowledged advantage of supervenience is its relative

neutrality; philosophers with very different views on the intelligibility and utility of concepts like grounding, metaphysical laws, and the like can nonetheless agree on various supervenience theses (cf. Kathrin Koslicki's chapter on skeptical doubts about grounding [Chapter 11] and Louis deRosset's about responses to them [Chapter 12]). By accepting the Kment-inspired analysis, we risk tying the reasonably uncontroversial tool of supervenience to the much more contentious ideology of grounding and metaphysical explanation. More cautiously: even if the good standing of supervenience as a philosophical notion doesn't depend on the standing of the grounding-theoretic notions used to analyze it, one might still worry that specific (and relatively uncontroversial) supervenience theses shouldn't hang on the truth of more controversial claims about grounding and metaphysical explanation. Similar remarks apply to Glazier's view.

I leave it to the reader to weigh these concerns against the advantages of having a nonmodal analysis of supervenience. There may also be grounding-theoretic analyses other than the two mentioned. However, I won't dwell on exploring these here; instead, I will move on to a more widely discussed link between grounding and supervenience, that of metaphysical entailment.

3. Entailment

Does either of grounding and supervenience metaphysically entail the other? It doesn't seem that any kind of supervenience is by itself sufficient for grounding. The cases that spelled trouble for Bricker's and McDaniel's "naturalness-enhanced" sufficient conditions for grounding *a fortiori* frustrate any attempt to derive grounding from supervenience alone.

It has been much more common to assume that at least grounding entails supervenience, i.e., that no two possible worlds can differ with respect to a grounded fact unless they also differ with respect to its grounds. Indeed, grounding is often motivated by remarks to the effect that supervenience is 'too weak' for certain purposes or that grounding does everything supervenience does *and some more*, which implies that supervenience is a necessary (even if not sufficient) condition of grounding (see, e.g., Fine 2012: 41; Kment 2014: 14; and Schaffer 2016: 61 ff10). Yet most attempts to formulate a precise link between grounding and supervenience face difficulties. To see why, start with the following thesis:

Simple Supervenience (SimSup): If the facts in Γ ground ϕ , then any two possible worlds that are indiscernible with respect to Γ are also indiscernible with respect to ϕ

As Leuenberger (2014a) points out, widely accepted cases of grounding violate SimSup. Grounding is normally thought to allow for multiple realization, which in turn gives rise to counterexamples to SimSup. Suppose (as standardly assumed) that disjunctions are fully grounded in their true disjuncts (see Michaela McSweeney's chapter on the relation between grounding and the truth-functional connectives [Chapter 32]). Take a contingent disjunctive fact $[AvB]$; suppose it's grounded by $[A]$, and that A and B are independent from each other. Since both disjuncts are contingent, there is a possible world w_1 in which A is false but B is true, and another one, w_2 , in which both are false. These two worlds falsify the supervenience of $[AvB]$ on $[A]$: w_1 and w_2 are indiscernible with respect to $[A]$ but not with respect to $[AvB]$, yet $[A]$ grounds $[AvB]$. Similar counterexamples can be constructed using the general recipe of finding a fact with multiply realizable grounds and two possible worlds that differ with respect to the grounded fact but not with respect to its actual-world grounds (absent from both worlds but replaced with an alternative ground in only one of them). For example, facts about determinable properties are often thought to be grounded by facts about their determinates (Schnieder 2006: 32–33; Rosen 2010: 126; Audi 2012: 686, 689; Schaffer 2012: 126–27, 2016: 54). If so, then the fact that the

beer bottle is green is grounded in the fact that it has a particular shade of green, say, green₁₄. But there is a pair of possible worlds, one where the bottle is green₂₁ and another where it's brown. These worlds differ with respect to whether the beer bottle is green but not with respect to whether it has shade green₁₄. Yet in the actual world, [the bottle is green₁₄] plausibly grounds [the bottle is green].

One reaction to these counterexamples is to reject them. A particularly interesting way of doing so has been proposed by Guigon (2018) against the backdrop of a general counterpart-theoretic account of grounding. Most of its details need not concern us here; what matters is that according to Guigon, facts are world-bound and modal truths about their grounding status are properly analyzed as truths about their *counterparts*. Importantly, Guigon also borrows Lewis's (2003) "qua" locution as a context-fixing device and maintains that if some fact ϕ is grounded by some set of facts, Γ , then any possible grounds of ϕ are "truths *qua* grounds": they are ways the facts in Γ could be. That is, if a set of facts Γ ground ϕ , then by taking any possible world in which some set of facts, Δ , ground a counterpart of ϕ , we thereby fix a context in which the facts in Δ are counterparts of the facts in Γ . For instance, while there is a possible world w in which [The bottle is green] is grounded by [the bottle is green₂₁], this latter fact is a counterpart of the actual fact [the bottle is green₁₄]. Moreover, all possible grounds of [The bottle is green] are ways its actual grounds could be. Thus, we can reconcile multiple realizability with the supervenience of the grounded on its grounds: in some worlds, facts are grounded differently, yet they are grounded by the same facts *qua* grounds as in the actual world.

A natural way to proceed for those who are not prepared to adopt Guigon's counterpart-theoretic approach is to weaken SimSup. This is Leuenberger's own strategy. As he notes, one common feature of counterexamples like those mentioned is that they involve a pair of possible worlds, neither of which contains the actual-world grounding fact. It's natural to react to such scenarios by restricting the scope of our supervenience thesis to pairs of worlds one member of which is the actual world. Leuenberger calls this "actuality-sensitive supervenience": fact ϕ actuality-sensitively supervenes on a set of facts, Γ , iff any world that differs from the actual world with respect to ϕ also differs from it with respect to Γ . The corresponding grounding thesis would then go as follows:

Actuality-sensitive Supervenience (AcSup): If the facts in Γ ground ϕ , then any possible world that is indiscernible from the actual world @ with respect to Γ is also indiscernible from @ with respect to ϕ

As Leuenberger points out, supervenience *per se* plays no distinctive role in AcSup, since in the case of any actually obtaining facts, AcSup turns out to be equivalent to Grounding Necessitarianism. For the supervenience-to-entailment direction, suppose that some fact, [A], actuality-sensitively supervenes on some set of facts Γ . Take a world w in which the facts in Γ obtain. Since [A] actuality-sensitively supervenes on Γ , it also obtains in w . So, Γ entails [A]. For the entailment-to-supervenience direction, suppose that all members of Γ obtain and Γ entails [A]. Take a world w in which all members of Γ obtain. Since Γ entails [A], [A] also obtains in w . But then if w and @ are indiscernible with respect to Γ , they are also indiscernible with respect to [A], and so [A] actuality-sensitively supervenes on Γ . Reading these two results together, we get the desired biconditional that grounding satisfies AcSup if and only if Grounding Necessitarianism is true.

Since AcSup is equivalent to Grounding Necessitarianism, I won't discuss it here; presumably, the arguments for and against them will be the same, and Grounding Necessitarianism is already discussed in Alex Skiles's separate chapter [Chapter 10]. Instead, I will consider another

fix suggested by Leuenberger. We originally considered supervenience as a relation between a set of facts and a particular fact. But in informal contexts, supervenience is often treated as a relation between *types* of facts, where types aren't simply individuated by their members (formally, they can be represented by functions from worlds to a set of facts). For example, we often say that mental facts supervene on physical ones or that general facts supervene on particular facts. But these statements are not plausibly interpreted as talking about the relation between the sets of those mental facts and those physical facts, or those general and particular facts, which happen to obtain in the actual world. Following Leuenberger, let 'TΓ' stand for the maximal type to which the members of Γ belong; that is, the type to which each member of Γ belongs and to which no further fact in @ belongs. Then we can formulate the following thesis:

Type Supervenience Thesis (TypSup): If the facts in Γ ground ϕ , then any two possible worlds that are indiscernible with respect to TΓ are also indiscernible with respect to ϕ

Our earlier counterexamples seem ineffective against TypSup. Take again the beer bottle example. As we have seen, the bottle's being green is grounded in but doesn't supervene on its being green₁₄. But the bottle's being green does appear to supervene on its determinate color properties (where "determinate color properties" count as a type). After all, although the pair of cases in our counterexample don't differ with respect to being green₁₄ (neither is of that shade), they do differ with respect to their determinate color properties (one of them is green₂₁ while the other isn't). Likewise for the disjunction case: if A is true in @ but false in w_1 and w_2 whereas B is true in w_1 but false in w_2 , then even though w_1 and w_2 don't differ with respect to [B], they still differ with respect to "the disjuncts of [AvB]".

Leuenberger notes two problems with TypSup, which he calls the *reference type problem* and the *problem of heterogeneous realizers*. Let's start with the reference type problem. The problem is simply that the notion of a type at issue is not entirely clear, since any set of facts can plausibly be associated with multiple types at the same time. This means that type supervenience theses are not well defined. One potential remedy Leuenberger suggests is to differentiate between natural and nonnatural fact types, analogously to Lewis's (1983) familiar distinction between natural and nonnatural properties. Naturalness is usually taken to be unanalyzable, but natural properties have a number of distinctive features. What's most important in the present context is that they make for similarity among their instances. The generalization of naturalness-theoretic ideology beyond properties is not unprecedented (see especially Sider 2011; also recall our earlier discussion of Bricker 2006), and Leuenberger takes it to be an important mark of natural fact types that they comprise objectively resembling facts. We can try to solve the reference type problem by narrowing down initially ambiguous type supervenience theses to ones phrased in terms of natural fact types (and perhaps in terms of the most natural fact types the respective facts belong to) that satisfy the following two further constraints: (i) the actual members of a type TΓ are all and only the members of Γ, and (ii) a fact's membership in a type is essential: if ϕ obtains in two possible worlds, w_1 and w_2 , then ϕ is $\in$ TΓ at w_1 iff $\phi \in$ TΓ at w_2 . To see how the introduction of naturalness could help us select the right type, take the following example. The type that involves the mass facts in the actual world and the mass facts in other possible worlds is more natural than the fact type that involves the mass facts in the actual world and facts about charge in some other possible world. Hence, 'T_{the mass facts in @}' plausibly refers to a type of fact to which other (nonactual) mass facts also belong.

In a recent paper, Samuele Chilovi (forthcoming) raises a number of objections to Leuenberger's treatment of the reference type problem. He begins by objecting to Leuenberger's first constraint that the actual members of a type TΓ are all *and only* the members of Γ. Facts of a

certain type T, Chilovi argues, might ground another fact without being the only facts of type T. For example, not all the physical facts are relevant to grounding some particular mental fact, yet it's intuitive to say that the mental facts are grounded in physical facts. This problem can be fixed by simply relaxing the 'and only' proviso and allowing for nonmaximal fact types in supervenience theses. A more serious worry, according to Chilovi, is that the naturalness constraint won't always deliver the intuitively appropriate fact type. Suppose we want to formulate in terms of fact types the idea that the general supervenes on the particular. Take a world, w , in which there are only two individuals, a and b , each instantiating some very natural chemical property F . Now, $[Fa]$ and $[Fb]$ belong to at least two different fact types: they are both *particular facts* as well as *chemical facts*. According to the criterion of similarity-making (which Leuenberger heavily relies on), the type *chemical fact* is more natural than the type *particular fact*, since on the whole, chemical facts are less miscellaneous than particular facts. Yet when we said that the general facts supervened on the particular facts, we didn't use the phrase 'particular facts' to pick out the type *chemical fact*; moreover, understood that way, the supervenience claim isn't even plausible (there may well be worlds chemically indistinguishable from ours that nonetheless differ with respect to the general facts obtaining in them). Chilovi concludes that there's no mechanical, context-invariant method of pinning down the type in terms of which a supervenience claim is best formulated; instead, we should settle for an intuitive understanding of TypSup and let contextual factors determine what the relevant type at issue is.

Let's move on to the problem of heterogeneous realizers. Here, the problem is that some facts could have been grounded by facts numerically distinct from the actual grounds that don't even belong to the same type. For example, perhaps the mental facts are grounded by physical facts in the actual world but alien "ectoplasmatic" facts in some other possible worlds. Take two such worlds, w_1 and w_2 ; suppose Jim is in pain in w_1 but not in w_2 . Since both w_1 and w_2 are ectoplasmatic, there doesn't have to be any physical difference between them. So it seems that the mental facts could be grounded in the physical facts without supervening even on physical types of facts.

To deal with the problem of heterogeneous realizers, Leuenberger proposes a weakened supervenience thesis that merely asserts necessary covariation between a fact's actual-world and merely possible types of grounds:

Actuality-sensitive Type Supervenience Thesis (AT-Sup): If the facts in Γ ground ϕ , then any possible world that is indiscernible from @ with respect to $T\Gamma$ is also indiscernible from @ with respect to ϕ

Unlike AcSup, AT-Sup requires only that if a world w differs from ours with respect to an actually obtaining grounded fact, it also differs with respect to the type of facts that ground it. That is, even if $[B]$ is a fact in @ but not in w , its actual-world grounds Γ might hold in w so long as w differs from @ in some of the *other* $T\Gamma$ -type facts. AT-Sup is also strictly weaker than Grounding Necessitarianism (this can be most easily seen by observing that it is entailed by but doesn't entail AcSup, which as we have seen is equivalent with Grounding Necessitarianism) and deserves to be discussed in its own right.

While AT-Sup is immune to the problem of heterogeneous realizers, it might falter on other puzzle cases. Leuenberger mentions one of these, the *problem of heterogeneous blockers*. Put in intuitive terms, blockers are non-actual facts that, when added to the full grounds of an actually obtaining fact ϕ , prevent ϕ from obtaining in another possible world. Blockers already pose a challenge to AcSup (which was formulated in terms of sets of facts and not discussed in detail). But *heterogeneous blockers* also threaten AT-Sup, since they appeal to alien types of extra facts doing the preventing work. Suppose, for instance, that the mental facts are grounded in physical

facts. Leuenberger asks us to imagine a possible world in which the same physical facts obtain, but there is also some alien substance (“chromoplasm”) that prevents our physical duplicates from undergoing pain. If such a scenario is possible, then grounding doesn’t even guarantee the supervenience of the grounded on the type of fact to which the actual grounds belong.

Now, as Leuenberger notes, the possibility of heterogeneous blockers is contentious. More cautiously, it doesn’t seem especially dogmatic to insist that if chromoplasm-style scenarios are possible, they are simply counterexamples to the original grounding theses rather than to the relevant instance of AT-Sup. This is exactly how Chilovi responds, who maintains that if chromoplasm really is possible, then the mental facts aren’t, after all, grounded in the physical facts. In place of the problem of heterogeneous blockers, Chilovi raises what he considers a more serious problem: the *problem of heterogeneous grounding bases*, which threatens both TypSup and AT-Sup as formulated earlier. The worry, in a nutshell, is that even in the actual world, we have no compelling reason to think that for every fact there is exactly one type to which all of its grounds belong. For example, facts about cities may be grounded in facts about geography and sociology, and legal facts may be grounded in social as well as moral facts. To overcome this problem, Chilovi proposes a revision of AT-Sup in which the supervenience base is a plurality of types:

Actuality-sensitive Plural Type Supervenience Thesis (PluSup): If the facts in Γ ground ϕ , then any possible world that is indiscernible from @ with respect to $\text{TF}_1 \dots \text{TF}_n$ is also indiscernible from @ with respect to ϕ (where two worlds, w_1 and w_2 , are indiscernible with respect to $\text{TF}_1 \dots \text{TF}_n$ iff TF_1 at $w_1 = \text{TF}_1$ at w_2 , TF_2 at $w_1 = \text{TF}_2$ at w_2 , ..., TF_n at $w_1 = \text{TF}_n$ at w_2)

This revised formulation is promising and seems resistant to many of the problems that beset simpler versions. Of course, questions still remain as to whether we can get a more rigorous characterization of types than Chilovi’s context-sensitive proposal and about whether PluSup could be generalized beside the actual world. Moreover, the jury is still out on how we should evaluate scenarios involving heterogeneous blockers. In comparison to Grounding Necessitarianism, the grounding-supervenience link is a relatively under-researched topic, possibly at least in part due to technical difficulties about proper formulation that don’t arise with Grounding Necessitarianism.

4. Grounding, Supervenience, and Their Theoretical Roles

In contemporary discussions, grounding is usually invoked to play theoretical roles that were often assigned to supervenience before the recent surge of interest in grounding. This being said, at least Leuenberger is skeptical about this overlap: if the counterexamples to the various proposed grounding-supervenience links are genuine, he says, then we should see grounding and supervenience as complementary tools rather than rival characterizations of the same pre-theoretical notion (2014a: 239). However, even if grounding and supervenience are not rival characterizations of the same notion, it’s worth asking whether their theoretical roles at least partially overlap. Two potential (and in the case of grounding, fairly uncontroversial) roles come to mind: a ‘fundamentality-tracking’ and an explanatory role. Can supervenience play either of these roles?

Let’s start with fundamentality. Grounding is commonly used to define absolute fundamentality: a fact is (absolutely) fundamental iff nothing grounds it. Moreover, there is often thought to be a tight conceptual link between grounding and relative fundamentality: if a fact, ϕ , partially grounds another fact, ψ , then ϕ is more fundamental than ψ . This is merely a sufficient not a necessary condition of relative fundamentality, since there may be facts that stand in the *more*

fundamental than relation without being ground-related at all. (See Bennett 2017: Chs. 5 and 6, respectively, for detailed discussion of these links. For more on the relation between grounding and fundamentality, see Ricki Bliss's contribution to this volume [Chapter 23].)

These conceptual links between grounding and fundamentality are useful, but they have a limitation: the grounding-theoretic definition gives us no guidance as to how we should evaluate the fundamentality status of things other than facts, for example properties. One way to go would be to assume that facts are structured and try to read off the fundamentality status of properties from that of the facts they are constituents of. For example, one could say that a property is fundamental just in case it's a constituent of at least one fundamental fact. However, this proposal faces problems. For one, it appears to rule out fundamental properties that are uninstantiated and thus don't feature in any facts. For another, it implies (perhaps objectionably) that if a property is a constituent of a fundamental fact only in some nonactual world, then that property is merely contingently nonfundamental. And even aside from these issues, one might want a handle on property fundamentality that is independent of the metaphysics of facts. (See also Tuomas Tahko's entry [Chapter 27] on structure for a discussion of related issues).

Supervenience might come in handy here. When characterizing perfectly natural properties (which many take to be the fundamental properties), one of the theoretical roles Lewis (1983) attaches to them is that they constitute a complete supervenience base for all other properties. That is, the pattern of instantiation of all properties supervenes on the pattern of instantiation of the perfectly natural (fundamental) properties. So if a property F is fundamental, then it's the member of a set of properties $\{F_1 \dots F_n\}$ such that all facts supervene on the distribution of $F_1 \dots F_n$. Obviously, this is not a sufficient condition of perfect naturalness, since the distribution of various sets of gerrymandered, nonnatural properties may also provide a complete supervenience base (cf. Sider 1996; Schaffer 2004; and Dorr and Hawthorne 2013); still, this partial characterization at least gives us some means of figuring out whether a certain set of properties is fundamental.

Needless to say, there is much more to be said about the relation between fundamentality and supervenience. But we can cautiously say at least that even if the grounding-based definition of fundamentality is correct, supervenience can serve as a useful extra tool in this area. By bearing different links to fundamentality that can't simply be read off each other, grounding and supervenience might usefully complement each other in giving us a fuller understanding of this notion.

What about explanation? The contemporary consensus is that supervenience is not itself an explanatory relation; indeed, one commonly cited justification for introducing grounding is that it can play an explanatory role supervenience is not suitable for. However, the exact reason for this commonly held view is not entirely clear, and the orthodoxy has lately been criticized in a recent paper by David Kovacs (2019). There is no doubt that part of the difficulty stems from the unclarity of the expressions 'explanatory relation' and 'backing'. In what follows, I will consider a few commonly used rationales for the supposed explanatory impotence of supervenience and mention some reasons for not taking them to be conclusive.

One popular complaint against supervenience is that it has the wrong formal properties to be an explanatory relation: explanation is irreflexive, asymmetric, and nonmonotonic, whereas supervenience is reflexive, nonsymmetric, and monotonic (cf. Schaffer 2009: 364; McLaughlin and Bennett 2018: §3, §5; and Raven 2012: 690). That is, everything supervenes on itself, sometimes two distinct facts or sets of facts supervene on each other, and for any fact ϕ and its supervenience base Γ , the base amended by some arbitrary extra fact ψ still serves as a supervenience base for ϕ . However, it is unclear how much of a wedge these differences put

between grounding and supervenience. First, the irreflexivity and asymmetry of grounding are not entirely uncontroversial; some authors take seriously the possibility of self-grounding and mutual grounding, yet they take grounding to be an explanatory relation. (See also Francesca Poggiolesi's [Chapter 14], Stephan Krämer's [Chapter 18], and Naomi Thompson's [Chapter 17] contributions to this volume for more on the formal features of grounding.) Second, as Berker (2018) showed, if the main issue with supervenience lied in its formal properties, we could easily fix this problem by adding those properties by fiat. That is, instead of supervenience, we could focus on "proper supervenience" (supervenience that is asymmetric and irreflexive) and insist that *that* relation is explanatory (though see Raven 2012: 690 for an objection to this move.) Third, as Kovacs notes, we can use a similar trick to ensure nonmonotonicity. So in the end, we can simply direct our attention to proper supervenience, the relation that holds between ϕ and Γ iff (i) ϕ supervenes Γ , (ii) Γ doesn't supervene ϕ , and (iii) there is no proper subset of Γ on which ϕ supervenes.

Another popular argument against the explanatoriness of supervenience is that it's an intensional relation: if ϕ supervenes on Γ , then any fact that holds in the same possible worlds as ϕ supervenes on any set of facts that hold in the same possible worlds as Γ . This is not thought to be true of explanation, which may vary between pairwise cointensional relata. However, Kovacs (2019) argues that this cannot be the right diagnosis of the supposed unexplanatoriness of supervenience, since proper supervenience (as defined earlier) is non-monotonic and therefore cannot be intensional: if ϕ properly supervenes on Γ , it doesn't properly supervene on any superset of Γ , including one that is co-intensional with Γ .

A third objection is that supervenience doesn't *guarantee* explanation: it's obviously not the case that if ϕ supervenes on Γ then the facts in Γ explain ϕ . Grounding, on the other hand, is always explanatory: if the facts in Γ ground ϕ then they also explain it (cf. Audi 2012: 687–88; Schaffer 2009: 364–65, 2016: f52; Rosen 2010: 110–14; and Fine 2012: 41).

This is a more promising justification of the supposed unexplanatoriness of supervenience, but it still raises questions. For one, even if grounding is always explanatory, this is not clearly true of all paradigmatic explanatory relations. For example, there is a good case to be made that causes don't always explain their effects; in a deterministic world, the Big Bang is causally sufficient for John's being late for school on a certain day but doesn't explain it. For another, if grounding is indeed a relation that always by itself suffices for explanation, one may wonder why. It would be bad news if it turned out that our very concept of grounding is simply "that-which-ensures-metaphysical-explanation"; for just like the natural, worldly relation of causation is unlikely to answer all our intuitions concerning puzzle cases about high-level causation, so is a natural relation of grounding (assuming there is one) unlikely to exactly line up with our intuitions about metaphysical explanation. So whether the mere failure of supervenience to guarantee explanation is by itself sufficient to distinguish it from properly explanatory relations is at the very least an open question (Kovacs 2017: 2936–39; 2019: 1978–80).

Finally, one could take inspiration from Kim (1988/1993: 167), who famously claimed that supervenience was a "surface relation": in every case when supervenience tracks explanation, there is a deeper underlying relation ultimately responsible for whatever explanatory "oomph" we might have wanted to attribute to supervenience (cf. Horgan 1993; Kim 1998: 84ff; and Wilson 1999). But as before, the tricky question is whether we cannot say a similar thing about paradigmatic explanatory relations, including perhaps grounding itself. An increasingly influential strand of theorizing ties grounding to essence, and according to one version of this view, grounding obtains in virtue of links of essential connectedness between the grounding relata (or constituents thereof).⁴ If a view along these lines is true, then the relation ultimately responsible

for whatever explanatory connection is tracked by grounding is essential connectedness (cf. Glazier 2017). But grounding essentialists wouldn't on that account conclude that grounding is a "surface relation" in the objectionable sense supervenience is.

To be sure, this is hardly the last word on the issue, and there *is* something intuitive about the popular slogan that supervenience is not an explanatory relation. So Kovacs's challenge could also be read as an invitation to get more precise about what the slogan means and in what sense we should take it to be true. Exactly how grounding and supervenience are related to fundamentality and explanation is a vexed question, and we cannot rule out some overlap between the two relations' theoretical roles. Yet this isn't necessarily a bad thing. Even if supervenience is linked to fundamentality and explanation more closely than usually assumed these days, it's clearly linked very differently from the way grounding is. If so, then grounding and supervenience can usefully complement each other in helping us gain a better understanding of these adjacent notions.

Related Topics

For the modal consequences of grounding see Alexander Skiles's chapter on necessity and inter-nality [Chapter 10]; for the latter, see also Jon Erling Litland's chapter on meta-grounding [Chapter 9]. For the role of laws in grounding explanation, see Tobias Wilsch's chapter on laws [Chapter 30]. In connection to the extent to which grounding and supervenience can play explanatory and fundamentality-tracking roles, see Martin Glazier's chapter on explanation [Chapter 8], Ricki Bliss's chapter on fundamentality [Chapter 23], and Tuomas Tahko's chapter on structure [Chapter 27].

Notes

- 1 For many helpful comments and criticisms, I thank Samuele Chilovi, Kit Fine, Mike Raven, Alex Skiles, and participants of the 2018 Hamburg Workshop on Metaphysical Grounding.
- 2 The "relations between facts" view of grounding has been adopted by Rosen 2010, Audi 2012, Raven 2012, Skiles 2015, and many others. Contrast this with Fine 2001, 2012, Correia 2014, Dasgupta 2014, and Litland 2017, who express grounding with a sentential connective instead.
- 3 See Horgan 1982, Petrie 1987, Paull and Sider 1992, McLaughlin 1995, Stalnaker 1996, Sider 1999, Shagrir 2002, Bennett 2004, and Hoffman and Newen 2007 on local, global, and various "intermediate" notions of supervenience and the entailment relations (or lack thereof) among them.
- 4 Cf. Rosen 2010, Audi 2012, Correia 2013, Dasgupta 2014, Fine 2015, and Correia and Skiles 2019 for similar views.

Further Reading

- Duncan, Michael, Kristie Miller, & James Norton (2017), "Is Grounding a Hyperintensional Phenomenon?" *Analytic Philosophy*, 58: 297–329
(explores what it means for grounding to be a hyperintensional relation, and whether that thesis is true)
- Leuenberger, Stephan (2014b), "Grounding and Necessity," *Inquiry*, 57: 151–74
(a paper on grounding necessitarianism that complements Leuenberger's arguments against the grounding-supervenience link)
- Raven, Michael (2016), "Fundamentality without Foundations," *Philosophy and Phenomenological Research*, 93: 607–26
(distinguishes two notions of fundamentality, ineliminability and foundationality, which play different theoretical roles)

Steinberg, Alex (2013), "Supervenience: A Survey," in Miguel Hoeltje, Benjamin Schnieder, & Alex Steinberg (eds.), *Varieties of Dependence: Ontological Dependence, Grounding, Supervenience, Response-Dependence*, pp. 123–166, Munich: Philosophia Verlag

(a survey of various notions of supervenience and their relations to grounding and dependence)

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25

ONTOLOGY

Noël B. Saenz

1. Introduction

Ontology is the study of being. This is what ‘ontology’ means. But this is not too informative. After all, what do we mean by ‘being’ and what is it to study being? Let’s say, to the dismay of Meinongians (those who think that something can be even though it does not exist), that by ‘being’ we mean existence.¹ So ontology is the study of existence. But what is it to study existence? As practiced by ontologists, it is at least this:

To tell us what, in a very general sense, exists; to provide us with a list of what, generally speaking, is.²

It would be good to say a little bit on what is meant by ‘general’ and ‘generally’. Consider the question ‘what exists?’ As Quine (1948) famously points out, one answers truly by saying ‘everything’. One will also answer truly by giving a list of all the things there are. But neither will do for the ontologist. The former is not informative enough and the latter is too informative (the ontologist is not seeking to tell us how many Bichon Frises, or simples arranged Bichon Frise-wise, there are). Where, between these, does the ontologist want us to land? Perhaps the best thing to do here is to take a look at the kinds of things that ontologists, in their capacity as ontologists, are concerned with.³ Here are some of the most general: abstracta, concreta, composites, simples, events, facts, individuals, properties, particulars, universals, substances, and modes.⁴ We can expand by including less general, but still general enough, kinds. Some of these kinds involve things that ontologists regularly discuss: absences, the future, fictional objects, gunk, junk, hunk, holes, impossible objects, modal parts, necessary beings, numbers, the past, possible objects, possible worlds, propositions, sets, space-time, sums, and temporal parts. Others do not: deities, minds, souls, and value. For better or worse, I will in this chapter lay content with this ostensive account of what is meant by ‘general’ in the given characterization of ontology.⁵

Consider now grounding, that relation whereby something depends on other things in, as it is often put, a noncausal, constitutive way.⁶ What does grounding have to say about ontology—about what, generally speaking, exists? In this chapter, I will explore a number of ways the literature has said that grounding and ontology relate. To summarize: §2 concerns itself with grounding’s ability to *save* the ontology: to provide a safe and sane way of quantifying over

certain kinds of things in our theories. §3 with its ability to *price* the ontology: to show how we should measure ontological simplicity. And §4 with its ability to *restrict* derivative ontology: to restrict what can be grounded from what.⁷

2. Saving the Ontology

In giving an ontology, and so in saying what there is, one must also say what is not (which can be done by saying what is and then saying that there is nothing more). Here are just some of the many things ontologists have put in the trash bin: properties, material things, temporal parts, the past and the future, space-time, composites, and proper parts of the universe. But not only is it counterintuitive to deny that some of these things exist, there are also powerful reasons to think that some of them do (Hofweber 2005, 2007; Schaffer 2009: 356–62).

What does grounding have to do with any of this? Well, many seem to treat it as being able to *save the ontology* all while *preserving the primacy* of something else.⁸ Take properties. You can be a nominalist and still ontologically commit to them. How? By grounding them in things that are not properties. For instance, if you think that sets are grounded in their members, you can identify properties with sets of individuals and then ground these nonempty sets in such individuals.⁹ So there are individuals and properties, but individuals are more fundamental. So when nominalists say ‘The only real things are individuals’, it is open to them to understand ‘real’ as the ‘real’ of primacy or fundamentality and not the ‘real’ of existence. That is, it is open for them to think that properties are grounded in individuals and not vice versa and that nothing, other than perhaps other individuals, ground individuals.

Matter. Consider idealism. Guyer and Horstmann (2018: §1) describe it as “something mental (the mind, spirit, reason, will) is the *ultimate foundation of all reality*, or even exhaustive of reality” (italics mine). The second disjunct tells us that the first permits the ideal to play a role in grounding whatever the rest of reality is exhausted by. So when it comes to understanding idealism, we have two ways. One way says that matter does not exist; though there are apples, they are not material.¹⁰ This is extreme. Better, at least as regards to what exists, to opt for the other and say that there is matter; there are apples, apples are material, and that apples, being material, are grounded in idea. This saves the ontology (there is matter) while giving primacy to something else (the ideal). (If saving matter is not a better candidate for saving “the ontology” than is saving apples, we should instead say that in saving matter, the second way does a better job of saving apples than does the first. The first way identifies apples with ideas and so does away with their material nature. The second grounds matter in ideas and so preserves that apples are material.)

Temporal Parts. Consider a way, suggested by Hawthorne (2006: 99–100, 2008: 271–3) and Fine (2012b: 42), of understanding the dispute between three and four-dimensionalists. What is at issue here are questions of ground. Do temporal parts of persisting objects ground those objects, or is it rather that persisting objects, perhaps in conjunction with particular times, ground their temporal parts? This is a departure from the standard way of understanding this dispute, which hinges not on whether temporal parts ground or are grounded but, at least in part, on whether such parts exist.¹¹ This departure is also a way of preserving the spirit of three-dimensionalism (or enough of it, as both Fine and Hawthorne suggest) all while saving the ontology of temporal parts.

The Past and the Future. Baron (2015) has, in a manner friendly to the character of presentism, attempted to save the ontology of the past and the future by means of grounding. According to this new kind of presentism, the present grounds both the past and the future. So the past and the future are, but the present has primacy. Of course, this is to change the dispute between

presentists and eternalists from issues of existence to issues of grounding. Still, it may be a good one. After all, perhaps the most serious worry with presentism (as seen in the truthmaker objection and the problem of cross-temporal relations) is that it posits too small an ontology in positing only the present.

Space-time. Schaffer (2009: 363), Dasgupta (2011), and North (2018) all reconceive the dispute between relationalists and substantialists from issues of existence to issues of grounding.¹² North insert (2018: 4) says,

The traditional relational-substantial debate is about whether space—in modern terms, spacetime—exists. . . . The debate that I will present is not exactly the traditional one. But it is close enough in spirit that I think it is the best way of understanding that dispute . . . I will argue that, regardless of whether you are a relationalist or a substantialist, you can be a realist about spatiotemporal structure. I do this by framing the debate in terms of fundamentality and ground, notions that have gotten lots of press recently in metaphysics.

Given that, as she says, you can be a realist about spatiotemporal structure even if you are a relationalist, then we have a way of making sense of the dispute all while saving what to many seems obvious: that space-time (or spatiotemporal structure) exists. Indeed, as North (4) says, this way of putting the dispute allows us to “formulate the most plausible—if not entirely traditional—versions of the two main positions on it”.

Wholes. Moving to composite objects, Baker (2007: 192) says that mereological sums

come into existence automatically when their parts come into existence: there is a sum whose parts are your left eyebrow and Tony Blair’s favorite shirt, simply *in virtue of* the existence of your left eyebrow and Tony Blair’s favorite shirt. In this way, *sums are ontological free-riders.* (italics mine)¹³

The import of her statement is clear.¹⁴ There should be no ontological scruples with sums over and above any such scruples with their parts—they are ontological free-riders, their existence is automatic—since for any things, there are sums with those things as parts simply in virtue of those things existing.

Parts. Finally, consider existence monism, which says that there is only one concrete thing and that this thing is the universe (Horgan and Potrč 2000). Now this is a hard view to accept, banishing from existence the universe’s proper parts. But Schaffer (2010a) has it that we can save these parts all while giving primacy to the universe if we think that what there is one of is not concrete things *per se* but *fundamental* concrete things. Add to this that this fundamental concrete thing, the universe, grounds all other concrete things, and we have priority monism. Now this kind of monism is, like existence monism, radical. But its radicality lies in the direction it has grounding going in (from whole to part) and not in the ontology it posits. Unlike existence monism, it retains the ontology of proper parts of the universe.

Because of the work grounding can and does do, grounding is seen by some as providing us with a kind of ontological safe space. For things like properties, matter, temporal parts, the past and the future, space-time, sums, proper parts, and more, grounding is a shelter from the many (to some dubious) denials of existence ontologists have been prone to make. And grounding is seen as this shelter because although these things exist, they are not fundamental.¹⁵

It is worth relating the strategy of saving portions of ontology via grounding to an issue in metaontology. Schaffer (2009) has put forward a metaontology according to which the central

questions and issues of ontology are not existence questions and issues but grounding ones. For him, the existence questions that have dominated contemporary ontology are, unlike grounding questions, trivially answered in the affirmative. As he says “The deep questions about numbers, properties, and parts (*inter alia*) are not *whether* there are such things, but *how*.” (361–2)¹⁶

Schaffer’s metaontology is controversial. But it does go well with grounding’s ability to save the ontology. Why give up on the Fs, especially if we have independent and powerful reasons for accepting that there are Fs, if we can ground them in things we already accept?¹⁷ Still, there is nothing about grounding that mandates that we be permissive with respect to what exists (and Schaffer is not saying that it does). Making it easier to be a permissivist about what exists is one thing. Demanding that we be permissive is another.

3. Pricing the Ontology

Other relations have been put to the task of saving the ontology. Chief among these is reduction (others include supervenience and translatability). But attempted reductions of one kind of entity to another (for e.g., the mental to the physical, the moral to the natural, the whole to the parts) have struck many as implausible. One potentially nice thing about grounding’s ability to save the ontology is that it need not require reducing what we are trying to save to something else that, on the face of it, looks rather different. Not only can it save the ontology, but it can, at least in principle, do so while preserving what seems true of it. Still, some might not see in this much comfort. After all, is not a theory with just individuals simpler than a theory with individuals and properties? And if simpler, then should we not, on grounds of simplicity, prefer it?

The problem here is perhaps significant. If grounding can do saving work only at the expense of inflating the ontology, then perhaps this work of grounding is not worth it. But a number of grounding theorists disagree. They think that one can inflate the ontology without incurring a cost. Here is their thought.

Orthodoxy has it that *ontological* simplicity should be measured in terms of the number of entities or kinds of entities a theory quantifies over.¹⁸ We can express this in the following principle of ontological simplicity:

T_x is simpler than T_y if and only if T_x has less entities than T_y .¹⁹

Here, when it comes to ontological simplicity, the price of an ontology should be measured in, and only in, the number of entities. Given then that ontological simplicity is a guide to truth, this principle supports the following maxim:

The Razor. Do not multiply entities without necessity!

But a number of philosophers are now questioning the above principle and the maxim it supports (Schaffer 2009, 2015; Cameron 2010: 250, 2014: 100; Sider 2013: 240; Bennett 2017: 220–9). According to them, the following principle of ontological simplicity does better:

T_x is simpler than T_y if and only if T_x has less *fundamental* entities than T_y .

If we say that to be fundamental just is to be ungrounded, then this principle amounts to

T_x is simpler than T_y if and only if T_x has less *ungrounded* entities than T_y .²⁰

So when it comes to ontological simplicity, the price of an ontology should be measured in, and only in, the number of ungrounded entities. And this, coupled with the claim that simplicity is a guide to truth, supports another maxim (this is Schaffer's name for it):

The Laser. Do not multiply ungrounded entities without necessity!

The Laser is attractive. But we want reasons to accept it. Here are four:

Reason 1. Schaffer (2015: 648) argues that The Laser gets cases concerning ontological simplicity right. In order to convince us of this, he provides us with the following thought-experiment

Esther posits a fundamental theory with 100 types of fundamental particle. Her theory is predictively excellent and is adopted by the scientific community. Then Feng comes along and—in a moment of genius—builds on Esther's work to discover a deeper fundamental theory with 10 types of fundamental string, which in varying combinations make up Esther's 100 types of particle. This is intended to be a paradigm case of scientific progress in which a deeper, more unified, and more elegant theory ought to replace a shallower, less unified, and less elegant theory. Feng's theory is evidently better in every relevant methodological respect.

In spite of having more overall entities, Feng's theory does better when it comes to ontological simplicity than does Esther's *because it has less ungrounded ones*. We thus have reason to think that The Laser, and not The Razor, is the preferable maxim (see Bennett (2017: 220–1) for a similar argument).

Reason 2. Schaffer (2015: 648–51) gives us the following argument by analogy. When it comes to conceptual simplicity, all that matters are one's undefined notions. Once you have FEMALE and UNMARRIED in your conceptual toolbox, adding to it BACHELORETTE does not count against conceptual simplicity. Given that "it is defeasibly reasonable to treat ontological economy and conceptual economy as relevantly analogous" (649), we have reason, in the above fact about how to measure conceptual simplicity, to prefer The Laser over The Razor.

Reason 3. Schaffer (2015: 651–3) focuses again on conceptual simplicity. He begins by noting that the more useful concepts we can define from our stock of undefined ones, the more powerful our theory will be. We should therefore go in for

Conceptual Bang for the Buck. Optimally balance minimization of primitive (undefined) concepts with maximization of defined concepts (especially useful ones).

Given the analogy between conceptual and ontological simplicity, we should therefore accept

Ontological Bang for the Buck. Optimally balance minimization of fundamental (ungrounded) entities with maximization of derivative entities (especially useful ones).

Given this, which "pressures one to minimize fundamental entities" (653), we have reason to think that it is the number of ungrounded entities that matters when it comes to simplicity and not the number of entities *tout court*.

Reason 4. Bennett (2017: 223) provides us with an argument on the basis of considerations of modality and probability. Consider theories T⁻ and T⁺, which differ only in the number of grounded entities they posit: T⁺ posits some, while T⁻ posits none, and both posit the same ungrounded entities. Now if we make a modal assumption, it turns out that T⁻ is no more likely

to be true than is $T+$. Here is the assumption: grounds necessitate what they ground.²¹ So the grounded entities $T+$ posits are necessitated by those ungrounded entities that $T+$ and $T-$ share. Given the following theorem of the probability calculus

If $A \vdash B$, then $\Pr(A) = \Pr(A \& B)$,

it follows that the probability of $T+$ is the same as the probability of $T-$ on account of the entities $T-$ posits entailing, in $T+$, the entities $T+$ posits. Bennett concludes that “ $T+$ ’s extra ontological commitments do not tell against its simplicity in a way that makes it less likely to be true” (223). And since this notion of simplicity is the notion of simplicity at work, it follows that The Razor, but not The Laser, gets it wrong when it comes to ontological simplicity.

Worries have been raised with The Laser, some of which are anticipated by Schaffer (2015: 654–64) and Bennett (2017: 223–6). Some are also taken up and extended on by Turner (2016: 382–5), Baron and Tallant (2018), and Fiddaman and Rodriguez-Pereyra (2018).²² I will not describe these worries here. My aim has only been to show that according to grounding enthusiasts, saving the ontology need not come at a price. For they can say that the ontological price of a theory is measured in terms of the number of ungrounded entities. Since what is being saved is being grounded, then, unless saving these entities requires that we expand on our ungrounded base, what is being saved will come for free.

4. Restricting the Ontology

Up to this point, we have said next to nothing about what it takes to ground something, and so to ground something that we hope to save. What I want to do in this section is discuss a number of principles of grounding that restrict, in some manner or other, what can ground what.²³

Modal Restrictions. Where the necessity at work is metaphysical necessity, consider the following much-talked-about modal principle

Necessitation. $x_1, \dots, x_n$ ground $x \rightarrow \Box(x_1, \dots, x_n \text{ exist} \rightarrow x \text{ exists})$.

That is, if some things ground something, then necessarily, if the former exist, then so does the latter.²⁴

Necessitation tells us that we cannot posit in our ontology grounded things that are not necessitated by what grounds them. Though not usually made explicit, this thought plays an important role in arguments against nonreductive physicalism from the possibility of zombies. Borrowing from Kripke (1972), consider what Kirk (2015: §2) says

Imagine God creating the world and deciding to bring into existence the whole of the physical universe. Having created this purely physical universe, did he have to do any more work to provide for consciousness? . . . Physicalists . . . are committed to answering no. *They have to say that by fixing the purely physical facts, God did everything that was needed to fix the mental facts about the organisms thereby created, including their thoughts, feelings, emotions, and experiences.* (italics mine)

In saying “God did everything that was needed to fix the mental facts”, it is assuming that if the physical grounds the mental, then once we have the physical, we have the mental. In other words, if the physical grounds the mental, then given Necessitation, there can be no worlds just like this one in all physical respects with no accompanying mental

phenomena.²⁵ But if zombie worlds are possible, then there are. And if there are, then we cannot ground the mental by means of the physical. We must, if we are to have the mental at all, have it at the fundamental level (as seen in physicalism's chief rivals: property and substance dualism).²⁶

Consider the following modal principle, advanced by Saenz (2018), which requires making explicit the structure of the facts related by grounding:

Sensitivity. $F(x_1, \dots, x_n)$ grounds $G(x_1, \dots, x_n) \rightarrow$ if the $x_1, \dots, x_n$ were to exist but not be G , they would not be F .

That is, if that some things are F grounds that they are G , then if these things were to exist but not be G , they would not be F . Saenz argues in favor of Sensitivity by showing that without it, we cannot infer (or infer in the right kind of way) certain plausible counterfactuals.²⁷

Now Saenz claims that Sensitivity rules out mereological sums given the following:

1. If γ is a sum of some things, then that these things compose γ is grounded in their existence.

This is supposed to express the thought that for mereological sums, once you have got the parts, you have got all you need in order to explain why those parts compose a whole. Now putting the consequent of 1 in the antecedent of Sensitivity yields:

2. If these things were to exist but not compose γ , then these things would not exist.

This, however, is false. If these things were to exist but not compose γ , then of course they would exist!²⁸ So given 1 and Sensitivity, 2 follows given the existence of sums. But since we should keep Sensitivity, then either sums or 1 must go. Either way, composition cannot be had so cheaply. That the parts exist does not, on its own, give us composition and so does not give us wholes composed of these parts. If we want composition, we need something more than that the parts exist.

Determination Restrictions. Consider a view of the world according to which reality comes in levels with the things at higher levels existing and having the natures they do because the things at lower levels exist and have the natures they do. To put this in terms of grounding, consider a view according to which the existence and nature of various kinds of things (macroscopic) is completely grounded in the existence and nature of other kinds of things (microscopic). There is something intuitive about this hierarchical picture. But deRosset (2010) raises a problem with it. Here is a constraint on explaining some phenomenon:

Determination. Fx because $\phi(x_1, \dots, x_n) \rightarrow \forall y_1, \dots, y_n \forall \gamma (\phi(y_1, \dots, y_n) \rightarrow F\gamma)$.²⁹

That is, if x is F because $x_1, \dots, x_n$ collectively satisfy ϕ , then for any objects $y_1, \dots, y_n$ and any object γ , if $y_1, \dots, y_n$ collectively satisfy ϕ (that is, if they collectively satisfy the explanans clause), then γ is F . The thought here is that something is a bad explanation of x 's being F if there is some γ that is just like x with respect to what explains that x is F but where γ is not F .

Now suppose that we have a transparent raindrop r , a tectonic plate t , and that r is F in virtue of facts about its microstructure $m_1, \dots, m_n$. So where R stands for some relation, r is F because $m_1, \dots, m_n$ satisfy R . Now Determination has it that this explanation is good only if

3. $\forall y_1, \dots, y_n \forall \gamma (R(y_1, \dots, y_n) \rightarrow F\gamma)$,

which say that for any things and any thing, if the former things collectively satisfy R, then the latter thing is F. But this is logically equivalent to

$$4. \quad \exists y_1, \dots, y_n R(y_1, \dots, y_n) \rightarrow \forall y Fy,$$

which says that if some things collectively satisfy R, then everything is F. Now if r is F because $m_1, \dots, m_n$ satisfy R, then some things satisfy R. Assuming then that r is F because $m_1, \dots, m_n$ satisfy R, some things satisfy R. So the antecedent of 4 is satisfied. But then everything is F. Assuming that 'F' denotes being transparent, it follows falsely that our tectonic plate t is transparent! Something must go. Since we should not reject Determination, then what we should reject is that r is F because $m_1, \dots, m_n$ satisfy R. But that r is F because $m_1, \dots, m_n$ satisfy R follows from the above hierarchical view that the existence and nature of certain things (in our case, macroscopic things) is completely grounded by the existence and nature of other things (microscopic things). So this view must go.

It is an implication of this argument from Determination that certain facts about r must be fundamental. As deRosset (83–4) says,

not every qualitative fact involving r can be completely explained solely in terms of the properties of other things . . . the determination argument . . . shows that some of the facts involving r are fundamental.

For anyone attracted to a view of grounding according to which macroscopic things like raindrops are not part of fundamental reality, something must be said in response to the present restriction on grounding.³⁰

Wilson (2012) has raised a problem for thinking that determinables are grounded in determinates. She (2012: 15) says the following

So if determinables are to be grounded in determinates—if all facts about (instances of) determinable properties are to be fixed by facts about (instances of) determinate properties—some more complex determinate must be doing the grounding.

Now what principle is being assumed here? Perhaps it is the following

If x grounds y , then all facts about y are to be fixed (that is, grounded) by facts about x .

But most will reject this. Suppose that p is true but q is false. Then p grounds (p or q). But it's false that all facts about (p or q) are fixed by facts about p alone. Perhaps we should restrict the quantifiers in this principle to entities (individuals and properties)? In trying to undermine that determinate *properties* ground determinable ones, this fits well with Wilson's aim. We thus have

If entity x grounds entity y , then all facts about y are to be fixed (that is, grounded) by facts about x .

Though controversial, there are others who accept something like this principle (Schaffer 2010b: 345; deRosset 2013: 4). Assuming then that this principle is what Wilson has in mind, we now have reason to think that determinates such as crimson do not ground determinables such as red. Here is why. It is a modal fact about red that it is of a type whose instances might be differently determined (even if red is determined by crimson, it need not have been). However, no

determinate of red is such that facts about it are suited to ground the above modal fact about redness. After all, how could a fact about crimson ground that red is of a type whose instances might be differently determined given that crimson necessarily excludes other possible determinates of red (such as maroon)?³¹ According to Wilson (2012: 12) then,

it may be that determinate instances metaphysically ground certain nonmodal facts about their associated determinable instances. But modal facts and features about entities at a world are also part of the world's inventory! A supposed 'ground' for some entity which failed to provide a metaphysical basis for modal facts constitutive of the entity would be an *incomplete* ground, at best.

Given that there is no fact about crimson that could ground why it is that red is of a type that could have had been differently determined (since many other determinate instances of red are such that crimson necessarily excludes them), it follows, by means of the given principle, that crimson does not ground redness.

The implications of this style of argument are far from trivial. Notice that the principle makes saving the ontology by means of grounding it very hard. If we are idealists who want to save matter by grounding it in the ideal, then it had better be that all facts about material objects are grounded in facts about ideal objects. Or if the priority monist wants to save the parts from the existence monist by grounding them in the universe, then it had better be that all facts about the parts are grounded in facts about the whole. But that these claims are true is by no means obvious. Saying, and even providing arguments in favor of thinking, that matter is grounded in the ideal is one thing. Ensuring that every fact about matter is grounded in facts about the ideal is something else (as we saw with Wilson, modal facts here seem to be particularly problematic).

In light of this, perhaps the right thing to do is to reject the principle under consideration. Maybe, as Wilson mentions, what should be claimed is that if entity x grounds entity y , then all *nonmodal* facts about y are to be fixed by facts about x . Or as Schaffer says, if entity x grounds entity y , then all *nonrelational* facts about y are to be fixed by facts about x .³² But if either of these is the way to go, then we need reason to think that modal or nonrelational facts should be excluded. Barring this, these restrictions on the principle seem ad hoc.

5. Closing

This chapter described three ways grounding has been used when it comes to ontological matters. The first was to retain portions of controversial ontology by means of grounding. The second was to provide an account of ontological simplicity in terms of grounding. And the third was to restrict, by means of principles of grounding, what portions of ontology can be grounded.

Let me close this chapter by suggesting a promising direction for future research. When it comes to grounding and ontology, more attention needs to be given to the kind of work we saw being done in §4—work that restricts, by means of principles of grounding, what can be grounded from what. What kind of grounding claims are appropriate to make is not always clear, and yet a cursory glance at the literature reveals that it is filled with all sorts and kinds of grounding claims. Now on account of this liberalism, some think that we ought to abandon grounding (see especially Hofweber 2009).³³ But the solution to unbridled use of grounding is not to abandon it but to put constraints on it. We need to know what kinds of grounding claims we can make. Like theories of mereology, which involve a whole host of principles that significantly constrain what models of proper parthood we can accept, we need theories of grounding that appeal to principles that significantly constrain what models of grounding we can accept. Then

and only then will we be in a position to know, or know better, what grounds what. Then and only then will we be in a position to know whether or not we can ground properties in individuals, matter in idea, temporal part in persisting object, the past and the future in the present, space-time in facts about material bodies, sums in their summands, and parts in wholes. I would like to thank Mike Raven for helpful comments. I would also like to thank my wife, Amy Saenz, for her love and always present support and encouragement in my work.

Notes

- 1 Meinongianism is an important position in ontology and metaontology (Meinong 1904; Parsons 1980; Zalta 1983; Fine 2009; von Solodkoff and Woodward 2013). In spite of this, I must limit myself here and stick to the more orthodox view that being and existence cannot come apart.
- 2 It is perhaps also about this: To tell us what the general nature of the general kinds of things is. van Inwagen (2001: 1–2) calls much of what goes under this ‘B-ontology’, and Fine (2017: 98) calls it ‘meta-physics proper’. But whatever one wants to call it, I mention it only because ontologists *qua* ontologists are very much concerned with it and often times use ‘ontology’ to refer to it. Be that as it may, in this chapter, I will not concern myself with this aspect of ontology (though issues concerning how things are do play a role in this chapter).
- 3 As I tell my students, if one wants to know what the subject matter of ontology is, one should simply do and read some ontology. For someone who goes further and tries to define what this subject matter is, see van Inwagen (2012).
- 4 These kinds are not mutually exclusive. There is much overlap. It may also be that some of these kinds are extensionally equivalent. Perhaps individuals, particulars, and substances are extensionally equivalent, as are, perhaps, properties, universals, and modes.
- 5 Note that physicists also want to know what, in general, exists. What this shows us is that there is more to doing ontology than asking and answering questions of a sufficiently general type. There is also an epistemic component (Fine 2012a: 9). Whether or not there are fictional objects can be investigated *a priori*. Whether or not there are electrons should not be.
- 6 This description is controversial in at least three respects. Not everyone thinks of grounding as a relation (Correia 2010; Fine 2012b; Dasgupta 2014; Litland 2017). Not everyone thinks of it in terms of dependence, favoring instead its explanatory aspect (Fine 2001; Litland 2013; Dasgupta 2014). And not everyone wants to distinguish it entirely from causation (Schaffer 2016; Wilson 2018). To the first, we can, in most cases and with some modifications, substitute talk of grounding in terms of a predicate (‘grounds’) with talk of it in terms of an operator (‘because’). To the second, at least when what grounding relates are facts, no harm will be done in speaking of it as an explanatory relation (see §4, where I express a grounding principle in terms of ‘because’). And to the third, even those who think that grounding and causation are closely related grant either that it is distinct from causation or that it is a type of causation to be contrasted with the type typically referred to in cases involving the breaking of windows and the hitting of billiard balls.
On how grounding relates to dependence, see Schnieder’s contribution [Chapter 7]. On how grounding relates to explanation, see Glazier’s [Chapter 8]. And on how grounding relates to causation, see Wang’s [Chapter 20].
- 7 Notice that we can also ask what ontology has to “say” about grounding. This can be divided into two. We can ask about the ontology of grounding: does grounding exist? But we can also ask about what an ontology has to say about grounding: given a preferred ontology, how should we think about grounding? Since we need to assume that grounding exists in order to discuss its import on ontology, and since this chapter is devoted to applying grounding to ontology and not ontologies to grounding, I will not address these questions here.
- 8 In what follows, I will be neither endorsing nor repudiating the various proposed grounding claims. My aim here is only to describe what some have said with respect to what grounds what.
- 9 This involves making two moves. First is a *reduction* from properties to sets and then a *grounding* from sets to members. For work on the relationship between reduction and grounding, see Rosen (2010: 122–6), Audi (2012: 704), and Dorsey (2016).
- 10 Berkeley (1710) seemed to have held this view. He says in §1 of the *Principle of Human Knowledge*:

It is evident to any one who takes a survey of the objects of human knowledge, that they are either ideas actually imprinted on the senses, or else such as are perceived by attending to the passions and

- operations of the mind, or lastly ideas formed by help of memory and imagination, either compounding, dividing, or barely representing those originally perceived in the aforesaid ways. This, coupled with the belief that things like apples are the objects of human knowledge by means of being imprinted on the senses (something Berkeley believed), entails that things like apples are not material but ideal. See also Bradley (1893: 144) and McTaggart (1934: 273), whom Guyer and Horstmann (2018: §8) call ‘spiritual monists’ on account of their thinking that reality is exclusively spirit.
- 11 For example, van Inwagen (2000: 442–8) argues against four-dimensionalism by arguing that there are no such things as temporal parts. And Wiggins (1980: 25) says that three-dimensional continuants are “things with spatial parts and no temporal parts”. Even Sider (2001: 64–7), who argues that three-dimensionalism cannot be defined as the denial of persisting objects having proper temporal parts, grants that three-dimensionalism permits the possibility of persisting objects with no proper temporal parts.
 - 12 To be strict, North frames it in terms of spatiotemporal facts about the world, or spatiotemporal structure, and not in terms of space–time.
 - 13 Armstrong (1997: 120) says something similar when he says of sums that “they are no increase of being beyond that of their parts” and that they have “no real metaphysical consequences”. But whether he has in mind grounding here or something else is unclear. For a discussion on understanding what Armstrong may have in mind that does not appeal to grounding, see Sider (2015).
 - 14 In spite of the import of her statement being clear, her statement is not. Is it saying that these things are parts of the sum in virtue of existing or that the sum exists in virtue of these things existing? In the first case, what is being grounded is a relational fact—that these are parts of that—and in the second case, an existential fact—that the sum exists. Now either way, sums come for free, and this is the part her statement makes clear. Still, we have a difference that makes a difference. For some focus their attack on the ontology of sums (Koslicki 2008), while others focus on the ontology of parthood (Sider 2013, though he phrases it in term of the ideology, and not ontology, of parthood).
 - 15 It is an implication of this that grounding cannot help to save, in the manner presently discussed, entities deemed to be fundamental (unless, of course, you give up on that entity’s being fundamental). On how grounding relates to fundamentality, see Bliss’s contribution [Chapter 23].
 - 16 Grounding also relates to views of ontological commitment, another metaontological issue. On grounding and a Quinean view of ontological commitment, see Schaffer (2009: 366–73). On discussion of grounding and ontological commitment by way of truthmaking, see Cameron (2008, 2010) and Schaffer (2008). And for a nice overview of various metaontological positions one can take on grounding, fundamentality, and ontological commitment, see von Solodkoff and Woodward (2013).
 - 17 Of course, this is a big ‘if’. It might turn out that grounding something in something else will prove quite hard (see §4). It might also turn out that even if we can ground something in something we already accept, doing so incurs a cost (see §3).
 - 18 Ontological simplicity is to be contrasted with theoretical elegance (Baker 2016; Bennett 2017, 227–8) and ideological simplicity (Sober 2001, 14; Cowling 2013). The former has to do with the complexity of the explanations being given and the latter with the primitive notions a theory makes use of.
 - 19 Whether we should count in terms of entities or kinds of entities is a matter of some dispute (Lewis 1973: 87; Nolan 1997). Though I will talk in terms of the number of entities, feel free to interpret what I say in term of kinds. Nothing of import hangs on this.
 - 20 For more on fundamentality, see Bliss’s contribution to this volume [Chapter 23].
 - 21 On grounding and necessity, see Skiles’s contribution to this volume [Chapter 10].
 - 22 Baron & Tallant & Fiddaman & Rodriguez-Pereyra also seek to show that Schaffer’s arguments for The Laser fail and, in Baron & Tallant’s case, that two common reasons to accept The Razor do not amount to reasons to accept The Laser.
 - 23 I have expressed a number of these principles using standard logical notation. However, in order not to require that the reader be familiar with such notation, I have also expressed these principles in a way that requires little to no familiarity with logical notation.
 - 24 For discussion of this principle, see Skiles’s contribution to this volume [Chapter 10].
 - 25 The kind of saving done by physicalism here is not so much saving what others have banished. It is rather saving what almost everyone accepts in such a way that no ontological costs are incurred (see §3).
 - 26 In a similar vein, Necessitation tells us that we cannot posit truths that are not necessitated by what grounds them, their truthmakers (on how grounding and truthmaking relate, see Trogdon’s contribution to this volume). For more on this when it comes to making true past–tense truths, see Caplan and

- Sanson (2011) and Ingram and Tallant (2018: §6). For more on this when it comes to universal and negative truths, see Russell (1972: chapter 5) and Armstrong (2004: chapters 5 & 6). And on whether truthmakers ground what they make true, see Saenz (forthcoming).
- 27 But see Kappes (2019) for a reason to think that we can secure these counterfactuals without appealing to Sensitivity.
- 28 Saenz (2018, 108–9) assumes that counterpossibles have nontrivial truth-conditions.
- 29 There are some syntactic differences here between how deRosset formulates this principle and how I formulate it. But these differences are, for our purposes, immaterial.
- 30 One very natural response is to require that some intimate relation hold between $x_1, \dots, x_n$ and x such that the consequent of Determination holds only when this requirement is met (deRosset (2010: 83) considers such a response but argues that it will not do). For another kind of response, see von Solodkoff (2012).
- 31 Wilson here seems to be relying on this principle of grounding:
- If a fact about entity x grounds that entity y could have been F , then this fact about x is consistent with y being F .
- Is this principle true? I will leave deciding this to the reader.
- 32 Schaffer says this by way of personal correspondence with deRosset. See deRosset (2010: 74).
- 33 But see Raven (2012) for a response to Hofweber.

Cross-References

Dependence [Chapter 7]; Explanation [Chapter 8]; Fundamentality [Chapter 23]; Modality [Chapter 24]; Necessity [Chapter 10], Physicalism [Chapter 35]; Truthmaking [Chapter 28]

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26

REALISM

Olla Solomyak

I. Introduction: Questions of Realism and Ground

Questions of ground and questions of realism appear to be tightly connected in a number of ways, but there has not been consensus on precisely how these connections should be understood nor a full explication of the various approaches one might take on this issue. My aim in this chapter will be to spell out several ways in which we might see these questions as connected and thereby clarify the role for questions of ground in metaphysical inquiry more generally.

Before we attend to the connections between them, we must first clarify what is meant by ‘questions of ground’ and ‘questions of realism’ in this context. Questions of ground can be spelled out and conceptualized in a number of ways, but what I’ll focus on here are questions of ground for *facts*: Given some fact [p], what *grounds* this fact? That is, *in virtue of what* does [p] obtain?^{1, 2} My aim here will be to clarify how questions of this form relate to questions of realism.

Now to clarify ‘questions of realism’: there are at least four different *kinds* of questions we might think of as questions of realism, and it will be important to distinguish between them here. I’ll thus start by spelling out these four questions and briefly clarifying the connections between them; this will lay the groundwork that will then allow us to examine how questions of ground might fit in.

The first kind of question of realism is a question of the following form: **(1)** Given some (purported) (kind of) object, property, or phenomenon A, we can ask: Is A *real*? For example, we might ask: Are mathematical objects real? Are moral properties real? Is time real? It’s important to note that while some questions of this form are straightforwardly ontological questions—asking about the existence of some purported *object(s)*—others are not obviously questions of ontology. The question of the reality of time, for example, is not, at least on the face of it, a question of ontology. Rather, it’s a question of whether reality has a certain (temporal) *structure* or whether there are *facts* of a certain kind. So questions of this first form will be understood differently depending on the *kind* of thing (object, property, or phenomenon) whose reality is in question.

The second kind of question of realism is a question about language and factuality: **(2)** Given some domain of discourse Y, we can ask: Are the claims made in Y to be understood as having

objective, realist import? Or, are the claims of *Y factual*? For example, we might ask whether moral, mathematical, or aesthetic claims have objective, realist import or whether there really are moral, mathematical, or aesthetic facts. Questions of this form are of course closely related to the first kind of question of realism: perhaps factual domains of discourse are those whose central notions pick out real objects, properties, or phenomena. Or perhaps real objects, properties, and phenomena are those that figure in the real facts. (Meta)metaphysicians will differ as to how exactly the relationship between these two questions of realism should be understood, as well as on how best to understand and formulate each of these questions themselves.³

A third kind of question of realism delves precisely into these meta-metaphysical issues and asks: **(3)** how should questions of type (1) and (2) be understood? What is it to ascribe *reality* to an object, property, or phenomenon or to be a realist about a particular domain of discourse? More generally, this is the meta-metaphysical question: *What is realism*? Answering this meta-metaphysical question thus involves clarifying precisely how questions of type (1) and (2) should each be understood, and in the process, also clarifying the relationship between them.

Finally, a fourth question that might also be termed a ‘question of realism’ takes another step back, and asks: **(4)** should we be ‘realists’ about the answers to questions of type (3)? That is, we can ask whether there is a real, deep, or objective distinction between the realist and the anti-realist stance—either with respect to a particular domain, or more generally: Can we make sense of the metaphysically weighty notion of *reality* that is supposed to be at issue in realist versus anti-realist disputes? Interestingly, this very question is also a question of type (2): We ask about some domain of discourse—in this case the dispute between realists and anti-realists or simply metaphysics as a whole—whether it is *factual* or concerns real, objective distinctions in the world.⁴

Following Kit Fine (2001), I set aside ‘anti-realist’ answers to this fourth sort of question of realism—i.e., a kind of quietism or deflationism about realist metaphysical inquiry as a whole.⁵ So my focus will be on questions of type (1) through (3) and their connection to the notion of ground, on the assumption that we can make sense of a realist approach to metaphysical inquiry in general. But providing satisfying answers to questions (1) through (3)—perhaps with the help of the notion of ground—is ultimately a way of addressing question (4) as well, by presenting a viable example and understanding of realist metaphysical inquiry. I’ll thus return briefly to question (4) at the end of the chapter, once the potential role for questions of ground in this connection has been clarified.

Assuming, then, that we can make sense of a realist approach to metaphysical inquiry as a whole, we can turn back to our central question of interest and ask how questions of ground relate to questions of realism—more specifically now, to questions of type (1) through (3). Here too, there are a number of more specific questions we might ask. I’ll focus in particular on three: First, **[A]** how do questions of ground about a particular domain relate to questions of realism about that domain? More specifically, how do questions of type (1) and (2) relate to questions of ground concerning the objects or facts in question? For example, how does the question of ground for moral facts—i.e., the question of how moral facts are *grounded*—relate to the question of realism about moral facts or properties? Does answering the question of ground immediately entail, or perhaps simply guide us to, an answer to the question of realism? More generally, how might questions of ground inform the debate between a realist and an anti-realist in a particular domain?

Second, **[B]** how might the notion of ground help us answer the meta-metaphysical question of realism, i.e., question (3)? That is, how might the notion of ground help clarify what it *is* to be a realist or an anti-realist with respect to some domain, or in general? For example, might the question of realism be understood in *terms* of ground?

Finally, we can ask [C], should we be realists about ground itself? This question can take two forms, corresponding to questions of type (1) and (2): First, (1_g) is there some real structure or relation in the world corresponding to the notion of ground? And second, (2_g) should grounding claims be taken to have objective, realist import?

With this web of questions on the table, I'll now turn to a brief overview of the various responses to questions [A] through [C] that have been suggested or implicit in recent literature. I'll then tie these ideas together and present what I take to be the upshot for connections between realism and ground.

II. Realism by Way of Ground?

It will be helpful to start with question [B] and ask first whether the questions of realism might be understood *by way of* the notion of ground. That is, might realism be explicitly defined or understood in terms of ground? In his (2009) "On What Grounds What," Jonathan Schaffer suggests that *ontological* questions—i.e., questions about the existence of particular (kinds of) entities—be replaced in metaphysical inquiry by questions of how the relevant entities are *grounded*.⁶ For example, rather than ask whether mathematical objects *exist*—because of course, in a trivial sense, they do—we should, on Schaffer's picture, ask how they are *grounded*: are they grounded in (e.g.) the intentions and activities of mathematicians or in a Platonic realm of abstract objects? On this approach, then, the only substantive questions of concern to the metaphysician are first, whether the purported entities are *fundamental*, and second, *how* they are grounded if not. The traditional questions of ontology thus become 'cheap'—their answers taken to be a trivial 'yes,' with the real work left to the questions of ground and fundamentality.

One might wish to generalize Schaffer's picture and take it to apply to questions of realism beyond ontology. More specifically, one might propose to understand realism in terms of fundamentality: Perhaps to be *real* is just to be *fundamental*⁷—i.e., to be at the fundamental, ungrounded layer of reality—and there is no further substantive question of realism to be asked once an object's status with respect to fundamentality has been determined. On this approach, the questions of realism would thus be replaced with questions of ground: We would ask of any fact, object, or phenomenon whether and how it was grounded, and metaphysical inquiry into its 'reality' would end there.

Despite its simplicity and initial appeal, however, two difficulties can be raised for this proposal. The first concerns the way the notion of fundamentality is itself to be understood. In order for the suggestion above to amount to a definition of realism in terms of *ground*, fundamentality must be understood only in its negative sense, as ungroundedness. If to be fundamental is just to be ungrounded, then defining realism in terms of fundamentality is a way of defining realism in terms of ground: to be real is, on this version of the proposal, to be ungrounded.

The trouble with this is twofold: First, while some metaphysicians do understand fundamentality in this way, this is far from uncontroversial, and positive conceptions of fundamentality (which we will turn to in what follows) are quite common.⁸ Second and more problematically, the proposed definition of realism arguably loses its appeal when fundamentality is understood only in its negative sense. Why should we think that to be real is just to be *ungrounded*? Intuitively, we seem to be attributing something positive and substantial to a fact or entity by calling it *real*. To attribute *reality* to something sounds like more than stating that it lacks a ground. Perhaps something could be ungrounded and nevertheless not *real* in some more weighty positive sense of the term. These intuitions suggest that the appeal of the proposed definition relies on an implicit understanding of fundamentality in its *positive* sense. We find it natural to identify the *real* with the *fundamental* because we have some positive understanding of fundamentality that goes beyond (and perhaps comes apart from) ungroundedness.

This brings us to positive conceptions of fundamentality, on which fundamentality is not understood in terms of ground. Fine (2001), for example, takes fundamentality to be a basic notion of *reality*—the notion of how reality is *in itself*—which is not definable in other terms (and, for Fine, in fact comes apart from ungroundedness). This conception of fundamentality of course fits perfectly with the proposal that questions of realism (at least of a certain kind) are to be understood as questions of fundamentality, but this is no reduction or definition of realism in terms of ground (or anything else)—fundamentality, on this understanding, is just a particular *notion* of reality.

The second worry that can be raised for the suggestion that realism be understood in terms of fundamentality—which applies both to the reductive and non-reductive versions of the suggestion distinguished earlier—concerns the non-fundamental, ‘higher-levels’ of reality. We seem to have a conception of realism that allows for something to be *real* but not fundamental.⁹ And at least in some cases, there seems to be a substantive question to be asked about the reality of some nonfundamental aspect of reality. For example, we might agree that fairies, tables, and nation-states are all non-fundamental and still want to ask whether they’re *real*. If there is room to be a realist about some non-fundamental aspects of reality but not others, the relevant realist stance cannot be understood in terms of ground and fundamentality—at least not in the straightforward way we considered earlier. If some grounded non-fundamental entities are real and others are not, their reality must be something beyond their status in the grounding hierarchy. This worry suggests that there is another notion of reality that comes apart from fundamentality as well as from groundedness but nevertheless plays an important role in realist metaphysical inquiry.

Fine (2001) explicates another notion of reality that we might appeal to for these purposes. In particular, Fine distinguishes between fundamentality and *factuality* and takes the latter to also be a basic notion of reality—undefinable in grounding or other more basic terms.¹⁰ For Fine, then, the notion of ground cannot help provide a definition for realism; realism must be understood on its own terms. But Fine maintains that questions of ground can nevertheless help settle questions of realism—questions of realism can be seen as *turning* on certain questions of ground. I turn to the details of Fine’s proposal next.

III. Fine’s (2001) Proposal: Ground as a *Guide* to Realism

Turning to consider how ground might be seen as a guide to realism, we now move to question [A] from our series of questions: How do questions of ground about a particular domain relate to questions of realism about that domain? We should note that had we succeeded in understanding realism in *terms* of ground, there would be nothing left for question [A] to address—the questions of realism could simply be reformulated in terms of questions of ground. But if we maintain that realism is to be understood independently of the notion of ground, there is much left to be worked out regarding the connections between them. Fine’s proposal is a way of addressing question [A], in that it provides a way of thinking about the connection between specific questions of realism and specific questions of ground. In particular, Fine’s focus here is on questions of factuality (rather than fundamentality) and how those relate to surrounding questions of ground.

The central thought behind the proposal is that a disagreement between a realist and an anti-realist about the factuality of some claim or domain of discourse will generally *come along with* a disagreement about a related question of ground. Identifying this question of ground could thus be a way of clarifying what is at issue and how the debate might be resolved. Consider, for example, the moral domain and claims about wrongness—e.g., that torturing people for

fun is wrong. A realist and an antirealist about morality will disagree about the factuality of this claim—the realist maintaining that it's factual, the anti-realist maintaining that it's not. Fine suggests that such a disagreement can turn on another disagreement, about the ground for a claim that both parties agree is *factual*. In this case, that claim might be "Clara *believes* that torturing people for fun is wrong." Both the realist and the anti-realist take this claim to be factual, but the realist takes it to be partially grounded in facts about *wrongness*, while for the antirealist no such property is a constituent of the claim's ground.

More generally, Fine provides a kind of recipe for arriving at a disagreement about the ground for an agreed-to-be-factual claim, given a disagreement about factuality. The procedure relies on a number of important assumptions about factuality and ground which I won't spell out explicitly here; rather, I'll present a rough and intuitive formulation of the proposed methodology and then turn to some objections in the literature which pinpoint some of the assumptions behind the proposal that might be questioned.

Roughly speaking, then, Fine's proposed methodology proceeds as follows: One begins by identifying the grounds for the claim whose factuality is in question—i.e., identifying the *basic* propositions which ground the relevant claim. In case the realist and the anti-realist disagree here, this already constitutes a disagreement about what grounds what, which can be used to clarify and potentially progress the debate. But there needn't be a disagreement between the realist and the antirealist about grounds at this point: both might agree on the basic grounds for the claim whose factuality is in question. What they will disagree on, however, is the factuality of these basic grounds. There will presumably be at least one proposition among the agreed-upon grounds which the anti-realist takes to be non-factual and the realist takes to be factual (since the factual is grounded in the factual and the non-factual cannot be grounded solely in what is factual). Given the basic proposition whose factuality is now in question, Fine then argues that there will (plausibly) be a *constituent* of this basic proposition which the anti-realist takes to be nonfactual and the realist takes to be factual.¹¹ Having identified this constituent (which in the example sketched earlier would be *wrongness*), one can now find an agreed-to-be *factual* claim which essentially contains this constituent.¹² Finally, Fine argues that the realist and the anti-realist will disagree about the *grounds* for such an agreed-to-be factual claim: The realist will take it to be grounded in a proposition containing the disputed constituent, while the antirealist will not, given that she takes the constituent to be non-factual.

This procedure thus takes a disagreement about factuality and uncovers a corresponding disagreement about ground, which can clarify what is at stake between the two positions, and pinpoint where progress on the issue might be made. A methodology of this sort would thus be of great use in debates between realists and anti-realist where—as is often the case—the issues at stake have not been fully clarified, and the two sides are accused of speaking past each other without identifying a precise point of disagreement.^{13, 14}

Nevertheless, a number of objections have been raised for Fine's methodological proposal. In particular, these objections have taken the form of apparent counter-examples to Fine's criteria for distinguishing between realist and anti-realist views by way of ground. Paul Horwich (2007) and Emad H. Atiq (2018) have raised several apparent counter-examples of this sort, particularly targeting Fine's assumption about how the realist and anti-realist will ground an agreed-to-be-factual proposition with a disputed constituent (i.e., a constituent that the realist takes to be factual and the antirealist takes to be non-factual). Fine assumes that a realist with respect to a given *constituent* of a proposition will take that constituent to play a role in the grounds for the proposition, while an anti-realist will not—e.g., that for a realist about wrongness, a claim essentially containing wrongness will be grounded in facts about *wrongness*, while for an antirealist,

the claim will be grounded without appeal to any such facts. But both Horwich and Atiq raise possible ways in which this assumption might fail to hold.

Horwich (2007) brings up the possibility of what he calls ‘holistic grounding’—whereby each constituent of a proposition is not grounded individually but together with other components of the proposition.¹⁵ In such cases, it may not be right to assume that a disagreement about the factuality of a particular constituent will give rise to a disagreement about the ground of the overall proposition. For example, a realist and anti-realist about wrongness might *agree* on the grounds for a claim such as “Clara believes that torturing people for fun is wrong” if the claim is grounded holistically and not via each of its constituents individually.¹⁶ In that case, the dispute could no longer be taken to turn on a question of ground in the way Fine suggests.

Targeting the same assumption, Atiq (2018) raises some cases of positions which we would intuitively classify as realist but which don’t qualify as such according to Fine’s criterion. One such case is that of a logical Platonist, who takes logical operators to be fundamental entities but accepts the standard grounding claim that disjunctions are grounded in their true disjuncts.¹⁷ On such a view, any disjunction essentially contains a fundamental constituent—namely, the logical operator—which nevertheless needn’t figure in its grounds. So the Platonist would be taking disjunctive propositions to be grounded in a way that doesn’t appeal to disjunction itself. But this, Atiq argues, doesn’t thereby make her an anti-realist about disjunction—especially given that she likely takes some propositions about disjunction itself to be fundamental. It appears, then, that one can be a realist about some (essential) constituent of a (factual) proposition and nevertheless not take it to figure in that proposition’s grounds. This could again give rise to a case in which a realist and anti-realist agree on the grounds for a claim containing a disputed constituent, and thus Fine’s methodology will not have drawn a clear line between realism and anti-realism as we’d hoped.

Fine may be able to get around some or all of the apparent counter-examples that have been raised and has responded to the details of Horwich’s critique in Fine (2007). But I will leave aside the details of how these particular cases might be treated and focus on Fine’s (2007) more general response to the kind of criticism that has been raised here: that independently of whether the precise proposal he has offered is successful as it stands, the general *approach* of the methodology might still be right. That is, even if some details of Fine’s proposal must be tweaked, there may be a successful way in the vicinity, which could provide an alternate route to seeing questions of realism as turning on questions of ground.

We can thus take the proposal as a more general suggestion for our purposes here, in thinking about the kinds of methodological and conceptual links there may be between realism and ground. The heart of Fine’s suggestion is that questions of realism can be seen as *turning* on questions of ground, despite the fact that they are understood on independent terms. A realist stance in a given domain will typically come along with certain commitments about what grounds what, and clarifying these commitments can limn the underlying issues at the heart of the debate. Even if we don’t understand realism in *terms* of ground, then, questions of ground can be seen as central and even essential to making progress in realist versus anti-realist disputes.¹⁸

One might worry though that there is something deeper behind the criticisms of Fine’s proposal; perhaps there is reason to think that *in principle* questions of ground can’t help us get to the bottom of questions of realism. There are two kinds of reasons that might stand behind such a worry. The first is a kind of skepticism or quietism about questions of realism in general, along the lines of which Horwich (2007) concludes his critique—if one is skeptical that we can make sense of questions of realism altogether, one will be skeptical that there is any particular methodology involving questions of ground that can help.¹⁹ As I said at the outset, I’m putting this sort of concern aside, at least for the time being. But there’s another kind of concern that

could be behind a skepticism about the prospects of a successful methodology along the lines of Fine's proposal, and that's a concern about the questions of ground themselves. One might worry that although there are meaningful questions of realism, the questions of ground are not sufficiently 'reality oriented' to be of help. This brings us to our last question [C]: Should we be realists about ground *itself*, and what are the implications of each stance?

IV. Realism About Ground

As we saw at the outset, the question of realism with respect to ground can take two forms. First, we can ask whether the notion of ground picks out some distinctive structure or relation in the world.²⁰ And relatedly, *must* it pick out some such real structure in order to be of use in metaphysics? Second, we can ask whether claims of ground are *factual* or should be understood as having objective, realist import. And (relatedly, and perhaps more importantly) *must* they be factual in order to be of use as the grounding theorist maintains? I'll briefly address each of these issues in turn.

First, does the notion of ground pick out some distinctive structure or relation in the world? It's important to distinguish between two prevalent ways of thinking about ground here, which I sidestepped in my introduction to the issues. On one approach, the notion of ground is taken to be (primarily) an *explanatory* one—to say that A is grounded in B is to provide a metaphysical or constitutive *explanation* of A and not (necessarily) to ascribe any particular structure to reality.²¹ On this approach, it appears that the question of whether there is a genuine *relation* in the world that underlies the grounding explanation is left open: there may or may not be genuine grounding structure in reality corresponding to the explanations the metaphysician provides in terms of ground. On the second way of thinking about ground, however, the notion is primarily a *worldly* one: ground is a *relation* between facts or objects that genuinely structures reality and that underlies the corresponding grounding explanations.²²

At least in principle though, both of these approaches might be accompanied by either realism or anti-realism about ground. Some (e.g., Audi 2012) have argued that there must be a worldly grounding relation that underlies grounding explanations;²³ others might speak in terms of ground as though it is a worldly relation between facts and nevertheless not interpret these claims in an ontologically committal way.²⁴

Further, it's not obvious what it would *take* for there to be distinctive grounding structure in the world, i.e., what kind of relation or relation such a metaphysical commitment would involve. Determining the kind of structure that realism about ground demands of reality requires substantial (meta-)metaphysical inquiry—both in clarifying precisely what kind of structure is involved and in clarifying what it would mean to genuinely attribute this structure to reality. This of course goes back to the meta-metaphysical question of realism discussed earlier: What exactly does it *mean* to attribute reality to a fact, object, or phenomenon?

The issue is also closely related to that of meta-ground: What *grounds* ground—that is, how are the claims of ground themselves grounded?²⁵ If they are grounded rather than fundamental, and we understand the relevant question of realism in terms of fundamentality, the answer to this question of realism will be *no* (i.e., anti-realism about ground will follow from its non-fundamentality). But there are various approaches to the question of meta-ground, and one might wish to take (some) facts about how things are grounded to be fundamental.²⁶ We've also seen that the question of realism might come apart from that of fundamentality—perhaps ground *really* structures reality, even though it's not fundamental. Thus the question of realism about worldly ground is left open and depends in part on how questions of realism and fundamentality are themselves ultimately understood.

What unifies grounding theorists, then, is not the metaphysical commitment to worldly grounding structure or any particular metaphysical treatment of ground. Rather, it is a methodological commitment to the *usefulness* of the notion of ground in (meta-)metaphysical inquiry.²⁷ Shamik Dasgupta (2017) helpfully distinguishes between this methodological commitment and any kind of realism about ground, arguing that one needn't be a realist about ground in any substantial sense in order to take the notion of ground to be useful or even central to metaphysical inquiry.

This brings us to the question of factuality: should, or must, grounding claims be taken as *factual* or as having objective, realist import? Dasgupta (2017) argues that factuality is also not a requirement of methodologically useful ground—i.e., that grounding claims needn't be taken as factual in order to play the role that grounding theorists have wanted them to play. In particular, Dasgupta argues that claims of ground might not be mind or culture independent in any robust sense, and nevertheless be central and useful in metaphysical inquiry and debate. One can be committed to the methodological usefulness of the notion of ground, and use it to clarify metaphysical issues as well as provide metaphysically illuminating explanations, all the while not taking these claims to have objective realist import.

One might find this position surprising and wonder whether it also entails a kind of anti-realism about metaphysics as a whole: if the central work of metaphysics is done through non-factual discourse, how can metaphysics (more generally) be factual? Perhaps the stance reflects a position that falls somewhere between full-fledged realism about metaphysics on the one hand and quietism or total skepticism on the other. One might take metaphysical inquiry to be a useful and meaningful endeavor that helps us reach important truths which are nevertheless not ultimately as objective or factual as the realist maintains.

But one needn't reject full-fledged realism about metaphysics in order to be an anti-realist about ground in the In the sense just described” 蔞. That is, one can take claims of ground to be useful but non-factual and nevertheless take metaphysical inquiry as a whole to be concerned with ultimate reality, i.e., with how reality is in itself, independently of our interests or concerns. But this, it seems, would mean understanding realism as only and ultimately concerning the *fundamental*, or taking Fine's notion of Reality—reality as it is in itself—to be the metaphysician's ultimate concern. Questions of how exactly things are grounded in this fundamental reality and of what there is in *addition* to the fundamental would ultimately be rendered non-factual.²⁸

Interestingly, such a position would take us back to a variant of the original suggestion we considered in Section II—on which the questions of realism are to be understood in terms of fundamentality, with no further substantive questions (of realism) to be asked. But here, not only is fundamentality not understood in *terms* of ground, the questions of ground themselves (i.e., of how non-fundamental aspects of reality are grounded in the fundamental) are taken in an anti-realist vein: they may be important and useful, even essential to our concerns, and nevertheless ultimately non-factual. Further, even this statement of non-factuality may be taken in an anti-realist spirit! If the questions of realism are ultimately understood in terms of fundamentality, then there is no further substantive question of how the relationship between the non-fundamental and the fundamental should *really* be understood.²⁹ The only deep, substantive questions are about fundamental reality, *reality as it is in itself*, and—somewhat paradoxically, though also quite naturally—our attempts to explain anything else in *terms* of the fundamental fall short of capturing how things *really* (fundamentally) are.

It appears, then, that one can be an anti-realist about ground while being a realist about metaphysics more generally and nevertheless take questions of ground to be useful or even central to metaphysical inquiry and debate. But clarifying precisely what methodological role ground would play on such an approach is beyond the scope of this chapter. For our purposes here, we

can conclude that a number of positions are left open for the grounding theorist: A grounding theorist—who is methodologically committed to the usefulness of ground in metaphysical inquiry—can be a realist, an anti-realist, or neutral about the reality or factuality of ground.

V. Conclusion

We've seen, then, that there are various ways in which we might see questions of ground and questions of realism as fitting together. While it is clear that these questions are tightly connected, the precise nature of the relationship between them is still far from obvious and will take significant further inquiry to fully clarify. Perhaps we can embrace a meta-version of Fine's methodological proposal: While there is no necessary or conceptual connection between questions of realism and questions of ground that will automatically clarify the relationship between them, it appears that each way of understanding realism will come along with certain possible ways of understanding ground and rule out others. The converse holds as well: Particular conceptions of ground will come along with certain ways of understanding realism. Clarifying each (meta)metaphysical theory's commitments with respect to one of these notions will thus involve, and can progress, via, a clarification of its commitments with respect to the other. While this is short of a precise methodological recipe, the spirit of the proposal is clear—perhaps even unavoidable—given everything we've seen above: the notions of ground and realism are tightly connected in ways that each position in (meta-)metaphysics must clarify out for itself.

Related Topics

- Bliss, R. (Chapter 23, this volume) "Fundamentality."
Glazier, M. (Chapter 8, this volume) "Explanation."
Saenz, N. B. (Chapter 25, this volume) "Ontology."

Notes

- 1 Grounding claims can also be formulated in terms of a sentence operator, rather than in terms of a relation between objects or facts. In formulating grounding claims in terms of a relation between facts as I do here, one needn't be committed to a genuinely layered reality of facts or objects or to a worldly relation of ground. The discussion to follow will be neutral on this issue until I return to discuss it explicitly in Section IV.
- 2 For an introduction to the notion of ground, see Fine (2001, 2012), Rosen (2010), and Schaffer (2009).
- 3 In particular, this has changed substantially over the last years—contemporary ways of understanding realism differ from those of the 'linguistic turn' of the mid-20th century, which took realism to be primarily concerned with linguistic meaning, as e.g., in Dummett (1958, 1976). Contemporary metaphysics has turned away from the focus on thought and language to a focus on the world, and conceptions of realism have changed accordingly. My focus in this chapter will be on contemporary understandings of realism (as this is the context where questions of ground most naturally arise and fit in), but it's important to note that these approaches have substantially evolved over the last half century. See, for example, Papineau (1987) and Wright (1992) for earlier understandings of realism, different from the more contemporary approaches I focus on here.
- 4 This raises a question about the conceptual coherence of the 'anti-realist' stance about questions of factuality: how can it be a factual matter that questions of factuality are to be deflated? One can take a general quietist or deflationary stance on these issues, but this will include a kind of deflated understanding of the stance itself. See Fine (2001: 12–13) for a discussion of this tension.
- 5 Fine (2001) addresses the concern of this fourth sort of 'anti-realist,' which he calls the 'quietist challenge.' Fine distinguishes between what he calls the conceptual and the methodological quietist: The conceptual quietist questions whether there is a real, deep distinction to be drawn between realism and

- anti-realism in metaphysics, while the methodological quietist worries that the answers to questions of realism simply cannot be determined. Fine raises some considerations against the conceptual quietist but primarily focuses on responding to the methodological worry by proposing a methodology (involving ground) for making progress in resolving realist versus anti-realist disputes. Fine's proposal will be discussed in depth in Section III.
- 6 See Saenz (Chapter 25, this volume) on the more general question of how ground might be related to questions of ontology.
 - 7 See Bliss (Chapter 23, this volume) for a discussion of the various ways in which the notion of fundamentality itself can be understood, only a subset of which I focus on here.
 - 8 This is also not the only way of understanding fundamentality in terms of ground. See, e.g., Raven (2016) for an alternative, on which fundamentality comes apart from ungroundedness.
 - 9 One might think that with Fine's notion of fundamentality, or with any conception of fundamentality on which fundamentality comes apart from ungroundedness (such as Raven's 2016), this worry is avoided—if the grounded can also be fundamental, this allows for grounded but nevertheless *real* entities and/or facts. But to some extent, the difficulty can be seen as applying to these views as well: Whatever one's conception of fundamentality, there is an intuitive sense in which the non-fundamental can also be *real*. One might nevertheless take these views as steps towards addressing the worry, in that the intuition is arguably weaker once the assumption that only the ungrounded can be fundamental is rejected. As we'll see in what follows, however, Fine himself does accept an additional notion of reality that admits of the non-fundamental in his sense.
 - 10 There are some complications which must be clarified here: On Fine's picture, factuality can be understood in terms of *reality* (Fine's positive conception of fundamentality) together with ground—as Fine explains, “any true factual proposition is real or grounded in what is real” (2001: 26). But Fine's notion of reality also comes apart from other familiar conceptions of fundamentality. Not only is this notion not understood in *terms* of ungroundedness (as we saw earlier), it can also in fact come apart from ungroundedness: A factual proposition can be grounded and nevertheless also *real*. So some (but not all) of what is grounded in the real is also itself real, and all of what is grounded in the real is factual. The non-factual, on the other hand, is grounded in what is *basic* but not real. I won't focus on these details of Fine's picture in what follows; what's important for our purposes is just the idea that realism is not to be defined (solely) in terms of grounding structure and that what is real might itself also be grounded.
 - 11 As Fine explains, “Intuitively, a nonfactual constituent is one which can be a source of non-factuality in a proposition to which it belongs” (2001: 17, note 25).
 - 12 For a proposition to contain a constituent essentially is for its truth-value to depend (in part) on the constituent in question; i.e., such that replacing the constituent with another could result in a change in truth-value (see Fine 2001: 17–18).
 - 13 This is precisely where the quietist's deflationary worries about realist versus anti-realist disputes often arise. See, e.g., McDowell (1994), Rosen (1994), and Horwich (2006, 2007).
 - 14 For an attempt to apply Fine's proposed methodology, see Dreier (2004), who utilizes Fine's proposal in distinguishing between realism from a kind of quasi-realism in meta-ethics.
 - 15 It's important to distinguish between this kind of holistic grounding and what Dasgupta (2014a) calls *plural* grounding—whereby a proposition is not grounded individually but as part of a plurality. The possibility of plural ground may also raise difficulties for Fine's methodological proposal as it stands, but it doesn't obviously threaten the assumption that Horwich targets here.
 - 16 I won't go further into the complexities of Horwich's argument here; he develops more abstract and formal examples in which this kind of holistic grounding may be plausible.
 - 17 See McSweeney (Chapter 32, this volume) on the grounding of logically complex facts.
 - 18 There is a related question I haven't asked here, which is the question of whether debates about ground could in turn or instead be illuminated by first settling questions of realism. Of course, if we had an independent methodology for settling questions of realism, this could help settle questions of ground: What's real must be ungrounded or grounded in the fundamental (at least on Fine's picture, as well as according to an intuitive conception of these notions.) The unreal, on the other hand, must be grounded in a way that makes clear why it's unreal. But the assumption behind Fine's proposal, as well as behind my focus on the other direction of inquiry (i.e., going from ground to realism), is that questions of ground are in some sense easier to settle than questions of realism. As Fine (2001) explains, we can see questions of realism as being the *end* of metaphysical inquiry with respect to some domain, with questions of ground coming in earlier and playing a role in our getting to that end.

- 19 Of course, one might also be skeptical about the questions of realism *because* one is skeptical about the prospect of developing a methodology that would resolve them; but one's reasons for rejecting a particular methodological proposal would then likely be specific to the details of the methodology in question.
- 20 See Tahko (Chapter 27, this volume) on the notion of structure and its relation to ground.
- 21 For explanatory conceptions of ground, see Dasgupta (2014b), Fine (2001, 2012), and Litland (2017).
- 22 For conceptions of ground as a worldly relation, see Audi (2012), Rosen (2010), and Schaffer (2009, 2012).
- 23 But see, e.g., deRosset (2017) for a way of taking something close to the explanatory notion of ground as describing the relationship between truths but not as picking out a genuine relation that holds between facts.
- 24 See Glazier (Chapter 8, this volume) for a more thorough discussion of explanatory versus worldly conceptions of ground.
- 25 See Litland (Chapter 9, this volume) for an in-depth treatment of meta-ground.
- 26 See, e.g., Glazier (2016) for a view on which there are fundamental 'laws of metaphysics,' which account for the connections between fundamental and non-fundamental aspects of reality. See also deRosset (2013) and Sider (2012, ch. 8) for some concerns about views of this sort, and Wilsch (Chapter 30, this volume) for further discussion.
- 27 Of course there are also grounding skeptics, who theorize about ground but ultimately reject this methodological claim (see, e.g., Wilson 2014; Koslicki 2015). See Koslicki (Chapter 11, this volume) for an overview of the skeptical challenges for the grounding theorist and deRosset (Chapter 12, this volume) for responses on the grounding theorist's behalf.
- 28 I'm setting aside grounding skeptics and other non-grounding-theorists here; this is meant to be a way for the *grounding* theorist (who is committed to the methodological centrality of ground) to be an anti-factualist about grounding claims.
- 29 I'm equating substantivity with a realist conception of factuality here; one might resist this, but the terminology is not essential to the picture I've sketched.

Suggestions for Further Reading

- Dasgupta, S. (2017) "Constitutive Explanation," *Philosophical Issues* 27(1): 74–97.
 Fine, K. (2001) "The Question of Realism," *Philosophers' Imprint* 1(1): 1–30.
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 Bliss, R. (this volume) "Fundamentality."
 Dasgupta, S. (2014a) "On the Plurality of Grounds," *Philosophers' Imprint* 14(20): 1–28.
 ——— (2014b) "The Possibility of Physicalism," *Journal of Philosophy* 111(9–10): 557–592.
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 deRosset, L. (2013) "Grounding Explanations," *Philosophers' Imprint* 13: 1–26.
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STRUCTURE

Tuomas E. Tahko

1. What Is Structure?

The notion of *structure* is often invoked in connection to ground. By way of example, it is telling that the subtitle of Fabrice Correia's and Benjamin Schnieder's 2012 volume on *Metaphysical Grounding* is *Understanding the Structure of Reality*, and the 2018 volume on fundamentality by Ricki Bliss and Graham Priest is called *Reality and Its Structure*. In general, grounding is often understood to impose constraints on the 'structure of reality'. Specifically, grounding is thought to impose an asymmetric, transitive, and irreflexive order on reality, giving rise to a hierarchical structure.¹

However, structure, as it is used in the titles of the mentioned volumes and also more generally in the grounding literature, is not always intended as a technical notion. Rather, structure simply refers to the idea that reality has heterogeneous content, which can be systematised in terms of grounding. There is another, technical sense of structure, which is known especially from Ted Sider's 2011 book *Writing the Book of the World*. Sider thinks that metaphysics is about the fundamental structure of reality and discerning that structure means discerning *patterns*. The idea is now very often captured with reference to the notion of *carving reality at its joints* or *joint-carving*, which originates in Plato. Following this line of thought, we are looking for the most fundamental distinctions that we may use to categorise reality.²

In this chapter, I will discuss both of these senses of structure as well as their potential connections. I will first briefly outline a potential connection between ground, structure, and fundamentality. Next we take up the idea that reality has a hierarchical structure and we will then examine whether there could be a systematic connection between ground and structure via the Lewisian notion of *naturalness*. This leads us to a question about the relationship between reality and representation, before concluding with a more detailed discussion about a famous principle regarding fundamentality, namely, the principle of *purity*.

2. Ground, Structure, and Fundamentality

It's clear that the notion of ground is somehow related to the idea of structure: the most fundamental facts in nature are presumably those that cannot be grounded in other, more fundamental facts.³ On this view, structure is fundamental or describes what reality is fundamentally

like; it concerns ungrounded facts about reality.⁴ While the idea that structure is about what is *absolutely fundamental* or ungrounded is common, we should also note that the notion of *relative fundamentality* is of importance. If reality is hierarchically structured, it means that some things are less fundamental than others, and it is often thought that grounding gives us an ordering of this hierarchical structure—an ordering of relative fundamentality. So the notion of structure could also be used to refer to the idea that there are more and less fundamental facts in the world. Sometimes this is put in terms of *levels of reality*, whereby the ungrounded things are at the absolutely fundamental level and the less fundamental things are on higher levels and ultimately grounded in the absolutely fundamental.

There is an initial challenge that may be raised for the idea of joint-carving facts as the most fundamental ones (this is considered by Sider 2011: 3). Say that we have discovered that fermions and bosons are both fundamental objects. Assuming that facts constructed from fundamental constituents are themselves fundamental, it follows that the fact that fermions exist and the fact that bosons exist both carve at the joints. But one might insist that there is nevertheless something that fermions and bosons share, namely, they both have the feature of *being either a fermion or a boson*. If this disjunctive property is a genuine feature of reality, then the facts about the existence of fermions and bosons are not, after all, fundamental ones, because there appears to be a more fundamental fact that they have in common: the fact that either fermions exist or bosons exist (or both). The worry is that it is not clear whether we should privilege the disjunctive property or the disjuncts. Indeed, by using logical operators such as disjunction, we can postulate many other combinations of properties, and if we have no principled reason to favour one way of carving at the joints over another, then we will never find the joint-carving facts.

A possible reply is that since the problematic feature in question, *being either a fermion or a boson*, is disjunctive, one might think that disjunction and other logical operators are just features of *language*, and we are here bringing two fundamental features of reality together at the level of *representation* rather than at the ‘worldly’ level, the level of *reality*.⁵ But what reasons do we have for thinking that the difference between *being a boson* and *being either a boson or a fermion* is objective in the sense that one describes reality’s structure and the other one doesn’t? The difficulty, then, is to pick out the genuinely fundamental features of reality, those that are genuinely part of the world’s structure. There is a natural analogy here with grounding, since it is commonly thought that the disjuncts *f* and *g* are more fundamental than the disjunction (*f* $\vee$ *g*); in other words, disjunctions are grounded in their disjuncts. But what is the role of disjunction in this? Is it also part of what grounds the disjunction or not? We will return to this issue in Section 5, but let us first examine the underlying idea of a hierarchical structure of reality in more detail.

3. Hierarchically Structured Levels of Reality

The idea that reality is *layered* or has a hierarchical structure of *levels* predates the contemporary grounding literature. It was, until quite recently, common to conceive of this structure in terms of *supervenience* (see, e.g., Schaffer 2003, 2004), whereby psychological properties supervene on biological properties, which in turn supervene on chemical properties, and ultimately everything supervenes on physical properties.⁶ The ultimate basis of all reality—a *supervenience base*—are the physical properties. The notion of a fundamental supervenience base was popularised by David Lewis (1986), but the idea of a hierarchical structure of levels is familiar already from the old ideal of a *Unity of Science* (Oppenheim and Putnam 1958).

In their famous paper, Oppenheim and Putnam suggested that by properly employing the notion of reduction, we can reach a *Unity of Science*. This was thought to be an empirical hypothesis, aiming for a full ontological reduction of all the laws of the special sciences into the most

elementary elements. However, this ideal turned out to be too strong. It aimed for a transitive, irreflexive, and asymmetric *micro-reduction* from the higher levels to the lower levels, producing a system of reductive levels, which is finite and has a unique lowest level, a common denominator for each level.

More recent work on levels tends to eschew strong versions of reductionism, at least in the more extreme form that Oppenheim and Putnam recommended. But this means that the question about the relationship between the different levels as well as the status of the properties on those levels becomes acute. In particular, and related to the question that emerged in the previous section, do we draw the joint-carving properties from all the levels or just from the fundamental level? Jonathan Schaffer (2004: 92) distinguishes two ways to go here, the *scientific* conception and the *fundamental* conception, where the first draws the properties from all 'scientific' levels and the second only from the fundamental level. On the latter view, *being a fermion* would likely qualify as a joint-carving property, but *being a proton* might not (since it supervenes on its composite parts).

Schaffer argues that we should draw the joint-carving properties from all scientific levels. He gives two primary reasons. First, the competing, fundamental conception would face insurmountable problems if it turned out that there is *no* fundamental level. On the Lewisian view of *naturalness*, the important criteria involve the ability to provide a basis for objective similarities and causal powers, which ultimately form a fundamental supervenience base. This fundamental supervenience base—the famous Humean mosaic—provides a minimal, non-redundant ontological basis for everything else. Second, Schaffer argues that the scientific conception is a better fit with Lewis's (1986) original theory of *natural* or *sparse* (vs. *abundant*) properties. This is relevant because the idea of a fundamental supervenience base was originally formulated by Lewis in terms of natural properties. We will discuss this theory and its link to structure in the following section.⁷

One of Schaffer's worries with the fundamental conception of naturalness, reflected in the idea of a fundamental supervenience base, is that the minimal basis needs to be found at the fundamental level. But what if there could be an *infinite descent* of the supervenience structure? In that case, there could be no such minimal basis.⁸ So if it is possible that there is an infinite descent of properties in such a way that they are 'endlessly supervenient upon lower-level properties' (Schaffer 2004: 99), then we need to revise the Lewisian requirement of minimality accordingly.

The main upshot of this section is that there seems to be a tension of some sort between the Lewisian project of providing a minimal, fundamental supervenience base consisting of joint-carving natural properties and the idea that the world is hierarchically structured. The details depend at least partly on how we conceive of the idea of joint-carving and whether we think that infinite descent is possible. One possible way forward here is to introduce *degrees of naturalness*, as Lewis (1986: 61) himself does. The thought is that properties could then be ordered according to their relative degree of naturalness, with *perfectly* natural properties at the fundamental end.

4. Ground, Structure, and Naturalness

We have now seen that there is a close connection between the well-known Lewisian notion of naturalness and the notion of structure, so it could be helpful to approach structure via naturalness.⁹ Naturalness is often thought to come in degrees, and *perfectly natural properties* would seem to be good candidates for fundamental, ungrounded entities. The previous examples of *being a boson* and *being a fermion* are likely candidates for perfectly natural properties. Alternatively,

we could put this in terms of ungrounded *facts* as we did in the beginning of this Chapter: the fact that electrons have unit negative charge might be fundamental in this sense. I will remain neutral about the issue regarding whether ground relates facts or any entities whatsoever, but for laying out a connection between grounding and naturalness, the latter approach, as used by Schaffer (2009) is somewhat more promising. This is because if grounding relates only facts and if naturalness applies only to properties, then they would clearly come apart. Indeed, to draw this connection, we will likely have to adopt a liberal understanding of grounding, whereby it can relate any entities whatsoever, and also a liberal understanding of naturalness or structure, so that it can apply also to other entities besides properties.¹⁰

So if structure is connected to naturalness and naturalness could be analysed in terms of grounding, then we would have a clear-cut connection between ground and structure. Sider's notion of structure is in fact 'a generalization and extension of Lewisian naturalness' (Sider 2011: vii). However, it should be noted that it is not Sider's intention to provide a definition of structure or naturalness in terms of grounding or fundamentality. Rather, for him, 'structure' is a primitive, theoretical term, and its intended role is to capture the fundamental structure of reality.

There are some good reasons to think that structure/naturalness and ground must come apart even on a liberal conception of both notions, because the two notions may not be able to do all the same jobs (see Bennett 2017: Chs. 5.7, 5.8). There are two potential connecting principles to consider (cf. Bennett 2017: 127):

- (1) All perfectly natural entities are ungrounded.
- (2) All ungrounded entities are perfectly natural.

Our previous examples would seem to be compatible with (1). Fermions, such as electrons, are thought to have no internal structure, so the fact that electrons exist is a good candidate for being fundamental, ungrounded. What about things like a hydrogen atom? The fact that hydrogen atoms exist is presumably not fundamental, since hydrogen atoms are composed of electrons and protons, and protons as well have internal structure. But it is not entirely clear that *being a hydrogen atom* couldn't be understood as a perfectly natural property. Here we return to Schaffer's scientific conception of naturalness that we discussed in the previous section. Scientific properties like *being a hydrogen atom* or *being a proton* do appear to be objective in the required sense, but their ultimate status will depend on how we conceive of structure/naturalness. As we have already observed, on the traditional, Lewisian understanding of naturalness, *being a hydrogen atom* seems like a good candidate for a perfectly natural property, as hydrogen atoms have distinctive causal powers, and they may also feature in laws of nature. Yet since naturalness comes in degrees and causal powers are features that help us distinguish natural properties from non-natural, abundant properties, it's not clear that *being a hydrogen atom* is indeed *perfectly* natural.

There are trickier cases though. Consider sets (as discussed by Bennett 2017: 128). For Lewis, sets can be perfectly natural; after all, according to Lewis's version of class nominalism, properties are just sets or classes (this is admittedly a rather controversial view). But are sets ungrounded? This doesn't seem to be the case, at least if we understand sets in terms of their members. The set that contains Socrates as its only member could not exist if Socrates did not exist, so a set is plausibly grounded in its members. The same will be true of all sets that have members: the members of the set are, in some sense, prior to the set itself. So this might be one good reason to think that structure/naturalness and ground come apart, at least if one agrees with Lewis's conception of sets.

Regarding (2), one might point out (as Bennett 2017: 128 does) that the Lewisian notion of naturalness is clearly going to fail to respect the condition, insofar as entities other than properties, predicates, and sets can be ungrounded. For Lewis, the domain of naturalness only contains these types of entities, so entities such as electrons could not be natural (only properties such as *being an electron* are candidates).

However, this doesn't necessarily apply to Sider's notion of structure. Recall that Sider's notion of structure is a generalisation of Lewisian naturalness, with the aim of capturing those *expressions* that carve at the joints (Sider 2011: vii). The question, then, is how we should understand this idea. Are the expressions themselves joint-carving, or do they refer to something in reality—structure—that could contain things other than just properties, predicates, and sets? The answer may seem obvious: of course it's reality itself that we are interested in here rather than just the expressions. But the issue may be more complicated than it first seems, as will become clear in the next section.

A related issue, as discussed in the previous section, is Schaffer's (2004: 100) proposal to replace the minimality qualification for naturalness with the primacy qualification, whereby we should only require that natural properties serve as the ontological basis for propositions. We must ask whether we wish to allow for redundant elements and infinite descent among the joint-carving structure of the world. To do so, we must be able to distinguish between *reality* and *representation*.

5. Reality and Representation

The question about the relationship between reality and expression or representation was a central topic in a book symposium on Sider's *Writing the Book of the World*. Kit Fine has made these two interpretations explicit. The issue can be illuminated via Fine's distinction between what he calls the *E-project* (for 'expression') and the *D-project* (for 'description'):

The E-project is concerned with *saying* what can be said in the most fundamental terms, while the D-project is concerned with *describing* what can be described in the most fundamental terms.

(Fine 2013: 730)

The focus of the E-project is on sub-propositional constituents of propositions, whereas the D-project is concerned with propositions which we take to describe reality. Consider the disjunction $(f \vee g)$ of the facts f and g . Presumably, the *worldly constituents* of this fact will always be either f or g or both, but it will not involve the disjunction operator, $\vee$, itself. So, from the point of view of the D-project, the disjunction might very well be dispensable. We might compare this to a discussion familiar from the grounding literature, on so-called worldly facts. This type of view is not uncommon among grounding theorists, here's Paul Audi's take on the view:

[A] fact is something's or some things' having properties or standing in relations [. . .] Facts are particulars, and are individuated by their worldly constituents (objects, properties, relations) and the manner of their combination.

(Audi 2012: 686)

The worldly constituents of $(f \vee g)$ could be things such as the object electron, the property of having a unit negative charge, or relations like being more massive than or being 2 miles

apart. But they do not obviously involve disjunction. Or do they? This is a question where the E-project and the D-project may get different results. Similarly, when the conjunction ($f \& g$) is stripped down to its worldly constituents, we may just end up with the facts f and g but not the conjunction operator, $\&$. The underlying issue, once again, is how to interpret structure. It may be the case that the disjunction operator is indispensable in the E-project, but does that make it part of reality's structure? As Fine himself puts it:

The E-fundamentals are likely to include the logical constants, for how else is one to say something negative or disjunctive or existential?

(Fine 2013: 730)

So the question is whether the notion of structure is to be understood in terms of the E-project or the D-project, and Fine's main complaint is that it's not at all clear which project Sider is involved with. Fine even claims that Sider sometimes seems to take it for granted that logical constants such as $\vee$ or $\&$ are D-fundamental without giving any reasons to think that they are necessary in the relevant sense, i.e., for describing (rather than just expressing) the mind- and language-independent world (Fine 2013: 731).

Sider has provided a reply to Fine's concerns. Sider (2013: 739) explicitly states that his approach to fundamentality is sub-propositional (whereas Fine's is propositional). This would appear to put Sider's project within the remit of the E-project, but he doesn't seem to be entirely happy with this result. Regarding Fine's distinction between the E-project and the D-project, Sider acknowledges that it is illuminating but considers one of his core assumptions to be the rejection of this distinction, or at least 'denying that it has the significance Fine thinks it has' (Sider 2013: 742). Sider attempts to put pressure on Fine's idea by insisting that, say, if disjunction is indeed E-fundamental, then 'a complete description of the world' must also include the true disjunctions.

To summarise, we are again faced with the same problem that we observed already in previous sections (compare with Schaffer 2004), namely, how do we decide what the minimal basis of reality is, what are the minimal elements required for 'a complete description of the world'? The debate between Sider and Fine, then, turns on how we should understand 'a complete description of the world'. Fine's idea seems to be that only the D-project is concerned with 'a complete description of the world'. It may be that there are things that we cannot express, say, without the use of disjunction, but this does not make disjunction D-fundamental.

Finally, we can introduce the question of *redundancy* at the fundamental level. Sider (2011: 259–260) discusses this condition. Relevant examples of redundant properties include relations and their converses, such as 'earlier' and 'later', as well as the fact that several subsets of the logical operators are functionally complete (e.g., conjunction and negation, or the Sheffer stroke). A potential motivation to avoid redundancy is the theoretical cost of redundant properties. However, Sider seems to think that redundancy may be acceptable at least in some cases. Explicit discussions of this issue in the literature are still scarce, but Michaela McSweeney (2019) puts forward a clear defence of non-redundancy. However, she weakens the principle to what she calls *Weak Non-Redundancy*: 'There are no unexplained necessary connections between fundamental facts'.¹¹ So one issue that may influence our interpretation of just what counts as 'a complete description of the world' is the question of which, if any, redundant elements we are prepared to accept.

Let us now move on to one final issue that may shed light on the relationship between ground and structure: the principle of purity.

6. Ground, Structure, and Purity

There is an important principle concerning structure, discussed in detail by Sider (2011: Ch. 7.2), that might give rise to reasons for distinguishing ground and structure:

PURITY: Fundamental facts involve only fundamental notions.

For Sider, ‘fundamental’ means ‘joint-carving’. Sider thinks that on a Finean account of what is fundamental, we end up violating PURITY. That’s because facts such as the following would seem to be perfectly acceptable for the Finean grounding theorist (where C^* is a predicate giving a complete microstructural account of the city of Hamburg, cf. Sider 2011: 108):

(C) The proposition that there is a C^* grounds the proposition that there is a city.

Since the notion of city is presumably not fundamental, PURITY dictates that (C) is not fundamental. The question here concerns whether something needs to ground things like (C) or whether they are themselves fundamental. There is a debate about this issue within the grounding literature. Since (C) is a further fact—part of reality—it must either be fundamental itself or something needs to ground it. If it is fundamental, it violates PURITY, so to maintain a connection between ground and structure, one should deny that (C) is fundamental. But what could possibly ground (C)? This is the problem of *Meta-Ground*, as discussed, e.g., by Shamik Dasgupta (2014) and Jon Erling Litland (2017, 2018).¹² Unless a satisfactory answer can be given, one must either abandon PURITY or admit that grounding and structure are competing notions, each with their own problems.

There is another related challenge, also put forward by Sider (2011: 145–147). This has to do with the idea of infinite descent, which we encountered in Section 3. This time, the question is whether there could be an infinite descent of grounds, such that the proposition p_0 is grounded in p_1 , p_1 is grounded in p_2 , and so on, but with no proposition grounding the whole sequence.¹³ It has been a relatively common idea in the grounding literature that such infinite chains of ground should be dismissed (see, e.g., Cameron 2008), even though this may no longer be the most popular view (consider Bliss 2013). Be that as it may, Sider (ibid.) argues that infinite descent of ground is commonplace, whereas we can satisfy the intuition that such descent should be denied if we adopt a *structure-theoretic* description of reality. One of his examples involves topological considerations:

A friend of ground should describe a continuous curve as being continuous *because* its left and right halves are continuous (and continuously connected). Those halves are themselves continuous because their parts are continuous, and so on. [. . .] [E]ven if space is atomic, provided it is not discrete there is no end to this descent of ground, because there are no “atoms of continuity”—there are no smallest continuous paths.

(Sider 2011: 146)

The thought here is that if we wish to take considerations like this seriously, they seem to require an unacceptable, infinite descent of grounds, whereas Sider thinks that a structure-theoretic account of such phenomena does *not* require this type of problematic infinite descent. This is because, for instance, topological facts may be represented quantificationally by using perfectly fundamental notions (ranging over an infinite collection of points).

It is, however, worth noting that the strength of these types of arguments involving infinite descent depends heavily on whether we consider infinite descent of grounds to be unacceptable in the first place—and even if we do, this might not apply to *all* types of infinite descent. For one thing, if infinite *ascent* is not considered problematic, then it is clearly something about the *direction* of the chain that is problematic. Moreover, infinite chains that nevertheless *terminate* would not seem to be similarly problematic. There has been important progress on these questions in recent work on grounding and fundamentality (see, e.g., Dixon 2016; Rabin and Rabern 2016; Tahko 2018).

7. Concluding Remarks

In conclusion, there are still several open questions about the connections between ground and structure, as well as the related issues of fundamentality, reality versus representation, infinite descent, and principles such as purity. Based on our survey, we can conclude that a straightforward connection with the Lewis-inspired technical notion of structure as used by Sider faces some important challenges, although there clearly is a connection between ground and structure understood more loosely, in the sense of a hierarchical conception of reality.¹⁴

Related Topics

Explanation & Determination: Meta-ground [Chapter 9]; Logic & Structure: Granularity [Chapter 15], Infinite Descent [Chapter 16], Strict Partial Order [Chapter 17]; Connections: Fundamentality [Chapter 23], Modality [Chapter 24]

Notes

- 1 This is known as a *strict partial order*; for discussion, see Naomi Thompson's chapter in this volume [Chapter 17].
- 2 Sider also talks about the structure of 'the world', which we may take to be interchangeable with 'reality' in this context.
- 3 See Ricki Bliss's [Chapter 23] and Scott Dixon's [Chapter 16] chapters in this volume as well as Tahko 2018 for further discussion about fundamentality.
- 4 I have put this in terms of 'facts' for convenience, but the idea of fundamentality itself does not imply a commitment to facts.
- 5 See Fabrice Correia's chapter [Chapter 15] in this volume for a discussion about the distinction between 'worldly' and 'representational' grounding. See also McSweeney 2019 for a discussion about logical constants and joint-carving.
- 6 On the notion of supervenience, see David Mark Kovacs's chapter in this volume [Chapter 24]. As outlined by Kovacs, supervenience may be understood as necessary co-variation: A-entities supervene on B-entities if and only if the A-entities cannot vary without some variation in the B-entities.
- 7 Schaffer (2004: 92) pitches the distinction between sparse or natural and abundant properties as follows: 'The abundant properties provide the semantic values of meaningful predicates, while the sparse properties carve out the joints of nature on which the causal powers hinge'.
- 8 The problem of infinite descent is further discussed in Ricki Bliss's [Chapter 23] and Scott Dixon's [Chapter 16] chapters in this volume, as well as in Tahko 2014 and Raven 2016.
- 9 As we saw in the previous section, the notion of *sparseness* is sometimes used synonymously with *naturalness*; see for instance Schaffer 2004.
- 10 It should be noted though that there may be ways to find a connection also under a fact-based conception of ground, e.g., by taking the fundamental entities to be those that appear in some fundamental fact.
- 11 For cases in favour of some forms of redundancy, see also Sider 1996; Eddon 2013; Raven 2016; Wang 2016.
- 12 See also Jon Erling Litland's entry in this volume [Chapter 9].

- 13 Again, for further discussion on infinite descent, see Ricki Bliss's [Chapter 23] and Scott Dixon's [Chapter 16] chapters in this volume, as well as Tahko 2014 and Raven 2016.
- 14 Thanks to Kit Fine and Mike Raven for comments on earlier drafts of this entry.

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TRUTHMAKING

Kelly Trogdon

1. Introduction

My dog Gomer is friendly. The truth of the claim that Gomer is friendly doesn't 'float free' from how the world is. The claim is somehow made true. There are many questions to ask about this phenomenon, what has come to be called *truthmaking*. For example, does every truth have a truthmaker? What is the connection between truthmakers and the nature of truth? How are truthmaking and ontological commitment related? Let's focus, however, on another question that is arguably conceptually prior to others: just what is truthmaking?

The friendliness of Gomer is also a highly derivative aspect of the world. It's somehow determined and explained by more fundamental phenomena. I assume that talk of determination in this context points to *grounding*, which is the topic of this handbook. My interest in this chapter is how appealing to grounding might help us answer the what-is-truthmaking question.

Before getting into the details, I should note some of the assumptions I'm going to make about grounding and truthmaking. While each of these assumptions has been challenged in the literature, they strike me as reasonable starting points. As for regimentation, I assume that grounding and truthmaking claims are to be formulated with the predicates "grounds" and "makes true" rather than sentential connectives such as "because". As for the ontological status of grounding and truthmaking, I assume that they're relations 'out there' in the domain of our ontology rather being primitive pieces of ideology. As for their relata, I assume that something is a ground, a grounded entity, or a truthmaker only if it's a *fact*, and something is made true only if it's a *proposition*. I assume that facts and propositions are structured entities, where the former are worldly in that they lack concepts or modes of presentation as constituents, and the latter are representational in that they do have such constituents. Collections of one or more facts ground other facts, and collections of one or more facts make true propositions.¹

Following convention, let square brackets indicate facts and angle brackets propositions. Given these assumptions, here are two simple examples of what grounding and truthmaking claims look like: [the ball is burgundy] grounds [the ball is red], and [the ball is burgundy] makes true <the ball is red>. Much of what I say in this entry can be recast with different starting assumptions about grounding and truthmaking, but I leave this as an exercise for the reader.

2. Three Accounts

Here is a common informal characterization of truthmaking: a fact makes a proposition true when the latter is true *in virtue of* the former. Different accounts of the truthmaking relation interpret the in-virtue-of talk here differently. Some, such as Armstrong (2004: Ch. 2), think that it points to metaphysical necessitation. Where Δ is a collection of one or more facts, here is one way of formulating the idea:

Δ makes true $\langle p \rangle$ just in case Δ obtains and it's metaphysically necessary that if Δ obtains then $\langle p \rangle$ is true.

There is, however, a straightforward problem for the necessity account, one concerning necessary truths. Suppose that it's metaphysically necessary that $\langle 1 + 1 = 2 \rangle$ is true. In this case, $\langle 1 + 1 = 2 \rangle$ is true no matter what. So it's true if, say, Beijing is a city. So it's metaphysically necessary that if [Beijing is a city] obtains then $\langle 1 + 1 = 2 \rangle$ is true. In this case the necessity account says that [Beijing is a city] is a truthmaker for $\langle 1 + 1 = 2 \rangle$. But this seems wrong—the facts that anchor the truth of this proposition to the world, whatever they turn out to be, don't have cities as constituents. Why not? Intuitively, truthmakers are relevant to the truth of what they make true—call this the *relevance condition* for truthmaking. [Beijing is a city], however, seems irrelevant to the truth of $\langle 1 + 1 = 2 \rangle$.

Some, such as Lowe (2006: Ch. 12), claim that the in-virtue-of talk in the informal characterization of truthmaking instead points to a sort of dependence involving essences, where the operative notion of essence isn't purely modal in character (see Fine 1994). Here is one way of formulating the idea:

Δ makes true $\langle p \rangle$ just in case Δ obtains and it's essential to $\langle p \rangle$ that if Δ obtains then $\langle p \rangle$ is true.

The essence account is designed to satisfy the relevance condition. In this case, Δ makes true $\langle p \rangle$ only if a connection between Δ and the truth of $\langle p \rangle$ is built into the very nature of the proposition. Relatedly, the essence account doesn't face the same problem with necessary truths. While it's metaphysically necessary that if [Beijing is a city] obtains then $\langle 1 + 1 = 2 \rangle$ is true, the essence of this proposition doesn't involve Beijing—you might know everything there is to know about the nature of $\langle 1 + 1 = 2 \rangle$ without knowing anything about Beijing in particular. So it isn't essential to $\langle 1 + 1 = 2 \rangle$ that if [Beijing is a city] obtains then $\langle 1 + 1 = 2 \rangle$ is true. Hence, the essence account doesn't say that [Beijing is a city] is a truthmaker for $\langle 1 + 1 = 2 \rangle$.

A problem for the essence account, however, concerns the essence claims that it commits us to. Consider a proposition to the effect that something is a city. The fact that Beijing is a city is, we will suppose, a truthmaker for this proposition. So a consequence of the essence account is this: it's essential to $\langle \text{something is a city} \rangle$ that if [Beijing is a city] obtains then $\langle \text{something is a city} \rangle$ is true. But this seems wrong. The essence of this proposition, like the essence of $\langle 1 + 1 = 2 \rangle$, doesn't involve Beijing—you might know everything there is to know about the nature of this proposition without knowing anything about Beijing in particular.

As a potential fix, we might revise the essence account so that it appeals to the natures of truthmakers in addition to the natures of what they make true. Here is one approach along these lines:

Δ makes true $\langle p \rangle$ just in case Δ obtains and it's essential to either $\langle p \rangle$ or facts among Δ that if Δ obtains then $\langle p \rangle$ is true.

Here the thought is that, while it isn't essential to $\langle \text{something is a city} \rangle$ that if [Beijing is a city] obtains then $\langle \text{something is a city} \rangle$ is true, this is essential to [Beijing is a city]. If this is right, then the revised essence account says that [Beijing is a city] is a truthmaker for $\langle \text{something is a city} \rangle$, as it should.

This revised essence account, however, also brings in its train questionable essence claims. Suppose that the liquid in the flask is boiling. And suppose that thus-and-so molecules compose the liquid; they're behaving in thus-and-so way; and their evolution is governed by thus-and-so laws—call this trio of facts MOLECULE. MOLECULE is presumably a truthmaker for $\langle \text{the liquid is boiling} \rangle$. So a consequence of the revised essence account is this: it's essential to either $\langle \text{the liquid is boiling} \rangle$ or facts among MOLECULE that if MOLECULE obtains then $\langle \text{the liquid is boiling} \rangle$ is true. But, on the face of it, the essence of $\langle \text{the liquid is boiling} \rangle$ doesn't involve laws governing the evolution of molecules. And, on the face of it, no sub-collection of MOLECULE is such that its essence involves boiling. So it's apparently essential to neither $\langle \text{the liquid is boiling} \rangle$ nor facts among MOLECULE that if MOLECULE obtains then $\langle \text{the liquid is boiling} \rangle$ is true.²

Perhaps the way to fix this problem is to be even more liberal about what the relevant essential truths might characterize. Returning to the liquid example, one thought is that we should focus on the collection that results from combining $\langle \text{the liquid is boiling} \rangle$ and MOLECULE—it's essential to this collection of entities that if MOLECULE obtains then $\langle \text{the liquid is boiling} \rangle$ is true. But things are getting pretty baroque here, and this essence claim lacks any intuitive appeal. Moreover, pursuing this strategy might lead to overgenerating truthmakers in any case—you might think, for example, that it's essential to the collection consisting of $\langle 1 + 1 = 2 \rangle$ and [Beijing is a city] that if [Beijing is a city] obtains then $\langle 1 + 1 = 2 \rangle$ is true. So we would do well to see what alternative approaches to truthmaking might be available.

Many, including Correia (2014), Fisher (2015), Jago (2018: Ch. 6), Rodríguez-Pereyra (2005, 2015), Rosen (2010), Schnieder (2006), and Tahko (2013), suggest that the in-virtue-of talk in the informal characterization points not directly to necessity or essential dependence but instead to grounding. With this in mind, consider the following simple grounding-theoretic account of truthmaking:

Δ makes true $\langle p \rangle$ just in case Δ grounds the fact that $\langle p \rangle$ is true.³

The grounding account, like the essence account, speaks to the aforementioned relevance condition, as grounds are relevant to what they ground. Relatedly, the grounding account doesn't seem to have a problem with necessary truths, at least in the absence of further argument. While there are many plausible grounding claims, the claim that [Beijing is a city] grounds $\langle 1 + 1 = 2 \rangle$ is true isn't among them. Grounds contribute to explaining what they ground, and clearly the former doesn't contribute to explaining the latter. And it doesn't commit us to implausible essence claims either, at least in the absence of further argument.⁴ Is the grounding account otherwise plausible? There are various potential problems for the account that I don't have the space to discuss here. One concern is that the account vitiates the transitivity of grounding—see Tahko (2013) and Griffith (2014) for discussion. In the next section, I focus on a potential problem that concerns the theoretical role of truthmaking in particular.

Before proceeding, I should say something about an even simpler grounding-theoretic account truthmaking that might have occurred to you, viz. Δ makes $\langle p \rangle$ true just in case Δ grounds [p]. Why not work with something like this instead? Well, the proposal as formulated is difficult to evaluate, provided that propositions have concepts as constituents and facts do

not. For suppose that $[p]$ has property P as a constituent. As there are multiple concepts whose semantic value is P , it's unclear just which conceptually structured entity is supposed to be made true by Δ when Δ grounds $[p]$. There is, however, a similar proposal worth considering that addresses this concern, viz. for any proposition $\langle p \rangle$ that represents $[p]$, Δ makes $\langle p \rangle$ true if and only if Δ grounds $[p]$. But this proposal too faces a difficult question: for any given proposition, just which facts does that proposition represent? Let's therefore continue to focus on the original grounding-theoretic account of truthmaking, one that doesn't face these problems.

3. A Potential Problem

Proponents of truthmaking think that the notion is theoretically useful. So, if an account of truthmaking has the effect of undermining its theoretical utility, this counts against the account from the perspective of the truthmaking theorist. What theoretical role is truthmaking supposed to play? Two potential theoretical applications of truthmaking are helping characterize ontological commitment (Cameron 2008a; Rettler 2016) and truth (Jago 2018). Let's focus, however, on a different potential application, one Sider memorably glosses as that of catching "cheaters", where a cheater is someone who "is unwilling to accept an ontology robust enough to bear the weight of the truths he feels free to invoke" (2001: 41).

How should we understand this evaluative role that truthmaking is supposed to play? Here's a straightforward take on the matter: supposing that the true propositions concerning some subject matter have truthmakers, if a metaphysical thesis about this subject matter leaves us with no plausible story to tell about what makes these propositions true, this counts against the thesis. Armstrong (2004: Ch. 1) claims that truthmaking considerations undermine behaviorism in particular, so let's work through this example. True propositions about behavioral dispositions, we will suppose, have truthmakers. Returning to my friendly dog, functionalists about the mental claim that the property of being friendly is the property of having some property or other that plays a certain causal role, one characterized in terms of internal mental states, sensory inputs, and behavioral outputs. Functionalists point to facts concerning the physical properties that actually play this role as a truthmaker for $\langle \text{Gomer is friendly} \rangle$. In rejecting the existence of internal mental states, however, you might think that the behaviorist is left with no plausible story to tell about what makes this proposition true. With respect to Ryle, who defended a view similar to behaviorism, Armstrong writes, "What is there in the world in virtue of which these [dispositional] truths are true? Ryle had no answer" (2004: 3). If functionalism but not behaviorism has recourse to truthmakers for true propositions concerning behavioral dispositions, then the former is preferable to the latter, all other things being equal. (Armstrong himself endorses a version of the identity theory rather than functionalism as I've characterized it above, identifying mental properties with what the functionalist claims are their physical realizers—see Armstrong 1968: Ch. 6.)

You might think that, given the grounding account, truthmaking isn't up to playing the role of helping us evaluate the plausibility of metaphysical theses. Various facts such as $[\text{Gomer is friendly}]$ ground $[\langle \text{Gomer is friendly} \rangle \text{ is true}]$. Given the grounding account, these facts are truthmakers for the proposition, and the behaviorist is free to appeal to them. Hence, given our take on the evaluative role of truthmaking, truthmaking doesn't help us adjudicate between behaviorism and functionalism after all. And you might think that the same applies to other debates that truthmaking is supposed to help us with, such as the debate between idealists about the physical and physicalists about the mental and the debate between presentists and eternalists about the temporal extent of reality. The thought is that no matter how

implausible a metaphysical thesis about some subject matter might be, the advocate of that thesis won't be without truthmakers for the true propositions concerning that subject matter. Trivial truthmakers abound.⁵

4. Appealing to Fundamentality

How to proceed? One option is to revise the grounding account. The idea is to add further conditions that rule out the sort of trivial truthmaking described earlier. One such approach appeals to the notion of fundamentality, where a fact is fundamental in the relevant sense just in case it isn't grounded:

Δ makes true $\langle p \rangle$ just in case the facts among Δ are fundamental and Δ grounds the fact that $\langle p \rangle$ is true.⁶

Given this proposal, while [Gomer is friendly] grounds [$\langle$ Gomer is friendly $\rangle$ is true], the former isn't a truthmaker for $\langle$ Gomer is friendly $\rangle$, as [Gomer is friendly] isn't fundamental. Both Schaffer (2010) and Cameron (2018) propose grounding-theoretic accounts of truthmaking that appeal to fundamentality so understood. Schaffer in particular claims that all truthmakers are fundamental, as truthmaking is "the relation of grounding between substance and truth", where substances are fundamental entities (310).⁷ And Schaffer's central rationale for this claim is that it resolves the problem of trivial truthmakers outlined above—he claims that "the restriction to fundamental entities is needed if 'cheaters' [such as the behaviorist] are to be caught" (319).

Given our characterization of the evaluative role that truthmaking is supposed to play in metaphysics, it's unclear, however, that the fundamentality account of truthmaking does any better than the original simple grounding account in showing how truthmaking might play this role. Returning to the debate between the behaviorist and functionalist, the fundamentality account likewise undermines the idea that truthmaking is potentially useful here. The facts that the functionalist cites as truthmakers for $\langle$ Gomer is friendly $\rangle$ concern his cognitive economy, which consists of ordinary mental states like his belief that he will get to play fetch soon. The functionalist (and most of everyone else as well) denies that such facts are fundamental.

Moreover, if all truthmakers are fundamental, then it seems that we need to have a good sense of what the fundamental facts are like before truthmaking considerations can provide any guidance in adjudicating between competing theories. Consider, for example, the debate between idealists about the physical and physicalists about the mental. Suppose we have a good grip on what the fundamental facts are like—let's say that we have reason to think that none are mental in nature. The problem is that in this case, we already know which thesis is the more plausible (physicalism), so truthmaking considerations are otiose in this context. Merricks (2007: 35) makes a similar observation about accounts of truthmaking formulated so as to rule out truthmakers that appeal to "suspicious" properties.

Aside from not helping show how truthmaking has an important evaluative role to play in metaphysics, there are other concerns for the fundamentality account. The most obvious problem is implicit in our discussion of functionalism—on the face of it, some propositions have among their truthmakers nonfundamental facts. It seems, for example, that MOLECULE is a truthmaker for $\langle$ the liquid is boiling $\rangle$, yet the facts among MOLECULE aren't fundamental.

Another potential problem for the fundamentality account is that it undermines potential theoretical applications of truthmaking outside the domain of metaphysics. Consider semantic theories cast in terms of truthmaking. The relevant truthmakers in this case are typically non-fundamental in nature—the meaning of "There is a chair over there", for example, is thought to

consist of truthmakers involving ordinary macroscopic objects. So the fundamentality account in effect says that truthmaker semantics is wrong even before we consider any of the details.⁸

Still another potential problem for the fundamentality account concerns its underlying assumptions. For any fact to the effect that a particular proposition is true, let's say that it's a *truthmade* fact just in case the proposition it concerns has a truthmaker. A consequence of the fundamentality account is that all truthmade facts have fundamental grounds. Why think that all such facts are grounded in this way?

Let *metaphysical foundationalism* be the view that all grounded facts are grounded by fundamental facts.⁹ It seems that in the actual world every semantic fact is grounded—there are no fundamental semantic facts. If this is right, then every truthmade fact is grounded. Provided that every truthmade fact is grounded, you might think that every truthmade fact has a fundamental ground because you think that metaphysical foundationalism is true. Indeed, it's unclear how else we might go about motivating this claim.

So what? Well, the fundamentality account aims to specify the nature of truthmaking so it carries metaphysical necessity—the proposal is if true then true in all metaphysically possible worlds. So it looks like we have reason to believe that every truthmade fact has a fundamental ground only if both metaphysical foundationalism and the claim that every semantic fact is grounded are if true then true in all metaphysically possible worlds. It's not absolutely clear, however, why we should think that worlds in which fundamental semantic facts obtain are metaphysically impossible. Compare: while the physicalist claims that no fundamental mental facts actually obtain, she typically grants that worlds in which such facts obtain (e.g., worlds in which our minds are roughly as Descartes conceives of them) are metaphysically possible.¹⁰ And Cameron (2008b) argues that we at most have reason to believe that metaphysical foundationalism is contingently true—while the thesis is actually true, there are metaphysically possible worlds in which it's false.¹¹

5. Resolving the Problem

So far we've considered two grounding-theoretic accounts of truthmaking. And we've seen that truthmaking on neither proposal plays the evaluative role that it's supposed to, given our take on what this role comes to. What are our options for addressing this issue? One option is to revise the grounding account of truthmaking in some way we've yet to consider. For example, Griffith (2014) and Saenz (forthcoming) reject the grounding account but defend grounding-theoretic accounts of truthmaking that don't appeal to fundamentality. Griffith argues that truthmaking is to be defined in terms of a species of grounding rather than grounding per se, and Saenz argues that the grounding account needs to be supplemented with considerations involving the idea that true propositions correspond to their truthmakers. (Neither of their discussions, however, directly engages with the problem concerning the evaluative role of truthmaking that has guided our discussion.)

Another option is to reject the idea that truthmaking is to be defined in terms of grounding in the first place. If you're sympathetic with truthmaking but skeptical of grounding, this option is for you. Heil (forthcoming), for example, articulates a pro-truthmaking, anti-grounding view.¹² Note, however, that the proponent of grounding might also reject grounding theoretic accounts of truthmaking. Audi (2012) and (forthcoming), for example, is a proponent of grounding but proposes a grounding-free take on truthmaking that appeals to the idea that true propositions correspond to obtaining facts. Suppose that A makes true $\langle p \rangle$. Audi suggests that, provided that there are conjunctive facts, the fact that $\langle p \rangle$ is true is conjunctive in nature—its conjuncts are A and the fact that $\langle p \rangle$ corresponds to A. In this case, A grounds the alethic fact,

given the standard grounding principle that conjunctive facts are (partially) grounded by each of their conjuncts. Interestingly, Audi responds by arguing that, if there are conjunctive facts, they *aren't* grounded by their conjuncts.

Still another option is to accept the grounding account of truthmaking (or the fundamental-ity account, provided that its other potential problems can be addressed satisfactorily) but reject the idea that truthmaking has an important evaluative role to play in metaphysics. For example, you might argue that grounding *tout court* is important to metaphysics, and truthmaking as a special case of grounding is useful only outside the domain of metaphysics. While Fine doesn't think that truthmaking is a special case of grounding, he endorses a similar view. Fine (2012) sees grounding and truthmaking theorists as disagreeing about which relation should take pride of place in metaphysics, as they offer competing takes on the general phenomenon of what accounts for what. He sides with the grounding theorist, joking that "truthmaking is fine as a guide to metaphysics as long as we junk the relata on the left, the things whose existence makes true, the relata on the right, the things made true, and the relation of making true" (2017: 556). But at the same time, he argues that truthmaking has important theoretical applications outside the domain of metaphysics.¹³

A more radical option is to reject the idea that there is truthmaking altogether. Tallant (2018) defends a striking version of this approach. He argues that truthmaking is a grounding theoretic notion, and as there is no grounding, there is no truthmaking! (Surprisingly, Tallant elsewhere conceives of metaphysics as "... the quest for truthmakers" (2017: 3). So perhaps metaphysics is the quest for nothing?)

Let's explore a different approach. Assay (2017) argues that grounding and truthmaking theorists are best interpreted as having different, compatible theoretical objectives within the domain of metaphysics. So perhaps we've simply mischaracterized the evaluative role that truthmaking plays in metaphysics. What might a different take on this role look like?

To get us started, here's a simple proposal: supposing that the true propositions concerning some subject matter have *grounded* truthmakers, if a thesis about this subject matter leaves us with no plausible story to tell about what these truthmakers are like, this counts against the thesis. Returning to the debate between behaviorists and functionalists, true propositions about behavioral dispositions presumably have grounded truthmakers. So it's not the case that any truthmaker for <Gomer is friendly> is fundamental. Assuming that true propositions are about their truthmakers, the idea is that at least part of what this proposition is about doesn't concern how the world is fundamentally speaking. Now, behaviorists and functionalists can agree that the fact that [Gomer is friendly] is a truthmaker for <Gomer is friendly>. We've seen that functionalists claim that there is a further truthmaker for this proposition, one concerning the physical properties that actually play the relevant causal role definitive of being friendly. And functionalists reasonably claim that these facts ground [Gomer is friendly]. So functionalists have a plausible story to tell according to which there are grounded truthmakers for <Gomer is friendly>. Behaviorists, by contrast, don't have such a story to tell. With respect to [Gomer is friendly], apparently part of their view is that this fact doesn't have grounds, as behavioral dispositions are brute.¹⁴

There is, however, a potentially more interesting way we might revise our initial take on the evaluative role that truthmaking plays in metaphysics. This will require some stage setting, so bear with me. How do we evaluate grounding claims? One way to do so is to specify the modal consequences of grounding and then ask whether the facts involved in a given grounding claim are modally connected in the right way. Let physicalism about the mental be the thesis that the mental facts are grounded by nonmental physical facts (perhaps together with facts that are neither mental nor physical such as indexical and totality facts). Many claim that grounding carries

necessity in the following sense—if Δ (fully) grounds some fact, then it's metaphysically necessary that if Δ obtains then the grounded fact obtains. Some argue that the nonmental physical facts, however supplemented with additional nonmental facts, don't metaphysically necessitate the mental facts. If they're right, then physicalism is false, provided that grounding carries metaphysical necessity.¹⁵

There is another way to evaluate grounding claims. Many implausible claims to the effect that Δ grounds A seem implausible in part because we have no sense of how Δ is supposed to ground A. This is the case, for example, with respect to the claim that [Gomer is friendly] grounds [Socrates is a philosopher]. A comparison to causal mechanisms will be helpful here. Suppose we're dealing with a subject matter like biochemistry in which causal relations have underlying causal mechanisms, and suppose I make a claim about what causes a neurochemical event such as the release of neurotransmitters. If we have no idea what sort of underlying causal mechanism might be operative in this case, this counts against my causal claim.

The physicalist claims that mental facts like [Gomer is friendly] are grounded by physical facts. It seems that we should endorse such claims only if we have a general sense of how the physical facts might ground the mental facts. At the very least, it counts against a grounding-theoretic physicalist thesis if that thesis doesn't speak to how the physical is supposed to ground the mental. And note that we've already seen in broad outline one way things might go in this case. Let's say that property P *realizes* property Q on an occasion just in case P plays thus-and-so causal role on that occasion and Q is a functional property, one defined in terms of this very role.¹⁶ The functionalist claims that, when neurochemical facts ground mental facts, property constituents of the former realize property constituents of the latter. Returning to Gomer, the idea is that, whatever neurochemical facts ground the fact that Gomer is friendly, some fact among the former has as a constituent some physical property that realizes the property of being friendly.

Some, of course, object to this story about how physical facts ground the mental facts, arguing that mental properties aren't functional properties. Indeed, it seems that the point of departure for the so-called explanatory gap challenge to physicalism is our general impression that there just is no plausible story to tell about how physical facts might ground the mental facts. But whether or not functionalism and physicalism more generally speaking are true, the important point for our purposes is this: there are cases in which considerations about how Δ grounds A are relevant to whether we should think that Δ grounds A in the first place.

What does this have to do with truthmaking and its evaluative role in metaphysics? The idea is that various relations in addition to realization make similar contributions to the evaluation of grounding claims, and truthmaking is among these relations. The overall suggestion is that pointing to truthmaking relations can help us assess the plausibility of grounding claims in a way that is analogous to how pointing to causal mechanisms can help us assess the plausibility of causal claims.¹⁷

There is a potential concern, however, with the package of views consisting of the grounding account of truthmaking and this broadly mechanistic proposal about its evaluative role. Suppose you ask, "How does Δ ground A?" and I respond by pointing out that Δ grounds A. It's clear that my simple response doesn't really answer your question. Generally speaking, establishing *that* Δ grounds A doesn't show *how* Δ grounds A. According to the grounding account of truthmaking, truthmaking is a special case of grounding. So you might think that a more complicated response to your question according to which truthmaking is part of what yokes Δ and A together isn't much of an improvement on my simple response. By contrast, you might think that relations like realization are potentially relevant to understanding how grounding takes place precisely because they aren't merely special cases of grounding.

I concede that there are cases in which Δ grounds A that involve truthmaking where the latter doesn't speak to how Δ grounds A. Suppose you ask, "How does MOLECULE ground [$\langle$ the liquid is boiling $\rangle$ is true]?" And suppose I respond by pointing out that MOLECULE is a truthmaker for $\langle$ the liquid is boiling $\rangle$. Given the grounding account of truthmaking, my response doesn't add much of interest, as I've in effect just reiterated the initial grounding claim. Relatedly, Audi (forthcoming) observes that, while grounding relations between facts are typically underwritten by certain connections between their constituent properties, there apparently aren't such underwriting connections in the case of alethic facts and truthmakers.

There are, however, cases in which Δ grounds A and underlying truthmaking relations do speak to how Δ grounds A. Let's consider a simple case. Let $\langle s \rangle$ be a proposition according to which the singleton of the statue (the set whose sole member is the statue) exists. Consider the following grounding claim: [the clay exists], together with other facts, grounds [$\langle s \rangle$ is true]. Is this claim plausible? Well, if there is nothing to say about how the grounding connection might run here, this counts against the grounding claim.

It seems, however, that we can point to underlying relations that yoke the relevant facts together, and by doing so, we specify how the grounding connection runs in this case. And one of the relations that we point to in this case is truthmaking. Here's how the story goes: there is a constituent of [the clay exists] (the clay) that stands in the material constitution relation to a constituent of [the statue exists] (the statue); the statue stands in the set formation relation to a constituent of [the singleton of the statue exists] (the singleton of the statue); and the fact that this set exists itself stands in the truthmaking relation to a constituent of [$\langle s \rangle$ is true] ($\langle s \rangle$). Provided that the relations of material constitution, set formation, and truthmaking are indeed instantiated in this way on this occasion, it seems that we have a fairly good grip on how [the clay exists] grounds [$\langle s \rangle$ is true].¹⁸

6. Conclusion

We began by considering a simple proposal about the connection between grounding and truthmaking according to which truthmaking is a special case of grounding. Then we considered a potential problem for this proposal concerning the evaluative role of truthmaking. We considered one way of revising the proposal—the fundamentality account of truthmaking—in an effort to resolve the problem, but we saw that this proposal has a similar problem as well as various other problems. Then we considered other options for responding to the problem. We focused on one option in particular, that of revising our conception of the evaluative role of truthmaking. We ended up with what I take to be a fairly attractive package of views—a simple grounding account of truthmaking and a broadly mechanistic conception of the evaluative role of truthmaking in metaphysics.

I'll close by noting two potential directions of future research concerning the connection between grounding and truthmaking (provided that what I've said in this entry is on the right track). The first concerns the various relations we can appeal to in specifying how some facts ground another fact. I suggested earlier that truthmaking is such a relation. But what if anything unifies these relations beyond the fact that they potentially contribute to our understanding of how grounding takes place? Addressing this question may shed light on what truthmaking has in common with other familiar relations from metaphysics, as well as what's distinctive about it. A related question: in addition to the ones we've mentioned, what further relations speak to how the connection runs between grounds and what they ground?

The second issue concerns the evaluation of truthmaking claims rather than grounding claims more generally speaking. Provided that truthmaking is a special case of grounding, what

sorts of considerations are relevant to the evaluation of truthmaking claims themselves? If it makes sense to speak of how some fact makes a proposition true, then perhaps we can reiterate the strategy discussed herein for evaluating grounding claims. In this case, we would need to look for underlying relations that yoke truthmakers together with either the propositions they make true or facts to the effect that those propositions are true.¹⁹

Related topics

Bliss, R. (Chapter 23, this volume)
Bryant, A. (Chapter 35, this volume)
deRosset, L. (Chapter 12, this volume)
Dixon, S. (Chapter 16, this volume)
Koslicki, K. (Chapter 11, this volume)
Kovacs, D. (Chapter 24, this volume)
Raven, M. (Introduction, this volume)
Skiles, A. (Chapter 10, this volume)
Zylstra, J. (Chapter 22, this volume)

Notes

- 1 See Raven (this volume) for more on the predicate vs. connective approaches to grounding and the relata of grounding; Melia (2005) for corresponding considerations with respect to truthmaking; Trogdon and Cowling (2019) and Kment (2014: Ch. 6) for ontological versus ideological treatments of grounding; and Audi (forthcoming) for corresponding issues with respect to truthmaking.
- 2 Thanks to Ken Aizawa for proposing a case along these lines in conversation.
- 3 Among those who think that there is a definitional connection between grounding and truthmaking, it's generally agreed that the former is definitionally prior to the latter—Cameron (2010), however, is an exception.
- 4 See Glazier (Chapter 8, this volume) for more on the connection between grounding and explanation, and Zylstra (Chapter 22, this volume) for connections between grounding and essence.
- 5 See Daly (2005) for more on the idea that behaviorists, idealists, and other misfits have recourse to truthmakers (though his discussion isn't framed in terms of grounding).
- 6 See Bliss (Chapter 23, this volume) for more on grounding-theoretic conceptions of fundamentality.
- 7 Rettler (2016) claims that all truthmakers are fundamental, though he doesn't explicitly endorse (or reject) the view according to which a fact is fundamental just in case it lacks grounds. Barnes (2012) also sees a connection between truthmaking and (a primitive notion of) fundamentality.
- 8 For more on truthmaker semantics, see Fine (2017 and Chapter 36, this volume).
- 9 See Bliss (Chapter 23, this volume) and Dixon (Chapter 16, this volume) for more on metaphysical foundationalism.
- 10 See, however, Levine and Trogdon (2009) for critical discussion of this idea.
- 11 See Fisher (2015), Jago (2018: Ch. 5), Merricks (2008), Mulligan (2007), and Schipper (forthcoming) for further critical discussion of the project of tying truthmaking to fundamentality.
- 12 See Koslicki (Chapter 11, this volume) and deRosset (Chapter 12, this volume) for discussion of skepticism about grounding.
- 13 See MacBride (2019) for critical discussion of the idea that a central theoretical role of truthmakers is to "catch cheaters".
- 14 Perhaps the evaluative role of truthmaking can also be reconceived in terms of minimal truthmaking—see O'Conaill and Tahko (2015) for discussion.
- 15 See Bryant (Chapter 35, this volume) for more on the connection between grounding and physicalism, Skiles (Chapter 10, this volume) and Kovacs (Chapter 24, this volume) for the connection between grounding and necessity, and Chalmers (2009) for necessitation-based objections to physicalism.
- 16 See Baysan (2015) for more on realization.

- 17 For more on this broadly mechanistic approach to specifying how grounding connections run, see Trogdon (2018). For related approaches, see Litland (2017) on what he calls “explanatory arguments” and Schaffer (2016) on structural equation models for grounding claims.
- 18 See Wasserman (2015) for more on material constitution and Fine (2010) and Lewis (1991: Ch. 1) for set formation.
- 19 Thanks to Sam Cowling, Stephan Krämer, Mike Raven, and Robin Stenwall for helpful comments on drafts of this entry.

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PART V

Applications



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INTRODUCTION

Michael J. Raven

One way of approaching what ground is proceeds *from below*. The idea is to extrapolate what ground is from its potential applications to various topics. An approach from below begins by collecting examples of ground's potential applications. Each application on its own may be of interest. But the collection itself may be of interest for its bearing on ground. Any given collection might contain patterns or recurring themes. If the collection is suitably broad and diverse, it might serve as a representative sample. And if so, these patterns and themes might then serve as indicators of what ground is. Exploring applications of ground therefore promises not only to illuminate the topics to which ground is applied but also ground itself.

But which topics should we focus on for potential applications of ground? There is an endless supply of possible topics, at least in principle. So there is no reasonable hope of a comprehensive survey of them here. The range of pertinent topics, however, is more manageable in practice. In the literature, one finds certain topics coming up again and again. These topics have attracted and sustained the interest of those exploring ground's potential applications. A partial list of these topics includes: identity, laws of metaphysics, laws of nature, logic, mind, normativity, physicalism, semantics, and social entities.

The essays in Part V survey potential applications of ground to these and related topics.

29. Erica Shumener's "Identity" discusses whether we can use ground to formulate identity criteria. Philosophers often understand identity criteria in terms of necessary and sufficient conditions. For example, set $x =$ set y iff x and y have all and only the same members. If we appeal to ground instead, we can construct identity criteria that tell us *in virtue of what* set $x =$ set y . This chapter explores proposals for stating identity criteria in terms of ground. It also addresses whether and when to take identity facts to be ungrounded.
30. Tobias Wilsch's "Laws of Metaphysics" argues that ground explanations require the backing of metaphysical laws. The chapter's main focus is on the question of whether metaphysical laws are primitive or reducible. It sketches three arguments in favor of primitive metaphysical laws and discusses three problems that such laws would face.
31. Nina Emery's "Laws of Nature" explores the ways in which the notion of metaphysical ground has proven increasingly important in understanding the possible positions that one might take with respect to the metaphysics of laws of nature and in giving rise to novel argumentative strategies in defense of one position or another.

32. Michaela M. McSweeney's "Logic" surveys the literature on philosophical motivations for principles governing logical ground. It also discusses some challenges to those principles that arise from certain metaphysical theories. And it examines what it is that these principles really commit us to, metaphysically speaking.
33. Alyssa Ney's "Mind" describes how concepts of ground may be used to formulate answers to the mind-body problem, surveying concerns about the appropriateness, adequacy, and indispensability of concepts of ground for addressing questions about the status of mind in a physical world. A central lesson will be that any adequate assessment of the usefulness of ground frameworks for formulating views in the philosophy of mind must be sensitive to distinctions between the different concepts of ground that have been introduced.
34. Stephanie Leary's "Normativity" concerns the role that ground plays in normative ethics and metaethics. First, the chapter discusses whether first-order normative ethical theories are in the business of making claims about metaphysical ground or a distinctively normative ground relation. Second, it discusses whether the naturalism versus nonnaturalism debate in metaethics is best characterized as a dispute about what metaphysically grounds normative facts.
35. Amanda Bryant's "Physicalism" considers potential applications of ground to the formulation of physicalism. She begins by surveying a number of metaphysical physicalist theses. She then considers whether ground physicalism overcomes well-known and seemingly fatal problems with supervenience physicalism. She concludes that while ground physicalism improves upon supervenience physicalism in certain respects, it arguably falls victim to some of the same difficulties that plague supervenience physicalism.
36. Kit Fine's "Semantics" explores the connection between the truthmaker approach to semantics and the theory of ground. It shows how one might derive the standard clauses of truthmaker semantics from corresponding principles concerning ground; and it also shows how one might employ the methods of the truthmaker approach, in the reverse direction, in providing a semantics for the logic of ground.
37. Asya Passinsky's "Social Entities" surveys applications of ground to central debates and topics in the study of the social world. Three main applications are examined: how the notion of social construction may be analyzed in ground-theoretic terms; how debates over the nature of social facts may be fruitfully recast as debates concerning ground; and how the doctrine of ontological individualism may be elucidated using the notion of ground. The chapter concludes by briefly discussing skepticism about such applications of ground.

29

IDENTITY

Erica Shumener

With good identity criteria, we can make progress in a variety of philosophical debates. With a criterion for event identity, we can approach the question of whether mental events are identical with or distinct from physical events, knowing what types of features would render them identical or distinct. A diachronic identity criterion for persons can tell us whether you would survive a journey through a transporter machine that disassembles your atoms completely and then puts them back together again. And when considering entities at a single time, we want to know whether distinct objects can share the same qualitative properties: can two objects be alike in every respect except for the fact that there are two of them? A synchronic identity criterion for objects (a criterion for the identity of objects *at a time*) would address this question.

This chapter concerns the nature of identity criteria and the relationship between ground and facts of identity or distinctness. After some preliminaries in Section I, we turn to formulations of identity criteria in terms of ground in Section II. Section III explores reasons for and against taking identity and distinctness facts to be fundamental. Section IV tackles specific proposals for grounding identity and distinctness facts.

I. Preliminaries

Identity facts are ones like World War I = World War I, and distinctness facts are ones like The Louvre $\neq$ The Prado. Most identity facts discussed here take the form of ' $x = y$ ' (where x and y are objects). Facts of the form ' $\sim x = y$ ' (often written as $x \neq y$) counts as distinctness facts for our purposes. In what follows, I discuss proposals for grounding both identity and distinctness facts. It is controversial whether we should always group identity and distinctness facts together: we may take distinctness facts to be grounded while taking identity facts to be ungrounded (or *vice versa*). Or we may take both identity and distinctness facts to be grounded but in radically different ways. I will highlight points where a separate treatment of identity and distinctness facts may be desirable.

I make a series of assumptions that, while not necessary, help streamline the discussion. I employ a fact-based account of ground where ground is taken to be a relation holding among facts. I also assume the grounding relation is transitive, asymmetric and (hence) irreflexive and that ground is factive (see Thompson's contribution to this volume for further discussion).

I primarily discuss two varieties of identity and distinctness facts.

1. *General (or Quantificational) identity and distinctness facts:* These involve quantification over entities. Examples include, $(\forall x)(x = x)$, $(\exists x)(x = \text{Angela Merkel})$, and $(\exists x)(\exists y)(x \neq y)$.
And
2. *Individual identity and distinctness facts:* These concern the identity or distinctness of individual entities (they can be objects, properties, relations, etc.). Examples include, Angela Merkel = Angela Merkel, and Angela Merkel $\neq$ Emmanuel Macron.

II. Identity Criteria and Ground

We often provide identity criteria in terms of necessary and sufficient conditions: $x = y$ if and only if condition P obtains. Following Kit Fine (2016), we call identity criteria of this form “material criteria”.¹ Fine (2016) formulates identity criteria in terms of ground as well: if $x = y$, then some fact P grounds $x = y$. We call identity criteria of this form, “grounding criteria”. An example will help us see the difference between material and grounding criteria. Let’s consider a material criterion for set identity. For sets x and y :

Set-Identity_{mat}: $x = y$ iff $(\forall z)(z \in x \equiv z \in y)$

In other words, set x is identical with set y if and only if x and y have exactly the same members. We can also formulate a grounding criterion for set-identity:

Set-Identity_g: if $x = y$, then $x = y$ is fully grounded in the fact: $(\forall z)(z \in x \equiv z \in y)$.

Fine (2016) maintains that we should seek grounding criteria if we want to pinpoint *in virtue of* what entities are identical or distinct. Since *in virtue of* is plausibly an asymmetric notion, material criteria cannot tell us in virtue of what objects are identical or distinct. Material criteria do not establish any direction of dependence. Set-Identity_{mat} merely tells us that if $(\forall z)(z \in x \equiv z \in y)$ obtains then $x = y$, and if $x = y$, then $(\forall z)(z \in x \equiv z \in y)$ obtains. Alternatively, since ground strives to capture a metaphysical notion of *in virtue of*, it is better suited for capturing asymmetric dependence. If ground is an asymmetric relation, Set-Identity_g tells us that if $x = y$, $(\forall z)(z \in x \equiv z \in y)$ grounds $x = y$, in which case $x = y$ does not ground $(\forall z)(z \in x \equiv z \in y)$.

Given that the grounding theorist typically thinks that if P grounds Q then P metaphysically explains Q (see Glazier’s contribution to this volume [Chapter 8] for discussion of the connection between ground and metaphysical explanation), grounding criteria will be attractive to those who want a metaphysical explanation of why entities are identical or distinct. We often seek a metaphysical explanation when tackling puzzling cases about identity. For instance, when encountering the thought experiment in which Man A enters a transporter and two psychological and physical duplicates of him, Man B and Man C, emerge, we want to explain in virtue of what Man A is identical with or distinct from Man B. To be fair, first we want to know *whether* Man A is identical with or distinct from Man B, and similarly for the relation between Man A and Man C. But to support our verdict, we want to know *why* Man A is identical with or distinct from Man B. We want a metaphysical explanation of this identity or distinctness fact. Thus, a grounding criterion of personal identity would be valuable for us if we could find one.

Let’s turn to the relationship between grounding and material criteria: material criteria do not entail grounding criteria as we have seen, but do grounding criteria entail material criteria?

It depends on how we formulate grounding criteria. Set-Identity_g will establish that if $x = y$ then $(\forall z)(z \in x \equiv z \in y)$, which is the necessity condition of Set-Identity_{mat} (the necessity condition follows from other principles as well, like Leibniz's Principle of the Indiscernibility of Identicals). Since Set-Identity_g states that if $x = y$, then $(\forall z)(z \in x \equiv z \in y)$ grounds $x = y$, it follows that $(\forall z)(z \in x \equiv z \in y)$ obtains if $x = y$ does. What about sufficiency? If $(\forall z)(z \in x \equiv z \in y)$, then will $x = y$ hold? The formulation of Set-Identity_g does not yield this result automatically. The criterion does not rule out that $(\forall z)(z \in x \equiv z \in y)$ obtains yet x is distinct from y . Set-Identity_g says only that if $x = y$, then $(\forall z)(z \in x \equiv z \in y)$ grounds $x = y$. While we antecedently believe it is metaphysically impossible for distinct sets to share all their members, the formulation of Set-Identity_g itself does not prohibit this.

However, we can construct a grounding criterion that would explicitly entail the sufficiency condition of the material criterion. One way of doing so is as follows. For sets x and y :

Set-Identity_{g2}:

- (1) If $x = y$, then $x = y$ is fully grounded in the fact $(\forall z)(z \in x \equiv z \in y)$.
And
- (2) If $x \neq y$, then $x \neq y$ is fully grounded in the fact $\sim(\forall z)(z \in x \equiv z \in y)$.

If $x \neq y$, then $x \neq y$ is fully grounded in $\sim(\forall z)(z \in x \equiv z \in y)$. So $\sim(\forall z)(z \in x \equiv z \in y)$ obtains if $x \neq y$ does. By contraposition and double negation elimination, if $(\forall z)(z \in x \equiv z \in y)$ then $x = y$. The sufficiency condition of Set-Identity_{mat} is thereby established.

We have now examined differences between grounding criteria and material criteria. Grounding criteria capture a direction of dependence, they back metaphysical explanations, and certain formulations of grounding criteria are strong enough to entail corresponding material criteria. If we want identity criteria to back metaphysical explanations while generating both necessary and sufficient conditions for the identity of the entities in question, formulations of identity criteria with the form of Set-Identity_{g2} will be preferable over formulations like Set-Identity_g and Set-Identity_{mat}.

There are still other varieties of grounding identity criteria to consider. Fine (2016) discusses two. First, a grounding identity criterion may tell us, for *any* objects x and y , in virtue of what they are identical. Fine calls this a “general” criterion and describes it thusly: “[the general criterion] tells us, for any two particular objects of the sort in question, what makes them the same” (Fine 2016: 4). We can clarify the form of general criteria using universal quantifiers as follows:

Set-Identity_{g-general}: $(\forall x)(\forall y)[\text{Set}(x) \ \& \ \text{Set}(y) \ \& \ x = y \supset (x = y \text{ is fully grounded in } (\forall z)(z \in x \equiv z \in y))]$

A second alternative is to formulate identity criteria in terms of arbitrary objects. Following Fine, we call this a “generic” criterion. Instead of stating that for any sets x and y , if $x = y$ then $x = y$ is grounded in $(\forall z)(z \in x \equiv z \in y)$, we claim that for the *arbitrary sets* x and y , if $x = y$ then that fact is grounded in $(\forall z)(z \in x \equiv z \in y)$. The generic criterion for sets would then answer the question: “in virtue of what are these two sets the same, i.e., what is it about the two arbitrary sets (considered as representative individual sets, not as objects in their own right) that would make them the same?” (Fine 2016: 13).

Set-Identity_{g-generic}: For arbitrary sets x and y , if $x = y$ then $x = y$ is fully grounded in the fact: $(\forall z)(z \in x \equiv z \in y)$.

Fine favors generic criteria over general criteria.² General criteria demonstrate how to ground individual identity and distinctness facts, the instances of the universal generalization. Fine thinks the question of what grounds the fact that individual sets are identical with themselves is just a “pseudo-problem—one that we cannot take seriously as answering to any real issue about the identity of sets” (Fine 2016: 12). When providing a metaphysical explanation of set-identity, we do not care about explaining in virtue of what is it the case that $\{\text{Socrates}\}$ is identical with $\{\text{Socrates}\}$. We do not care about $\{\text{Socrates}\}$ in particular. Instead, we want a generic criterion that will tell us in virtue of what any two arbitrary sets are identical. For further discussion of generic identity criteria and its appeal, see Fine (2016), and for further discussion of arbitrary objects, see Fine (1985).

II. When (if Ever) Are Identity or Distinctness Facts Fundamental?

Is it truly just a pseudo-problem to consider in virtue of what $\{\text{Socrates}\}$ is identical with itself? If such facts do not hold in virtue of other facts, are we pressured to take them to be fundamental? In this section, we turn to identity and distinctness facts that do not involve arbitrary objects. We will examine three sources of motivation for taking at least some identity and distinctness facts of this form to be fundamental.

One may take the identity or distinctness of objects to serve as a “precondition” of their standing in other kinds of relations to each other. The distinctness of The Louvre and The Prado, for example, is a precondition for their being spatially separated from one another. Fiocco (forthcoming) articulates an idea in this vicinity, and he maintains that the “individuation of a thing is inexplicable” (although it is not clear that Fiocco himself would conclude from this that identity and distinctness facts are metaphysically fundamental).

We need to better understand what it is for one fact to be a “precondition” of another to see whether this inclines us to take identity and distinctness facts involving individuals to be metaphysically ungrounded. One option is to characterize preconditions modally: a fact Φ is a precondition of another fact Ψ when Φ is a necessary condition of Ψ . On this characterization, Φ may still be metaphysically grounded—and even metaphysically grounded in Ψ . The conjunctive fact, $2 + 2 = 4 \ \& \ 3 + 3 = 6$, is a modal precondition of $2 + 2 = 4$: It is necessary that the former obtains in order for the latter to obtain. Nevertheless, $2 + 2 = 4$ grounds the conjunctive fact. Thus, it cannot be identity and distinctness facts serving as modal preconditions for other facts that establishes that identity and distinctness facts are metaphysically ungrounded.

The lesson may be that we cannot think of preconditions purely modally. We can import a notion of ground, essence, or real definition into our characterization of a precondition to see if that will help. I will appeal to ground. Let’s assess whether identity and distinctness facts serve as ‘grounding preconditions’ for other facts—where a fact Φ is a grounding precondition of fact Ψ iff whenever Ψ obtains, Φ partially grounds Ψ . Perhaps the fact that objects x and y are distinct serves as a grounding precondition for x and y ’s standing in other relations. For example, the fact that a is distinct from b may be a grounding precondition of the fact that a is more massive than b .

This is one point at which we may wish to treat identity and distinctness facts separately: Perhaps only distinctness facts serve as grounding preconditions in this way. It is odd, for example, to claim that $a = a$ is a grounding precondition of the fact that a is as massive as a , especially since distinct objects can be as massive as each other. Nevertheless, I will continue to package identity and distinctness facts together for ease of exposition.

Even if identity and distinctness facts are grounding preconditions for objects standing in polyadic relations, this does not establish that they are ungrounded. Identity and distinctness

facts may still be grounded in objects' instantiating certain monadic properties. In response, we can broaden the proposal: maybe identity and distinctness facts are grounding preconditions for objects instantiating any properties/standing in any relations whatsoever (other than the identity and distinctness relations themselves). Identity and distinctness facts may then be ungrounded: we are running out of facts to potentially ground them!

However, this suggestion leads to our treating many intuitively metaphysically fundamental facts as nonfundamental; such facts will have identity/distinctness facts as grounding preconditions. For instance, if $e = e$ (where e is an electron) is a grounding precondition of e 's having -1 charge, then we cannot claim that the fact that e has -1 charge is ungrounded. It is also not clear what the full grounds of e has -1 charge would be if $e = e$ serves as a partial (but not full) ground of that fact. This issue arises for every fact involving objects instantiating intuitively fundamental physical properties/relations.

A second motivation for taking certain identity and distinctness facts to be fundamental arises if you think that identity is "joint-carving." Theodore Sider (2011) takes certain quantificational identity and distinctness facts or truths to be fundamental; although he does not understand fundamentality in terms of ground (see Tahko's "Structure" in this volume [Chapter 27] for more on Sider's notion of fundamentality and joint-carvingness). Sider treats the following notions as joint-carving: "first-order quantification theory (with identity), plus a predicate symbol $\in$ for set-membership, plus predicates adequate for fundamental physics, plus the notion of structure" (2011: 292–293). Sider characterizes facts as fundamental when they involve purely joint-carving notions. Fundamental facts will include ones like $(\exists x)Rx$, $(\exists x)(Rx \vee Lx)$, etc. where R and L are joint-carving predicates, and $\exists$ is a joint-carving existential quantifier. Since the identity predicate appears to be joint-carving, identity and distinctness facts like $(\exists x)(\exists y)(x = y)$, $(\exists x)(\exists y)(x \neq y)$ and $(\exists x)(\exists y)((Px \ \& \ Qy) \ \& \ (x \neq y))$ will count as fundamental if P and Q are joint-carving predicates and $\exists$ is a joint-carving existential quantifier.

Does the grounding theorist also have this basis for treating these identity facts as fundamental? Grounding theorists typically ground logically complex facts in their simpler components. For instance, grounding theorists do not commonly take disjunctions, conjunctions, or existential generalizations to be fundamental. Conjunctions and disjunctions are grounded in their conjuncts and disjuncts, respectively. Existential generalizations are commonly grounded in their instances (see McSweeney's [Chapter 32] and Poggiolesi's [Chapter 14] entries in this volume for further discussion). Sider's rationale for treating certain identity and distinctness facts as fundamental carries does not straightforwardly carry over into a grounding context. The identity and distinctness facts Sider treats as fundamental, such as $(\exists x)(\exists y)(x \neq y)$ and $(\exists x)(\exists y)(x = y)$, will be nonfundamental for the grounding theorist who claims these facts are grounded in their instances, $a \neq b$ and $a = a$, respectively. Thus, even if facts contain only joint-carving notions in Sider's sense, the grounding theorist may still ground them.

Of course, the grounding theorist can reject that quantificational facts are grounded in their instances, in which case she may be more sympathetic to Sider's view.³ But if the grounding theorist is not willing to go this route, the question becomes: do we have good reasons to treat individual identity and distinctness facts, like $a \neq b$ and $a = a$ —the ones which ground $(\exists x)(\exists y)(x \neq y)$ and $(\exists x)(\exists y)(x = y)$ —as fundamental?

There are different options here. A grounding theorist sympathetic with Sider's perspective may maintain that identity and distinctness facts involving intuitively nonfundamental objects are nonfundamental but ones involving intuitively fundamental objects are fundamental. This would be a mixed view upon which some individual identity and distinctness facts are grounded and others are ungrounded. We may like such a view if we think there should be no fundamental facts involving nonfundamental objects, but we have no problem with taking identity and

distinctness facts involving only fundamental objects to be fundamental. A second option would be to take all individual identity and distinctness facts to be fundamental, and a third option would be to take them all to be grounded. We will explore this third option in the next section.

A third reason to posit at least some fundamental distinctness facts (this reasoning does not necessarily extend to identity facts) is to account for certain kinds of metaphysical possibilities. For example, we will consider a scenario popularized by Max Black (1952), the “sphere world”. The sphere world contains two qualitatively identical spheres, Castor and Pollux, in an otherwise empty universe. It is difficult to determine on what basis Castor and Pollux are distinct. They are both silver, they both have a mass of 5 kg, they are 10 meters from each other, and so on.

If we take distinctness facts to be fundamental, we can sidestep the question of what makes Castor distinct from Pollux. They just *are* distinct; it is a brute fact. We can posit a fundamental distinctness fact in the sphere world to accommodate this. Shamik Dasgupta (2009) recommends this approach. The fundamental distinctness fact in the sphere world could take the following form:

$$(\exists x)(\exists y)((Px \ \& \ Py) \ \& \ x \neq y)$$

where P is a predicate that picks out the full qualitative profile of each of the spheres. The fan of individual identity and distinctness facts could also adapt his proposal and posit a fundamental distinctness fact of the form: Castor $\neq$ Pollux.

The attractiveness of taking facts like Castor $\neq$ Pollux to be fundamental hinges on the idea that we have no basis upon which to distinguish the spheres in Max Black’s famous scenario. It also relies upon the idea that we do not always need grounding criteria of identity: We do not need grounding criteria for the distinctness of material objects like spheres on this proposal. In the next section, we consider opposing proposals. We will investigate grounding criteria that attempt to distinguish Castor and Pollux.

IV. Proposals for Grounding Identity and Distinctness Facts

We have looked at three sources of motivation for treating at least some identity and distinctness as fundamental as well as issues potentially undermining these motivations. Now we will consider four options for taking identity and distinctness facts to be grounded. I focus on grounding criteria for object identity and distinctness, as this has been the most discussed in the literature. Burgess (2012), Donaldson (2017), and Fine (2012) all discuss proposals for grounding individual identity facts involving objects. It is a substantive question which (if any) of these grounding proposals can be extended to accommodate the identity of entities in other ontological categories (such as properties, relations, facts, etc.).

While I discuss grounding identity and distinctness facts involving objects in general, we may only try to ground *certain* identity and distinctness facts involving objects. For instance, as mentioned, we may take identity and distinctness facts involving nonfundamental objects to be grounded and ones involving only fundamental objects to be ungrounded. If one believes that only identity and distinctness facts involving certain objects (say, the nonfundamental objects) should be grounded, then grounding criteria like those that follow will not be desirable.

One way to ground identity and distinctness facts is by appealing to the properties objects share. More specifically, we can appeal to one half of Leibniz’s Law, the Principle of the Identity of Indiscernibles, in order to offer a metaphysical explanation of identity and distinctness facts (see Della Rocca (2005)). While this proposal has trouble accommodating the sphere world

from the previous section, it will be worthwhile to examine it more closely to see why it is problematic.

The Properties Proposal:

- (1) If $x = y$, then $x = y$ is fully grounded in the fact that $(\forall F)(Fx \equiv Fy)$. And,
- (2) If $x \neq y$, $x \neq y$ is fully grounded in the fact $(\exists F)(Fx \ \& \ \sim Fy) \vee (\exists F)(\sim Fx \ \& \ Fy)$

$\forall F$ ranges over properties. We should restrict the class of properties to qualitative ones—properties that do not involve the identities of individual objects. If we invoke facts involving nonqualitative properties (properties like *is identical with b* or *is distinct from a*) to ground individual identity and distinctness facts, we could ground the fact that $a \neq b$ in the fact that b has the monadic property *is identical with b* while a does not have that property. This would render the account trivial. We will also violate irreflexivity if the fact that b has the monadic property *is identical with b* is grounded in b 's standing in the binary identity relation to b .

While we can just restrict the class of properties to 'qualitative' ones that do not involve the identities of particular objects, this is problematic: The resulting proposal cannot accommodate the metaphysical possibility of distinct yet qualitatively identical objects in an otherwise empty world, i.e., the sphere world discussed in the previous section. We lack the grounds for the distinctness of the two qualitatively identical spheres. In fact, since $(\forall F)(F(\text{Castor}) \equiv F(\text{Pollux}))$ obtains, the grounds for $\text{Castor} = \text{Pollux}$ obtains. And if we accept a grounding necessitation principle (see Skiles's entry [Chapter 10]), this should establish that $\text{Castor} = \text{Pollux}$, which conflicts with the set-up of the scenario.

A second proposal appeals to facts about the existence of objects in order to ground identity and distinctness facts. Burgess (2012: 90) suggests that identity facts at first "seem to be nothing over and above the relevant existential facts." Epstein (2015: 169–181) expresses sympathy for this proposal. Perhaps we can ground identity and distinctness facts in the existence of objects.

The Existence Proposal:

- (1) If $x = y$, then $x = y$ is fully grounded in the fact x exists.
And
- (2) If $x \neq y$, $x \neq y$ is fully grounded in the plurality of facts: x exists, y exists.

One advantage of the Existence Proposal over the Properties Proposal is that the former accounts for the distinctness of the Max Black spheres. In the possible world where only Castor and Pollux exist, $\text{Castor} \neq \text{Pollux}$ is grounded in the two facts: Castor exists, Pollux exists. $\text{Castor} = \text{Castor}$ is grounded in the single fact, Castor exists.

Burgess (2012) explores a version of the Existence Proposal and points out a troubling feature. If the fact that Castor exists has the logical form $(\exists x)(x = \text{Castor})$ then $\text{Castor} = \text{Castor}$ is fully grounded in the fact that $(\exists x)(x = \text{Castor})$. And if existentially quantified facts are grounded in their instances, $(\exists x)(x = \text{Castor})$ is grounded in $\text{Castor} = \text{Castor}$, yielding a violation of irreflexivity.

The proponent of the Existence Proposal can avoid this result in a few ways: (1) They can deny the transitivity or irreflexivity of ground (See Thompson's "Strict Partial Order" for discussion), (2) they can deny that existential generalizations are grounded in their instances (see chapters in this volume by McSweeney [Chapter 32], Poggiolesi [Chapter 14], and Krämer [Chapter 18] for discussion of options in line with 2), or (3) they can deny that existence facts are always existential generalizations.

I will set aside options 1 and 2 and explore option 3. Instead of treating existence quantificationally, we can understand existence as a monadic property of objects. In this case, Castor exists will have the form of an atomic fact, $E(\text{Castor})$. As we have no reason to think $E(\text{Castor})$ will be grounded in the fact that $\text{Castor} = \text{Castor}$, we can avoid a potential violation of circularity by claiming that identity facts like $\text{Castor} = \text{Castor}$ are grounded in facts like $E(\text{Castor})$.

The advocate of this alternative should say more about the grounds of facts like $E(\text{Castor})$. After all, if $E(\text{Castor})$ is at least partially grounded in $(\exists x)(x = \text{Castor})$ we will face the same circularity. If $E(\text{Castor})$ is not grounded in this way, we should explain how existential-property facts and existential generalizations relate to one another (see Fine (2012) for discussion).

The Existence Proposal faces another issue: it cannot accommodate identity and distinctness facts (if there are any) involving nonexistent entities. You may think that Santa Claus is identical with Santa Claus even though Santa Claus does not exist. If left unrestricted, the Existence Proposal predicts that if ‘Santa Claus = Santa Claus’ picks out a genuine identity fact, the ground of $\text{Santa Claus} = \text{Santa Claus}$ is the existence of Santa Claus. But Santa Claus does not exist.

The proponent of the Existence Proposal should presumably deny that there are identity facts involving nonexistent objects. The plausibility of this may depend upon whether one has a fact-based or sentential operator-based account of ground. We can deny that $\text{Santa Claus} = \text{Santa Claus}$ picks out a genuine identity fact on a fact-based account of ground, especially if we take facts of the form $a = a$ to involve objects instantiating properties/relations. We would presumably deny that $\text{Santa Claus} = \text{Santa Claus}$ picks out a genuine fact on this view because there is no object to stand in the identity relation. However, if we have a sentential-operator account of ground, we may want to uphold that ‘Santa Claus = Santa Claus’ is a true sentence (even if it does not correspond to a worldly fact). Thus, it is not clear what a proponent of the Existence Proposal who subscribes to the sentential-operator approach should say about the grounds of ‘Santa Claus = Santa Claus’.

A third alternative is to ground identity and distinctness facts in facts concerning parthood (see Burgess 2012⁴). Perhaps x is identical with y when x is part of y and y is part of x . This approach requires us to take the notion of part to be more fundamental than that of identity. Here is one way to formulate the parthood proposal, where the predicate P picks out the relation *is part of*.

The Parthood Proposal:

- (1) If $x = y$, $x = y$ is fully grounded in the plurality of facts: Pxy , Pyx .
And
- (2) If $x \neq y$ then $x \neq y$ is fully grounded in the fact $\sim Pxy$, or $x \neq y$ is grounded in the fact $\sim Pyx$.

When x is part of y and y is part of x , then x is what is often called an ‘improper part’ of y . One common way to understand the *improper parthood* relation is in terms of identity: x is an improper part of y iff x is identical to y . We contrast the *improper parthood* relation with the more intuitive *proper parthood* relation, where x stands in the proper parthood relation to z when x is part of z and x is not identical to z . To avoid circularity, it is important that we do not define P in terms of improper parthood and then define improper parthood in terms of identity. For example, we cannot think of P as picking out the relation *is an improper part of* or the disjunctive relation, *is either a proper part of or an improper part of*, and then go on to define *improper parthood* in terms of *being identical with*. Instead, the proponent of the Parthood Proposal would more likely leave the *is part of* relation undefined or primitive.⁵

This proposal can distinguish everyday objects we encounter: the Coca-Cola bottle is distinct from the cheeseburger because neither is part of the other. They do not even share any parts in

common. And the fact that the cheeseburger is an improper part of itself will ground the fact the cheeseburger is identical with itself. While the bottle and the cheeseburger are intuitively mereological fusions made up of proper parts, the Parthood Proposal is supposed to work for mereological atoms as well. While mereological atoms have no proper parts, they have themselves as improper parts. This proposal can also distinguish Castor and Pollux in the sphere world: Castor is not a proper or improper part of Pollux, nor is Pollux a proper or improper part of Castor.

The Parthood Proposal is not for everyone. Sider resists defining identity in terms of parthood. While he does not discuss ground, his concerns may carry over into a discussion of grounding criteria. Sider states:

[C]onsider the objection that adopting parthood in fundamental theories allows the elimination of identity from ideology via the definition ' $x = y =_{df} x$ is part of y and y is part of x '. The savings in ideological parsimony would be outweighed by increased complexity in the laws, which I take to include laws of logic and metaphysics. The logical laws governing '=' must now be rewritten in terms of the proposed definition, making them more complex; and further, the laws of mereology will be needed.

(Sider 2013: fn. 10)

Sider thinks if we understand identity in terms of parthood, we must rewrite the logical laws in terms of mereological notions, and this revision will be much more complex than the versions we have involving identity. This added complexity is problematic if we favor simpler theories.

The Parthood Proposal will also face resistance from some (but not all) mereological nihilists.

The mereological nihilist denies that objects have proper parts. Under one version of mereological nihilism, only mereological atoms exist, and they are improper parts of themselves. This view appears to be compatible with the Parthood Proposal: every atom is an improper part of itself and not a part of any distinct atoms. Other versions of mereological nihilism cannot accept the Parthood Proposal. Consider a version of mereological nihilism that denies the existence of parthood relations from the outset. This mereological nihilist will also claim that the only objects that exist, fundamentally speaking, are mereological atoms; yet they will deny that the atoms are parts of themselves because they deny that anything stands in the parthood relation. The proponent of this version of mereological nihilism cannot use The Parthood Proposal to generate the correct verdicts about identity and distinctness. Finally, the Parthood Proposal may be too limited in scope if we are seeking a general account of object-identity. Insofar as there are objects that do not stand in any parthood relations—such as abstract objects, like numbers—the Parthood Proposal cannot ground identity and distinctness facts concerning them.⁶

The fourth and final proposal we will consider is whether identity and distinctness facts are zero-grounded. A fact is zero-grounded when it is not grounded in further facts, but it is not ungrounded either. Fine describes the distinction between being ungrounded and being zero-grounded:

There is a . . . distinction to be drawn between being zero-grounded and ungrounded. In the one case, the truth in question simply disappears from the world, so to speak. What generates it . . . is its zero-ground. But in the case of an ungrounded truth . . . the truth is not even generated.

(Fine (2012): 48)

Fine then considers taking identity facts to be zero-grounded:

But in other cases—as with Socrates being identical to Socrates or with Socrates belonging to singleton Socrates—it is not so clear what the contingent truths might be; and a plausible alternative is to suppose that they are somehow grounded in nothing at all.

(Fine (2012): 48)

Tom Donaldson (2017) explores taking certain mathematical identity facts to be zero-grounded as well. We can formulate a version of the Zero-Ground Proposal as follows:

The Zero-Ground Proposal₁:

- (1) If $x = y$, then $x = y$ is zero-grounded.

Identity facts are grounded on this proposal even though there are no facts that ground them. The distinction between being ungrounded and being zero-grounded is significant because were we to take identity facts to be ungrounded, we would be pressured to treat them as fundamental. Since the Zero-Ground Proposal₁ maintains that identity facts are grounded, they are nonfundamental.

This is another point in which our treatment of identity and distinctness facts may come apart: it is not clear that distinctness facts should be zero-grounded as well. While Fine and Donaldson are primarily concerned with grounding identity facts, we could try to extend the Zero-Ground Proposal₁ to accommodate distinctness facts as follows.

The Zero-Ground Proposal₂:

- (1) If $x = y$, then $x = y$ is zero-grounded.
(2) If $x \neq y$, then $x \neq y$ is zero-grounded.

With the added clause for distinctness, the Zero-Ground Proposal₂ can accommodate the sphere world. $\text{Castor} \neq \text{Pollux}$ is zero-grounded. Unlike the other proposals, the Zero-Ground Proposal₂ maintains that identity facts and distinctness facts have the same grounds. $\text{Castor} = \text{Castor}$ and $\text{Pollux} = \text{Pollux}$ are also zero-grounded. This is not necessarily problematic, but it is not yet clear that this proposal yields a satisfactory metaphysical explanation of why objects are distinct. If we are looking for a basis upon which to distinguish objects like Castor and Pollux, then I am not sure the Zero-Grounding Proposal₂ provides it. The distinctness of Castor and Pollux is metaphysically explained on the Zero-Grounding Proposal₂—but not on the basis of any facts.

This proposal maintains that distinctness facts are grounded in the same way identity facts are grounded: they have the null ground. Perhaps different kinds of facts should admit of different grounds. For example, Dasgupta (2014) objects to grounding facts in their grounds because it would render ' P grounds $P \vee P$ ' as grounded in the same way as ' P grounds $P \ \& \ P$ '. (Both would be grounded in P). He thinks there should be a difference in the grounds of ' P grounds $P \vee P$ ' and ' P grounds $P \ \& \ P$ ': the former fact concerns disjunction whereas the latter concerns conjunction. The grounds should reflect this difference, so the thought goes. Likewise, we may object to the Zero-Ground Proposal₂ because it is implausible that identity and distinctness facts would have exactly the same grounds. It is strange for facts of the form $x = y$ and of the form $x \neq y$ to have exactly the same grounds.

All of these proposals for grounding identity and distinctness facts have issues in need of further examination. There is a lot of philosophical room left to explore when questioning whether

and how to ground identity and distinctness facts. The aim in this chapter is not to advocate formulating identity criteria in terms of ground, nor is it to convince the reader that identity and distinctness facts must be grounded. Instead, I hope to have provided an array of options open for investigation and that the reader has gained a sense of which avenues she can pursue.

Notes

- 1 We also often witness a modal analogue of material criteria: $\Box[x = y \text{ if and only if condition } P \text{ obtains}]$. We will only discuss nonmodal material criteria in what follows.
- 2 As a presentational note: I have not taken a stand as to whether Set-Identity_g, Set-Identity_{g2}, and Set-Identity_{mat} should be understood generally or generically, and I leave further formulations of grounding criteria similarly ambiguous.
- 3 See Rosen (2010) for discussion of reasons to take certain universal generalizations as fundamental.
- 4 See Smid (2017) as well; although he is not concerned with ground in his paper.
- 5 We should note that if we adopt extensional mereology with a strong supplementation principle, we cannot have distinct objects with the same proper parts on this view.
- 6 However, some philosophers argue that entities other than concrete objects stand in parthood relations. See Paul (2010) and Fine (2010).

Cross-References

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Further Readings

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30

LAWS OF METAPHYSICS

Tobias Wilsch

1. Are There Metaphysical Laws?

A central metaphysical question in the philosophy of explanation concerns the status of explanations: are there explanation-facts of the form ‘p explains q’, and if so, are these facts accounted for by anything else? Setting irrealist views aside, which consider explanations mere speech acts expressive of the speaker’s attitudes, the world does contain various kinds of causal and non-causal explanation-facts. It is moreover a standard assumption that causal explanation-facts are explained by underlying laws of nature: the rock impact explains the window shattering because of physical laws about rocks and windows.

If causal explanations are supported by general laws, non-causal explanation might be supported by general laws as well. Consider *metaphysical explanations* like ‘Peter the elephant is colored because he is grey’ and ‘{Socrates} exists because Socrates exists’. These explanations might arise from general laws which entail that any grey object is also colored and that any object is the member of its singleton set (see Glazier’s entry “Ground and Explanation”, Chapter 8 in this volume). The role of such *laws of metaphysics* is to guide the bottom-up development of facts, much like the role of laws of nature is to govern facts along the temporal axis.

There is a way to resist this analogy between causal and metaphysical explanation (see Wang’s entry “Cause”, Chapter 20 in this volume). Metaphysically explained facts might be ‘already contained’ in the explaining facts, in which case no law is required to account for metaphysical explanations. Consider Peter the elephant again, whose greyness *grounds*, i.e. metaphysically explains, his being colored. If Peter’s being colored is already contained in his being grey, no metaphysical law is required to take the explanans to the explanandum; the explanation-fact *Peter is colored because he is grey* is simply accounted for by the fact that Peter is grey (Bennett 2011; deRosset 2013).

But without laws it will remain mysterious why similar grounds have similar ‘metaphysical effects’ and why dissimilar grounds have dissimilar effects. Why is *every* grey object colored because of its being grey? This sort of generality of metaphysical explanations can perhaps only be accounted for with metaphysical laws (Wilsch 2015).

Jonathan Schaffer (2018), moreover, argues that metaphysical explanations are supported by laws because explanation involves *unification* and *manipulation*. Unification requires that p explains q only if p and q belong to a unifying pattern with the status of a general law;

manipulation requires that p explains q only if manipulations on p would yield systematic changes with respect to q . Since such counterfactual variation requires modally robust laws, both the unification and the manipulation constraint on metaphysical explanation entail that there are metaphysical laws.

2. What Are Metaphysical Laws?

We have been understanding metaphysical laws as general principles that support metaphysical explanations.¹ But what could such principles be? This section asks whether the notion of a metaphysical law is *sui generis* or reducible.

For the sake of concreteness, consider the following putative *singleton law* (where the “if . . . then . . .” expresses the material conditional):

SL For all objects x , if x exists, then the singleton set $\{x\}$ exists.

It is easy to see what the explanatory import of the singleton law is supposed to be. Instances of the antecedent, ‘ x exists’, explain associated instances of the consequent, ‘ $\{x\}$ exists’:

SE That Socrates exists explains that $\{\text{Socrates}\}$ exists.

An important but subtle point is that SL does not account for explanations like SE, because SL is a universal generalization, a ‘mere regularity’. Since regularities are grounded in their instances (Fine 2012), SL is partly grounded in *if Socrates exists, then $\{\text{Socrates}\}$ exists*, which in turn is partly grounded *$\{\text{Socrates}\}$ exists*. By the transitivity of ground, it follows that SL is partly grounded in *$\{\text{Socrates}\}$ exists*. Therefore, SL does not in turn help to explain ‘ $\{\text{Socrates}\}$ exists’, on pain of circularity, and so SL does not account for SE.

In order to avoid any circularity, we should say that not SL, but the fact that SL is a law of metaphysics, supports SE:

SL★ It’s a law of metaphysics that for all x , if x exists, then $\{x\}$ exists.

If SL★ is not grounded in instances of SL, it may explain such instances without circularity. Since the notion of a metaphysical law features in explanatory principles such as SL★, we must either regard it as a primitive notion or else understand it in different terms. We will see that some metaphysical laws can be understood through *truth-maker rules* and *essences*, whereas other may need to be considered *sui generis* metaphysical laws. (Compare the discussion in Glazier’s entry “Ground and Explanation”.)

Assume that nothing corresponds to disjunctive representations: there are disjunctive sentences and propositions but no disjunctive states of affairs. We should then say that disjunctive representations are ‘made true’ by non-disjunctive facts. Logical explanations of disjunctions— *p or q because p* —are supported by the following *truth-maker rule*: “ p or q ” is true if p ($/q$) holds in reality. We could also appeal to truth-maker rules for other areas of discourse. On the extreme end of the spectrum, even talk about ordinary objects is made true by a reality that contains only atoms in the void (Williams 2010). Whether or not we accept such an extremist position, it is plausible that some metaphysical explanations are backed by truth-maker rules; some metaphysical laws are truth-maker rules.²

Another plausible candidate for supporting metaphysical explanations is the essence or nature of things (see Zylstra’s entry “Essence”, Chapter 22 in this volume, and also Glazier’s entry,

“Ground and Explanation”). Boris Kment (2014: ch. 5) refers to certain essence-truths as laws of metaphysics, and Shamik Dasgupta (2014a, 2014b) offers essences as supporting metaphysical explanations in examples such as the following: the presence of the conference is metaphysically explained by various activities of the attendees, because it’s essential to conferences that they obtain if such activities occur.

Perhaps all the metaphysical laws we ever need are truth-maker rules and essence-truths. But if truthmaking and essence do not suffice to support all metaphysical explanations, we must appeal to some laws of metaphysics that are not reducible to these other notions. Let us call such unreducible laws of metaphysics *sui generis laws*. The question of whether there are *sui generis* laws is currently among the most pressing questions about metaphysical laws.

If all metaphysical laws were truth-maker rules, reality would be flat as a pancake, and the world’s layers would be mere artefacts of our. Since this view is at odds with a rather sober form of realism, call it ‘realism about layers,’ we should expect that some metaphysical laws are either essences or *sui generis*. But should we side with Dasgupta (2014a), who holds that no *sui generis* laws are needed, or with Schaffer (2018), who rejects the need for essence? Or do we need both pieces of ideology? The following discussion will be framed as a competition between essences and *sui generis* laws. Sections 3–5 discuss three arguments for *sui generis* laws, and sections 6–8 raise three problems for such laws. The reader should note that the following sections are meant to stimulate further work rather than to settle any debates.

3. Pinning Questions and Idle Wheels

This section and the following two introduce reasons to go beyond essences to *sui generis* metaphysical laws. The key consideration of this section is that essences contain more information than needed for supporting metaphysical explanations. Consider again the explanation that Peter the elephant is colored because he is grey. We would expect the underlying law to look something like the following (‘ML’ is short for ‘it’s a metaphysical law that’):

Color ML (for all x, if x is grey then x is colored)

Essence-claims of the form ‘it’s essential to entity x that p’ require not only a content clause (p) but also a bearer (x). If Color is a veiled essence-truth, we must therefore supply it with a bearer. To use Jonathan Schaffer’s (2018) expression, we would need to *pin* the essence-claim on an entity (or some entities, in the case of collective essences). Any essence that supports a metaphysical explanation raises a corresponding *pinning-question*: is the law essential to entities from the explanans, from the explanandum, or both?

Some pinning-questions may appear problematic because it’s unclear how we could answer them and because nothing seems to hinge on how we answer them. Consider, for instance, the committed physicalist, who claims that psychological facts are grounded in physiological facts. Should she pin the relevant law on psychological or on physiological entities? One might worry that there is no way to tell which one is the better option and that this question is one which the physicalist shouldn’t have to settle.

These considerations are more compelling from the point of view of an essence-skeptic. Schaffer (Ibid.: 312), for instance, argues that the bearer-aspect of essences is “explanatorily inert” because it has no bearing on metaphysical explanations. As Schaffer observes, if the only reason for positing essences is to support metaphysical explanations, we should rather posit *sui generis* metaphysical laws, which have less structure than essences but seem just as fit to do the job.

Many proponents of essence, however, will deny that essences earn their keep merely by accounting for metaphysical explanations. There is, after all, a live practice of distinguishing accidental from essential properties and of supplying real definitions for relatively simple entities such as sets or events. These practices might justify accepting the ideology of essence. Moreover, proponents of essence will point out that many pinning-questions have obvious answers: it is essential to {Socrates}, and not to Socrates, that Socrates is a member of {Socrates} (Fine 1994), and it is essential to you, and not to your parents, that you are your parents' child.

But even proponents of essence may agree that pinning-questions for some metaphysical laws appear so esoteric and difficult to answer that they better be avoided. To develop this sort of argument further, one could try to isolate a class of putative essence-claims, whose associated pinning-questions seem in principle unanswerable. This class might include essences whose sole purpose is to connect entities from different layers of reality.

4. Anselmian Essences and Substantive Ontology

The second motivation for *sui generis* metaphysical laws is based on Gideon Rosen (2006), who argues that there aren't enough essences to ground all metaphysical necessities. If successful, his argument also shows that essences do not support all metaphysical explanations, and hence that we require *sui generis* metaphysical laws.

Rosen rejects so-called *Anselmian essences*, which in the strictest case entail the existence of their bearers. The most famous Anselmian essence-claim is Anselm's own proposition that God's existence is included in her essence. God's essence thus conceived would be Anselmian because it entails the existence of its bearer. According to the Kantian response to Anselm's ontological argument, existence is not an essential property because it is not a property. Rosen's response is different from Kant's; he proposes a general ban on existence-entailing essences.

God's essential existence is an extreme case of Anselmian essence. Less extreme cases involve essences that entail the existence of their bearers conditionally. To illustrate, consider standard set theories and mereological theories, which entail that *if* certain objects exist (and have certain features), *then* the associated set or mereological fusion also exists. If these theories are essential to sets and fusions, the relevant essences are conditionally existence entailing and therefore Anselmian. To illustrate, consider the following Anselmian Essence:

AE It is essential to {Socrates} that if Socrates exists, then {Socrates} exists

AE entails the existence of {Socrates} conditional on the existence of Socrates. Conditional Anselmian essences such as AE 'generate' their bearers from more fundamental entities. AE, for instance, generates Socrates's singleton set from Socrates.

Rosen's argument for a general ban on Anselmian essences rests on the following *intuitive claim*: we could full-well understand what sets and mereological fusions are supposed to be and yet deny their existence. IF AE was in fact a correct characterization of the essence of {Socrates}, then a complete understanding of what that singleton set is would require accepting that entity on the assumption that Socrates exists. But it seems unfair to charge nominalists, who reject sets altogether, with a lack of understanding. From Rosen's perspective, it should make sense to say: "I reject your sets and fusions for what they are!" The nominalist might be wrong, but she is not confused.

If Rosen is right about Anselmian essences, there are no essences to support metaphysical explanations of derivative objects, as the metaphysical laws supporting such explanations are conditionally existence entailing (SL[★] is a case in point). *Sui generis* metaphysical laws are the

natural alternative: we can posit laws to the effect that every object forms a set without making that principle part of what sets are. Correspondingly, a failure to know these laws would not reveal a lack of understanding what sets are. To push back against *sui generis* laws, therefore, proponents of Anselmian essence must deny Rosen's intuitive claim.

The question of Anselmian essences also bears on ontological debates about derivative entities. Essences may be determined by real definitions, just like meanings are determined by nominal definitions (Fine 1994). Assume that Rosen's ban on Anselmian essences and corresponding real definitions is lifted. Then there could be definitions that conditionally entail the existence of all sorts of derivative entities. There could, for instance, be definitions not only of mereological fusion but also of schmusion, which are like fusions but which are composed only by things arranged in a circle.

The possibility of the Anselmian real definition of schmusion is problematic if it entails that this real definition is 'in effect' and, hence, that there is an existence-entailing essence of schmusion. If we don't rule out such an essence with a general ban on Anselmian essences, then what reason could there be for rejecting it? After all, any phenomena we can think of, including unicorns, dodos, or platonic universals, have definitions with associated essences, independent of whether it exists or whether anyone has defined it. Since real definitions are cheap in this way, Anselmian essences run the risk of deflating derivative ontology: the threat is that every derivative entity that can be consistently defined also exists. *Sui generis* laws of metaphysics, in contrast, are robust principles that enable substantive debates about which derivative entities there are.

5. The Governance of Metaphysical Laws

A third argument in favor of *sui generis* laws of metaphysics runs as follows. Some metaphysical explanations are productive in the sense that the explained facts are 'entirely new'. Productive explanations, however, require the support of governing laws. We need *sui generis* metaphysical laws because only such laws have the power to govern.

To unpack this argument, let us first look at the notion of *productive explanation*. Familiar causal explanations appear productive. When the rock-impact causes the window to shatter, the resulting shattering is entirely new to reality and was brought about with causal 'oomph'. Productive explanations contrast with *systematizing* explanations, which subsume already existing facts and events under general laws.

For instance, philosophers in the Humean tradition take causal explanations to be systematizing rather than productive. On their view, the four-dimensional manifold—"the vast mosaic of local matters of particular fact, just one little thing and then another" (Lewis 1986: ix)—is metaphysically prior to any causal explanation. Humean explanations are lines drawn between facts that were already given; Humean laws, therefore, don't produce but systematize the facts.

Contrast productive causal explanations also with mathematical explanations. The arithmetical truth $2 + 3 = 5$, for instance, is explained by the Peano axioms, as $2 + 3 = 5$ can be derived from the axioms and suitable definitions. But it sounds odd to say that axioms 'produce' arithmetical truths. Axioms rather systematize arithmetical truths in light of the entire 'arithmetical manifold'.

Are metaphysical explanations productive or systematizing? As Karen Bennett (2011: 33) points out, broadly reductive views, such as physicalism about the mental and naturalism about the normative, require that mental and normative facts are "entirely new" upon their explanations. If metaphysical explanations merely systematized a mosaic of facts that were

already given, they would not be suitable for sustaining such views. Hence, some metaphysical explanations are productive.

How do we account for the productive nature of metaphysical explanations? The productivity of causation is (assuming that actual causation is productive) due to the governing operations of the laws of nature. For laws to govern is for them to generate output-facts based on input-facts: laws of nature govern the shattering of the window, just in case they take the rock-impact and generate the shattering on its basis. Governing laws must therefore be understood fundamentally as input–output mechanisms (Maudlin 2007: ch.1).

The problem for a purely essence-based understanding of metaphysical laws is that essences don't seem to operate in an input–output sort of way. Essences seem to directly explain the truths in their scope. For instance, your essence explains that you are the child of your actual parents, and the essence of {Socrates} explains that Socrates is its sole member. As Glazier (2017) explains, 'it is essential to *x* that *p*' directly explains *p*. If essences directly explain the truths in their scope, then they are not input output operations and are therefore not apt to govern.

To further illustrate this point, consider the essence-claim 'It's essential to color that everything grey is colored'. How does this essence-claim support the metaphysical explanation 'Peter is colored because he is grey'? If essences directly explain the truths in their scope, then the assumed essence of color directly explains the general truth 'Every grey object is colored'. The explanation 'Peter is colored because he is grey' then comes about derivatively by subsuming the facts 'Peter is grey' and 'Peter is colored' under the general truth.

If this is how essences give rise to explanation, then essence-based explanations are not productive. In the example, 'Everything grey is colored' partly grounds the explanation of 'Peter is colored', while 'Peter is colored' is also a partial ground of 'Everything grey is colored'. 'Peter is colored' is thus a partial ground of its own explanation. However, if a fact is prior to its own explanation, that explanation is not productive.

This reasoning shows that essences don't govern if they directly explain the essential truths in their scope. It is quite natural, in contrast, to conceive of laws as input–output mechanisms, which don't establish the truths in their scope directly fundamentally take input–facts to output facts.³ A law of nature which says that all hit windows shatter, for instance, might, in the first instance, take rock-impacts to window-shatterings. The explanation of the regularity that all hit windows shatter is then derivative upon the law's work on its instances. Since essences don't act as input–output mechanisms but as direct explainers of the truths in their scope, it is hard to see how essences could govern and hence how metaphysical explanations supported by essence could be productive.

6. Purity and Fundamental Laws

We saw over the past three sections that *sui generis* metaphysical laws may avoid idle complexity, allow for substantive ontological debates, and account for the productive aspect of metaphysical explanations. But there are also difficulties and open questions arising for these laws. The following three sections present three of the most pressing difficulties.

Schaffer (2018) argues that some laws of metaphysics are fundamental. Assume that 'it's a law of metaphysics that *p*' is grounded in *q*. Then there is a second law that supports this grounding explanation. Since it cannot be the same law ('it's a law of metaphysics that *p*') on pain of circularity, a new law is required. If this new law is metaphysically explained,

yet another law is needed, and so on *ad infinitum*. Only fundamental metaphysical laws terminate the regress.

Fundamental metaphysical laws, however, conflict with the claim that nothing fundamental involves anything derivative. Ted Sider has dubbed a similar claim ‘the purity principle’ or *purity* for short (Sider 2012: ch. 7). Since metaphysical laws connect entities of different ontological levels, they involve derivative objects and properties. If Schaffer’s regress argument is persuasive, we must therefore give up either metaphysical laws or the purity principle. But purity is attractive, especially if one wants to define a ‘fundamental entity’ as an entity that occurs in some fundamental fact (deRosset 2013).

Purity does not only conflict with fundamental *sui generis* metaphysical laws but also with fundamental essences. Psychophysical laws, for instance, feature derivative entities, whether they are construed as *sui generis* metaphysical laws or as essentialist truths. Shamik Dasgupta (2014a, 2014b), however, argues that essences are *autonomous*, in the sense that essences are not apt for metaphysical explanation. The autonomy of essences is motivated in part by the relationship between essence and real definition. It makes little sense, Dasgupta argues, to explain a definition.

The autonomy of essence supposedly removes the tension with purity. For the purity constraint understood properly, says Dasgupta, is the claim that all ungrounded truths *which are apt for explanation* mention only fundamental phenomena. Since essences are autonomous, ungrounded essences are not problematic, even if they involve derivative entities. Essences are therefore preferable to *sui generis* laws.

Aside from rejecting purity or exempting metaphysical laws from its jurisdiction (Schaffer 2017), the proponent of *sui generis* laws might try to push back against Dasgupta’s use of autonomy (Glazier 2017; Raven forthcoming) or else claim that their laws are also autonomous. But there are two other strategies worth considering, which don’t invoke the notion of autonomy: first, we might reduce all metaphysical laws (pace Schaffer’s regress) and, secondly, we could develop a conception of fundamental metaphysical laws that don’t mention derivative entities.

Consider first the attempt to reduce *sui generis* metaphysical laws to regularities in an antecedently given mosaic: metaphysical laws are regularities that are important to us in metaphysical theorizing. Gideon Rosen was the first to sketch such a regularity theory. On his view, a regularity has the status of a metaphysical law if it is “governing how objects in general are put together” (Rosen 2006: 35). Regularity theories of laws of metaphysics avoid the conflict with purity by eschewing fundamental metaphysical laws.

Regularity theories face the following three difficulties. First, they must specify a recipe, perhaps analogous to Lewis’s (1983) best system account of laws of nature, which details the reduction of laws to regularities. Secondly, just like causal regularity theories, the regularity theoretic approach renders metaphysical explanation non-productive (section 5). And thirdly, regularity theorists must address the regress-worry: if laws are grounded in regularities, don’t they require further laws, and so on *ad infinitum*?

These difficulties do not knock regularity theory off the table. There should be some way to devise the relevant best system, and if Humeans can live without productive causal explanations, they should be able to live without productive metaphysical explanations as well. Moreover, we might try to avoid the regress if we deny that the explanation of metaphysical-law facts requires further *sui generis* laws. Instead, these explanations might appeal to the essence of laws or to truth-maker rules: law-talk might be made-true by facts about the ‘metaphysical mosaic’. As so often, it looks as though the committed reductionist can build a fortress.

A less radical response to purity worries offers fundamental laws that don't involve derivative phenomena. One implementation of this strategy is the *constructional conception* of metaphysical laws. Following Kit Fine's (1991, 2010) discussion of constructional ontology, the constructional conception presupposes that every derivative object, property, or any other kind of derivative entity is *constructed* from more basic entities. Some entities, like a mereological sum, a constituted artifact, or a complex property, are constructed from their constituents; and some entities, like a smile or a derivative part, are the constructed aspects of more expansive entities like faces and organic wholes. According to the constructional conception (Wilsch 2015), metaphysical laws characterize how construction operations apply to entities and how the constructed entities 'go together' based on their constructional profiles.

On that conception, metaphysical laws do not name derivative objects and properties but only refer to them by means of quantification and fundamental construction operations. To illustrate, consider a conjunctive-property law, which takes facts of the form of Fa and Ga to facts involving the conjunctive property $F \& G$: For all x , if Fx and Gx , then $F \& G(x)$. Does a law of this sort violate purity constraints? It doesn't if the property-constructor ' $\&$ ' is considered a fundamental ingredient of reality. We can construe metaphysical laws in a purity-friendly way if they only concern objects and properties that are constructed via fundamental constructors from the fundamental ontology (Wilsch 2016).

7. Metaphysical Laws and the Intimacy of Ground

A second difficulty for *sui generis* laws of metaphysics arises from the dictum that metaphysical explanations are the "tightest" of all explanations (Fine 2001: 15). In this vein, it is sometimes said that the metaphysically explained fact is 'nothing over and above' the explaining facts or that metaphysical explanations are *reductive*. The challenge for proponents of *sui generis* laws is to explain how metaphysical laws manage to establish explanations of this intimate sort.

To further clarify this challenge, a comparison to laws of nature and to emergence laws might be helpful. Laws of nature establish causal explanations, and emergence laws establish non-causal explanations, which are, however, supposed to leave the explained facts unreduced and independent to a degree from the facts in the emergence base. We can debate over whether there is room for the notion of an emergence-law, as the phenomenon of emergence is not well understood (see Leuenberger's entry "Emergence", Chapter 21 in this volume). But the challenge remains: what separates laws of metaphysics from other laws, which establish explanations less tight than metaphysical explanations?⁴

We could answer this question with brute force by stipulating that this is just what a law of metaphysics is: metaphysical laws are those laws that establish maximally intimate explanations. The problem with this primitivist answer, apart from leaving intimacy unexplained, is that it makes reductive enterprises look too easy. If you wish to ground the normative in the descriptive or the mental in the physical, simply postulate metaphysical laws connecting various input-states to associated output-states and be done. The intimacy-inducing nature of the postulated laws determines that the mental (or normative) is nothing over and above the physical (or natural). Doesn't that make vindicating physicalism and normative naturalism too easy?⁵

To avoid this consequence, we can constrain the contents of metaphysical laws: not every function from input- to output-states may qualify as metaphysical law. We can impose constraints on the contents of laws that make it harder to postulate them and that also explain their connection to the tightness of metaphysical explanations. One kind of constraint that

may play the desired role comes from the constructional conception (section 6). On that conception, metaphysical laws take physical states to mental states, for instance, by specifying how the resultant state is constructed. Of the following examples, only LAW 2 is an acceptable metaphysical law on the constructional conception, as LAW 1 does not mention construction-operations:

LAW 1 When S's c-fibers fire, then S is in pain

LAW 2 When S's c-fibers fire, then S has property F, where F is constructed via operation O from c-fiber stimulation.

The first law is easy to postulate and does not render intelligible why resulting explanations of pain are reductive. The second law is harder to postulate, as it requires us to make sense of a construction operation that generates psychological from physiological properties or states. As we will see next, LAW 2 also explains why metaphysical explanations of pain are especially tight.

The constructional conception is naturally combined with the claim that talk of 'nothing over and above' does not really apply to truths but rather to entities: the composite entity is nothing over and above its parts, the statue is nothing over and above its constituting matter, and the smile is nothing over and above the smiling face. Generalizing from these cases, *x* is nothing over and above the *ys*, just in case *x* is constructed from the *ys*. Since being nothing over and above is an ontological relationship and not one between truths, we should say that two truths, *p* and *q*, bear an intimate relationship to one another only insofar as some of the entities featuring in *p* are nothing over and above the entities in *q*.⁶

On this view, there is nothing about the laws of metaphysics *per se* which renders metaphysical explanations especially tight. Rather, the constructional relationships between the entities featuring in metaphysical explanations is what makes these kinds of explanation intimate. No explanation of mental truths in terms of physical ones makes for a reduction; what is needed for physicalism, for instance, are reductive ontological relationships between mental and physical entities like mental and physical states or properties.

To sum up, the reductive nature of metaphysical laws could come either from the notion of a metaphysical law or from constraints on their content. The first option puts hardly any constraints on a reductive agenda. On the second option, the content of the laws might explain both why metaphysical laws don't come for free and why metaphysical explanations are especially intimate. The constructional conception of the laws is one possible implementation of the second option.

8. Are Metaphysical Laws Metaphysically Necessary?

Metaphysical laws carry necessity in a variety of ways. If *p* is a metaphysical law, then *p* is necessary, and so is the fact that *p* is a metaphysical law. Moreover, if a metaphysical law establishes a metaphysical explanation '*q* because *p*', then the conditional '*if p then q*' is also necessary (although see Skiles 2015 and Skiles's entry "Necessity", Chapter 10 in this volume, for critical discussions). But how do metaphysical laws give rise to necessity, and what kind of necessity do such laws give rise to? We will see that these questions spell trouble for *sui generis* metaphysical laws.

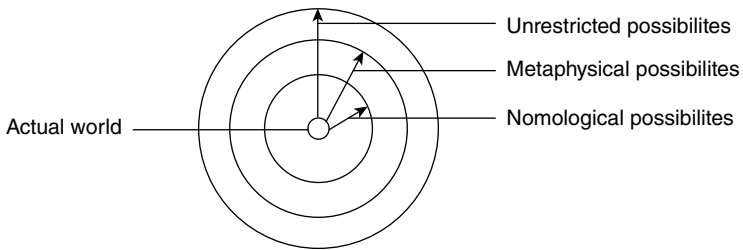
On the received view, there is one *absolute* kind of necessity or necessity *tout court*, which is often referred to as 'logical' or 'metaphysical' necessity, and there are several *relative* kinds of necessity, which are defined from the absolute one. The most common example of a relative

kind of necessity is *nomological necessity*, which applies to a proposition, p , just in case it is necessary in the absolute sense that if the actual laws of nature hold, then p . Since necessity and possibility are duals, the notion of a relative kind of necessity can also be understood as a restriction on a space of possibility. The space of nomological possibilities, for instance, is gained by restricting the space of metaphysical possibilities to those possibilities at which the actual laws of nature hold (see Fine 2002 for disagreement).

The problem for *sui generis* laws of metaphysics is now as follows. On the one hand, metaphysical laws seem necessary *tout court*. For instance, it just seems absolutely impossible that Peter is grey and yet not colored or that Socrates exists without his singleton. But the case of laws of nature suggests, on the other hand, that the necessity of laws is a relative kind. Since *sui generis* laws of metaphysics are laws, they thus don't seem to secure absolute necessity.

If metaphysical laws are essences, this difficulty is easily avoided, because essences are associated with absolute necessity. But there are also two responses available to proponents of *sui generis* metaphysical laws. The first one denies that the necessity of metaphysical laws is absolute, and the second response denies that laws only give rise to relative kinds of necessity. As we will see, both responses raise difficult questions.

The first response insists that the necessity of metaphysical laws is a relative sort of necessity. Gideon Rosen (2006), for instance, suggests that there is an order of strength running from the necessity of essence to metaphysical necessity to the necessity of laws of nature. On his view, the necessity of essence is absolute, but the other two kinds of necessity are relative because metaphysical and natural possibility are defined as restrictions of essence-possibility: p is metaphysically possible iff_{def} p is consistent with essences and metaphysical laws, and p is naturally possible iff_{def} p is consistent with essences as well as metaphysical and natural laws. Putting these definitions together, we get a nested-spheres picture of possibility:



On Rosen's view, metaphysical and natural necessity are relative. But as Rosen points out, relative kinds of necessity run the risk of deflation. For we can use essence-necessity together with any arbitrary set of truths to define a relative kind of necessity. We could define, for instance, that a truth p has *Peter-necessity* just in case p is essence-necessitated by truths about Peter. The definition of Peter-necessity is completely analogous to the definition of metaphysical necessity as whatever is essence-necessitated by the metaphysical laws. This analogy is problematic for Rosen's account. If the only relevant difference between metaphysical necessity and Peter-necessity consists in the utility of those notions to us, the difference between them may not be sufficiently objective. (Rosen 2006 and Leech 2016 disagree. See also Fine 2002; Wilsch 2017.)

If these difficulties stick, no genuine kind of necessity should be understood as a relative kind. It follows that the necessity of metaphysical laws is absolute, and so is the necessity of essence and the necessity of laws of nature. The view that essences, metaphysical laws, and

laws of nature exert absolute kinds of necessity is surprisingly underdeveloped. To show that it is a workable view, one would need to address the following questions. First, is there one or are there multiple absolute kinds of necessity? Secondly, what determines whether two sources exert the same or different kinds of necessity? And what does it take for one absolute kind of necessity to be stronger than another absolute kind of necessity? If these questions can be addressed, proponents of *sui generis* laws must explain whether the necessity of metaphysical laws is distinctive or whether it is the same as the necessity of essence or the necessity of natural laws.⁷

Related Topics

A more detailed discussion of essence can be found in Justin Zylstra's contribution to this volume (Chapter 22).

For a more thorough discussion of Humean reductions of laws, the reader should look at Nina Emery's contribution on the laws of nature (Chapter 21).

For a general discussion of the attempt to ground metaphysical explanations, see Jon Litland's entry on meta-ground (Chapter 9).

Notes

- 1 See Wilsch (2016) for an account of how metaphysical laws support explanations.
- 2 Ted Sider (2012: ch. 7) develops the 'metaphysical semantics' framework for truth-maker rules; not to be conflated with truth-maker theory.
- 3 Glazier's (2015) account of metaphysical laws as a variable-binding connective seems particularly well-suited for a construal of metaphysical laws as input-output mechanisms.
- 4 There is a tension between the intimacy of metaphysical explanations and the claim from section 5 that some metaphysically explained entities are 'entirely new'. While this tension may perhaps be resolved, one could use it to argue against the productivity or the intimacy of metaphysical explanation.
- 5 'Grounding physicalism' as developed in Schaffer 2017 is a potential target of this argument.
- 6 An interesting exception to this ontological construal of 'nothing over and above' occurs with applications of truth-maker rules: the disjunctive truth is nothing over and above the disjunct-truth because it is made true by the same bit of reality.
- 7 Many thanks to Nina Emery, Kit Fine, Stephanie Leary, Mike Raven, Jonathan Schaffer, and Erica Shumener for feedback on earlier drafts.

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31

LAWS OF NATURE

Nina Emery

Distinguish between two questions about laws of nature. On the one, hand we might wonder: what are the laws of nature? On the other hand, we might wonder: what are laws of nature?

That one little ‘the’ makes a big difference. While the first question usually falls within the purview of science, the second is squarely within the realm of metaphysics. Scientists collect data, analyze it, and try to determine what the laws of nature are. (Consider Schrödinger’s equation. Does that equation describe a law? Does it describe the only law? If not, what other laws are there?)¹ Metaphysicians, meanwhile, consider the range of possible laws of nature and attempt to discern what they have in common—what laws of nature are. (Suppose physicists reach a consensus that Schrödinger’s equation does in fact describe a law of nature. What does that mean? What sort of thing is it that that equation describes?)

In answering the second question—the question of what laws of nature are—the notion of *metaphysical grounding* (hereafter just *grounding*) has proved increasingly important. This is true both when it comes to understanding the possible positions that one might take in response to the question ‘what are laws?’ and in giving rise to novel argumentative strategies in defense of one position or another. This application of the notion of metaphysical grounding should be of interest in its own right, but it will perhaps be of particular note to those who are otherwise skeptical of the notion of grounding and of the sorts of debates in contemporary metaphysics where grounding plays an obvious role. For if grounding also plays a role in helping us understand what laws of nature are, that is an important indication that investigations into grounding cannot be purely speculative or wholly esoteric. They make contact with our best science in interesting and important ways.

Here is a plan for what follows. In Section 1, I will set out some groundwork with respect to the notion of laws of nature. I will then turn to two central questions in the metaphysics of laws: what (if anything) grounds the laws (Section 2) and what (if anything) the laws ground (Section 3). To keep things (relatively) simple, I will focus on these questions as they apply to deterministic laws that show up in fundamental physics. In Section 4, I say a bit about how the discussion might extend to laws that are not deterministic.

The reader will note that I plan to say little here by way of introduction to the notion of grounding. For those who are interested, I recommend the introduction to this volume and the earlier chapters. There is, however, one important clarification regarding the notion of grounding that is worth discussing at the outset. As will be familiar to those who have read

the introduction to this volume, grounding theorists are divided on the question of what sorts of entities are the relata of grounding relations. Perhaps the most standard view is what we might call *factualism*—the view that grounding relations only hold between facts. But at least some prominent philosophers endorse *ecumenicism*—the view that grounding relations can hold between many types of entities. According to the ecumenicist, grounding relations can hold between Socrates and {Socrates} for instance, or between two hydrogen atoms and an oxygen atom, on the one hand, and a water molecule, on the other.²

The discussion that follows will appear to endorse ecumenicism: it will focus on the viability of claims like “the Humean mosaic grounds the laws”, or “laws ground causal relations”, in which the relata of the grounding relations are not facts. But this appearance is misleading. As far as I can tell, all of the substantive claims under discussion can easily be reinterpreted in language that is acceptable to the factualist. When some Humeans claim that the Humean mosaic grounds the laws, for instance, that claim can be reinterpreted as the claim that certain facts about the Humean mosaic ground certain facts about the laws. So although the discussion appears to endorse ecumenicism, this appearance is misleading. A factualist about grounding can and should engage with the debates taken up here, albeit in slightly different terms than those in which they are presented.

Finally, let me note that there has been important work expressing skepticism about both the notion of laws of nature and the notion of metaphysical grounding.³ I will not engage with any of that work in what follows. Instead I will assume that there are such things as laws of nature and there is such a relation as metaphysical grounding and focus on how work on the latter might inform work on the former.

1. Laws of Nature

What are laws of nature? A particularly simple account starts from observing how scientists go about discovering the laws: they pay attention to patterns, or *regularities*, in the data that they collect. They pay attention, for instance, to the fact that, in their observations, net force is always equal to mass times acceleration. (Let’s assume that this *is* a fact for the time being.) Under certain circumstances, scientists then conclude that the observed regularity corresponds to a law. They conclude, for instance, that it is a law that net force is equal to mass times acceleration.

This suggests the following account:

The simple regularity account. It is a law that all Fs are Gs if and only if all Fs are Gs.

Unfortunately, the simple regularity account faces serious issues. For one, being a regularity is not sufficient for being a law. Every road in Rhode Island has potholes. But it is surely not a law that every road in Rhode Island has potholes. Or consider the following pair of propositions:

- (G) All spheres of gold are less than 1 mile in diameter.
- (U) All spheres of uranium are less than 1 mile in diameter.⁴

Both (G) and (U) are regularities. Indeed, they seem to be regularities of precisely the same sort. But only the latter corresponds to a law. The former, while true, is merely an accident.

In addition, it does not seem as though regularities are capable of fully playing the role that laws play in scientific inquiry. Scientists use the regularities they observe as a guide to discovering the laws. But they also use laws to *explain* those regularities. Why is it that net force is always equal to mass times acceleration? Because it is a law that net force is equal to mass times

acceleration. But if law L just is regularity R, and we use law L to explain regularity R, then we have used something to explain itself. And nothing can explain itself.

So the simple regularity account of laws fails. Several other potential accounts have been suggested in its place. Among these, there is an important distinction between accounts that are *Humean* and those that are *non-Humean*.

Humeanism about laws of nature. Laws of nature are, in some important sense, nothing over and above the actual distribution of nonmodal entities throughout spacetime.⁵

Non-Humeanism about laws of nature. Laws of nature are, in some important sense, something over and above the actual distribution of non-modal entities throughout spacetime.⁶

Humeanism about laws of nature is a part of a broader Humean program according to which, in some sense, all there is is the *Humean mosaic*—the actual distribution of nonmodal entities throughout spacetime.⁷ For the Humean, it isn't just laws that are nothing over and above the mosaic; causal facts and modal facts and dispositional facts and chancy facts—indeed, all of those facts that you might have naively thought of as at least in some sense modal—are all nothing over and above the mosaic.

Why be a Humean? Think again of the regularity account. One significant advantage of that view, had it been viable, would have been that it identified laws with patterns of events and, in so doing, made laws entirely prosaic features of our metaphysics. In particular, it would have removed any concern that laws are, in some sense, modal entities. Whereas it is natural to describe laws using modal language—laws determine what *can* happen or what *would* have happened or what *must* happen—the simple regularity analysis said that this language does not have any deep modal implications. Laws are just certain kinds of patterns in the distribution of actual events. Even as metaphysicians have largely given up on the simple regularity analysis, many of them have tried to preserve this aspect of that view. Perhaps we cannot give quite such a prosaic account of laws as the simple regularity account promised, these metaphysicians think, but let's try to at least avoid any commitment to the sort of necessary connections in nature about which Hume was so famously concerned.

Of course, in order for the Humean/non-Humean distinction to be made precise, one needs to be more specific about the relevant sense in which the laws might or might not be *something over and above* the Humean mosaic. Following David Lewis (1986), the traditional way to spell out this distinction is in terms of supervenience.

Supervenience-based Humeanism about laws. The laws of nature supervene on the Humean mosaic.

Supervenience-based Non-Humeanism about laws. The laws of nature do not supervene on the Humean mosaic.

As will be familiar to many, if the Fs *supervene* on the Gs, then any two worlds that are identical with respect to the Gs must also be identical with respect to the Fs. So according to supervenience-based Humeanism about laws, any two worlds that are identical with respect to the Humean mosaic are identical with respect to the laws. According to supervenience-based non-Humeanism, two worlds may be identical with respect to the Humean mosaic and still differ with respect to the laws.

Humeans about laws also need to say something more about what sort of entities laws are, beyond just that they are entities that supervene on the Humean mosaic. Currently, the most

prominent Humean account of laws, which was the account endorsed by Lewis, is the *Best Systems Analysis*. According to the BSA, the laws of nature are the true statements that, when taken together as a set, convey as much information as possible about the Humean mosaic without sacrificing too much by way of simplicity. In slogan form: according to the BSA, laws are the axioms of the deductive system that best balances simplicity and strength.⁸

Advocates of the BSA tout two main advantages of their theory.⁹ First, they think that their theory effectively encodes the criteria that practicing scientists use when determining what the laws are: scientists look for true propositions that will allow them to convey large amounts of information about actual events in a relatively simple way. Second, advocates of the BSA think that in making sure that laws are nothing over and above the Humean mosaic, they avoid allowing suspect entities into their metaphysics.

At the same time, many philosophers reject the BSA and Humeanism about laws in general and instead endorse various kinds of non-Humean analyses. Some non-Humeans, like Carroll (1994) and Maudlin (2007), are primitivists about laws of nature. They think laws admit of no further analysis. Others attempt to give an analysis of laws, but not one that has laws supervening on the Humean mosaic. According to Armstrong (1983), for instance, laws are necessitation relations between universals. Some of these non-Humeans are motivated by concerns about the supervenience claim to which Humeans are committed—is it really true, they wonder, that any two worlds that are the same with respect to the mosaic will have the same laws?¹⁰ Others are motivated by the same sort of explanatory concern that was leveled against the simple regularity account from earlier.¹¹ What all these non-Humeans have in common is that laws, on their accounts, are something over and above the Humean mosaic.

The Humean/non-Humean distinction remains one of the most divisive debates in the metaphysics of science.¹² And as we will see over the course of the next sections, the notion of metaphysical ground may help make clear what exactly the Humean/non-Humean distinction amounts to, as well as opening new avenues of argument for and against views on either side of the distinction.

2. What Grounds the Laws?

Let's focus first on the question, "what (if anything) grounds the laws?" And let's restrict our focus specifically to the laws of physics, and even more specifically to deterministic laws of physics. To have a particular example to hand, suppose that we live in a Newtonian world. So it is in fact a law that net force is equal to mass times acceleration. It is a law that $f=ma$.

Perhaps the most significant application of the question "what (if anything) grounds the laws?" is in providing a novel way of understanding the Humean/non-Humean distinction set out in Section 1. According to many contemporary metaphysicians, the correct way to understand that distinction is not, in fact, in terms of supervenience as Lewis proposed. Instead, the distinction should be understood as follows.

Grounding-based Humeanism. The laws of nature are fully grounded by the Humean mosaic.

Grounding-based Non-Humeanism. The laws of nature are not fully grounded by the Humean mosaic.

So, for instance, Beebe (2000: 572) writes that the difference between the Humeans and the Non-Humeans is that Non-Humeans want to "ground the distinction between laws and accidentally true generalizations in some metaphysically substantive feature of the world—something

irreducibly nomic.” And Schaffer (2008: 82) says that the view that laws of nature “are nothing over and above the pattern of events, just like a movie is nothing over and above the sequence of frames,” can be expressed in slogan-form as “what must be is grounded in what is.” And Loewer (2012: 116) says that the key feature of Humean accounts is that they “eschew fundamental nomological modalities.”¹³

Whether Lewis himself would have been amenable to putting the Humean/non-Humean distinction in terms of grounding is a tricky question.¹⁴ But in any case, this way of interpreting the Humean/non-Humean distinction is at least a somewhat natural development. After all, it seems plausible to think that what Humeans were hoping to capture all along was the idea that the laws depend in some important way on the mosaic. But as has often been noted in the literature on grounding, supervenience doesn’t seem to be the right sort of relation to capture any substantive notion of dependence.¹⁵

Of course, whether one finds this way of formulating the Humean/non-Humean distinction helpful will depend at least in part on whether one finds the introduction of grounding relations helpful in other areas of metaphysics. One can imagine, for instance, a version of Wilson’s 2016 argument against understanding the physicalist/nonphysicalist distinction in terms of grounding,¹⁶ adapted as an argument against understanding the Humean/non-Humean distinction in terms of grounding. But it is worth emphasizing that, in addition to being potentially useful in understanding the distinction between Humeanism and non-Humeanism about laws, the notion of metaphysical grounding has helped defend Humeanism against an important challenge. Indeed, the notion has proved so promising in this light that, at least anecdotally, it seems that Humeans who are otherwise skeptical of the notion of grounding are tempted to set their skepticism aside.

In order to understand the challenge that is relevant here, think back to the explanatory challenge for the simple regularity account of laws as described in Section 1. On the one hand, the simple regularity account says that laws are regularities. On the other hand, scientists seem to use laws to explain regularities. But nothing can explain itself.

This challenge is a specific version of a more general *explanatory challenge* for Humeanism about laws. This general version of the challenge starts by noting that, according to the Humean, the laws are nothing over and above the mosaic. It seems plausible, then, to think that the mosaic, in some sense, explains the laws. Why are the laws what they are? Surely for the Humean, the answer to this question must be: because the mosaic is the way it is. But again, one of the key roles of laws in science is to explain both particular features of and patterns across the mosaic. So it seems that the Humean is committed to an explanatory circle: the laws explain features of the mosaic and the mosaic explains the laws. As Maudlin (2007: 172) puts it: “If the laws are nothing but generic features of the Humean Mosaic, then there is a sense in which one cannot appeal to those very laws to explain the particular features of the Mosaic itself: the laws are what they are in virtue of the Mosaic rather than vice versa.”¹⁷

One way for the Humean to avoid this sort of explanatory challenge would be to give up on the idea that laws play any robust explanatory role. But that seems scientifically revisionary in a problematic way. So instead it has become increasingly standard for Humeans to endorse a strategy put forward by Loewer 2012, according to which the explanatory challenge can be avoided by disambiguating two different notions of explanation. To see how this strategy works, start with a grounding-based formulation of the Humean/non-Humean distinction. This formulation commits us to a very specific sense in which the laws are nothing over and above the mosaic: the laws are grounded in the mosaic. In so doing, it commits us to a very specific sense in which the mosaic explains the laws.¹⁸ The mosaic *metaphysically explains* the laws. And, according to Loewer at least, metaphysical explanation is something importantly distinct from the kind of

explanatory relation that scientists have in mind when they use laws to explain particular events in and patterns across the mosaic. Calling the latter sort of explanatory relation a *scientific explanation*, we get the following result: the mosaic metaphysically explains the laws, while the laws play a role in scientifically explaining the mosaic. And it is perfectly fine, so says Loewer, at least, for A to metaphysically explain B while B plays a role in scientifically explaining A.

There's no clear consensus as to whether this move on behalf of the Humean is successful.¹⁹ What is clear is that the notion of metaphysical grounding has opened up a response to the explanatory challenge to Humeanism about laws that was previously unappreciated. Moreover, further work on metaphysical grounding could have important further implications for this debate. In particular, Loewer's maneuver should make us curious about the similarities and differences between metaphysical explanation and the grounding relations that back such explanations, on the one hand, and scientific explanation and the dependence relations—like causal relations—that back those explanations, on the other. Take for instance the claim found in Kment 2015 that metaphysical explanation is “exactly analogous” to scientific explanation or the contention in Schaffer 2016 and Wilson 2018 that there are few, if any, important differences between grounding and causation.²⁰ Presumably these sorts of views make Loewer's maneuver more difficult to get off of the ground.²¹

In this section I have been focused on the question, “what grounds the laws?” I have suggested that understanding Humeanism as the claim that the laws are grounded in the mosaic, has already been, and promises to continue to be, philosophically fruitful. Is the question of what grounds the laws also helpful for the non-Humean? Although there is less explicit work to date on this topic, it certainly seems as though it might be. In particular, the question “what grounds the laws?” seems as though it might be helpful in allowing us to differentiate among varieties of non-Humeanism. Non-Humeans don't think that the laws are fully grounded in the mosaic. Do they think that the laws are fully grounded in anything at all? Some say yes. A natural way to understand Bird's (2007) view, for instance, is that laws are grounded in the nature of the fundamental, dispositional properties. And according to Lange (2009), laws are grounded in counterfactuals.²² Others say no. According to Carroll (1994) and Maudlin (2007), laws are primitives, admitting of no further analysis. Plausibly this means that they think laws are not even partially grounded in anything at all. Still others may say that laws are partially but not wholly grounded.²³

Getting clear on whether and how non-Humean laws might be grounded will surely be helpful in assessing the costs and benefits of non-Humeanism. We might, for instance, wish to extend Hildebrand's (2013) argument against “unanalyzable” non-Humean laws as an argument against ungrounded non-Humean laws and thus conclude that non-Humeanism is plausible only insofar as non-Humean laws are grounded in something (though not, of course, in the Humean mosaic). Or suppose we follow Schaffer (2015) in thinking that Occam's razor only requires us to be concerned with minimizing those entities that we posit at the fundamental level. Then Occam's razor will not cut against non-Humean laws as long as those laws are fully grounded. Of course, there may be other reasons to be suspicious of non-Humean laws or to take such accounts as costly, but mere concerns about simplicity would not be relevant.

3. What, if Anything, Do the Laws Ground?

So much for a discussion of the various ways in which laws might be grounded. Can laws also be grounds?

One reason for thinking that laws are grounds is that many philosophers have a strong intuition that laws in some sense govern (or guide) features of the world, and it may be that the

best way to understand this notion of governance is in terms of grounding.²⁴ But the notion of governance has been subject to significant scrutiny,²⁵ and there is no need to start from such a controversial starting place.

Instead, let's start from the (relatively) uncontroversial claim that laws play an important explanatory role in standard scientific practice. As we saw in Section 2, even many Humeans are eager to find a way to accommodate this claim.

In fact, closer attention to scientific practice suggests that laws play two distinct sorts of explanatory role. First, laws are the reason why a certain set of initial conditions causally explains a certain set of final conditions.²⁶ So, for instance, the reason why dropping a rock of a certain mass and from a certain height causes it to hit the ground traveling a certain speed is that it is a law that $f=ma$. Second, laws explain events in the world directly. Sometimes the explanandum for which laws serve directly as explanans are individual events. For instance, a plausible answer to the question, "why is that meteor traveling slower than the speed of light?" is that it is law that nothing travels faster than the speed of light.²⁷ But, as pointed out in Section 1, laws also are used to explain patterns of events. If you ask a scientist "why is it the case that net force always equal mass times acceleration?", a natural response would be "because it is a law that $f=ma$." This, recall was one of the main reasons we didn't want to endorse the simple regularity account of laws. Laws can't be regularities because laws are supposed to explain regularities.

So laws seem to play two important explanatory roles—they explain causal relations between events, and they explain events—both individual events and patterns of events—directly. Now notice that explanatory relations are paradigmatically backed by dependence relations. In particular, most uncontroversial cases of explanation are either backed by a causal relation or by a grounding relation. Zeke's throwing the rock explains the window breaking, for instance, because Zeke's throwing the rock causes the window to break. Or the physical properties of water ground the chemical properties of water, to take another example, because the physical properties of water ground the chemical properties of water. A natural thing to think, therefore, is that *all* explanations are backed by either a causal relation or a grounding relation. It is natural, in other words, to adopt the following assumption:

The simple backing assumption. If A explains B then either A causes B or A grounds B.

Finally, note that laws do not seem like the right sorts of things to themselves stand in causal relations. Paradigmatic cases of causation involve causal relations holding between events. And laws are not events.²⁸ Taken together, these observations suggest that laws are grounds. Specifically, laws both ground causal relations between events and ground events in the world directly.

One thing that is interesting about the idea that laws ground particular events in the world is that it threatens to undercut Loewer's maneuver in response to the explanatory challenge discussed in Section 2. Loewer's maneuver, remember, was to say that it was not a problem for Humeans to claim that the mosaic explains the laws while the laws also explain features of the mosaic, because there are two different types of explanation involved. The mosaic metaphysically explains the laws, while the laws scientifically explain the mosaic. But what are metaphysical and scientific explanation? Loewer is explicit that metaphysical explanations are those explanations that are backed by grounding relations. But if the argument presented earlier in this section is correct, then at least some cases in which the laws explain the mosaic are also backed by grounding relations. So there is, in at least some cases, a genuine circle of metaphysical explanation after all. This suggests that Humeans about laws who like Loewer's maneuver will need to reject some component of the foregoing argument. Either they need to (i) allow that laws can be causes, (ii)

reject the simple backing assumption, or (iii) reject the claim that laws explain events—either individual events or patterns of events—directly.

Note, however, that even those Humeans who take route (iii) may still remain committed to the view that laws explain causal relations between events in the way described above. And insofar as they are so committed, and they find the simple backing assumption plausible, they will still be committed to the view that laws are grounds—laws ground causal relations.²⁹

4. What if the World Is Indeterministic?

Let's broaden our scope to consider the possibility that we live in an indeterministic world. That is, let's broaden our scope to consider the possibility that the complete state of the world at one time, plus the laws of nature, does not determine the complete state of the world at every other time. Some indeterministic worlds plausibly have no laws at all—events happen at random. But other indeterministic worlds surely do have laws. These laws do not entail that one state follows another. Instead, they assign objective probabilities to one state following another.

It is an open possibility, given our best science, that our world is indeterministic. Some interpretations of quantum mechanical phenomena say that it is, while others do not. So presumably we should have a metaphysics of laws of nature that is compatible with both scenarios. And taking into account indeterministic laws alongside deterministic laws has the potential to significantly impact the role that grounding plays in the metaphysics of laws.

First, consider the question of what grounds the laws. Famously, the shift to indeterministic laws gives rise to a serious problem for the Best Systems Analysis. This problem, which David Lewis famously dubbed the *Big Bad Bug*, is due to the fact that on the most obvious way of extending the Best Systems Analysis to indeterministic laws, the laws will sometimes assign a non-zero chance (at some time) to a series of events (after that time) that would result in the laws being different than what they in fact are. This is surprising. The laws shouldn't say that it is physically possible that the laws be different.³⁰

The literature on the Big Bad Bug is too large to survey here; the key point is just that the BSA—which is by far the most prominent Humean account—faces serious issues with indeterministic laws. Once one shifts one's focus to indeterministic laws, the claim that the laws are grounded in the mosaic is significantly more difficult to establish. But some non-Humean accounts—especially those that are not primitivist—may also face issues here. As an example, consider Armstrong's account of laws, according to which laws are necessitation relations between universals. Clearly this account is inadequate if the laws are indeterministic—if the laws are indeterministic, then in at least some cases the instantiation of one universal leads in a lawlike way to the instantiation of another universal without the one necessitating the other. Perhaps we can endorse a straightforward extension of the Armstrong line according to which indeterministic laws are probabilifying relations between universals. But what is a probabilifying relation? Or perhaps we can say that indeterministic laws are necessitation relations between the instantiation of a certain set-up and the instantiation of a certain chance. But then what account will we give of chance?³¹ This is not to say that something like Armstrong's account cannot be extended to indeterministic laws but only that there is work yet to be done here.

Perhaps the most interesting consequence of leaving open the possibility of indeterministic laws arises when one shifts from thinking about what grounds the laws to thinking about what the laws ground. In particular, think back to the argument, presented in Section 3, that laws ground events (either individual events or patterns of events) in the world. If the laws are indeterministic and the laws ground events, then grounding may turn out to be indeterministic as well. That is, there are at least some grounding relations where the grounds do not

metaphysically necessitate the things that they ground—indeed, they don’t even physically necessitate those things.³² This is a surprising result, given that the standard view is that grounding relations are metaphysically necessary, and even those who dissent from the standard view usually still allow that grounding relations are at least physically necessary. Indeed, this result is surprising enough that many of those with previous commitments regarding the nature of grounding will probably take it as a reason to reject the argument in Section 3 that laws ground events. But it is worth noting that insofar as one approaches the notion of grounding with fewer antecedent commitments—perhaps, say, because one is hopeful that the notion of grounding will help them understand the notion of governing as it has played an important but somewhat mysterious role in the metaphysics of laws—one ends up with these sorts of surprising consequences.

Conclusion

The notion of metaphysical grounding has important applications to the metaphysics of laws of nature. It provides new ways of understanding important distinctions within the literature on laws and novel arguments for and against the views that arise on each side of those distinctions. And as the metaphysics of laws of nature is generally assumed to be a fairly naturalized area of metaphysics, these applications seem especially important. Insofar as the notion of metaphysical ground has applications to the metaphysics of laws of nature, that notion cannot be dismissed as entirely abstract or esoteric or uninteresting.

That is a promising result for the fan of metaphysical grounding. But here is a note of caution. Several philosophers, including, Wilsch (2015), Schaffer (2018), and Kment (2015), have suggested that we give a nomological account of metaphysical explanation, according to which, just as scientific explanation is mediated by laws of nature, metaphysical explanation is mediated by laws of metaphysics.³³ For those who might otherwise have been skeptical of the notion of metaphysical explanation, this sounds encouraging. After all, laws of nature, though by no means unproblematic, are at least relatively familiar, and presumably the extant analyses of laws of nature can prove helpful in understanding laws of metaphysics. But the earlier discussion shows that the notion of grounding itself plays an important role in understanding what laws of nature are. As a result, there is no theoretical standpoint here that is independent of considerations of metaphysical grounding that we can use as a neutral starting point from which to try to understand that notion. As in so many areas of philosophy, we instead find ourselves working with a group of interrelated concepts and tools, each of which depends in important ways on the others.

In closing, let me mention one aspect of the relationship between laws and grounding that, due to considerations of space, has been overlooked so far. The foregoing discussion has focused entirely on fundamental laws. Although it is controversial, many philosophers also think that there are non-fundamental laws.³⁴ If so, these laws are presumably grounded in the fundamental laws or perhaps in other fundamental entities. Little of what has been written to date about nonfundamental laws has explicitly engaged with and made use of recent work on metaphysical grounding.³⁵ I leave it to the reader to explore these issues further.³⁶

Related Topics

Bryant, “Physicalism” [Chapter 35]; Glazier, “Explanation” [Chapter 8]; Kovacs, “Modality” [Chapter 24]; Skiles, “Necessity” [Chapter 10]; Wang, “Cause” [Chapter 20]; and Wilsch, “Laws of Metaphysics” [Chapter 30].

Notes

- 1 When there is substantive empirical equivalence, the question of what the laws of nature are might be left to philosophers of science. This is plausibly what has happened in nonrelativistic quantum theory.
- 2 Schaffer 2009 is an example of an ecumenicist. Rosen 2010 and Audi 2012 are factualists.
- 3 Prominent skeptics about laws include Cartwright 1983 and van Fraassen 1989. Prominent skeptics about grounding include Hofweber 2009 and Wilson 2014.
- 4 This example is from van Fraassen 1989: 27.
- 5 Prominent Humeans include Lewis 1986; Loewer 1996; Beebe 2000; Albert 2000.
- 6 Prominent non-Humeans include Tooley 1977; Armstrong 1983; Carroll 1994; Maudlin 2007; Lange 2009.
- 7 As Lewis famously put it, the mosaic is just “one little thing after another” (1986: ix).
- 8 See Lewis 1983: 41–42, 1986: ix–x. Lewis himself cites Mill and Ramsey as inspiration. For a non-Humean version of the Best Systems Analysis, see Demarest 2015.
- 9 See, for instance, Loewer 2007: 313, for a pithy summary of both of these advantages.
- 10 See Tooley 1977; Menzies 1993; Carroll 1994; Beebe 2000 for discussion.
- 11 See especially Armstrong 1983: 40; Maudlin 2007: 172.
- 12 According to Hall ms, p. 1, it is, “by far the most central and important question about laws of nature”.
- 13 See also Bhogal 2017. Note that even though this analysis is becoming mainstream, it is far from consensus. Just as an example, Carroll’s *Stanford Encyclopedia of Philosophy* article on philosophy of time, revised in 2016, treats Humeanism solely in terms of the supervenience account. See Miller 2015 and Kovacs ms for further discussion of why it might be a mistake to think of the Humean/non-Humean distinction in terms of grounding.
- 14 On the one hand, Lewis occasionally used language suggesting that the Humean was committed to some kind of reduction. On the other, he may have thought that in some cases, asymmetric supervenience—such as the asymmetric supervenience of Humean laws on the Humean mosaic—was itself sufficient for reduction. The discussion in Miller 2015, especially footnote 8 is helpful here.
- 15 For more on this, see Kovacs, “Modality” [Chapter 24], in this volume.
- 16 For more, see Bryant, “Physicalism” [Chapter 35], in this volume.
- 17 See also Armstrong 1983: 40; Bird 2007: 86.
- 18 See Glazier, “Explanation” [Chapter 8], in this volume for more on the connection between grounding and explanation.
- 19 See Lange 2013; Hicks and van Elswyk 2014; Marshall 2015; Miller 2015; Roski 2018; Emery 2019, for further discussion.
- 20 See Wang, “Cause” [Chapter 20], in this volume.
- 21 In a different sort of response on behalf of the Humean to the explanatory challenge, Marshall 2015 distinguishes between a law *L* and the metanomological fact that *L* is a law and suggests that Humeans should think that the mosaic grounds the metanomological fact that *L* is a law, but does not ground *L* itself.
- 22 Carroll 2008 puts forward a view according to which laws are those universal generalizations that are not coincidences; they are what they are because of “nature itself” as opposed to being what they are because of something in nature—like some series of unlikely events or some very specific set of initial conditions. This view, too, can be understood in terms of grounding: laws are those universals that are grounded in nature itself.
- 23 Note that other, especially older non-Humean views may be somewhat more difficult to classify in terms of the extent to which they say that laws are grounded. For instance, Armstrong (1983) says that laws are necessitation relations between universals. Once the notion of metaphysical grounding is in play, it seems at least in principle possible to distinguish between two versions of this view. On the one, laws just are certain kinds of necessitation relations. On the other, laws are grounded in certain kinds of necessitation relations.
- 24 Beebe 2000: 257, characterizes the governing conception of laws as one on which “It isn’t just that the laws plus current facts entail future facts; rather the laws ‘make’ the future facts the way they will be: the laws are the ontological ground of the future facts.” Rosen 2010 and Bhogal 2017 also suggest that governance is best understood in terms of grounding.
- 25 See especially Beebe 2000.
- 26 This claim is defended in detail in Skow 2016. Note that Skow himself shies away from talk of explanation and thus would disagree with my characterization of the relation here as explanatory.
- 27 See Woodward 2003: 209 and Skow 2014 for discussion of this example.

- 28 See Emery 2019 for further discussion of a version of this argument.
- 29 The emphasis on causation here may sound distinctly non-Humean, but it need not be. See Hicks forthcoming for details.
- 30 For discussion of the Big Bad Bug, see Lewis 1994: 478ff.
- 31 This question is especially pressing in the current context because philosophers tend to give an account of chance that is of a kind with their account of laws.
- 32 See Skiles, “Necessity” [Chapter 10], in this volume and the discussion in Emery 2019.
- 33 See Wilsch, “Laws of Metaphysics” [Chapter 30], in this volume.
- 34 Insofar as there are special science laws, they are, presumably, nonfundamental laws. Such laws—if they are laws—are usually considered *ceteris paribus* laws, since they only hold under certain conditions. See Earman and Roberts 1999 and Lange 2002 for discussion.
- 35 Prominent work on nonfundamental laws includes Albert and Loewer’s work on the *mentaculus* (see Albert 2000; Loewer 2004, 2009), which attempts to reduce many higher-level patterns, e.g., the second law of thermodynamics, to certain facts about the initial conditions of the universe, and Cohen and Callender’s (2009) *Better Best Systems Analysis*, which provides a recipe for identifying laws at multiple scientific levels.
- 36 Thanks to Tobias Wilsch and Mike Raven for helpful comments.

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32

LOGIC

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Introduction

Some of the paradigmatic examples of grounding (that are often used to motivate, or help us latch onto, the notion of grounding itself) are relations between logically complex facts and the logically simpler facts that entail them. For example:

[The grass is green] grounds [Either the grass is green or the moon is made of cheese].
[The grass is green], [The sky is blue] ground [The grass is green and the sky is blue].
[The chair is orange] grounds [Something is orange].

Either implicitly or explicitly, these are usually (but not always) taken as instances of variations of the following principles:

Conjunctive grounding ('CG'): If each of p , q is true, then $[p]$, $[q]$ together ground $[p \ \& \ q]$.
Disjunctive grounding ('DG'): If p is true, then $[p]$ grounds $[p \vee q]$.
Existential grounding ('EG'): If Fa is true, then $[Fa]$ grounds $[\exists x \ Fx]$.

This entry surveys some things that have been said in favor of these principles (and about logical grounding in general) and raises (but does not resolve) some questions about why we should accept these principles, and, if we should, what it *means* to accept these principles.

I will also discuss the relationship between facts that are instances of universal generalizations and the universally quantified facts that they bear a similar relationship to. Universal generalizations are at least superficially more complicated than other cases. But the difficulty they pose might tell us something about how to think about other cases.

Before I get into the details, let me address some preliminaries.

First, there is not very much literature directly challenging either these principles or the purportedly paradigmatic examples. Wilson (2014: 549–50 & note 46) raises some of the issues I will discuss here, though it is not her central focus. Turner (2016) argues against many of the too-quick moves that he takes pro-grounders to have made, one of which is the assumptions I will discuss here about the principles governing logical ground. Jansson (2017: §4) argues that we do not need to posit ground to account for the felt asymmetry present in logical cases. And

Kovacs (2019) provides an account of metaphysical explanation (not ground) that resists the principles. Many have argued that these cases of *logical ground* involve a different kind of grounding than paradigmatically metaphysical cases of ground, such as: [Michaela's c-fibers are firing] grounds [Michaela is in pain]. *How* different these two kinds of cases are, though, is a matter of disagreement.

Koslicki (2015: 317) points out one difference between cases of logical ground and other cases: logical ground seems to allow for systematic *grounding overdetermination*. DG tells us that if p is true, $[p]$ grounds $[p \vee q]$. If we read 'grounds' here as 'fully grounds' (which most authors seem to want us to), then in every case in which both disjuncts of a disjunction are true, there are multiple distinct full grounds for the disjunctive fact.

If we want to reject that ground allows for systematic overdetermination, it seems that we either need to weaken DG somehow or deny that disjuncts ground their disjunctions at all. (There is actually a third option here: Poggiolesi (2016) gives an account of 'robust conditions' for grounding on which the falsity of a disjunct gets involved, along with the true disjunct, in grounding the disjunction.) But many don't think that systematic overdetermination is problematic. Another option (and Koslicki's point here) is not to take the systematic overdetermination as problematic but rather to take it to indicate that there is a distinctive kind of dependence relation in logical cases, and it is perhaps not best lumped together with (e.g.) distinctively *metaphysical* grounding (for which overdetermination may be more problematic).

Correia suggests a tripartite distinction between metaphysical, conceptual, and logical grounding. Unlike Koslicki, Correia thinks that there are important relationships between his three different kinds of grounding: "It is indeed plausible to hold that every case of logical grounding is a case of conceptual grounding (but not vice versa), and that every case of conceptual grounding is a case of metaphysical grounding (but not vice versa)" (2014: 32). The discussion in §3 touches on whether this is the case.

I should note two choices I have made for ease of explication. First, I will use a predicational notion of grounding throughout. This is in part because some of the discussion is framed around Rosen (2010), so it is easiest to follow his notation. Second, I will, for the most part, only focus on the simplest cases of interaction between grounding and classical logic. This should not mislead readers into thinking that there are not hard questions about more complex cases or about the relationship between choice of logic and grounding. Indeed, part of the point of this chapter is to suggest that these questions are very hard in even the simplest cases and so are likely much harder in more complex cases.

§1 surveys a few places in the grounding literature in which these issues are discussed, and examines four defenses of the principles. §2 discusses particular challenges with grounding universally quantified facts and discusses a related principle. §3 examines EG and raises some problems for the (perhaps strawmanned but still useful for the purposes of explication) claim that these principles are (a) theoretically harmless and (b) easy answers to questions of logical ground. I do not discuss problems with CG and DG, largely because there are no challenges to these principles in the literature (that I know of).¹ However, I suggest that there are likely similar challenges to these principles.

§1. Discussions of and Motivations for CG, DG, and EG

While there is not a huge amount of discussion of the motivation for principles like CG, DG, and EG, there is a fair amount of work that attempts to lay out a formal logic of ground, e.g., Fine (2012), Correia (2014, 2015), Litland (2016; forthcoming), Poggiolesi (2016, 2018), Schnieder (2011), among others. (For discussion of the logic of ground, see Poggiolesi's entry

[Chapter 14] in this handbook.) I want to emphasize that in some ways these logics of ground *themselves* might be taken to motivate certain principles, in that the principles follow from the logics. Poggiolesi (2016, 2018), for example, develops a definition of ‘complete and immediate’ formal (logical) grounding and shows that certain principles follow from that definition. And some motivation is provided, in particular in (2016), for thinking this is the correct account of logical ground. But in most of the works listed, such motivation is fairly minimal and relies on appeal to our intuitions. Further, when questions of e.g., logical complexity are raised, it is often simply assumed that in logical cases, the grounds must be less complex (or at least cannot be more complex) than what is grounded.

Suppose that we had a plethora of well-developed logics of ground to choose from. In that case, the question I am asking in this section could be construed as ‘on what philosophical grounds do we choose between these logics?’ While a logical framework might internally motivate a choice of principles, here I am looking for general philosophical motivation for adopting a given framework, principle, or set of principles.

That being said, there is a bridge between the question of choice of a framework and the question of what internally follows from a framework. The motivation for adopting CG, DG, and EG might come from them helping to give us a clean, simple, or theoretically fruitful theory of the logic of ground. That is, if we develop logics of ground that are themselves theoretically virtuous in various respects, that might help motivate the adoption of the particular principles that those theories put forth. The only place I know of in which an explicit defense that looks anything like this is given in Schnieder (2016: §3), which I discuss in what follows.

Rosen (2010: §6) proposes all of CG, DG, and EG. More generally, he suggests that these are instances of the following principle:

ENTAILMENT: if [p] is grounded in [q], then q entails p.

(He also puts things this way: “The facts that ground [p] together ensure as a matter of metaphysical necessity that [p] obtains” (118), which is closer to other formulations, e.g., Leuenberger (2014).)

This principle is sometimes also called *necessitation* in the grounding literature. One interesting thing to note here, though, is that it does not actually look like CG, DG, and EG are instances of ENTAILMENT. In the case of conjunction, for example, ENTAILMENT would only suggest the following principle:

CG★: If [p & q] is grounded in [p], [q], then [p & q] entails [p] and entails [q].

Granted, Rosen is not suggesting that ENTAILMENT is the *basis* for adopting CG, DG, and EG. Still, it is interesting that ENTAILMENT does not seem to motivate any of our principles. In §2, I’ll suggest that there are reasons to reject ENTAILMENT. (Leuenberger (2014) and Skiles (2014) argue against it, but only one of Skiles’ arguments are relevant to this chapter. For further discussion of ENTAILMENT, see Skiles’ entry in this handbook.)

Fine employs versions of two of our three principles in motivating his “puzzles of ground”. However, he uses the notion of “helping to ground” (partial ground). Fine says that “given the truth of a disjunct, that disjunct will help ground its disjunction” (2010: 100) and “given that y exists and that Ay, then y’s being an A helps ground that something is an A” (2010: 101). Correia (2014) produces ‘rules of logical grounding’ that look quite similar.

One interesting feature of almost all of the work discussing logical cases of grounding is that it is rare to find explicit *motivation* or *argument* for the claim that logical grounding works in the way it does and in the direction that it does. There are four notable exceptions. The first is to appeal to intuitions and is advocated by Schnieder (2016: §2) and Correia (2014). Both Correia and (in a more developed argument) Schnieder appeal to the legitimacy of intuitions in determining the relevant principles about the logic of ground. And indeed, in some sense, all of the canonical cases of ground seem to be motivated via intuitions—e.g., [that Socrates exists] grounds [that {Socrates} exists] seems to be motivated by an intuition that, despite the necessary coextension of Socrates and his singleton set, the concrete members of singleton sets are somehow *prior* to, or *more fundamental than*, their singleton sets. So, one might ask, why not appeal to intuitions in the logical cases? Especially given that there seems to be widespread agreement about these intuitions? Correia says of his logical grounding rules: “they all strike me as correct once a fine-grained conception of grounding is taken for granted . . . there would actually seem to be something like a consensus, among the friends of a fine-grained conception, that the basic rules plausibly record genuine grounding ties, and I guess that those who do share that view would be happy to say that the ties in question are properly logical in character” (2014: 37). (For more on the coarse-grained vs. fine-grained conceptions of ground, see Correia’s entry [Chapter 15] in this handbook.)

So intuitions are one way to motivate canonical cases of logical ground. I won’t take a stand here about whether this is the right way to go, but one possibility is that this motivates canonical cases (e.g., [the ball is red] and [the ball is round] together ground [the ball is red and round]) without motivating perfectly general principles like the three principles discussed above. (One reason to think this might be the case is to compare this to, e.g., methodology in the moral realm. Very often, applied ethics questions or questions about thought experiments are ‘settled’ by intuitions; but it is less common for us to think that we can simply immediately intuit which normative theory is true, except by piecemealing together our intuitions about individual cases.) Or perhaps our intuitions are meant to be direct intuitions of the principles themselves. One question that arises from this issue is the following: even supposing we assume that principles like CG, DG, and EG are true, what is their status? Are they completely inviolable? Might they instead be general rules that admit of exceptions? (Turner (2016) raises some similar worries about why we should think there are any universal ‘laws’ of logical ground.)

A second way to motivate the principles is discussed by Fine (2010: 105–106). Fine suggests that perhaps the reason these principles seem right to us is because of their relationship to truth conditions—but then points out that, given a deflationary reading of truth, there is no substance to truth conditions (where ‘T’ means ‘it is true that’, a statement of the truth conditions for conjunction, $T(p \ \& \ q)$ IFF $(Tp \ \& \ Tq)$, given deflationism about truth, says the same thing as $(p \ \& \ q)$ IFF $(p \ \& \ q)$). One way to recover the directionality that seems to have been lost here is to adopt the ground-theoretic principles. It’s not that the truth of $(p \ \& \ q)$ somehow is grounded in the truth of p and the truth of q ; if we adopt deflationism about truth, this would become a reflexive case of ground. Rather, it’s that p and q ground $p \ \& \ q$. I suspect, then, that Fine is trying to suggest that the principles are the best way to recover this feeling of directionality given deflationism about truth. This may, however, be a more spelled-out version of the first way of motivating the principles: what it seems to boil down to is that we have an intuition about the direction of dependence in either specific or generic cases of these kinds of logical relationships.

A third way to motivate the principles—and the most extensive positive argument for them—can be found in Schnieder (2016: §3). Schnieder’s argument is as follows:

- (P. 1) The rejection of my inference rules leads to a situation where at least some truth-functionally complex statements are regarded as fundamental truths.

- (P.2) (i) The view that a truth-functionally complex statement can be classed as a fundamental truth is, however, counter-intuitive;
- (ii) The view is also theoretically unattractive, given the common principles of theory choice.
- (C) Hence, there are two reasons (that are independent of . . . intuitions) for accepting my rules. (Schnieder (2016: 165)).

While intuitions are invoked in P2, these intuitions are different than those we consult when we just ask whether particular logical principles of ground are true. What is crucial to Schnieder's argument is this: he suggests that (a) rejecting that there are any systematic grounding relations at stake when it comes to things like disjunctions, conjunctions, and existentially quantified claims results in us having to accept truth-functionally complex statements (such as disjunctions) as fundamental and that (b) the only possible alternative principles to (his versions of) CG, DG, and EG are things like "an explanation of a true disjunction proceeds in terms of whatever explains its true disjunct(s)." (Schnieder 2016: 165). (i.e., something like the reverse of DG: in our lingo, if $p \vee q$, p are both true, then $[p \vee q]$ grounds $[p]$.) In both cases, Schnieder argues, we end up with fundamental truths that are disjunctive, conjunctive, etc.

Schnieder's defense of P2 (ii) is worth mentioning: he argues that if we have to posit truth-functionally complex fundamental truths, we will end up with a bloated fundamental base. And this violates a principle of theory choice that tells us to favor theories with smaller fundamental bases. (This issue is related to but not identical to the question of whether parsimony only matters at the fundamental level. Schaffer (2015) argues that it does.) A different line of defense for P2 might be to argue that positing fundamental truth-functionally complex truths will end up violating some sort of neo-Humean principle about the independence of and/or free recombination of fundamental truths.

A final way to motivate the principles is to claim that (e.g.) it is *in the nature of* or *in the essence of* disjunction that true disjuncts ground their disjunctions. This is the strategy that Rosen (2010: §13) uses. One reason that this strategy might seem appealing is that it corresponds to the way entailment works: Fa entails $(\exists x)Fx$, and not vice versa; Fa entails $Fa \vee Gb$, and not vice versa, and so on. A second reason that it is appealing is that it corresponds to the way that the *semantics* for disjunction, the existential quantifier, and conjunction work.

But there are problems for this approach. One immediate problem is that it is unclear what it is for a quantifier or conjunction or disjunction to have an essence. I will focus only on conjunction and disjunction here (with the promise that quantifiers present even more difficult issues). I suspect that the following are roughly the most common views about what conjunction and disjunction are: (a) truth functions and (b) this question is misguided; '&' and 'v' are *nonreferential, syncategorematic* expressions; conjunction and disjunction aren't anything, or, at most, they are the *meanings* of these expressions. If (b) is true, then the only candidate entity that could have an essence is either the expression, '&', or the meaning (but not the referent!) of '&'. If it is the *expression* that has an essence, then it seems to me that we make a mistake if we claim that it is part of the essence of the expression '&' that [snow is white and grass is green] is grounded in [snow is white] and [grass is green]. For this grounding claim is about facts, not linguistic entities. If instead it is the *meaning* that has an essence, but '&' has no referent, then, at the least, more work needs to be done to spell out how this is supposed to work.

If (a) is true, many questions remain. If we think truth functions have essences, then it is probably reasonable to think that it is essential to conjunction that, for any p, q , if p and q are both true, then $p \& q$ is true. But this claim can't be a grounding claim, nor, by itself, is it enough to somehow generate a grounding claim; notice that it should also be essential to conjunction that, for any $p \& q$ that is true, then p is true and q is true. But to my knowledge, no one thinks that

when p, q are true: $[p \ \& \ q]$ grounds $[p]$ and $[p \ \& \ q]$ grounds $[q]$. The person who thinks that it is essential to a truth function hence needs to either explain why it is *not* essential to conjunction that, for any $p \ \& \ q$ that is true, then p is true and q is true, *or* they need to explain why the grounding facts here are asymmetrical while the essence facts are symmetrical.

One option is to claim that it is something about our commitment to compositionality that explains why the essence claim only goes in one direction (and hence why the grounding claim only goes in one direction): because we think of our semantics for conjunction as working by assuming that the conjuncts have truth values and spitting out a truth value for their conjunction, we think of this function as moving from conjuncts to conjunction. But note that this kind of justification seems to be committed to the idea that functions are somehow ‘productive’ or ‘determining’: they, by their nature, take their inputs and *produce* a new entity, claim, truth value, etc. as an output. This is a highly contentious view of functions. More generally: if functions have these kinds of directional essences—if it is essential to them that their inputs somehow ground their outputs—then it seems to me that we need an account of either what is special about a truth function or of what a function is, generally speaking, that makes sense of this, given that standard accounts of functions as, e.g., sets or classes of ordered pairs will not get us directional essences. (I develop an argument similar to this one in McSweeney (forthcoming).)

Further, notice that the proponent of predicational grounding (rather than operational grounding—see the introduction of this handbook for a discussion of this distinction) might have some additional work to do here. Some predicational grounders (e.g., Rosen) take facts to be (something like) true Russellian propositions. If propositions are the relata of truth functions, then the fact grounder does not have to do additional work. But if facts are something else—e.g., obtaining Armstrongian states of affairs, that is, individuals actually instantiating properties—then it is mysterious why the essence of a truth function would produce a principle of grounding between facts, unless the truth function is literally a part of the obtaining state of affairs. (But this would seem to be confused, since states of affairs are not the kinds of things which truth functions take as inputs and outputs—states of affairs are not true or false.)

§2. Universal Quantification and Entailment

Is there a principle like CG, DG, and EG for universally quantified facts? Rosen doesn’t think so. Fine, on the other hand, appeals to a fairly weak principle about the universal quantifier: “Given that everything is an A and that y exists, then y ’s being an A helps ground that everything is an A ” (Fine 2010: 100). Should we accept Fine’s principle? Universally quantified facts pose a challenge to both the “essence” and the “intuition” approaches to motivating a principle, for the universal quantifier, that would correspond to CG, DG, and EG. This is, at least in part, because of the interaction between universally quantified facts and laws.

Rosen argues that there are three kinds of cases of ground between universally quantified facts and their instances. First, universally quantified facts may be grounded in nonquantified facts about *essences*; to use Rosen’s example, [it lies in the nature of triangles that for all x , if x is a triangle then x has three sides] grounds $[\forall x$: if x is a triangle, x has three sides] (2010: 119). Second, universally quantified facts may be grounded in strong laws of nature. Again to use Rosen’s example: that “[any two bodies attract one another with a force inversely proportional to their square distance and proportional to their masses]” is grounded in [it is a law of nature that all bodies act in this way] (2010: 119–120). (Some philosophers—Humeans about laws of nature—should reject this.)

And finally, when a universally quantified fact is an accidental generalization—i.e., when there is no reason that the fact must obtain, but it just happens that it has no counterinstances,

Rosen claims that it is grounded in its instances together with a *totality fact*. What this means is best seen by example. Suppose it is an accidental generalization that all of the five people in this room were born in May. This is grounded by (a) the facts about each individual, e.g., [Michaela was born in May]; and (b) the fact that the exhaustive list of people in the room are *all* the people in the room; there are no other people in the room. (b) is a (restricted) totality fact (Rosen 2010: 120–121).

It is important to note about this last claim that if we deny ENTAILMENT, we don't need to supplement the instances of the universally quantified claim with a totality fact. This is not on its own a reason to deny ENTAILMENT, but doing so might be motivated in two different ways. First, if one was generally squeamish about ungrounded, basic, universally quantified facts, one might want to deny ENTAILMENT rather than admit that we needed an ungrounded, basic, universally quantified fact. (Rosen claims that the totality fact is the only such fact one is left with, if one adopts his disjunctive story about grounding universals).

Second, one might want to deny ENTAILMENT if one had a particular conception of what grounding is: one that adopts as a rough metaphor the idea that the fundamental facts are “what god would have to fix” in order for everything else to be “gotten for free”. Skiles (2014: §4.1) argues, partly based on this kind of conception of ground, that the totality fact does not enter into the grounds of accidental generalizations. I won't examine the details of his proposal, but it is easy to see its intuitive motivation. (God didn't have to lay down a totality fact in addition to everything else she did to make accidental generalizations true.) (For more discussion of questions of entailment/necessitation, see Skiles's entry in this handbook [Chapter 10].)

There is an additional issue with ENTAILMENT in the context of cases of logical ground. Metaphysical necessitation is not the same thing as logical entailment, and indeed there is no obvious relationship between the two. I won't explore this issue further here, but in §3, I will show that there is reason to think that if CG, DG, and EG are true principles, they are principles about our concepts (of conjunction, disjunction, and the existential quantifier) and not about mind-and-language-independent reality. If this is true, and if there is an important relationship between CG, DG, and EG, then it might be that the kind of logical entailment that CG, DG, and EG seem to be based on is unrelated to metaphysical necessitation, which seems to be mind-and-language independent.²

Even if Rosen is not right about the details, it seems that universally quantified facts don't elicit the same strong intuitions about there being a single principle of ground that relates these facts and their instances. Why are universally quantified facts so different in this respect? One possibility is that they aren't—that, instead, we should take the diversity of ways that universally quantified facts can be grounded (or at least, the general sense that there is no obvious consistent grounding relationship at play when it comes to universally quantified facts) to be indicative that there should be more disagreement and diversity at play when it comes to grounding other logically complex facts.

§3. Challenges to EG

In this section, I describe two challenges that arise for EG. (One could produce related challenges to CG and DG, but I do not discuss them here.) The two challenges come from Sider (2011) and Dasgupta (2009), though neither would describe their arguments as providing a challenge to EG, for two different reasons: Sider rejects the relevant notion of grounding, and Dasgupta seems to think that it's at least possible that EG holds, but that this is a reason to theorize about the world without either conjunction or the existential quantifier.

Sider argues that the fundamental facts are best expressed by quantified sentences and argues against the opposing ‘Tractarian’ view, on which fundamental facts are individualistic rather than general (2011: ch. 9.13). Though they would disagree with the details here, the arguments in Sider (and Dasgupta, who I’ll discuss in a moment) can be construed as arguments in favor of a metaphysical picture of the world on which true sentences like $(\exists x)Fx$ (or facts like $[(\exists x)Fx]$ are more fundamental than true sentences like Fa (or facts like $[Fa]$), not vice versa.

I won’t go into the details of Sider’s argument here. But the general strategy is this: Sider is trying to describe reality in the most “joint-carving” terms possible. And he argues that the existential quantifier is indispensable to doing so. I’ll just mention two ingredients in this argument. First, given that Sider sees the metaphysical project as importantly related to and informed by the projects of physicists and mathematicians, one part of this argument is to point out that “no serious work on the foundations of physics and mathematics has been done in a quantifier-free setting” (2011: 184). (For further discussion of the relationship between “joint-carving” terms and fundamentality/grounding, see Tahko’s entry in this handbook [Chapter 27].)

Second, Sider is suspicious of intuitions about “distinctively metaphysical production or causation” except when they can be rephrased as being about *explanation*. He thinks that the Tractarian intuition just *is* the intuition that “the general fact that $\exists xFx$ holds *because* of the individualistic fact that Fa ” (2011: 204–205). And he argues that if we attempt to re-phrase this claim as being about explanation, it does not hold up—a theory of the world that involves individual names and no quantifiers is not more explanatory than a theory of the world that involves quantifiers. We can now see how this relates to the first point: if no scientific theories fail to use quantifiers, and none of them are Tractarian in nature, then this might in part be because a scientific theory that failed to use quantifiers would be less explanatory than one that used them.

I should note here that Fine (2010) provides a weakened version of EG that perhaps seems to avoid Sider’s concern here, if the concern is genuinely one about whether individual names appear in our fundamental theory. This weakened principle says that $\exists xAx$ implies that $\exists y (Ay \text{ helps ground } \exists xAx)$ (108). (Roughly, the difference between this weakened principle and EG is that the weakened principle just says that any existential truth is grounded by *an* instance, whereas EG says that any true instance of an existential helps ground the existential truth.) I think that there is no way, even if he wanted to do things in terms of ground, that Sider could adopt such a principle given his framework, but it is worth thinking through what work Fine’s weakened principle could do generally here.

Sider, I suspect, would reject that there is grounding in either direction between existential facts and their instances. So he would be suspicious of EG (and weakened EG) for independent reasons. However, it is easy to see how his arguments could supply a grounding theorist with the tools to resist EG, even if they wanted to maintain that there was grounding between an existential fact and its instances. My sense is that we already have reason to worry about the status of EG.

The second argument I want to examine is built out of ingredients from Dasgupta (2009). Dasgupta’s thesis, ‘generalism’, is that what is fundamental is a single, holistic, perfectly general fact. There are no fundamental individuals. We needn’t explore Dasgupta’s argument for this thesis here. What is important for our purposes is how we should understand what grounding claims will hold if this thesis is true. Dasgupta suggests that the right way to understand his view is to adopt a “predicate functorese” language, G , that involves very little overlap with standard first order logic. However, before he does this, he explores the idea that we might instead express/understand the view by adopting a first order logic with identity but with no individual constants. Call this, following Dasgupta, *PL*.

One reason that Dasgupta rejects the idea that *PL* should be used to express his view is that “we have been brought up to understand that quantifiers range over a domain of individuals. So our natural understanding of the facts listed above is that they hold in virtue of facts about individuals” (2009: 50). So facts like $[(\exists x)Fx]$, $[(\exists x)(\exists y)Rxy]$, and so on are understood to be grounded in facts like $[Fa]$ and $[Rab]$.

Note that it is still open to the generalist to use (at least the syntax of) *PL* to express her view. She simply would need to deny that facts like $[(\exists x)Fx]$, $[(\exists x)(\exists y)Rxy]$, and so on are grounded in facts like $[Fa]$ and $[Rab]$. She can claim that our ordinary understanding of how the existential quantifier works is in tension with this, but that our ordinary understanding of how the existential quantifier works is wrong.

When are we licensed to argue that we are wrong about our ordinary understanding of either the meaning of a word or a concept? One answer is: whenever that concept is supposed to correspond to something out there in the mind-and-language-independent world, or whenever that concept is supposed to *carve nature at its joints* (that is, while not corresponding to a worldly entity, still being required to explain the objective structure of reality). (There are other answers: for example, we might follow philosophers like Haslanger (2019) in thinking that conceptual choice and negotiation matters, even when [perhaps even more than when] none of the candidate concepts is taken to carve nature at its joints—for example, when those concepts are politically and socially important, e.g., *woman*.) For discussion of related issues, see Passinsky’s entry in this handbook [Chapter 37].)

So if the generalist thinks that logical concepts are like this—that it is not just ordinary use that determines how we should use these concepts, and which concepts we should adopt, but that instead, these concepts reflect something important about the mind-and-language-independent world—then she might adopt *PL* and insist on a revisionary understanding of the existential quantifier. After all, she is trying to express what are supposed to be mind-and-language-independent, objective facts about fundamental reality, and perhaps about how non-fundamental reality depends upon fundamental reality. Her logical terms should be bound by those facts and not by our practices.

Now return to what Rosen claims: that it is essential to the existential quantifier that EG holds. Suppose this is true. Then the generalist could not redefine the quantifier. Instead, if she wanted to use *PL*, she would have to replace the quantifier with a new one, call it $\exists^*$. EG * would not be true of this quantifier (indeed, if there are individualistic facts, according to Dasgupta, they are grounded by the general facts—so something like the inverse of EG * might be essential to $\exists^*$).

If this is right, what does it tell us about the status of EG? I think it tells us that EG gives us no insight whatsoever into what *mind-and-language-independent* reality is like. It tells us that in order to state metaphysical views that conflict with EG, we cannot use the existential quantifier. So if EG is essential to the existential quantifier, then EG is uninteresting from a metaphysical perspective: it has to do with how our ordinary concept of existential quantification works but nothing to do with what reality is like. Perhaps EG is a principle that is best understood as solely belonging in a *representational* rather than a *worldly* grounding framework.

What I take this to show is that there are questions about fundamental reality to which the truth of EG is irrelevant: is reality fundamentally individualistic? Is it fundamentally general? Does (some kind of) quantification “carve at (its) joints”?

But there also may well be questions of *ground* that EG (and, I suspect, CG and DG) obscure, even if they are true (but regard conceptual or representational matters). To see this, consider the following: suppose it is a fact that p . It follows that $[\sim p]$ and $[p \vee q]$. Now focus on a *worldly* understanding of these facts: facts are something like obtaining Armstrongian states of affairs.

According to such an understanding of facts, are [snow is white], [$\sim$ snow is white], and [snow is white $\vee$ Shaquille O’Neal is a ballet dancer] all distinct? Perhaps. But perhaps what is contained between each pair of square brackets is simply a *representationally* different way of picking out the same worldly fact. You might think that they are obviously distinct, given that the sentences in the square brackets all have different truth conditions. In a possible world in which Shaq is a ballet dancer and snow is grey, one of these things will obtain and the others won’t. But this is to think about the question in the wrong way. If facts are just actually obtaining states of affairs, then what appears in between the square brackets is simply a name for an actually obtaining state of affairs. Perhaps all of these sentences name a single actually obtaining state of affairs, despite *meaning* different things qua sentences and having different truth conditions qua sentences. This may be a reason to prefer operational accounts of ground; but it may simply be a reason to be careful about what, exactly, we are talking about when we talk about the relata of ground.

If [snow is white], [$\sim$ snow is white], and [either snow is white or Shaq is a ballet dancer] are not identical and they are worldly states of affairs, then there are important grounding questions that we must ask about them. And these grounding questions may well come apart from questions of whether it follows from our concept of (e.g.) disjunction that true disjuncts ground their disjunctions.

The main upshot I take from this section is that EG (and, I think, CG and DG, though I have not discussed them in detail here) should be part of a broader conversation about the status of logical constants, logical entailment, and their relationship to metaphysics. This conversation is alive and well (see e.g., Tahko (2009, 2019), the papers in Rush (2014), McSweeney (2017, 2019)), and would also be well-served by more interaction with the grounding literature.³

Related Topics

Logics [Chapter 14], Puzzles [Chapter 18], Necessity [Chapter 10], Structure [Chapter 27], Laws of Metaphysics [Chapter 30], Varieties [Chapter 13], Realism [Chapter 26], Bolzano [Chapter 5], Skeptical Doubts [Chapter 11].

Notes

- 1 While I do not know of any challenges to DG in the *grounding* literature, Michael J. Raven has pointed out to me that there may be an important connection here to the (an?) Aristotelian view about future contingents that accepts LEM but denies bivalence (‘there will or won’t be a sea battle’ is true because it is such an instance, but neither of its disjuncts is true). If this is right, then this is a case where disjuncts (or the truth of disjuncts) can’t possibly ground a disjunction (or the truth of the disjunct). This kind of view is often attributed to Aristotle (*De Interpretatione* 9), but a related view that perhaps most closely approaches denying DG is also developed in Thomason (1970).

Additionally, while I’ve tried to focus on principles that are restricted to classical logic, an obvious connection here is to discussion of quantum logic, which can be interpreted in such a way as to get a similar result. (See Putnam 1965, 1994, 2005; Hughes 1980.)

- 2 Two related questions, which I won’t discuss here but is important to thinking through logical questions of ground, are whether grounding is *internal* (see e.g., Litland 2015) and whether grounding is *superintendental* (see e.g., Bennett (2011), deRosset (2013), Dasgupta (2014), Clark (2018).
- 3 Thanks to Mark Balaguer, Francesca Poggiolesi, Michael J. Raven, and participants of the 2018 Workshop on Metaphysical Grounding at the University of Hamburg for helpful suggestions.

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33

MIND

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1. Introduction

The concept of grounding is typically introduced in order to formulate and address questions about metaphysical relationships. Sometimes, these are metaphysical relationships between entities: parts and wholes, individuals and sets. In other cases, these are metaphysical relationships between facts: conjunctive and atomic, normative and descriptive, mathematical and physical. Either way, this would seem to make grounding concepts apt for use in the philosophy of mind, where one central question concerns the metaphysical relationship between mental and physical phenomena. Yet the use of grounding concepts has been contentious in the philosophy of mind, where specialists have questioned what the introduction of grounding concepts could add to philosophical discussions of the mind–body problem. This chapter will (i) describe how some metaphysicians have proposed the introduction of grounding concepts in order to formulate and provide answers to the mind–body problem and (ii) survey concerns about the appropriateness, adequacy, and indispensability of grounding concepts for addressing questions about the status of mental phenomena in a physical world. Finally, this chapter will (iii) consider replies to these concerns. A central lesson will be that any adequate assessment of the usefulness of grounding frameworks for formulating issues and positions in the philosophy of mind must be sensitive to distinctions between the different grounding concepts that have been introduced.

2. The Mind–Body Problem as a Question of Grounding

Traditionally, the mind–body problem has been understood as a problem about the relationship between two kinds of substances: mental substances (minds) and material or physical substances (bodies). Are our minds simply our bodies or parts of our bodies? Or are they, as Descartes thought, substances of a different kind, distinct from our bodies? And if the latter, how are changes in minds so well synchronized with changes in bodies so that it may seem that changes in the one may cause changes in the other?

Gilbert Ryle (1949) showed that this construal of the problem may rest on a category mistake. Perhaps minds aren't substances after all, things of which we may sensibly ask whether they

are or are not identical to bodies. Recognizing this possibility, in his classic textbook on the philosophy of mind, Jaegwon Kim characterized the mind–body problem in the following way:

The task here is to clarify and make intelligible the relation between our mentality and the physical nature of our being—or more generally, the relationship between mental and physical properties.

(2011: 3)

Although they reject the existence of distinct mental *substances*, neither Ryle nor Kim rejects the reality of mentality, of mental phenomena. As Ryle says, “doing long division is a mental process and so is making a joke” (1949: 22). And so, one may ask, as Kim does, how it is that things that are physical may also instantiate mental phenomena. What is the relationship between these physical and mental properties? For the purposes of this essay, we shall construe the mind–body problem in a way that is neutral between the Cartesian and Rylean positions, as a question about the relationship between mental and physical *phenomena*, whether these be substances, properties, or entities of another ontological category. In the past several decades, philosophers of mind have debated the correct way to capture such relationships, formulating positions according to which the relationship between mental and physical phenomena is, for example, one of identity, supervenience, realization, or the determinable–determinate relation.

Jonathan Schaffer emphasizes that this “debate . . . over the mind is not a dispute over whether mind or matter exists, but rather whether mind is based in matter” (2009: 363). This is an adequate characterization of one part of the mind–body problem literature. (I will return in following sections to what it leaves out.) Physicalists argue that minds, or more broadly mental phenomena, are based in matter, or physical phenomena. Dualists argue they are not. Schaffer draws from this point the conclusion that to make sense of the mind–body problem, one “must make sense of claims of grounding” (2009: 363). For Schaffer, when one entity is based in or ontologically dependent upon another, grounding is the relation that binds them.

Kit Fine has also suggested that the mind–body problem be construed as a question of grounding:

What is it to explain the appearance of a world with minds in terms of a mindless world . . . My own view is that what is required is that we somehow *ground* all of the facts which appear to presuppose the reality of the mental . . . in terms of facts which do not presuppose their reality. Nothing less and nothing else will do.

(Fine 2012: 6, *italics in original*)

Although Fine shares with Schaffer the view that philosophers of mind should add to their toolbox an additional concept of grounding, one may note in Fine a different framing of the mind–body problem. Grounding is not introduced as a relation, one that may or may not obtain between mental and physical entities. Officially for Fine, grounding is a two-place sentential operator acting on a sequence of sentences $\langle Q, R, \dots \rangle$ (the grounds) and a target sentence P (the grounded) where: $\langle Q, R, \dots \rangle$ *grounds* P . Although Fine does not analyze this notion, offering it rather as an ideological primitive, he clarifies grounding thus:

If the truth that P is grounded in other truths, then they *account* for its truth; P ’s being the case holds *in virtue of* the other truths’ being the case.

(2001: 15)

And so, for Fine, the mind–body problem is a challenge to find physical facts that can account for the truth of the mental facts.

In addition to proposing *formulations* of the mind–body problem in terms of a concept of grounding, metaphysicians have also proposed formulating *solutions* to it using such notions. For example, Gideon Rosen (2009) argues that metaphysical naturalism should be formulated as a position about the grounding of facts, such as mental facts, in terms of the natural facts. Shamik Dasgupta (2014) and Thomas Kroedel and Moritz Schulz (2016) propose interpreting physicalism as a position about grounding, that the physical facts are sufficient to ground all other facts (Dasgupta 2014: 560) or “Necessarily, all instances of mental properties are grounded in instances of physical properties” (Kroedel and Schulz 2016: 1911), where grounding is understood by these authors as a relation between facts that obtains when some facts explain in virtue of what some others obtain. Donnchadh O’Conaill (2017) proposes that grounding physicalists must utilize a conception of grounding for which there exists an essential connection between the grounding (in this case, physical) and grounded (in this case, mental) properties (cf. Zylstra, Chapter 22, this volume).

It should be noted that philosophers of mind have for a long time (if not always) been explicit that they are interested in whether or how mental phenomena may ontologically depend on physical (or “natural”) phenomena, and how the mental facts may be explained in terms of underlying physical facts (cf. Schnieder, Chapter 7, this volume). Here again is Kim:

[T]his dependency thesis, as we might call it, is also important, and we can set it down as follows . . . What mental properties a given thing has depends on, and is determined by, what physical properties it has . . . The dependence thesis is important in that it is an explicit affirmation of the *ontological primacy*, or *priority*, of the physical in relation to the mental, thereby opening the possibility of *explaining* the mental in terms of the physical . . .

(1996: 11, *italics in original*)

Kim uses this dependency thesis to partially characterize a position he refers to as “minimal physicalism” (1996: 12–13). Thus, the appeal to dependence and metaphysical explanation in characterizing the mind–body problem and its solutions did not originate with the proponents of grounding. The main novelty is that Kim and other philosophers of mind thought that logical notions like identity or other defined notions like realization could adequately characterize mind–body dependence. The proponents of grounding concepts are new in advocating the introduction of new *primitive* (at least provisionally primitive) and *hyperintensional* notions of grounding, notions that go beyond those made use of (at least self-consciously) by the majority of philosophers of mind. As Schaffer says, “the notion of metaphysical dependence is needed but unanalyzable, and hence best treated as a primitive” (2016a: 145).

Jessica Wilson (2014) has labeled these new primitives big ‘G’ grounding concepts, marking them as distinct from others commonly used in the philosophy of mind: concepts of identity, reduction, composition, and realization, for example. Wilson labels the latter little ‘g’ grounding concepts. Karen Bennett (2011, 2018) calls them building relations. And so proponents of grounding are recommending not merely an elucidation of how others have understood the mind–body problem but a transformation of the way it ought to be framed.

These proposals that we reinterpret the mind–body problem as a problem about (big-G) grounding of one sort or another have prompted critiques by philosophers of mind working on the mind–body problem. Two sorts of questions have been raised. First, are these concepts of ground needed? Do they actually add anything to the debate as it has been framed by

philosophers of mind, or are they *dispensable*? And second, are grounding concepts *adequate* to characterizing the mind–body problem and its solutions?

3. The Disposability of Grounding Concepts

One of the main arguments that is typically made in defense of introducing a primitive concept of ground is that concepts like supervenience are inadequate to capturing the sense in which mental phenomena may be dependent upon or explained by physical phenomena (e.g., Fine 2001; Schaffer 2009; Kroedel and Schulz 2016).

Depending on their preferred framework for addressing the mind–body problem, philosophers of mind critical of grounding have tended toward various ways of rebutting this argument. Some argue that grounding advocates are wrong to think that anything more than supervenience is needed to capture the link between the mental and the physical. For example, David Mark Kovacs (2018) argues that advocates of grounding are mistaken in their arguments that supervenience principles fail to capture interesting senses in which physical phenomena may explain mental phenomena. This follows on earlier defenses of supervenience as a way of capturing physicalism, for example by Robert Howell (2009).

A more common reply is that it has been known for a long time that supervenience is inadequate for capturing a metaphysical relation of dependence. In 1984, Kim defined the notion of strong supervenience, emphasizing that different interpretations of the modal terms will generate different concepts of supervenience and, hence, physicalism:

A strongly supervenes on B just in case, necessarily, for each x and each property F in A, if x has F, then there is a property G in B such that x has G, and *necessarily* if any y has G, it has F.

(1984: 65)

But even this strong sense of supervenience in the strongest modality does not entail a sense of dependence of F on G. Of this, Kim is explicit: “all that we have is that the base property entails the supervening property. This alone does not warrant us to say that the supervening property is *dependent on*, or *determined by*, the base, or that an object has the supervening property *in virtue of* having the base property” (1984: 67).

This is a point argued later in greater detail by Terry Horgan (1993) and Jessica Wilson (2005). Both note that mental properties could be supervenient on physical properties and yet fundamental, i.e., not determined or explainable by those properties. Wilson argued that one could spell out a notion of overlapping causal powers that would more adequately express the sense in which the physicalist sees the mental as dependent on the physical (1999). In his 2016 critique of grounding formulations of physicalism, Andrew Melnyk makes a similar point, arguing that grounding is not needed because one can instead use a “carefully spelled-out” concept of realization to formulate physicalism and nothing-over-and-aboveness, as he did already in his 2003 book.

Either way, one could question whether grounding proponents’ criticisms of supervenience provide adequate motivation for introducing novel grounding concepts into the philosophy of mind (cf. Koslicki, Chapter 11, this volume and deRosset, Chapter 12, this volume). In defense of grounding, one might try to argue that it is not just supervenience, but any of the “little-g” notions used by philosophers of mind that are inadequate to formulating the mind–body problem and its most promising solutions. But, although it is clear their concern stretches beyond supervenience to these other notions as well, up until now, grounding proponents have not

explicitly articulated what is lacking in these notions philosophers of mind have most frequently used for these purposes, notions such as identity, realization, and the determinable–determinate relation.

Schaffer especially (2009) stresses the hyperintensionality of grounding notions, which would serve to distinguish them from those relations most discussed by philosophers of mind. But he does not state why the mind–body problem or its solutions would need to be formulated in terms of anything finer than intensional notions. He doesn’t provide an explicit critique, for example, of the most ubiquitous notion of realization. There is reason to suspect, given other discussions (in e.g., Schaffer 2010, 2016a) that his concern with using realization alone might be something like the following. When we specify that a given mental event *m* is realized by a physical event *p*, we say that there is a certain causal role whose playing is sufficient for the occurrence of *m*, and that *p* as a matter of fact plays that causal role. But to say that *p* in this sense realizes *m* fails to specify any direction of ontological priority. It could be that *p* is more fundamental than the realized event *m*. But it could also be that *m* is more fundamental than the realizer event *p*. The claim that *p* grounds *m* (or vice versa, whatever the case may be) is needed in order to capture this direction of ontological priority (cf. Raven 2017).

This argument for including an additional, hyperintensional notion of ground raises further questions. Primarily, is it really ever an open question whether the realizer or realized is more fundamental? Some have argued that the relation between realizer and realized is one of identity (e.g., Lewis 1972; Kim 1992). In this case, neither is more fundamental than the other, since relative fundamentality is an irreflexive relation. On the other hand, if one finds reason to consider the realized as numerically distinct from its realizer (as in e.g., Fodor 1974; Shoemaker 2007), it is still not clear that there is an open question whether it is the realizer or the realized that is more fundamental. Schaffer once noted (2010) that when some entities compose another entity, there is always an open question about which is more fundamental, the composite whole or the entities out of which it is composed. He adduces reasons why one might take the stance that in such cases, it is the composite that is more fundamental. However, is there such an open question in the case of realization? On standard accounts of realization, the reason *p*’s playing a certain causal role *R* is sufficient for the occurrence of *m* is that *R* analytically characterizes what it is for *m* to occur (Kim 1998: 98–9). This is because it is a matter of definition that if *R* is instantiated, then *m* exists. In such a case then, the occurrence of *m* is analytically nothing over and above this behavior of *p*. If *m* is nothing over and above *p* (or *p* behaving in this way in these circumstances), then by the meaning of ‘nothing over and above’, it is *p*, not *m*, that is more fundamental. This is reason to think that facts about realization settle questions of ontological priority, thus precluding the need for a primitive grounding relation.¹

4. The Inadequacy of Grounding Frameworks

A second concern raised against grounding frameworks concerns whether they are in fact adequate to formulate the mind–body problem and its solutions. Recall that, according to Schaffer, the mind–body problem concerns whether the mental is based in or dependent upon the physical. Physicalists say yes; dualists say no. The claim that the mental is grounded in the physical is supposed to capture the physicalist position.

But Melnyk (2016) argues that the claim that the mental is grounded in the physical fails to capture physicalism. For physicalists are supposed to believe that the mental is “nothing over and above” the physical. Yet since grounding is a primitive relation, there can be no positive justification for thinking that grounding provides a connection between the mental and physical adequate to capturing this over-and-aboveness. That would require there being some conceptual

truth relating grounding to over-and-aboveness. And, Melnyk argues, there is positive reason to think there is no such conceptual truth:

[I]t's untrue (not just unwarranted) that anything that is grounded in a narrowly physical entity would be bound, metaphysically, to be nothing over and above that entity in the sense required for physicalism. For, precisely because the relation of grounding is primitive, and therefore not constituted even in part by facts about its relata, nothing rules out the metaphysical possibility of its holding between a narrowly physical entity, on the one hand, and, on the other, something that *isn't* nothing over and above it in the sense required for physicalism.

(2016: 256)

Moreover, Melnyk argues that there is no good reason to think the instantiation of grounding relations are even compatible with physicalism, since grounding isn't a relation that is postulated by physics (so physical in his 'narrow' sense), nor is it a relation explainable in terms of the instantiation of narrowly physical entities, since it is postulated to be a primitive (2016: 260–8).

To see a second kind of inadequacy concern, recall a remark I made in Section 2. There I noted Schaffer's claim that the mind–body problem concerns not whether minds and bodies exist but whether minds depend on bodies. As I noted there, this captures some of the mind–body debate, but not all of it. For much of the debate concerns not *whether* minds depend on bodies—most philosophers of mind are physicalists—but *how* they do. And it is not initially clear how an appeal to grounding can help frame answers to this latter question.

This is what Wilson (2014: 244–8) calls the metaphysical underdetermination problem for theories of grounding. This is that knowing that the physical grounds the mental leaves very much open the nature of the relation between mental and physical entities. For example, Wilson argues, the claim that a physical entity *p* grounds a mental entity *m* leaves open the question of whether (i) *m* is real, (ii) *m* reduces to *p*, (iii) *m* emerges from *p*, (iv) *m* has causal powers in addition to *p*, and so on. And so if the argument for introducing a new grounding primitive was that the relations philosophers of mind were previously using for capturing the metaphysical relations between (say) mental phenomena and physical phenomena were inadequate for their purposes, then an appeal to grounding doesn't offer an improvement in this respect and so is unmotivated. Wilson notes:

[It] is not just that Grounding (failure of Grounding) claims leave some interesting questions open; rather, it is that such claims leave open questions that *must be answered* to gain even basic illumination about or allow even basic assessment of claims of metaphysical dependence, or associated theses such as naturalism.

(2014: 544–5)

Schaffer (2016a) responds to this critique of Wilson's. He notes that saying that one entity grounds another does not answer all questions one may have about the relation between the grounded and the ground. It articulates the fact that there exists an ontological dependence relation between the two that backs metaphysical explanation. But one might want to know more about this linkage. One way of saying more would be to spell out the grounding relationships that obtain in finer detail. Schaffer describes how this might go for a case in which one would claim a grounding relation obtains between the natural and normative:

One should . . . embed [the] grounding claim in a deeper framework that posits not just an on–off connection (ground or no ground?) but a function relating a range of values

for the more basic option . . . to a range of values for the derivative option . . . The grounding framework would then include information about ‘how exactly normative goings-on are related to naturalistic goings-on’.

(Schaffer 2016a: 150)

Schaffer fills in how these grounding networks might be articulated, drawing on an analogy with the directed acyclic graphs appealed to in the causal modeling literature (see also Schaffer 2016b).

But I think there is more to the underdetermination concern that needs to be addressed. For, as Wilson notes, grounding proponents themselves have been divided on many of the questions that have vexed philosophers of mind. In particular, grounding proponents give different answers to the question of whether if *p* grounds *m*, whether *m* must exist, and to the question of whether if *p* grounds *m*, then *m* is thereby reducible to *p*. And so, it can seem, a claim about grounding does not tell us everything the philosopher of mind wants to know, even if we settle on answers to the more fine-grained grounding questions Schaffer points to.

What needs to be emphasized here, to adequately respond to Wilson’s concern, is that different grounding proposals have different implications for questions about the existence or reality of the grounded and for questions about reduction. Schaffer’s grounding framework, as noted, takes ground to be a relation between entities (of any ontological category), one that may back metaphysical explanations. Fine’s framework, on the other hand, does not posit a fundamental grounding relation. Instead it introduces a concept of ground that may be used to formulate metaphysical explanations of one fact in terms of another. In the next section, we will see how these differences manifest themselves in different proposals concerning the mind–body debate. But it is worth pointing out that it is not necessary for a positive claim of grounding to answer all of the questions the philosopher of mind may have about the relation between mental and physical phenomena, even if it may answer some of them. It is no flaw of the grounding framework that it leaves some open questions for the metaphysician of mind.

6. Grounding and Reduction

One of the main topics of concern since about the 1970s in the philosophy of mind has been whether the relationship between the mental and the physical should be viewed as one of reduction. Although as physicalism became more popular among philosophers, it was initially construed in purely a reductive version, in the 1960s and 1970s, Hilary Putnam (1967), Jerry Fodor (1974), and others proposed nonreductive versions of physicalism. This quickly became more or less the received view in the philosophy of mind. One might then ask whether a grounding physicalism is a reductive or nonreductive version of physicalism.

As indicated, the answer to this question depends on which grounding framework one considers, but it also depends on how one understands reduction. Let’s turn to the latter issue first. In much of the current philosophy-of-mind literature, reduction is construed ontologically. One may say that one (putative) entity *m* reduces to another entity *p* if one of two situations obtains:

Eliminative reduction: *m* is shown not to exist; only *p* exists.

Retentive reduction: *m* is shown to be numerically identical to *p*.

This construal of reduction as an ontological matter is revealed in many discussions of nonreductive physicalism, where anti-reductionism is construed simply as a failure of identity between mental and physical phenomena (e.g., in Bennett 2003; Kim 2005).

Understanding reduction in this way, it is straightforward that a grounding physicalism interpreted as the claim that all entities (including mental entities) are grounded in physical entities, where ground is understood in the way Schaffer proposes, is a nonreductive version of physicalism. Since Schaffer's grounding relation is irreflexive, it will never be the case mental entities are grounded in themselves. The grounded and the ground must be numerically distinct. In addition, Schaffer's framework will require that the grounded mental properties exist and are as real as their grounds, for a relation cannot link what does not exist.

On the other hand, a grounding physicalism framed in terms of some other grounding proposal will not necessarily be nonreductive.² For example, suppose one uses Fine's framework to formulate a version of grounding physicalism. This will most naturally be formulated as the view that all truths (including mental or psychological truths) have grounding explanations in physical truths.³ Even if Fine adopted an analogous irreflexivity of grounding principle to Schaffer's, that no truth may ground itself, this would entail nothing about the numerical identity (or distinctness) of mental and physical phenomena. All mental entities may be identical to physical entities, and yet the mental truths may be distinct from the physical truths, so that a mental truth may be grounded in a physical truth. (Just as water may be identical to H_2O and yet the water truths may fail to be identical to the H_2O truths.) Grounding physicalism would then seem compatible with reductive as well as nonreductive physicalism.

Both Fine and Rosen are explicit that in their frameworks, grounding does not entail reduction but is consistent with it (Fine 2001: 25; Rosen 2010: 130–3).⁴ In 2001, Fine defined a notion of reduction using his notion of ground and an additional primitive notion of *real* (cf. Solomyak this volume [Chapter 26]). Fine gave a positive characterization of this notion of real as follows:

One might think of the world and of the propositions by which the world is described as each having its own intrinsic structure; and a proposition will then describe how things are in themselves when its structure corresponds to the structure of the world. Thus it is this positive idea of the intrinsic structure of reality . . . that should be taken to inform the relevant conception of what is fundamental or real.

(2001: 25)

The real propositions are those that describe the intrinsic structure of reality. This isn't to say that propositions that do not describe the intrinsic structure of reality may not be true. They may be. But it is the goal of the metaphysician (at least some of the time) to make claims that are not just true but also real that do describe the intrinsic structure of reality. With the notions of grounding and real at hand, we may see Fine's account of reduction:

The true proposition *P* *reduces* to the propositions *Q*, *R*, . . . iff (i) *P* is not real; (ii) *P* is grounded in *Q*, *R*, . . .; and (iii) each of *Q*, *R*, . . . is either real or grounded in what is real.

(2001: 26)

It is Fine's view that when a proposition is grounded in some others, there is a "general presumption" in favor of the former's being not real (2001: 27). However, it is not ruled out that the grounded may be real just as its ground is. This is useful. For it gives one the ability to use Fine's grounding framework to formulate new positions in the philosophy of mind that can resolve longstanding debates between reductionists and nonreductive physicalists.

To see this, consider the main arguments that have been used on one or the other side of the reductionist/nonreductive physicalist debate. Nonreductive physicalists, such as Fodor (1974) and Putnam (1975), emphasized that the special sciences make use of specific terms in their explanations that fail to denote univocal physical kinds. On the other hand, reductionists, such as Kim (1998), note that given physicalism, to be physically efficacious, any kinds must be identical to physical kinds. If one wants to accept both points, one would seem led either to (a) the consequence that the psychological kinds referred to in the special sciences are causally ineffacious, epiphenomena, or, if one wanted to avoid epiphenomenalism, to (b) the consequence that special science explanations attempting to refer to kinds that are not identical to physical kinds are universally false.⁵ Fine's grounding framework, however, provides an additional option. One can say that the explanations given in the special sciences are true, but they are not real; i.e., they do not track the intrinsic structure of reality, picking out distinctive special science kinds. Instead these special science claims may be true in virtue of being grounded in some other truths that are real, truths that only make reference to physical kinds. What the grounding framework permits then is a way of embracing both the metaphysical insights of Kim and other reductionists and the metascientific insights of Fodor and Putnam.⁶

So far in this section, we have discussed two conceptions of reduction: the simple ontological one favored by many philosophers of mind (where reduction is a matter of identity or elimination) and Fine's 2001 conception. However, there is one more conception of reduction worth mentioning. In this sense, grounding physicalism may be construed more generally as a species of reductive physicalism. This is the sense of reduction as a kind of explanation, one that leaves no gaps in understanding between the reduced and its explanatory base, outlined, for example, in Chalmers and Jackson's 2001 account of reductive explanation:

[I]n showing how any instance of the phenomenon is itself implied by microphysical phenomena, we show that there is a sort of transparent epistemic connection between the microphysical and macroscopic phenomena.

(2001: 32)

A salient difference is that Chalmers and Jackson want to understand reductive explanation in terms of a priori entailment, and Fine and others are explicit that in the case of grounding explanations, there need be no a priori connection between the grounded and its ground (Fine 2001: 25; Raven 2012: 699). However, one might consider that (as noted), for Fine, where there is grounding, there are no more open questions as to how or why the grounded obtains (cf. Block 2015). This, then, would seem to close off the existence of explanatory gaps.⁷ This is an explicit difference between Fine's way of conceiving ground and Schaffer's. Schaffer is explicit (2016b) that grounding does not close explanatory gaps:

I claim that, in all concrete transitions from more to less fundamental, the dependence functions involved provide substantive information. An ideal mind, informed only of the more fundamental aspects of concrete reality, could not yet infer anything at all about the less fundamental aspects of concrete reality, without the added information as to *how the less fundamental is determined by the more fundamental*.

(2016b: 10)

Thus, for Schaffer, even in the presence of a grounding relation, one cannot know from the facts about the grounds why or even whether the grounded obtains.

Contra Wilson, I don't think the fact that there is no straightforward answer to the question of whether grounding entails reduction is a liability for the grounding proponent. There are many different conceptions of ground, as there are conceptions of reduction. Different such conceptions will entail different answers as to whether grounding entails reduction and so whether grounding physicalism is a kind of reductive physicalism. One should adopt the notions that play a useful role in one's philosophical projects. However, as Kevin Morris (forthcoming) emphasizes, to the extent that one fills in one's notion of ground in more detail, so that it has concrete consequences for questions about grounding and the other sorts of metaphysical issues raised e.g., by Wilson (2014) in her underdetermination critique, the more controversial it becomes that mental phenomena are grounded in physical phenomena. As he puts it, "while increasingly restricting the application of the Grounding idiom makes Grounding increasingly substantive, it makes establishing such theses correspondingly difficult."

7. Conclusion

As we have seen, it is still contested whether and how grounding may be of use in stating and resolving the mind-body problem. When evaluating these issues, however, one should remain sensitive to distinctions in the various grounding frameworks that have been proposed.

Related Topics

Essence [Chapter 22], Dependence [Chapter 7], Skeptical Doubts [Chapter 11], Anti-Skeptical Rejoinders [Chapter 12], Realism [Chapter 26]

Notes

- 1 It is possible that one may try to offer a similar argument to rebut those concerning composition in Schaffer (2010), but I am skeptical and so will not try to do so here.
- 2 See Morris (forthcoming: Chapter 5) for a more thorough discussion of how different grounding proposals entail different reductive consequences.
- 3 One may dispute whether physicalism must be interpreted as a claim about the grounding of 'everything' or 'all truths', or some more limited class, e.g., the contingent truths (Poland 1994), the truths apt to be grounded (Dasgupta 2014), etc. For brevity, I will bypass these issues.
- 4 Rosen argues that the reverse entailment does obtain. Reduction entails grounding. This he calls the "grounding-reduction" link (2010: 122). For further discussion of this principle, see Dorsey (2016).
- 5 There is also the possibility of rejecting one or other of the mentioned points. Nonreductive physicalists may reject the pull of Kim's 1998 exclusion problem and say that the distinctive kinds of the special sciences may have causal powers over and above those of physical kinds. The challenge then is to say how this may be compatible with physicalism. Reductionists may insist, as Kim (1992) does, that special science explanations never make reference to distinctive kinds but only to univocal physical kinds.
- 6 This proposal is developed in greater detail in Ney (2016). A similar reconciliation has been proposed by Heil (2012) and Morris (forthcoming), although using the concept of truthmaking instead of grounding. See Wilson (2016) for a critique of the proposal made in Ney (2016) and Morris (forthcoming) for a rebuttal.
- 7 This sense of explanatory gap is inherited from Levine (1983).

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34

NORMATIVITY

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1. Introduction

One of the ways in which grounding earns its keep as a respectable bit of ideology is by being useful for understanding long-standing philosophical debates, including debates in normative ethics and metaethics. For example, many grounding enthusiasts claim that grounding is indispensable to understanding first-order normative theories like utilitarianism: the utilitarian's view is that an action is right if and only if, and *because*, it maximizes happiness, where this because-claim is a claim about grounding. And many also claim that the naturalism versus non-naturalism debate in metaethics is best characterized as a debate about whether all normative facts are fully grounded in natural, non-normative facts.

But this common story about grounding's applications in normative inquiry is rife with controversy. First, even if normative ethicists are in the business of making because-claims, it's controversial whether these because-claims are about *metaphysical grounding* or a related but distinctly normative relation—*normative grounding*.

Second, this grounding characterization of the naturalism versus non-naturalism debate begins to look problematic once we ask whether on this characterization, for example, the fact that an act's maximizing happiness fully grounds that it's right is itself a normative fact that must be fully grounded in natural, non-normative facts in order for naturalism to be true. Very general metaphysical considerations suggest that such grounding-facts cannot be so grounded. So if this is what naturalism requires on the grounding characterization of the debate, this way of understanding the debate fails to capture that this is a substantive, local debate about normativity. But if the only normative facts that we take to be relevant to the debate are facts like the fact that a particular thing *x* is good or that a particular action *A* is right, then the grounding characterization fails to squarely place G.E. Moore on the non-naturalist side of this debate. This is because Moore (1942) seems to clarify that while he takes *goodness* to be an indefinable normative property, he nonetheless takes facts about what is good to be fully grounded in natural, non-normative facts.

This chapter surveys these two main issues in §2 and §3, respectively, while highlighting what's at stake in these disputes for both normative and metaphysical theorizing.

2. Grounding in Normative Ethics: Normative or Metaphysical?

Most proponents of grounding insist that grounding is simply a new label for an old, familiar idea in first-order metaphysics: that certain facts obtain *because*, or *in virtue of*, other facts, where this “because” and “in virtue of” talk is not about anything causal, semantic, or conceptual. For example, when we claim that Michelangelo’s David has an innie belly button because of, or in virtue of, the fact that the hunk of marble that constitutes David has an S-shaped dent in location L, presumably what we mean is not that the marble’s dent caused David to have an innie or that what “David has an innie” means is that the marble has such a dent or anything about the concept of David’s innie. Instead, we mean that the dent *makes* or *determines* that David has an innie in a non-causal, metaphysical sense. Grounding is meant to capture this non-causal kind of metaphysical determination.

So, one way of showing that grounding has applications in a certain area of philosophical inquiry is to show that this area of inquiry is rife with “because,” “making,” or “in virtue of” talk, and that such talk is not plausibly understood as being about anything causal, semantic, or conceptual. This is precisely what Berker (2018) does in order to argue that grounding has been pervasive throughout the history of normative ethics. Indeed, many paradigm normative theories aim to specify what *makes* acts right, or what it is *in virtue of which* right acts are right. For example, utilitarians claim not just that, necessarily, an act A is right if and only if it maximizes happiness but that, necessarily, an act A is right if and only if, and *because*, A maximizes happiness. Whereas W.D. Ross (1930) claims that there are a few different features that an act may have (e.g., breaking a promise, promoting self-improvement, or making reparations) *in virtue of which* it is prima facie right or prima facie wrong, and that what *makes* acts all-things-considered right, is the overall balance of prima facie rightness and wrongness of the act. In fact, the very debate between consequentialists and deontologists is often understood as being about whether, necessarily, an action A is right *in virtue of* the fact that A promotes the good. All this “making,” “because,” and “in virtue of” talk is not plausibly understood as being about anything causal, semantic, or conceptual but is instead best understood as talk of grounding (Berker 2018).

But is the relevant grounding relation here the same grounding relation that pervades first-order metaphysics? Fine (2012) says “no”: he claims that the grounding relation that the ethicist’s “making,” “because”, and “in virtue of” talk concerns is *normative grounding*, which is entirely distinct from *metaphysical grounding* in the sense that neither relation can be defined in terms of the other, nor can they be defined in terms of some more generic grounding relation.¹ According to Fine, normative grounding is a primitive relation, and while it shares many of the same formal features as metaphysical grounding, it concerns a distinct and weaker modality: whereas the metaphysical grounds of p make p metaphysically necessary, the normative grounds of p only make p *normatively necessary*, but not metaphysically necessary.

Fine thus takes all necessity claims and in-virtue-of claims made in normative ethics that involve both normative and non-normative properties to only state normative necessities (and not metaphysical necessities). This is because Fine (2002, 2012) takes metaphysically necessary connections, including metaphysical grounding connections, to derive from the *essences* of things, and he claims that the essences of normative and non-normative properties do not involve one another. For example, according to Fine, an act’s maximizing happiness may normatively ground that it’s right and thus make it normatively necessary that acts that maximize happiness are right. But Fine would insist that it’s not metaphysically necessary that acts that maximize happiness are right, since the essence of maximizing happiness doesn’t involve rightness, and the essence of rightness does not involve maximizing happiness.

So, as Rosen (forthcoming) points out, Fine's essentialist view of metaphysical modality together with his claim that the essences of normative and non-normative properties do not involve one another commits him to denying what Rosen calls "the least controversial claim in metaethics":

SUPERVENIENCE For any two entities x_1 and x_2 (e.g., actions, states of affairs, people, etc.) in any metaphysically possible worlds w_1 and w_2 , if x_1 and x_2 are non-normatively identical, x_1 and x_2 are normatively identical.

For example, SUPERVENIENCE implies that, necessarily, if two metaphysically possible killings K_1 and K_2 have exactly the same non-normative features and K_1 is wrong, K_2 must be wrong too. SUPERVENIENCE thus implies that there are some non-normative properties that metaphysically necessitate normative properties. Since Fine denies that there are any metaphysically necessary connections between normative and non-normative properties, he's committed to denying SUPERVENIENCE.

Of course, Fine may endorse a weaker supervenience claim: that, for any two entities x_1 and x_2 in any *normatively possible* worlds (i.e., worlds in which the normative laws of our world hold), if x_1 and x_2 are non-normatively identical, x_1 and x_2 are normatively identical. But the reason SUPERVENIENCE is so widely accepted is that it seems like we can't even conceive of two entities that are non-normatively identical but normatively distinct. Endorsing only the weaker supervenience claim and denying SUPERVENIENCE runs afoul of this intuition. This is what makes denying SUPERVENIENCE seem like such a big cost of Fine's view. (Although Hat-tiangadi (2018), Roberts (2017), and Rosen (forthcoming) argue that denying this intuition is not as radical as one might think.)

Like Fine, Bader (2017) takes there to be a normative grounding relation that is distinct from metaphysical grounding, but he provides an alternative framework for thinking about these grounding relations and their corresponding modalities that allows SUPERVENIENCE to be true. On Bader's view, what makes normative grounding distinct from metaphysical grounding is that they are governed by different laws: just like the laws of nature govern causal relations between causes and effects, *metaphysical laws* govern metaphysical grounding relations between facts, and *normative laws* govern normative grounding relations between non-normative and normative facts (Bader 2017: 117–18). Normative and metaphysical necessities, then, are simply whatever facts obtain in every world in which the corresponding laws hold, just like the nomological necessities are those facts that obtain in all the worlds in which the laws of nature hold. And unlike Fine, Bader takes the normative laws to be metaphysically necessary, because otherwise serious epistemological problems would arise: if the normative laws are contingent and brute, then we can't know them a priori, but we also can't know them a posteriori on the basis of non-normative facts, so we would need to have a mysterious "quasi-perceptual faculty" that gives us direct access to normative properties (Bader 2017: 110–11).

Since Bader takes the normative laws to be metaphysically necessary, his view of normative grounding is compatible with there being metaphysically necessary connections between normative and non-normative properties and is thereby compatible with SUPERVENIENCE. However, since Bader only provides an epistemic argument for why we should believe that the normative laws are metaphysically necessary rather than a metaphysical explanation for why this is so, his view does not provide a metaphysical explanation for SUPERVENIENCE.

Regardless of their differences, though, both Bader and Fine take particular normative facts (like the fact that action A is right) to be only normatively grounded, and not metaphysically grounded. Rosen (2017a) takes this to be unpalatable because it implies that the universe

fundamentally consists of, “along with (say) the quantum state of the universe and the basic principles of mathematics, facts about the normative features of particular human actions” (Rosen 2017a: 138). So, Rosen proposes an alternative view of normative grounding on which normative grounding is distinct from but definable in terms of metaphysical grounding. Specifically, Rosen proposes that to say that *Fa normatively grounds Ga* is to say that *Fa metaphysically grounds Ga* in conjunction with a moral law connecting *G* and *F* (Rosen 2017a: 139). For example, on Rosen’s account, to say that *A’s maximizing happiness normatively grounds that A is right* is to say that the fact that *A maximizes happiness together with the moral law that maximizing happiness is right metaphysically grounds that A is right*. Unlike Fine and Bader, then, Rosen takes all particular normative facts to be metaphysically grounded. Rosen’s view is also compatible with SUPERVENIENCE, so long as the moral laws are metaphysically necessary and not just normatively necessary.

Bader and Fine are thus *extremists* about normative grounding: they hold that normative grounding is entirely distinct from metaphysical grounding in the sense that these two grounding relations cannot be defined in terms of one another or in terms of a more generic grounding relation. Whereas Rosen is a *moderate*: he holds that there is a normative grounding relation, but that it’s defined in terms of, and thus not entirely distinct from, metaphysical grounding.

One might instead be a *nihilist* about normative grounding: someone who holds that there is no distinctly normative grounding relation and that the in-virtue-of claims made by normative ethicists are just about metaphysical grounding. One reason one might be drawn toward nihilism is simply that there are no intuitive examples that show an obvious need for positing a normative grounding relation like there are in the case of metaphysical grounding. That is, one of the main motivations for adopting metaphysical grounding into our ideology is that certain examples show that we need to posit a non-causal metaphysical determination relation in order to make sense of certain in-virtue-of claims made in first-order metaphysics. For example, we need to posit such a relation to make sense of the claim that Michelangelo’s David has an innie in virtue of the marble having an S-shaped dent in location *L*, since it’s obvious that this can’t be about causation, semantics, or conceptual analysis. But the same thing cannot be said of examples from first-order normative ethics: when the utilitarian says that an act *A* is right in virtue of the fact that *A* maximizes happiness, it’s not obvious that this *can’t* be about metaphysical grounding.

This debate about normative grounding is significant because it has implications regarding the relation (or divide) between normative ethics and metaethics. The important dividing line here is between the extremists on one side and the moderates and nihilists on the other. According to the extremists, the in-virtue-of claims made in first-order normative inquiry are only about normative grounding, which is entirely distinct from metaphysical grounding and thus do not involve any metaphysical grounding claims. But for moderates and nihilists, the in-virtue-of claims made in first-order normative ethics are either directly (according to nihilists) or indirectly (according to moderates) about metaphysical grounding. Since metaethics concerns, in part, the metaphysics of moral matters, the moderate and nihilist’s views thus suggest that many of the paradigm views and debates in normative ethics actually fall within the purview of metaethics (Berker 2018). The extremist’s view, on the other hand, allows for a more traditional view according to which disputes in first-order normative ethics are orthogonal to metaethical inquiry.

Berker (2018) argues that the fact that normative grounding and metaphysical grounding are intimately linked in certain ways tells against the extremist’s view and in favor of a moderate or nihilist one. First, he points out that in most cases, both metaphysical grounding and normative grounding are transitive and asymmetric: if *p* grounds *q* and *q* grounds *r*, *p* also grounds *r* (transitivity), and if *p* grounds *q*, then *q* does not ground *p* (asymmetry). Then Berker notes that

“mixed” transitivity and asymmetry principles also usually hold: if *p* metaphysically (or normatively) grounds *q*, and *q* normatively (or metaphysically) grounds *r*, then *p* grounds “in some non-rigged-up sense” *r*; and if *p* metaphysically (or normatively) grounds *q*, then *q* does not normatively (or metaphysically) ground *p* (Berker 2018: 760).

If normative grounding and metaphysical grounding are entirely distinct relations, as the extremist claims, it would be mysterious why these two grounding relations are typically linked up in these ways (Berker 2018). The moderate and nihilist views about normative grounding, on the other hand, explain why these mixed principles hold (Berker 2018). First, consider Rosen’s moderate view: that what it is for *p* to normatively ground *q* is for *p*, together with some moral law, to metaphysically ground *q*. On this view, if *p* normatively grounds *q*, *p* also partially metaphysically grounds *q*. So *q* cannot metaphysically ground *p*, given the asymmetry of metaphysical grounding; and if *q* metaphysically grounds *r*, then *p* metaphysically grounds *r*, given the transitivity of metaphysical grounding. Rosen’s moderate view thus explains why mixed transitivity and asymmetry principles hold. And the nihilist has an even simpler explanation for why these principles hold: it’s because normative grounding *just is* metaphysical grounding.²

Regardless of this debate, though, some grounding relation seems indispensable to understanding first-order normative theories and debates. This is significant because it provides some justification for accepting grounding into our ideology in the first place. The following section, however, suggests that it’s much more contentious whether grounding also earns its keep by being useful for understanding the naturalism versus non-naturalism debate in metaethics.

3. Grounding in Metaethics: the Naturalism Versus Non-Naturalism Debate

Moore (1903) began one of the most long-standing debates in metaethics: whether normative properties are so-called “non-natural” properties. But specifying what this debate amounts to is a notoriously tricky task. The core pre-theoretical claims of Moorean non-naturalism are (i) that normative properties are entirely of their own kind (*sui generis*), and consequently, (ii) that countenancing normative properties is thereby incompatible with a purely scientific worldview (Leary 2017). The Moorean non-naturalist’s opponents are thus not only *naturalists*, who take normative properties to be of the same kind as paradigmatic scientific properties (i.e., *natural* properties), but also *supernaturalist non-naturalists* like Divine Command Theorists, who take normative properties to be of the same kind as paradigmatic *supernatural* properties like *being commanded by God* (McPherson 2013). But how do we specify what this debate amounts to in more precise metaphysical terms? What exactly does it mean for normative properties to be *entirely of their own kind* or *of the same kind* as some other set of properties?

One might think that whether normative properties are entirely of their own kind just depends on whether they supervene on non-normative properties. After all, the parallel debate in the philosophy of mind was long understood as being about whether the mental supervenes on the physical. But unlike the philosophy-of-mind debate, virtually all parties to the naturalism vs. non-naturalism debate agree that SUPERVENIENCE is true (McPherson 2013; Rosen 2017b). Moreover, many grounding enthusiasts now suggest that physicalism about the mind is best understood not in terms of supervenience but in terms of metaphysical grounding: that the physical ultimately fully grounds the mental (Schaffer 2017).³ (In the rest of this section, any use of “grounding” should be taken to mean metaphysical grounding unless otherwise specified.)

Analogously, then, one might take the central claim at stake in the naturalism versus non-naturalism debate to be as follows:

FULL GROUND All normative facts are either identical to or fully grounded in the non-normative facts.

On this characterization of the debate, naturalists endorse FULL GROUND because they claim that all normative facts are either identical to or ultimately fully grounded in paradigmatic scientific facts. Whereas supernaturalist non-naturalists endorse FULL GROUND because they claim that all normative facts are either identical to or ultimately fully grounded in supernatural facts or that the normative facts are partially grounded in supernatural facts together with scientific facts. The Moorean non-naturalist, on the other hand, denies FULL GROUND. On this taxonomy, then, what it is for normative properties to be entirely of their own kind is for their instantiations to not be fully grounded by instantiations of any other kinds of properties.

This taxonomy renders Moorean non-naturalism compatible with SUPERVENIENCE. For example, some non-naturalists attempt to explain SUPERVENIENCE by claiming that all particular, contingent normative facts are partially grounded in particular, contingent non-normative facts together with fundamental, general normative principles that connect the normative and the non-normative (Enoch 2011; Scanlon 2014). This explanation for SUPERVENIENCE is compatible with denying FULL GROUND. (However, without an explanation for why these normative principles are metaphysically necessary, this view fails to offer a fully satisfying non-naturalist explanation of SUPERVENIENCE (Leary 2017).)

The problem with this taxonomy, though, is that very general metaphysical considerations suggest that FULL GROUND is false, and so, this characterization of the debate fails to capture that this is a substantive, local debate about normativity. To see the problem, suppose one endorses the following claim:

(U) The fact that A maximizes happiness fully grounds that A is right.

Since (U) involves the normative property *rightness*, (U) itself is a normative fact. So in order for FULL GROUND to be true, (U) must be fully grounded in non-normative facts.⁴

But the most plausible views about what grounds grounding-facts like (U) seem to imply that (U) is ultimately fully grounded in some further fact that also involves *rightness*. For example, on Fine's (2012) *essentialist view*, grounding facts are explained by the essences of the grounded properties. So, (U) is fully grounded in:

(E) It's essential of *rightness* that if A maximizes happiness, A is right.

Or on Wilsch's (2015) *nomological account*, grounding-facts are fully grounded in metaphysical laws concerning various *construction relations* such as constitution, realization, set formation, and so on. The nomological account thus suggests that (U) may be grounded in:

(L) It's a metaphysical law that *maximizing happiness* realizes *rightness*, and that if Fa and F realizes G, then Ga.⁵

Since both (E) and (L) involve the property *rightness*, both of these views imply that what grounds (U) is a further normative fact.

There is one view about what grounds grounding-facts, however, that entails that (U) is not grounded in a further normative fact. On Bennett (2011) and deRosset's (2013) "bootstrapping" view, if Fa grounds Ga, the fact that Fa grounds Ga itself is grounded in Fa. This view thus implies that (U) is grounded in:

(BS) A maximizes happiness.

But (BS) seems to offer a worse explanation of (U) than (E) or (L). The fact that A maximizes happiness does not seem to explain why there is a connection between A's maximizing happiness and A's being right, whereas the fact that it's part of the very nature of *rightness* that if A maximizes happiness, A is right, or the fact that it's a metaphysical law that maximizing happiness realizes rightness does explain this connection (Dasgupta 2014). This seems to be a general feature of explanatory relations: the explanation for why there is an explanatory relation between being F and being G must involve both F and G. So, in order to explain why A's maximizing happiness grounds A's being right, we must appeal to some fact like (E) or (L) that involves both *maximizing happiness* and *rightness*. But this implies that whatever grounds (U) must be a further normative fact.

The truth of FULL GROUND thus requires, again, that whatever fact grounds (U) is also fully grounded in non-normative facts. But, again, this seems implausible. For example, consider (E): either (E) must just be a brute fact about the nature of rightness and is thus not grounded at all, or (E) must be grounded in some further normative fact that involves *rightness*. After all, how could any fact that doesn't involve *rightness* explain why *rightness* has the nature that it does? Likewise, if (L) grounds (U), then either (L) must just be a brute law of metaphysics or (L) must be grounded in some further normative fact that involves *rightness*, given the general feature of explanatory relations just mentioned. We can keep repeating this line of reasoning ad infinitum or until we ultimately bottom out in some brute fact that will inevitably still involve *rightness*. Either way, FULL GROUND is false—not all normative facts are fully grounded in non-normative facts. (This argument is offered in Leary (manuscript); it's the normative analogue of an argument against using grounding to characterize the physicalism vs. non-physicalism debate, which Dasgupta (2014) explicates and attributes to Sider (2011).)

This argument relies on very general considerations about metaphysical explanation and has nothing to do with normativity. We can use the very same argument to show that the biological facts are not fully grounded in the micro-physical facts. For any biological property B and micro-physical property P, if Pa grounds Ba, the fact that Pa grounds Ba is itself a biological fact, since it involves the biological property B. And since whatever fact q that grounds the fact that Pa grounds Ba must involve both P and B, q must also be a further biological fact, and so on ad infinitum or until we ultimately bottom out in some brute fact that will inevitably still involve the biological property B. So not all biological facts are fully grounded in the micro-physical facts. What this argument shows, then, is that the naturalism vs. non-naturalism debate cannot be about FULL GROUND, since it is supposed to be a local debate about normativity.

We might rescue this characterization of the debate, though, by restricting the range of normative facts that are relevant to FULL GROUND. Specifically, one might think that the only facts with which parties to the debate are concerned are facts like that a particular state of affairs is good or that a particular action is right or that some consideration is a reason, and not metaphysically "heavy duty" normative facts like facts about what grounds facts about what's right, the essence of rightness, or metaphysical laws involving rightness. So, as long as all the former kinds of normative facts are fully grounded in paradigmatic scientific facts, then even if what explains these grounding connections are some fundamental metaphysical laws that involve

normative properties, this may seem to suffice for naturalism (Schaffer 2017). We may then take the debate to be about the following restricted grounding claim:

FULL GROUND_R For all normative properties F and for all entities a: if Fa, then Fa is identical to or fully grounded in non-normative facts (Rosen 2017b: 157).

FULL GROUND_R offers a similar taxonomy as FULL GROUND, but whether FULL GROUND_R is true cannot be settled by general metaphysical considerations. It must be settled by considering what grounds normative facts in particular.

But this revised grounding characterization of the debate faces a new worry: it doesn't fit very well with how Moore himself thought of the issue at stake. This is because, while Moore (1903) argues that *goodness* is indefinable, Moore (1942) later clarifies that he nonetheless thinks that facts about what is good are fully grounded in natural, non-normative facts:

I should never have thought of suggesting that goodness was “non-natural,” unless I had supposed that it was “derivative” in the sense that, whenever a thing is good (in the sense in question) its goodness (in Mr. Broad's words) “depends on the presence of certain non-ethical characteristics” possessed by the thing in question: I have always supposed that it did so “depend,” in the sense that, if a thing is good (in my sense), then that it is so *follows* from the fact that it possesses certain natural properties, which are such that from the fact that it is good it does *not follow* conversely that it has those properties.

(Moore 1942: 588)

In claiming that it follows from the fact that x has certain non-normative properties that x is good, Moore cannot mean that the fact that x has certain non-normative properties logically or semantically entails that x is good. This is because Moore (1903) claims that it's always an open question whether something is good, given that it has certain non-normative features. Nor does it seem plausible to interpret Moore as claiming that a thing's non-normative properties cause it to be good. Rather, the best interpretation of what Moore is claiming here seems to be that he takes the fact that something has certain non-normative features to ground the fact that it's good. And since Moore doesn't claim that it follows from those non-normative features together with any other facts, Moore's claim seems to be about full grounding.

One might think the claim that all facts about what's good are nonetheless fully grounded in non-normative facts seems plain incompatible with the Moorean's commitment that normative properties are *sui generis* and thus suggest that a more charitable interpretation of Moore is that he is claiming that non-normative facts only partially ground facts about what's good, together with general normative principles (assimilating Moore's view to Enoch's (2011)). Or one might suggest we interpret Moore as claiming, instead, that facts about what is good are fully normatively grounded in non-normative facts (while assuming that normative grounding is distinct from metaphysical grounding). These are both interpretations of Moore's view that deny FULL GROUND_R.

But these interpretations of Moore's (1903) view are a stretch without textual evidence to support them. And there is a way of charitably understanding the view that *goodness* is a *sui generis* normative property, even though all facts about what is good are fully metaphysically grounded in non-normative facts, that more closely resembles Moore (1903). It just requires an alternative way of thinking about what it is for normative properties to be *sui generis* (one that doesn't have to do with grounding). In fact, Moore (1903) does not seem to be at all concerned

with what *makes* something good but is instead concerned with what the property *goodness* is and whether it can be given a definition:

Definitions of the kind that I was asking for, definitions which describe the real nature of the object or notion denoted by a word, and which do not merely tell us what the word is used to mean, are only possible when the object or notion in question is something complex . . . “Good,” then, if we mean by it that quality which we assert to belong to a thing, when we say that the thing is good, is incapable of any definition.

(Moore 1903: 7–9)

The sort of definition that Moore has in mind here seems to be a *real definition*: a real definition of x states the very nature of x in the form, “To be x is to be ϕ ,” where ϕ is a “complex condition not containing $[x]$ ” (Rosen 2017b: 154). So what Moore seems to take the central issue to be is whether, for any normative property F , there is some non-normative condition ϕ that defines F . And Moore claims there is no such condition for *goodness*.

Understanding the naturalism vs. non-naturalism debate in terms of whether all normative properties have a real definition, though, would rule out the potential for a non-reductive naturalist view about the normative, analogous to non-reductive physicalist views in the philosophy of science (Rosen 2017b). One might think that such a view about the normative is at least an intelligible position and that this is sufficient reason to understand the debate as being about FULL GROUND_R rather than real definition (Rosen 2017b).⁶

But we could instead use the notion of *essence* to characterize the debate in a way that is very close to how Moore thought about the issue and makes room for a non-reductive naturalist position (Leary manuscript). According to Fine (1994), the *essence* of x is the set of true propositions that directly state the very nature of x .⁷ Essence is thus broader than real definition because a real definition of x is a true proposition stating the very nature of x with a particular form—“To be x is to be ϕ ”—while the essence of x may contain true propositions that state the very nature of x but not in this form. So, if x has a real definition, its real definition is at least part of x ’s essence, but not all essential truths need be real definitions. And x may have an essence without having a real definition at all. Consider disjunction: we may not be able to define what it is for p or q to obtain in terms not containing “or,” but it may nonetheless lie in the nature of disjunction that disjunctive facts are grounded in their disjuncts (Rosen 2010: 131).

Since essence is broader than real definition, this opens up conceptual space for two different non-reductive views about normative properties. First, suppose that some normative properties cannot be given a real definition in natural terms but that their essences are exhausted by propositions that specify natural sufficient conditions for their instantiation. For example, suppose that *goodness* has no real definition but that it’s nonetheless essential of *goodness* that x ’s being pleasant makes x good and that there’s nothing more to the nature of *goodness* than this. This seems to capture the spirit of non-reductive naturalism (Leary manuscript). Alternatively, suppose that some normative properties cannot be given a real definition in any non-normative terms but also that their essences do not specify non-normative sufficient conditions for their instantiation. This seems to capture the Moorean non-naturalist’s core claims that the normative is *sui generis* and incompatible with a purely scientific worldview. If the essence of *goodness*, for example, does not ultimately involve non-normative properties in this way, then the very nature of *goodness*—*what it is* to be good—is something distinctively normative that is “over and above” non-normative properties; and consequently, there is some feature of reality that cannot be fully described in scientific terms (Leary 2017).

So we may characterize the debate between Moorean non-naturalists and her opponents as concerning the following:

ESSENTIALLY NORMATIVE For some normative property N, the essence of N
(i) cannot be specified entirely in non-normative terms, and (ii) does not specify non-normative sufficient conditions for being N.

(Leary 2017: 96)

On this construal of the debate, the Moorean non-naturalist affirms ESSENTIALLY NORMATIVE, while the naturalist and supernaturalist deny it. And, importantly, ESSENTIALLY NORMATIVE is compatible with FULL GROUND_R (Leary 2017). This taxonomy thus suggests not only that naturalists and non-naturalists need not disagree about grounding but also that Moorean non-naturalists may appeal to FULL GROUND_R to explain SUPERVENIENCE. (Although, in order to give a fully satisfying explanation of SUPERVENIENCE, the non-naturalist must also explain why FULL GROUND_R is true even though the essences of some normative properties do not specify non-normative sufficient conditions (Leary 2017).)

So we have two competing characterizations of the naturalism versus non-naturalism debate: one in terms of grounding and one in terms of essence. The essence characterization of the debate is a closer fit with the way Moore (1903) thought about the issue than the grounding characterization. Of course, some might be warier of the ideology of essence than the ideology of grounding. But if essence is best suited for capturing what's at stake in this debate, that in itself would be reason to adopt it in the first place. In fact, that is precisely what's at stake in this dispute over how to characterize this metaethical debate. It's not just a dispute about how to use the labels "naturalism" or "non-naturalism"; it's a dispute about what metaphysical ideology we should adopt for metaphysical theorizing more generally.

But in order to decide which ideology is best for characterizing the debate, we should also ask which characterization makes the debate substantive and tractable. That is, how would we adjudicate whether ESSENTIALLY NORMATIVE is true or whether FULL GROUND_R is true and what hangs on these two issues? If there's no way to settle a dispute about ESSENTIALLY NORMATIVE, but there is a way to settle the dispute over FULL GROUND_R, then even if FULL GROUND_R has not been the central issue in this metaethical debate (historically speaking), perhaps it should be. Or perhaps whether FULL GROUND_R is true and whether ESSENTIALLY NORMATIVE is true are just two equally substantive, tractable questions. In that case, even if ESSENTIALLY NORMATIVE best captures what the metaethical debate has been about, FULL GROUND_R may nonetheless be a different, equally interesting issue in the neighborhood.⁸ Further work needs to be done here in order to decide which characterization of the debate to adopt.

3. Conclusion

This chapter has explored the extent to which grounding is applicable to both normative ethics and metaethics. What we've seen is that while grounding is definitely crucial for understanding paradigm theories and disputes in normative ethics (whether it's metaphysical grounding or normative grounding), it's much more contentious whether grounding is crucial for understanding the naturalism versus non-naturalism debate in metaethics.

Related Topics

Varieties [Chapter 13], Mind [Chapter 33], Meta-Ground [Chapter 9], Laws of Metaphysics [Chapter 30], Essence [Chapter 22]

Notes

- 1 See Chapter 13: “Varieties.”
- 2 See Litland (2018) for a response to Berker’s challenge: he argues that the extremist should deny the mixed asymmetry principles and can explain the mixed transitivity principle.
- 3 See Chapter 33: “Mind.”
- 4 See Chapter 9: “Meta-Ground.”
- 5 See Chapter 30: “Laws of Metaphysics.”
- 6 McPherson (2013) takes real definition to be apt for characterizing the naturalism versus non-naturalism debate and attempts to make room for a non-reductive naturalist position, but the notion of “elite” properties is indispensable to his characterization as well.
- 7 Fine (1994) actually distinguishes between many different senses of essence. This is what he calls *immediate constitutive essence*. See also Chapter 22: “Essence.”
- 8 One way of thinking about these two different issues might be what Fine (2013) calls the D-project versus the E-project: the D-project is focused on the fundamentality of propositions, whereas the E-project is focused on the fundamentality of “sub-propositional constituents.” See Fine (2013) and Sider (2013) for discussion.

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35

PHYSICALISM

Amanda Bryant

Philosophers have long labored to articulate conceptions of physicalism adequate to their various dialectical purposes. As a result, *physicalism* denotes a rich and intricately related family of views, each with its own deep and dense philosophical literature. Painstaking work on supervenience has allowed the physicalist to attempt comparatively sophisticated formulations of the view. However, supervenience physicalism arguably faces serious unresolved challenges. A relatively new entry onto the metaphysical scene—*grounding*—enables an alternative conception of the key relation at the heart of some versions of physicalism. This chapter will consider the prospects for an account of physicalism with grounding at its core.

Before determining whether grounding may be fruitfully applied to physicalism, we will need some sense of what physicalism is supposed to be. To that end, Section 1 will lay out some key metaphysical physicalist theses. Section 2 will discuss supervenience and some central challenges to supervenience physicalism (see also Kovacs, Chapter 24, this volume). Section 3 will introduce the notion of grounding and use it to formulate physicalism. Finally, Section 4 will consider the relative merits and demerits of grounding physicalism. We will see that while grounding physicalism arguably improves on supervenience physicalism in some respects, serious work remains if grounding is to enable a clearly viable formulation of physicalism.

1. Physicalism

Physicalism is, like many philosophical doctrines, notoriously difficult to define. We often encounter it in the context of the mind–body problem. There, in opposition to substance dualism, the physicalist denies that the mental and the physical are distinct basic kinds of substance and integrates both under a common category. Against idealist metaphysical monism, which claims mentality to be the one monolithic substance, the physicalist claims the common category to be physical substance. Several standard views of the mind–body relation are considered physicalist, including behaviorism (at least in its ontological form), token and type identity theory, reductionism, and eliminativism.

Physicalism is generally thought to say more than simply that the mind is physical. While one could endorse domain-specific versions of physicalism (see Yates 2009), physicalism is often meant to be an overarching philosophical doctrine—a *Weltanschauung*. The trouble is, the precise nature of the doctrine is subject to significant controversy. I will shortly turn to the

task of surveying some of the controversy by distinguishing a number of competing physicalist theses. Note, however, that some see physicalism not as a contentful view but as a stance or attitude (Ney 2008b; van Fraassen 2002). Those who do see physicalism as a contentful view sometimes formulate it as a linguistic thesis, as in the case of its progenitors, Carnap (1959) and Neurath (1931). Others formulate it as a partly epistemic thesis about whether the necessitation of the mental by the physical is known *a priori* or *a posteriori* (Howell 2015; Jackson 2005; Levin 2019; Witmer 2006). Since my interest here is in applications of grounding—a metaphysical relation—to physicalism, I will bracket such formulations and discuss some standard metaphysical theses. I will not attempt to give an exhaustive survey (but for a more extensive one, see Stoljar 2010).

The first metaphysical thesis captures the anti-dualist sentiment mentioned earlier.

(1) *Negative thesis*: There are no non-physical things.

This thesis is meant to capture the physicalist's insistence that there are no non-physical minds, spirits, souls, gods, and so forth (Kim 1997; Poland 1994: 15). In its negative form, this thesis reflects a relatively focused dialectical motivation. The physicalist who avows it primarily aims to distinguish herself from those she regards as holding a spooky or otherwise suspect metaphysical view. The challenge she faces is that of delimiting the category of non-physical things in a non-*ad-hoc* manner. A natural move would be to define the category contrastively, as whatever falls outside the category of *physical* things. This naturally leads us to an alternative version of our first thesis—its logically equivalent but positively stated obverse.

(1') *Positive thesis*: Everything is physical.

Some important features of this thesis need unpacking. First, we need to know what falls within the scope of the quantifier—that is, what is meant by the claim that every 'thing' is physical. 'Thing' should be construed broadly to include entities, properties, kinds, relations, events, processes, and facts. With 'thing' understood in this broad sense, this positive thesis is sometimes presented as the core idea of physicalism (Papineau 2001; Pettit 1993; Elpidorou 2018; Stoljar 2010: 28). Moreover, having understood 'thing' broadly, we can further discriminate between token and type things. On the *token physicalist* interpretation, every token thing is a physical token. On the *type physicalist* interpretation, every type of thing (for instance, every mental property or every kind of mental event) is a physical type. Since types are instantiated by tokens, type physicalism entails token physicalism.

Next, we need to know what it means to say that everything is 'physical'. That is, which account of the physical does the physicalist assume or endorse? This is a thorny and controversial issue, and I can consider it only briefly (for suggested desiderata on conceptions of the physical, see Witmer 2018). On one major view, that which is physical is in some sense *countenanced* by science—or more specifically by natural science (Stoljar 2010: 26), physical science (Brown and Ladyman 2009: 34), physics (Elpidorou 2018; Kirk 1994), or fundamental physics (Loewer 2001; Wilson 2006). Call this the *scientific conception* of the physical.¹ Let 'science' be a placeholder for whatever the physicalist deems to be the appropriate science.

The phrase 'countenanced by science' is unclear. It could mean *mentioned, explicitly posited, or described* in the language of science (Hellman and Thompson 1975). We might characterize this roughly in Quinean fashion, in terms of the things (or kinds of things) quantified over by our scientific theories. Alternatively, 'countenanced' by science could mean *investigated* by it. In that case, the things countenanced by science are the targets of scientific inquiry, which we

sometimes successfully intervene on and manipulate, posit for the purposes of explanation and prediction, mathematically describe, or speculate about. Where the physicalist means ‘science’ to denote the natural sciences, this latter interpretation would make physicalism equivalent to the view Papineau calls *ontological naturalism*, according to which “reality involves nothing more than the entities studied in the natural sciences” (2014: 166).

Regardless of how we cash out ‘to countenance’, the scientific conception faces what is known as *Hempel’s Dilemma* (Hempel 1969, 1980). The dilemma emerges when, in response to the proposal that science guide ontology, we ask *which science, from when* (see Crane and Mellor 1990; Dowell 2006a; Melnyk 1997; Montero 1999; Ney 2008a; Stoljar 2010; Wilson 2005, 2006)? If we index our conception of the physical to *past science*, we grant physicality to things whose existence we now deny. That route is a non-starter. The dilemma, then, is as follows. On the first horn, if we index our conception of the physical to *current science* (Loewer 2001; Melnyk 1997), we face a few related difficulties, including that current science is incomplete, likely to change, and false. On the second horn, if we index our conception of the physical to ideal or complete *future theories* (Poland 1994), arguably we render physicalism vague, lacking determinate content, or trivial (Chomsky 1968; Crook and Gillett 2001). At the very least, we leave ourselves epistemically in the dark about which things count as physical. Those who endorse the scientific conception of the physical must choose a horn and resolve it.

However, an alternative conception of the physical has emerged in an attempt to avoid Hempel’s Dilemma. On the *via negativa* conception, that which is physical is non-mental. There is some debate as to whether the *via negativa* conception in fact avoids Hempel’s dilemma (see Gillett and Witmer 2001; Montero and Papineau 2005). Moreover, while I characterize the *via negativa* as a conception of the physical (following Montero and Papineau 2005; Crook and Gillett 2001; Stoljar 2010; Tiehen forthcoming; Worley 2006), some characterize it as a way of altogether *avoiding* the substantive difficulty of defining the physical (Levine and Trogdon 2009; Montero 1999, 2001; Spurrett and Papineau 1999). Indeed, the *via negativa* appeals most to those who see dim prospects for an adequately worked-out conception of the physical (Crane and Mellor 1990; Montero 1999, 2001; Stoljar 2010).

It is not clear which of the discussed conceptions, if either, is adequate or best relative to physicalist purposes. On the one hand, the scientific conception uncontroversially faces Hempel’s Dilemma. On the other hand, the *via negativa* must be relativized to dialectical context—after all, it is only apt to characterize the physical as *non-mental* if we are concerned with minds in the first place—and such relativization might strike one as *ad hoc*. What’s more, the *via negativa* clearly leaves something to be desired for those who think a satisfying or complete physicalist story requires a positive account of the physical. Of course, whether these are bugs or features is among the bones of contention. Ultimately there is no easy answer to the question of what the ‘physical’ in ‘physicalist’ means.

Setting aside conceptions of the physical, let us return to the positive thesis we were attempting to cash out—the thesis that everything is physical. It might be thought that this thesis, however interpreted, will lead to an overly restrictive ontology. That is because we can identify entities, properties, and phenomena that seem not quite physical in the senses discussed. For instance, if we adopt a brand of physicalism that assumes physical things are mentioned in the language of physics, we must contend with the fact that physics tends not to mention lots of the things we believe in, such as “mothers, baseballs, nations, or even rocks” (Ney 2008a: 1036).

In response, the defender of the positive thesis can either eliminate or reduce such things. The eliminativist move is revisionary in the extreme. As for the reductionist move, setting aside conceptual, theoretical, and explanatory reductions, we may distinguish two kinds of metaphysically reductionist views: *narrowly reductive* and *broadly reductive*. For the narrow reductionist, to

reduce something is to establish its (token or type) identity with something at a lower level of organization. The narrow reductionist move requires a promissory note, which we may regard with some dubiousness, to the effect that all the ontological items we are interested in saving will be shown to be identical with fundamental physical phenomena. Identity theory has been so closely associated with reduction that it is common to deem brands of physicalism that reject it *non-reductionist* (as in List and Menzies 2009; Wilson 2010). However, Lewis identified a broader sense in which physicalist theories can be reductionist, in which to reduce something is to show it to be *nothing over and above* something at a lower level of organization (1983: 358). This broad-sense reductionism can figure into physicalism in the following way.

(2) *Free lunch thesis*: There is nothing over and above the physical.

I call this the ‘free lunch’ thesis (borrowing Armstrong’s well-known 1997 phrase) because it suggests that things like mental events and properties add nothing to the furniture of the world. As it were, they come for free with our physical commitments. So this version of physicalism is suitably ontologically conservative from the perspective of those who wish to save mothers, baseballs, nations, and rocks while at the same time minimizing their ontological status.

The pressing question for the free lunch physicalist is, naturally, what does it mean for something to be *over and above* the physical (or not)? Rather than grappling with that particular question, I wish to consider the more general issue of what sort of relation the physicalist needs at the heart of her view. The final thesis I will discuss prompts us in that direction. It shares the spirit of the free lunch thesis, but is slightly more informative or contentful in virtue of its disjunctive form.

(3) *Disjunctive thesis*: Everything is either physical or appropriately related to the physical.

In cashing out this disjunctive thesis, the crucial question now becomes: what does it mean to be *appropriately related* to the physical? Since this ‘appropriate relation’ is highly sought after and physicalists have put such effort into identifying it, I will call it the *million-dollar relation*. A number of candidates for the million-dollar relation have been suggested. Here I again bracket non-metaphysical relations, such as entailment. The following list is not exhaustive, and there may be some overlap among its items, depending on how one cashes out the terms. Candidates include constitution (Baker 2000; Johnston 1992), composition (Boyd 1980), determination (Hellman and Thompson 1975; MacDonald and MacDonald 1986; Yablo 1992), fixing (Elpidorou 2018), necessitation (Stoljar 2010), truthmaking (Morris 2016), and realization (Melnik 2003, 2006, 2018; Shoemaker 2007).

Importantly, the disjunctive thesis provides the clearest entry point for grounding, since grounding may be thought to be a comparatively attractive candidate for the physicalist’s million-dollar relation. I will explore this thought in Sections 3 and 4. Since supervenience has had the most attention devoted to it, in the following section, I will treat supervenience physicalism as the benchmark against which to compare the prospects for a grounding-based formulation of physicalism. This is not to suggest that the other candidates are not interesting or promising in their own right.

2. Supervenience Physicalism

On the standard conception of supervenience, A supervenes on B when there cannot be an A-difference without a B-difference. The relata of the supervenience relation are generally

thought to be properties or facts (Leuenberger 2008). So for instance, to say the mental supervenes on the physical is to say there cannot be a difference in mental properties or facts without a difference in physical ones. Moreover, the formal properties of the supervenience relation include reflexivity (there can be no A-difference without an A-difference) and transitivity (one thing may supervene on another via intermediate supervenience relations). Supervenience is also non-symmetric (Kim 1984).

That it is impossible to have an A-difference without a B-difference is a modal claim, the force of which can be characterized in different ways. For instance, with *natural supervenience*, an A-difference without a B-difference would involve a violation of physical law. With *metaphysical supervenience*, the B-facts metaphysically necessitate the A-facts. With *logical supervenience*, an A-difference without a B-difference would involve a contradiction.²

Importantly, supervenience is a mere covariation relation. It builds in no substantive metaphysical commitment as regards the reason for the covariation. For instance, there is no assumption that the supervenience-base somehow *produces* supervenient things, nor that the supervenient things *metaphysically depend* on the supervenience-base. Because the supervenience relation is metaphysically minimal, reflexive and non-symmetric, it also lacks explanatory import (Blackburn 1984; Kim 1993).

It is worth noting at this juncture that Horgan (1993) advances the notion of *superdupervenience*—supervenience that is amenable to physicalistically acceptable explanation (i.e., supervenient properties or facts can be explained in terms of a physical supervenience base). While *prima facie* superdupervenience may be thought to overcome the explanatory limitations of supervenience, Horgan admits that explaining supervenience relations in a physicalistically acceptable way is a “daunting task” (1993: 581). Lynch and Glasgow argue that, worse, it is impossible (2003). Even if it were possible, Jessica Wilson (1999) argues that the explanatory constraint Horgan places on supervenience “is neither sufficient nor necessary to ensure that properties supervening on presumed physicalistically acceptable base properties are themselves physicalistically acceptable” (1999: 34).

Returning to garden-variety supervenience, the supervenience of mental properties on fundamental physical properties is typically thought to be at least a necessary condition for physicalism (though against this see Montero 2013; Montero and Brown 2018). If we plug supervenience into the disjunctive thesis, we get the following formulation of physicalism.

Supervenience physicalism: Everything either is physical or supervenes on the physical.

‘Thing’ may still be construed broadly, though assuming the standard view of supervenience, anything satisfying the right disjunct will be a property or fact. Nevertheless, some things—like concrete particulars—will satisfy the left disjunct and not the right, since they are not the sorts of things that can be related in supervenience relations (characterizing them as such would be a category mistake). Moreover, there is some question as to which modal profile the physicalist intends to assign to supervenience in this context—but the physicalist often takes the type of supervenience at issue to involve metaphysical necessity (Wilson 2005: 436).

There are different ways of cashing out supervenience physicalism. First, it can be cashed out in terms of individuals (Kim 1987) or regions (Horgan 1982). For instance, one might hold that there cannot be a mental difference in an individual or region without some physical difference in that individual or region. Call this *local* supervenience physicalism. Alternatively, supervenience physicalism can be cashed out in terms of possible worlds. For instance, one might hold

that there cannot be a mental difference at a world without some physical difference at that world. Call this *global* supervenience physicalism. Since physicalism is meant to be an overarching metaphysical doctrine, it is often formulated as a global thesis. Jackson and Lewis articulate some classic global formulations:

Any world which is a *minimal* physical duplicate of our world is a duplicate *simpliciter* of our world.

(Jackson 1998: 12)

Among worlds where no natural properties alien to our world are instantiated, no two differ without differing physically; any two such worlds that are exactly alike physically are duplicates.

(Lewis 1983: 364)

For further discussion of the global thesis, see Bennett (2004); McLaughlin (1995, 1997); Paull and Sider (1992); Shagrir (2002); Sider (1999); and Stalnaker (1996).

While supervenience physicalism faces many challenges (see Leuenberger 2008 for a survey), I will highlight two main problems. First, it is usually thought that the physicalist's million-dollar relation should be a metaphysical dependence relation of one form or another. However, as I have already noted, supervenience is not a metaphysical dependence relation but a mere covariation relation. As such, supervenience formulations of physicalism fail to capture or illuminate the common physicalist contention that the mental metaphysically depends on the physical. Since supervenience is not a metaphysically or explanatorily robust dependence relation, we might think that it somewhat impoverishes physicalism. As Koslicki puts it, supervenience is a "purely modal relation . . . too explanatorily coarse-grained to capture and illuminate the nature of the connections at issue" (2015: 308). So the first problem is that supervenience seems to be rather less metaphysically and explanatorily robust than the physicalist wants her million-dollar relation to be.

The most serious problem facing supervenience physicalism is that supervenience fails to be sufficient for physicalism (Heil 1992; Horgan 1993; Kim 1990, 1993; Schiffer 1987). The problem is that "any necessary facts globally supervene on any facts" (Leuenberger 2008: 755). In other words, with necessary facts, we get supervenience on the cheap. That is because no matter what kind of fact B is, if A is a necessary fact, there cannot be an A-difference. So, trivially, we satisfy the supervenience dictum that there cannot be an A-difference without a B-difference. Switching from facts to entities, Montero and Brown explain the problem in the following way: "any necessarily existing thing supervenes on the physical since any world that duplicates the nitty-gritty of our world will also contain this necessarily existing thing; even God, if she goes in for necessary existence, supervenes on the physical" (2018: 524). Alternatively, Wilson imagines a god that sets things up so that emergent mental states covary, in a metaphysically necessary fashion, with physical states (2014: 543). Since non-physical gods and emergent mental phenomena can fit into the supervenience mould, supervenience formulations of physicalism fail to meet a minimal condition of physicalism—namely, that it rule out the existence of things like non-physical gods and emergent mental states.

We have seen in my survey of the available conceptions of physicalism that the physicalist who wishes to avoid an overly restrictive ontology will characterize her view using the disjunctive thesis, according to which everything is either physical or appropriately related to the physical. However, we have seen that the standard candidate for the million-dollar

relation—supervenience—is both metaphysically and explanatorily impoverished, as well as insufficient for physicalism. This suggests that we should consider alternatives. This is the natural entry point for the discussion of grounding. Several philosophers have suggested the possibility of a grounding-based formulation of physicalism (Bennett 2011a; Fine 2001; Kroedel and Schulz 2016; Schaffer 2009; Rosen 2010). After introducing the notion of ground, the remainder of the chapter will sketch grounding physicalism and assess its prospects.

3. Grounding Physicalism

Grounding is a metaphysical dependence relation that tracks the ‘in virtue of’ locution (Fine 2012), so that whenever we say that x exists or obtains in virtue of y , we may also say that y grounds x .³ The relation is generally thought to involve necessitation (though against this, see Skiles 2015). The nature of the dependence is also sometimes thought to be productive in some sense. For example, Bennett characterizes grounding as a *building relation*—a directed relation, the relata of which differ in how fundamental they are and share a connection, where ‘connection’ is left as a primitive (2011b: 90–92; see also her 2017). At any rate, the thing doing the grounding is thought to be more fundamental ontologically than the thing being grounded. Since some prominent grounding theorists (Schaffer 2009; Rosen 2010) take *grounding* to be a primitive term, examples often play a key role in its elucidation. Putative examples of grounding claims include:

Categorical facts ground dispositional facts.
Non-aesthetic facts ground aesthetic facts.
Social and psychological facts ground semantic facts.

(Audi 2012)

The fact that A and the fact that B ground the fact that A&B.
Socrates’ existence grounds the existence of singleton Socrates.

(Fine 2015)

Natural features ground moral features.
Sparse properties ground abundant properties.
Truthmakers ground truths.

(Schaffer 2009)

Some grounding theorists take the relata of the grounding relation to be facts (Audi 2012; Fine 2012; Rosen 2010); some take them to be propositions (Cameron 2015; Rosen 2010); others take them to be objects (Schaffer 2009).

Grounding is also similar to causation in a number of interesting ways (Schaffer 2016b; Wang, Chapter 20, this volume, A. Wilson 2018). Like causation, grounding is thought to be closely connected with explanation—it is indispensable to explanation (Audi 2012: 104), or itself explanatory (Dasgupta 2014: 558; Fine 2001: 15; Rosen 2010: 117), or at least backs explanation (Schaffer 2016b: 50). On the connection between grounding and explanation, see also Glazier (Chapter 8, this volume) and Raven (2015); see J. Wilson (2018) on the conflation of metaphysical dependence with explanation. Moreover, just as we can distinguish causal relevance from sufficiency, we can distinguish partial from full grounds, where partial grounds might contribute to grounding some fact or entity without being alone sufficient. For instance, *the fact that A* and *the fact that B* are each partial grounds of *the fact that A&B* (Fine 2012). Finally, like

the causal relation, the formal properties of grounding are typically thought to be asymmetry (no two things mutually ground one another), transitivity (one thing may ground another via intermediate grounding relations), and irreflexivity (nothing grounds itself). Note, however, that there is some disagreement about these formal properties (see Bliss 2014; Jenkins 2011; Tahko 2013; Thompson 2016, Chapter 17, this volume; Wilson 2014).

Some key differences between grounding and supervenience have emerged. First, the relations may have different relata—if the grounding relata are objects or propositions and if the supervenience relata are properties (though it is also possible that the relata of both are facts). In terms of formal properties, grounding is standardly thought to be irreflexive and asymmetric, while supervenience is reflexive and non-symmetric. Moreover, grounding is a metaphysical dependence relation, while supervenience is not. In particular, grounding is arguably what Bennett calls a building relation; supervenience is not. Finally, grounding is in some important sense connected to explanation, while supervenience is not. See Kovacs (Chapter 24, this volume) for further discussion of grounding in relation to supervenience.

Having spelled out the notion of ground and its differences from supervenience, we are now in a position to consider its application to physicalism. Recall that the disjunctive physicalist thesis declares everything to be physical or appropriately related to the physical. A grounding-based formulation of physicalism will plug grounding into the thesis as the million-dollar relation.

Grounding physicalism: Everything is either physical or grounded in the physical.

Here, the domain of quantification should again be understood broadly, though (trivially) only those sorts of things that can be grounding relata can satisfy the right disjunct. This formulation of grounding physicalism accords with Loewer's characterization of physicalism as the view that "the fundamental properties and facts are physical and everything else obtains *in virtue* of them" (2001: 31). On this view, the mental metaphysically derives from a more ontologically fundamental physical basis, in the sense that the one exists or obtains in virtue of the other.

One might wonder how grounding physicalism compares with physicalism cashed out in terms of superdupervenience. Superdupervenience physicalism would claim that everything is either physical or superdupervenient on the physical (that is, covaries with and is explainable in terms of it). *Prima facie*, the two are similar in that both involve an explanatorily loaded necessitation relation, and both build in a physical base. However, superdupervenience still differs from grounding in the ways that supervenience does: superdupervenience is not directional, nor is it productive. In virtue of this key difference between the two relations, grounding physicalism differs importantly from superdupervenience physicalism.

What might grounding physicalism end up looking like as an overarching metaphysical view? It may come as a surprise to learn that grounding physicalism could (*though need not*) be combined with ontological permissivism. For instance, Schaffer argues that the answer to most existence questions (at least the controversial ones) is a trivial affirmation. In his view, numbers, properties, mereological composites, and fictional characters exist, and trivially so (2009: 357). Schaffer even claims, notwithstanding his atheism, that the existence of God is a trivial truth (2009: 359). It is just that God exists derivatively rather than fundamentally, just as Sherlock Holmes, the numbers, and all the rest of the ontological misfits (2009: 357). If we import this permissive attitude into grounding physicalism, *permissive grounding physicalism* might say that, trivially, there are abundantly many non-physical things, but those things are grounded in a relatively more sparse physical base.

It is important to note, however, that this kind of permissivism is incidental to grounding physicalism, not a consequence of it. One could equally well adopt an ontologically austere

grounding physicalism. I consider the possibility of permissive grounding physicalism simply because it is an interestingly weird kind of view—weird because the spirit of physicalism seems anti-permissive, and interesting because it raises the question of whether such permissivism could actually square with the physicalist's broadly reductive tendencies. For his part, Schaffer takes derivative posits to be harmless. In his view, norms prescribing simplicity should apply at the level of fundamentality or grounds (2009: 361). Whether one sees this as a significant departure from the ontologically austere intentions of the physicalists depends on whether one agrees with Schaffer that it is harmless to posit abundantly many derivative entities (see Audi 2012 for an opposing view).

At any rate, we now have a flavor of what grounding physicalism amounts to and how it differs from nearby views. We have seen that grounding physicalism takes the disjunctive physicalist thesis and offers up grounding to fill the formidable role of the physicalist's million-dollar relation. I will now turn to my final task of the chapter, which will be to consider whether grounding can really make good on the demands of the role.

4. Grounding Physicalism Assessed

4.1 *Relative Improvement*

The first thing to consider is whether any of grounding's characteristic features make grounding physicalism an improvement over supervenience physicalism. In response to general skeptical worries regarding grounding, its defenders have argued that grounding has distinctive merits. We may look to those suggested merits for a sense of how grounding might enrich the physicalist story.

First off, Cameron (2016) argues that the distinctive merit of grounding is its directionality. He claims that grounding, in being inherently directional, illuminates more of the world's structure than other relations. He argues that it helps us describe ontic priority (2016: 386), including priority differences among non-fundamental things (2016: 389). He also claims it to be essential for accounting for the dependence of determinable properties on determinate properties (206: 390–91). Likewise, Raven (2017) argues that fixing the direction of priority is part of the crucial work that grounding accomplishes.

Importantly, Cameron and Raven are responding to Jessica Wilson (2014), who argues that we don't need grounding to fix the direction of priority. So whether or not grounding has a unique or indispensable role in fixing the direction of priority is a matter of debate. However, we can safely set this debate aside. Our current task is to see whether and how grounding physicalism improves on supervenience physicalism—and to that end, the relevant question here is just whether the grounding relation can be useful to the physicalist in a way that the supervenience relation can't. Regardless of whether grounding is *required* to fix the direction of priority, the fact remains that grounding *can* do it and supervenience can't. As we saw in the comparison of grounding to supervenience, supervenience is a non-symmetric modal relation, while grounding is indeed directional and productive. It is therefore plausible that compared with supervenience, grounding would increase the physicalist's descriptive capacity with regard to the hierarchical relation between the physical and non-physical. This may be considered a comparative merit.

Raven (2017) also argues that grounding's work includes helping us to analyze what he calls *ineliminable* fundamentality. Something is ineliminably fundamental when a fundamental account of reality must refer to it (2017: 649). Grounding enters the analysis of this notion in

the following way: “an entity is ineliminable just in case not all facts about it are grounded in facts about other entities” (2017: 650). The relevance of this for my purposes here is that the physicalist will likely claim the physical to be ineliminable in Raven’s sense. If so, then grounding may be crucial for analyzing the ultimate metaphysical status of the physical. So grounding’s directionality and its ability to account for the ineliminable status of the physical arguably make it an improvement over supervenience.

4.2 Grounding the Grounding Facts

At the same time, grounding faces a difficulty of its own. The difficulty concerns the status of the grounding facts themselves. Following Sider (2011), Dasgupta (2014) identifies a problem for the articulation of grounding physicalism, stemming from the question of what grounds the grounding facts (see also Litland, Chapter 9, this volume). Suppose that grounding physicalism claims that all non-physical facts are grounded in physical ones (Dasgupta 2014: 581). Take an example of a token grounding claim: *conscious state x is grounded in brain state y* . The problem arises when we ask about the status of that token grounding claim. Dasgupta points out that such grounding facts are not themselves physical facts (2014: 560; see also Melnyk 2016). If they are not physical facts, then grounding physicalism seems to require that they be grounded in physical facts. This doesn’t seem to be the case. If so, then grounding facts seem to present a counterexample to grounding physicalism.

A number of possible resolutions have been explored. Dasgupta resolves the problem by arguing that grounding facts belong to a category of *autonomous* facts—facts which are not apt for being grounded. Just as pure arithmetical facts are not apt for causal explanation, grounding facts are not apt for grounding explanation. The question of what grounds them “does not legitimately arise in the first place” (2014: 563). Rather, it rests on a category mistake.

Alternatively, in her treatment of the problem, Bennett argues that the grounding fact $\langle A \text{ grounds } B \rangle$ is grounded in A (2011a; see also her 2017). Then we have a further fact, which itself requires grounding: the fact $\langle A \text{ grounds } \langle A \text{ grounds } B \rangle \rangle$. On her view, A likewise grounds this further fact, and so on to infinity. The regress is, she argues, benign, since for each fact in the chain, there is an answer to the question of what grounds it. Similarly, DeRosset (2013) argues that facts stated by grounding explanations are grounded in their explanans and that this does not generate vicious regress because each fact in the series is grounded. So the problem of what grounds the grounding facts may be resolved by looking to the grounding relata themselves.

So far we have seen that grounding physicalism has some relative merits compared with supervenience physicalism, and while the grounding framework brings with it the problem of what grounds the grounding facts, there are some potential resolutions available. However, in assessing the relative attractiveness of grounding physicalism, a crucial step remains. If grounding physicalism is to be a viable form of physicalism, it must overcome the problems that arguably prove fatal to supervenience physicalism. I now return to those problems.

4.3 Coarse-Grainedness

The first problem was that supervenience is a purely modal relation, not a metaphysically or explanatorily robust dependence relation. Grounding one-ups supervenience in this regard, since it is explicitly a metaphysical dependence relation that its proponents believe to be importantly

connected to explanation. However, Koslicki argues that, like supervenience, grounding “fails to be sufficiently fine-grained to do its intended explanatory work” (2015: 309; see also her Chapter 11 in this volume). In her view, to learn that something physical grounds something mental is only to learn that “this entity functions as a minimal element in a domain over which a certain well-founded partial order is defined” (2015: 315). But this is not illuminating, especially since such a relation is “formally indistinguishable from other well-founded partial orderings which are presumably not numerically identical with the grounding relation” (2015: 315; see also J. Wilson 2018).

Jessica Wilson also argues that “Grounding, like supervenience, is too coarse-grained to do the work of appropriately characterizing metaphysical dependence on its own” (2014: 540). She argues that it is severely underdetermined with respect to importantly different theoretical options. Grounding claims say nothing, for example, about “whether the dependent goings-on exist, are reducible or rather distinct from the base goings-on, are efficacious, and so on” (2014: 540). In that regard, they wash over significant distinctions between eliminativist, reductionist, and non-reductionist accounts of dependence. As a result, compared with our rich array of pre-existing theoretical options, we might think that grounding physicalism is relatively contentless. Wilson puts the point this way:

[S]uppose someone claims that the mental is Grounded in the physical. Am I in position to know whether I should agree with them? Not at all . . . the bare assertion of Grounding is compatible with both reductive and non-reductive versions of physicalism—indeed, perhaps even with anti-realist eliminativism about the mental . . . Absent further information . . . I am stuck: I am not in position to assess, much less endorse, the claim that the mental is Grounded in . . . the physical. (2014: 549)

Wilson concludes that grounding claims leave open a number of critical questions, the answers to which are required for even basic understanding and assessment of dependence claims (2014: 545).⁴ Koslicki’s and Wilson’s arguments may be taken to raise serious doubts as to whether grounding fares better than supervenience with respect to its metaphysical and explanatory import.

Grounding’s arguable coarse-grainedness problem is one of several general skeptical worries about grounding. DeRosset (Chapter 12, this volume) responds to some of these worries. In particular, he considers the worry that grounding’s coarse-grained logical features make it theoretically unhelpful or uninformative relative to other metaphysical dependence relations. He replies that even if grounding claims are uninformative, that does not make them false—and neither does it motivate global skepticism with respect to grounding. Rather, it motivates the grounding theorist to move beyond the logic of ground to develop a more positive substantive theory of ground.⁵ With regard to the present concern that grounding is too coarse-grained to improve on supervenience in the formulation of physicalism, a response of this kind gives us only a promissory note for a harder-working conception of ground.

Since the coarse-grainedness worry is a general skeptical worry, my discussion of it may have had a kind of preaching-to-the-choir effect—those who were already convinced by the skeptical challenges will not have been optimistic about grounding’s prospects in the domain of physicalism anyway. As we turn to the sufficiency problem, we will see a distinctive difficulty that doesn’t similarly arise from general skeptical worries, but that arises *specifically* for grounding’s application to physicalism and in response to physicalism’s unique demands.

4.4 Insufficiency

The second problem facing supervenience physicalism was the insufficiency of supervenience for physicalism. How does grounding fare in this regard? Melnyk (2016) argues that grounding physicalism allows for a failure of broad reduction. That is, like supervenience, the grounding relation permits entities over and above the physical. However, as we saw, physicalism is sometimes thought to require nothing over and above the physical. If so, then grounding physicalism violates a minimal condition of physicalism.

You may recall that I earlier shifted away from the language of over-and-aboveness. So we don't yet have any sense of what 'nothing over and above' means. Melnyk uses a placeholder phrase, "nothing over and above *in the intuitive sense required for physicalism*" (my emphasis, 2016: 251). Can the phrase be made more contentful? Sider (2015) devotes significant effort to precisifying it. Without going down that particular rabbit hole, my rough intuitive gloss of 'nothing over and above' is 'not ontologically additional to'. We may cash this out in terms of numerical distinctness. For instance, for a grounded entity to be nothing over and above its grounding entity (or entities) would be for it to be *not* numerically distinct from it (or them). Holding this gloss fixed, is Melnyk right that grounding physicalism permits entities over and above the physical? Clearly so. For the grounding theorist, the relata of the grounding relation are indeed distinct. Even Fine, for whom the grounding relation is in some important sense *constitutive*, denies that the grounded fact *just is* the grounding fact(s) (2012: 39). Indeed, the grounding relation seems to guarantee that grounded things are over and above their grounds in my sense.

However, one might think that 'not additional to' misses an important connotation of 'nothing over and above'. The aboveness metaphor may suggest that the physicalist means to rule out are things additional to the physical *in some more metaphysically weighty or significant sense*. Recall from the discussion of permissivism that Schaffer claims derivative entities have little import relative to the ontological roster—they are, he says, an ontological free lunch (2009: 361). Perhaps that could be the sense of nothing-over-and-aboveness the physicalist is getting at. Applied to grounding physicalism, to say grounded items are nothing over and above their physical grounds would be to say they constitute nothing *metaphysically significant* in addition to their physical grounds. If Schaffer is right that grounded entities are an ontological free lunch, then *contra* Melnyk, grounding physicalism does not permit entities over and above the physical. However, with regard to whether Schaffer *is* right, it is unclear how we would go about settling the matter (especially for those with naturalist inclinations).

Further regarding the insufficiency of grounding in the formulation of physicalism, Jessica Wilson (2018) argues that grounding physicalism cannot distinguish itself from strong emergence. Strongly emergent properties or phenomena are (in some sense) over and above their underlying physical properties or phenomena and are not deducible from statements describing them. Wilson argues that nothing prevents strong emergence from being, like grounding, a strict partial order, and so grounding's status as a nothing-over-and-above relation is "just a matter of brute stipulation" (2018: 506).

Previously, we saw doubts raised regarding the metaphysical and epistemic import of grounding. Now, on the question of whether grounding fares better than supervenience with respect to its sufficiency for physicalism, we see further doubt. If these doubts are well founded, then while grounding physicalism may be thought to improve upon its standard supervenience rival in certain respects, it doesn't overcome the problems that seem to defeat supervenience physicalism—and so is not yet a viable formulation of physicalism.

Supposing grounding physicalism does indeed fail to resolve the problems that are fatal to supervenience physicalism, by failing to illuminate the nature of the metaphysical dependence

between the non-physical and the physical and by failing to rule out things over-and-above the physical, what shall the physicalist do? The issue regarding the contentless character of grounding, as well as its alleged insufficiency for physicalism, may stem from the fact that in practice, grounding is typically left a “thinly described primitive” (J. Wilson 2018: 506), even by those who claim it to be primitive only provisionally. Perhaps some of the challenges to grounding may be resolved if, as deRosset (Chapter 12, this volume) also suggests, we develop a more positive substantive account. Alternatively, Ney (2016, Chapter 33, this volume) suggests that grounding need not be adopted *to the exclusion* of other relations—so perhaps a hybrid view is possible. A final possibility would be to turn our attention elsewhere, to a range of alternate dependence relations, which some regard as more promising candidates for the physicalist’s million-dollar relation.

Acknowledgements

This research was supported by the Fundação para a ciência e a tecnologia, award #PTDC/FER-HFC/30665/2017. My thanks to Barbara Montero, David Papineau, Michael J. Raven, and David Yates for their invaluable feedback on this chapter.

Related Topics

Mind [Chapter 33], Modality [Chapter 24]

Notes

- 1 If the scientific conception is a bona fide account of the meaning of ‘physical’, then it should give us necessary and sufficient conditions for physicality. If so, then with suitable refinement, it should be amenable to biconditional formulation. However, I do not claim that it succeeds in providing necessary and sufficient conditions on any of its interpretations.
- 2 There are many important and sophisticated further refinements of supervenience, which I must bracket for reasons of length. For instance, on weak and strong supervenience, see Kim (1984) and (1987). On multiple domain supervenience, see Kim (1998).
- 3 While I characterize grounding as a dependence relation, McLaughlin and Bennett (2018) note that in cases of ‘grounding overdetermination’, you can have grounding without ontological dependence. See also Schnieder’s contribution to this volume [Chapter 7].
- 4 In response, Ney (2016) argues that Wilson’s concerns don’t apply to Fine’s grounding framework.
- 5 Compare Schaffer’s (2016a) claim that *more* formalism is the answer. He argues that structural equation models resolve the problem of uninformativeness.

Further Reading

1. Dasgupta, S. (2014) ‘The Possibility of Physicalism’, *Journal of Philosophy*, 111(9–10), pp. 557–92. (A treatment of the problem of whether token grounding facts are themselves grounded in physical facts.)
2. Melnyk, A. (2016) ‘Grounding and the Formulation of Physicalism’, in Aizawa, K. and Gillett, C. (eds) *Scientific Composition and Metaphysical Ground*. London: Palgrave Macmillan, pp. 249–69. (An articulation of problems for grounding formulations of physicalism, as compared with realization formulations.)
3. Ney, A. (2016) ‘Grounding in the Philosophy of Mind: A Defense’, in *Scientific Composition and Metaphysical Ground*. London: Palgrave Macmillan. (A defence of the usefulness of grounding in articulating key views in philosophy of mind.)
4. O’Conaill, D. (2018) ‘Grounding, Physicalism, and Necessity’, *Inquiry*, 61(7), pp. 713–30. (A defence of grounding physicalism against difficulties arising from metaphysical necessitation.)

5. Wilson, J. (2018) 'Grounding-Based Formulations of Physicalism', *Topoi*, 37, pp. 495–512. (An articulation of problems for grounding formulations of physicalism.)

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36

SEMANTICS

Kit Fine

There is an intimate connection between ground and truth-making. Suppose I make a statement of ground, such as:

- (1) it is raining < it is raining or it is snowing.¹

I may then subject this statement to ‘ontic’ reification on the left and to ‘semantic’ reification on the right, thereby giving:

- (2) the fact that it is raining makes-true the sentence ‘it is raining or it is snowing’.

Thus the sentence ‘it is raining’ in (1) is replaced with a term for a fact on the left, and the sentence ‘it is raining or it is snowing’ is replaced with a term for a sentence on the right; and since we now have two terms rather than two sentences, we should replace the operator ‘<’ for ground with the corresponding predicate for truth-making. It should be clear that, in this way, any ground-theoretic statement flanking two sentences can be seen to correspond to a truth-making statement flanking two terms and vice versa.²

Under such a correspondence, the truth-making statement will inherit many of the features of the ground-theoretic statement. Thus it may be thought, given that (1) holds, that it will be necessary that if it is raining then it is raining or snowing. It may therefore likewise be thought to be necessary that if the fact that it is raining obtains then the sentence ‘it is raining or it is snowing’ will be true (though perhaps relative to the sentence still meaning what it does). Or again, it may be thought, given that (1) holds, that its raining should be wholly relevant to its raining or snowing. It may therefore likewise be thought that there should be a relevant connection of this sort between a fact and the sentence it makes true.³

It has often been supposed that there need only be a modal connection between a truth-maker and the sentence it makes true or that the truth-maker need only be partially relevant to the sentence it makes true, so that the fact that it is raining and windy, for example, would then be a truth-maker for the sentence ‘it is raining or snowing’.⁴ It is therefore important to note that the notion of ground gives rise to a quite distinctive notion of truth-making, which requires not merely a modal connection but also a very strong relevant connection.

Truth-making has been used for two quite distinct ends, one metaphysical and the other semantical. By attempting to discern the truth-makers of sentences, it has been thought that we might achieve a better understanding of the world via an understanding of *what* makes the sentences true and also that we might achieve a better understanding of language via an understanding of *how* the sentences are made true. Thus we might think of statements such as (2) as telling us something about the world—that it is not really ‘disjunctive’—or we might think of it as telling us something about the meaning of ‘or’—that or-statements are always made true through their disjuncts.

It has been questioned whether reification is either necessary or helpful in regard to the metaphysical project.⁵ But whether or not this is so, it is certainly helpful in regard to the semantical project. For one thing, we need reification on the right, with reference to sentences, in order for a semantics done in terms of truth-making to hook up to language in the right way. We also need reification on the left, with reference to facts or the like. For we would like to be able to represent the meanings or contents of sentences as mathematical objects so that the semantics can itself be regarded as a piece of mathematics to which mathematical methods will apply. But treating truth-makers as facts and contents as sets of facts provides us with the most natural and straightforward way in which this might be done.

If semantical truth-making is conceived in the image of ground, then it is likely to have certain other features. If facts had to obtain, i.e., to be *actual* facts, then false sentences would all have the same truth-makers, viz. none, and so false sentences could not be distinguished in terms of their truth-makers (though they might still be distinguished in terms of their falsity-makers). However, for many semantic purposes, we *will* want to distinguish between false sentences in terms of their truth-makers. Consider, for example, the two counterfactual statements ‘if I had taken the poison I would have died’ and ‘if I had taken the poison and the antidote I would have died’ and suppose I did not take the poison, so that both antecedents are false. Then I might well want to attribute the difference in truth-value in the two counterfactuals to a difference in the truth-makers for their antecedents.⁶

Thus it is important to allow the truth-makers to be possible facts and not merely actual facts, so that the truth-making relation in semantics should be seen to correspond to the nonfactive as opposed to the factive notion of ground.⁷ Indeed, one should probably go even further and allow truth-makers to be impossible facts. We may again use counterfactuals to illustrate the point, since one might well want to distinguish between the counterfactuals ‘if I had been Adolph Hitler I would have been a better man’ and ‘if I had been Albert Schweitzer I would have been a better man’ in terms of the truth-makers for their antecedents, even though both antecedents are impossible (I cannot be someone other than who I am).⁸

The attentive reader will have noticed that the grounding statement (1) above is one-one. But grounding statements can also be many-one as in:

- (3) it is raining, it is windy < it is raining and it is windy.

This then raises the question of how such statements should be reified for the purpose of semantics.

One possibility is to suppose that two facts, the fact that it is raining and the fact that it is windy, are *joint* truth-makers for the sentence ‘it is raining and windy’. Following through on this suggestion, we should then in general allow a sentence to have any number of joint truth-makers, and we should also allow the individual truth-makers in a joint set of truth-makers to themselves be joint truth-makers. Thus if *r* is a truth-maker for *R* (‘it is raining’), *w* a truth-maker

for W ('it is windy') and c a truth-maker for C ('it is cold'), then the set $\{r, w\}$ will be a joint truth-maker for $R \wedge W$ while the set $\{\{r, w\}, c\}$ will be a joint truth-maker for $(R \wedge W) \wedge C$.

This hierarchical conception of truth-making is not without its interest. But another possibility is to fuse the plurality of truth-makers into a single truth-maker. Thus instead of having r and w as a joint set $\{r, w\}$ of truth-makers for the sentence $R \wedge W$, we might take their fusion $r \sqcup w$ (rain-and-wind) to be a truth-maker for the sentence. The truth-maker for $R \wedge (W \wedge C)$ would then be $(r \sqcup w) \sqcup c$ ((rain-and-wind)-and-cold) But $(r \sqcup w) \sqcup c = r \sqcup (w \sqcup c)$; and so the truth-maker $(r \sqcup w) \sqcup c$ for $(R \wedge W) \wedge C$ would be the same as the truth-maker $r \sqcup (w \sqcup c)$ for $R \wedge (W \wedge C)$, in contrast to the previous proposal, under which the truth-makers for the respective sentences would be $\{\{r, w\}, c\}$ and $\{r, \{w, c\}\}$.

The 'flat' approach has the great advantage of simplicity; all of the truth-makers will be at the same level, so to speak, and the complex hierarchical process by which truth-makers are generated can be ignored. Moreover, for the majority of semantic purposes, we will have no interest in this hierarchical structure; there is, for example, no reason to distinguish between the semantic content of the left-to-right and the 'right-to left bracketing of 'it is raining and it is windy and it is cold'. Thus another advantage of the flat approach is that it enables us to ignore distinctions in the syntactic structure of a sentence that, for most purposes, are irrelevant to its meaning.

The flat approach leads to a conception of truth-makers as constituting a 'state space'. This is a set of truth-makers or 'states' that are closed under fusion; given any states whatever, they will have a fusion that contains each of the states as parts and is part of any other state that contains each of the states as parts.⁹ State spaces, in one form or another, can be seen to underlie much of the recent work in truth-maker semantics.

Given that any number of states have a fusion in a state space, it will follow, in particular, that zero states will have a fusion; and this fusion will be the null state, which is a part of every state and which, as a rule, will be devoid of content. It is arguable that some sentences, such as 'if it rains then it rains', will be made true by the null state; and its being made true by the null state will correspond to the case of zero ground, as in:

(4) $<$ it if rains then it rains

with zero sentences to the left of ' $<$ '.¹⁰

Given that any number of states have a fusion, it will follow that the fusion of incompatible states, such as the state of its being hot and the state of its being cold, will also belong to the state space. Thus the mere existence of incompatible states in a state space will guarantee the existence of impossible states; and there is no reason in general to suppose that these impossible states will all be the same. The state of its being both hot and cold, for example, might be taken to be different from the state of its being both wet and dry, with the first having hot and cold as parts, but not wet and dry, and the second having wet and dry as parts, but not hot and cold.

There are two other modifications that need to be made if we are to obtain a viable form of truth-maker semantics. In the first place, we may want the fact that it rains to be a truth-maker for the sentence 'it rains'. This corresponds to the following self-reflexive statement of ground:

it rains $<$ it rains.

But it would not normally be supposed that this truth is capable of grounding itself.¹¹ This suggests that the notion of ground corresponding to truth-making is not the usual strict notion but a weak notion, under which each truth will indeed serve as a ground for itself. The truth-makers

for logically complex sentences will normally correspond to strict grounds, while the truth-makers for atomic sentences may only correspond to weak grounds.¹²

The second modification arises from the desire to have a compositional semantics, one in which the content of a complex sentence or expression is determined on the basis of the content of its parts. Negation raises a problem in this regard, since we cannot in general determine the truth-makers of the negation $\neg S$ from the truth-makers of S .¹³ We may get round this problem by adopting a *bilateral* semantics, in which we separately specify the truth- and falsity-makers for each logically complex sentence.

Putting all these pieces together, we are led to adopt the following clauses for negation, conjunction and disjunction (where the ‘if’ can also be read as an ‘if and only if’):

- T(i)⁺ s is a truth-maker for $\neg A$ if s is a falsity-maker for A
- T(i)⁻ s is a falsity-maker for $\neg A$ if s is a truth-maker for A
- T(ii)⁺ s is a truth-maker for $A \wedge B$ if s is the fusion of a truth-maker for A and a truth-maker for B
- T(ii)⁻ s is a falsity-maker for $A \wedge B$ if s is a falsity-maker for A or a falsity-maker for B
- T(iii)⁺ s is a truth-maker for $A \vee B$ if s is a truth-maker for A or a truth-maker for B
- T(iii)⁻ s is a falsity-maker for $A \vee B$ if s is the fusion of a falsity-maker for A and a falsity-maker for B .

We can also formulate corresponding clauses for the quantifiers. Thus if $i_1, i_2, \dots$ are the distinct individuals in the range of the quantifier and $i_1, i_2, \dots$ are names for $i_1, i_2, \dots$, then we may say:

- T(iv)⁺ s is a truth-maker for $\forall x A(x)$ if it is the fusion of truth-makers for each of $A(i_1), A(i_2), \dots$
- T(iv)⁻ s is a falsity-maker for $\forall x A(x)$ if it is a falsity-maker for some one of $A(i_1), A(i_2), \dots$
- T(v)⁺ s is a truth-maker for $\exists x A(x)$ if it is a truth-maker for some one of $A(i_1), A(i_2), \dots, A(i_k)$
- T(v)⁻ s is a falsity-maker for $\exists x A(x)$ if it is a fusion of falsity-makers for each of $A(i_1), A(i_2), \dots$

However, these clauses are not adequate if it is possible that there are individuals other than the individuals $i_1, i_2, \dots$ that there actually are for, in that case, the truth of $A(i_1), A(i_2), \dots$ will not be sufficient to guarantee the truth of $\forall x A(x)$. This is a familiar problem in the metaphysics of ground, since in such a case, we cannot say that a universal truth is grounded in its instances.¹⁴ One solution to the metaphysical problem is to say that the universal truth is grounded in its instances and a totality truth to the effect that $i_1, i_2, \dots$ are all the individuals that there are. We might adopt an analogous solution to the semantical problem. Thus it may be supposed that there is a totality fact in regard to the individuals $i_1, i_2, \dots$ and the truth-maker for $\forall x A(x)$ will then be the fusion of truth-makers for its instances along with the totality fact. One nice feature of the semantical solution over the metaphysical solution is that we do not need to go on and ask what grounds the totality truth (or fact) since, for semantical purposes, it can simply be taken as given.

Each of the clauses can be seen to have its basis in some corresponding claims about ground. Thus, in the case of T(iii)⁺, either of A or B will be a ground of $A \vee B$, and so any truth-maker for A or for B will be a truth-maker for $A \vee B$. Moreover, if C is a strict ground of $A \vee B$ then it must be a weak ground for A or for B , and so any truth-maker for $A \vee B$ must be a truth-maker for A or for B .¹⁵

We can, in this way, see truth-maker semantics as the natural result of the attempt to convert the metaphysical notion of ground into a semantical notion of truth-making.

The main alternative to truth-maker semantics is the possible worlds semantics.¹⁶ This alternative approach differs both in its conception of the truth-makers and in its conception of truth-making. The truth-makers are whole worlds, which settle every question under consideration. And truth-making is a purely modal notion. For a world to make a sentence true is simply for it to be impossible that the world obtain and the sentence be false.

We then have the following semantical clauses under the possible worlds approach, with w a world and $i_1, i_2, \dots$ the individuals of that world:

- W(i) w is a truth-maker for $\neg A$ if w is not a truth-maker for A
- W(ii) w is a truth-maker for $A \wedge B$ if w is a truth-maker for both A and B
- W(iii) w is a truth-maker for $A \vee B$ if w is a truth-maker for A or for B
- W(iv) w is a truth-maker for $\forall x A(x)$ if it is a truth-maker for each of $A(i_1), A(i_2), \dots$
- W(v) w is a truth-maker for $\exists x A(x)$ if it is a truth-maker for some one of $A(i_1), A(i_2), \dots$

These clauses are much simpler than the corresponding clauses under the truth-maker approach: the semantics is unilateral rather than bilateral; the clauses involve no shift in the world w of evaluation as we move from left to right (in contrast to clauses T(ii)⁺, T(iii)⁻, T(iv)⁺ and T(v)⁻); and there is no need for a totality fact.

However, the truth-maker approach provides us with much more information about the content of a sentence, for instead of simply telling us *when* a sentence is true in a world, it tells us what it is *in* the world (including an impossible world) that makes the sentence true. Thus whereas the possible worlds semantics will assign the same content to ‘it is raining or it is not raining’ and ‘it is windy or it is not windy’, viz. the set of all worlds, the truth-maker semantics will assign a different content to these sentences, since what it is in each world that makes these sentences true will be different. We have here, if you like, a semantical reflection of the metaphysical distinction between ground and the purely modal notion of supervenience.¹⁷

Philosophers and linguists have in recent times made many applications of the truth-maker approach to topics within formal logic and the semantics of natural language, paying special attention to linguistic or logical phenomena which cannot be explained under the possible worlds approach. The first important application was made by van Fraassen (1969). He in effect suggested that a sentence A should be taken to *entail* the sentence B if:

- (5) every truth-maker for A contained a truth-maker for B , and he showed that this gave rise to the familiar notion of first-degree tautological entailment (Anderson & Belnap 1961). The sentence $R \wedge \neg R$, for example, would not entail W under this conception of entailment since an impossible truth-maker for $R \wedge \neg R$ could not be expected to contain a truth-maker for W .

Another important application is to the notion of containment (Gemes 1994; Yablo 2014; Fine 2016). One might be interested in the idea that the content of one sentence is part of the content of another. Thus the content of ‘it is raining and it is windy’ would contain the content of ‘it is raining’, in this sense, though the content of ‘it is raining’ would not contain the content of ‘it is raining or it is windy’. We might obtain a semantical account of containment by adding to (5) the stipulation that:

- (6) each truth-maker for B is contained in a truth-maker for A .

The containment relation would not then hold between R and $R \vee W$, for example, since a truth-maker for W could not be expected to be contained in a truth-maker for R .

Other applications of truth-maker semantics include scalar implicature, conditionals, imperative and deontic statements, belief revision, partial truth, verisimilitude, confirmation, the is-ought gap. It is not my aim to review these applications here.¹⁸ However, of special note in the present context is the application of truth-maker semantics to the logic of ground.

Truth-maker semantics has been conceived in the image of ground; and this suggests that we might ‘reverse engineer’ and see ground in the image of truth-making. An early attempt of this sort was made in Fine (2012c), who laid out a pure logic of ground, provided it with a truth-maker semantics, and showed that the logic was sound and complete for the semantics. What is distinctive about the *pure* logic is that it deals with statements of ground without regard for the logical complexity of the statements that ground or are grounded. The logic contains an operator $<$ for strict ground, an operator $\leq$ for weak ground and corresponding operators $<$ and $\leq$ for strict and weak partial ground.¹⁹ Among the logical principles are the following subsumption thesis:

$$\frac{A_1, A_2, \dots < B}{A_1, A_2, \dots \leq B}$$

which says that strict ground implies weak ground, and the following transitivity principle:

$$\frac{A < B \quad B < C}{A < C}$$

according to which if A is a strict partial ground of B and B of C then A is a strict partial ground of C .

In the semantics, the statement $A_1, A_2, \dots \leq B$ of weak ground is taken to be true if the fusion of truth-makers for $A_1, A_2, \dots$ is a truth-maker for B and a statement $A_1, A_2, \dots < B$ of strict ground is taken to be true if the corresponding statement $A_1, A_2, \dots \leq B$ of weak ground is true and cannot be reversed in the sense that for no statements $C_1, C_2, \dots$ does $B, C_1, C_2, \dots$ weakly ground one of $A_1, A_2, \dots$.²⁰ There are similar clauses for the statements of partial ground.

Developing a system for the impure logic of ground, in which the logical complexity of component statements is taken into account, raises further problems. In the first place, it will make a difference whether one adopts a worldly or representational conception of ground, whereas this makes no difference to the pure logic of ground when only universal logical principles are in question.²¹ Under the worldly conception, A will not be taken to be a strict ground for $A \wedge A$, since the fact that A is taken to be the same as the fact that $A \wedge A$ even if the way these facts are represented is not the same. On the representational conception, by contrast, A is taken to be a strict ground for $A \wedge A$ because of a difference in representation (or even, in fact, if the distinction between fact and representation is not respected).

It is relatively straightforward to provide a semantics for the impure logic of ground under the worldly conception, and for these purposes, we may adopt a flat conception of truth-makers, since it is merely the ground level-facts which will be relevant to whether one statement grounds another (Correia 2010). It is much harder to provide a semantics for the impure logic of ground under the representational conception since one will then need to adopt something closer to a hierarchical conception of truth-makers, and it is not clear how this might best be

developed. Some semantics along these lines have already been proposed by Correia (2017) and Krämer (2018). But it may be doubted whether their solutions are fully satisfactory; and much more work along these lines needs to be done.

One interesting prospect opened up by the current work on truth-making is the possibility of developing a general account of ground-theoretic content under which various more specific accounts might be subsumed. If we reflect on the previous clauses $T(i)^+ - T(v)^-$ for the semantics of sentential and quantificational logic, we see that there are two basic operations by which semantic contents might be obtained. Two items might be combined into a single item, as when we form a truth-maker for $A \wedge B$ from a truth-maker for A and a truth-maker for B ; and two items might be presented as a choice, as when we take either a truth-maker for A or a truth-maker for B to be a truth-maker for $A \vee B$. Thus there are two basic operations of combination and choice, and semantic contents are obtained by applying these operations to some underlying material.

We might think of a complex that is formed by applying these operations as a menu, since it offers a choice between options, where these options may themselves be combinations of options (as in fish and chips) or a choice of options (as in a menu that offers different seasonal menus). Ground, from this perspective, is essentially a matter of making a selection from a menu. Thus if I am offered a choice $A \vee B$ of A and B , then a selection is either A or B , whereas if I am offered a combination $A \wedge B$ of A and B then a selection is A and B (i.e., A, B).

We might now think of different semantics as the result of adopting different principles governing the behavior of these two operations. Thus if combination is analogous to fusion then we would not expect to distinguish between the content of $(A \wedge B) \wedge C$ and $A \wedge (B \wedge C)$, while if it is analogous to set formation then we would expect to be able to distinguish between their contents. The raw material from which we construct the content can also make a difference. If the raw material is just possible worlds, then combination of two distinct worlds, given their incompatibility, could be taken to result in a crash, and so the semantic content of a statement would essentially be given by a choice of worlds. If the raw material is given by the fundamental facts, then we would obtain a metaphysical semantics from which the underlying structure of the world might be discerned.²² On the other hand, if the raw material consists of the states or facts that lie at the level of the atomic sentences of the language, then we would obtain a semantics of a usual sort in which purely linguistic considerations determine what the truth-makers should be.²³

Notes

- 1 I here adopt an operational rather than a relational conception of ground. The connection is somewhat different under a relational conception of ground (see Introduction).
- 2 This interplay between operators and predicates is very familiar in other areas of philosophy. A case in point concerns the proper expression of modal claims, as considered in Quine (1953). A somewhat different approach to the connection between ground and truth-making is taken in Correia (2014).
- 3 See the chapter “Necessity” [Chapter 10].
- 4 As in Holliday (2015), Humberstone (1981) and Rumfitt (2015) in the first case and in much of the work on situation semantics, as typified by Barwise and Perry (1983), in the second case.
- 5 Some criticisms along these lines have been made by Horwich (2008), Schnieder (2006) and Fine (2012b). Truth-making from a metaphysical perspective is further discussed in the chapter “Truth-Making” [Chapter 28]; and for an opposed point of view, see Asay (2017).
- 6 The possible worlds semantics for counterfactuals was first developed in Lewis (1973) and Stalnaker (1968). A truth-maker semantics for counterfactuals is given in Fine (2012a).
- 7 See the Introduction for the factive/non-factive distinction.

- 8 The issue of developing a semantics for counterfactuals with impossible antecedents—using the possible worlds framework, though, rather than the truthmaker framework—is discussed in Nolan (2013). I should point out that there are other reasons for wanting to admit impossible truth-makers into our semantics. They are necessary, for example, to obtain a satisfactory truth-maker semantics for intuitionistic logic or the logic of containment (Fine [2014], Fine [2016]).
- 9 Under a more formal approach, we might take a state space to be an ordered pair (S, Ξ) , where S (states) is a nonempty set and Ξ is a partial order on S subject to the condition that any subset of S has a least upper bound. I shall eschew technicalities in what follows.
- 10 See the Introduction and the chapter “Meta-Ground” [Chapter 9] for further discussion of zero ground.
- 11 The general issue is discussed in the chapter “Strict Partial Order” [Chapter 17]. Even if the irreflexivity of ground is challenged in other cases, it is hardly likely to be challenged in the present case.
- 12 See the chapter “Logics” [Chapter 14] for further discussion of the distinction between weak and strict ground.
- 13 At least for classical logic. We can get by with a unilateral semantics for intuitionistic logic (Fine [2014]).
- 14 See the chapter “Logics” [Chapter 14] for more on this case.
- 15 A more rigorous deduction of this sort is given in §10 of Fine (2012b).
- 16 The reader might like to consult Stalnaker (1984) for a general introduction to possible worlds semantics. It has become the dominant paradigm in developing a formal semantics for natural language.
- 17 See the chapter “Modality” [Chapter 24] for more on the distinction between ground and supervenience.
- 18 A survey of some of them is to be found in Fine (2017). Related work has been done within the school of inquisitive semantics (Ciardelli, Groenendijk, and Roelofsén [2019]).
- 19 See the chapter “Logics” [Chapter 14] for these distinctions and principles.
- 20 See the chapter “Logics” [Chapter 14] for more on the logic of ground.
- 21 See the chapter “Logics” [Chapter 14] for this distinction.
- 22 As in Sider (2011: 112ff.), although a semantics of this sort need not state necessary and sufficient conditions.
- 23 I am grateful for many valuable comments from the participants at the Hamburg Conference on Ground in 2018, the discussants in the Academia forum on the paper and the students in my MA class at Lugano in 2019; and I am also grateful to Mike Raven for his encouragement.

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SOCIAL ENTITIES

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In recent years, there has been an increased interest in applying the tools and methods of analytic metaphysics to the study of social phenomena. This chapter examines how one such tool—the notion of *metaphysical ground*—may be used to elucidate some central notions, debates, and positions in the philosophy of race and gender, social ontology, and the philosophy of social science. Three main applications are examined: how the notion of social construction may be analyzed in ground-theoretic terms (§1); how debates over the nature of social facts may be recast as grounding debates (§2); and how the doctrine of ontological individualism may be formulated using the notion of ground (§3). The chapter concludes by considering a skeptical challenge concerning the usefulness of the grounding framework for social metaphysics (§4).

1. Social Construction

The notion of *social construction* figures prominently in discussions of race and gender, as well as in discussions of other phenomena such as disability and mental illness. For instance, many feminists have maintained that gender is not a biological or anatomical phenomenon but a social construct.¹ Race theorists argue over whether human races are biologically real or socially constructed.² And some critics of the scientific mainstream have claimed that mental illnesses such as schizophrenia are socially constructed.³ In order to understand these various positions and debates, we need to understand what it is for something to be socially constructed. Yet there is little consensus in the literature on how this notion (or cluster of notions) is to be understood.⁴

Grounding theorists have proposed that the notion of ground may be used to analyze one important notion of social construction (Schaffer 2017; Griffith 2018). The relevant notion is that of *constitutive* social construction, which is to be distinguished from *causal* social construction.⁵ Roughly, something is causally socially constructed just in case it is caused by social factors, whereas something is constitutively socially constructed just in case it is constituted by social factors. To illustrate, consider the fact that males are on average taller than females. Some feminists have argued that this fact is causally socially constructed insofar as it is the product of a long history of differing social norms for males and females when it comes to nutrition and exercise.⁶ However, the fact is not constitutively socially constructed, since differences in height are not constituted by differences in social features but rather differences in physical features.

The basic ground-theoretic proposal is that the notion of constitutive social construction may be analyzed in terms of the notion of ground. The two specific proposals in the literature, due to Jonathan Schaffer and Aaron Griffith, differ over the details. Schaffer suggests that “to be socially constructed is to be grounded in distinctive social patterns,” leaving open what those distinctive social patterns are (2017: 2454–2455). Since Schaffer is working with a notion of entity grounding, his proposal pertains to entities of all sorts including facts, objects, properties, etc. Griffith is specifically concerned with the construction of social identities such as gender, race, class, and disability. He suggests that a person *S* is non-causally socially constructed as a member of human social kind *K* if and only if the fact that *S* is a *K* is at least partly grounded in particular features of social reality (2018: 395). For example, Sally is non-causally socially constructed as a woman if and only if the fact that Sally is a woman is at least partly grounded in particular features of social reality. Griffith adds that the relevant particular features of social reality are “specific, repeated patterns of human interaction” that involve “individual or collective responses to the subject’s body (or perception thereof),” and are “not inevitable, natural, or fixed” (2018: 395).

One central motivation for a ground-theoretic analysis of social construction along these lines is that many social constructionists wish to maintain that socially constructed entities are real, despite being socially constructed (cf. Griffith 2018: §§1–2). But as Ian Hacking notes, “‘Social construct’ and ‘real’ do seem terribly at odds with each other” (1999: 101). The ground-theoretic analysis gives social constructionists a way to resolve this apparent tension. Since the grounded exists, social constructionists may maintain that the socially constructed is ‘real’ in the sense that it exists. At the same time, since the grounded is derivative of more fundamental features of reality, social constructionists can acknowledge that the socially constructed is not part of the fundamental furniture of reality.

Other considerations that favor a ground-theoretic analysis of social construction are of a broader theoretical character. First, the ground-theoretic analysis successfully integrates social construction into a more general metaphysics (Schaffer 2017: §2.3; Griffith 2018: §3). Second, the analysis is able to account for the inferential role of social construction. In particular, it is able to account for the inferences that we draw from something’s being socially constructed to its being nonfundamental and generated by, dependent upon, and explicable on the basis of social factors (Schaffer 2017: §2.3). Third, the analysis provides social constructionists with a rigorous set of conceptual tools that are useful for articulating various further social constructionist claims and commitments (Griffith 2018: §3).⁷

Since the literature on this topic is at a nascent stage, there are many open questions that merit further investigation. I will mention three here. First, how do we distinguish cases of social construction from other cases of grounding in the social? After all, not every case of grounding in the social is a case of social construction, at least as this notion is typically understood. For example, the existence of the singleton of the Supreme Court is grounded in the social, but it is not socially constructed (cf. Schaffer 2017: 2455). The grounding theorist has at least four options for dealing with this issue. Option 1 is to say that what distinguishes cases of social construction is something about the *grounds* (Griffith 2018: 395; cf. Schaffer 2017: §2.2). But it is unclear whether a view of this sort can avoid the problem raised by the singleton Supreme Court example. For suppose it is agreed that being grounded in the existence of social pattern *P* suffices for being socially constructed. The existence of the singleton of *P* is grounded in the existence of *P*, yet it is not socially constructed. Option 2 distinguishes cases of social construction by reference to the *grounded*. For instance, it might be maintained that an object *x* is socially constructed just in case its essence is grounded in the social. Provided that the essence of singleton Supreme Court is not grounded in the social, singleton Supreme Court would

not count as socially constructed on this proposal. Option 3 appeals to the *variety of grounding* that is involved. On this proposal, cases of social construction involve a distinct variety of social grounding, whereas cases like that of singleton Supreme Court do not. Finally, Option 4 distinguishes cases of social construction via the *grounding connection*. The basic idea here is that in the case of singleton Supreme Court, the existence of the Supreme Court grounds the existence of singleton Supreme Court on the basis of set-theoretic principles. In contrast, in cases of social construction, the grounding connection has its basis in social principles, rules, or mechanisms. This idea could be implemented in various ways. For instance, the relevant principles, rules, or mechanisms might figure in laws of metaphysics, essentialist truths, or meta-grounds, which in turn account for the grounding connection.

Another important question is whether a ground-theoretic analysis of social construction is superior to alternative analyses. A salient competitor is Sally Haslanger's definitional analysis, which says that "something is constitutively constructed iff in defining it we must make reference to social factors" (Haslanger 1995: 98). One way to understand this proposal is in terms of the notion of *real definition*. Real definitions are definitions of things themselves, as opposed to definitions of words or concepts. For instance, the number 1 itself might be defined as the successor of 0 (Fine 1994: 14). Using this notion of real definition, we may restate Haslanger's proposal as follows: something is constitutively socially constructed iff its real definition makes reference to social factors.⁸ Such a real definition analysis of social construction shares many of the virtues of a ground-theoretic analysis. For instance, it successfully integrates social construction into the more general metaphysical framework of real definition; and it is able to account for the inferences that we draw from something's being socially constructed to its being explicable on the basis of social factors. One question for future research is whether there are considerations that favor a ground-theoretic analysis over a real definition analysis. Another question is whether the ground-theoretic analysis and the real definition analysis are even genuine competitors. If, for instance, real definition itself is to be analyzed in terms of ground (cf. Rosen 2015), then the analyses may not be genuine competitors after all.⁹

A more general question inspired by the ground-theoretic analysis of social construction is whether other central notions of social theory can also be analyzed in ground-theoretic terms. As an example, I will consider the notion of *intersectionality*, which plays an important role in discussions of categories such as race, gender, sexuality, class, and disability. The basic idea put forward by intersectionality theorists is that the oppression associated with membership in some such category can 'intersect' with the oppression associated with membership in another such category, thereby producing a new and different form of oppression that cannot be understood in simple additive terms. For example, black women may face a form of oppression that cannot be understood by simply analyzing the oppression faced by women and the oppression faced by black people. Sara Bernstein (2020: §3) proposes that intersectionality may be understood in terms of the ontological and explanatory priority of the intersectional category over its constituent categories. For instance, the intersectionality of black womanhood may be understood in terms of the ontological and explanatory priority of the category *black woman* over the categories *black* and *woman*. Since grounds are ontologically prior to the grounded and metaphysically explain the grounded, Bernstein's account of intersectionality may naturally be construed in grounding terms: what it is for a category φ to be intersectional is for φ 's existence to partly ground the existence of its constituent categories. A question for future research is whether, and to what extent, the considerations adduced earlier in support of a ground-theoretic analysis of social construction also favor a ground-theoretic analysis of intersectionality or ground-theoretic analyses of other notions in social theory.

2. Social Facts

A central debate in social ontology concerns the nature of social facts and how these facts are ‘built up’ from more basic facts. This debate may be fruitfully recast in ground-theoretic terms (cf. Epstein 2015: ch. 6, 2016; Schaffer 2019). By way of illustration, let us consider the dispute between Ásta and John Searle over the nature of institutional social facts such as the fact that Trump is president of the United States, that I am a registered voter in Vermont, and that Obama is a U.S. citizen.

On Searle’s account, such social facts are created and maintained through our collective acceptance of constitutive rules of the form *X counts as Y in C*, where ‘X’ picks out a particular object or a class of objects that meet certain specified conditions; ‘Y’ names a status with an associated function; and ‘C’ names a context (Searle 1995: 27–29, 43–46). For example, the fact that Obama is a U.S. citizen is created and maintained through collective acceptance of a constitutive rule that says (roughly), ‘Anyone who was born in the United States or is the child of a U.S. citizen or has been naturalized as a U.S. citizen counts as having the status function of U.S. citizen in the context of the United States.’ On Ásta’s account, such social facts are instead created and maintained by authorities conferring social statuses onto individuals that are taken to have the relevant ‘base property,’ which is the property that the authorities are attempting to track in their conferral (2018: §1.4; cf. Ásta 2013). Thus, the fact that Obama is a U.S. citizen is created and maintained by authorities conferring the status of U.S. citizen onto Obama, who is taken to have the relevant base property of having been born in the United States or being the child of a U.S. citizen or having been naturalized as a U.S. citizen.

Since Ásta and Searle formulate their accounts in terms of their own distinctive notions, it is not easy to pinpoint the main locus of disagreement. By recasting the debate as one over the grounds of institutional social facts, we can clarify what is in dispute. Letting ‘*p*’ denote the fact that *p*, here is a natural way to recast the debate in grounding terms:

On Searle’s account, [Obama is a U.S. citizen] is grounded in the following two facts:

S-GROUND-1 [The American people collectively accept a constitutive rule that says, ‘Anyone who was born in the United States or is the child of a U.S. citizen or has been naturalized as a U.S. citizen counts as having the status function of U.S. citizen in the context of the United States’]; and

S-GROUND-2 [Obama was born in the United States or is the child of a U.S. citizen or has been naturalized as a U.S. citizen].

Whereas on Ásta’s account, [Obama is a U.S. citizen] is grounded in the following two:

A-GROUND-1 [When authorities are conferring the status of U.S. citizen, they are attempting to track the base property of having been born in the United States or being the child of a U.S. citizen or having been naturalized as a U.S. citizen]; and

A-GROUND-2 [The authorities take Obama to have the base property of having been born in the United States or being the child of a U.S. citizen or having been naturalized as a U.S. citizen].

When the debate is recast in this way, it becomes clear that one of the main points of contention is whether Obama’s being a U.S. citizen depends upon his *actually* meeting the conditions laid down for U.S. citizenship, as per S-GROUND-2, or upon his being *taken* to meet these conditions, as per A-GROUND-2. These are two importantly different views of

citizenship, with differing real world implications. To illustrate, suppose that the relevant authorities mistakenly take person S to meet the conditions laid down for U.S. citizenship. On Ásta's account, S is a U.S. citizen, and may therefore be entitled to various benefits and privileges that come with U.S. citizenship. On Searle's account, S is not a U.S. citizen and has no such entitlements.

There are other ways in which one might try to recast the debate between Ásta and Searle in terms of grounding. I will discuss one prominent alternative, which is due to Brian Epstein (2015: ch. 6, 2016). Epstein proposes a grounding-framing-anchoring framework for social ontology, on which social facts are to be understood not only in terms of their grounds but also in terms of associated *frame principles* and *anchors*. On his view, the grounds of a social fact are the “metaphysical reason” that the fact obtains (2015: 76). A frame principle is a general principle that describes how facts of a certain form are grounded in possible worlds that are governed by the principle (2015: 77–80). In the case of U.S. citizenship facts such as [Obama is a U.S. citizen], the relevant frame principle will describe how facts of the form [x is a U.S. citizen] are grounded in worlds that are governed by the principle. Frame principles themselves are “put in place” by a set of facts (2015: 81). Epstein calls this putting-in-place relation ‘anchoring,’ and the facts that put in place a frame principle ‘anchors.’ He maintains that anchoring is distinct from grounding, but notes that the two relations are similar insofar as the anchors of a frame principle are the “metaphysical reason” that the frame principle is the case (2015: 82). On Epstein's framework, the debate between Ásta and Searle may be recast as follows:¹⁰

On Searle's account:

S-GROUND [Obama is a U.S. citizen] is fully grounded in [Obama was born in the United States or is the child of a U.S. citizen or has been naturalized as a U.S. citizen];

S-FRAME The relevant frame principle is: $(\forall x)(x \text{ was born in the United States or is the child of a U.S. citizen or has been naturalized as a U.S. citizen} \rightarrow [x \text{ was born in the United States or is the child of a U.S. citizen or has been naturalized as a U.S. citizen}] \text{ fully grounds } [x \text{ is a U.S. citizen}])$;

S-ANCHOR The relevant frame principle is anchored in [The American people collectively accept a constitutive rule that says, ‘Anyone who was born in the United States or is the child of a U.S. citizen or has been naturalized as a U.S. citizen counts as having the status function of U.S. citizen in the context of the United States’].

Whereas on Ásta's account:

A-GROUND [Obama is a U.S. citizen] is fully grounded in [The authorities take Obama to have the base property of having been born in the United States or being the child of a U.S. citizen or having been naturalized as a U.S. citizen];

A-FRAME The relevant frame principle is: $(\forall x)(\text{The authorities take } x \text{ to have the base property of having been born in the United States or being the child of a U.S. citizen or having been naturalized as a U.S. citizen} \rightarrow [\text{The authorities take } x \text{ to have the base property of having been born in the United States or being the child of a U.S. citizen or having been naturalized as a U.S. citizen}] \text{ fully grounds } [x \text{ is a U.S. citizen}])$;

A-ANCHOR The relevant frame principle is anchored in [When authorities are conferring the status of U.S. citizen, they are attempting to track the base property of having been born in the United States or being the child of a U.S. citizen or having been naturalized as a U.S. citizen].

This alternative ground-theoretic reconstrual of the dispute between Ásta and Searle prompts a number of interesting questions that warrant further consideration. First, do we need to posit a relation of anchoring in addition to the relation of grounding in order to adequately represent such debates in social ontology? Or can grounding alone do the job? (cf. Hawley 2019; Mikola 2019; Schaffer 2019) This question largely turns on whether anchoring can be construed as a *form* of grounding. The following two considerations favor treating anchoring as a form of grounding. First, the two relations play a very similar role. As Epstein himself acknowledges, anchors are the ‘metaphysical reason’ for why the anchored obtains, just as grounds are the ‘metaphysical reason’ for why the grounded obtains. Second, by treating anchoring as a form of grounding, we avoid positing a *sui generis* relation that obtains only in the social domain (Schaffer 2019: §2.3).¹¹

If anchoring is taken to be a form of grounding, then a further choice point is whether to treat anchors as *partial grounds* of the associated social facts or as *meta-grounds* of the associated grounding facts. To illustrate, consider again Searle’s account of [Obama is a U.S. citizen]. On the partial ground view, this fact is partly grounded in the collective acceptance of the relevant constitutive rule (as per my initial proposal). On the meta-ground view, this fact is fully grounded in [Obama was born in the United States or is the child of a U.S. citizen or has been naturalized as a U.S. citizen]. But the fact that [Obama is a U.S. citizen] is grounded in [Obama was born in the United States or is the child of a U.S. citizen or has been naturalized as a U.S. citizen] is itself grounded in the collective acceptance of the relevant constitutive rule.

The following two arguments support the partial ground view. First, the meta-ground view violates grounding necessitation, which is a widely endorsed principle (cf. Schaffer 2019: 757).¹² Grounding necessitation says that the full grounds of a fact necessitate that fact. But on the meta-ground view, the full grounds of [Obama is a U.S. citizen] clearly do not necessitate that fact. Sticking with Searle’s account, there are metaphysically possible worlds in which [Obama was born in the United States or is the child of a U.S. citizen or has been naturalized as a U.S. citizen] obtains, but [Obama is a U.S. citizen] does not obtain. For instance, a world in which the one and only condition laid down for U.S. citizenship is having been born in Washington, DC. The partial ground view does not likewise violate grounding necessitation, as it includes facts concerning the conditions laid down for U.S. citizenship in the full grounds. Second, the meta-ground view is at odds with some of our intuitions about what grounds what. For example, suppose that persons A and B agree that whatever they point to is to be the border between their territories. They then point to a particular river, R. On the meta-ground view, [A and B pointed to R] is the full ground of [R constitutes the border], whereas the fact of A and B’s agreement is the meta-ground of the relevant grounding fact. But intuitively, A and B’s having pointed to R does not by itself metaphysically explain why R constitutes the border between their territories. The partial ground view avoids this clash with intuition by identifying a further ground for [R constitutes the border], namely the fact of A and B’s agreement.

However, the partial ground view faces the following challenge, which is raised by Epstein (2015: 123–124).¹³ For some social kinds, it can be a meaningful and substantive question whether something or someone is a member of the kind even at times and worlds at which the relevant anchors do not obtain. For example, consider the kind *war criminal*. It can be a meaningful and substantive question whether Genghis Khan was a war criminal, even though the relevant anchors—facts concerning certain statutes of the International Criminal Court (ICC), say—did not obtain at the time that Genghis Khan committed his crimes. But arguably, the partial ground view cannot account for this. For if the relevant anchors are partial grounds of facts of the form [x is a war criminal], then the question of whether Genghis Khan was a war

criminal seems trivial: he was not, since the ICC did not even exist at the time that he committed his crimes.

I will mention two possible lines of response to this challenge. The first says that social kinds such as *war criminal* are fundamentally different from social kinds such as *U.S. citizen*. Whereas the latter have associated anchors that partly ground the relevant social facts, the former do not. What appear to be anchors in the former cases are actually the grounds of real definitions. For example, facts concerning the relevant statutes of the ICC do not anchor facts of the form [*x* is a war criminal]. Rather, these facts ground the real definition of the kind *war criminal*. Thus, *what it is* to be a war criminal is to have committed any of the crimes listed in the statutes. In contrast, it is not the case that *what it is* to be a U.S. citizen is to have been born in the United States or be the child of a U.S. citizen or have been naturalized as a U.S. citizen. Rather, *what it is* to be a U.S. citizen is to play a certain social role and have certain rights, duties, and obligations. Since on this proposal the partial ground view only applies to ‘anchored’ social kinds such as *U.S. citizen* and not ‘defined’ social kinds such as *war criminal*, the challenge is averted.

The second response denies that the partial ground theorist is committed to saying that Genghis Khan was not a war criminal because the ICC did not exist at the time that he committed his crimes. The idea here is that Genghis Khan’s being a war criminal *back then* may be partly grounded in the goings-on of *today*. Such cases of facts being grounded in temporally later facts are not unfamiliar. For example, suppose that Sally scores the winning goal in the second quarter of the game. Then [Sally scored the winning goal] is partly grounded in facts that obtained after Sally scored the winning goal, for instance, [Sally’s team won the game]. Of course, the property of being a war criminal is not obviously tied to the goings-on of today in the way that the property of scoring the winning goal is obviously tied to the outcome of the game. But this apparent difference might be explained away by construing the property of being a war criminal in such a way that it is more obviously tied to the goings-on of today. For instance, it might be maintained that strictly speaking, the property in question is that of being a war criminal by the standards of the ICC (cf. Schaffer 2019: 763–764). Since on this proposal Genghis Khan’s being a war criminal may be partly grounded in present-day facts concerning the ICC, the partial ground theorist can allow that it is a meaningful and substantive question whether Genghis Khan was a war criminal.

If the partial ground view is adopted, then a further question is whether to draw some alternative theoretical distinction between what we have been calling ‘anchors’ and ‘grounds.’ Here the grounding theorist may be able to draw on discussions in the metaphysics of causation, where distinctions have been drawn between different kinds of causal conditions. For instance, background causal conditions are sometimes distinguished from foreground causal conditions. Consider, for example, the striking of a match. The striking of the match is the foreground causal condition of the burning of the match, whereas the presence of oxygen is a background causal condition. Similarly, a proponent of the partial ground view could take ‘grounds’ to be foreground grounds and ‘anchors’ to be background grounds.¹⁴

To sum up: there are a number of outstanding questions regarding how to best recast debates over the nature of social facts in ground-theoretic terms. First and foremost, do we need to posit a relation of anchoring in addition to grounding, as Epstein suggests? If not, should anchors be construed as partial grounds of the relevant social facts or as meta-grounds of the relevant grounding facts? And if anchors are construed as partial grounds, then should some alternative theoretical distinction be drawn between ‘anchors’ and ‘grounds’? As yet, there is no consensus on the answers to these questions.

3. Ontological Individualism

The last application of grounding that I will consider pertains to the philosophy of social science. A central and longstanding debate in this area is the debate between methodological individualists and holists.¹⁵ Historically, this debate centered on methodological issues in the social sciences, but it was deeply enmeshed with ontological issues. In more recent discussions, the methodological issues are usually disentangled from the ontological issues. The methodological debate centers on the question of whether explanations in the social sciences should only make reference to individuals and their interactions, or whether these explanations can also legitimately make reference to ‘higher-level’ social entities such as nations, political parties, classes, ethnic groups, and cultures. The explanatory individualist insists on individualistic explanations, whereas the explanatory holist allows for non-individualistic explanations. The ontological debate, on the other hand, centers on the question of whether social facts are fully determined by and dependent on ‘individualistic facts,’ viz. facts about individuals and their interactions. The ontological individualist says yes, whereas the ontological holist says no.

The notion of ground may be used to elucidate the doctrine of ontological individualism (and its negation, ontological holism). In particular, ontological individualism may be construed as a thesis about the individualistic grounds of social facts (Epstein 2015: ch. 8).¹⁶ There are various possibilities for how to formulate this thesis more precisely in ground-theoretic terms (cf. Epstein 2015: 109–110), but here is one natural formulation:

OI-GROUND Necessarily, any social fact f is fully grounded in some individualistic facts.

Since grounding is a relation of both dependence (the grounded depends upon the grounds) and determination (the grounds determine the grounded), OI-GROUND captures the idea that social facts are fully determined by and dependent on individualistic facts.

In contrast, standard formulations of ontological individualism in the contemporary literature do not adequately capture the relevant dependence and determination (cf. Epstein 2014: §1.1). These standard formulations are cast in terms of supervenience. For instance, Philip Pettit says that individualism “insists on the supervenience claim that if we replicate how things are with and between individuals, then we will replicate all the social realities that obtain in their midst: there are no social properties or powers that will be left out” (2003: 191). And Christian List and Kai Spiekermann formulate the relevant doctrine of individualism thus: “The individual-level facts fully determine the social facts; i.e., any possible worlds that are identical with respect to all individual-level facts will necessarily be identical with respect to all social facts” (2013: 632). But as many have noted, the relation of supervenience is too course-grained to adequately capture dependence and determination. Consider, for example, the mathematical Platonist. She would presumably agree that any possible worlds that are identical with respect to all physical facts will necessarily be identical with respect to all mathematical facts. Nevertheless, she would deny that the mathematical facts are determined by, or dependent on, the physical facts. Since a ground-theoretic formulation of ontological individualism does adequately capture the relevant dependence and determination, it is preferable to a supervenience-based formulation.

4. Skeptical Doubts

As we have seen, a number of philosophers have championed applications of grounding to the study of social phenomena. Others, however, have raised skeptical doubts about the usefulness of the grounding framework for social metaphysics (Barnes 2014, 2017; Mikkola 2015, 2017). In

this concluding section, I consider one important skeptical challenge raised by Elizabeth Barnes (2014). The challenge is that the grounding framework cannot adequately represent many of the central debates and positions within social ontology and feminist metaphysics, because these debates are not about what grounds what. As an example, Barnes considers the dispute between Sally Haslanger and Katharine Jenkins about the nature of gender. On Haslanger's view:

S is a woman iff_{df} S is systematically subordinated along some dimension (economic, political, legal, social, etc.), and S is 'marked' as a target for this treatment by observed or imagined bodily features presumed to be evidence of a female's biological role in reproduction.
(Haslanger 2000: 39)

Jenkins (2016) argues that this definition is inadequate because it excludes some *prima facie* women, namely trans women who don't regularly 'pass' as cis women. Her competing proposal is that in addition to Haslanger's category of *gender as class*, we need to countenance another category of *gender as identity*. The category *gender*, on her view, is comprised of these two subcategories. This proposal does not exclude any trans women from the category *woman*, since all trans women will belong to this category on account of having a female gender identity. As Barnes understands this dispute, Haslanger and Jenkins disagree over "what gender is," despite agreeing on what it is that grounds the existence of genders, namely "a complex network of interpersonal social hierarchies" (Barnes 2014: 344). Thus, the dispute cannot be adequately recast in terms of grounding.

One possible line of response for the grounding theorist is to insist that Haslanger and Jenkins do disagree about what grounds what. In particular, the grounding theorist may maintain, pace Barnes, that Haslanger and Jenkins disagree over the relevant *grounds*: on Jenkins's view but not on Haslanger's, the existence of genders is partly grounded in certain facts about the inner experiences of individuals as they navigate the social world (cf. Barnes 2014: 344 n. 10; Schaffer 2017: 2460–2461; Griffith 2018: 400). Alternatively, the grounding theorist might maintain that Haslanger and Jenkins agree about certain grounds but disagree about what these grounds give rise to (Griffith 2018: 400). That is, they disagree about what gets *grounded* by the grounds in question: on Haslanger's view, it is the existence of *gender as class*, whereas on Jenkins's view, it is the existence of both *gender as class* and *gender as identity*.

But even if one or both of these replies can successfully deal with the case at hand, it is unclear whether replies of this sort will work in the case of other debates in social ontology that center on questions of the form 'What is *x*?' For instance, consider the debate over what a social group is. Here are two possible views:

SET A social group is the set of its members.

AGG A social group is the aggregate of its members.

SET and AGG are plausibly taken to agree on what grounds the existence of a social group—namely, the existence of its members. Moreover, they are plausibly taken to agree on what these grounds give rise to—namely, the existence of the group. Nevertheless, they disagree about what a social group *is*. It is open to the grounding theorist to insist that SET and AGG disagree over some *other* grounding claims concerning social groups. For instance, they may be taken to disagree about what grounds the fact that the existence of a social group is grounded in the existence of its members. That being said, such a ground-theoretic reconstrual of the debate does not seem to perspicuously represent what's at issue.

There is, however, another line of response available to the grounding theorist. Namely, to concede that *some* debates in social ontology and feminist metaphysics are not about what

grounds what. What follows from this is that grounding may not be the only metaphysical tool that we need when studying the social world. We may also need other tools—for instance, essence, real definition, or metaphysical laws—to adequately represent certain positions and debates (cf. Passinsky forthcoming: §3). This conclusion, though, does not warrant a general skepticism about the usefulness of grounding for social metaphysics. It only warrants a modesty concerning the limits and reach of the grounding framework.¹⁷

Related Topics

Schnieder, B. “Dependence” [Chapter 7]
 Glazier, M. “Ground and Explanation” [Chapter 8]
 Litland, J. “Meta-Ground” [Chapter 9]
 Skiles, A. “Necessity” [Chapter 10]
 Koslicki, K. “Skeptical Doubts” [Chapter 11]
 deRosset, L. “Anti-Skeptical Rejoinders” [Chapter 12]
 Richardson, K. “Varieties” [Chapter 13]
 Zylstra, J. “Essence” [Chapter 22]
 Kovacs, D. “Modality” [Chapter 24]
 Saenz, N. “Ontology” [Chapter 25]
 Wilsch, T. “Laws of Metaphysics” [Chapter 30]

Notes

- 1 See, for instance, Haslanger (2000, 2003) and Ásta (2013, 2018: chs. 1–4).
- 2 See, for instance, Haslanger (2012) and Spencer (2015).
- 3 See Hacking (1999: ch. 4).
- 4 See Haslanger (1995), Hacking (1999: ch. 1), Boghossian (2001), Mallon (2007), and Ásta (2013, 2018: ch. 2) for some analyses of social construction that do not appeal to the notion of metaphysical ground.
- 5 On the distinction between causal and constitutive social construction, see Haslanger (1995: §2, 2003: 316–318).
- 6 See Haslanger (2003: 317).
- 7 For instance, Griffith suggests that the tools of the grounding framework are useful for articulating gender anti-essentialism and the context-sensitivity of gender and race (2018: §3).
- 8 I believe that this restatement of Haslanger’s proposal is reasonably faithful to the original. See, in particular, Haslanger (1995: 120 n. 12).
- 9 See Griffith (2018: 404–406) for another argument that the ground-theoretic analysis and Haslanger’s definitional analysis are not genuine competitors.
- 10 I do not claim that Epstein himself would recast the debate in exactly this way. In particular, my formulation of frame principles differs somewhat from Epstein’s own formulation (cf. 2015: 77–80). Moreover, Epstein might wish to formulate S-ANCHOR in terms of collective acceptance of a frame principle rather than collective acceptance of a constitutive rule (cf. 2015: 75–77, 82–83).
- 11 The proponent of anchoring could maintain that anchoring also obtains in nonsocial domains. But such an application of anchoring to other domains has not yet been attempted.
- 12 Though see Skiles (2015) for some putative counterexamples to grounding necessitation.
- 13 See Epstein (2015: ch. 9) for additional arguments against the partial ground view (which he calls ‘conjunctivism’). See Hawley (2019), Mikkola (2019), and Schaffer (2019) for replies to Epstein’s arguments.
- 14 Alternatively, the partial ground theorist could take ‘anchors’ to be structuring grounds and ‘grounds’ to be triggering grounds, on analogy with structuring and triggering causes (Schaffer 2019: 752).
- 15 For a nice discussion of the history of this debate, see Epstein (2015: chs. 1–2).

- 16 Epstein also proposes that the notion of anchoring may be used to articulate a new version of individualism concerning the anchors of frame principles associated with social facts. See Epstein (2014, 2015: ch. 8).
- 17 Many thanks to Mike Raven, Jonathan Schaffer, and the participants of the Hamburg Workshop on Metaphysical Grounding for helpful comments and discussion.

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THE ESSENTIAL GLOSSARY OF GROUND

*Compiled by Kit Fine
with assistance from Michael J. Raven*

ground—a foodstuff, considered by some to be the most basic form of nutrition and by others to be a deadly poison.

groundhog—a creature that inhabits intricately constructed burrows and survives largely on a diet of ground.

groundling—a fledgling groundhog who consumes large quantities of ground in the hope of one day becoming a fully mature groundhog.

groundfather—a term of affection used by groundhogs for their founder, Bernard Bolzano.

grund—an appreciative sound emitted by German groundhogs whilst gorging on their favorite food.

ground zero—a mythical land believed by some groundhogs to provide an endless supply of ground from nothing at all.

go to ground—a characteristic maneuver of groundhogs when in a sticky situation and not knowing what else to do.

breeding ground—a department or institution, to be found throughout Europe and the USA, especially devoted to the nurture of groundlings.

ground buster—a sinister and menacing creature who makes it her mission to destroy all groundhogs and to deny them the ground that provides them with their means of sustenance.

ground flaw—(a) the bottom rung of ground, thought by some to support all other forms of ground; (b) a character trait of groundhogs which leads, in the opinion of ground busters, to their unhealthy addiction to ground.

ground breaking—the characteristic destructive activity of ground busters when faced with a groundhog.

ground down—the feeling of surfeit and desuetude that sets in after excessive consumption of ground, leading the victim to be especially vulnerable to attack from ground busters.

ground up—the state of a groundhog who has been devastated by ground busters; the resulting ground meal is hungrily devoured by the ground busters until no trace of the original groundhog remains.

ground cover—a defensive posture adopted by groundhogs when under attack from ground busters.

middle ground—a fertile area lying between the territories inhabited by groundhogs and ground busters to which neither side is willing to go.

high ground—the territory occupied by groundhogs when engaged in the time-honored ritual of hurling insults at ground busters.

low ground—the territory in which groundhogs take cover when subject to counterattack from ground busters and in which self-congratulatory slogans are hurled from one side to the other, as in ‘One for all, all for one!’ or ‘Unity in diversity!’

groundless—the state of a groundhog who has been deprived of its daily diet of ground and is thereby unable to function.

ground-swell—an unduly proud groundhog who has become bloated on an excessive diet of ground and is unwilling to acknowledge any other form of nutrition.

groundwater—a miraculous substance which, according to lore, renders ground digestible without itself being any part of what is digested.

groundhog day—a meeting of groundhogs, devoted to the ritualistic worship of ground, which endlessly repeats itself with subtle but barely perceptible variation from one day to the next.

grounded—the unfortunate state a groundhog may find itself in when prevented by ground busters from consuming ground or engaging in any other ground-related activities.

ground rules—a defiant slogan chanted by jubilant groundhogs when ungrounded, as in ‘ground rules, OK?’

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